

Supplementary material for *Insertions, deletions, and exchangeable couplings: a Dirichlet process over TKF Potts models*

Annabel Large and Ian Holmes

Department of Bioengineering, University of California, Berkeley
{annabel_large, ihh}@berkeley.edu

Abstract

This supplement collects the long technical derivations referenced by the main paper. It is organised into five appendices that mirror the conceptual ladder of the main text. Appendix A develops the finite-state CTMC and the linear birth–death–immigration process, derives the closed-form bridge-expectation and Fisher-score sufficient statistics for TKF91 and TKF92, resolves the L’Hôpital singularity at $\lambda = \mu$, and discusses the relationship between TKF92 and the latent-boundary-free General Geometric Indel (GGI) model. Appendix B collects the closed-form substitution M-steps for TKF91 / TKF92 / MIXDOM in their many GTR specialisations, the stochastic-variational Baum–Welch loop with its convergence theorem and linearised analysis at stationarity, Maraschino (the TKF92 cherry-distilled generalisation of CherryML) and its tree-level inference algorithms (FSA, BeamASR, VarAnc, svi-VarAnc), the mixture-of-trees variational ancestral presence/absence inference, and the structural bias of the BP cumulant under a column-factorised variational field. Appendix C develops the recursive TKF family: MIXDOM, the hierarchical-mixture-of-domains generalisation of TKF92, with its exact closed-form M-step via six-step chain restoration through a fully exploded null-state model; the order-1 Maraschino adjacency distillation; algebraic distillation of MIXDOM; the MIXDOM-specific SVI-BW convergence considerations; the tree-level VEM and ancestral-reconstruction algorithms; the generalised phylo-HMM; the labeled-MIXDOM Singlet and WFST; and the recursive-grammar-elaboration rules together with worked recursive examples (L-TKF, TKFST, TKFStack, TKF-Genome). Appendix D develops the TKF-DP generative model, the class-level path-measure variational likelihood with pairwise bridge expectations, the time-indexed gravestone-augmented pair SCFG, the SVI inference loop, and the pairwise alignment postprocessing landscape. Appendix E develops the infinite Pair HMM as the principled fixed point and the Gibbs+MH+replica-exchange MCMC sampler that draws from it.

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Common notation

Several important symbols and their cross-appendix conventions are collected below.

Symbol	Meaning	First defined
λ, μ	TKF insertion / deletion rates (per link)	§A
r	TKF92 fragment-extension probability	§A
$\mathcal{A}$	residue alphabet ($A = \mathcal{A} = 20$ for proteins)	§A
π, Q, S	GTR stationary, asymmetric, exchangeability matrices	§A
R	substitution rate matrix (generator)	§A
n, f, c	domain, fragment, and substitution-class indices (MixDOM)	§C
η_s	per-site gamma rate multiplier	§D
$z_s, c_s, h_{cc'}$	key, site-class, Potts-atom indices (TKF-DP)	§D
$\alpha_z, \alpha_c, \alpha_H$	key/class/atom Dirichlet-process concentrations	§D
Q, H	generator and Potts coupling tensor	§D
τ	TKF91 Pair HMM transition matrix	§A
v	exploded 7×7 MixDOM top-level transitions with null states	§C
$\mathcal{T}$	effective 5×5 MixDOM top-level matrix (nulls summed out)	§C
χ	final nested MixDOM Pair HMM transition matrix	§C
N_a, N_g	alive / gravestone counts (TKF-DP)	§D
F_0, F_1, F_2	TKF92 Pair HMM 0/1/2-cell marginal partition functions	§E
π_M	CRP partition of Match cells (infinite Pair HMM)	§E

A BDI and TKF foundations

This appendix builds the foundations on which the rest of the supplement rests. We develop the finite-state continuous-time Markov chain (CTMC) sufficient statistics; the linear birth–death–immigration (BDI) process and its endpoint-conditioned sufficient statistics via the Fisher score identity; the TKF91 model as a factorisation of these; its finite-state machine representations; the TKF92 generalisation; the explicit 6×6 WFST derivation by Pair-HMM-over-Singlet division; the L’Hôpital limits at $\lambda=\mu$ that close the 0/0 singularity in the score identity; and a discussion of how the latent-boundary TKF92 model relates to the latent-boundary-free General Geometric Indel (GGI) model via moment matching.

A.1 The TKF91 Model

A.1.1 Finite-State Continuous-Time Markov Chain (CTMC)

A finite-state continuous-time Markov chain (CTMC) on alphabet $\mathcal{A}$ has rate matrix (generator) R with off-diagonal entries $R_{ij} \geq 0$ (instantaneous rate of substitution from i to j) and diagonal entries $R_{ii} = -\sum_{j \neq i} R_{ij}$. The stationary distribution π satisfies $\pi R = 0$ and $\sum_i \pi_i = 1$.

We restrict our analysis to time-reversible CTMCs, which satisfy detailed balance $\pi_i R_{ij} = \pi_j R_{ji}$ for all i, j . The symmetrized rate matrix $S_{ij} = R_{ij} \sqrt{\pi_i / \pi_j}$ is real symmetric, so it has a spectral decomposition $S = \sum_k \xi^{(k)} v^{(k)} v^{(k)\top}$ with real eigenvalues $\xi^{(k)} \leq 0$ and orthonormal eigenvectors $v^{(k)}$. Conjugating back, the transition matrix is

$$P(X(T) = j \mid X(0) = i, R) = (e^{RT})_{ij} = \sum_k e^{\xi^{(k)} T} \sqrt{\frac{\pi_j}{\pi_i}} v_i^{(k)} v_j^{(k)}$$

A.1.2 Sufficient Statistics for Finite-State CTMCs

The sufficient statistics for a continuously-observed path $\mathbb{X} = \{X(t)\}_{0 \leq t \leq T}$ in the finite-state CTMC are the dwell times W_i and transition counts U_{ij} . The complete-data log-likelihood is

$$\log P(\mathbb{X} | X(0), R) = \sum_{i \neq j} U_{ij} \log R_{ij} + \sum_i W_i R_{ii} = \sum_{i \neq j} U_{ij} \log R_{ij} - \sum_i W_i \sum_{j \neq i} R_{ij}. \quad (\text{A.1})$$

This is an exponential-family form with natural parameters $\log R_{ij}$ and $-R_{ii}$, and sufficient statistics U_{ij} and W_i .

The *a posteriori* expectations of these sufficient statistics for an endpoint-conditioned path in the finite-state CTMC are (25, 26, 31, 58)

$$\mathcal{C}_i^W(a, b, T) \equiv \mathbb{E}[W_i | a, b] = \frac{1}{M_{ab}(T)} I_{ii}^{ab}(T) \quad (\text{A.2})$$

$$\mathcal{C}_{ij}^U(a, b, T) \equiv \mathbb{E}[U_{ij} | a, b] = \frac{R_{ij}}{M_{ab}(T)} I_{ij}^{ab}(T) \quad (\text{A.3})$$

where $M(T) = e^{RT}$ is the matrix exponential and

$$I_{ij}^{ab}(T) = \int_0^T M_{ai}(t) M_{jb}(T-t) dt. \quad (\text{A.4})$$

Writing $M_{ij}(t)$ in the eigenbasis of the symmetrized rate matrix $S_{ij} = R_{ij} \sqrt{\pi_i/\pi_j}$ (with eigenvalues $\xi^{(k)}$ and orthonormal eigenvectors $v^{(k)}$):

$$I_{ij}^{ab}(T) = \sqrt{\frac{\pi_i \pi_b}{\pi_a \pi_j}} \sum_k v_a^{(k)} v_i^{(k)} \sum_l v_j^{(l)} v_b^{(l)} J^{kl}(T) \quad (\text{A.5})$$

where $J^{kl}(T) = T e^{\xi^{(k)} T}$ if $\xi^{(k)} = \xi^{(l)}$, and $(e^{\xi^{(k)} T} - e^{\xi^{(l)} T})/(\xi^{(k)} - \xi^{(l)})$ otherwise.

Using the GTR parameterization $R_{ij} = Q_{ij} \pi_j$, where Q is a symmetric exchangeability matrix (so $S_{ij} = Q_{ij} \sqrt{\pi_i \pi_j}$), and including the prior $P(X(0) = i) = \pi_i$, the complete data log-likelihood is

$$\log P(\mathbb{X}, X(0) | Q, \pi) = \sum_i \left(V_i + \sum_{j \neq i} U_{ji} \right) \log \pi_i + \sum_{j > i} (U_{ij} + U_{ji}) \log Q_{ij} - \sum_{j > i} Q_{ij} (W_i \pi_j + W_j \pi_i)$$

where V_i is the indicator/count for the initial state $X(0) = i$.

A.1.3 Linear Birth-Death-Immigration Process (BDI)

The linear birth-death-immigration (BDI) process (36) has per-capita birth rate λ , per-capita death rate μ , and constant immigration rate ν , so that the total event rates from state k are $\lambda_k = k\lambda + \nu$ and $\mu_k = k\mu$.

The transition distribution for this process is well-studied in the general case. We confine our analysis to the specific case $\nu = \lambda, \mu > \lambda$ where the immigration rate is equal to the birth rate and the process is positive recurrent and has an equilibrium distribution, a zero-based geometric with parameter λ/μ . This is the regime relevant to the TKF91 and related models. However, many of the formulas we derive will hold more generally, and the methods we use can be applied to other parameter regimes as well.

Let Y_f denote the number of founder deaths and Y_o the number of youngest orphans (i.e. the number of times a lineage goes extinct while leaving behind at least one surviving descendant). Then the finite-time transition probability is obtained by summing the joint probability

$$P(X(T) = j, Y_f = m, Y_o = n \mid X(0) = i) = \binom{i}{m} \binom{m}{n} \binom{j}{i-m+n} \times \alpha^{i-m} (1-\alpha)^m \beta^{j-i+m-n} (1-\beta)^{i+1-m+n} \gamma^n (1-\gamma)^{m-n}$$

over all m, n consistent with i and j ($0 \leq m \leq i$ and $0 \leq n \leq \min(m, j-i+m)$), where

$$\alpha(\lambda, \mu, T) = \exp(-\mu T) \quad (\text{A.6})$$

$$\beta(\lambda, \mu, T) = \frac{\lambda \exp(-\lambda T) - \lambda \alpha}{\mu \exp(-\lambda T) - \lambda \alpha} \quad (\text{A.7})$$

$$\gamma(\lambda, \mu, T) = 1 - \frac{\mu \beta}{\lambda(1-\alpha)}. \quad (\text{A.8})$$

$$\gamma(\lambda, \mu, T) = 1 - \frac{\mu \beta}{\lambda(1-\alpha)}. \quad (\text{A.9})$$

A.1.4 Sufficient Statistics for Linear BDI Process

The sufficient statistics for an augmented continuously-observed path $\{X(t)\}_{0 \leq t \leq T}$ in the linear BDI process wherein immigration events are labeled as such (i.e. we observe the full path and know which births are due to immigration and which are due to reproduction) are the individual-birth count B_{ind} , the immigration count B_{imm} (with $B = B_{\text{ind}} + B_{\text{imm}}$ the total), the death count D , and the time-integrated population size $S = \int_0^T X(t) dt$. The complete-data log-likelihood is

$$\log P(\mathbb{X} \mid X(0), \lambda, \mu, \nu) = B_{\text{ind}} \log \lambda + B_{\text{imm}} \log \nu + D \log \mu - (\lambda + \mu) S - \nu T + c \quad (\text{A.10})$$

where c does not depend on (λ, μ, ν) . The natural parameters are $(\log \lambda, \log \mu, \log \nu, -(\lambda + \mu))$ and the sufficient statistics $(B_{\text{ind}}, D, B_{\text{imm}}, S)$. This is a curved exponential-family form because the natural parameters are not independent.

Score function identity. Let $P_{ij}(T; \theta)$ denote the observed likelihood, i.e. the transition probability $P(X(T) = j \mid X(0) = i; \theta)$, with $\theta = (\lambda, \mu, \nu)$. The *score function identity* states that the gradient of the observed log-likelihood equals the posterior expectation of the complete-data score:

$$\nabla_{\theta} \log P_{ij}(T; \theta) = \mathbb{E}[\nabla_{\theta} \log P(\mathbb{X} \mid X(0), \theta) \mid X(0) = i, X(T) = j; \theta]. \quad (\text{A.11})$$

No approximation is involved; (A.11) holds for any parametric family and any choice of complete/incomplete data.

Specializing to $\nu = \lambda$ and writing $B = B_{\text{ind}} + B_{\text{imm}}$ for the total birth count, the complete-data log-likelihood (A.10) becomes $\log P = B \log \lambda + D \log \mu - (\lambda + \mu) S - \lambda T + c$ with two free parameters (λ, μ) . Differentiating and applying (A.11):

$$\begin{aligned} \frac{\partial}{\partial \lambda} \log P_{ij}(T; \theta) &= \frac{\mathbb{E}[B \mid i, j, T]}{\lambda} - \mathbb{E}[S \mid i, j, T] - T \\ \frac{\partial}{\partial \mu} \log P_{ij}(T; \theta) &= \frac{\mathbb{E}[D \mid i, j, T]}{\mu} - \mathbb{E}[S \mid i, j, T] \end{aligned}$$

Each equation involves two unknowns because the natural parameters $(\log \lambda, \log \mu, \lambda + \mu)$ are not independent. The conservation law $i + B - D = j$ provides the missing equation:

$$\mathbb{E}[B \mid i, j] - \mathbb{E}[D \mid i, j] = j - i \quad (\text{A.12})$$

Solving this system (using $\lambda \neq \mu$):

$$\begin{aligned}\mathbb{E}[S \mid i, j] &= \frac{j - i + \mu \frac{\partial}{\partial \mu} \log P_{ij} - \lambda \frac{\partial}{\partial \lambda} \log P_{ij} - \lambda T}{\lambda - \mu} \\ \mathbb{E}[B \mid i, j] &= \lambda \frac{\partial}{\partial \lambda} \log P_{ij} + \lambda \mathbb{E}[S \mid i, j] + \lambda T \\ \mathbb{E}[D \mid i, j] &= \mu \frac{\partial}{\partial \mu} \log P_{ij} + \mu \mathbb{E}[S \mid i, j]\end{aligned}$$

Inclusion of an initial distribution $P(X(0) = L) = \kappa^L(1 - \kappa)$ brings the complete-data log-likelihood into the form

$$\log P(\mathbb{X}, X(0) \mid \lambda, \mu) = B \log \lambda + D \log \mu + L \log \kappa + M \log(1 - \kappa) - (\lambda + \mu) S - \lambda T + c$$

where $M = 1$ represents the number of BDI trajectories observed, introduced for later generalization.

A.1.5 The TKF91 Model: Linear BDI + Finite-State CTMC

Thorne, Kishino, and Felsenstein’s 1991 model (TKF91) was introduced to describe the molecular evolution of a biological sequence (DNA, RNA, or protein) subject to insertions, deletions, and point substitutions (62).

The parameters $\theta = (\lambda, \mu, Q, \pi)$ consist of the insertion rate λ , deletion rate μ , exchangeability matrix Q , and equilibrium distribution π .

The BDI part of the model, known as the *links model*, describes the evolution of an “immortal link” followed by zero or more “mortal links”. Each link evolves according to an independent linear BDI process; births and immigrations correspond to insertions, which generate new mortal links, and deaths correspond to deletions.

In the TKF91 model, each mortal link is associated with an observed character that evolves according to an independent finite-state CTMC with rate matrix Q and stationary distribution π . Inserted characters are drawn from the equilibrium distribution π .

In later TKF91-derived models, a more elaborate structure may be associated with each link; for example, in the TKF92 model, a fragment of characters of geometrically-distributed length (63).

To distinguish between these models we introduce the notation $\text{Links}(M; \lambda, \mu)$ for the links model where an independent stationary stochastic process $x(t) \sim M$ is associated with every mortal link. This defines a stationary stochastic process over sequences $\mathcal{S}(t) \sim \text{Links}(M; \lambda, \mu)$ where $\mathcal{S}$ is a sequence over the alphabet defined by the state space of M .

Let $\text{Subst}(Q, \pi)$ denote the time-reversible continuous-time Markov chain with stationary distribution π and generator $R = Q \text{diag}(\pi)$, where Q is the exchangeability matrix. Then the TKF91 model is just the links model with a point substitution process transpiring at each mortal link

$$\mathcal{S}(T) \sim \text{Links}(\text{Subst}(Q, \pi); \lambda, \mu)$$

A.1.6 Finite State Machines of the TKF91 Model

There are three distinct state machines associated with nodes and branches in a tree of TKF-related sequences.

TKF91 Stationary HMM The stationary distribution of the TKF91 model can be represented as a Hidden Markov Model (HMM) that emits $n \sim \text{Geometric}(\kappa)$ characters drawn i.i.d. from π .

TKF91 Transducer The conditional distribution of a descendant sequence $\mathcal{S}(T)$ given ancestor sequence $\mathcal{S}(0)$ can be represented as a Weighted Finite-State Transducer (WFST) that consumes an ancestor sequence and emits a descendant sequence, with transition weights determined by the TKF91 parameters

$$\mathbb{T}(\lambda, \mu, T) = \left(\begin{array}{c|ccccc} & \text{S} & \text{M} & \text{I} & \text{D} & \text{E} \\ \hline \text{S} & 0 & (1-\beta)\alpha & \beta & (1-\beta)(1-\alpha) & 1-\beta \\ \text{M} & 0 & (1-\beta)\alpha & \beta & (1-\beta)(1-\alpha) & 1-\beta \\ \text{I} & 0 & (1-\beta)\alpha & \beta & (1-\beta)(1-\alpha) & 1-\beta \\ \text{D} & 0 & (1-\gamma)\alpha & \gamma & (1-\gamma)(1-\alpha) & 1-\gamma \\ \text{E} & 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

with **M** aligning pairs of characters (i, j) with probability $\exp(RT)_{ij}$, **I** emitting unaligned descendant characters (ϵ, i) with $i \sim \pi$, and **D** consuming unaligned ancestral characters (i, ϵ) without any additional weight penalty.

This transducer can be interpreted as a decomposition of each mortal-link BDI trajectory into founder survival/death, possible survival of the youngest orphan (which, by insertion order, is the leftmost surviving descendant when the founder dies), and a geometric tail of further offspring. This yields the parameters α, β, γ appearing in the WFST transition matrix and determines which transition counts correspond to birth and death events.

TKF91 Pair HMM The joint distribution of ancestor and descendant is given by the transducer composition product of the stationary HMM and the transition WFST. This product can itself be represented as a Pair HMM with states $\{\text{S}, \text{M}, \text{I}, \text{D}, \text{E}\}$ that models the joint distribution, as opposed to the conditional distribution represented by the WFST. The transition matrix τ of the Pair HMM is similar to the transition matrix $\mathbb{T}$ of the transducer, with the differences that incoming transitions to **M** and **D** states acquire a factor of κ for the extension of the ancestral sequence and transitions to **E** states acquire a factor of $1 - \kappa$ for the termination of the ancestral sequence.

For concreteness, because we will refer to it frequently, the transition matrix of the TKF91 Pair HMM is

$$\tau(\lambda, \mu, T) = \left(\begin{array}{c|ccccc} & \text{S} & \text{M} & \text{I} & \text{D} & \text{E} \\ \hline \text{S} & 0 & (1-\beta)\alpha\kappa & \beta & (1-\beta)(1-\alpha)\kappa & (1-\beta)(1-\kappa) \\ \text{M} & 0 & (1-\beta)\alpha\kappa & \beta & (1-\beta)(1-\alpha)\kappa & (1-\beta)(1-\kappa) \\ \text{I} & 0 & (1-\beta)\alpha\kappa & \beta & (1-\beta)(1-\alpha)\kappa & (1-\beta)(1-\kappa) \\ \text{D} & 0 & (1-\gamma)\alpha\kappa & \gamma & (1-\gamma)(1-\alpha)\kappa & (1-\gamma)(1-\kappa) \\ \text{E} & 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

The emission weights of the Pair HMM are similarly related to the input/output emission weights of the WFST, with **M**- and **I**-state emissions acquiring a factor of π_i for the ancestral character i .

For the indel score identities associated with birth, death, and exposure statistics, it is convenient to work with the conditionally normalized TKF91 WFST representing $P(y|x, \theta)$, since its transition weights correspond directly to the finite-time BDI factors α, β, γ . However, the M-step for (λ, μ) under the stationary TKF91 model must also include the prior contribution of the ancestral sequence length, which belongs to the joint distribution $P(x, y|\theta) = P(x|\theta)P(y|x, \theta)$. Accordingly, we use the conditional WFST to obtain posterior expectations of B, D, S and then augment the M-step objective with the joint-model terms involving L and M .

Appendix A.5 considers the limits and implications of these formulas in the biologically-relevant regime of long sequences, $\lambda \rightarrow \mu$.

A.1.7 Sufficient Statistics for the TKF91 Model

For observed ancestral and descendant sequences x, y under TKF91 with parameters $\theta = (\lambda, \mu, Q, \pi)$, and a particular hidden Pair HMM/WFST path z , the (λ, μ) -dependent part of the complete-data log-likelihood is linear in the transition and emission counts,

$$\log P(y, z \mid x, \theta) = \sum_{i,j} n_{ij}(z) \log \mathbb{T}_{ij}(\lambda, \mu, t) + \sum_b e_{(\epsilon,b)}^I(z) \log \pi_b + \sum_{a,b} e_{(a,b)}^M(z) \log \exp(Rt)_{ab}.$$

Let

$$\begin{aligned} \hat{n}_{ij} &= \mathbb{E}[n_{ij} \mid x, y; \theta] \\ \hat{e}_{(\epsilon,b)}^I &= \mathbb{E}[e_{(\epsilon,b)}^I \mid x, y; \theta] \\ \hat{e}_{(a,b)}^M &= \mathbb{E}[e_{(a,b)}^M \mid x, y; \theta] \end{aligned}$$

denote the posterior expectations of these counts returned by the Forward-Backward algorithm. By the score function identity,

$$\begin{aligned} \lambda \frac{\partial}{\partial \lambda} \log P(y \mid x, \theta) &= \sum_{i,j} \hat{n}_{ij} \mathcal{C}_{ij}^\lambda(\lambda, \mu, t) \\ \mu \frac{\partial}{\partial \mu} \log P(y \mid x, \theta) &= \sum_{i,j} \hat{n}_{ij} \mathcal{C}_{ij}^\mu(\lambda, \mu, t) \end{aligned}$$

where $\mathcal{C}_{ij}^\xi(\lambda, \mu, t) = \xi \frac{\partial}{\partial \xi} \log \mathbb{T}_{ij}(\lambda, \mu, t)$ for $\xi \in \{\lambda, \mu\}$. (See Sections A.5.2 and A.5.3 for explicit formulae.) The conservation law is likewise linear in the same counts:

$$\mathbb{E}[B \mid x, y] - \mathbb{E}[D \mid x, y] = \hat{n}_{SI} + \hat{n}_{MI} + \hat{n}_{II} - \hat{n}_{SD} - \hat{n}_{MD} - \hat{n}_{DD} - \hat{n}_{ID} \quad (\text{A.13})$$

$$= \sum_{i,j} \hat{n}_{ij} \mathcal{C}_{ij}^{\text{cons}} \quad (\text{A.14})$$

where $\mathcal{C}_{ij}^{\text{cons}} = \delta(j = I) - \delta(j = D)$ counts whether the WFST transition $i \rightarrow j$ corresponds to a birth, death, or neither. Substituting into the BDI score identities gives

$$\mathbb{E}[S \mid x, y] = \frac{\sum_{i,j} \hat{n}_{ij} (\mathcal{C}_{ij}^{\text{cons}} - \mathcal{C}_{ij}^\lambda + \mathcal{C}_{ij}^\mu) - \lambda T}{\lambda - \mu} \quad (\text{A.15})$$

$$\mathbb{E}[B \mid x, y] = \lambda \sum_{i,j} \hat{n}_{ij} \mathcal{C}_{ij}^\lambda(\lambda, \mu, t) + \lambda \mathbb{E}[S \mid x, y] + \lambda T \quad (\text{A.16})$$

$$\mathbb{E}[D \mid x, y] = \mu \sum_{i,j} \hat{n}_{ij} \mathcal{C}_{ij}^\mu(\lambda, \mu, t) + \mu \mathbb{E}[S \mid x, y] \quad (\text{A.17})$$

Thus the posterior expectations of the BDI sufficient statistics $\mathcal{C}^B(\mathbf{n}, T) = \mathbb{E}[B \mid x, y]$, $\mathcal{C}^D(\mathbf{n}, T) = \mathbb{E}[D \mid x, y]$, and $\mathcal{C}^S(\mathbf{n}, T) = \mathbb{E}[S \mid x, y]$ are affine functions of the Forward-Backward expected transition counts $\hat{n}_{ij}$.

The TKF91 CTMC sufficient statistics take the form

$$\begin{aligned} \mathbb{E}[W_i \mid x, y] &= \sum_{a,b} \hat{e}_{(a,b)}^M \mathcal{C}_i^W(a, b, T) + \mathcal{G}_i^W(x, y) \\ \mathbb{E}[U_{ij} \mid x, y] &= \sum_{a,b} \hat{e}_{(a,b)}^M \mathcal{C}_{ij}^U(a, b, T) + \mathcal{G}_{ij}^U(x, y) \end{aligned}$$

where the first term is the contribution from observed endpoint-matched lineages, obtained from CTMC bridge expectations conditional on $a \rightarrow b$ over time T . The correction terms $\mathcal{G}_i^W, \mathcal{G}_{ij}^U$ account for unobserved partial CTMC trajectories on lineages that are deleted before time T , inserted after time 0, or born and deleted within $(0, T)$. Exact evaluation of these terms would require genealogical information about the latent BDI history, including birth and death times and lineage lifetimes. In the reduced Pair-HMM EM algorithm used here, we optimize the endpoint alignment likelihood rather than the complete-data likelihood of the full birth-death-substitution process. Consequently, the reduced substitution update for Q retains only matched-lineage bridge contributions. For the equilibrium distribution π , we additionally retain observable endpoint composition counts from insertion and deletion states, while omitting genealogical correction terms from nonterminal birth and death times and from fully transient hidden lineages.

Considering now the joint distribution $P(x, y | \theta)$ and including the sufficient statistics for the stationary composition and length distribution of ancestral sequences, as well as the compositional statistics for inserted sequences that are already included in the conditional case, the expectations are

$$\begin{aligned}\mathbb{E}[L \mid x, y] &= \sum_{i \in \{\mathbf{S}, \mathbf{M}, \mathbf{I}, \mathbf{D}\}} (\hat{n}_{i\mathbf{M}} + \hat{n}_{i\mathbf{D}}) = |x| \\ \mathbb{E}[M \mid x, y] &= 1 \\ \mathbb{E}[V_i \mid x, y] &= \sum_b \hat{e}_{(i,b)}^{\mathbf{M}} + \hat{e}_{(\epsilon,i)}^{\mathbf{I}} + \hat{e}_{(i,\epsilon)}^{\mathbf{D}} + \mathcal{G}_i^V(x, y)\end{aligned}$$

where the three observable terms count, respectively, the expected number of ancestral sequence extensions L , ancestral sequence terminations M (always one for a single training alignment), and positions V_i initially having residue i (including ancestral, descendant, and transiently inserted residues); $\mathcal{G}_i^V$ is another genealogical correction term accounting for these transients, the omission of which again corresponds to assuming (for the purposes of the CTMC parameter update) that insertion and deletion events are synchronized with the trajectory endpoints, and that no transient characters are born and die within the trajectory. Thus V_i counts every observable equilibrium π -draw of character i in the trajectory, regardless of whether the residue subsequently underwent a match-position substitution, was deleted, or never had an ancestor (i.e. was an insertion). The conditional $P(y|x, t)$ analogue drops the two ancestor-derived terms (matches and deletions), since the ancestral sequence is given.

A.1.8 Baum-Welch Algorithm for TKF91

Given a collection of ancestral and descendant sequences $\{x^{(n)}, y^{(n)}\}$, the Baum-Welch algorithm iterates between an E-step and an M-step.

E-step. For each of the sequence pairs, run the Forward-Backward algorithm on the TKF91 WFST yielding the expected transition counts $\hat{n}_{ab}$ for $a, b \in \{\mathbf{S}, \mathbf{M}, \mathbf{I}, \mathbf{D}, \mathbf{E}\}$ and the expected emission counts $\hat{e}_{(a,b)}^{\mathbf{M}}, \hat{e}_{(\epsilon,b)}^{\mathbf{I}}, \hat{e}_{(a,\epsilon)}^{\mathbf{D}}$. Use these to accumulate expectations for the sufficient statistics S, B, D, W, U, L, M, V as described in the previous section. In this section we will simply refer to the accumulated expectations for these statistics using the variable names for the statistics themselves. Thus $S \equiv \mathbb{E}[S \mid \{x^{(n)}, y^{(n)}\}]$ and so forth.

M-step. The goal is to maximize $\ell(\theta) = \ell_1(\lambda, \mu) + \ell_2(Q, \pi)$ where

$$\ell_1(\lambda, \mu) = (B + L) \log \lambda + (D - L - M) \log \mu + M \log(\mu - \lambda) - (\lambda + \mu) S - \lambda T \quad (\text{A.18})$$

$$\ell_2(Q, \pi) = \sum_i V'_i \log \pi_i + \sum_{j>i} (U_{ij} + U_{ji}) \log Q_{ij} - \sum_{j>i} Q_{ij} (W_i \pi_j + W_j \pi_i) \quad (\text{A.19})$$

using the reversible parameterization $\kappa = \lambda/\mu$ and $R_{ij} = Q_{ij}\pi_j$ and letting $V'_i = V_i + \sum_{j \neq i} U_{ji}$ for notational convenience. Differentiating and setting to zero yields equations the MLEs must satisfy. For λ, μ these are simultaneous quadratics

$$\partial_\lambda \ell_1 = \frac{B + L}{\lambda} - \frac{M}{\mu - \lambda} - S - T = 0 \quad (\text{A.20})$$

$$\partial_\mu \ell_1 = \frac{D - L - M}{\mu} + \frac{M}{\mu - \lambda} - S = 0 \quad (\text{A.21})$$

which reduce to a single quadratic in κ . Setting $a = B + L$ and $b = D - L - M$

$$\kappa^2(S + T)b - \kappa(S(a + b + 2M) + T(b + M)) + Sa = 0. \quad (\text{A.22})$$

For the coefficients $A_\kappa = (S + T)b$, $B_\kappa = -(S(a + b + 2M) + T(b + M))$, and $C_\kappa = Sa$, the root $\kappa \in (0, 1)$ will be the smaller of the two roots

$$\hat{\kappa}(B, D, L, M, S, T) = \frac{-B_\kappa - \sqrt{B_\kappa^2 - 4A_\kappa C_\kappa}}{2A_\kappa} \quad (\text{A.23})$$

$$\hat{\mu}(B, D, L, M, S, T) = \frac{1}{S} \left(b + \frac{M}{1 - \hat{\kappa}} \right) \quad (\text{A.24})$$

$$\hat{\lambda}(B, D, L, M, S, T) = \hat{\kappa} \hat{\mu} \quad (\text{A.25})$$

For Q, π the MLE equations are

$$\partial_{Q_{ij}} \ell_2 = \frac{U_{ij} + U_{ji}}{Q_{ij}} - (W_i \pi_j + W_j \pi_i) = 0 \quad (\text{A.26})$$

$$\partial_{\pi_i} \ell_2 = \frac{V'_i}{\pi_i} - \sum_{j \neq i} Q_{ij} W_j + \eta = 0 \quad (\text{A.27})$$

where in the second equation η is a Lagrange multiplier for the constraint $\sum_i \pi_i = 1$. These equations are nonlinearly coupled and do not, in general, have a closed-form solution, nor even a closed-form iterative solution. There are several practical alternatives:

1. Set π from insertions and ancestral composition, $\pi_i = V_i / \sum_j V_j$, and then solve for Q in closed form

$$Q_{ij} = \frac{U_{ij} + U_{ji}}{W_i \pi_j + W_j \pi_i}.$$

2. Fixing π , solve for Q using the above closed form; then solve for η from $\sum_i V'_i / (\sum_{j \neq i} Q_{ij} W_j - \eta) = 1$, and set $\pi_i \leftarrow V'_i / (\sum_{j \neq i} Q_{ij} W_j - \eta)$, iterating until convergence. For fixed Q , the optimization over π is strictly concave on the simplex since $V'_i > 0$ for all i and the Hessian is negative definite.
3. Eliminate Q to yield a self-consistency map for π . Banach fixed-point guarantees are non-obvious for this approach.

4. Use the unconstrained MLE for $R_{ij} = Q_{ij}\pi_j$ and then project or reparameterize it into reversible GTR form. If the underlying process is reversible, this can provide a useful approximation or initialization, though finite-sample estimates need not satisfy detailed balance exactly.

A.1.9 Extending TKF91 to Phylogenetic Trees

Hein (24) showed how to compute a multi-sequence Forward likelihood for TKF91 on a binary tree. Holmes and Bruno (30) used a similar approach to compute posterior marginals for Gibbs sampling of ancestral sequences and alignments. Lunter *et al* showed how to calculate alignment likelihoods efficiently (45). Suchard and Redelings (52) extended the MCMC sampling approach to jointly sample over MSAs and trees. Westesson et al. (67) showed how this approach can be seen as a generalization of Felsenstein’s pruning algorithm for substitution-only models, where the WFST plays the role of a matrix exponential defined on whole sequences, with the WFST’s Forward algorithm computing the entries of this matrix exponential. The linear algebraic interpretation was further remarked upon by Bouchard-Côté (?).

A.2 The TKF92 Model

In sequence-alignment terms, TKF92 is the classical fragment-based extension of TKF91 and induces gap behavior analogous to an affine gap penalty (63). Instead of each mortal link getting a single character from $\mathcal{A}$, each link is associated with a fragment consisting of $K \geq 1$ characters from $\mathcal{A}$, where $K \sim \text{Geometric}(r)$. We write this as

$$\mathcal{S}'(T) \sim \text{TKF92}(\text{Subst}(Q, \pi); \lambda, \mu, r)$$

where $\mathcal{S}' \in (\mathcal{A}^+)^*$ is a sequence of multi-character fragments. Specifically

$$\text{TKF92}(\text{Subst}(Q, \pi); \lambda, \mu, r) \equiv \text{Links}(\text{Frag}(\text{Subst}(Q, \pi); r); \lambda, \mu)$$

where $\text{Frag}(M, r)$ denotes a tuple of K i.i.d. random variables, each governed by M , and K is geometrically-distributed with parameter r

$$x \sim \text{Frag}(M, r) \Leftrightarrow K \sim \text{Geometric}(r), x = (x_1 \dots x_K), x_k \sim M$$

The TKF92 model thus has the same BDI process as TKF91, but with a more complex substitution process that emits fragments instead of single characters.

A.2.1 Latent Information in TKF92

The mean length of an insertion in TKF92 is $1/(1-r)$, i.e. the expected fragment length. For deletions, things are slightly more complicated. Superficially, it looks like deletions are also fragments so they should have the same mean length. However, if we do not have knowledge of the fragment boundaries, but are conditioning just on the current sequence length, we have to weight this by the posterior probability that the subsequence being deleted does in fact constitute a single fragment.

For a subsequence of length k , this posterior probability is proportional to $\left(\frac{r}{r+(1-r)\frac{\lambda}{\mu}}\right)^{k-1}$ (indels at the end of the sequence have a slightly different correction factor since they are more likely to constitute a fragment, but the k -dependence is the same). The rate of starting an insertion or deletion *at a particular site* is subject to a similar correction factor.

This illustrates how the derivation of the WFST from the Pair HMM of such models becomes more complicated as the models carry greater amounts of latent information in their state space.

A.2.2 Singlet HMM, Pair HMM, and WFST Representations

The TKF92 Singlet HMM, Pair HMM, and WFST have the same state spaces $\{\mathbf{S}, \mathbf{M}, \mathbf{I}, \mathbf{D}, \mathbf{E}\}$ as TKF91, but with transition probabilities modified to account for fragment extension. Within a link's fragment, each character beyond the first extends the fragment with probability r , so the fragment terminates with probability $1 - r$.

Singlet HMM (stationary distribution). The total sequence length in TKF92 is a geometric sum of geometric fragment lengths. Since $N \sim \text{Geometric}(\kappa)$ links and each fragment has length $K \sim \text{Geometric}(r)$, the total length $L = \sum_{i=1}^N K_i$ is distributed as $P(L = 0) = 1 - \kappa$ and $P(L = \ell \mid L \geq 1) = (1 - p)p^{\ell-1}$ where $p = r + \kappa(1 - r)$ is the effective continuation probability. The HMM emits no characters with probability $1 - \kappa$, emits a first character with probability κ , and then continues with probability p after each emitted character. This is equivalent to a zero-inflated geometric distribution with parameters κ and p .

Pair HMM (finite-time joint distribution). Each emitting state has a fragment extension self-loop with probability r . On fragment termination (probability $1 - r$), the TKF91 link-level transitions apply. The Pair HMM transition matrix is

$$\tau'(\lambda, \mu, T, r) = \begin{pmatrix} & \mathbf{S} & \mathbf{M} & \mathbf{I} & \mathbf{D} & \mathbf{E} \\ \mathbf{S} & 0 & \tau_{\mathbf{SM}} & \tau_{\mathbf{SI}} & \tau_{\mathbf{SD}} & \tau_{\mathbf{SE}} \\ \mathbf{M} & 0 & r + (1 - r)\tau_{\mathbf{MM}} & (1 - r)\tau_{\mathbf{MI}} & (1 - r)\tau_{\mathbf{MD}} & (1 - r)\tau_{\mathbf{ME}} \\ \mathbf{I} & 0 & (1 - r)\tau_{\mathbf{IM}} & r + (1 - r)\tau_{\mathbf{II}} & (1 - r)\tau_{\mathbf{ID}} & (1 - r)\tau_{\mathbf{IE}} \\ \mathbf{D} & 0 & (1 - r)\tau_{\mathbf{DM}} & (1 - r)\tau_{\mathbf{DI}} & r + (1 - r)\tau_{\mathbf{DD}} & (1 - r)\tau_{\mathbf{DE}} \\ \mathbf{E} & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

where $\tau_{ab}(\lambda, \mu, T)$ are the TKF91 Pair HMM transition probabilities. Each row sums to 1: for $a \in \{\mathbf{M}, \mathbf{I}, \mathbf{D}\}$, $r + (1 - r)\sum_b \tau_{ab} = r + (1 - r) = 1$.

Gap probabilities summed over indel orderings. For any Pair HMM M whose states include $\mathbf{X}, \mathbf{I}, \mathbf{D}, \mathbf{Y}$ (where $\mathbf{X}, \mathbf{Y}$ are arbitrary, possibly equal to one of $\mathbf{I}, \mathbf{D}$, or each other), define $G_{\mathbf{XY}}^M(i, j)$ as the sum of path probabilities over all paths of the form $\mathbf{X} \rightarrow (\text{any sequence of } i \text{ D's and } j \text{ I's}) \rightarrow \mathbf{Y}$. Let the relevant transition probabilities ($i \rightarrow j$ in row i , column j) be

	$\mathbf{Y}$	$\mathbf{I}$	$\mathbf{D}$
$\mathbf{X}$	a	b	c
$\mathbf{I}$	f	g	h
$\mathbf{D}$	p	q	r

The path can be thought of as making i horizontal steps and j vertical steps. By conditioning on the number k of zig-zags on the path (i.e., cycles $\mathbf{I} \rightarrow \mathbf{D}^+ \rightarrow \mathbf{I}$ or $\mathbf{D} \rightarrow \mathbf{I}^+ \rightarrow \mathbf{D}$), counting the

positions at which they can occur, and summing out, we obtain (for $i > 0$ and $j > 0$)

$$\begin{aligned}
G_{\mathbf{XY}}^M(0, 0) &= a \\
G_{\mathbf{XY}}^M(i, 0) &= c r^{i-1} p \\
G_{\mathbf{XY}}^M(0, j) &= b g^{j-1} f \\
G_{\mathbf{XY}}^M(i, j) &= g^{j-1} r^{i-1} \left(\sum_{k=1}^{\min(i,j)} \left(\frac{hq}{gr} \right)^k \left[\binom{i-1}{k-1} \binom{j-1}{k} brf + \binom{i-1}{k} \binom{j-1}{k-1} cgp \right] \right. \\
&\quad \left. + \sum_{k=0}^{\min(i,j)} \left(\frac{hq}{gr} \right)^k \binom{i-1}{k} \binom{j-1}{k} (bhp + cqf) \right) \\
&= g^{j-1} r^{i-1} [z(j-1) brf \cdot {}_2F_1(1-i, 2-j; 2; z) + z(i-1) cgp \cdot {}_2F_1(2-i, 1-j; 2; z) \\
&\quad + (bhp + cqf) \cdot {}_2F_1(1-i, 1-j; 1; z)] \\
z &= \frac{hq}{gr}
\end{aligned}$$

where ${}_2F_1$ is the Gauss hypergeometric function

$${}_2F_1(a, b; c; z) = \sum_{k=0}^{\infty} \frac{(a)_k (b)_k}{(c)_k} \frac{z^k}{k!}, \quad (a)_k = a(a+1) \cdots (a+k-1) = \frac{\Gamma(a+k)}{\Gamma(a)},$$

with $(a)_k$ the Pochhammer symbol (rising factorial) and Γ the gamma function; we have used the identities

$$\begin{aligned}
\sum_{k=0}^{\infty} \binom{a}{k} \binom{b}{k} z^k &= {}_2F_1(-a, -b; 1; z) \\
\sum_{k=1}^{\infty} \binom{a}{k-1} \binom{b}{k} z^k &= -bz \cdot {}_2F_1(-a, 1-b; 2; z)
\end{aligned}$$

Applied to the TKF92 Pair HMM τ' above, this lets one evaluate gap-block sums (summed over all I/D orderings) in closed form, which is the canonical likelihood factor for pairwise alignments under TKF92.

WFST (finite-time conditional distribution). Forming a WFST for TKF92 is complicated by the latent information associated with fragment boundaries. If those boundaries are known, one can construct a WFST that respects them exactly. If they are not known, one can still construct a plausible WFST that imputes them on the fly. This trick, however, ceases to be straightforward for more highly decorated descendants of TKF91. Alternatively, one can expose the fragment boundary information in the transducer alphabet, at the cost of reifying indivisibility of fragments over time. The choice between these approaches is non-obvious and presumably application-dependent. The explicit TKF92 WFST construction, obtained by dividing the Pair HMM by the Singlet HMM, is given in Appendix A.4. That construction requires a 6×6 state space ($\mathbf{S}, \mathbf{M}, \mathbf{I}_0, \mathbf{I}_1, \mathbf{D}, \mathbf{E}$) rather than the 5×5 form one might naively borrow from TKF91, because the singlet's outgoing-emission weight differs between the start state (κ for the first ancestor character) and any subsequent emit state ($p = r + (1-r)\kappa$), so the insert state must be split into a pre-immortal-link variant ($\mathbf{I}_0$) and a post-immortal-link variant ($\mathbf{I}_1$). The resulting WFST is conditionally normalised in the global sense: for any input ancestor sequence x , the total weight of all paths from $\mathbf{S}$ to $\mathbf{E}$ that read x exactly equals

1, summing over all alignments and descendant outputs. Equivalently, $\text{Singlet} \circ \mathbb{T}' = \tau'$ is row-stochastic on the 6-state space. Local single-step row sums and per-state ε -closure normalisation hold for the $\mathbf{S}, \mathbf{M}, \mathbf{I}_0, \mathbf{D}$ rows but not for the $\mathbf{I}_1$ row, which is improper in isolation but accumulates the correct κ/p factor whenever it is reached from a proper source state along an $\mathbf{S}$ -rooted path (Appendix A.4.7). The same trade-off recurs for MixDom WFSTs in Section C.1.5, and is discussed more carefully in Appendix A.4.7.

These issues are not unique to TKF92, but are a general feature of models with latent information that is not directly observable in the sequence data. They are dealt with at greater length in the section on MixDom WFSTs (Section C.1.5), where the same issues arise in a more complex setting. To state the general principle: as TKF91-derived models acquire additional latent decoration, that decoration must either be promoted to explicit symbols in the interface alphabet of the transducer construction, or integrated out, in which case the resulting transducer is only approximate.

A.2.3 Baum-Welch Algorithm for TKF92

The Baum-Welch algorithm for TKF92 is similar to that for TKF91, but we must now resolve the Pair HMM's self-looping transition counts onto their separate components (fragment extension vs. new link), and accumulate fragment-level sufficient statistics in addition to the substitution-level sufficient statistics.

E-step. Run Forward-Backward on the TKF92 Pair HMM. This yields expected transition counts $\hat{n}'_{ab}$. For $a \in \{\mathbf{M}, \mathbf{I}, \mathbf{D}\}$, the self-loop count $\hat{n}'_{aa}$ combines fragment extension (probability r) and new-link-same-state (probability $(1-r)\tau_{aa}$). The expected number of fragment extensions from state a is $\hat{n}'_{aa} \cdot r / (r + (1-r)\tau_{aa})$ and the expected number of new-link self-transitions is the remainder. Thus, we accumulate new expectations for the additional fragment-level sufficient statistics, F the number of fragment extensions and E the number of fragment ends, and then recover the TKF91 link-level counts by removing the fragment extensions

$$\begin{aligned} F_a &= \frac{\hat{n}'_{aa} \cdot r}{r + (1-r)\tau_{aa}}, \quad a \in \{\mathbf{M}, \mathbf{I}, \mathbf{D}\} \\ F &= \sum_{a \in \{\mathbf{M}, \mathbf{I}, \mathbf{D}\}} F_a \\ E &= \sum_{a \in \{\mathbf{M}, \mathbf{I}, \mathbf{D}\}} \sum_b \hat{n}'_{ab} - F \\ \hat{n}_{ab} &= \hat{n}'_{ab} - \delta_{ab} F_a \end{aligned}$$

We then proceed to accumulate S, B, D, W, U, L, M, V as for TKF91. Note that L counts *links*, which are now fragments rather than simply residues: $L = \hat{n}_\kappa = \sum_i (\hat{n}_{i\mathbf{M}} + \hat{n}_{i\mathbf{D}})$ where $\hat{n}$ is the resolved (TKF91-level) count matrix. The character-level statistics W, U, V are accumulated per character exactly as in TKF91, because fragment extension affects the indel process estimates but not the substitution process estimates.

M-step. This is identical to TKF91, except we also set $r \leftarrow F / (F + E)$.

A.2.4 Maraschino: Distilled Cherries

The TKF92 Baum-Welch algorithm, as with other instances of the Expectation Maximization algorithm, is most useful if there is hidden information that we need to marginalize during parameter

estimation; e.g. an unknown alignment that needs to be summed over, or (as in Section C.1.1) latent site or fragment classes that are not directly observed.

For many applications, we can take the alignment data (and potentially site classes) as a given, and then a more efficient estimation strategy becomes available. This is the CherryML approach: a composite likelihood consisting of a product of alignments likelihoods for nearest-neighbor sibling pairs, a.k.a. “cherries” (50). The CherryML developers observe that, given cherry branch lengths, the alignment substitution counts are sufficient to estimate substitution rates and can be performed efficiently using autograd, independently of data size except for a fast preprocessing step. We observe that this can be generalized to TKF92 by including the alignment transition statistics i.e. the sixteen transition counts $n_{X \rightarrow Y}$ where $X \in \{S, M, I, del\}$ and $Y \in \{M, I, D, E\}$ states. The CherryML approach can then be applied: 1) discretize times, yielding a fixed-shape tensor of summary counts; 2) write down the observed data likelihood in terms of the summary counts; 3) maximize using autograd and standard gradient-based optimizers.

We call this algorithm *Maraschino*, as it involves distilling cherries to state machines.

A.3 The TKF92 Model with Fragment Mixtures (MixFrag)

MixFrag generalises TKF92 (Section A.2) by promoting the single fragment-extension parameter r to a categorical latent variable. When a mortal link’s fragment is created, it is independently assigned a *fragtype* $f \in \{1, \dots, \mathcal{F}\}$ with $f \sim \text{Categorical}(w_1, \dots, w_{\mathcal{F}})$ (weights $w_f \geq 0$, $\sum_f w_f = 1$), and the fragment then consists of $K \sim \text{Geometric}(r_f)$ characters from $\mathcal{A}$, all governed by the same substitution process $\text{Subst}(Q, \pi)$. In the notation of Section A.2,

$$\begin{aligned} \mathcal{S}'(T) &\sim \text{MixFrag}(\text{Subst}(Q, \pi); \lambda, \mu, \{r_f\}, \{w_f\}) \\ &\equiv \text{Links}(\text{Frag}^{\mathcal{F}}(\text{Subst}(Q, \pi); \{r_f\}, \{w_f\}); \lambda, \mu) \end{aligned}$$

where the fragment process first draws a fragtype and then a fragtype-specific geometric run,

$$\begin{aligned} x \sim \text{Frag}^{\mathcal{F}}(M; \{r_f\}, \{w_f\}) &\Leftrightarrow f \sim \text{Categorical}(w_1, \dots, w_{\mathcal{F}}), \\ K &\sim \text{Geometric}(r_f), \quad x = (x_1 \dots x_K), \quad x_k \sim M. \end{aligned}$$

The only parameters beyond TKF91 are the $\mathcal{F}$ fragtype-specific extension probabilities r_f and the $\mathcal{F}$ fragtype weights w_f ; the indel (BDI) process and the substitution process are both shared across fragtypes. With $\mathcal{F} = 1$ (so $w_1 = 1$) MixFrag reduces to TKF92. It uses a single substitution model and a single, non-nested indel process, and so introduces no null states.

A.3.1 Singlet HMM and Pair HMM Representations

We do not give a WFST for MixFrag. The fragtype is latent information that is not visible in the sequence, so a transducer that respects fragment indivisibility over evolutionary time must expose the fragtype in its interface alphabet—in order to keep branch-to-branch composition consistent—making the construction even more elaborate than the TKF92 WFST of Appendix A.4. The Singlet and Pair HMMs, by contrast, follow systematically from TKF91, exactly as for TKF92.

The same construction yields both machines from their TKF91 counterparts: a transition *out of* an emitting state of fragtype f acquires a $(1 - r_f)$ multiplier (the current fragment terminates); a transition *into* an emitting state of fragtype g acquires a w_g multiplier (the new fragment is assigned fragtype g); and each emitting state acquires an added self-loop term $r_f \delta_{fg}$ that extends the current fragment within its own fragtype.

Singlet HMM (stationary distribution). The Singlet HMM has $2 + \mathcal{F}$ states $\{\mathbf{S}, \mathbf{I}_f, \mathbf{E}\}$. Applying the construction to the TKF91 stationary HMM (which emits $n \sim \text{Geometric}(\kappa)$ characters, $\kappa = \lambda/\mu$) gives the transition matrix

$$\begin{pmatrix} & \mathbf{S} & \mathbf{I}_g & \mathbf{E} \\ \mathbf{S} & 0 & \kappa w_g & 1 - \kappa \\ \mathbf{I}_f & 0 & r_f \delta_{fg} + (1 - r_f) \kappa w_g & (1 - r_f)(1 - \kappa) \\ \mathbf{E} & 0 & 0 & 0 \end{pmatrix}$$

where $\mathbf{I}_f$ emits a character from π . Each $\mathbf{I}_f$ row sums to $r_f + (1 - r_f)(\kappa \sum_g w_g + (1 - \kappa)) = r_f + (1 - r_f) = 1$, using $\sum_g w_g = 1$.

Pair HMM (finite-time joint distribution). The Pair HMM has $2 + 3\mathcal{F}$ states $\{\mathbf{S}, \mathbf{M}_f, \mathbf{I}_f, \mathbf{D}_f, \mathbf{E}\}$; for convenience we write $\mathbf{S}_f \equiv \mathbf{S}$ and $\mathbf{E}_f \equiv \mathbf{E}$, so that any state may be written $\mathbf{X}_f$ with $\mathbf{X} \in \{\mathbf{S}, \mathbf{M}, \mathbf{I}, \mathbf{D}, \mathbf{E}\}$. Writing $\tau_{\mathbf{X}\mathbf{Y}} \equiv \tau_{\mathbf{X}\mathbf{Y}}(\lambda, \mu, T)$ for the TKF91 Pair HMM transition probabilities (Section A.1.6), the joint Pair HMM transition matrix is

$$\tau^{\mathcal{F}}(\lambda, \mu, T, \{r_f\}, \{w_f\}) = \begin{pmatrix} & \mathbf{S} & \mathbf{M}_g & \mathbf{I}_g & \mathbf{D}_g & \mathbf{E} \\ \mathbf{S} & 0 & w_g \tau_{\mathbf{S}\mathbf{M}} & w_g \tau_{\mathbf{S}\mathbf{I}} & w_g \tau_{\mathbf{S}\mathbf{D}} & \tau_{\mathbf{S}\mathbf{E}} \\ \mathbf{M}_f & 0 & r_f \delta_{fg} + (1 - r_f) w_g \tau_{\mathbf{M}\mathbf{M}} & (1 - r_f) w_g \tau_{\mathbf{M}\mathbf{I}} & (1 - r_f) w_g \tau_{\mathbf{M}\mathbf{D}} & (1 - r_f) \tau_{\mathbf{M}\mathbf{E}} \\ \mathbf{I}_f & 0 & (1 - r_f) w_g \tau_{\mathbf{I}\mathbf{M}} & r_f \delta_{fg} + (1 - r_f) w_g \tau_{\mathbf{I}\mathbf{I}} & (1 - r_f) w_g \tau_{\mathbf{I}\mathbf{D}} & (1 - r_f) \tau_{\mathbf{I}\mathbf{E}} \\ \mathbf{D}_f & 0 & (1 - r_f) w_g \tau_{\mathbf{D}\mathbf{M}} & (1 - r_f) w_g \tau_{\mathbf{D}\mathbf{I}} & r_f \delta_{fg} + (1 - r_f) w_g \tau_{\mathbf{D}\mathbf{D}} & (1 - r_f) \tau_{\mathbf{D}\mathbf{E}} \\ \mathbf{E} & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Emissions are inherited unchanged from TKF92 and are independent of fragtype: $\mathbf{M}_f$ aligns an ancestral–descendant residue pair under $\exp(RT)$, while $\mathbf{I}_f$ and $\mathbf{D}_f$ emit a single inserted or deleted residue from π . Each emitting row sums to 1: for $\mathbf{X} \in \{\mathbf{M}, \mathbf{I}, \mathbf{D}\}$, since $\sum_g w_g = 1$ the fragtype weights collapse and the row sum reduces to $r_f + (1 - r_f) \sum_{\mathbf{Y}} \tau_{\mathbf{X}\mathbf{Y}} = r_f + (1 - r_f) = 1$, exactly as for TKF92. The Pair HMM has no null states, so no null-elimination (collapse) step is required.

A.3.2 Baum-Welch Algorithm for MixFrag

The Baum-Welch algorithm follows that for TKF92 (Section A.2.3), but the fragment-extension and fragment-end statistics are now accumulated per fragtype.

E-step. Run Forward-Backward on the MixFrag Pair HMM, yielding expected transition counts $\hat{n}'_{a_f b_g}$ over the $2 + 3\mathcal{F}$ states. For an emitting state a_f ($a \in \{\mathbf{M}, \mathbf{I}, \mathbf{D}\}$), the self-loop count $\hat{n}'_{a_f a_f}$ combines fragment extension within fragtype f (probability r_f) with a new same-state link that is again assigned fragtype f (probability $(1 - r_f) \tau_{aa} w_f$). We resolve this self-loop and recover the TKF91 link-level counts, accumulating the per-fragtype fragment-extension count F_f and fragment-end count E_f (the latter is also the expected number of type- f fragments, since each fragment terminates exactly once):

$$\begin{aligned} F_{a_f} &= \frac{\hat{n}'_{a_f a_f} r_f}{r_f + (1 - r_f) \tau_{aa} w_f}, \quad a \in \{\mathbf{M}, \mathbf{I}, \mathbf{D}\} \\ F_f &= \sum_{a \in \{\mathbf{M}, \mathbf{I}, \mathbf{D}\}} F_{a_f} \\ E_f &= \sum_{a \in \{\mathbf{M}, \mathbf{I}, \mathbf{D}\}} \sum_{b_g} \hat{n}'_{a_f b_g} - F_f \\ \hat{n}_{a_f b_g} &= \hat{n}'_{a_f b_g} - \delta_{ab} \delta_{fg} F_{a_f} \end{aligned}$$

The denominator of F_{af} is the full weight of the $a_f \rightarrow a_f$ self-loop: its fragment-extension branch (r_f) and its new-link-same-state branch (which must re-draw the same fragtype, hence the w_f factor). Summing the resolved counts over fragtypes, $\hat{n}_{ab} = \sum_{f,g} \hat{n}_{afbg}$, recovers the TKF91 link-level transition counts, from which S, B, D, W, U, L, M, V are accumulated exactly as for TKF91 (the indel and substitution processes being shared across fragtypes); in particular $L = \sum_i (\hat{n}_{iM} + \hat{n}_{iD})$ counts links (fragments of any type) and the character-level statistics W, U, V are accumulated per character as in TKF91. Because the fragtype is latent, the Forward-Backward E-step marginalises it together with the alignment, so the per-fragtype counts F_f, E_f are posterior expectations rather than counts read directly off a given alignment.

M-step. Identical to TKF91, except that the fragtype-specific parameters are set from the per-fragtype fragment statistics,

$$r_f \leftarrow \frac{F_f}{F_f + E_f}, \quad w_f \leftarrow \frac{E_f}{\sum_{f'} E_{f'}}.$$

Here E_f is the expected number of fragments of fragtype f (each fragment terminates exactly once), so w_f is the categorical MLE over fragtypes, while r_f is the geometric MLE, giving mean fragment length $1/(1 - r_f) = (F_f + E_f)/E_f$ for fragtype f .

A.4 TKF92 WFST by Singlet Division

This appendix derives the TKF92 conditional WFST $T'(\lambda, \mu, T, r)$ representing $P(\text{descendant} \mid \text{ancestor}, \theta)$ by dividing the TKF92 Pair HMM by the TKF92 Singlet HMM.

The TKF91 WFST (Section A.1.6) is a 5×5 matrix obtained from the TKF91 Pair HMM τ by stripping κ from ancestor-consuming columns (**M**, **D**) and $1 - \kappa$ from the end column (**E**). The TKF92 WFST cannot be expressed as a 5×5 matrix in the same state space. As we show in Section A.4.2, the singlet's outgoing-emission weight differs between its **S** state and its emit state, so the two cases must be carried as distinct insert states (*Ins0* and *Ins1*) in the WFST. The natural object is therefore 6×6 , with state space $\{\mathbf{S}, \mathbf{M}, \mathbf{I}_0, \mathbf{I}_1, \mathbf{D}, \mathbf{E}\}$. A 5×5 presentation that lumps $\mathbf{I}_0$ and $\mathbf{I}_1$ into a single **I** state and uses the emit-state divisor p throughout is incorrect for paths that traverse $\mathbf{S} \rightarrow \mathbf{I} \rightarrow \dots \rightarrow \{\mathbf{M}, \mathbf{D}\}$; this is discussed in Section A.4.6.

A.4.1 TKF92 Singlet HMM Transitions

The TKF92 singlet at stationarity emits sequences whose length L follows a zero-inflated geometric distribution (Section A.2.2). The singlet HMM has two transition contexts:

- *From S (before any character):* $P(\text{emit first character}) = \kappa$, $P(\text{end}) = 1 - \kappa$.
- *From an emitting state (after character $i \geq 1$):* $P(\text{emit next character}) = p \equiv r + (1-r)\kappa$, $P(\text{end}) = (1-r)(1-\kappa)$.

The effective continuation probability p decomposes as: fragment extension with probability r , or fragment termination $(1-r)$ followed by a new nonempty link (κ). Each emitted character is drawn i.i.d. from π .

The key consequence for the WFST construction below is that $\kappa \neq p$ whenever $r > 0$.

A.4.2 Why an Ins0 / Ins1 Split is Necessary

Transducer composition $\text{Singlet} \circ \text{WFST} = \text{Pair HMM}$ couples each ancestor-consuming Pair HMM transition to one of the two singlet emission events:

- the $S \rightarrow \text{emit}$ event (weight κ), used *exactly once*, for the first ancestor character; or
- the $\text{emit} \rightarrow \text{emit}$ event (weight p), used for every subsequent ancestor character.

The Pair HMM's I state corresponds to “the descendant is in the middle of a descendant-only fragment.” This state is reached either *before* the first ancestor character has been consumed (entered from S , possibly via $I \rightarrow I$ self-loops) or *after* ≥ 1 ancestor characters have been consumed (entered from M, D, I). In the first case, the next ancestor-consuming transition out of I corresponds to the $S \rightarrow \text{emit}$ event in the singlet (weight κ); in the second case, it corresponds to $\text{emit} \rightarrow \text{emit}$ (weight p). The two divisors differ unless $r = 0$.

The 5×5 Pair HMM $\tau'(\lambda, \mu, T, r)$ of Section A.2.2 marginalises over the “has any ancestor character been consumed yet?” bit and is correct as a generator of the joint distribution. But because that bit controls which singlet divisor applies, the WFST construction must expose it. We therefore split the Pair HMM I state into

- I_0 — reachable only from S (via $S \rightarrow I_0$) and from itself (via $I_0 \rightarrow I_0$): the singlet has not yet emitted.
- I_1 — reachable from M, D, I_1 : the singlet has emitted at least one character.

The 6×6 Pair HMM in this state space is row-stochastic (verified below), and the 5×5 Pair HMM is recovered by lumping I_0 and I_1 into a single I row. The lumping preserves the joint distribution because the outgoing transitions from I_0 and I_1 have identical Pair HMM structure once weighted by their singlet contributions; what fails to lump is the WFST.

A.4.3 The 6×6 TKF92 Pair HMM

Writing $\alpha, \beta, \gamma, \kappa$ for the TKF link-level parameters (as in Section A.1.6), and $r = r$ for the fragment-extension probability:

$$\tau''(\lambda, \mu, T, r) = \begin{pmatrix} & S & M & I_0 & I_1 & D & E \\ \begin{matrix} S \\ M \\ I_0 \\ I_1 \\ D \\ E \end{matrix} & \begin{matrix} 0 & r + (1-r)(1-\beta)\kappa\alpha & 0 & (1-r)\beta & (1-r)(1-\beta)\kappa(1-\alpha) & (1-r)(1-\beta)(1-\kappa) \\ 0 & (1-r)(1-\beta)\kappa\alpha & r + (1-r)\beta & 0 & (1-r)(1-\beta)\kappa(1-\alpha) & (1-r)(1-\beta)(1-\kappa) \\ 0 & (1-r)(1-\beta)\kappa\alpha & 0 & r + (1-r)\beta & (1-r)(1-\beta)\kappa(1-\alpha) & (1-r)(1-\beta)(1-\kappa) \\ 0 & (1-r)(1-\gamma)\kappa\alpha & 0 & (1-r)\gamma & r + (1-r)(1-\gamma)\kappa(1-\alpha) & (1-r)(1-\gamma)(1-\kappa) \\ 0 & 0 & 0 & 0 & 0 & 0 \end{matrix} \end{pmatrix} \quad (\text{A.28})$$

The structural zeros are: S cannot reach I_1 in one step (no ancestor has yet been emitted), and M, D, I_1 cannot reach I_0 in one step (an ancestor has already been emitted). The I_0 and I_1 rows differ only in their self-loop column ($I_0 \rightarrow I_0$ vs $I_1 \rightarrow I_1$); their entries to M, D, E are identical.

Row-stochasticity. For each non- E source row, the row sum equals

$$r + (1-r) \left[\beta + (1-\beta)\kappa\alpha + (1-\beta)\kappa(1-\alpha) + (1-\beta)(1-\kappa) \right] = r + (1-r) [\beta + (1-\beta)] = 1$$

(replacing β with γ for the D row), confirming that τ'' is a valid Markov transition matrix.

Recovery of the 5×5 Pair HMM. Lumping I_0 and I_1 into a single I state preserves all four non- I -column entries (which are identical between the two rows by construction). In the self-loop column, $I_0 \rightarrow I_0$ and $I_1 \rightarrow I_1$ both equal $r + (1-r)\beta$, so the lumped self-loop is also $r + (1-r)\beta$. The result is the 5×5 Pair HMM of Section A.2.2, recovered exactly.

A.4.4 The 6×6 TKF92 WFST

Each Pair HMM transition consumes either one ancestor character ($\rightarrow M$ or $\rightarrow D$), zero ancestor characters ($\rightarrow I_0$ or $\rightarrow I_1$), or signals termination ($\rightarrow E$). The WFST T'' is obtained by dividing each Pair HMM entry by the corresponding singlet weight:

- Ancestor-consuming entries from S, I_0 (the next ancestor character is the *first*): divide by κ .
- Ancestor-consuming entries from M, I_1, D (the next ancestor character is a *subsequent* one): divide by $p = r + (1-r)\kappa$.
- E entries from S, I_0 (the ancestor sequence has length zero): divide by $1-\kappa$.
- E entries from M, I_1, D (the ancestor sequence has length ≥ 1): divide by $(1-r)(1-\kappa)$.
- I_0 and I_1 entries (no ancestor consumed): divide by 1.

The emission factor π_a is also stripped from the emission weights, exactly as in TKF91, so that M -state emission becomes $\exp(RT)_{ab}$ and D -state emission becomes 1; we focus on the transition matrix below.

$$T''(\lambda, \mu, T, r) = \begin{pmatrix} & S & M & I_0 & I_1 & D & E \\ S & 0 & (1-\beta)\alpha & \beta & 0 & (1-\beta)(1-\alpha) & 1-\beta \\ M & 0 & \frac{r + (1-r)(1-\beta)\kappa\alpha}{p} & 0 & (1-r)\beta & \frac{(1-r)(1-\beta)\kappa(1-\alpha)}{p} & 1-\beta \\ I_0 & 0 & (1-r)(1-\beta)\alpha & r + (1-r)\beta & 0 & (1-r)(1-\beta)(1-\alpha) & (1-r)(1-\beta) \\ I_1 & 0 & \frac{(1-r)(1-\beta)\kappa\alpha}{p} & 0 & r + (1-r)\beta & \frac{(1-r)(1-\beta)\kappa(1-\alpha)}{p} & 1-\beta \\ D & 0 & \frac{(1-r)(1-\gamma)\kappa\alpha}{p} & 0 & (1-r)\gamma & \frac{r + (1-r)(1-\gamma)\kappa(1-\alpha)}{p} & 1-\gamma \\ E & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad (A.29)$$

The I_1 row uses the p divisor on its M and D exits and $1-\beta$ on its E exit, matching the post-immortal-link regime. The I_0 row uses κ on its M, D exits (absorbed into the numerator: $(1-r)(1-\beta)\kappa\alpha/\kappa = (1-r)(1-\beta)\alpha$, etc.) and $1-\kappa$ on its E exit (giving $(1-r)(1-\beta)$). These are exactly the TKF91 WFST $S/M/I$ -row outputs scaled by $(1-r)$, reflecting the leading-extension fragment's chance of terminating before any ancestor is consumed.

A.4.5 Structure and Verification

Sanity check 1: S row. The S row inherits the TKF91 form because the first character of the immortal-link descendant fragment has no extension self-loop. Multiplying by the singlet emission weights from S (κ for M, D , 1 for I_0 , $1-\kappa$ for E) recovers the Pair HMM S row of (A.28).

Sanity check 2: I_0 row. Multiplying $\mathbb{T}''$ entries by the appropriate singlet weights (“from $\mathbf{S}$ ” weights: κ for $\mathbf{M}/\mathbf{D}$, 1 for I_0 , $1-\kappa$ for $\mathbf{E}$):

$$\begin{aligned}\kappa \cdot (1-r)(1-\beta)\alpha &= (1-r)(1-\beta)\kappa\alpha &= \tau''_{I_0\mathbf{M}}, \\ 1 \cdot [r + (1-r)\beta] &= r + (1-r)\beta &= \tau''_{I_0I_0}, \\ \kappa \cdot (1-r)(1-\beta)(1-\alpha) &= (1-r)(1-\beta)\kappa(1-\alpha) &= \tau''_{I_0\mathbf{D}}, \\ (1-\kappa) \cdot (1-r)(1-\beta) &= (1-r)(1-\beta)(1-\kappa) &= \tau''_{I_0\mathbf{E}}.\end{aligned}$$

All four entries match the Pair HMM, so $\text{Singlet} \circ \mathbb{T}'' = \tau''$ on the I_0 row.

Sanity check 3: $\mathbf{M}$ row. Multiplying by emit-state weights (p for $\mathbf{M}/\mathbf{D}$, 1 for I_1 , $(1-r)(1-\kappa)$ for $\mathbf{E}$):

$$\begin{aligned}p \cdot \frac{r + (1-r)(1-\beta)\kappa\alpha}{p} &= r + (1-r)(1-\beta)\kappa\alpha &= \tau''_{\mathbf{M}\mathbf{M}}, \\ 1 \cdot (1-r)\beta &= (1-r)\beta &= \tau''_{\mathbf{M}I_1}, \\ p \cdot \frac{(1-r)(1-\beta)\kappa(1-\alpha)}{p} &= (1-r)(1-\beta)\kappa(1-\alpha) &= \tau''_{\mathbf{M}\mathbf{D}}, \\ (1-r)(1-\kappa) \cdot (1-\beta) &= (1-r)(1-\beta)(1-\kappa) &= \tau''_{\mathbf{M}\mathbf{E}}.\end{aligned}$$

The I_1 and $\mathbf{D}$ rows verify identically (with γ for β in the $\mathbf{D}$ row).

TKF91 limit ($r = 0$). At $r = 0$, $p = \kappa$ and $r = 0$. The I_0 and I_1 rows become:

$$\begin{aligned}\mathbb{T}''_{I_0*|r=0} &= (0, (1-\beta)\alpha, \beta, 0, (1-\beta)(1-\alpha), 1-\beta), \\ \mathbb{T}''_{I_1*|r=0} &= (0, (1-\beta)\alpha, 0, \beta, (1-\beta)(1-\alpha), 1-\beta).\end{aligned}$$

With the I_0 and I_1 self-loops both equal to β and all other columns identical, the two rows reduce to the same TKF91-Ins row when I_0, I_1 are merged. The TKF91 WFST (Section A.1.6) is recovered.

Latent fragment-boundary posterior. The $\mathbf{M}$ row’s $\mathbf{M} \rightarrow \mathbf{M}$ entry decomposes as

$$\mathbb{T}''_{\mathbf{M}\mathbf{M}} = \underbrace{\frac{r}{p} \cdot 1}_{\text{same frag}} + \underbrace{\frac{(1-r)\kappa}{p} \cdot (1-\beta)\alpha}_{\text{new frag}},$$

exactly as in the original construction; under same-fragment continuation the descendant certainly stays in $\mathbf{M}$ (the link persists within the fragment), and under new-fragment start the descendant must survive $(1-\beta)$ and align (α) . The fragment-boundary posterior $r/p + (1-r)\kappa/p = 1$ holds by definition of p . The I_0 row has *no* same-fragment ancestor-consuming term — since the singlet has not yet emitted, the fragment-boundary posterior is degenerate at “new fragment” — which is why the $I_0 \rightarrow \mathbf{M}$ entry has the simple $(1-r)(1-\beta)\alpha$ form rather than a $1/p$ -divided form.

A.4.6 Comparison with the 5×5 Form

A 5×5 presentation that replaces the two rows I_0, I_1 of (A.29) by a single $\mathbf{I}$ row equal to the I_1 row is correct only at $r = 0$. For $r > 0$, that 5×5 form misweights paths $\mathbf{S} \rightarrow \mathbf{I} \rightarrow \dots \rightarrow \{\mathbf{M}, \mathbf{D}, \mathbf{E}\}$ with one or more leading inserts before the first ancestor consumption: the divisor on the leading-insert

exit should be κ (or $1-\kappa$ for E) rather than p (or $(1-r)(1-\kappa)$). For a single-leading-insert path $\mathbf{S} \rightarrow \mathbf{I} \rightarrow \mathbf{M} \rightarrow \mathbf{E}$, the multiplicative discrepancy is $(1-r)\kappa/p < 1$ when $r > 0$, so the 5×5 WFST *underweights* such paths.

For benchmarks dominated by alignments without leading inserts (typical when the ancestor’s N -terminus is well-conserved), the numerical effect of the discrepancy is small per pair. However, the 5×5 form does not represent the true conditional distribution $P(\text{descendant} \mid \text{ancestor}, \theta)$, and the 6×6 WFST in (A.29) is the construction we recommend.

A.4.7 Normalization Structure

The WFST is conditionally normalised in the global sense: for every input sequence $x \in \Sigma^*$,

$$\sum_{(y, \text{alignment})} [\text{WFST path-weight}] = 1,$$

where the sum is over all paths from $\mathbf{S}$ to $\mathbf{E}$ that read x exactly. This is a direct consequence of $\text{Singlet} \circ \mathbb{T}'' = \tau''$ (verified row by row in Section A.4.5) combined with the fact that τ'' is row-stochastic and the Singlet HMM defines a proper distribution on x .

The WFST is *not* stochastic in the local single-step sense: its rows do not sum to 1, and the divisor structure (ancestor-consuming columns divide by κ from $\mathbf{S}, \mathbf{I}_0$ and by p from $\mathbf{M}, \mathbf{I}_1, \mathbf{D}$; the $\mathbf{E}$ column divides by $1-\kappa$ from $\mathbf{S}, \mathbf{I}_0$ and by $(1-r)(1-\kappa)$ from $\mathbf{M}, \mathbf{I}_1, \mathbf{D}$; insert columns divide by 1) means a fixed source row’s outgoing weights are not constrained to sum to 1.

ε -closure conditional normalisation. A more useful local property is the following. Define w_X^L as the total WFST weight of paths starting at state X that consume the next non- ε input symbol L (a character or end-of-input sentinel $\$$), where each path is a (possibly empty) geometric chain of insert self-loops followed by a single input-consuming exit. For our 6×6 WFST:

X	w_X^c	$w_X^\$$
$\mathbf{S}$	1	1
$\mathbf{M}$	1	1
$\mathbf{I}_0$	1	1
$\mathbf{I}_1$	κ/p	$1/(1-r)$
$\mathbf{D}$	1	1

(A.30)

The $\mathbf{S}, \mathbf{M}, \mathbf{I}_0, \mathbf{D}$ rows are conditionally normalised: each sums to exactly 1. The $\mathbf{I}_1$ row alone is not, because its destination divisors mix the r (extension) and $1-r$ (exit) weights asymmetrically into the single-step edges.

This local non-normalisation does *not* affect the WFST’s correctness as a representation of $P(\text{descendant} \mid \text{ancestor}, \theta)$: every WFST path begins at $\mathbf{S}$, and from $\mathbf{S}$ the chain-through- $\mathbf{I}_1$ contribution accumulates exactly the right κ/p factor at the appropriate step to bring the total back to 1. Concretely, starting from $\mathbf{M}$ the ε -closure weight to consume the next character is

$$w_{\mathbf{M}}^c = (\mathbb{T}_{\mathbf{M}\mathbf{M}}'' + \mathbb{T}_{\mathbf{M}\mathbf{D}}'') + \mathbb{T}_{\mathbf{M}\mathbf{I}_1}'' \cdot \frac{\mathbb{T}_{\mathbf{I}_1\mathbf{M}}'' + \mathbb{T}_{\mathbf{I}_1\mathbf{D}}''}{1 - \mathbb{T}_{\mathbf{I}_1\mathbf{I}_1}''} = \frac{r + (1-r)(1-\beta)\kappa}{p} + (1-r)\beta \cdot \frac{\kappa}{p} = \frac{r + (1-r)\kappa}{p} = 1,$$

so the κ/p factor that the $\mathbf{I}_1$ row alone undershoots is exactly recovered when $\mathbf{I}_1$ is reached from a “proper” source state. The $\mathbf{I}_1$ row is improper only if the WFST were started mid-stream from $\mathbf{I}_1$ in isolation, which never happens in normal use.

Per-input single-step row sums. The simpler invariant $\sum_b \mathbb{T}''_{Xb} = 1$ does *not* hold for any of the body rows. This is a state-encoding artefact of folding the Bernoulli- r extension-vs-exit decision into the same outgoing edges that carry the destination-singlet factor. An alternative WFST that exposes the extension event as a silent “decision” state would be locally row-stochastic at the cost of more states. The compact 6×6 form trades local stochasticity for a smaller graph; the global property $\text{Singlet} \circ \mathbb{T}'' = \tau''$ (with τ'' row-stochastic) is preserved either way.

A.4.8 Testable Invariants

The following invariants hold for any valid parameters $0 < \lambda, \mu, T > 0$, $0 \leq r < 1$ and can be verified numerically from the WFST tensor code.

1. **Fragment-boundary posterior:** $r/p + (1-r)\kappa/p = 1$.
2. **Pair HMM reconstruction:** For each state a in $\{\mathbf{S}, \mathbf{M}, \mathbf{I}_0, \mathbf{I}_1, \mathbf{D}\}$, reconstruct the Pair HMM row by multiplying WFST entries by their singlet factors (per the rules above), and verify $\sum_b \tau''_{ab} = 1$.
3. **Mixture decomposition:** $\mathbb{T}''_{\mathbf{M}\mathbf{M}} = r/p + [(1-r)\kappa/p](1-\beta)\alpha$ (and analogously for $\mathbf{D} \rightarrow \mathbf{D}$ with γ). Both mixture components are non-negative.
4. **TKF91 limit:** At $r = 0$: $p = \kappa$, the $\mathbf{I}_0$ and $\mathbf{I}_1$ rows become identical (and equal the TKF91 $\mathbf{I}$ row), and the 6×6 WFST reduces to the TKF91 5×5 WFST.
5. **E-column dichotomy:** $\mathbb{T}''_{XE} = 1-\beta$ for $X \in \{\mathbf{S}, \mathbf{M}, \mathbf{I}_1\}$ and $1-\gamma$ for $X = \mathbf{D}$, but $\mathbb{T}''_{\mathbf{I}_0\mathbf{E}} = (1-r)(1-\beta)$ scales linearly in $1-r$ because the $\mathbf{I}_0$ exit weight is divided by the smaller $\mathbf{S} \rightarrow \mathbf{E}$ singlet weight $1-\kappa$ rather than $(1-r)(1-\kappa)$.
6. **$\mathbf{I}_1$ column scaling:** $\mathbb{T}''_{\mathbf{M}\mathbf{I}_1} = (1-r)\beta$ and $\mathbb{T}''_{\mathbf{D}\mathbf{I}_1} = (1-r)\gamma$ scale linearly in $(1-r)$, reflecting that inserts can only begin at fragment boundaries.
7. **Structural zeros:** $\mathbb{T}''_{\mathbf{S}\mathbf{I}_1} = \mathbb{T}''_{\mathbf{M}\mathbf{I}_0} = \mathbb{T}''_{\mathbf{I}_1\mathbf{I}_0} = \mathbb{T}''_{\mathbf{D}\mathbf{I}_0} = 0$, reflecting that $\mathbf{I}_0$ is reachable only before any ancestor has been consumed and $\mathbf{I}_1$ only after.

A.4.9 Boundary-sampler factorisation of $\mathbb{T}''$

The 6×6 WFST admits a clean factorisation into a boundary sampler, a TKF91 WFST on fragment-level links, and a fragment-expansion projection. This makes the “TKF91 hiding inside TKF92” structure explicit and exposes a degenerate Doob h -transform that is worth recording.

Boundary sampler B . A row-stochastic WFST that reads ancestor residues (without fragment markers) and outputs a per-residue mark in $\{N, E\}$ (N = “new fragment,” E = “extends current fragment”):

- From start $\mathbf{S}_B$ on the first residue: emit N with weight 1.
- From the working state $\mathbf{M}_B$ on subsequent residues: emit E with weight r/p , or N with weight $(1-r)\kappa/p$. Row sum on a residue input: $r/p + (1-r)\kappa/p = p/p = 1$.
- EOS transitions to $\mathbf{E}_B$ with weight 1 from either state.

Trivial Doob h -transform. In TKF92 the residue identities are i.i.d. $\sim \pi$ *independent of* the boundary indicators, so the posterior $P(\mathbf{b}|\mathbf{x})$ equals the prior $P(\mathbf{b})$ residue-by-residue. The Doob h -transform $h(b, x_i) = P(x_i|b)/P(x_i)$ that would in general be required to make a posterior boundary-sampler locally row-stochastic reduces to $h \equiv 1$, leaving B a strict prior sampler. For models where the per-residue emission distribution does depend on fragment-context (some MixDom variants), B would carry a non-trivial h and would no longer be locally row-stochastic without explicit normalisation.

Factorisation. Let W_{91}^{link} denote the TKF91 conditional WFST (Section A.1.6) acting on fragment-level links (one symbol per ancestor fragment-start), and let F denote the deterministic expand-and-erase projection that inflates each TKF91 link into a $1+\text{Geom}(r)$ -length descendant fragment of residues (drawn from $\exp(QT)$ for match links, from π for insert links, or empty for delete links) and discards the boundary marks from the output. Then

$$\mathbb{T}'' = B \circ W_{91}^{\text{link}} \circ F.$$

Row-by-row verification. The M-row entries of $\mathbb{T}''$ decompose as

$$\begin{aligned} \mathbb{T}_{\text{MM}}'' &= \underbrace{r/p \cdot 1}_{B:E} + \underbrace{(1-r)\kappa/p \cdot (1-\beta)\alpha}_{B:N} \\ &= (r + (1-r)(1-\beta)\kappa\alpha)/p \quad \checkmark \\ \mathbb{T}_{\text{MI}_1}'' &= (1-r)\beta \quad (\text{new descendant link inserted between ancestor fragments}) \quad \checkmark \\ \mathbb{T}_{\text{MD}}'' &= (1-r)\kappa/p \cdot (1-\beta)(1-\alpha) = (1-r)(1-\beta)\kappa(1-\alpha)/p \quad \checkmark \\ \mathbb{T}_{\text{ME}}'' &= 1-\beta \quad \checkmark \end{aligned}$$

The D-row verifies identically with γ in place of β . The I_1 -row is the same TKF91 link-level row reached after some non-empty ancestor prefix, weighted by the same B structure as M. The S and I_0 rows are the boundary case $b_1 = N$ (the first residue is forced new), which removes the r/p self-extend branch and recovers the no- r entries given in the table.

Detailed balance. Detailed balance holds at the augmented $(x, \mathbf{b}, y, \mathbf{b}')$ level: W_{91}^{link} is reversible under π (Holmes-Bruno), B is a proper conditional sampler with $h \equiv 1$, and marginalising the boundary marks $\mathbf{b}, \mathbf{b}'$ recovers the TKF92 joint pair HMM τ'' exactly.

Consequence. For MCMC inference under TKF92 one can augment the state with per-residue fragment-boundary indicators $\mathbf{b}$ at each tree node (sampled cheaply from $\text{Bernoulli}(r/p)$, *independently of the substitution matrix*) and run TKF91 moves at the link level. This decomposition is the operational realisation of the “fragments-as-TKF91-links” view originally outlined in Thorne, Kishino, and Felsenstein (1992). At $r = 0$ the B step is trivial and the construction reduces to plain TKF91.

A.5 Equal-Rate Limits for TKF Parameters

For biologically realistic equilibrium lengths under TKF91/TKF92, one is typically driven into the near-critical regime $\lambda \approx \mu$ ($\kappa \approx 1$). Under TKF91, the expected equilibrium sequence length is $\kappa/(1-\kappa)$ characters. Under TKF92 with fragment extension probability r , it is $\kappa/((1-r)(1-\kappa))$ characters (over $\kappa/(1-\kappa)$ fragments). The average protein in UniProtKB/Swiss-Prot is approximately 360 amino acids long (61). Even under TKF92 with a mean indel length of $1/(1-r) \approx 3$ residues (matching the empirically observed predominance of 1–5 residue indels in protein evolution (1, 11)), the required κ is $120/121 \approx 0.992$. Under TKF91, $\kappa = 360/361 \approx 0.997$. For DNA sequences (a typical prokaryotic ORF or eukaryotic intron-free coding sequence is ~ 1000 bp (68); intron-bearing eukaryotic genes and syntenic genomic regions can be much longer), $\kappa > 0.999$. Since $1-\kappa = (\mu-\lambda)/\mu$, these biological sequence lengths force λ very close to μ .

The TKF91 transition probabilities α, β, γ and the BDI score/sufficient statistics all contain factors of $(\mu-\lambda)$ in denominators. When $\lambda \rightarrow \mu$ ($\kappa \rightarrow 1$), these become 0/0 indeterminate forms.

This section derives the L'Hôpital limits, providing numerically stable expressions in the equal-rate regime.

Let $\varepsilon = \mu - \lambda \rightarrow 0$ and $s = \mu T$. We parameterize $\lambda = \mu - \varepsilon$ and take $\varepsilon \rightarrow 0^+$.

Throughout, we write $\Phi = 1 - \alpha$ for the complement of the survival probability (Φ is well-defined at any (λ, μ)). In the equal-rate regime, where $s = \mu T$ and $\alpha = e^{-s}$, we additionally define:

$$\phi = \Phi/s = (1 - e^{-s})/s$$

Note $\phi(0) = 1$ and $\phi(s) \rightarrow 0$ as $s \rightarrow \infty$.

A.5.1 TKF91 Transition Parameters

α (survival probability). $\alpha = e^{-\mu T}$ has no dependence on λ ; no singularity.

β (insertion continuation probability).

$$\beta = \frac{\lambda(e^{-\lambda T} - e^{-\mu T})}{\mu e^{-\lambda T} - \lambda e^{-\mu T}} \quad (\text{A.31})$$

Write $e^{-\lambda T} = \alpha e^{\varepsilon T}$ and let $u = \varepsilon T$. Then:

$$\begin{aligned} \text{Numerator} &= (\mu - \varepsilon) \alpha (e^u - 1) \\ \text{Denominator} &= \alpha (\mu e^u - \mu + \varepsilon) \end{aligned}$$

Cancelling α and expanding $e^u = 1 + u + u^2/2 + \dots$ with $u = \varepsilon T$:

$$\begin{aligned} \text{Numer} &= (\mu - \varepsilon)(\varepsilon T + \varepsilon^2 T^2/2 + \dots) = \varepsilon(\mu T + O(\varepsilon)) \\ \text{Denom} &= \mu \varepsilon T + \varepsilon(1 + O(\varepsilon)) = \varepsilon(\mu T + 1 + O(\varepsilon)) \end{aligned}$$

After cancelling ε :

$$\boxed{\beta \xrightarrow{\lambda \rightarrow \mu} \frac{s}{1+s}} \quad (\text{A.32})$$

γ (orphan insertion probability). Using $\gamma = 1 - \mu\beta/(\lambda(1 - \alpha))$:

$$\boxed{\gamma \xrightarrow{\lambda \rightarrow \mu} 1 - \frac{1}{(1+s)\phi} = \frac{(1+s)\phi - 1}{(1+s)\phi}} \quad (\text{A.33})$$

Complementary parameters.

$$1 - \beta \rightarrow \frac{1}{1+s} \quad (\text{A.34})$$

$$1 - \gamma \rightarrow \frac{1}{(1+s)\phi} \quad (\text{A.35})$$

A.5.2 Score Derivatives: General Case

Define $\eta = e^{-\lambda T}$ and $\delta = \mu\eta - \lambda\alpha$ (the denominator of β). The score of the observed log-likelihood decomposes into 8 log-parameter groups. Below we list the derivatives of each log-factor with respect to λ and μ , from which the log-elasticities follow by multiplication:

$$\frac{\partial \log \xi}{\partial \log \lambda} = \lambda \frac{\partial \log \xi}{\partial \lambda}, \quad \frac{\partial \log \xi}{\partial \log \mu} = \mu \frac{\partial \log \xi}{\partial \mu}, \quad \frac{\partial \log \xi}{\partial \log(\lambda+\mu)} = \frac{\partial \log \xi}{\partial \log \lambda} + \frac{\partial \log \xi}{\partial \log \mu}$$

The third identity holds because differentiating with respect to $\log(\lambda+\mu)$ at fixed ratio $\kappa = \lambda/\mu$ is equivalent to scaling both rates by a common factor c and differentiating with respect to $\log c$, which gives $\lambda \partial_\lambda + \mu \partial_\mu$.

log α and $\log(1-\alpha)$ Since $\alpha = e^{-\mu T}$ depends only on μ :

$$\frac{\partial \log \alpha}{\partial \lambda} = 0, \quad \frac{\partial \log \alpha}{\partial \mu} = -T \quad (\text{A.36})$$

$$\frac{\partial \log(1-\alpha)}{\partial \lambda} = 0, \quad \frac{\partial \log(1-\alpha)}{\partial \mu} = \frac{T\alpha}{1-\alpha} = \frac{T\alpha}{\Phi} \quad (\text{A.37})$$

log β and $\log(1-\beta)$ From $\log \beta = \log \lambda + \log(\eta - \alpha) - \log \delta$:

$$\frac{\partial \log \beta}{\partial \lambda} = \frac{1}{\lambda} - \frac{T\eta}{\eta - \alpha} + \frac{T\mu\eta + \alpha}{\delta} \quad (\text{A.38})$$

$$\frac{\partial \log \beta}{\partial \mu} = \frac{T\alpha}{\eta - \alpha} - \frac{\eta + T\lambda\alpha}{\delta} \quad (\text{A.39})$$

Both $(\eta - \alpha) \rightarrow 0$ and $\delta \rightarrow 0$ as $\lambda \rightarrow \mu$, so each term individually diverges; the combination is finite (Section A.5.3).

For $\log(1-\beta)$, using the chain rule $\partial \log(1-\beta) = -\frac{\beta}{1-\beta} \partial \log \beta$:

$$\frac{\partial \log(1-\beta)}{\partial \lambda} = -\frac{\beta}{1-\beta} \cdot \frac{\partial \log \beta}{\partial \lambda} \quad (\text{A.40})$$

$$\frac{\partial \log(1-\beta)}{\partial \mu} = -\frac{\beta}{1-\beta} \cdot \frac{\partial \log \beta}{\partial \mu} \quad (\text{A.41})$$

log $(1-\gamma)$ and $\log \gamma$ From $1 - \gamma = \mu\beta/(\lambda(1-\alpha))$, so $\log(1-\gamma) = \log \mu + \log \beta - \log \lambda - \log(1-\alpha)$:

$$\frac{\partial \log(1-\gamma)}{\partial \lambda} = \frac{\partial \log \beta}{\partial \lambda} - \frac{1}{\lambda} \quad (\text{A.42})$$

$$\frac{\partial \log(1-\gamma)}{\partial \mu} = \frac{1}{\mu} + \frac{\partial \log \beta}{\partial \mu} - \frac{T\alpha}{1-\alpha} \quad (\text{A.43})$$

For $\log \gamma$:

$$\frac{\partial \log \gamma}{\partial \lambda} = -\frac{1-\gamma}{\gamma} \cdot \frac{\partial \log(1-\gamma)}{\partial \lambda} \quad (\text{A.44})$$

$$\frac{\partial \log \gamma}{\partial \mu} = -\frac{1-\gamma}{\gamma} \cdot \frac{\partial \log(1-\gamma)}{\partial \mu} \quad (\text{A.45})$$

Summary Table: General Case

	$\log \alpha$	$\log(1-\alpha)$	$\log \beta$	$\log(1-\beta)$	$\log \gamma$	$\log(1-\gamma)$
$\frac{\partial}{\partial \log \lambda}$	0	0	$1 - \frac{s_\lambda \eta}{\eta - \alpha} + \frac{s_\lambda \mu \eta / \lambda + \lambda \alpha / \lambda}{\delta / \lambda}$	$-\frac{\beta}{1-\beta} \cdot [\cdot]_\beta^\lambda$	$-\frac{1-\gamma}{\gamma} \cdot [\cdot]_{1-\gamma}^\lambda$	$[\cdot]_\beta^\lambda - 1$
$\frac{\partial}{\partial \log \mu}$	$-s$	$\frac{s\alpha}{\Phi}$	$\frac{s\alpha}{\eta - \alpha} - \frac{\mu\eta + s\lambda\alpha}{\delta}$	$-\frac{\beta}{1-\beta} \cdot [\cdot]_\beta^\mu$	$-\frac{1-\gamma}{\gamma} \cdot [\cdot]_{1-\gamma}^\mu$	$1 + [\cdot]_\beta^\mu - \frac{s\alpha}{\Phi}$
$\frac{\partial}{\partial \log(\lambda+\mu)}$	(sum of above two rows)					

where $s_\lambda = \lambda T$, $s = \mu T$, and $[\cdot]_\xi^\theta$ denotes the $\partial \log \xi / \partial \log \theta$ entry.

The table is dense because each $\log \beta$ and $\log \gamma$ entry involves the singular terms $(\eta - \alpha)^{-1}$ and δ^{-1} . These singularities cancel in each entry, as we show next.

A.5.3 Score Derivatives: L'Hôpital Limits

We systematically expand each derivative to first order in $\varepsilon = \mu - \lambda$ and extract the finite limit. With $u = \varepsilon T$, the key expansions are:

$$\eta - \alpha = \alpha(e^u - 1) = \alpha(u + u^2/2 + \dots) \quad (\text{A.46})$$

$$\delta = \mu\eta - \lambda\alpha = \alpha\varepsilon(1 + s + O(\varepsilon)) \quad (\text{A.47})$$

and the Laurent expansion $1/(e^u - 1) = 1/u - 1/2 + u/12 - \dots$, giving:

$$\frac{T\eta}{\eta - \alpha} = \frac{Te^u}{e^u - 1} = \frac{1}{\varepsilon} + \frac{T}{2} + O(\varepsilon) \quad (\text{A.48})$$

$$\frac{T\alpha}{\eta - \alpha} = \frac{T}{e^u - 1} = \frac{1}{\varepsilon} - \frac{T}{2} + O(\varepsilon) \quad (\text{A.49})$$

$$\frac{T\mu\eta + \alpha}{\delta} = \frac{1}{\varepsilon} + \frac{sT}{2(1+s)} + O(\varepsilon) \quad (\text{A.50})$$

$$\frac{\eta + T\lambda\alpha}{\delta} = \frac{1}{\varepsilon} - \frac{sT}{2(1+s)} + O(\varepsilon) \quad (\text{A.51})$$

Derivation of (A.50): the numerator is $T\mu\eta + \alpha = \alpha(se^u + 1) = \alpha((1+s) + s\varepsilon T + O(\varepsilon^2))$ and the denominator is $\delta = \alpha\varepsilon((1+s) + s\varepsilon T/2 + O(\varepsilon^2))$ from (A.47). So:

$$\frac{T\mu\eta + \alpha}{\delta} = \frac{(1+s) + s\varepsilon T + \dots}{\varepsilon((1+s) + s\varepsilon T/2 + \dots)} = \frac{1}{\varepsilon} \cdot \frac{1 + \frac{s\varepsilon T}{1+s} + \dots}{1 + \frac{s\varepsilon T}{2(1+s)} + \dots}$$

The $O(\varepsilon)$ coefficients differ: $sT/(1+s)$ in the numerator vs. $sT/(2(1+s))$ in the denominator. Their difference gives the $O(1)$ correction: $1/\varepsilon + sT/(2(1+s)) + O(\varepsilon)$. Similarly for (A.51).

$\partial \log \beta / \partial \lambda$ and $\partial \log \beta / \partial \mu$ Using (A.38) with (A.48) and (A.50):

$$\begin{aligned} \frac{\partial \log \beta}{\partial \lambda} &= \frac{1}{\lambda} - \left(\frac{1}{\varepsilon} + \frac{T}{2} \right) + \left(\frac{1}{\varepsilon} + \frac{sT}{2(1+s)} \right) + O(\varepsilon) \\ &= \frac{1}{\mu} - \frac{T}{2} + \frac{sT}{2(1+s)} = \frac{1}{\mu} - \frac{T}{2(1+s)} \end{aligned}$$

Using (A.39) with (A.49) and (A.51):

$$\begin{aligned}\frac{\partial \log \beta}{\partial \mu} &= \left(\frac{1}{\varepsilon} - \frac{T}{2} \right) - \left(\frac{1}{\varepsilon} - \frac{sT}{2(1+s)} \right) + O(\varepsilon) \\ &= -\frac{T}{2} + \frac{sT}{2(1+s)} = -\frac{T}{2(1+s)}\end{aligned}$$

Converting to log-elasticities:

$$\boxed{\frac{\partial \log \beta}{\partial \log \lambda} \rightarrow \frac{s+2}{2(1+s)}, \quad \frac{\partial \log \beta}{\partial \log \mu} \rightarrow -\frac{s}{2(1+s)}} \quad (\text{A.52})$$

Check. As a consistency check, we verify that the sum of the two elasticities (which gives the response to uniformly scaling both rates at fixed $\kappa = \lambda/\mu$) matches the direct derivative of the limit formula. The sum is $(s+2)/(2(1+s)) + (-s)/(2(1+s)) = 1/(1+s)$. At the limit, $\beta = s/(1+s)$ where $s = \mu T$, so $d \log \beta / d \log s = 1/(1+s)$. ✓

$\partial \log(1-\beta)/\partial \log \lambda$ and $\partial \log(1-\beta)/\partial \log \mu$ Using $\beta/(1-\beta) \rightarrow s$ at equal rates:

$$\boxed{\frac{\partial \log(1-\beta)}{\partial \log \lambda} \rightarrow -\frac{s(s+2)}{2(1+s)}, \quad \frac{\partial \log(1-\beta)}{\partial \log \mu} \rightarrow \frac{s^2}{2(1+s)}} \quad (\text{A.53})$$

Check. Summing the two elasticities (uniform rate scaling): $(-s^2-2s+s^2)/(2(1+s)) = -s/(1+s)$. Direct: $d \log(1-\beta)/d \log s = -s/(1+s)$. ✓

$\partial \log(1-\gamma)/\partial \log \lambda$ and $\partial \log(1-\gamma)/\partial \log \mu$ From (A.42): $\partial \log(1-\gamma)/\partial \lambda = \partial \log \beta / \partial \lambda - 1/\lambda \rightarrow 1/\mu - T/(2(1+s)) - 1/\mu = -T/(2(1+s))$.

From (A.43): $\partial \log(1-\gamma)/\partial \mu = 1/\mu + \partial \log \beta / \partial \mu - T\alpha/\Phi \rightarrow 1/\mu - T/(2(1+s)) - T\alpha/\Phi$.

Converting:

$$\boxed{\frac{\partial \log(1-\gamma)}{\partial \log \lambda} \rightarrow -\frac{s}{2(1+s)}, \quad \frac{\partial \log(1-\gamma)}{\partial \log \mu} \rightarrow \frac{s+2}{2(1+s)} - \frac{s\alpha}{\Phi}} \quad (\text{A.54})$$

Check. Summing the two elasticities (uniform rate scaling): $-s/(2(1+s)) + (s+2)/(2(1+s)) - s\alpha/\Phi = 1/(1+s) - s\alpha/\Phi$. Direct: at the limit, $1-\gamma = s/((1+s)\Phi)$, so $\log(1-\gamma) = \log s - \log(1+s) - \log \Phi$. Under $s \rightarrow cs$: $d/d \log c|_{c=1} = 1 - s/(1+s) - s\alpha/\Phi = 1/(1+s) - s\alpha/\Phi$. ✓

$\partial \log \gamma / \partial \log \lambda$ and $\partial \log \gamma / \partial \log \mu$ With $R = (1-\gamma)/\gamma$ at equal rates. Since $1-\gamma = 1/((1+s)\phi)$:

$$R = \frac{1}{(1+s)\phi - 1}$$

$$\boxed{\frac{\partial \log \gamma}{\partial \log \lambda} \rightarrow \frac{Rs}{2(1+s)}, \quad \frac{\partial \log \gamma}{\partial \log \mu} \rightarrow -R \left(\frac{s+2}{2(1+s)} - \frac{s\alpha}{\Phi} \right)} \quad (\text{A.55})$$

Summary Table: L'Hôpital Limits At $\lambda = \mu$, with $s = \mu T$, $\alpha = e^{-s}$, $\Phi = 1 - \alpha$, $\phi = \Phi/s$, $R = 1/((1+s)\phi - 1)$:

	$\log \alpha$	$\log(1-\alpha)$	$\log \beta$	$\log(1-\beta)$	$\log \gamma$	$\log(1-\gamma)$
$\frac{\partial}{\partial \log \lambda}$	0	0	$\frac{s+2}{2(1+s)}$	$-\frac{s(s+2)}{2(1+s)}$	$\frac{Rs}{2(1+s)}$	$-\frac{s}{2(1+s)}$
$\frac{\partial}{\partial \log \mu}$	$-s$	$\frac{s\alpha}{\Phi}$	$-\frac{s}{2(1+s)}$	$\frac{s^2}{2(1+s)}$	$-R\left(\frac{s+2}{2(1+s)} - \frac{s\alpha}{\Phi}\right)$	$\frac{s+2}{2(1+s)} - \frac{s\alpha}{\Phi}$
$\frac{\partial}{\partial \log(\lambda+\mu)}$	$-s$	$\frac{s\alpha}{\Phi}$	$\frac{1}{1+s}$	$-\frac{s}{1+s}$	$R\left(\frac{s}{2(1+s)} - \frac{s+2}{2(1+s)} + \frac{s\alpha}{\Phi}\right)$	$\frac{1}{1+s} - \frac{s\alpha}{\Phi}$

All entries are finite. The third row equals the sum of the first two (as it must by construction). The $\log \alpha$ and $\log(1-\alpha)$ columns are trivially continuous (no λ dependence). The $\log \beta$ and $\log(1-\beta)$ entries have a particularly clean form involving only $s/(1+s)$. The $\log \gamma$ entries additionally involve $\alpha/\Phi = e^{-s}/(1 - e^{-s}) = 1/(e^s - 1)$ and the ratio $R = 1/((1+s)\phi - 1)$.

A.5.4 BDI Sufficient Statistics

The EM-relevant quantities are the posterior expectations $\mathbb{E}[B|i, j, T]$, $\mathbb{E}[D|i, j, T]$, $\mathbb{E}[S|i, j, T]$, recovered from the score via the conservation law $i + B - D = j$:

$$\mathbb{E}[S] = \frac{j - i + \mu \partial_\mu \log P_{ij} - \lambda \partial_\lambda \log P_{ij} - \lambda T}{\lambda - \mu} \quad (\text{A.56})$$

$$\mathbb{E}[B] = \lambda \partial_\lambda \log P_{ij} + \lambda \mathbb{E}[S] + \lambda T \quad (\text{A.57})$$

$$\mathbb{E}[D] = \mu \partial_\mu \log P_{ij} + \mu \mathbb{E}[S] \quad (\text{A.58})$$

The score $\partial_\lambda \log P_{ij}$ itself is computed from the transition count groups:

$$\partial_\lambda \log P_{ij} = \sum_{\xi \in \{\alpha, 1-\alpha, \beta, \dots\}} n_{\log \xi} \cdot \frac{\partial \log \xi}{\partial \lambda}$$

Each derivative $\partial \log \xi / \partial \lambda$ has a finite L'Hôpital limit (Section A.5.3), so the score itself has a finite limit. The denominator $\lambda - \mu$ in (A.56) is what creates the 0/0 for $\mathbb{E}[S]$.

The $\mathbb{E}[S]$ singularity. The numerator of (A.56) also vanishes at $\lambda = \mu$, since the complete-data log-likelihood for $\nu = \lambda$ is $\ell = B \log \lambda + D \log \mu - (\lambda + \mu)S - \lambda T + c$, and at $\lambda = \mu$ the numerator becomes $j - i + \mathbb{E}[D] - \mathbb{E}[B] = 0$ by conservation.

To resolve the 0/0, write $f(\lambda) = j - i + \mu \partial_\mu \log P - \lambda \partial_\lambda \log P - \lambda T$ and $g(\lambda) = \lambda - \mu$. By L'Hôpital ($g' = 1$):

$$\lim_{\lambda \rightarrow \mu} \mathbb{E}[S] = f'(\mu) = \mu \partial_{\lambda\mu}^2 \log P - \partial_\lambda \log P - \mu \partial_{\lambda^2}^2 \log P - T \quad (\text{A.59})$$

where all derivatives are evaluated at $\lambda = \mu$.

The first and second derivatives of $\log P$ that appear in (A.59) can be expressed in terms of the transition count groups and the L'Hôpital-limited score derivatives from Section A.5.3, together with their first-order corrections in ε . Specifically, the second derivatives of $\log P$ require the $O(\varepsilon)$ terms in the Laurent expansions (A.48)–(A.51).

Alternative: direct gradient on μ . At equal rates, the BDI has a single parameter μ . The gradient is:

$$\left. \frac{\partial \log P}{\partial \mu} \right|_{\lambda=\nu=\mu} = \sum_{\xi} n_{\log \xi} \left(\frac{\partial \log \xi}{\partial \lambda} + \frac{\partial \log \xi}{\partial \mu} \right) = \frac{\mathbb{E}[B+D]}{\mu} - 2\mathbb{E}[S] - T$$

This is well-defined using the L'Hôpital limits, and gives the combined quantity $\mathbb{E}[B+D] - \mu(2\mathbb{E}[S] + T)$ without needing to separate $\mathbb{E}[B]$, $\mathbb{E}[D]$, $\mathbb{E}[S]$ individually. The M-step (A.60) sets this to zero: $\hat{\mu} = (\mathbb{E}[B] + \mathbb{E}[D]) / (2\mathbb{E}[S] + T)$.

To *break out* of equal rates (determine whether λ should differ from μ), we need the individual $\partial \log P / \partial \lambda$ and $\partial \log P / \partial \mu$, which the table in Section A.5.3 provides, or we need to use additional information from the ancestral length distribution.

A.5.5 Direct BDI Rate EM at Equal Rates

If one ignores the stationary root-length prior and considers only the conditional BDI bridge model with $\lambda = \nu = \mu$, the complete-data log-likelihood (A.10) reduces to:

$$\ell = (B + D) \log \mu - 2\mu S - \mu T + c$$

This has a single free parameter μ . Setting $\partial \ell / \partial \mu = 0$:

$$\boxed{\hat{\mu} = \frac{B + D}{2S + T}} \tag{A.60}$$

where B , D , S are understood as posterior expectations.

Implication. At equal rates, the M-step collapses to estimating a single rate from the total event count and total sojourn time.

A.5.6 $\log \kappa$ and $\log(1-\kappa)$ Derivatives

The $\kappa = \lambda/\mu$ and $(1-\kappa)$ groups are not among the six (α, β, γ) factors but appear in the transition matrix. At $\kappa = 1$:

$$\frac{\partial \log \kappa}{\partial \lambda} = \frac{1}{\lambda} \rightarrow \frac{1}{\mu}, \quad \frac{\partial \log \kappa}{\partial \mu} = -\frac{1}{\mu}$$

No singularity. However, $\log(1-\kappa)$ diverges:

$$\frac{\partial \log(1-\kappa)}{\partial \lambda} = \frac{-\kappa}{(1-\kappa)\lambda} \rightarrow -\infty$$

Unlike the conditional WFST factors α, β, γ , the stationary-length factors κ and $1-\kappa$ do not admit a regular equal-rate limit in the full joint TKF model. As $\kappa \rightarrow 1$, the equilibrium length distribution ceases to be normalizable, and the $\log(1-\kappa)$ contribution diverges. Accordingly, the L'Hôpital limits derived above apply to the conditional bridge/WFST quantities, whereas the full stationary joint model must either remain in the subcritical regime $\kappa < 1$ or be reparameterized with a nonstationary root-length model.

A.5.7 Joint vs Conditional Likelihoods

The TKF91 transition matrix includes factors of $\kappa = \lambda/\mu$ and $(1-\kappa)$ arising from the geometric(κ) prior on ancestor length: $\Pr(|\text{anc}| = i) = \kappa^i(1-\kappa)$. This prior is part of the *joint* likelihood $P(x, y)$ but can be factored out with respect to the ancestor-length prior to yield an *ancestor length-conditioned* likelihood $P_{\text{cond}}(x, y \mid |x| = i) = P(x, y)/[\kappa^i(1-\kappa)]$.

The four regimes. Of the four combinations of $\kappa < 1$ vs $\kappa = 1$ and joint vs ancestor length-conditioned, only one is degenerate:

	$\kappa < 1$	$\kappa = 1$
Joint $P(x, y)$	well-defined	degenerate ($\rightarrow 0$)
Ancestor length-conditioned P_{cond}	well-defined	well-defined

At $\kappa = 1$, the geometric prior places vanishing mass on finite sequences, so $P(x, y) \rightarrow 0$ and $\log P \rightarrow -\infty$. The ancestor length-conditioned WFST removes the κ and $(1-\kappa)$ factors corresponding to the stationary length distribution from the transition matrix, leaving only (α, β, γ) -dependent terms. These have finite L'Hôpital limits (Sections A.5.3–A.5.4), so P_{cond} is finite and well-defined even at $\kappa = 1$.

BDI sufficient statistics. The BDI sufficient statistics $\mathbb{E}[B]$, $\mathbb{E}[D]$, $\mathbb{E}[S]$ are most naturally recovered from the ancestor length-conditioned WFST, whose score depends only on the six TKF bridge groups $(\alpha, 1-\alpha, \beta, 1-\beta, \gamma, 1-\gamma)$. In the joint model, additional contributions from the ancestral length prior appear through the κ and $1-\kappa$ factors. These must be included in the M-step when optimizing the full stationary TKF likelihood, but they are conceptually distinct from the bridge statistics themselves.

Equations (A.15)–A.17 give the correct expectations for the BDI sufficient statistics in terms of the counts $\hat{n}_{ij}$ regardless of whether those counts were computed using the joint Pair HMM or the conditionally-normalized transducer, as long as the WFST transition matrix $\mathbb{T}$ is used for the counts $\mathcal{C}_{ij}^\xi = \xi \partial_\xi \log \mathbb{T}_{ij}$ (e.g. using the tables in Section A.5.2 or Section A.5.3). The M-step for the joint likelihood includes additional terms from the κ and $1-\kappa$ factors, leading to a coupled system of equations that can be solved in closed form (Section A.1.8).

Closed-form M-steps. Both formulations yield closed-form M-steps:

- *Ancestor-length conditioned:* pure BDI exponential family. For the conditional model away from the $\kappa = 1$ singularity of the joint prior, setting $\partial Q_{\text{cond}}/\partial \lambda = 0$ gives

$$\hat{\lambda} = \frac{\max(0, \mathbb{E}[B] + \alpha_\lambda - 1)}{\mathbb{E}[S] + T + \beta_\lambda}, \quad \hat{\mu} = \frac{\max(0, \mathbb{E}[D] + \alpha_\mu - 1)}{\mathbb{E}[S] + \beta_\mu} \quad (\text{A.61})$$

where (α, β) are Gamma prior parameters.

- *Joint* ($\kappa < 1$): includes $L \log \kappa + M \log(1-\kappa)$. Since, for a single sequence-pair, $L = i$ (ancestor length, exact) and $M = 1$ (one End transition, exact), with corresponding accumulated values across multiple sequence-pairs, the optimality conditions $\partial Q/\partial \lambda = 0$ and $\partial Q/\partial \mu = 0$ form a coupled system in (λ, μ) . These can be reduced to a single quadratic equation in κ with a closed-form solution, which can be substituted back to get $\hat{\lambda}$ and $\hat{\mu}$, as described in Section A.1.8.

In both cases, the M-step maximizes the correct Q -function, so EM monotonicity is guaranteed for the corresponding tracked objective ($\log P_{\text{cond}}$ for the ancestor length-conditioned model, $\log P$ for the full joint model).

A.5.8 Irreversible Models and the $\lambda > \mu$ Regime

The $\kappa > 1$ regime. When $\lambda > \mu$, the ratio $\kappa = \lambda/\mu > 1$ and the geometric(κ) prior on ancestor length is not normalizable: $\sum_{i=0}^{\infty} \kappa^i (1 - \kappa)$ diverges. Therefore the joint likelihood $P(x, y)$ does not exist. However, the ancestor length-conditioned likelihood $P_{\text{cond}}(x, y \mid |x|)$ is well-defined for all $\kappa > 0$. The transition parameters α, β, γ are continuous functions of (λ, μ, T) with no singularity at $\lambda = \mu$ or $\lambda > \mu$. The BDI sufficient statistics $\mathbb{E}[B]$, $\mathbb{E}[D]$, $\mathbb{E}[S]$ remain well-defined, and the conditioned M-step (A.61) applies unchanged: it is unconstrained in λ and μ and permits $\hat{\lambda} > \hat{\mu}$ if the data support it.

Irreversible substitution models. For completeness, our implementations support dropping the reversibility assumption for the substitution CTMC. The integrals described in Section A.1.2 and in (31) exploit the symmetric factorization of R using π -weighted similarity, which yields real eigenvalues. In the irreversible case, R is a general rate matrix with no detailed-balance constraint. The eigendecomposition $R = V \text{diag}(d) V^{-1}$ may have complex eigenvalues. Assuming R is diagonalizable, the spectral formulae for the integrals of Section A.1.2 carry over, with V^{-1} replacing $V^{\top}$ and complex arithmetic throughout:

$$I_{ij}^{ab}(T) = \sum_{k,l} V_{ak} V_{ki}^{-1} \frac{e^{d_k T} - e^{d_l T}}{d_k - d_l} V_{jl} V_{lb}^{-1}$$

with the convention $(e^{d_k T} - e^{d_l T})/(d_k - d_l) = T e^{d_k T}$ when $d_k = d_l$. The result is real (complex eigenvalues come in conjugate pairs).

Fixed-point iteration for π . In the reversible case, the equilibrium distribution π is constrained to be the stationary distribution of R by detailed balance, and may either be fixed a priori or updated jointly with the reversible rate parameters during the M-step. In the irreversible case, π must be recomputed as the left eigenvector of R (with eigenvalue 0) after each M-step update of R from the bridge-expectation sufficient statistics. This gives an ECM (Expectation-Conditional-Maximization) scheme (46): the E-step uses the current (R, π) ; the M-step updates R , then π is recomputed from the updated R . Standard monotonicity results apply providing each conditional maximization step is well-defined.

Differentiability. Standard autograd libraries provide general eigendecomposition and matrix-exponential primitives (with Fréchet derivatives). The eigendecomposition route is convenient for EM-style closed-form updates, while gradient-based optimization is often more naturally based on the matrix exponential. We do not rely on differentiating through eigenvectors here.

A.5.9 Summary of Limits

Quantity	Formula	Limit as $\lambda \rightarrow \mu$
α	$e^{-\mu T}$	e^{-s} (continuous)
β	$\frac{\lambda(\eta-\alpha)}{\mu\eta-\lambda\alpha}$	$\frac{s}{1+s}$
$1 - \beta$		$\frac{1}{1+s}$
γ	$1 - \frac{\mu\beta}{\lambda(1-\alpha)}$	$1 - \frac{1}{(1+s)\phi}$
κ	λ/μ	1
$\partial_{\log \lambda} \log \beta$	(A.38)	$\frac{s+2}{2(1+s)}$
$\partial_{\log \mu} \log \beta$	(A.39)	$-\frac{s}{2(1+s)}$
$\hat{\mu}_{\text{MLE}}$		$\frac{\mathbb{E}[B] + \mathbb{E}[D]}{2\mathbb{E}[S] + T}$

where $s = \mu T$ throughout.

A.6 TKF91 Score Function

We derive the explicit partial derivatives needed to compute the observed-data score from Pair HMM transition counts, as described in Section A.1.8.

A.6.1 Derivatives of TKF parameters

Define $\eta = e^{-\lambda T}$ (so that $\alpha = e^{-\mu T}$ and $\eta = e^{-\lambda T}$) and $\Delta = \mu\eta - \lambda\alpha$ (the denominator of β).

$$\alpha = e^{-\mu T}.$$

$$\frac{\partial \log \alpha}{\partial \lambda} = 0, \quad \frac{\partial \log \alpha}{\partial \mu} = -T$$

$\beta = \lambda(\eta - \alpha)/\Delta$. Using $\log \beta = \log \lambda + \log(\eta - \alpha) - \log \Delta$:

$$\begin{aligned} \frac{\partial \log \beta}{\partial \lambda} &= \frac{1}{\lambda} - \frac{T\eta}{\eta - \alpha} + \frac{T\mu\eta + \alpha}{\Delta} \\ \frac{\partial \log \beta}{\partial \mu} &= \frac{T\alpha}{\eta - \alpha} - \frac{\eta + T\lambda\alpha}{\Delta} \end{aligned}$$

$\gamma = 1 - \mu\beta/(\lambda(1 - \alpha))$. Using $\log(1 - \gamma) = \log \mu + \log \beta - \log \lambda - \log(1 - \alpha)$:

$$\begin{aligned} \frac{\partial \log(1 - \gamma)}{\partial \lambda} &= \frac{\partial \log \beta}{\partial \lambda} - \frac{1}{\lambda} \\ \frac{\partial \log(1 - \gamma)}{\partial \mu} &= \frac{1}{\mu} + \frac{\partial \log \beta}{\partial \mu} - \frac{T\alpha}{1 - \alpha} \end{aligned}$$

and $\frac{\partial \log \gamma}{\partial \theta} = -\frac{1-\gamma}{\gamma} \frac{\partial \log(1-\gamma)}{\partial \theta}$ for $\theta \in \{\lambda, \mu\}$.

$$\kappa = \lambda/\mu.$$

$$\frac{\partial \log \kappa}{\partial \lambda} = \frac{1}{\lambda}, \quad \frac{\partial \log \kappa}{\partial \mu} = -\frac{1}{\mu}$$

Complementary parameters. For any parameter ξ : $\frac{\partial \log(1-\xi)}{\partial \theta} = -\frac{\xi}{1-\xi} \frac{\partial \log \xi}{\partial \theta}$.

Table A.1: Correspondence between transition counts, complete-data log-likelihood terms, and lineage fates.

Log-likelihood term	Coefficient	Interpretation
$\log \alpha$	$n_{SM} + n_{MM} + n_{DM} + n_{IM}$	Founder survival (match)
$\log(1 - \alpha)$	$n_{SD} + n_{MD} + n_{DD} + n_{ID}$	Founder death (delete)
$\log(1 - \gamma)$	$n_{DM} + n_{DD} + n_{DE}$	Lineage extinction (no $D \rightarrow I$)
$\log \gamma$	n_{DI}	Youngest orphan ($D \rightarrow I$)
$\log \beta$	$n_{SI} + n_{MI} + n_{II}$	Further offspring (insertions continue)
$\log(1 - \beta)$	$n_{IM} + n_{ID} + n_{IE}$	No further offspring (insertions end)

A.6.2 Observed-data score

Using the transition count decomposition from Table A.1, the Pair HMM log-likelihood under the *conditional* model $P(y|x)$ is

$$\ell_{\text{cond}} = n_{\log \alpha} \log \alpha + n_{\log(1-\alpha)} \log(1-\alpha) + n_{\log \beta} \log \beta + n_{\log(1-\beta)} \log(1-\beta) + n_{\log \gamma} \log \gamma + n_{\log(1-\gamma)} \log(1-\gamma)$$

where we have suppressed the κ and $(1 - \kappa)$ terms since they cancel in the conditional likelihood (they appear symmetrically in the transition probabilities and the ancestral sequence prior).

The score with respect to λ is

$$\frac{\partial \ell_{\text{cond}}}{\partial \lambda} = \sum_{\xi} n_{\xi} \frac{\partial \xi}{\partial \lambda}$$

where ξ ranges over $\{\log \alpha, \log(1 - \alpha), \log \beta, \log(1 - \beta), \log \gamma, \log(1 - \gamma)\}$, and similarly for μ .

A.6.3 Joint likelihood correction

For the *joint* model $P(x, y)$, the log-likelihood includes the ancestral sequence prior

$$\ell_{\text{joint}} = \ell_{\text{cond}} + |x| \log \kappa + \log(1 - \kappa) + \sum_i n_i \log \pi_i$$

where $|x|$ is the ancestor length, n_i counts character i in the ancestor, and the $\kappa/1 - \kappa$ terms come from the geometric length distribution. The additional score contributions are

$$\frac{\partial}{\partial \lambda} [|x| \log \kappa + \log(1 - \kappa)] = \frac{|x|}{\lambda} - \frac{1}{\mu - \lambda}, \quad \frac{\partial}{\partial \mu} [|x| \log \kappa + \log(1 - \kappa)] = -\frac{|x|}{\mu} + \frac{1}{\mu - \lambda}$$

A.6.4 Recovering sufficient statistics and M-step

The gradients $\partial \ell / \partial \lambda$ and $\partial \ell / \partial \mu$ relate to the BDI sufficient statistics via

$$\frac{\partial \ell}{\partial \lambda} = \frac{\mathbb{E}[B]}{\lambda} - \mathbb{E}[S] - T, \quad \frac{\partial \ell}{\partial \mu} = \frac{\mathbb{E}[D]}{\mu} - \mathbb{E}[S]$$

Combined with the conservation law (A.12), the closed-form M-step updates are $\lambda_{\text{new}} = \mathbb{E}[B] / \mathbb{E}[S]$ and $\mu_{\text{new}} = \mathbb{E}[D] / \mathbb{E}[S]$.

Alternatively, gradient ascent can be performed directly using the score, without needing to recover the individual sufficient statistics.

A.7 General BDI Sufficient Statistics

We derive the endpoint-conditioned sufficient statistics for the general linear BDI process with birth rate λ , death rate μ , and immigration rate ν (not necessarily equal to λ), conditioned on $X(0) = i$ and $X(T) = j$.

A.7.1 Complete-data log-likelihood

The complete-data log-likelihood for a path is

$$\log P(\mathbb{X}) = B_{\text{ind}} \log \lambda + B_{\text{imm}} \log \nu + D \log \mu - (\lambda + \mu)S - \nu T + c$$

with sufficient statistics B_{ind} (individual births), B_{imm} (immigrations), D (deaths), and $S = \int_0^T X(t) dt$ (time-integrated population).

A.7.2 Score equations

Differentiating with respect to (λ, μ, ν) :

$$\begin{aligned} \frac{\partial}{\partial \lambda} \log P_{ij} &= \frac{\mathbb{E}[B_{\text{ind}}]}{\lambda} - \mathbb{E}[S] \\ \frac{\partial}{\partial \mu} \log P_{ij} &= \frac{\mathbb{E}[D]}{\mu} - \mathbb{E}[S] \\ \frac{\partial}{\partial \nu} \log P_{ij} &= \frac{\mathbb{E}[B_{\text{imm}}]}{\nu} - T \end{aligned}$$

A.7.3 Conservation law

Since population changes only by births and deaths, $i + B_{\text{ind}} + B_{\text{imm}} - D = j$, giving

$$\mathbb{E}[B_{\text{ind}}] + \mathbb{E}[B_{\text{imm}}] - \mathbb{E}[D] = j - i \tag{A.62}$$

A.7.4 Closed-form solution

Solving the system of four equations (three score equations plus conservation) in four unknowns, using $\lambda \neq \mu$:

$$\begin{aligned} \mathbb{E}[B_{\text{imm}}] &= \nu \frac{\partial}{\partial \nu} \log P_{ij} + \nu T \\ \mathbb{E}[S] &= \frac{j - i + \mu \frac{\partial}{\partial \mu} \log P_{ij} - \lambda \frac{\partial}{\partial \lambda} \log P_{ij} - \nu \frac{\partial}{\partial \nu} \log P_{ij} - \nu T}{\lambda - \mu} \\ \mathbb{E}[B_{\text{ind}}] &= \lambda \frac{\partial}{\partial \lambda} \log P_{ij} + \lambda \mathbb{E}[S] \\ \mathbb{E}[D] &= \mu \frac{\partial}{\partial \mu} \log P_{ij} + \mu \mathbb{E}[S] \end{aligned}$$

A.7.5 Transition probability for general BDI

For the general linear BDI process with per-capita birth rate λ , per-capita death rate μ , and constant immigration rate ν (where ν is a free parameter, not necessarily 0 or λ), the transition probability $P_{ij}(T) = P(X(T) = j \mid X(0) = i)$ is known in closed form (36).

Using the α and β from Section A.1.3, define $\rho = \nu/\lambda$ (the immigration-to-birth ratio). The transition probability factorizes as:

$$P_{ij}(T) = \underbrace{P_{ij}^{(0)}(T)}_{\text{no-immigration}} \cdot \underbrace{R_{ij}(T)}_{\text{immigration factor}} \quad (\text{A.63})$$

The no-immigration factor is

$$P_{ij}^{(0)}(T) = \sum_{k=0}^{\min(i,j)} \binom{i}{k} \binom{j-1}{k-1} \alpha^k (1-\alpha)^{i-k} (1-\gamma)^{i-k} \gamma^{j-k} (1-\beta)^{k+1} \beta^{j-k-1} \cdot \delta_{j>0} \quad (\text{A.64})$$

(with the $k=0$ term understood as $(1-\alpha)^i(1-\gamma)^i$ when $j=0$, and the convention $\binom{-1}{-1} = 1$).

For the immigration factor, when $\nu \neq \lambda$ (i.e. $\rho \neq 1$):

$$R_{ij}(T) = (1-\beta)^\rho \cdot {}_2F_1\left(-j, \rho; 1; \frac{\beta}{1-\beta} \cdot \frac{1}{\beta/(1-\beta)}\right) \quad (\text{A.65})$$

More explicitly, the full $P_{ij}(T)$ with immigration can be written using the negative binomial form (36):

$$P_{ij}(T) = \sum_{k=0}^{\min(i,j)} \binom{i}{k} \binom{j-1}{k-1} \alpha^k (1-\alpha-\gamma+\alpha\gamma)^{i-k} \gamma^{j-k} (1-\beta)^{k+1} \beta^{j-k-1} \cdot (1-\beta)^\rho \sum_{m=0}^{j-k} \binom{\rho+m-1}{m} \beta^m \quad (\text{A.66})$$

where the inner sum is a truncated negative binomial in β with parameter ρ . When $\rho = 1$ (i.e. $\nu = \lambda$), the immigration factor simplifies to a factor of $(1-\beta)$, recovering the TKF91 transition probabilities. When $\rho = 0$ (no immigration), $R = 1$ and we recover $P_{ij}^{(0)}$.

A.7.6 Score derivatives for ν

The score with respect to ν is

$$\frac{\partial}{\partial \nu} \log P_{ij} = \frac{1}{P_{ij}} \frac{\partial P_{ij}}{\partial \nu}$$

Since ν enters through $\rho = \nu/\lambda$ and the factor $(1-\beta)^\rho$ times the negative binomial sum, we have

$$\frac{\partial}{\partial \nu} \log P_{ij} = \frac{1}{\lambda} (\log(1-\beta) + \psi(\rho) \cdot [\text{correction from the sum}]) \quad (\text{A.67})$$

where ψ is the digamma function (arising from $\partial/\partial \rho$ (ρ^{+m-1})). The full expression is:

$$\frac{\partial}{\partial \nu} \log P_{ij} = \frac{1}{\lambda} \frac{\sum_k (\dots) \sum_m \binom{\rho+m-1}{m} \beta^m \left[\log(1-\beta) + \sum_{\ell=0}^{m-1} \frac{1}{\rho+\ell} \right]}{\sum_k (\dots) \sum_m \binom{\rho+m-1}{m} \beta^m} \quad (\text{A.68})$$

where $(\dots)$ abbreviates the terms from the outer sum in (A.66).

A.7.7 Chain rule

In practice, automatic differentiation of (A.66) is the preferred approach for computing $\partial \log P_{ij}/\partial \lambda$, $\partial \log P_{ij}/\partial \mu$, and $\partial \log P_{ij}/\partial \nu$, since the nested sums and special functions make manual differentiation unwieldy.

For completeness, the chain rule decomposes as:

$$\frac{\partial \log P_{ij}}{\partial \lambda} = \frac{\partial \log P_{ij}}{\partial \alpha} \frac{\partial \alpha}{\partial \lambda} + \frac{\partial \log P_{ij}}{\partial \beta} \frac{\partial \beta}{\partial \lambda} + \frac{\partial \log P_{ij}}{\partial \gamma} \frac{\partial \gamma}{\partial \lambda} + \frac{\partial \log P_{ij}}{\partial \rho} \frac{\partial \rho}{\partial \lambda}$$

with the α, β, γ derivatives given in Appendix A.6, and the additional term

$$\frac{\partial \rho}{\partial \lambda} = -\frac{\nu}{\lambda^2}, \quad \frac{\partial \rho}{\partial \nu} = \frac{1}{\lambda}$$

The derivative $\partial \log P_{ij}/\partial \rho$ is the quantity in (A.68) times λ . The μ derivative has no ρ term since ρ does not depend on μ .

The formulas above hold for any $(i, j, \lambda, \mu, \nu)$ with $\lambda \neq \mu$ and $P_{ij}(T) > 0$. In the TKF91 regime ($\nu = \lambda, i \in \{0, 1\}$), they specialize to the results of the main text.

A.8 The General Geometric model

Ideally, the underlying evolutionary model we'd use for indels would be the General Geometric Indel (GGI) model (16, 29). The GGI model is, in some sense, the maximum-entropy indel model for a given expected rate and length of indels. Unfortunately, the probability distributions of aligned ancestor-descendant sequences at finite times under this model are hard to solve. It turns out that the pairwise alignment distributions under the GGI model are well approximated by the TKF92 model (63), so we use that model for inference. In this section we discuss the relationship between the models and, in particular, how we can map parameters between them in a principled way.

The GGI model has parameters (λ_0, μ_0, x, y) . In this model, the instantaneous rate of insertion of k residues at any available site is $\lambda_0 x^{k-1}(1-x)$ (with the residues themselves sampled from a stationary distribution), while the instantaneous rate of deletion of a stretch of exactly k contiguous residues is $\mu_0 y^{k-1}(1-y)$. For reversibility (or, indeed, the existence of an equilibrium) we require that $\lambda_0 y(1-x) = \mu_0 x(1-y)$. The mean insertion and deletion lengths are $1/(1-x)$ and $1/(1-y)$. The rate at which any given residue is deleted, counting deletions that start at or before that residue (in an infinitely long sequence), is $\mu_0/(1-y)$. The probability that a sequence has length n at equilibrium is $(x/y)^n(1-x/y)$, so the expected sequence length at equilibrium is $\ell_{\text{GGI}} = \frac{x/y}{1-x/y} = \frac{1}{y/x-1}$.

The TKF92 model has parameters (λ, μ, r) . In this model, k -residue fragments (where $k > 0$ and $P(k) = r^{k-1}(1-r)$) are inserted and deleted at rates λ, μ . The rate at which any given residue is deleted is just the rate at which its fragment is deleted, μ . The mean number of fragments at equilibrium is $\frac{\lambda/\mu}{1-\lambda/\mu} = \frac{1}{\mu/\lambda-1}$ and the mean fragment length is $1/(1-r)$, so the mean sequence length at equilibrium is $\ell_{\text{TKF}} = \frac{1}{(\mu/\lambda-1)(1-r)}$. The precise distribution of sequence lengths at equilibrium is a zero-altered geometric distribution

$$P(n) = \begin{cases} 1 - \frac{\lambda}{\mu} & \text{if } n = 0 \\ \frac{\lambda}{\mu} \left(r + (1-r)\frac{\lambda}{\mu} \right)^{n-1} (1-r) \left(1 - \frac{\lambda}{\mu} \right) & \text{if } n > 0 \end{cases}$$

The mean length of an insertion in TKF92 is $1/(1-r)$, i.e. the expected fragment length. For deletions, things are slightly more complicated. Superficially, it looks like deletions are also

fragments so they should have the same mean length. However, if we do not have knowledge of the fragment boundaries, but are conditioning just on the current sequence length (as in GGI), we have to weight this by the posterior probability that the subsequence being deleted does in fact constitute a single fragment. For a subsequence of length k , this posterior probability is proportional to $\left(\frac{r}{r+(1-r)\frac{\lambda}{\mu}}\right)^{k-1}$ (indels at the end of the sequence have a slightly different correction factor since they are more likely to constitute a fragment, but the k -dependence is the same). The rate of starting an insertion or deletion *at a particular site* is subject to a similar correction factor.

Given the equilibrium constraint on GGI, each model has three free parameters. In order to find a correspondence, it is useful to match moments (expectations). In view of the above discussion, the most natural expectations to equate are (in the form “GGI expectation = TKF92 expectation”)

- The mean deletion rate of a given residue: $\mu_0/(1-y) = \mu$
- The mean length at equilibrium: $\frac{1}{y/x-1} = \frac{1}{(\mu/\lambda-1)(1-r)}$
- The mean length of an insertion event: $1/(1-x) = 1/(1-r)$ so $x = r$.

These equations, together with the reversibility constraint $\lambda_0 y(1-x) = \mu_0 x(1-y)$, fully specify a one-to-one mapping between GGI and TKF92. In what follows, we will simply use the TKF92 parameters (λ, μ, r) and likelihoods (including the zero-adjusted equilibrium length distribution), but with the understanding that we are approximating GGI. In particular, when we jointly sample the pairwise alignments around any given node in the phylogenetic tree, we will *not* require the fragment boundaries to coincide (as would be the case in TKF92). Technically speaking, any indel model parameter priors should be on the GGI parameters rather than the TKF92 ones, but since we are using uninformative priors this may be an acceptable compromise.

One way in which the TKF92 model, where the process state is augmented with latent fragment boundaries, deviates from the sequence-only GGI is that when we use the TKF92 Pair HMM (with the fragment boundaries marginalized out) as an approximation to $P(X_t|X_0)$, we cannot expect it perfectly to obey the Chapman-Kolmogorov equation $P(X_{t+u} = x''|X_0 = x) = \sum_{x'} P(X_t = x'|X_0 = x)P(X_{t+u} = x''|X_t = x')$. This has consequences for any MCMC moves that split branches using a Pair HMM (which models exactly the Chapman-Kolmogorov equation), since it means that those moves will be miscalibrated in probability, but the approximation should be mild. In fact, it seems possible that no five-state ($\{\text{S}, \text{M}, \text{I}, \text{D}, \text{E}\}$) Pair HMM exactly obeys the Chapman-Kolmogorov equation for a GGI model, except in the special case where the model reduces to the TKF91 model. The best approximations known are obtained by matching moments of the underlying counting process and thereby deriving ODEs for the transition probabilities (29). However, the TKF92 model is a close runner-up (and actually appears to fit real data better).

A.9 Composition, Relative-Entropy Fit, and the GGI–TKF92 Matched Flow

This appendix gives the minimum setup required to write down the matched-flow ODE for the best TKF92 surrogate of a GGI long-indel process as a function of branch length T . The objective is the conditional expected KL; the flow is realised by an m-projection of the 13-state *Triad HMM* $a(\theta, T) \circ b(\theta_{\text{GGI}}, dT)$. Only the indel process is analysed (substitutions factor out and are suppressed by working over a one-symbol alphabet).

A.9.1 The Triad HMM and the $U + V dT$ expansion

Compositional form. Let $a = a(\lambda, \mu, r, T)$ be the TKF92 joint Pair HMM transition matrix at surrogate parameters $\theta = (\lambda, \mu, r)$ and branch length T ; let $b = b(\theta_{\text{GGI}}, dT)$ be the GGI conditional

Pair HMM transition matrix for one infinitesimal step dT at GGI parameters $\theta_{\text{GGI}} = (\lambda_0, \mu_0, x, y)$

$$\begin{aligned}
b_{\text{SM}} &= b_{\text{MM}} = 1 - (\lambda_0 + \mu_0) dT + o(dT), & b_{\text{SI}} &= b_{\text{MI}} = \lambda_0 dT, \\
b_{\text{SD}} &= b_{\text{MD}} = \mu_0 dT, & b_{\text{SE}} &= b_{\text{ME}} = 1 - \lambda_0 dT, \\
b_{\text{II}} &= x, & b_{\text{IM}} &= b_{\text{IE}} = 1 - x, \\
b_{\text{DD}} &= y, & b_{\text{DM}} &= 1 - y, \\
b_{\text{DE}} &= 1, & b_{\text{ID}} &= b_{\text{DI}} = 0.
\end{aligned}$$

The Triad is the composite transducer $a(\theta, T) \circ b(\theta_{\text{GGI}}, dT)$ with the state space $\mathcal{S}'_{\text{triad}} = \mathcal{S}_{\text{triad}} \cup \{\text{ID}\}$ where $\mathcal{S}_{\text{triad}} = \{\text{SS}, \text{sI}, \text{MM}, \text{MI}, \text{MD}, \text{IM}, \text{II}, \text{Ds}, \text{Dm}, \text{Di}, \text{DD}, \text{EE}\}$. The cases in the state tuples $ik \in \mathcal{S}'_{\text{triad}}$ indicate which component(s) are updated on entry: $\{\text{sI}, \text{MI}, \text{II}\}$ update the b -component (k), $\{\text{Ds}, \text{Dm}, \text{Di}, \text{DD}\}$ update the a -component (i), and $\{\text{MM}, \text{MD}, \text{IM}, \text{ID}, \text{EE}\}$ update both components.

The resulting Triad transition matrix takes the form $Q'(\theta, \theta_{\text{GGI}}, T, dT) =$

$$\begin{pmatrix}
& \text{SS} & \text{sI} & \text{MM} & \text{MI} & \text{MD} & \text{IM} & \text{II} & \text{ID} & \text{Ds} & \text{Dm} & \text{Di} & \text{DD} & \text{EE} \\
\text{SS} & 0 & b_{\text{SI}} & a_{\text{SM}}b_{\text{SM}} & 0 & a_{\text{SM}}b_{\text{SD}} & a_{\text{SI}}b_{\text{SM}} & 0 & a_{\text{SI}}b_{\text{SD}} & a_{\text{SD}} & 0 & 0 & 0 & a_{\text{SE}}b_{\text{SE}} \\
\text{sI} & 0 & b_{\text{II}} & a_{\text{SM}}b_{\text{IM}} & 0 & a_{\text{SM}}b_{\text{ID}} & a_{\text{SI}}b_{\text{IM}} & 0 & a_{\text{SI}}b_{\text{ID}} & 0 & 0 & a_{\text{SD}} & 0 & a_{\text{SE}}b_{\text{IE}} \\
\text{MM} & 0 & 0 & a_{\text{MM}}b_{\text{MM}} & b_{\text{MI}} & a_{\text{MM}}b_{\text{MD}} & a_{\text{MI}}b_{\text{MM}} & 0 & a_{\text{MI}}b_{\text{MD}} & 0 & a_{\text{MD}} & 0 & 0 & a_{\text{ME}}b_{\text{ME}} \\
\text{MI} & 0 & 0 & a_{\text{MM}}b_{\text{IM}} & b_{\text{II}} & a_{\text{MM}}b_{\text{ID}} & a_{\text{MI}}b_{\text{IM}} & 0 & a_{\text{MI}}b_{\text{ID}} & 0 & 0 & a_{\text{MD}} & 0 & a_{\text{ME}}b_{\text{IE}} \\
\text{MD} & 0 & 0 & a_{\text{MM}}b_{\text{DM}} & b_{\text{DI}} & a_{\text{MM}}b_{\text{DD}} & a_{\text{MI}}b_{\text{DM}} & 0 & a_{\text{MI}}b_{\text{DD}} & 0 & 0 & 0 & a_{\text{MD}} & a_{\text{ME}}b_{\text{DE}} \\
\text{IM} & 0 & 0 & a_{\text{IM}}b_{\text{MM}} & 0 & a_{\text{IM}}b_{\text{MD}} & a_{\text{II}}b_{\text{MM}} & b_{\text{MI}} & a_{\text{II}}b_{\text{MD}} & 0 & a_{\text{ID}} & 0 & 0 & a_{\text{IE}}b_{\text{ME}} \\
\text{II} & 0 & 0 & a_{\text{IM}}b_{\text{IM}} & 0 & a_{\text{IM}}b_{\text{ID}} & a_{\text{II}}b_{\text{IM}} & b_{\text{II}} & a_{\text{II}}b_{\text{ID}} & 0 & 0 & a_{\text{ID}} & 0 & a_{\text{IE}}b_{\text{IE}} \\
\text{ID} & 0 & 0 & a_{\text{IM}}b_{\text{DM}} & 0 & a_{\text{IM}}b_{\text{DD}} & a_{\text{II}}b_{\text{DM}} & b_{\text{DI}} & a_{\text{II}}b_{\text{DD}} & 0 & 0 & 0 & a_{\text{ID}} & a_{\text{IE}}b_{\text{DE}} \\
\text{Ds} & 0 & 0 & a_{\text{DM}}b_{\text{SM}} & 0 & a_{\text{DM}}b_{\text{SD}} & a_{\text{DI}}b_{\text{SM}} & 0 & a_{\text{DI}}b_{\text{SD}} & a_{\text{DD}} & 0 & 0 & 0 & a_{\text{DE}}b_{\text{SE}} \\
\text{Dm} & 0 & 0 & a_{\text{DM}}b_{\text{MM}} & 0 & a_{\text{DM}}b_{\text{MD}} & a_{\text{DI}}b_{\text{MM}} & 0 & a_{\text{DI}}b_{\text{MD}} & 0 & a_{\text{DD}} & 0 & 0 & a_{\text{DE}}b_{\text{ME}} \\
\text{Di} & 0 & 0 & a_{\text{DM}}b_{\text{IM}} & 0 & a_{\text{DM}}b_{\text{ID}} & a_{\text{DI}}b_{\text{IM}} & 0 & a_{\text{DI}}b_{\text{ID}} & 0 & 0 & a_{\text{DD}} & 0 & a_{\text{DE}}b_{\text{IE}} \\
\text{DD} & 0 & 0 & a_{\text{DM}}b_{\text{DM}} & 0 & a_{\text{DM}}b_{\text{DD}} & a_{\text{DI}}b_{\text{DM}} & 0 & a_{\text{DI}}b_{\text{DD}} & 0 & 0 & 0 & a_{\text{DD}} & a_{\text{DE}}b_{\text{DE}} \\
\text{EE} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{pmatrix}$$

Eliminating the ID state gives the 12×12 transition matrix for the null-eliminated triad, $Q_{ij} = Q'_{ij} + Q'_{i \text{ ID}} Q'_{\text{ID } j} / (1 - Q'_{\text{ID ID}})$.

The null-eliminated transition matrix is linear in dT :

$$Q(\theta, \theta_{\text{GGI}}, T, dT) = U(\theta, T) + V(\theta, \theta_{\text{GGI}}, T) dT + O(dT^2), \quad (\text{A.69})$$

with U collecting the entries of Q that are dT -independent and V collecting the dT coefficients. Defining $C := (I - U)^{-1}$, to first order

$$(I - Q)^{-1} = C + (CVC) dT + O(dT^2). \quad (\text{A.70})$$

Coarse-graining. The coarse-graining $C_{\text{proj}} : \mathcal{S}_{\text{triad}} \rightarrow \{\text{S}, \text{M}, \text{I}, \text{D}, \text{E}\}$ projects Triad states $(\text{SS}, \text{MM}, \text{EE})$ to $(\text{S}, \text{M}, \text{E})$; $\{\text{sI}, \text{MI}, \text{IM}, \text{II}\}$ to I ; and $\{\text{Ds}, \text{Dm}, \text{Di}, \text{DD}\}$ to D . With Triad state ID marginalized, the map yields coarse-grained counts m_{XY} . In matrix form, with $\Sigma_{ab}^{XY} = 1$ iff $C_{\text{proj}}(a) = X, C_{\text{proj}}(b) = Y$,

$$m_{XY}(\theta, \theta_{\text{GGI}}, T, dT) = \sum_{a,b} [(I - Q)^{-1}]_{\text{SS } a} Q_{ab} \Sigma_{ab}^{XY} [(I - Q)^{-1}]_{b \text{ EE}}. \quad (\text{A.71})$$

A.9.2 Objective: conditional expected KL

We minimise the conditional expected KL between descendant distributions, with the expectation taken over the surrogate's ancestor marginal:

$$\bar{D}(\theta, T) = \mathbb{E}_{P_T(s; \theta)} \left[D(P_{\text{GGI}}(s' | s; T) \parallel P_T(s' | s; \theta, T)) \right]. \quad (\text{A.72})$$

The matched flow is the self-composition m-projection

$$\theta(T + dT) = \arg \min_{\theta'} \bar{D}(a(\theta(T), T) \circ b(\theta_{\text{GGI}}, dT) \parallel a(\theta', T + dT)). \quad (\text{A.73})$$

A.9.3 Three equivalent ODE systems

The Δ operator. For any function $f(\theta, \theta_{\text{GGI}}, T, dT)$ that is regular at $dT = 0$, write its dT expansion $f = f^{(0)} + \Delta[f] dT + O(dT^2)$, so

$$\Delta[f] := \left. \frac{\partial f}{\partial (dT)} \right|_{dT=0}. \quad (\text{A.74})$$

Applied along the matched flow $T \mapsto \theta(T)$, this is the total T -derivative of f : the m-projection condition (A.73) forces $df/dT = \Delta[f](\theta(T), T)$, since the source at (T, dT) becomes the surrogate at $T + dT$.

Transition counts. Expand (A.71) via (A.70). The surrogate's own counts at (θ, T) and the leading dT perturbation are

$$m_{XY}^{(0)}(\theta, T) = \sum_{a,b} C_{\text{SS},a} U_{ab} \Sigma_{ab}^{XY} C_{b,\text{EE}}, \quad (\text{A.75})$$

$$\Delta m_{XY}(\theta, \theta_{\text{GGI}}, T) = \sum_{a,b} \Sigma_{ab}^{XY} \left[(CVC)_{\text{SS},a} U_{ab} C_{b,\text{EE}} + C_{\text{SS},a} V_{ab} C_{b,\text{EE}} + C_{\text{SS},a} U_{ab} (CVC)_{b,\text{EE}} \right]. \quad (\text{A.76})$$

The count flow ODE is then

$$\frac{dm_{XY}}{dT} = \Delta m_{XY}(\theta(T), T), \quad m_{XY}(0) = 0, \quad (\text{A.77})$$

with $\theta(T)$ recovered at each step from $m(T)$ via the TKF92 M-step (see below). At every T , the flowing counts $m_{XY}(T)$ coincide with the surrogate's self-consistent value $m_{XY}^{(0)}(\theta(T), T)$ from (A.75).

BDI sufficient statistics. Let $X \in \{B, D, S, F, \bar{F}\}$ index the five TKF92 BDI sufficient statistics (insertions B , deletions D , time-integrated fragment count S , fragment extensions F , fragment terminations $\bar{F}$). Each is a linear functional of the coarse-grained counts m_{XY} under the standard TKF92 BDI map, so $\Delta[X]$ is the same map applied to Δm_{XY} . The flow is

$$\frac{dE_T[X]}{dT} = \Delta[X](\theta(T), T), \quad E_T[X](0) = 0, \quad X \in \{B, D, S, F, \bar{F}\}. \quad (\text{A.78})$$

TKF92 parameters. Differentiating the TKF92 M-step formulas $\lambda = E[B]/(E[S] + T)$, $\mu = E[D]/E[S]$, and $r = E[F]/(E[F] + E[\bar{F}])$ along the flow gives the equivalent system in the natural parameters:

$$\boxed{\begin{aligned}\dot{\lambda} &= \frac{\Delta[B] - \lambda(\Delta[S] + 1)}{E_T[S] + T} \\ \dot{\mu} &= \frac{\Delta[D] - \mu\Delta[S]}{E_T[S]} \\ \dot{r} &= \frac{(1-r)\Delta[F] - r\Delta[\bar{F}]}{E_T[F] + E_T[\bar{F}]}\end{aligned}} \quad (\text{A.79})$$

where the $+1$ in the λ numerator is the immortal-link clock ($dT/dT = 1$), and $T = T$ is the total observation time.

The three systems (A.77), (A.78), and (A.79) are equivalent (related by the linear BDI map and the TKF92 M-step). Form (A.78) is the numerically convenient one; (A.79) relates most directly to the TKF92 surrogate one actually uses; and (A.77) makes the algebraic structure (rational in C , linear in V) most visible.

A.9.4 Boundary conditions at $T \rightarrow 0^+$

At $T = 0$ the surrogate's conditional descendant distribution is the identity, so the conditional KL (A.72) vanishes for any θ ; the boundary value $\theta(0^+)$ is therefore pinned by the leading- $O(T)$ KL rate. In the per-residue (large- L) limit the insertion KL alone is minimised by $r = y$ and the deletion KL alone by $r = x$. The joint per-residue minimum is the weighted average

$$r^*(0) = \frac{\lambda_0 y(1-x) + \mu_0 x(1-y)}{\lambda_0(1-x) + \mu_0(1-y)}, \quad (\text{A.80})$$

with the corresponding rate boundaries

$$\lambda^*(0) = \frac{\lambda_0}{1 - r^*(0)}, \quad \mu^*(0) = \frac{\mu_0}{1 - r^*(0)}. \quad (\text{A.81})$$

Eq. (A.80)–(A.81) are the principled $T \rightarrow 0^+$ initial conditions for (A.79); the $E_T[X]$ in (A.78) start at zero and the parameters are recovered by the L'Hôpital limits $\lambda(0^+) = \Delta[B](\theta(0), 0)/(\Delta[S](\theta(0), 0) + 1)$, and similarly for $\mu(0^+)$, $r(0^+)$, evaluated at $\theta(0)$ from (A.80)–(A.81).

Validity of $r^*(0) \in (0, 1)$. Eq. (A.80) is the per-residue minimiser of the leading KL term, but for arbitrary (λ_0, μ_0, x, y) it need not lie in $(0, 1)$. Writing $1 - r^*(0) = (D - N)/D$ with $D = \lambda_0(1 - y) + \mu_0(1 - x)$ and $N = \lambda_0 y(1 - x) + \mu_0 x(1 - y)$, a short calculation gives

$$1 - r^*(0) = \frac{\lambda_0(1 - 2y + xy) + \mu_0(1 - 2x + xy)}{D}. \quad (\text{A.82})$$

Either bracket is negative when the corresponding length parameter exceeds $1/(2 - \text{other})$ (e.g. $1 - 2x + xy < 0$ iff $x > 1/(2 - y)$), and the combined numerator can become negative when one bracket dominates. In particular, the GGI reversibility condition $\lambda_0 y(1 - y) = \mu_0 x(1 - x)$ permits asymmetric solutions with $x > 1/2$, $y < 1/2$ (or vice versa), and for sufficiently extreme combinations even $\lambda_0 \leq \mu_0$ does not save the bound: the leading-order picture predicts $r^*(0) > 1$, i.e. no physical TKF92 boundary value exists. The closed-form trajectory (A.86) below then has $k < 0$ and $r(T)$ diverges; the closed form must be regarded as extrapolating outside its region of

validity. Practical fitters should either restrict to GGI shapes with $r^*(0) \in (0, 1)$ or refuse to score in the out-of-region tail (we observed this when fitting Pfam: a $\rho \leq 1$ constraint on λ_0/μ_0 alone does not suffice).

A.9.5 Empirically, all three drift modestly; r moves most

Numerical integration of (A.79) at Pfam-fitted TKF92 scale $(\lambda, \mu, r) \approx (0.046, 0.047, 0.68)$, inferring the underlying GGI rates from the boundary as $(\lambda_0, \mu_0) = ((1-r)\lambda, (1-r)\mu) \approx (0.0145, 0.0148)$, and sweeping (x, y) over the one-parameter family of Pfam-consistent GGI shapes (those for which (A.80) yields $r^*=0.683$), gives the envelope across $T \in [0.05, 10]$:

$$\lambda(T) \in [0.0410, 0.0450], \quad \mu(T) \in [0.0416, 0.0460], \quad r(T) \in [0.580, 0.679], \quad (\text{A.83})$$

with relative spreads of 9.4%, 10.1%, and 15.7% respectively. All three drift, r most. Two practical consequences:

- Freezing (λ, μ) at the boundary values during the r -flow incurs at most a $\sim 10\%$ rate-parameter error over $T \in [0.05, 10]$, and reduces the three-parameter system (A.79) to a single r -ODE. This is a useful approximation when T is varied (e.g. over a phylogeny with branch lengths in the Pfam range) but a fixed TKF92 fit is desired for the rate parameters.
- The branch-length dependence that genuinely cannot be ignored is $r(T)$: a single CherryML fit at a representative T misses up to $\sim 17\%$ of r 's actual variation.

First-order fragment-counting closed form for $r(T)$. A first-order argument tracks how a single GGI event changes the surrogate's expected extension and termination counts $F, \bar{F}$ in the alignment-count view. A GGI insertion of length $\ell \sim \text{Geom}(y)$ at a random link adds an I-block, contributing on average $E[(1-r)\Delta F - r\Delta \bar{F}] = (1-r)/(1-y) - (1+r)$ per event. A GGI deletion of length $\ell \sim \text{Geom}(x)$ at a random residue contributes $(1-r)/(1-x) - (1+r)$ by the analogous accounting (with the self-loop $\text{I} \rightarrow \text{I}$ contributions of insertion blocks playing the role of the $\text{D} \rightarrow \text{D}$ contributions of deletion blocks). Combining with the per-link and per-residue event rates gives the leading slope

$$\dot{r} = (1-r) \left[\frac{\lambda_0}{1-y} + \frac{\mu_0}{1-x} \right] - (1+r)(\lambda_0 + \mu_0). \quad (\text{A.84})$$

Evaluated at the boundary $r = r^*(0)$, the bracket collapses by the boundary identity (using $(1-r^*) = (1-x)(1-y)(\lambda_0 + \mu_0)/[\lambda_0(1-x) + \mu_0(1-y)]$ to show that $(1-r^*)[\lambda_0/(1-y) + \mu_0/(1-x)] = \lambda_0 + \mu_0$), giving the strikingly simple initial slope

$$\dot{r}(0^+) = -r^*(0)(\lambda_0 + \mu_0), \quad \text{i.e.} \quad \left. \frac{d \log r}{dT} \right|_{0^+} = -(\lambda_0 + \mu_0). \quad (\text{A.85})$$

Equation (A.84) is a separable linear ODE $\dot{r} = (A - B) - (A + B)r$ with $A = \lambda_0/(1-y) + \mu_0/(1-x)$ and $B = \lambda_0 + \mu_0$, integrating to the closed form

$$r(T) \approx r_\infty + (r^*(0) - r_\infty) \exp\left[-(\lambda_0 + \mu_0) \frac{2-r^*(0)}{1-r^*(0)} T\right], \quad r_\infty = \frac{r^*(0)}{2-r^*(0)} \quad (\text{A.86})$$

where the fixed point r_∞ and decay rate $k = (\lambda_0 + \mu_0)(2 - r^*)/(1 - r^*)$ both follow from the same boundary identity.

Reparameterisation in terms of $(\lambda(0), \mu(0), r(0))$. In practice the user often has a TKF92 fit at a representative branch length (e.g. CherryML) but no GGI shape parameters (x, y) . The boundary relations $\lambda_0 = (1 - r(0))\lambda(0)$ and $\mu_0 = (1 - r(0))\mu(0)$ let us recast (A.86) entirely in terms of the TKF92 boundary values:

$$\boxed{r(T) \approx r_\infty + (r(0) - r_\infty) e^{-kT}, \quad r_\infty = \frac{r(0)}{2 - r(0)}, \quad k = (\lambda(0) + \mu(0))(2 - r(0)).} \quad (\text{A.87})$$

The leading slope is $dr/dT|_{0+} = -r(0)(1 - r(0))(\lambda(0) + \mu(0))$. Note that the rate parameters λ_0, μ_0 of the underlying GGI do not appear separately — only the TKF92 sum $\lambda(0) + \mu(0)$ — and the GGI shape (x, y) does not appear at all. The slaved rate trajectories follow directly: $\lambda(T) \approx \lambda(0)(1 - r(0))/(1 - r(T))$, $\mu(T) \approx \mu(0)(1 - r(0))/(1 - r(T))$. This is the most directly usable form when a TKF92 fit is available.

For Pfam-scale parameters ($r(0)=0.683$, $\lambda(0) + \mu(0)=0.0926$): $r_\infty=0.52$, $k=0.12$. Comparison with the full three-parameter flow (A.79): r values agree to $< 2\%$ for $T \lesssim 1$, to $\sim 10\%$ at $T=10$ (Figure A.1). At branch lengths $T \gtrsim 50$ the full flow develops a non-monotonic kink (a local minimum of r followed by a slow rise) that is a higher-order effect not captured by this leading-order linearisation; for typical phylogenies the kink is well outside the relevant branch-length range.

Fragment coherence half-life. The rate constant k in (A.87) sets a natural time-scale for fragment coherence under TKF92:

$$\boxed{t_{1/2}^{\text{frag}} \equiv \frac{\ln 2}{k} = \frac{\ln 2}{(\lambda(0) + \mu(0))(2 - r(0))} = \frac{\ln 2 (1 - r^*(0))}{(\lambda_0 + \mu_0)(2 - r^*(0))}} \quad (\text{A.88})$$

(The TKF92-parametrisation form, useful when a TKF92 boundary fit $(\lambda(0), \mu(0), r(0))$ is available, and the GGI-parametrisation form, useful when starting from the underlying (λ_0, μ_0, x, y) , are equivalent under the boundary slaving $\lambda(0) = \lambda_0/(1 - r^*(0))$, $\mu(0) = \mu_0/(1 - r^*(0))$, $r(0) = r^*(0)$.) This is the branch length over which $r(T) - r_\infty$ decays to half its initial value. For $T \ll t_{1/2}^{\text{frag}}$ the constant- r (vanilla TKF92) approximation is excellent; for $T \gtrsim t_{1/2}^{\text{frag}}$ the GGI-induced r flow is the dominant correction. The single-component Pfam aligned fit ($\lambda \approx 0.030$, $\mu \approx 0.030$, $r \approx 0.65$) gives $k \approx 0.081$, $t_{1/2}^{\text{frag}} \approx 8.6$, well above typical Pfam cherry times (median $T \approx 0.76$); a median cherry traverses only $\sim 6\%$ of $r(0) - r_\infty$, making the indivisible-fragment assumption near-exact across the Pfam corpus. A $K=20$ Pfam-mixture fit's components span $t_{1/2}^{\text{frag}} \in [3.4, 15.0]$ (weighted mean ≈ 9.5); even the fastest-flow component moves only $\sim 14\%$ of the available flow over a median cherry. This is the proximate reason that constant- r TKF92 fits to Pfam are competitive with GGI-steered TKF92 fits: at Pfam composition, the data simply lacks the temporal span needed to resolve the GGI r -flow signal. Datasets with larger $T/t_{1/2}^{\text{frag}}$, either through faster (λ, μ) or longer branches, would break the degeneracy.

For applications outside the phylogenetic regime the three-parameter ODE (A.79) is integrated numerically, either by a finite-difference m -projection or by analytically including the immortal-link clock $(\partial_T E[X])$ and parameter-drift $(\nabla_\theta E[X] \cdot \dot{\theta})$ contributions to $\dot{E}_T[X]$.

Single-parameter (fragment-length) flow versus triple-parameter (rates and fragment-length) flow. The matched-flow ODE (A.79) can be integrated for all three of (λ, μ, r) jointly — the *triple-parameter flow* — or under the truncation that holds λ, μ fixed at their boundary values $\lambda_T(0) = \lambda_0/(1 - r^*(0))$, $\mu_T(0) = \mu_0/(1 - r^*(0))$ and integrates only r :

$$\dot{r}(T) = \frac{(1 - r) \Delta[F] - r \Delta[\bar{F}]}{E_T[F] + E_T[\bar{F}]} \Big|_{\lambda=\lambda_T(0), \mu=\mu_T(0)}. \quad (\text{A.89})$$

We refer to this truncation as the *single-parameter (fragment-length) flow*. The two ODEs agree to leading order in T , and (A.87) is the leading-order closed-form solution of either. Beyond leading order the two flows diverge. The triple-parameter flow integrates a coupled three-dimensional system in which the (λ, μ) drift back-couples into the $\dot{r}$ formula, and the resulting r trajectory can become non-monotonic at moderate T ; the single-parameter flow remains monotonic by construction and stays well-fit by (A.87) throughout. At the matched-flow projection geometry the two flows define different cumulative projection-error rates, but their conditional KL to the underlying GGI process is comparable across phylogenetic branch lengths. We adopt the single-parameter flow as the recommended approximation: it is algebraically clean, captures the matched-flow geometry of r , and admits the closed-form solution boxed below.

Closed-form TKF92 approximation to GGI: summary. Collecting the recommended approximation for the matched-flow surrogate of a GGI process with rates λ_0, μ_0 and extension parameters x (deletion) and y (insertion):

Closed-form TKF92 surrogate for GGI.

Given GGI parameters (λ_0, μ_0, x, y) , the recommended constant-rate TKF92 surrogate is

$$\begin{aligned} r^*(0) &= \frac{\lambda_0 y (1-x) + \mu_0 x (1-y)}{\lambda_0 (1-x) + \mu_0 (1-y)}, \\ \lambda_T &\equiv \frac{\lambda_0}{1-r^*(0)}, \quad \mu_T \equiv \frac{\mu_0}{1-r^*(0)}, \\ r(T) &= r_\infty + (r^*(0) - r_\infty) e^{-kT}, \\ r_\infty &= \frac{r^*(0)}{2-r^*(0)}, \quad k = (\lambda_0 + \mu_0) \frac{2-r^*(0)}{1-r^*(0)}. \end{aligned}$$

The rates λ_T, μ_T are held constant; only r evolves with T . The Pair HMM transition probability is then $\tau'(\lambda_T, \mu_T, r(T), T)$. The branch-length scale on which $r(T)$ is changing is the coherence half-life $t_{1/2}^{\text{frag}} = \ln 2/k$ (A.88).

Numerical comparison on Pfam parameters. Figure A.2 compares the single-parameter flow (A.89) and the closed-form approximation (A.87) against per- τ ML fits of TKF92 (via the conditional WFST) to Gillespie simulations on the Pfam joint-native fit ($\lambda_0=0.01061$, $\mu_0=0.01065$, $x=0.641$, $y=0.638$). The top panel shows $r(T)$ under the single-parameter flow, the closed-form approximation, and the per- τ ML fit; the fragment coherence half-life $t_{1/2}^{\text{frag}}$ (A.88) is marked. The bottom panel shows the per- τ conditional KL $D(P_{\text{GGI}}(y | x) \| P_{\tau_T''}(y | x))$ in bits per cherry on a linear y axis; the ancestor singlet is factored out analytically as $D_{\text{cond}} = D(P_{\text{GGI}}(x, y) \| P_\tau(x, y)) - D(P_{\text{GGI}}(x) \| P_\tau(x))$, so the comparison reflects only the surrogate's error on the descendant-given-ancestor distribution. The per- τ Gillespie cherry budget is adapted to the small- T event-paucity regime: at branch length T the expected number of indel decisions per cherry is approximately RT where $R \approx (L_x+1)\lambda_0/(1-y) + L_x\mu_0/(1-x)$ is the per-cherry equilibrium decision rate, so for a target absolute SE of σ on the per- τ ML estimate of r we set $N_{\text{cherries}}(T) = \max(N_{\text{floor}}, r(1-r)/(\sigma^2 RT))$. At $\sigma = 0.005$, $N_{\text{floor}} = 3000$, this peaks at $\sim 3.8 \cdot 10^5$ cherries near $T = 0.005$ and floors at 3000 once RT saturates. Error bars are over 5 independent simulations. The two surrogates give comparable conditional KL throughout the phylogenetic range, with both growing monotonically as the constant-rate TKF92 approximation becomes increasingly imperfect at long branch lengths; the single-parameter flow tracks the per- τ ML fit on r to within sampling error.

B EM, composite likelihoods, and variational inference

Building on Appendix A, this appendix develops the M-step and variational-inference machinery for TKF91 / TKF92 / MixDOM at full generality. First, closed-form GTR substitution M-steps in their many specialisations (JC69, K80, F81, HKY85, GTR, GY94, plus rate-rescaling, tied-equilibria, and mixture-of-GTR variants); then the stochastic variational Baum–Welch (SVI-BW) loop with its convergence theorem (ELBO derivation, natural-gradient updates, minibatch variance bounds), the linearised analysis at stationarity, Maraschino as the cherry-count distillation of TKF92, the tree-level inference algorithms it composes with (FSA, BeamASR, VarAnc, svi-VarAnc), the mixture-of-trees variational ancestral presence/absence inference, and the structural-bias analysis of the BP cumulant under a column-factorised variational field.

B.1 Substitution M-Steps for Specific Models

This appendix specializes the general GTR substitution M-step (Section A.1.8) to specific DNA and codon substitution models. In each case, we state the rate matrix R in terms of the model parameters, derive the closed-form MLE (or MAP) update given the bridge expectations (W_i, U_{ij}, V_i) , and note the spectral structure when it simplifies the endpoint-conditioned integrals (A.5).

Throughout, we work with the complete-data log-likelihood (A.1)

$$\ell_2 = \sum_i V'_i \log \pi_i + \sum_{j>i} (U_{ij} + U_{ji}) \log Q_{ij} - \sum_{j>i} Q_{ij} (W_i \pi_j + W_j \pi_i)$$

where $V'_i = V_i + \sum_{j \neq i} U_{ji}$, under the GTR parameterization $R_{ij} = Q_{ij} \pi_j$ with symmetric exchangeability $Q_{ij} = Q_{ji}$.

The indel M-step (quadratic (A.22) for κ , with L'Hôpital limits in Section A.5) is orthogonal and unchanged across all substitution models.

For DNA models ($|\mathcal{A}| = 4$), the bridge-expectation integral $I_{ij}^{ab}(T)$ (A.4) involves only $4 \times 4 = 16$ matrix entries per endpoint pair, and the $J^{kl}(T)$ matrix (A.5) is 4×4 , making the endpoint-conditioned expectations cheap to compute.

B.1.1 JC69 (Jukes–Cantor)

Rate matrix. All off-diagonal rates are equal: $R_{ij} = \rho/3$ for $i \neq j$ (total rate out of any state is ρ). Equivalently, $\pi_i = 1/4$ for all i and $Q_{ij} = 4\rho/3$ for all $i \neq j$.

The single free parameter is the overall rate ρ .

M-step. By detailed balance, $Q_{ij} = (U_{ij} + U_{ji})/(W_i \pi_j + W_j \pi_i)$ is the GTR MLE. Under JC69, all exchangeabilities are forced equal, so we pool:

$$\hat{Q} = \frac{\sum_{j>i} (U_{ij} + U_{ji})}{\sum_{j>i} (W_i \pi_j + W_j \pi_i)} = \frac{U_\bullet}{3W_\bullet/4} = \frac{4U_\bullet}{3W_\bullet} \quad (\text{B.1})$$

where $U_\bullet = \sum_{i \neq j} U_{ij}$ is the total substitution count and $W_\bullet = \sum_i W_i$ is the total dwell time. (The denominator uses $\sum_{j>i} (W_i/4 + W_j/4) = 3W_\bullet/4$ since each of the 4 states appears in 3 of the $\binom{4}{2} = 6$ pairs.) The overall rate is then $\hat{\rho} = 3\hat{Q}/4 = U_\bullet/W_\bullet$, which is simply the total substitution count divided by total dwell time.

The equilibrium frequencies are fixed: $\hat{\pi}_i = 1/4$.

Spectral structure. $S = (\rho/3)J - (4\rho/3)I$ where J is the 4×4 all-ones matrix. Eigenvalues: 0 (once, eigenvector $\mathbf{1}/2$) and $-4\rho/3$ (triply degenerate). Thus $J^{kl}(T)$ has only two distinct diagonal values (T and $Te^{-4\rho T/3}$) and the off-diagonal J^{01} term $(1 - e^{-4\rho T/3})/(4\rho/3)$.

B.1.2 K80 (Kimura 2-Parameter)

Rate matrix. Equal equilibrium frequencies $\pi_i = 1/4$. Transitions (purine $\leftrightarrow$ purine, pyrimidine $\leftrightarrow$ pyrimidine) occur at rate κ_s and transversions at rate 1 (or vice versa, depending on convention). Labeling the states A, G, C, T and writing $i \sim j$ for a transition pair:

$$R_{ij} = \begin{cases} \kappa_s/4 & \text{if } i \sim j \text{ (transition)} \\ 1/4 & \text{if } i \not\sim j \text{ (transversion)} \end{cases}$$

with total rate out of any state $(\kappa_s + 2)/4$. The single free parameter beyond rate scaling is κ_s (the transition/transversion rate ratio).

M-step. Partition the substitution counts into transitions $U_s = \sum_{i \sim j} U_{ij}$ and transversions $U_v = \sum_{i \not\sim j} U_{ij} = U_\bullet - U_s$. Similarly partition the dwell-weighted opportunities. Under K80, the exchangeability for transition pairs is κ_s and for transversion pairs is 1 (up to a common scale). The MLE for each group, from the pooled GTR formula:

$$\hat{\kappa}_s = \frac{U_s/N_s}{U_v/N_v} \quad (\text{B.2})$$

where $N_s = \sum_{i \sim j, i < j} (W_i \pi_j + W_j \pi_i) = W_\bullet/4$ is the opportunity for transitions (2 pairs: A–G and C–T, each contributing $(W_i + W_j)/4$, summing to $W_\bullet/4$) and $N_v = \sum_{i \not\sim j, i < j} (W_i \pi_j + W_j \pi_i) = W_\bullet/2$ is the opportunity for transversions (4 pairs, each state appearing in 2 transversion pairs, summing to $2W_\bullet/4 = W_\bullet/2$). Simplifying:

$$\hat{\kappa}_s = \frac{2U_s}{U_v} \quad (\text{B.3})$$

Given $\hat{\kappa}_s$, the overall scale $\hat{\sigma}$ satisfies $\hat{\sigma}(\hat{\kappa}_s N_s + N_v) = U_\bullet$, giving overall rate $\hat{\rho} = \hat{\sigma}(\hat{\kappa}_s + 2)/4 = U_\bullet/W_\bullet$.

Spectral structure. Three distinct eigenvalues: 0, $-(\kappa_s + 1)/2$ (doubly degenerate, within-purine and within-pyrimidine contrasts), and -1 (purine-vs-pyrimidine contrast).

B.1.3 F81 (Felsenstein 1981)

Rate matrix. Equal exchangeabilities, variable equilibrium: $R_{ij} = \rho \pi_j$ for $i \neq j$ (equivalently, $Q_{ij} = \rho$ for all $i \neq j$).

M-step. Since $Q_{ij} = \rho$ is constant across all pairs, the exchangeability MLE pools all pairs:

$$\hat{\rho} = \frac{U_\bullet}{\sum_{j>i} (W_i \pi_j + W_j \pi_i)} = \frac{U_\bullet}{\sum_i W_i (1 - \pi_i)} \quad (\text{B.4})$$

using the identity $\sum_{j>i} (W_i \pi_j + W_j \pi_i) = \sum_i W_i \sum_{j \neq i} \pi_j = \sum_i W_i (1 - \pi_i)$.

For equilibrium frequencies:

$$\hat{\pi}_i = \frac{V'_i}{\sum_j V'_j} \quad (\text{B.5})$$

This is the standard empirical-frequency estimator (exact MLE when Q is held fixed; approximate when Q and π are jointly optimized, as discussed in Section A.1.8).

In practice, iterate: fix $\hat{\pi}$, solve for $\hat{\rho}$ (B.4), update $\hat{\pi}$ (B.5), repeat. The decoupling is exact after one iteration when $W_i \propto \pi_i$ (which holds approximately for long dwell times).

Spectral structure. $S = \rho(\sqrt{\pi}\sqrt{\pi}^\top - I)$ where $\sqrt{\pi}$ is the vector with entries $\sqrt{\pi_i}$. Eigenvalues: 0 (once) and $-\rho$ ($|\mathcal{A}| - 1$ times, degenerate). The J^{kl} matrix has the same simple structure as JC69.

B.1.4 HKY85 (Hasegawa–Kishino–Yano)

Rate matrix. Variable equilibrium frequencies π and a transition/transversion ratio κ_s :

$$R_{ij} = \begin{cases} \kappa_s \pi_j & \text{if } i \sim j \text{ (transition)} \\ \pi_j & \text{if } i \not\sim j \text{ (transversion)} \end{cases}$$

Equivalently, $Q_{ij} = \kappa_s$ for transition pairs and $Q_{ij} = 1$ for transversion pairs. This is K80 + F81: the κ_s structure of K80 with the variable- π structure of F81.

M-step. Pool exchangeabilities within each class:

$$\hat{\kappa}_s = \frac{U_s / N_s}{U_v / N_v} \quad (\text{B.6})$$

where now (unlike K80) the opportunities depend on π :

$$N_s = \sum_{i \sim j, i < j} (W_i \pi_j + W_j \pi_i) \quad (\text{B.7})$$

$$N_v = \sum_{i \not\sim j, i < j} (W_i \pi_j + W_j \pi_i) \quad (\text{B.8})$$

With U_s and U_v as in K80 (summed over directed transition and transversion pairs respectively).

For π , use $\hat{\pi}_i = V'_i / \sum_j V'_j$ as in F81, then iterate with κ_s if desired.

Spectral structure. Let $\pi_R = \pi_A + \pi_G$ and $\pi_Y = \pi_C + \pi_T$. The eigenvalues are 0, $-(\pi_R + \kappa_s \pi_Y)$ (within-pyrimidine contrast), $-(\pi_Y + \kappa_s \pi_R)$ (within-purine contrast), and -1 (purine-vs-pyrimidine contrast). Generically four distinct values; when $\pi_R = \pi_Y = 1/2$ (i.e. K80), the two within-class eigenvalues merge to $-(\kappa_s + 1)/2$.

B.1.5 GTR (General Time-Reversible)

For completeness, we restate the general case from Section A.1.8.

Rate matrix. $R_{ij} = Q_{ij} \pi_j$ with $Q_{ij} = Q_{ji} \geq 0$ for $i \neq j$. For DNA, Q has 6 free parameters (upper triangle) and π has 3 free parameters (4 frequencies summing to 1).

M-step. Fixing π :

$$\hat{Q}_{ij} = \frac{U_{ij} + U_{ji}}{W_i \pi_j + W_j \pi_i} \quad (\text{B.9})$$

Fixing Q , the π update requires solving a nonlinear system (Section A.1.8). In practice, set $\hat{\pi}_i \propto V'_i$ and iterate with (B.9).

Spectral structure. $S_{ij} = Q_{ij}\sqrt{\pi_i\pi_j}$ is real symmetric with eigenvalues $0 = \xi^{(0)} > \xi^{(1)} \geq \dots \geq \xi^{(|\mathcal{A}|-1)}$ (generically all distinct for DNA). No further simplification beyond the general formulas (A.5).

Alternative closed-form π update via secret-destination augmentation. The nonlinearity above arises because π_j appears both in the transition-count term $\propto \pi_j$ (a factor of $\log \pi_j$ per transition into j) and in the exit-rate term $\propto \sum_j S_{ij}\pi_j$ (a linear factor with π_j inside a sum). One can trade an approximate closed form for the fixed-point iteration by re-parameterising each substitution event as (*attempted destination*, *secret destination*) under an EM-style data augmentation with a specific variational distribution q .

Concretely, let each substitution attempt from i occur at a π -independent rate S_{ij} per candidate destination j (the exchangeability). To every attempt attach a secret destination $s \sim \pi(\cdot)$, drawn independently of the attempted destination; the attempt succeeds (a visible substitution $i \rightarrow j$) iff $s = j$, and is otherwise an invisible rejected attempt. The effective visible rate $i \rightarrow j$ is then $S_{ij}\pi_j$, as required.

Let $R_i := \sum_{j' \neq i} S_{ij'}$ be the π -independent total exchangeability out of i , and set $S_{ii} := 0$. Every attempt carries exactly one secret destination, so the augmented data are the counts N_a of attempts whose secret destination is a , and, with the Holmes–Rubin sufficient statistics $\{V_i, U_{ij}, W_i\}$,

$$\ell_{\text{aug}}(\pi) = \sum_a N_a \log \pi_a + \text{const}, \quad N_a = \underbrace{V_a + \sum_i U_{ia}}_{=: V'_a} + Z_a. \quad (\text{B.10})$$

An accepted attempt’s secret destination *is* its observed destination, so those tokens are the visible counts V'_a ; only the tokens of the rejected attempts, Z_a , are latent. The attempt rates S_{ij} do not involve π and the exit factor $\exp(-\sum_i W_i R_i)$ is likewise π -free, so all of the π -dependence sits in the token counts.

Given the trajectory statistics and the current π , the rejected attempts are a thinned Poisson process: from state i , attempts proposing some $j \neq a$ occur at total rate $R_i - S_{ia}$, and each carries secret destination a with probability π_a . Hence

$$\mathbb{E}[Z_a] = \pi_a \sum_i W_i (R_i - S_{ia}), \quad (\text{B.11})$$

and maximising (B.10) on the simplex gives $\hat{\pi} \propto \mathbb{E}[N]$, i.e.

$$\boxed{\hat{\pi}_a \propto V_a + \sum_i U_{ia} + \pi_a \sum_i W_i (R_i - S_{ia})} \quad (\text{B.12})$$

iterated to convergence. The update is a Dirichlet-conjugate count update given $\{V, U, W\}$, which is also convenient for MCMC on π .

The augmented iteration is exact. Equations (B.11)–(B.12) are a genuine E- and M-step for a valid augmentation, so the iteration increases the true conditional log-likelihood monotonically and its fixed points are that likelihood’s stationary points. Writing $T_a = \sum_i W_i S_{ia}$ and $G = \sum_i W_i R_i$, a fixed point with normaliser Z satisfies $\pi_a Z = V'_a + \pi_a(G - T_a)$, that is

$$\frac{V'_a}{\pi_a} - T_a = Z - G, \quad (\text{B.13})$$

which is exactly the Lagrange stationarity condition for maximising $\sum_a V'_a \log \pi_a - \sum_i W_i \sum_{j \neq i} S_{ij} \pi_j$ on the simplex, with the normaliser playing the role of the multiplier. That objective is strictly concave, so its maximiser is unique and the iteration recovers it for *any* exchangeability S : there is no F81 restriction and no bias from exchangeability heterogeneity.

Two nearby estimators are *not* exact, and the distinction matters. Dropping the π_a factor from (B.12) — which would make the augmented counts π -independent and turn the update into a single closed-form step rather than an iteration — replaces the E-step by an π -independent q ; the resulting estimator maximises a lower bound on the likelihood instead of the likelihood, and is biased for any S that is not off-diagonal-constant. And the naive $\hat{\pi} \propto V'$, which ignores the rejected attempts altogether, is biased whenever the dwell times W_a are unequal — including under F81, where the exact stationary point is $\pi_a = V'_a / (\lambda + \mu(W_{\text{tot}} - W_a))$ with $W_{\text{tot}} = \sum_i W_i$, rather than $\propto V'_a$. The π_a -weighted term in (B.12) is precisely the correction for both. The augmentation is originally due to Bruno (10).

B.1.6 GY94 (Goldman–Yang Codon Model)

Rate matrix. The state space is the 61 sense codons ($|\mathcal{A}| = 61$, excluding stop codons). The instantaneous rate from codon i to codon j is:

$$R_{ij} = \begin{cases} 0 & \text{if } i \text{ and } j \text{ differ at more than one position} \\ \pi_j & \text{if synonymous transversion} \\ \kappa_s \pi_j & \text{if synonymous transition} \\ \omega \pi_j & \text{if nonsynonymous transversion} \\ \omega \kappa_s \pi_j & \text{if nonsynonymous transition} \end{cases}$$

where ω is the nonsynonymous/synonymous rate ratio (dN/dS), κ_s is the transition/transversion ratio, and π_j is the equilibrium frequency of codon j . This is a reversible model with exchangeability

$$Q_{ij} = \begin{cases} 0 & \text{(multi-nucleotide change)} \\ 1 & \text{(synonymous transversion)} \\ \kappa_s & \text{(synonymous transition)} \\ \omega & \text{(nonsynonymous transversion)} \\ \omega \kappa_s & \text{(nonsynonymous transition)} \end{cases}$$

M-step. Partition the substitution counts into four classes indexed by (synonymous/nonsynonymous) $\times$ (transition/transversion):

$U_{S,s}$ = total synonymous transition counts $U_{S,v}$ = total synonymous transversion counts

$U_{N,s}$ = total nonsynonymous transition counts $U_{N,v}$ = total nonsynonymous transversion counts

and the corresponding dwell-weighted opportunities $N_{S,s}, N_{S,v}, N_{N,s}, N_{N,v}$ defined analogously to (B.7)–(B.8) but summing over single-nucleotide codon pairs in each class.

The exchangeabilities are $Q = 1$ (synonymous transversion), κ_s (synonymous transition), ω (nonsynonymous transversion), $\omega \kappa_s$ (nonsynonymous transition). The complete-data log-likelihood for (ω, κ_s) , with π fixed, is:

$$\begin{aligned} \ell_2(\omega, \kappa_s) = & U_{S,v} \log 1 + U_{S,s} \log \kappa_s + U_{N,v} \log \omega + U_{N,s} \log(\omega \kappa_s) \\ & - N_{S,v} - \kappa_s N_{S,s} - \omega N_{N,v} - \omega \kappa_s N_{N,s} \end{aligned}$$

Setting $\partial\ell_2/\partial\kappa_s = 0$ and $\partial\ell_2/\partial\omega = 0$:

$$\frac{U_{S,s} + U_{N,s}}{\kappa_s} = N_{S,s} + \omega N_{N,s} \quad (\text{B.14})$$

$$\frac{U_{N,v} + U_{N,s}}{\omega} = N_{N,v} + \kappa_s N_{N,s} \quad (\text{B.15})$$

These are two equations in two unknowns. Substituting (B.15) into (B.14) to eliminate ω :

$$\omega = \frac{U_{N,v} + U_{N,s}}{N_{N,v} + \kappa_s N_{N,s}}$$

$$\frac{U_{S,s} + U_{N,s}}{\kappa_s} = N_{S,s} + \frac{(U_{N,v} + U_{N,s}) N_{N,s}}{N_{N,v} + \kappa_s N_{N,s}}$$

Multiplying both sides by $\kappa_s(N_{N,v} + \kappa_s N_{N,s})$ and rearranging:

$$N_{S,s} N_{N,s} \kappa_s^2 + (N_{S,s} N_{N,v} + (U_{N,v} - U_{S,s}) N_{N,s}) \kappa_s - (U_{S,s} + U_{N,s}) N_{N,v} = 0 \quad (\text{B.16})$$

The positive root gives $\hat{\kappa}_s$, and then

$$\hat{\omega} = \frac{U_{N,v} + U_{N,s}}{N_{N,v} + \hat{\kappa}_s N_{N,s}} \quad (\text{B.17})$$

The codon equilibrium frequencies π are estimated as $\hat{\pi}_j \propto V'_j$.

Spectral structure. For 61 sense codons, the symmetrized rate matrix is 61×61 with at most 61 distinct eigenvalues. No closed-form eigendecomposition is known in general for arbitrary π . When π factors as $\pi_{abc} = \pi_a^{(1)} \pi_b^{(2)} \pi_c^{(3)}$ (the F3x4 model), the symmetrized matrix is a sum of Kronecker products and its eigenvectors factor as tensor products of 4×4 matrices, reducing the eigendecomposition to three 4×4 problems. In practice, for the full GY94, numerical eigendecomposition of the 61×61 symmetric matrix is inexpensive ($O(61^3) \approx 2 \times 10^5$ flops) and need only be performed once per M-step.

B.1.7 Summary and Practical Notes

Model	Free params	π	M-step for exchangeability
JC69	1 (ρ)	fixed 1/4	$\hat{\rho} = U_{\bullet}/W_{\bullet}$
K80	1 (κ_s)	fixed 1/4	$\hat{\kappa}_s = 2U_s/U_v$
F81	4 (ρ, π)	$V'_i / \sum V'_j$	$\hat{\rho} = U_{\bullet} / \sum_i W_i (1 - \pi_i)$
HKY85	5 (κ_s, π)	$V'_i / \sum V'_j$	$\hat{\kappa}_s = (U_s/N_s)/(U_v/N_v)$
GTR	9 (Q, π)	$V'_i / \sum V'_j$	$\hat{Q}_{ij} = (U_{ij} + U_{ji})/(W_i \pi_j + W_j \pi_i)$
GY94	62 (ω, κ_s, π)	$V'_i / \sum V'_k$	quadratic in κ_s (B.16)

For all models above, the indel M-step is identical: solve the κ quadratic (A.22) for (λ, μ) . The substitution and indel parameter blocks decouple completely in the M-step (they share no sufficient statistics).

MAP estimates. To obtain MAP estimates, augment the sufficient statistics before applying the MLE formulas: $U_{ij} \rightarrow U_{ij} + \alpha_Q - 1$, $W_i \rightarrow W_i + \beta_Q$, $V_i \rightarrow V_i + \alpha_\pi - 1$, as in Section C.1.4. The independent $\text{Gamma}(\alpha_Q, \beta_Q)$ priors on each Q_{ij} together with a Dirichlet on π are conjugate to the *irreversible* CTMC complete-data likelihood (independent R_{ij}) but not to the reversible GTR parameterization—the symmetric coupling $Q_{ij} = Q_{ji}$ together with the requirement that the joint complete-data likelihood include the stationary draw $X(0) \sim \pi$ means the proper conjugate prior is the cycle-corrected edge-flow density of Diaconis and Rolles (17), with a Kirchhoff matrix-tree determinant accounting for the cycle structure of the state graph. See the discussion in Section C.1.4 for the form of the Diaconis–Rolles prior and its modification to incorporate the initial-state contribution. The simpler $\text{Gamma} \times \text{Dirichlet}$ pseudocounts used here are non-conjugate regularizers and remain perfectly valid for MAP, with the closed-form augmented M-step above; we adopt them throughout for simplicity.

Degenerate eigenvalues. Models with symmetry (JC69, K80, F81) have degenerate eigenvalues in S , which means the $J^{kl}(T)$ matrix has repeated diagonal entries. This reduces the number of distinct J values that must be computed, but requires care to use the $\xi^{(k)} = \xi^{(l)}$ branch ($J^{kl} = T e^{\xi^{(k)} T}$) rather than the difference quotient, which is a removable 0/0 singularity.

B.1.8 Reversible Mixture with Per-Component GTR

We now consider a C -component reversible mixture in which each component is a full GTR model with its own exchangeability matrix $Q^{(c)}$ and equilibrium distribution $\pi^{(c)}$:

$$R_{ij}^{(c)} = Q_{ij}^{(c)} \pi_j^{(c)} \quad (i \neq j), \quad c \in \{1, \dots, C\}. \quad (\text{B.18})$$

Each site is assigned (latently) to one component, and conditional on c , the site evolves under the reversible CTMC with rate matrix $R^{(c)}$. This is the parameterisation used by the per-(domain, fragment-type) site-class emissions of MixDom (Section C.1.4), where c is the site-class drawn independently at each position from u_{nf} . Because each component is an independent GTR model, the overall rate scale is absorbed into $Q^{(c)}$ itself and no separate rate multiplier is needed; this makes the parameterisation a strict generalisation of per-domain GTR (the per-domain GTR case is recovered when $u_{nf,c} = \mathbf{1}[c = n]$) and of the discrete-gamma rate-across-sites model (69).

E-step counts. The component assignment is latent. From the standard E-step, each branch produces an endpoint pair (a, b) with marginal posterior weight w_{ab} ; conditional on (a, b) at a site with prior class distribution $P(c \mid \text{site}) = u_c$, the component posterior is

$$P(c \mid a, b, T, \text{site}) = \frac{u_c \pi_a^{(c)} M_{ab}^{(c)}(T)}{\sum_{c'} u_{c'} \pi_a^{(c')} M_{ab}^{(c')}(T)}, \quad (\text{B.19})$$

with $M^{(c)}(T) = \exp(R^{(c)}T)$. Per-component bridge-expectation sufficient statistics are then accumulated by weighting the standard expressions (A.3)–(A.2) by (B.19):

$$V_i^{(c)} = \sum_{\text{starts}} w_a P(c \mid a, \cdot) \mathbf{1}[X(0) = i] \quad (\text{B.20})$$

$$U_{ij}^{(c)} = \sum_{\text{branches}} w_{ab} P(c \mid a, b, T, \cdot) \mathcal{C}_{ij}^U(a, b, T; R^{(c)}) \quad (\text{B.21})$$

$$W_i^{(c)} = \sum_{\text{branches}} w_{ab} P(c \mid a, b, T, \cdot) \mathcal{C}_i^W(a, b, T; R^{(c)}) \quad (\text{B.22})$$

and we write $V_i'^{(c)} = V_i^{(c)} + \sum_{j \neq i} U_{ji}^{(c)}$ as before. Writing $U_\bullet^{(c)} = \sum_{i \neq j} U_{ij}^{(c)}$ for the total per-component substitution count.

Complete-data log-likelihood. Because each class is an independent GTR model under this parameterisation, the complete-data log-likelihood decomposes over classes:

$$\ell_2(\{Q^{(c)}\}, \{\pi^{(c)}\}) = \sum_c \ell_2^{(c)}, \quad (\text{B.23})$$

$$\begin{aligned} \ell_2^{(c)} = & \sum_i V_i'^{(c)} \log \pi_i^{(c)} + \sum_{j>i} (U_{ij}^{(c)} + U_{ji}^{(c)}) \log Q_{ij}^{(c)} \\ & - \sum_{j>i} Q_{ij}^{(c)} (W_i^{(c)} \pi_j^{(c)} + W_j^{(c)} \pi_i^{(c)}). \end{aligned} \quad (\text{B.24})$$

Identifiability. No gauge to fix: the rate scale of class c lives inside $Q^{(c)}$, and the $\{Q^{(c)}\}$ are uncoupled in both (B.18) and (B.24). Every class can be updated independently.

Per-class GTR M-step. Since ℓ_2 decomposes class by class, the update for class c is exactly the GTR M-step of Section A.1.8 applied to that class's sufficient statistics. Differentiating (B.24) w.r.t. $Q_{ij}^{(c)}$:

$$\frac{\partial \ell_2^{(c)}}{\partial Q_{ij}^{(c)}} = \frac{U_{ij}^{(c)} + U_{ji}^{(c)}}{Q_{ij}^{(c)}} - (W_i^{(c)} \pi_j^{(c)} + W_j^{(c)} \pi_i^{(c)}) = 0,$$

giving the closed-form update

$$\hat{Q}_{ij}^{(c)} = \frac{U_{ij}^{(c)} + U_{ji}^{(c)}}{W_i^{(c)} \pi_j^{(c)} + W_j^{(c)} \pi_i^{(c)}}. \quad (\text{B.25})$$

The equilibrium update uses the empirical-frequency approximation

$$\hat{\pi}_i^{(c)} \propto V_i'^{(c)}, \quad (\text{B.26})$$

which is exact when $Q_{ij}^{(c)} W_j^{(c)}$ is independent of i (F81 limit) and approximate otherwise, identical to Section A.1.8. No alternation or projection is needed; one pass through each class completes the M-step.

Connection to standard models. With $C = 1$, (B.25)–(B.26) reduce exactly to the GTR M-step (B.9). With $u_{nf,c} = \mathbf{1}[c = n]$ so each domain uses its own dedicated class, the model is the per-domain GTR ($Q^{(n)}$) special case; the per-class M-step here reduces to the per-domain M-step of Section A.1.8 applied once per domain. With $\pi^{(c)} \equiv \pi$ and $Q^{(c)} = \gamma_c Q$ for some shared Q , the model is the discrete-gamma rate model (69); a constrained M-step that enforces $Q^{(c)} \propto Q$ recovers the shared-exchangeability case as a special case. Free-rate variants (56) are the shared- Q / free- γ_c special case.

MAP estimates. Per-class regularising priors apply independently to each class. A Dirichlet pseudocount $\alpha_i^{(c)}$ is added to each $V_i^{(c)}$, and a $\text{Gamma}(a_S, b_S)$ prior on each $Q_{ij}^{(c)}$ (e.g. calibrated to LG via $a_S = b_S Q_{ij}^{\text{LG}}$) adds $a_S - 1$ to the numerator and b_S to the denominator of (B.25). As above, this independent-Gamma $\times$ Dirichlet structure is conjugate only in the irreversible parameterization; it is a non-conjugate regularizer for the reversible GTR mixture component, with the

proper conjugate being a per-class Diaconis–Rolles prior (17). We use the simpler form here for the same reasons as the single-component case (Section C.1.4). The MAP objective

$$\mathcal{L} = \ell_2 + \sum_{c,i} (\alpha_i^{(c)} - 1) \log \pi_i^{(c)} + \sum_{c,j>i} [(a_S - 1) \log Q_{ij}^{(c)} - b_S Q_{ij}^{(c)}] \quad (\text{B.27})$$

decomposes across classes and has a *unique* joint maximum at

$$\hat{Q}_{ij}^{(c)} = \frac{U_{ij}^{(c)} + U_{ji}^{(c)} + a_S - 1}{W_i^{(c)} \pi_j^{(c)} + W_j^{(c)} \pi_i^{(c)} + b_S}, \quad (\text{B.28})$$

with the $\pi^{(c)}$ update obtained by the same bisection as Section A.1.8. Each class is a standalone GTR M-step; no cross-class coupling.

Hyperparameter sizing under SVI. Under SVI-BW (B.52) the EMA sufficient statistics are scaled to the full-dataset size N , so prior pseudocounts must live on the same scale to retain their relative weight. Parameterising by *effective sample size*, set $\alpha_i^{(c)} = N_\pi \pi_i^{\text{LG}}$ and $b_S = N_S$ with $a_S = N_S Q_{ij}^{\text{LG}}$. Defaults $N_\pi, N_S \sim 10^2\text{--}10^3$ (data-independent) give priors that vanish in the data limit but identify the posterior in small batches.

B.1.9 Rate Rescaling: M-Step for a Global Scalar Multiplier

In some training scenarios the relative pattern of substitution rates is known (e.g. from a previously-estimated GTR matrix or from an external published matrix such as LG) and only the overall *rate* needs to adapt to the current data. Concretely, we hold both the exchangeability matrix Q and the equilibrium distribution π fixed, and seek a single scalar multiplier $\sigma > 0$ such that the rescaled rate matrix

$$R_{ij}^{\text{new}} = \sigma Q_{ij} \pi_j \quad (i \neq j) \quad (\text{B.29})$$

maximises the complete-data log-likelihood ℓ_2 . Equivalently, the new exchangeability is $Q_{ij}^{\text{new}} = \sigma Q_{ij}$, so all relative exchangeabilities are preserved and only the overall tempo of the model changes.

Closed-form σ . Substituting $Q_{ij} \rightarrow \sigma Q_{ij}$ into the GTR complete-data log-likelihood,

$$\ell_2(\sigma) = \text{const} + \log \sigma \sum_{j>i} (U_{ij} + U_{ji}) - \sigma \sum_{j>i} Q_{ij} (W_i \pi_j + W_j \pi_i).$$

The terms not involving σ (the $V_i' \log \pi_i$ part and $\sum_{j>i} (U_{ij} + U_{ji}) \log Q_{ij}$) are constants under rescaling. Differentiating and setting to zero yields the unique maximiser

$$\hat{\sigma} = \frac{U_\bullet}{D}, \quad U_\bullet = \sum_{i \neq j} U_{ij}, \quad D = \sum_{j>i} Q_{ij} (W_i \pi_j + W_j \pi_i), \quad (\text{B.30})$$

where $U_\bullet$ is the total expected substitution count and D is the total dwell-weighted opportunity for substitution under the fixed shape (Q, π) . The objective is strictly concave in σ on $(0, \infty)$ (the Hessian $-U_\bullet/\sigma^2 < 0$), so (B.30) is the global maximiser.

Mixture form (per-class). Applied class by class to the reversible mixture of Section B.1.8, each class c has its own scalar σ_c updated independently from its own per-class statistics:

$$\hat{\sigma}_c = \frac{U_{\bullet}^{(c)}}{D^{(c)}}, \quad D^{(c)} = \sum_{j>i} Q_{ij}^{(c)} (W_i^{(c)} \pi_j^{(c)} + W_j^{(c)} \pi_i^{(c)}). \quad (\text{B.31})$$

The shapes $Q^{(c)}$ and $\pi^{(c)}$ are held at their current values and only the per-class rate scale σ_c moves; the class-decomposition (B.23) ensures classes do not couple. A common-rate variant (a single shared $\hat{\sigma}$ across all classes) is recovered by pooling: $\hat{\sigma} = \sum_c U_{\bullet}^{(c)} / \sum_c D^{(c)}$. The per-class form (B.31) attains a strictly higher ℓ_2 unless all σ_c already coincide.

Aggregation of counts. The only sufficient statistics required are $U_{\bullet}^{(c)} = \sum_{i \neq j} U_{ij}^{(c)}$ (a scalar per class) and the dwell–frequency contraction $D^{(c)}$, which is itself a scalar inner product of the fixed quantity $Q_{ij}^{(c)} (W_i^{(c)} \pi_j^{(c)} + W_j^{(c)} \pi_i^{(c)})$ against the per-pair count. Both are linear in the underlying per-pair (W, U) statistics, so under SVI-BW they accumulate via the same EMA used for the standard M-step (no special handling).

MAP estimate. With a $\text{Gamma}(a_\sigma, b_\sigma)$ prior on σ , the augmented M-step is

$$\hat{\sigma} = \frac{U_{\bullet} + a_\sigma - 1}{D + b_\sigma}. \quad (\text{B.32})$$

The Gamma prior on the global rate is conjugate (the model is linear in σ), in contrast to the non-conjugate prior on the full-shape M-step. In the rate-rescaling regime the standard (N_π, N_S) pseudocounts on (Q, π) are inactive (those parameters are frozen); (a_σ, b_σ) centred at $\sigma = 1$ (e.g. $a_\sigma = b_\sigma = N_\sigma$ for some effective sample size N_σ) regularises the rate towards the current frozen scale.

When to use. Rate rescaling is a useful intermediate regime between fully-frozen substitution parameters (no M-step) and the full-shape mixture M-step of Section B.1.8. It is particularly natural when warm-starting from a published matrix and one wishes to absorb a calibration drift in the time units of the data *without* re-estimating the relative chemistry. In an alternating schedule (e.g. alternate σ_c updates with tied- π (B.46) updates) the rescaling step contributes a single guaranteed ℓ_2 ascent per class per iteration, and the alternation does not destabilise the joint objective because each step is an exact maximisation in its own coordinate.

B.1.10 Joint Rate Rescaling and Equilibrium

A natural extension of the rate-rescaling regime (Section B.1.9) frees the equilibrium distribution π jointly with the scale σ while keeping the exchangeability shape Q fixed. This matches the substitution degrees of freedom of a Maraschino-style fit with `--freeze-class-S-shape` (per-class σ_c and $\pi^{(c)}$ free, $Q^{(c)}$ shape locked at LG), and is the natural SVI–BW counterpart when warm-starting from such a fit: the previous `rescaling-rates` mode artificially froze $\pi^{(c)}$ at the warm-start values, dropping $A - 1$ degrees of freedom per class.

Joint complete-data log-likelihood. With $R_{ij} = \sigma Q_{ij} \pi_j$ and Q frozen, the complete-data log-likelihood as a function of (σ, π) is

$$\ell_2(\sigma, \pi) = \sum_i V'_i \log \pi_i + U_\bullet \log \sigma - \sigma \sum_{j>i} Q_{ij} (W_i \pi_j + W_j \pi_i) + \text{const}, \quad (\text{B.33})$$

where the constant collects the frozen $\sum_{j>i} (U_{ij} + U_{ji}) \log Q_{ij}$ term. Using $Q_{ij} = Q_{ji}$ and $Q_{ii} = 0$, the dwell penalty contracts as

$$\sum_{j>i} Q_{ij} (W_i \pi_j + W_j \pi_i) = \sum_b \pi_b r_b, \quad r_b := \sum_{a \neq b} Q_{ab} W_a = (QW)_b,$$

so r_b depends only on the frozen quantities (Q, W) and is constant under variation of (σ, π) . Equation (B.33) simplifies to

$$\ell_2(\sigma, \pi) = \sum_b V'_b \log \pi_b + U_\bullet \log \sigma - \sigma \mathbf{r}^\top \pi + \text{const}. \quad (\text{B.34})$$

Stationary conditions. Lagrangian for $\sum_b \pi_b = 1$ with multiplier λ :

$$\mathcal{L}(\sigma, \pi, \lambda) = \ell_2(\sigma, \pi) - \lambda \left(\sum_b \pi_b - 1 \right).$$

The first-order conditions are

$$\frac{\partial \mathcal{L}}{\partial \sigma} = \frac{U_\bullet}{\sigma} - \mathbf{r}^\top \pi = 0 \implies \sigma = \frac{U_\bullet}{\mathbf{r}^\top \pi}, \quad (\text{B.35})$$

$$\frac{\partial \mathcal{L}}{\partial \pi_b} = \frac{V'_b}{\pi_b} - \sigma r_b - \lambda = 0 \implies \pi_b = \frac{V'_b}{\lambda + \sigma r_b}. \quad (\text{B.36})$$

The Lagrange multiplier is fixed. A clean structural property obtains by multiplying (B.36) through by π_b and summing:

$$\sum_b V'_b = \lambda \sum_b \pi_b + \sigma \mathbf{r}^\top \pi = \lambda + \sigma \mathbf{r}^\top \pi.$$

The σ -FOC (B.35) sets $\sigma \mathbf{r}^\top \pi = U_\bullet$, so

$$N = \lambda + U_\bullet, \quad N := \sum_b V'_b = V_\bullet + U_\bullet,$$

which gives

$$\boxed{\lambda = V_\bullet}, \quad (\text{B.37})$$

independent of σ : the multiplier is fully determined by the boundary count $V_\bullet = \sum_i V_i$.

Closed-form $\pi(\sigma)$ and 1-D root in σ . Substituting (B.37) into (B.36),

$$\hat{\pi}_b(\sigma) = \frac{V'_b}{V_\bullet + \sigma r_b}. \quad (\text{B.38})$$

Note $\sum_b \hat{\pi}_b(\sigma) = 1$ at σ satisfying the σ -FOC, automatically. Plugging back into the σ -FOC (B.35), σ satisfies the single one-dimensional equation

$$g(\sigma) := \sum_b \frac{\sigma r_b V'_b}{V_\bullet + \sigma r_b} - U_\bullet = 0, \quad \sigma > 0. \quad (\text{B.39})$$

Existence and uniqueness. g is well-defined and smooth for $\sigma > 0$ (denominators are positive when $V_\bullet > 0$ or $r_b > 0$). Boundary behaviour:

$$g(0) = -U_\bullet \leq 0, \quad g(\sigma) \xrightarrow{\sigma \rightarrow \infty} \sum_b V'_b - U_\bullet = V_\bullet \geq 0,$$

and the derivative is strictly positive,

$$g'(\sigma) = \sum_b \frac{r_b V'_b V_\bullet}{(V_\bullet + \sigma r_b)^2} > 0 \quad \text{whenever some } r_b V'_b > 0.$$

Hence g is strictly increasing and crosses zero exactly once at a unique $\hat{\sigma} > 0$ (provided $V_\bullet > 0$ and $U_\bullet > 0$; otherwise the boundary $\hat{\sigma} = 0$ or $\hat{\sigma} \rightarrow \infty$ is the maximiser and the regime is degenerate). The joint stationary point $(\hat{\sigma}, \hat{\pi})$ is thus globally unique on the simplex.

Numerical solution: 1-D Newton. g is monotone with a closed-form derivative, so a few Newton steps on $\log \sigma$ (or bisection in σ) starting from the warm σ_{old} converges to machine precision in ≤ 10 iterations. Once $\hat{\sigma}$ is found, $\hat{\pi}$ follows from (B.38).

Equivalent: coordinate ascent. Both individual updates (B.35) and (B.36) are exact closed-form maximisations of ℓ_2 in their own coordinate (for fixed value of the other), so alternating

$$\begin{aligned} \sigma &\leftarrow U_\bullet / (\mathbf{r}^\top \pi), \\ \pi_b &\leftarrow V'_b / (V_\bullet + \sigma r_b) \quad (\text{automatically normalised once } \lambda = V_\bullet \text{ holds}) \end{aligned}$$

gives a monotone ℓ_2 ascent and converges to the unique fixed point identified above. In code, ~ 5 alternation rounds typically suffice; the 1-D Newton on (B.39) is slightly faster and gives the same answer.

Mixture form (per-class). Applied class by class to the reversible mixture of Section B.1.8, each class c has its own pair $(\sigma_c, \pi^{(c)})$ updated independently from its own per-class sufficient statistics $(W^{(c)}, U^{(c)}, V^{(c)})$ and its own fixed $Q^{(c)}$ shape:

$$\hat{\sigma}_c \text{ solves } \sum_b \frac{\hat{\sigma}_c r_b^{(c)} V_b'^{(c)}}{V_\bullet^{(c)} + \hat{\sigma}_c r_b^{(c)}} = U_\bullet^{(c)}, \quad \hat{\pi}_b^{(c)} = \frac{V_b'^{(c)}}{V_\bullet^{(c)} + \hat{\sigma}_c r_b^{(c)}}, \quad (\text{B.40})$$

with $r_b^{(c)} = \sum_{a \neq b} Q_{ab}^{(c)} W_a^{(c)}$. Classes do not couple, by the same class-decomposition (B.23) that justifies per-class M-steps in the standard mixture regime.

Aggregation of counts. The required sufficient statistics per class are $(U_\bullet^{(c)}, V_\bullet^{(c)}, V'^{(c)}, r^{(c)})$. All are linear contractions of the underlying per-pair (W, U, V) statistics against the fixed quantities $(Q^{(c)}, V_\bullet^{(c)} = \text{count of } V \text{ entries})$, so under SVI-BW they accumulate via the same EMA used for the standard M-step (no special handling).

Reduction to pure rescaling. Setting $\pi = \pi_{\text{old}}$ in (B.35) recovers the closed form $\hat{\sigma} = U_\bullet / D$ of (B.30). The joint mode strictly dominates pure rescaling whenever $\hat{\pi} \neq \pi_{\text{old}}$ at the joint optimum (i.e. whenever the Maraschino-fit warm π is not exactly the M-step optimum given the current W, U, V); in practice the gain is largest immediately after a warm start that fixed π away from its joint optimum.

When to use. Use the joint (σ, π) rescaling when warm-starting from a fit that already learnt per-class $(\sigma_c, \pi^{(c)})$ but the SVI–BW M-step should keep updating both with new evidence. Use the pure σ -only rescaling when the warm $\pi^{(c)}$ should be treated as a strong prior or when one wishes to attribute any substitution-rate drift purely to the time scale.

B.1.11 Tied Equilibria across Class Blocks

A second restricted regime ties the equilibrium distribution $\pi^{(c)}$ across blocks of site classes while leaving exchangeabilities $Q^{(c)}$ free. Partition the C classes $\{1, \dots, C\}$ into $B = C/N_{\text{tied}}$ disjoint blocks $\mathcal{B}_1, \dots, \mathcal{B}_B$ each of size N_{tied} (we require $N_{\text{tied}} \mid C$). Within each block b all classes share a common equilibrium distribution

$$\pi^{(c)} \equiv \tilde{\pi}^{(b)} \quad \text{for all } c \in \mathcal{B}_b. \quad (\text{B.41})$$

Block-pooled complete-data log-likelihood. Substituting (B.41) into (B.23), the log-likelihood decomposes across blocks rather than across individual classes:

$$\ell_2 = \sum_{b=1}^B \ell_2^{(b)}, \quad (\text{B.42})$$

$$\begin{aligned} \ell_2^{(b)} = & \sum_i \tilde{V}_i'^{(b)} \log \tilde{\pi}_i^{(b)} + \sum_{c \in \mathcal{B}_b} \sum_{j>i} (U_{ij}^{(c)} + U_{ji}^{(c)}) \log Q_{ij}^{(c)} \\ & - \sum_{c \in \mathcal{B}_b} \sum_{j>i} Q_{ij}^{(c)} (W_i^{(c)} \tilde{\pi}_j^{(b)} + W_j^{(c)} \tilde{\pi}_i^{(b)}), \end{aligned} \quad (\text{B.43})$$

with the pooled equilibrium-character count

$$\tilde{V}_i'^{(b)} = \sum_{c \in \mathcal{B}_b} V_i'^{(c)}. \quad (\text{B.44})$$

The exchangeability parts $Q^{(c)}$ remain class-specific; only the equilibrium term pools.

Block M-step. Within block b , alternate between the per-class $Q^{(c)}$ update (holding $\tilde{\pi}^{(b)}$ fixed) and the pooled $\tilde{\pi}^{(b)}$ update (holding all $Q^{(c)}, c \in \mathcal{B}_b$ fixed). The Q -step uses the standard mixture formula (B.25) with the shared $\tilde{\pi}^{(b)}$:

$$\hat{Q}_{ij}^{(c)} = \frac{U_{ij}^{(c)} + U_{ji}^{(c)}}{W_i^{(c)} \tilde{\pi}_j^{(b)} + W_j^{(c)} \tilde{\pi}_i^{(b)}} \quad (c \in \mathcal{B}_b). \quad (\text{B.45})$$

The $\tilde{\pi}^{(b)}$ -step is the same Lagrange-multiplier solve as the unconstrained mixture M-step, but with pooled coefficients. Differentiating (B.43):

$$\tilde{\pi}_i^{(b)} = \frac{\tilde{V}_i'^{(b)}}{\tilde{c}_i^{(b)} - \eta}, \quad \tilde{c}_i^{(b)} = \sum_{c \in \mathcal{B}_b} \sum_{j \neq i} Q_{ij}^{(c)} W_j^{(c)}, \quad (\text{B.46})$$

with η found by 1D bisection from $\sum_i \tilde{V}_i'^{(b)} / (\tilde{c}_i^{(b)} - \eta) = 1$, exactly as in Section A.1.8. Each coordinate update is an exact maximiser within its coordinate; the joint objective on $(Q^{(\cdot)}, \tilde{\pi}^{(b)})$ is strictly concave on $\mathbb{R}_+^{|\mathcal{B}_b| \cdot \binom{|A|}{2}} \times \Delta^{|A|-1}$ (the latter the simplex), so the coordinate ascent converges to the unique block MLE.

Connection to standard models. With $N_{\text{tied}} = 1$ (one class per block) the tying is trivial and (B.46) reduces to the per-class Lagrange solve of Section B.1.8. With $N_{\text{tied}} = C$ (one block containing all classes) all classes share a single $\tilde{\pi}$ — the discrete-rate-classes parameterisation of (69) when the $Q^{(c)}$ are also proportional to a shared shape. For intermediate N_{tied} the tying captures shared equilibrium chemistry across classes that nonetheless differ in their exchangeability (and hence in their substitution *pattern* and *rate*), without forcing every class to maintain its own equilibrium estimate.

Pseudocounts. The LG-informed Dirichlet pseudocount on $\tilde{\pi}^{(b)}$ adds $N_{\pi} \pi_i^{\text{LG}}$ to $\tilde{V}_i'^{(b)}$ once per block (not once per class within the block); the per-class Gamma(a_S, b_S) pseudocount on $Q_{ij}^{(c)}$ enters the class-specific update (B.45) unchanged.

Combination with rate rescaling. Tied- π and rate rescaling can also be *combined* within a single M-step: each class has its own scalar σ_c but classes within a block share the equilibrium $\tilde{\pi}^{(b)}$. The coordinate-ascent factorisation does dispatch in that joint regime; see Section B.1.12. The frozen- π mode (which fixes $\pi^{(c)}$ at its initialisation, with no M-step update) remains an alternative restriction. Alternation *across* M-steps (rate rescaling on odd iterations, tied- π on even iterations) is valid in addition because each step is an unconditional ascent on the joint ℓ_2 .

B.1.12 Tied Equilibria with Per-Class Rate Rescaling

The two restrictions of Sections B.1.9 and B.1.11 can be applied jointly to the same mixture M-step: each class c in block $\mathcal{B}_b$ has its own *scalar* rate multiplier σ_c , with the exchangeability shape $Q^{(c)}$ frozen, while all classes within the block share a common equilibrium $\tilde{\pi}^{(b)}$. This matches the substitution degrees of freedom of a Maraschino-style fit with `--freeze-class-S-shape` under a block-tied `class_pi` constraint and is strictly more constrained than the joint $(\sigma_c, \pi^{(c)})$ M-step of Section B.1.10.

Block-pooled complete-data log-likelihood. With $R_{ij}^{(c)} = \sigma_c Q_{ij}^{(c)} \tilde{\pi}_j^{(b)}$ for $c \in \mathcal{B}_b$ (with $Q^{(c)}$ frozen), the per-block log-likelihood is

$$\ell_2^{(b)}(\{\sigma_c\}_{c \in \mathcal{B}_b}, \tilde{\pi}^{(b)}) = \sum_b \tilde{V}_{b'}'^{(b)} \log \tilde{\pi}_{b'}^{(b)} + \sum_{c \in \mathcal{B}_b} \left[U_{\bullet}^{(c)} \log \sigma_c - \sigma_c \mathbf{r}^{(c)\top} \tilde{\pi}^{(b)} \right] + \text{const}, \quad (\text{B.47})$$

where the constant collects $\sum_c \sum_{j>i} (U_{ij}^{(c)} + U_{ji}^{(c)}) \log Q_{ij}^{(c)}$ (frozen), $\tilde{V}_{b'}'^{(b)} = \sum_{c \in \mathcal{B}_b} V_{b'}'^{(c)}$ is the block-pooled boundary count, and $r_{b'}^{(c)} = \sum_{a \neq b'} Q_{a,b'}^{(c)} W_a^{(c)}$ depends only on the frozen $(Q^{(c)}, W^{(c)})$.

Stationary conditions. Lagrangian for $\sum_{b'} \tilde{\pi}_{b'}^{(b)} = 1$ with multiplier $\lambda^{(b)}$:

$$\frac{\partial}{\partial \sigma_c} = \frac{U_{\bullet}^{(c)}}{\sigma_c} - \mathbf{r}^{(c)\top} \tilde{\pi}^{(b)} = 0 \implies \sigma_c = \frac{U_{\bullet}^{(c)}}{\mathbf{r}^{(c)\top} \tilde{\pi}^{(b)}}, \quad (\text{B.48})$$

$$\frac{\partial}{\partial \tilde{\pi}_{b'}^{(b)}} = \frac{\tilde{V}_{b'}'^{(b)}}{\tilde{\pi}_{b'}^{(b)}} - \sum_{c \in \mathcal{B}_b} \sigma_c r_{b'}^{(c)} - \lambda^{(b)} = 0 \implies \tilde{\pi}_{b'}^{(b)} = \frac{\tilde{V}_{b'}'^{(b)}}{\lambda^{(b)} + \sum_{c \in \mathcal{B}_b} \sigma_c r_{b'}^{(c)}}. \quad (\text{B.49})$$

Block Lagrange multiplier is fixed. Multiplying (B.49) by $\tilde{\pi}_{b'}^{(b)}$ and summing over b' ,

$$\sum_{b'} \tilde{V}_{b'}'^{(b)} = \lambda^{(b)} + \sum_{c \in \mathcal{B}_b} \sigma_c \mathbf{r}^{(c)\top} \tilde{\pi}^{(b)}.$$

Substituting (B.48) into the right-hand side collapses the σ_c -dependent term to $\sum_c U_{\bullet}^{(c)}$, giving

$$\boxed{\lambda^{(b)} = \tilde{V}_{\bullet}^{(b)}}, \quad \tilde{V}_{\bullet}^{(b)} := \sum_{c \in \mathcal{B}_b} V_{\bullet}^{(c)} = \sum_{c \in \mathcal{B}_b} \sum_i V_i^{(c)}, \quad (\text{B.50})$$

generalising the single-class result (B.37). The multiplier is fully determined by the block-pooled boundary count.

Closed-form coupled updates. Substituting (B.50) into (B.49),

$$\hat{\pi}_{b'}^{(b)} = \frac{\tilde{V}_{b'}'^{(b)}}{\tilde{V}_{\bullet}^{(b)} + \sum_{c \in \mathcal{B}_b} \sigma_c r_{b'}^{(c)}}. \quad (\text{B.51})$$

Equation (B.51) together with (B.48) forms a closed nonlinear system in the $|\mathcal{B}_b| + A$ unknowns $(\{\sigma_c\}, \tilde{\pi}^{(b)})$ per block. Two practical solvers, both monotone in $\ell_2^{(b)}$:

1. *Coordinate ascent.* Alternate $\sigma_c \leftarrow U_{\bullet}^{(c)} / (\mathbf{r}^{(c)\top} \tilde{\pi}^{(b)})$ for each $c \in \mathcal{B}_b$ and $\tilde{\pi}^{(b)} \leftarrow \tilde{V}'^{(b)} / (\tilde{V}_{\bullet}^{(b)} + \sum_c \sigma_c r^{(c)})$ (componentwise), with a final renormalisation safety guard. Each update is the exact closed-form maximiser in its own coordinate; convergence to the unique block-optimum follows in ~ 5 –20 rounds.
2. *Substitute and root-find.* Eliminate σ_c from (B.51) via (B.48); the result is a vector fixed-point equation in $\tilde{\pi}^{(b)} \in \Delta^{A-1}$ alone. When $|\mathcal{B}_b|$ is small (typical: 2–5), the resulting low-dimensional system is amenable to a damped Newton iteration on the $A - 1$ free $\tilde{\pi}$ components; in practice coordinate ascent is faster and simpler.

Both solvers reach the same fixed point because $\ell_2^{(b)}$ is strictly concave in each block of variables given the others (the log terms ensure positivity through the FOCs).

Mixture-wide M-step. The blocks decouple in the same way the classes do under Section B.1.11: the joint $\ell_2 = \sum_b \ell_2^{(b)}$ separates over b , so each block can be solved independently.

Aggregation of counts. Per block, the required sufficient statistics are the per-class $(U_{\bullet}^{(c)}, V'^{(c)}, r^{(c)})$ for $c \in \mathcal{B}_b$ plus the block totals $\tilde{V}'^{(b)}, \tilde{V}_{\bullet}^{(b)}$. All are linear in (W, U, V) and accumulate via the same EMA used by the standard M-step.

Reduction to special cases. Setting $|\mathcal{B}_b| = 1$ (each class its own block) recovers the joint $(\sigma_c, \pi^{(c)})$ M-step of Section B.1.10; setting all $\sigma_c = 1$ (no rate rescaling within block) recovers a degenerate version of the tied- π M-step with frozen exchangeabilities; freezing $\tilde{\pi}^{(b)}$ at its current value recovers the rate-only rescaling of Section B.1.9. Strict dominance holds in both directions: this regime improves on Section B.1.9 by adding the π degree of freedom and improves on Section B.1.11 (with $Q^{(c)}$ frozen at LG) by adding per-class σ_c .

When to use. Use this regime when the warm-start fit (e.g. Maraschino with `--freeze-class-S-shape`) ties classes into blocks with shared equilibrium chemistry but distinct rates, and the SVI–BW M-step should preserve that tying while continuing to update both $(\sigma_c, \tilde{\pi}^{(b)})$ from new evidence. The block tying is a strong inductive bias when the number of classes per block is small relative to A and the per-class equilibrium counts $V^{(c)}$ would otherwise be too sparse to estimate $\pi^{(c)}$ reliably.

B.1.13 Stochastic Variational Baum-Welch

For large datasets it is impractical to run the E-step over all training pairs every iteration. A natural stochastic extension is to sample a minibatch of pairs, run the E-step on those, and update parameters. Naively replacing the full sufficient statistics with each batch’s statistics discards information from previous batches and results in noisy, slowly converging parameter estimates.

The framework of stochastic variational inference (27) provides a principled resolution in the conditionally conjugate case. Each M-step parameter group in TKF91 (and TKF92 and MixDom) has an exponential-family complete-data log-likelihood, regular for the multinomial groups (mixture weights, fragment-type transitions, site-class distributions) and curved for the BDI rates and the reversible-CTMC (GTR) submodel (Sections A.1.3, A.1.2). For the multinomial groups we use Dirichlet priors that are strictly conjugate; for the BDI and GTR groups we use independent $\text{Gamma} \times \text{Dirichlet}$ pseudocounts that are non-conjugate regularisers in the reversible / curved-family parameterisation (Section C.1.4). For the conjugate groups, Hoffman et al.’s natural-gradient argument applies in full and the update (B.52) below is exact natural-gradient ascent on the variational ELBO. For the non-conjugate groups, the natural-gradient interpretation no longer holds verbatim, but the same exponential-moving-average update remains a sound stochastic-MAP-EM scheme: each M-step is affine in the augmented sufficient statistics, so an EMA of statistics commutes with the M-step and converges to the MAP fixed point of full-batch EM under the standard Robbins–Monro step-size conditions. In practice both groups are updated by the same rule:

$$\bar{T}^{(t+1)} = (1 - \rho_t) \bar{T}^{(t)} + \rho_t \left(\bar{T}_0 + \frac{N}{|B_t|} T_{B_t} \right), \quad (\text{B.52})$$

where $\bar{T}^{(t)}$ denotes the running sufficient statistics at iteration t (transition counts $\hat{n}_{ij}$, bridge-expectation dwell times W_i and jump counts U_{ij} , character counts V_i , and BDI expectations $\hat{B}, \hat{D}, \hat{S}$), $\bar{T}_0$ the prior pseudocounts, T_{B_t} the batch sufficient statistics from the E-step on minibatch B_t , N the total number of training pairs, $|B_t|$ the batch size, and $\rho_t = (t + \tau)^{-\kappa}$ the Robbins–Monro step size with delay $\tau \geq 0$ and forgetting rate $\kappa \in (0.5, 1]$.

The M-step—quadratic solve for indel rates (A.22), Dirichlet MAP for mixture weights, Beta MAP for extension rates, and the iterative coordinate ascent for (Q, π) —is then applied to $\bar{T}^{(t+1)}$ exactly as in full-batch EM. Each M-step maximizes a (penalised) Q-function that is linear in the sufficient statistics, so averaging the statistics before solving is equivalent to finding the MAP of the averaged Q-function. For the conjugate parameter groups this maximizer is also the natural parameter of the updated variational posterior; for the non-conjugate groups it is the MAP of the regularised Q-function and not a variational posterior, but the iteration is identical.

This retains the closed-form exactness of every M-step while enabling online processing of arbitrarily large pair collections, with each training pair visited only once. The two hyperparameters τ and κ have robust defaults ($\tau = 10$, $\kappa = 0.7$) that work across a range of dataset sizes without tuning (27).

Equivalent formulations of the prior pseudocounts. Equation (B.52) places the prior pseudocounts $\bar{T}_0$ inside the EMA update. An equivalent implementation accumulates only the data

contribution in $\bar{T}$ and adds $\bar{T}_0$ at M-step time:

$$\tilde{T}^{(t+1)} = (1 - \rho_t) \tilde{T}^{(t)} + \rho_t \frac{N}{|B_t|} T_{B_t}, \quad \theta^{(t+1)} = \text{M-step}(\tilde{T}^{(t+1)} + \bar{T}_0).$$

For every M-step considered in this paper (Dirichlet MAP, Gamma-augmented counts, pooled GTR formula), the M-step operator is affine in its sufficient statistics, so $M(\bar{T}^{(t)}) = M(\tilde{T}^{(t)} + \bar{T}_0)$ exactly at every iteration; the two prescriptions yield numerically identical estimates. This affineness is what we actually need for the EMA-of-pseudocounts formulation, and it holds whether or not the prior is strictly conjugate to the complete-data likelihood (Section C.1.4 discusses the distinction). The post-hoc form is slightly more convenient for implementation because each M-step keeps the prior encapsulated in its closed-form update; see the code comment at the SVI loop for details.

B.2 Stochastic Variational Baum–Welch Convergence

B.2.1 Setup

We consider the idealized infinite-data setting: each minibatch of B sequence pairs is drawn i.i.d. from the true generative model, so there is no model misspecification. Stochastic Variational Baum–Welch (SVB) alternates:

1. **E-step:** Run forward–backward on a minibatch of B pairs, accumulate expected sufficient statistics $\hat{T}_B$.
2. **M-step:** Update parameters $\theta_{k+1} = M(\hat{T}_B)$ using the closed-form M-step.

At convergence, the true parameters θ^* are a fixed point of $\theta \mapsto M(\mathbb{E}_\theta[T])$.

B.2.2 BDI sufficient statistics

For a single sequence pair (x, y) with ancestor length i , descendant length j , and evolutionary time t , the BDI sufficient statistics are $T = (\mathbb{E}[B \mid x, y], \mathbb{E}[D \mid x, y], \mathbb{E}[S \mid x, y])$, recovered via the score function identity. The M-step is (Section A.1.8):

$$\hat{\lambda} = \frac{\bar{B}}{\bar{S} + t}, \quad \hat{\mu} = \frac{\bar{D}}{\bar{S}}, \quad (\text{B.53})$$

where $\bar{B}, \bar{D}, \bar{S}$ denote averages (or sums) over the minibatch.

B.2.3 Variance of Minibatch Estimates

Per-pair variance Let $\sigma_B^2, \sigma_D^2, \sigma_S^2$ denote the variances of $\mathbb{E}[B \mid x, y], \mathbb{E}[D \mid x, y], \mathbb{E}[S \mid x, y]$ under the data-generating distribution (randomness over (x, y) pairs drawn from the model). For a minibatch of B independent pairs, the sample means satisfy

$$\text{Var}[\bar{B}] = \frac{\sigma_B^2}{B}, \quad \text{Var}[\bar{D}] = \frac{\sigma_D^2}{B}, \quad \text{Var}[\bar{S}] = \frac{\sigma_S^2}{B}.$$

Parameter variance via the delta method By the delta method applied to (B.53), the relative error of $\hat{\lambda}$ after one M-step on a minibatch of size B is

$$\text{Var}[\hat{\lambda}]/\lambda^2 \approx \frac{1}{B} \left[\frac{\sigma_B^2}{\mathbb{E}[B]^2} + \frac{\sigma_S^2}{(\mathbb{E}[S] + t)^2} - \frac{2 \text{Cov}[B, S]}{\mathbb{E}[B](\mathbb{E}[S] + t)} \right] \equiv \frac{v_\lambda}{B}, \quad (\text{B.54})$$

and similarly for $\hat{\mu}$:

$$\text{Var}[\hat{\mu}]/\mu^2 \approx \frac{1}{B} \left[\frac{\sigma_D^2}{\mathbb{E}[D]^2} + \frac{\sigma_S^2}{\mathbb{E}[S]^2} - \frac{2 \text{Cov}[D, S]}{\mathbb{E}[D] \mathbb{E}[S]} \right] \equiv \frac{v_\mu}{B}. \quad (\text{B.55})$$

The constants v_λ, v_μ depend on (λ, μ, t) and the sequence length distribution. They are the diagonal entries of the inverse Fisher information matrix (per observation), expressed in the (λ, μ) parameterization.

Scaling of per-pair variance with sequence length The dominant source of randomness is the ancestor/descendant length pair (i, j) . At stationarity with $\kappa = \lambda/\mu < 1$:

- $\mathbb{E}[i] = \kappa/(1 - \kappa)$, $\text{Var}[i] = \kappa/(1 - \kappa)^2$.
- $\mathbb{E}[B | i] \approx \lambda \mathbb{E}[S | i]$, $\mathbb{E}[S | i] \approx i/(\mu - \lambda)$ for $t \gg 1/(\mu - \lambda)$.
- Conditioned on i , the links evolve as independent sub-lineages, so $v_\lambda(i) = \Theta(1/i)$. The relative variance *does* fall as a pair gets longer, and the Fisher information for λ grows linearly in i . What is fixed is not the information per link but the equilibrium length, pinned at $\mathbb{E}[i] = \kappa/(1 - \kappa)$.

Thus the per-pair Fisher information grows linearly in i , as it does for substitution parameters. Indels differ only in the constant and in the fact that the stationary length is not a free variable, so at stationarity extra precision comes from processing more pairs rather than from lengthening a given one.

Proposition B.1 (Per-pair Fisher information, long-time limit). *For $\delta = \mu - \lambda > 0$ and $t \rightarrow \infty$:*

$$\mathbb{E}[B] \rightarrow \frac{\kappa}{1 - \kappa}, \quad \mathbb{E}[D] \rightarrow \frac{\kappa}{1 - \kappa}, \quad \mathbb{E}[S] \rightarrow \frac{2\kappa}{(1 - \kappa)\delta},$$

and the coefficient of variation of $\hat{\lambda}$ from a single pair is $O(1)$ (bounded away from zero).

This reflects the fact that, at stationarity, each pair provides a single “observation” of the BDI process endpoint distribution $P(j | i)$ — the information content is fundamentally per-pair, not per-residue.

B.2.4 Pseudocount representation and its advantages

The SVB iterates can be represented equivalently in two ways: as an EMA of data-only sufficient statistics s_k with priors α added inside each M-step, or as an EMA directly in *posterior pseudocount* space $\tilde{\alpha}_k = s_k + \alpha$. In the latter view, the update rule is

$$\tilde{\alpha}_k = (1 - \eta_k) \tilde{\alpha}_{k-1} + \eta_k (\alpha + (N/|B|) s_{\text{batch}_k}), \quad (\text{B.56})$$

matching (B.52) with the prior $T_0 \equiv \alpha$ absorbed into the EMA target. By induction, $\tilde{\alpha}_k = s_k + \alpha$ at every step given a matched initial state, so the two formulations produce identical parameter iterates.

For the Dirichlet-multinomial groups, $\tilde{\alpha}$ is the natural parameter of the conjugate posterior; for the BDI / GTR groups it is the augmented sufficient-statistic vector at which the regularised Q-function is maximised (the underlying priors there are non-conjugate regularisers; see Section C.1.4). In either case, the closed-form M-step maximises

$$J(\theta | \tilde{\alpha}) = \tilde{\alpha} \cdot \log \theta - Z(\theta), \quad (\text{B.57})$$

where Z is the log-partition function of the family. For a Dirichlet-multinomial component (transitions, mixture weights, class pis, classdist), $J = \sum_i (\tilde{\alpha}_i - 1) \log \theta_i$ up to a constant, with unique maximizer $\theta_i \propto \tilde{\alpha}_i - 1$. The BDI rates (λ, μ) follow the same pattern but the joint J couples them through the geometric fragment length prior and the M-step is obtained from the quadratic in $\kappa = \lambda/\mu$ of (A.22). The SVB iteration thus reduces to updating one pytree of pseudocounts per step, with M-step parameters computed on demand.

Variance via a single Jacobian. Let $f: \tilde{\alpha} \mapsto \theta$ denote the M-step map. The delta method gives

$$\text{Var}[\theta_k] \approx J_f(\tilde{\alpha}_k) \text{Var}[\tilde{\alpha}_k] J_f(\tilde{\alpha}_k)^\top. \quad (\text{B.58})$$

Equations (B.54)–(B.55) are the diagonal entries of (B.58) in the (λ, μ) parameterization; in pseudocount form, the same delta method is one derivative of f away regardless of parameter group.

Polyak–Ruppert averaging is built in. Expanding (B.56),

$$\tilde{\alpha}_K = \sum_{j=1}^K w_{j,K} (\alpha + (N/|B|) s_{\text{batch}_j}), \quad w_{j,K} = \eta_j \prod_{i>j}^K (1 - \eta_i).$$

The pseudocount state is itself a weighted average of batch-level pseudocounts, and the M-step applied to this average is the Rao–Blackwell improvement over averaging parameter iterates — which matters when f is nonlinear, as for the BDI rate M-step (§C.1.4, eq. (A.22)). An effective sample size

$$\text{ESS}_K = \frac{(\sum_j w_{j,K})^2}{\sum_j w_{j,K}^2}$$

quantifies how “warmed up” the EMA is and is strictly less than K whenever $\eta_k < 1$. For a parameter group receiving a non-negligible contribution in only a fraction ε of batches (a rare domain, a rarely-visited fragment state, a rarely-active class), the per-group effective sample size is εESS_K , not ESS_K — exposing the rare-parameter bottleneck of §B.3 as a direct quantity rather than an approximation.

Complete-data Fisher information. For a Dirichlet-multinomial component, the Hessian of J at its maximum is diagonal in natural-parameter coordinates with $[I_{\text{comp}}(\tilde{\alpha})]_{ii} = 1/\tilde{\alpha}_i$. Rare categories (small $\tilde{\alpha}_i$) therefore have large per-component variance, and the bottleneck for each parameter group can be read off directly as $\tilde{\alpha}_i/\text{ESS}$ — a diagnostic in natural-parameter space that replaces the CV-based convergence recommendation of item 6 in §B.2.

Practical consequence. An implementation that carries $\tilde{\alpha}_k$ as its primary state — rather than separately tracking s_k and re-adding α in each M-step — makes (B.58), the Polyak–Ruppert weights, and the per-group Fisher diagnostics all computable from the same pytree, with no additional bookkeeping.

B.2.5 SVB Convergence Rate

Stochastic approximation framework SVB is a stochastic fixed-point iteration. By the results of Cappé & Moulines (2009), under regularity conditions satisfied by exponential families, the parameter iterates θ_k satisfy

$$\sqrt{K}(\bar{\theta}_K - \theta^*) \xrightarrow{d} \mathcal{N}(0, \Sigma) \quad (\text{B.59})$$

where $\bar{\theta}_K = \frac{1}{K} \sum_{k=1}^K \theta_k$ is the Polyak–Ruppert average and Σ depends on the learning rate schedule.

Direct M-step (no averaging). With direct M-step updates (step size 1), the iterate θ_K has variance

$$\text{Var}[\theta_K] \approx \frac{\Sigma_{\text{per-pair}}}{B} \quad (\text{B.60})$$

after convergence of the transient, where $\Sigma_{\text{per-pair}}$ is the asymptotic covariance of the M-step estimator applied to a single pair. This does *not* decrease with K — the iterates fluctuate around θ^* with amplitude $O(1/\sqrt{B})$.

Polyak–Ruppert averaging. Averaging the iterates:

$$\text{Var}[\bar{\theta}_K] \approx \frac{\Sigma_{\text{per-pair}}}{BK}. \quad (\text{B.61})$$

This is optimal: the total number of pairs processed is $N = BK$, and we achieve variance Σ/N regardless of how N is split between B and K .

Exponential moving average (EMA). In practice, one often uses $\theta_{k+1} = (1 - \eta)\theta_k + \eta M(\hat{T}_B)$ with step size $\eta \in (0, 1]$. The asymptotic variance is

$$\text{Var}[\theta_\infty] \approx \frac{\eta}{2 - \eta} \cdot \frac{\Sigma_{\text{per-pair}}}{B} \approx \frac{\eta \Sigma_{\text{per-pair}}}{2B} \quad (\eta \ll 1).$$

This trades convergence speed for lower asymptotic variance.

Convergence transient The EM rate matrix governs the transient. For exponential families, the EM convergence rate is $\rho = 1 - I_{\text{obs}}/I_{\text{comp}}$, where I_{obs} and I_{comp} are the observed and complete-data Fisher information matrices. For BDI, the fraction of missing information is typically moderate ($\rho \approx 0.3$ – 0.7 for typical protein evolution times $t \sim 0.5$ – 2), so the transient dies out in $O(10)$ iterations.

B.2.6 Practical Recommendations

Minibatch size For target relative error ε on indel parameters after K steps with Polyak–Ruppert averaging:

$$BK \geq \frac{v_\theta}{\varepsilon^2}, \quad (\text{B.62})$$

where $v_\theta \sim O(1)$ is the per-pair relative variance from (B.54)–(B.55).

Target ε	BK needed	$B = 50, K = ?$	$B = 200, K = ?$
0.10	$\sim 100 v_\theta$	2	1
0.05	$\sim 400 v_\theta$	8	2
0.01	$\sim 10,000 v_\theta$	200	50

Rule of thumb. Take $v_\theta \approx 5$ as a conservative estimate for typical protein BDI parameters ($\kappa \approx 0.9$, $t \approx 1$). Then:

- $B \geq 50$ ensures that each M-step estimate has $\lesssim 30\%$ relative error on indel parameters.
- $B \geq 200$ brings single-step error below $\sim 15\%$.
- For rare domains ($w_d \approx 0.05$), multiply B by $1/w_d = 20$.
- After K steps with averaging, total error scales as $1/\sqrt{BK}$.

Natural gradient The BDI complete-data statistics form a curved exponential family with natural parameters $(\log \lambda, \log \mu, -(\lambda + \mu))$ and sufficient statistics (B, D, S) . The natural gradient replaces the M-step update $\Delta\theta = -I_F^{-1} \nabla_{\theta} Q(\theta)$ where I_F is the Fisher information matrix of the complete-data family.

For the BDI M-step, the closed-form solution (B.53) already implicitly incorporates the complete-data Fisher information structure. Stochastic natural gradient (SNatGrad) would precondition the *stochastic* update direction using I_F^{-1} , but since $\hat{\lambda} = \bar{B}/(\bar{S} + t)$ is already an unbiased estimator of λ (to first order), the benefit of explicit preconditioning is marginal.

Recommendation: Natural gradient is not worth the implementation complexity for SVB on TKF-family models. The closed-form M-step already provides the natural parameterization. Use Polyak–Ruppert averaging of the M-step outputs instead.

Summary of recommendations

1. **Minibatch size:** $B \geq 50$ for general parameters; $B \geq 200$ or use averaging if rare domains ($w_d < 0.1$) are present.
2. **Averaging:** Use Polyak–Ruppert averaging of parameter iterates. Total error $\propto 1/\sqrt{BK}$, optimal regardless of B/K split.
3. **Step size:** With direct M-step ($\eta = 1$), iterates oscillate with amplitude $O(1/\sqrt{B})$. EMA with $\eta = 2/(k + 2)$ (Polyak schedule) achieves $O(1/\sqrt{BK})$.
4. **Decoupled updates:** Substitution parameters converge $\sim \bar{L}$ times faster than indel parameters. Consider accumulating indel statistics over $\sim \bar{L}/B$ minibatches before updating.
5. **Natural gradient:** Not recommended; the closed-form M-step is already natural.
6. **Convergence diagnostic:** Monitor the running coefficient of variation of $\hat{\lambda}_d, \hat{\mu}_d$ across minibatches. Convergence when $CV < \varepsilon$.

B.2.7 Bias diagnostics: Hellinger, ESS, and Fisher readouts

The convergence diagnostics above answer “have the parameter iterates stabilised?” but are silent on a distinct question that arises in practice: when validation log-likelihood *regresses* during training, is the regression within the noise floor predicted by the minibatch variance, or is it evidence of a structural problem (class-assignment drift, over-concentration of a mixture component, over-fitting of the training likelihood)? The pseudocount view of §B.2.4 makes several readouts available that are cheap to compute from saved checkpoints and sharply discriminate among these hypotheses.

Hellinger distance on mixture distributions. For any probability vector p (e.g. a row of `classdist[d, f, :]`, a fragment-transition row $r_{f,\cdot}^{(d)}$, or a domain-weight vector), the Hellinger distance

$$H(p, q) = \frac{1}{\sqrt{2}} \|\sqrt{p} - \sqrt{q}\|_2 \in [0, 1]$$

is the natural metric: bounded, symmetric (unlike KL), and *locally linear in probability-mass movement*. A transfer of δ mass from one category to another produces $H \approx \sqrt{\delta}/\sqrt{2}$, so a 1% reassignment registers as $H \approx 0.07$ and a clean class-swap gives $H = 1$. The drift hypothesis — that a mixture component is discretely reassigned between iterations k and $k+1$ and that this reassignment accounts for the val-LL regression — predicts one or more iter-to-iter Hellinger values well above the

EM-concentration noise floor, where the noise floor is empirically ~ 0.005 on training runs seeded from an informed mixture profile (e.g. IQ-TREE’s C10). A “sharp” drift signature appropriate to a $\sim 10\%$ per-pair val-LL regression would be $H > 0.15$ at the responsible iter; a flat trajectory at $H \leq 0.02$ throughout the regression window falsifies the discrete-drift hypothesis and redirects investigation to continuous alternatives (over-fitting of the training likelihood, slow limit cycles, or systematic mis-specification).

Effective sample size per parameter group. As derived in §B.2.4, the EMA effective sample size $\text{ESS}_K = (\sum_j w_{j,K})^2 / \sum_j w_{j,K}^2$ is a first-class quantity in pseudocount space, strictly smaller than K whenever $\eta_k < 1$. For a parameter group receiving a non-negligible batch contribution in only a fraction ε of minibatches, the *group-specific* effective sample size is εESS_K , turning the rare-parameter bottleneck of §B.3 from a theoretical concern into a direct quantity that can be logged per iter.

Fisher readout of remaining uncertainty. Combining the two quantities, the per-component posterior uncertainty for a Dirichlet-multinomial parameter group is proportional to $\tilde{\alpha}_i / \text{ESS}$: rare categories with small posterior pseudocount and small per-group ESS are exactly those whose point estimates are least certain. This replaces the CV-of- $\hat{\theta}$ heuristic of item 6 above with a reading in natural-parameter space and lets one rank parameter groups by residual uncertainty without running a second pass of DP.

Suggested dashboard. At minimum, log per iter: (i) η_k and ESS_k ; (ii) for each mixture group (`classdist`, `ext` rows, `dom_weights`), the max Hellinger distance to the previous iter and the entropy of the current iterate; (iii) $\|\Delta \tilde{\alpha}_k\|_1 / \|\tilde{\alpha}_k\|_1$ per parameter group as an $\tilde{\alpha}$ -space convergence diagnostic; (iv) the Hungarian-matched distance of the current `class_pis` (or analogous mixture-profile tensor) to the initial informed seed, as a “drift-from-prior” signature independent of relabelling. All are computable from saved parameters alone; no DP is required. Taken together, these give a rapid read-out on whether a val-LL regression reflects minibatch noise, a discrete drift event, or systematic over-fitting — and thus whether the response should be a larger minibatch, a frozen-component diagnostic, or a stronger prior.

B.3 Expected Statistics and Linearized Convergence

The preceding sections characterized SVB convergence in terms of abstract per-pair variances $\sigma_B^2, \sigma_D^2, \sigma_S^2$. We now derive these quantities in closed form at stationarity, giving concrete convergence estimates for the BDI parameter group of any TKF-family model. The corresponding specialisation to the MixDom hierarchy of top-level, per-domain, and intra-fragment chains is collected in Appendix C.6.

B.3.1 BDI expected statistics at stationarity

Lemma B.2 (Expected BDI sufficient statistics). *Consider a single BDI process with insertion rate λ , deletion rate μ , $\kappa = \lambda/\mu < 1$, $\delta = \mu - \lambda > 0$, and define $\alpha = e^{-\mu T}$, $\Phi = 1 - \alpha$. For an ancestor–descendant pair (i, j) with ancestor length i drawn from the stationary distribution $i \sim \text{Geom}(\kappa)$*

(i.e. $P(i) = (1 - \kappa)\kappa^i$ for $i = 0, 1, 2, \dots$), the expected BDI sufficient statistics are:

$$\mathbb{E}_\pi[i] = L = \frac{\kappa}{1 - \kappa} \quad (\text{B.63})$$

$$\mathbb{E}_\pi[S] = LT \quad (\text{B.64})$$

$$\mathbb{E}_\pi[B] = \lambda(L + 1)T = \frac{\lambda T}{1 - \kappa} \quad (\text{B.65})$$

$$\mathbb{E}_\pi[D] = \mu LT = \frac{\mu \kappa T}{1 - \kappa} = \mathbb{E}_\pi[B] \quad (\text{B.66})$$

with $M = 1$ (one trajectory endpoint per pair) and $T = T$ (total observation time for a single process).

The key intermediate result is: for a BDI process starting from $N(0) = i$ mortal links, the expected number at time s is

$$\mathbb{E}[N(s) \mid N(0) = i] = i e^{-\delta s} + L(1 - e^{-\delta s}), \quad (\text{B.67})$$

so the time-integrated expected count is

$$\mathbb{E}[S \mid i] = \int_0^T \mathbb{E}[N(s) \mid i] ds = (i - L) \frac{1 - e^{-\delta T}}{\delta} + LT. \quad (\text{B.68})$$

At stationarity, $\mathbb{E}_\pi[i] = L$, so $\mathbb{E}_\pi[S] = LT$. The birth and death expectations follow from $\mathbb{E}[B \mid i] = \lambda \mathbb{E}[S \mid i] + \lambda T$ (the λT term is the immortal link's contribution) and $\mathbb{E}[D \mid i] = \mu \mathbb{E}[S \mid i]$. The conservation law $\mathbb{E}_\pi[B] = \mathbb{E}_\pi[D]$ confirms that stationarity is maintained.

Proof. Equation (B.67) is the standard result for a linear birth-death process with immigration at rate λ and per-capita death rate μ . The i initial links each survive for $\text{Exp}(\mu)$ time (giving the $i e^{-\delta s}$ decay), while the immigration generates new links at rate λ from the immortal link plus the mortal links themselves; the mean converges to the stationary value $L = \kappa/(1 - \kappa)$ as $s \rightarrow \infty$.

Integrating (B.67):

$$\begin{aligned} \mathbb{E}[S \mid i] &= \int_0^T [i e^{-\delta s} + L(1 - e^{-\delta s})] ds \\ &= i \frac{1 - e^{-\delta T}}{\delta} + L \left(T - \frac{1 - e^{-\delta T}}{\delta} \right) \\ &= (i - L) \frac{1 - e^{-\delta T}}{\delta} + LT. \end{aligned}$$

Taking $\mathbb{E}_\pi[i] = L$, the $(i - L)$ term vanishes, giving $\mathbb{E}_\pi[S] = LT$.

For births: $\mathbb{E}_\pi[B] = \lambda \mathbb{E}_\pi[S] + \lambda T = \lambda(L + 1)T = \lambda T/(1 - \kappa)$.

For deaths: $\mathbb{E}_\pi[D] = \mu \mathbb{E}_\pi[S] = \mu LT = \mu \kappa T/(1 - \kappa) = \lambda T/(1 - \kappa) = \mathbb{E}_\pi[B]$. $\square$

B.3.2 Per-pair variance of sufficient statistics

The randomness in (B, D, S) across pairs arises from the randomness in the ancestor and descendant lengths (i, j) . At stationarity, $i \sim \text{Geom}(\kappa)$ with $\text{Var}[i] = \kappa/(1 - \kappa)^2$.

Since $\mathbb{E}[S \mid i]$ is linear in i (eq. (B.68)), the variance of $\mathbb{E}[S \mid i]$ across pairs is:

$$\text{Var}_\pi[\mathbb{E}[S \mid i]] = \left(\frac{1 - e^{-\delta T}}{\delta} \right)^2 \cdot \frac{\kappa}{(1 - \kappa)^2} \equiv \sigma_{S, \text{outer}}^2. \quad (\text{B.69})$$

Similarly, since $\mathbb{E}[B \mid i] = \lambda \mathbb{E}[S \mid i] + \lambda T$:

$$\text{Var}_\pi[\mathbb{E}[B \mid i]] = \lambda^2 \sigma_{S,\text{outer}}^2, \quad (\text{B.70})$$

and $\mathbb{E}[D \mid i] = \mu \mathbb{E}[S \mid i]$:

$$\text{Var}_\pi[\mathbb{E}[D \mid i]] = \mu^2 \sigma_{S,\text{outer}}^2. \quad (\text{B.71})$$

By the law of total variance, $\text{Var}_\pi[S] = \text{Var}_\pi[\mathbb{E}[S \mid i]] + \mathbb{E}_\pi[\text{Var}[S \mid i]]$. The inner variance $\text{Var}[S \mid i]$ captures the randomness of the BDI trajectory given a fixed starting state i . This is harder to compute in closed form, but for the purpose of convergence estimates, the outer variance (B.69) provides a *lower bound* on $\text{Var}[S]$ and is the dominant term when κ is near 1 (long sequences), since both sources of variance scale as $O(\kappa/(1-\kappa)^2)$.

The covariance structure is also dominated by the outer term:

$$\text{Cov}_\pi[\mathbb{E}[B \mid i], \mathbb{E}[S \mid i]] = \lambda \sigma_{S,\text{outer}}^2. \quad (\text{B.72})$$

B.3.3 Relative error and per-pair coefficients of variation

Substituting into the delta-method formulas (B.54)–(B.55) with $\mathbb{E}[B] = \lambda T/(1-\kappa)$, $\mathbb{E}[S] = LT$, and the outer-variance approximation:

Define

$$\psi = \frac{1 - e^{-\delta T}}{\delta T} \in (0, 1)$$

(the ratio of the “regression to mean” time scale to the total time). Then:

$$\sigma_{S,\text{outer}}^2 = \psi^2 T^2 \frac{\kappa}{(1-\kappa)^2} \quad (\text{B.73})$$

$$\frac{\sigma_{S,\text{outer}}^2}{\mathbb{E}_\pi[S]^2} = \frac{\psi^2}{\kappa/(1-\kappa)^2 \cdot T^2} \cdot \frac{\kappa}{(1-\kappa)^2} = \frac{\psi^2 (1-\kappa)^2}{\kappa} \cdot \frac{1}{1}$$

The relative variance of $\hat{\lambda}$ from a single pair becomes (using the outer-variance approximation in (B.54), and noting that the B – S covariance term partially cancels the B and S terms):

$$v_\lambda \approx \frac{\psi^2 (1-\kappa)^2}{\kappa} \left[1 + \frac{\lambda^2}{(\lambda + 1/T)^2} - \frac{2\lambda}{\lambda + 1/T} \right] = \frac{\psi^2 (1-\kappa)^2}{\kappa (1 + \lambda T)^2} \quad (\text{B.74})$$

This uses $\mathbb{E}[S] + T = (L+1)T = T/(1-\kappa)$, so $\lambda/(\lambda + 1/T) = \lambda T/(1 + \lambda T) = \kappa \delta T/(\delta + \delta/(\kappa \dots))$ —more directly, substitute into (B.54):

$$\begin{aligned} v_\lambda &= \frac{\lambda^2 \sigma_{S,\text{outer}}^2}{\mathbb{E}[B]^2} + \frac{\sigma_{S,\text{outer}}^2}{(\mathbb{E}[S] + T)^2} - \frac{2\lambda \sigma_{S,\text{outer}}^2}{\mathbb{E}[B](\mathbb{E}[S] + T)} \\ &= \sigma_{S,\text{outer}}^2 \left(\frac{\lambda^2 (1-\kappa)^2}{\lambda^2 T^2} + \frac{(1-\kappa)^2}{T^2} - \frac{2\lambda (1-\kappa)^2}{\lambda T^2} \right) \\ &= \frac{\psi^2 \kappa}{(1-\kappa)^2} \cdot \frac{(1-\kappa)^2}{T^2} \cdot (1 + 1 - 2) \\ &= 0. \end{aligned}$$

The outer-variance contribution to v_λ vanishes exactly! This is because $\mathbb{E}[B \mid i] = \lambda (\mathbb{E}[S \mid i] + T)$, so the ratio $\mathbb{E}[B \mid i]/(\mathbb{E}[S \mid i] + T) = \lambda$ is constant in i —the delta-method perturbation is zero.

This means the **dominant source of variance in $\hat{\lambda}$ is the inner (trajectory) variance**, not the length distribution. The per-pair coefficient of variation of $\hat{\lambda}$ is determined by the fluctuations in (B, D, S) given the endpoint pair (i, j) , which depends on the stochastic path structure of the BDI process.

Proposition B.3 (Per-pair relative variance from inner fluctuations). *For $\hat{\lambda} = \bar{B}/(\bar{S}+T)$ estimated from a single pair with ancestor length i drawn from stationarity:*

$$v_{\lambda} \approx \frac{\mathbb{E}_{\pi}[\text{Var}[B \mid i, j]]}{\mathbb{E}_{\pi}[B]^2} + \frac{\mathbb{E}_{\pi}[\text{Var}[S \mid i, j]]}{(\mathbb{E}_{\pi}[S] + T)^2} - \frac{2 \mathbb{E}_{\pi}[\text{Cov}[B, S \mid i, j]]}{\mathbb{E}_{\pi}[B] (\mathbb{E}_{\pi}[S] + T)}. \quad (\text{B.75})$$

The inner variances $\text{Var}[B \mid i, j]$ and $\text{Var}[S \mid i, j]$ are the posterior variances of the BDI trajectory statistics given the observed endpoints. These are computable from the observed Fisher information (Hessian of the log-likelihood), but do not simplify to elementary closed forms.

However, we can bound the total per-pair relative variance using the observed Fisher information. For a single BDI process observed at one timepoint, the complete-data Fisher information for λ is $I_{\text{comp}}(\lambda) = \mathbb{E}[S + T]/\lambda^2 = T/(\lambda^2(1 - \kappa))$, and the observed (incomplete-data) information satisfies $I_{\text{obs}} \leq I_{\text{comp}}$. The per-pair relative variance is $v_{\lambda} \geq 1/(\lambda^2 I_{\text{obs}}) \geq 1/(\lambda^2 I_{\text{comp}})$, giving:

$$v_{\lambda} \geq \frac{1 - \kappa}{\lambda T} = \frac{1}{\mathbb{E}_{\pi}[B]}. \quad (\text{B.76})$$

The per-pair relative variance is at least $1/\mathbb{E}_{\pi}[B]$. When $\mathbb{E}_{\pi}[B]$ is small (few births per pair), the relative error is large.

Practical approximation. For moderate times ($\delta T \sim 1$), the missing-information fraction is $\rho \approx 0.3\text{--}0.7$, so $I_{\text{obs}} \approx (1 - \rho) I_{\text{comp}}$ and $v_{\lambda} \approx (1 - \kappa)/(\lambda T (1 - \rho))$. Taking $\rho \approx 0.5$ as a rough estimate:

$$v_{\lambda} \approx \frac{2}{\mathbb{E}_{\pi}[B]} = \frac{2(1 - \kappa)}{\lambda T}. \quad (\text{B.77})$$

Similarly:

$$v_{\mu} \approx \frac{2}{\mathbb{E}_{\pi}[D]} = \frac{2(1 - \kappa)}{\mu \kappa T}. \quad (\text{B.78})$$

B.4 Maraschino: Cherry-Counts for TKF92

We earlier introduced Maraschino as a TKF92 generalization of CherryML in Section A.2.4. We now develop it in full.

The CherryML composite likelihood replaces the joint phylogenetic likelihood with a product of pairwise sibling (“cherry”) likelihoods (50), exploiting the fact that for a fixed alignment, the alignment-substitution counts are sufficient statistics for the substitution rate parameters. This reduces the training data—which can be many gigabytes of alignments—to a fixed-shape counts tensor whose size depends only on the alphabet $\mathcal{A}$ and the number of discretized cherry-time bins, and is independent of the number of training alignments. The resulting compact tensor fits in GPU memory and may be optimized by standard autograd-based gradient methods.

Maraschino extends CherryML to TKF92 by additionally including the alignment *transition* counts as sufficient statistics for the indel rates and fragment-extension probability. Concretely, for each pair of consecutive (non-empty) alignment columns we record the source column type ($\mathbf{X} \in \{\mathbf{S}, \mathbf{M}, \mathbf{I}, \mathbf{D}\}$), the destination column type ($\mathbf{Y} \in \{\mathbf{M}, \mathbf{I}, \mathbf{D}, \mathbf{E}\}$), and the ancestor/descendant characters in each column. We refer to the resulting tensor of counts—binned by discretized pairwise

divergence time and aggregated over training cherries—as the *cherry-count tensor*, and the procedure for training TKF92 from it as Maraschino.

The Maraschino composite log-likelihood for TKF92 is the sum, over divergence-time bins b and adjacency types $\mathbf{X} \rightarrow \mathbf{Y}$, of the cherry-count multiplied by the model log-probability of that adjacency at that bin’s representative time:

$$\begin{aligned} \mathcal{L}_{\text{cherry}}^{\text{TKF92}}(\lambda, \mu, r, Q, \pi) = \sum_{b=1}^{n_\tau} \left[\sum_{\mathbf{X}, \mathbf{Y}} n_{\mathbf{X} \rightarrow \mathbf{Y}}^{(b)} \log \tau'_{\mathbf{X} \rightarrow \mathbf{Y}}(\lambda, \mu, \bar{T}_b, r) \right. \\ \left. + \sum_{a,b} n_{a,b}^{\text{M},(b)} \log [\pi_a \exp(R\bar{T}_b)_{ab}] + \sum_b n_b^{\text{I},(b)} \log \pi_b + \sum_a n_a^{\text{D},(b)} \log \pi_a \right] \quad (\text{B.79}) \end{aligned}$$

where $n_{\mathbf{X} \rightarrow \mathbf{Y}}^{(b)}$ are the binned transition counts, $n_{a,b}^{\text{M},(b)}$, $n_b^{\text{I},(b)}$, $n_a^{\text{D},(b)}$ are the binned per-state emission counts, τ' is the TKF92 Pair HMM transition matrix from Section A.2.2, and $\bar{T}_b$ is the representative time of bin b (typically the geometric mean of the bin endpoints). The first term scores indel/fragment behaviour; the remaining three score the substitution model. Boundary contributions ($\mathbf{S} \rightarrow \cdot$, $\cdot \rightarrow \mathbf{E}$, $\mathbf{S} \rightarrow \mathbf{E}$ for empty alignments) are scored against the corresponding $\mathbf{S}$ -row and $\mathbf{E}$ -column entries of τ' .

Why this is a sufficient-statistic compression. For fixed alignment and fixed discretized cherry times, the entire collection of training cherries enters equation (B.79) only through the binned counts. In particular, two cherry collections with identical counts yield identical $\mathcal{L}_{\text{cherry}}^{\text{TKF92}}$, identical gradients, and identical maximum-likelihood estimates. The counts tensor is therefore a sufficient statistic for the TKF92 parameters under the cherry-count composite likelihood.

Sharded count accumulation. For a corpus such as Pfam-Full, cherries are extracted from each MSA in parallel and contribute additively to the global counts tensor. A simple map-reduce over Pfam clans accumulates the counts in constant memory per worker; the global tensor is the elementwise sum of per-shard tensors and is independent of the order of accumulation.

B.4.1 Optimizing Maraschino: gradient methods vs. inner EM

The cherry-count log-likelihood $\mathcal{L}_{\text{cherry}}^{\text{TKF92}}$ is a smooth function of the unconstrained parameters $(\log \lambda, \log \mu, \log r, \log Q, \log \pi)$ (with the simplex and exchangeability constraints handled by softmax / log-Cholesky reparameterisation). Two distinct optimizers apply.

Maraschino-as-autograd. The objective is differentiable, so Adam, L-BFGS, and similar optimizers may be applied directly via autograd. Each step requires one matrix exponential per cherry-time bin, and a forward+backward pass through equation (B.79); both steps are highly parallel on GPU.

Maraschino-around-EM. Alternatively, the full-batch EM iterates of Section A.2.3 apply, with the cherry-count tensor playing the role of the E-step output. Concretely:

1. For each bin b , run a one-step expected-count update on the TKF92 Pair HMM at parameters $\theta^{(t)}$ and time $\bar{T}_b$, using the cherry-count tensor as the input “observed alignment.” This replaces the usual Forward–Backward E-step with a closed-form linear pass that resolves the $\mathbf{M}$, $\mathbf{I}$, $\mathbf{D}$ self-loop counts into fragment-extension counts and link-level counts (Section A.2.3).

2. Accumulate the BDI sufficient statistics (B, D, L, M, S, T) via the score identity (Section A.1.2), and the bridge-expectation substitution statistics (W, U, V) via the matched-pair contribution.
3. Apply the closed-form M-step (equations (A.22)–(A.25) for λ, μ ; $r \leftarrow F/(F + E)$ for fragment extension; Section A.1.8 item 1 for Q, π).

Because every M-step in Section A.2.3 is closed-form and linear in the sufficient statistics, the inner EM iterates are exact ascent on $\mathcal{L}_{\text{cherry}}^{\text{TKF92}}$, with no gradient-step or learning-rate hyperparameter. In practice, EM converges in ~ 10 – 30 iterations on protein cherry-count tensors and is competitive with Adam on the same objective.

Practical comparison. Adam handles arbitrary regularizers and is straightforward to combine with stochastic minibatching over time-bins or alphabet slices. Inner EM does not require a learning rate, has monotonic convergence guarantees, and produces the natural-gradient direction “for free.” When the cherry-count tensor fits in memory (it does, even for Pfam-Full at $|\mathcal{A}| = 20$ and $n_\tau = 20$), inner-EM Maraschino is the recommended default.

B.4.2 EM-around-Maraschino: mixtures of TKF92

A mixture of K TKF92 components, each with its own indel rates, fragment-extension probability, and substitution model, captures heterogeneity within and across protein families. We fit such a mixture by an outer EM loop with Maraschino as the inner optimizer of each component:

Algorithm 1 EM-around-Maraschino for a mixture of K TKF92 components.

- 1: Initialize component parameters $\theta_1^{(0)}, \dots, \theta_K^{(0)}$ and mixture weights $\pi_1^{(0)}, \dots, \pi_K^{(0)}$.
 - 2: **for** $t = 0, 1, \dots$ **do**
 - 3: **E-step.** For each cherry c in the training set, compute the responsibility $\gamma_{c,k}^{(t)} \propto \pi_k^{(t)} P(c \mid \theta_k^{(t)})$ (the per-component cherry-count likelihood normalised across components). Each $P(c \mid \theta_k^{(t)})$ is a single forward pass through the cherry-count likelihood at the cherry’s counts subtensor.
 - 4: **Component-weighted count accumulation.** For each component k , accumulate the responsibility-weighted cherry-count tensor $C_k^{(t+1)} = \sum_c \gamma_{c,k}^{(t)} C_c$.
 - 5: **Component M-step (Maraschino).** For each component k , run Maraschino (Section B.4.1) on $C_k^{(t+1)}$ to obtain $\theta_k^{(t+1)}$.
 - 6: **Mixture-weight M-step.** $\pi_k^{(t+1)} \propto \sum_c \gamma_{c,k}^{(t)}$.
 - 7: **end for**
-

The outer EM is monotone in the mixture log-likelihood $\sum_c \log \sum_k \pi_k P(c \mid \theta_k)$ because both the responsibilities (E-step) and the per-component Maraschino solves (M-step) are exact ascent on their respective Q-functions.

The decisive feature is that the outer M-step does *not* require re-extracting cherries or recomputing alignments per iteration: the per-component cherry-count tensor $C_k^{(t+1)}$ is a responsibility-weighted average of the same per-cherry tensors that were extracted once and for all at the start of training.

B.4.3 EM-around-CherryML: mixtures over site classes

A complementary class of mixture model places the latent variable on *MSA columns* rather than on whole cherries: each MSA column c draws an unobserved *site class* $z_c \sim \text{Categorical}(\pi_1, \dots, \pi_C)$ from a shared per-class substitution model $\text{Subst}(Q^{(k)}, \pi^{(k)})$, and every cherry in the family that has

a residue pair at column c evolves its match emission under the same class. Because no indel rates vary across mixture components, this is strictly a substitution-only mixture, and the corresponding fitter is CherryML rather than Maraschino: the cherry-count tensor of Section B.4 contributes only the matched substitution observations (a, b, t_p) and the indel-free composite likelihood reduces to that of (50).

Column-shared responsibilities. The defining observation is that within a single MSA, all cherries that have a match at column c inherit the *same* latent class z_c . This couples cherries across the same column: the column-level posterior

$$\gamma_c(k) \propto \pi_k \prod_{p: \mathbf{M} \text{ at } c} P(a_{p,c}, b_{p,c} \mid k, t_p), \quad P(a, b \mid k, t) = \pi_a^{(k)} \exp(R^{(k)} t)_{ab},$$

multiplies the per-cherry match-emission likelihoods over all cherries p that contribute a residue pair to column c . Cherries that have a gap at c contribute no factor for that column. The resulting composite log-likelihood

$$\mathcal{L}_{\text{site-mix}} = \sum_m \sum_{c \in \text{cols}(m)} \log \sum_{k=1}^C \pi_k \prod_{p: \mathbf{M} \text{ at } c} \pi_{a_{p,c}}^{(k)} \exp(R^{(k)} t_p)_{a_{p,c} b_{p,c}} \quad (\text{B.80})$$

is a sum, over MSAs m and over columns c within each MSA, of a log-mixture in which cherries share the column's class. This is in direct contrast to EM-around-Maraschino (Section B.4.2), whose responsibilities $\gamma_{c,k}^{(t)}$ are per *cherry*: there each cherry is assigned to a single component independently of the columns it spans.

Outer / inner EM. We fit (B.80) by alternating an MSA-wide column-responsibility update with a per-class bridge-expectation M-step. At iteration t , the column responsibilities are

$$\gamma_{m,c}^{(t)}(k) \propto \pi_k^{(t)} \prod_{p: \mathbf{M} \text{ at } c} \pi_{a_{p,c}}^{(t,k)} \exp(R^{(t,k)} t_p)_{a_{p,c} b_{p,c}}, \quad (\text{B.81})$$

normalised across k for each (m, c) , and the responsibility-weighted bridge-expectation sufficient statistics for class k are

$$\begin{aligned} \hat{W}_i^{(t+1,k)} &= \sum_{m,c} \gamma_{m,c}^{(t)}(k) \sum_{p: \mathbf{M} \text{ at } c} \mathcal{C}_i^W(a_{p,c}, b_{p,c}, t_p), \\ \hat{U}_{ij}^{(t+1,k)} &= \sum_{m,c} \gamma_{m,c}^{(t)}(k) \sum_{p: \mathbf{M} \text{ at } c} \mathcal{C}_{ij}^U(a_{p,c}, b_{p,c}, t_p), \\ \hat{V}_i^{(t+1,k)} &= \sum_{m,c} \gamma_{m,c}^{(t)}(k) \sum_{p: \mathbf{M} \text{ at } c} \delta(a_{p,c} = i), \end{aligned} \quad (\text{B.82})$$

where $\mathcal{C}_i^W, \mathcal{C}_{ij}^U$ are the standard endpoint-conditioned bridge expectations (Section A.1.2, eqs. (A.2)–(A.3)) at the class- k rate matrix $R^{(t,k)}$. The full outer iteration is then Algorithm 2.

The outer iteration is monotone in $\mathcal{L}_{\text{site-mix}}$: the column responsibilities are the exact posterior over z_c and the per-class GTR M-step is exact ascent on its (responsibility-weighted) Q-function. The inner GTR EM is itself iterative because the GTR M-step couples $(Q^{(k)}, \pi^{(k)})$ nonlinearly (Section A.1.8, item 2 of the M-step list); a few inner sweeps per outer iteration suffice in practice.

Algorithm 2 EM-around-CherryML for C site classes.

- 1: Initialize per-class GTR parameters $\theta_k^{(0)}$ and class weights $\pi_k^{(0)}$.
 - 2: **for** $t = 0, 1, \dots$ **do**
 - 3: **Outer E-step.** Compute column responsibilities $\gamma_{m,c}^{(t)}(k)$ via (B.81).
 - 4: **Sufficient-statistic accumulation.** Compute $(\hat{W}^{(t+1,k)}, \hat{U}^{(t+1,k)}, \hat{V}^{(t+1,k)})$ via (B.82).
 - 5: **Per-class GTR inner EM.** For each class k , run the closed-form GTR coordinate-ascent M-step of Section A.1.8 on $(\hat{W}^{(t+1,k)}, \hat{U}^{(t+1,k)}, \hat{V}^{(t+1,k)})$ to obtain $\theta_k^{(t+1)}$. Because the responsibility weights are fixed, this step is a standard CherryML rate fit on a single GTR model with weighted observations.
 - 6: **Class-weight update.** $\pi_k^{(t+1)} \propto \sum_{m,c} \gamma_{m,c}^{(t)}(k)$.
 - 7: **end for**
-

Why CherryML, not Maraschino. No indel rates appear in (B.80): the model assumes the alignment is given and that all alignment columns share a single TKF92 indel process whose parameters are held fixed (or estimated elsewhere, e.g. by Maraschino on the same cherries). Mixing *only* the substitution model means the (λ, μ, r) -related row-normalisation and fragment-extension responsibility issues of TKF92 do not arise; the relevant sufficient statistics are exactly those of CherryML. The only structural change relative to plain CherryML is the per-MSA, per-column responsibility weighting in (B.82), which couples cherries that share columns of the same MSA.

Relation to per-class GTR mixtures elsewhere. The per-class GTR M-step of equation (B.82) is exactly the reversible-mixture update of Appendix B.1.8; the only difference between this appendix and Algorithm 2 is the provenance of the responsibilities (here: cross-cherry column sharing within an MSA; in the appendix: any user-supplied per-site posterior). Site-mixture parameters fitted by EM-around-CherryML can therefore be plugged into mixture-aware downstream methods — including Maraschino with a fixed indel-process backbone — without further re-derivation.

B.4.4 Expected sufficient statistics as fast custom VJPs

The Maraschino objective (equation (B.79)) is concretely a sum of log-probabilities, each of which has a closed-form gradient via the score identity and the BDI / bridge-expectation sufficient statistics. Specifically, the (λ, μ) -gradient of $\log \tau'_{\mathbf{x} \rightarrow \mathbf{y}}(\lambda, \mu, t, r)$ is given exactly by the score-derivative tables of Sections A.5.2–A.5.3, and the (Q, π) -gradient of $\log[\pi_a \exp(Rt)_{ab}]$ is given by the bridge-expectation dwell times and jump counts (equations (A.2)–(A.3)).

For training pipelines built around stochastic-gradient optimizers such as Adam, this observation enables a substantial speedup. Rather than relying on autograd to back-propagate through a matrix exponential and an eigendecomposition, we register a *custom vector-Jacobian product (VJP)* that returns the score-derivative-and-bridge-expectation closed forms directly:

- Forward pass: compute and cache the matrix exponential $P(t) = e^{Rt}$ and the BDI link-level transition factors (α, β, γ) at each cherry-time bin.
- Backward pass: contract the upstream cotangent with the expected sufficient statistics (bridge expectations and BDI score expressions) instead of with the autograd-computed Jacobian.

The forward and backward costs are both $O(|\mathcal{A}|^2)$ per bin per emission state for the substitution part and $O(1)$ for the indel part, matching the cost of evaluating the closed-form M-step itself.

This pattern generalises beyond Maraschino. Any model whose M-step is closed-form linear in expected sufficient statistics admits a corresponding fast custom VJP that exposes those expectations as the gradient of the observed log-likelihood (this is, again, the score identity (A.11)).

Concretely, this enables Adam and other gradient-based optimizers to train TKF91, TKF92, and mixture-of-TKF92 models at a per-step cost matching exact EM, while retaining Adam’s flexibility to incorporate arbitrary differentiable regularizers (e.g., weight decay on the BDI rates, KL-divergence priors on the GTR exchangeability), to interleave updates with non-EM-amenable parameters (e.g., the per-component mixture weights of Section B.4.2 when fit jointly with neural-network embeddings), and to work with the mini-batch SVI-BW pseudocount stream of Section B.1.13.

B.5 Selected Inference Algorithms for TKF92

This section describes four inference algorithms that operate on TKF91 / TKF92 models and on phylogenetic trees: the *Fast Statistical Alignment* (FSA) framework with a Newton-step time-optimization on the per-cherry expected log-likelihood (Section C.2.1); the *Beam Search Ancestral Reconstruction* (BeamASR) algorithm (Section C.2.2); the *Variational Ancestral Reconstruction* (VarAnc) algorithm that operates on a fixed MSA with known gap structure (Section B.5.3); and its stochastic-variational training extension *svi-VarAnc* (Section B.5.4).

B.5.1 Fast Statistical Alignment (FSA)

Given a set of sequences and a phylogenetic tree with branch-specific TKF91 or TKF92 pair HMMs, we construct a multiple sequence alignment using the *sequence annealing* approach of (9), to which we refer the reader for a full description of the algorithm.

Briefly, the method proceeds as follows. For each pair of sequences (x, y) in a selected subset (either all $\binom{N}{2}$ pairs or an $O(N \log N)$ Erdős–Rényi sample), we compute pairwise residue alignment posteriors $P(x_i \sim y_j)$ by running the Forward-Backward algorithm on the pair HMM at an optimized evolutionary time $\hat{\tau}$.

Per-pair time optimization (NR step). The time $\hat{\tau}$ is found by Newton–Raphson optimization of the expected log-likelihood (the “NR step”):

$$\hat{\tau} = \underset{\tau}{\operatorname{argmax}} \mathbb{E}_{P(\pi|x,y,\tau_0)} [\log P(x, y, \pi \mid \tau)] \quad (\text{B.83})$$

where π ranges over alignment paths and τ_0 is an initial estimate. This expectation is computed from Forward–Backward expected counts at τ_0 , and—because the TKF92 expected log-likelihood is quadratic-like in τ in a neighbourhood of the maximum, with analytic derivatives via the BDI score identities of Section A.1.3—the Newton iteration typically converges in 3–5 steps. This time-maximisation differs from the original FSA approach of (9), which attempts to optimize all model parameters via unregularized EM for every pair, and consequently must terminate the EM recursion early to avoid instability. Restricting the optimization to the per-pair time τ , while keeping the model parameters fixed at the population estimate, is both cheaper and more stable.

Sequence annealing. The pairwise posteriors are then assembled into a multiple alignment by the greedy sequence annealing procedure of (9), which iteratively merges alignment columns to maximize a sum-of-pairs posterior objective.

B.5.2 Beam Search Ancestral Sequence Reconstruction (BeamASR)

We now describe an alternative progressive reconstruction method that finds the maximum-likelihood ancestral sequence at each internal node by beam search, without materializing the full composite automaton.

At each internal node v with children l, r and observed descendant sequences c_l, c_r , we seek

$$\hat{a}_v = \underset{a}{\operatorname{argmax}} [\log P(a, c_l \mid B_l) + \log P(a, c_r \mid B_r) - \log P(a \mid R)] \quad (\text{B.84})$$

where $P(a, c_k \mid B_k)$ is the pair HMM forward probability on branch k and $P(a \mid R)$ is the singlet probability under the root generator, subtracted to avoid double-counting the prior on a .

Incremental forward profiles Since the branches are conditionally independent given the ancestor, we can evaluate (C.9) by maintaining *incremental forward profiles*: for each branch k , a 1D forward table $F_k[i, q]$ giving the log-probability that descendant positions $1, \dots, i$ have been emitted and the branch machine is in state q , given ancestor positions $1, \dots, j$ processed so far.

Each ancestor character extends both profiles independently in $O(L_k)$ time per branch, where $L_k = |c_k|$.

Beam search The ancestor sequence $\hat{a}_v$ is built left-to-right by beam search. At each position j , the beam maintains B candidate partial ancestors. For each candidate and each alphabet character σ :

1. Extend both branch profiles by one ancestor character σ , comprising a *match/delete phase* (the ancestor emits σ , descendant positions advance via M or D transitions) and an *insertion phase* (descendant-only insertions following the ancestor emission).
2. Update the singlet forward score for σ .
3. Score the extension: $\Delta(j, \sigma) = \Delta F_l + \Delta F_r - \Delta_{\text{singlet}}$.

The top B extensions (by cumulative score) are retained. Total cost per node is $O(K \cdot B \cdot A \cdot (L_l + L_r))$ where $K = |\hat{a}_v|$ and A is the alphabet size.

Insertion phase via associative scan The insertion recurrence within each branch profile has the form

$$x_{i+1} = \operatorname{logsumexp}(A_{II} x_i, b_i) + e_i \quad (\text{B.85})$$

where A_{II} is the I-to-I log-transition submatrix, b_i collects transitions into insertion states from M and D, and e_i is the emission score. This is a log-semiring affine recurrence, parallelizable via an associative scan with operator

$$(A_1, b_1) \oplus (A_2, b_2) = (A_2 \otimes A_1, \operatorname{logsumexp}(A_2 b_1, b_2))$$

where $\otimes$ denotes log-semiring matrix multiplication. This reduces the insertion phase from $O(L)$ sequential depth to $O(\log L)$.

Supported model types The beam search interface is generic over the pair HMM used on each branch. For the present (TKF) paper we specialize to:

1. **TKF91** — order-0, 5-state pair HMM.
2. **TKF92** — order-0, 5-state pair HMM with fragment self-loops (the standard model of Section A.2.2).

The same beam-search framework extends without algorithmic change to hierarchical TKF extensions in which each branch carries a larger latent-state pair HMM (38); we do not pursue that direction here.

B.5.3 Variational Ancestral Reconstruction (VarAnc)

Given an MSA with known gap structure—i.e., for each internal node v and MSA column c , we know whether v is present (ungapped) or absent (gapped)—but unknown ancestral residue identities, we optimize a variational lower bound (ELBO) on the marginal likelihood. Our approach is a “product of trees” structured variational approximation (34).

Model structure The joint distribution over observed leaf sequences $\mathbf{y}$ and ancestral sequences $\mathbf{a}$ factorises over tree edges. With the alignment structure fixed, the log-likelihood decomposes into contributions from consecutive alignment-column pairs along each edge $e = (u, v)$:

$$\log P(\mathbf{y}, \mathbf{a}) = \sum_{e \in \text{edges}} \sum_{k=1}^{K_e} \log w_e(\text{type}_{k-1}, \text{type}_k, \text{ctx}_{k-1}, \text{ctx}_k) \quad (\text{B.86})$$

where w_e is the TKF92 pair-HMM transition weight on edge e (depending on the previous and current column types and context characters, plus the branch length). The first and last columns contribute start (S) and end (E) transition factors.

The key computational challenge is that even the order-0 TKF92 transitions couple adjacent alignment columns through the fragment-extension self-loops: the pair-HMM weight at column c depends on the ancestral and descendant characters at both column c and its predecessor. This makes the full graphical model a tree (phylogeny) $\times$ chain (sequence) grid with undirected cycles, and exact inference is intractable.

Product-of-trees variational distribution Following (34), we factorize the variational posterior over MSA columns while retaining the full tree joint at each column:

$$q(\mathbf{a}) = \prod_{c=1}^L q_c(\mathbf{h}_c) \quad (\text{B.87})$$

where $\mathbf{h}_c = \{a_{v,c} : v \in \text{ungapped}(c)\}$ denotes the set of all ancestral characters at MSA column c , and each q_c is a tree-structured distribution over the characters at the internal nodes that are ungapped at column c . Leaf characters are observed (clamped).

This approximation decouples columns in q but preserves the full parent-child correlation structure within each column. In particular, the pairwise joint marginal $q_c^{(u,v)}(a_u, a_v)$ on each tree edge is available exactly from the Felsenstein peeling/unpeeling pass used to represent q_c (Section B.5.3).

Free energy and ELBO The evidence lower bound is

$$\mathcal{L}(q) = \mathbb{E}_q[\log P(\mathbf{y}, \mathbf{a})] + H(q) \geq \log P(\mathbf{y}) \quad (\text{B.88})$$

where $H(q) = -\sum_c \sum_{\mathbf{h}_c} q_c(\mathbf{h}_c) \log q_c(\mathbf{h}_c)$ is the entropy of the product-of-trees distribution. Since each q_c is tree-structured, its entropy decomposes as

$$H(q_c) = \sum_{v \in \text{ungapped}(c)} H(q_c^{(v)}) - \sum_{e=(u,v)} \text{MI}_c(u, v) \quad (\text{B.89})$$

where $H(q_c^{(v)})$ is the marginal entropy at node v and $\text{MI}_c(u, v)$ is the mutual information between parent u and child v at column c , both computable from the Felsenstein marginals.

Potentials from neighboring columns The inter-column coupling enters through the TKF92 pair-HMM transitions. For each edge $e = (u, v)$ and MSA column c where e has an event of type τ_c , let c^- denote the predecessor column (the previous column where e had an event) and c^+ the successor column. The *potential* at column c for edge e receives two contributions:

As-child term (from c^-). The transition from column c^- to c on edge e depends on the characters at both columns. Using the pairwise marginal from q_{c^-} :

$$\log \phi_e^{\text{child}}(a_{u,c}, a_{v,c}) = \sum_{a', b'} q_{c^-}^{(u,v)}(a', b') \log w_e(\tau_{c^-}, \tau_c, a', b', a_{u,c}, a_{v,c}) \quad (\text{B.90})$$

where the sum over (a', b') uses the *joint* pairwise marginal $q_{c^-}^{(u,v)}(a', b')$ —not the product of independent marginals. This is the key advantage over mean-field: the within-tree parent-child correlation at the predecessor column is preserved exactly.

As-parent term (from c^+). Symmetrically, column c acts as the predecessor for column c^+ :

$$\log \phi_e^{\text{parent}}(a_{u,c}, a_{v,c}) = \sum_{a'', b''} q_{c^+}^{(u,v)}(a'', b'') \log w_e(\tau_c, \tau_{c^+}, a_{u,c}, a_{v,c}, a'', b'') \quad (\text{B.91})$$

For insert transitions (only the descendant is present at c), the potential reduces to a per-node function $\psi_v(a_{v,c})$. For delete transitions (only the ancestor is present), it becomes $\psi_u(a_{u,c})$. For match transitions, it contributes a per-edge potential $\phi_e(a_{u,c}, a_{v,c})$. The start and end transitions contribute analogous per-node or per-edge terms.

Felsenstein coordinate ascent Each coordinate ascent step updates q_c for a single MSA column c , holding all other columns fixed. We accumulate, for each edge e and node v in the tree at column c :

- Per-edge log-potentials $\log \phi_e(a_u, a_v) = \log \phi_e^{\text{child}} + \log \phi_e^{\text{parent}}$ (for match transitions where both endpoints are present).
- Per-node log-potentials $\log \psi_v(a_v)$ (from insert/delete transitions on incident edges, plus the root prior $\log \pi(a)$ at the root node).

The optimal q_c , given the potentials, is the Gibbs distribution on the tree at column c :

$$q_c(\mathbf{h}_c) \propto \prod_v \psi_v(a_v) \prod_{e=(u,v)} \phi_e(a_u, a_v) \prod_{\ell \in \text{leaves}} \delta(a_\ell = y_\ell) \quad (\text{B.92})$$

Since this is a tree-structured MRF, the normalising constant and all node and edge marginals can be computed *exactly* by Felsenstein peeling (postorder) and unpeeling (preorder) in $O(|E| \cdot |\mathcal{A}|^2)$ time.

Peeling (postorder). For each node v in postorder, compute the conditional likelihood:

$$\text{CL}_v(a) = \psi_v(a) \prod_{\text{children } c} \left[\sum_{a_c} \phi_{(v,c)}(a, a_c) \text{CL}_c(a_c) \right] \quad (\text{B.93})$$

with $\text{CL}_\ell(a) = \delta(a = y_\ell)$ for observed leaves. The log-partition function is $\log Z_c = \log \sum_a \pi(a) \text{CL}_{\text{root}}(a)$.

Unpeeling (preorder). Propagate top-down to obtain the posterior marginal at each node:

$$q_c^{(v)}(a) \propto \text{CL}_v(a) \cdot \text{msg}_{\text{parent} \rightarrow v}(a) \quad (\text{B.94})$$

and the pairwise marginal on each edge:

$$q_c^{(u,v)}(a_u, a_v) \propto \text{msg}_{\text{above } u}(a_u) \cdot \phi_{(u,v)}(a_u, a_v) \cdot \text{CL}_v(a_v) \quad (\text{B.95})$$

where the “message from above u ” combines the top-down message to u with u ’s conditional likelihood excluding child v .

Sweep. One iteration sweeps through all MSA columns $c = 1, \dots, L$: for each column, recompute the potentials from the current neighbor marginals, run peeling/unpeeling, and store the updated node and edge marginals.

Properties The product-of-trees approximation enjoys the same monotonic convergence guarantee as mean-field coordinate ascent (each column update minimises the free energy in its coordinate), with the additional guarantee that the ELBO is at least as tight as the fully-factored mean-field bound. This follows because the product-of-trees family *contains* the mean-field family as a special case (where each q_c is itself fully factored).

Computational cost. The dominant cost per sweep is the potential computation: $O(|E| \cdot L \cdot |\mathcal{A}|^4)$ for Match→Match transitions (contracting the $(|\mathcal{A}|, |\mathcal{A}|)$ pairwise marginal against the $(|\mathcal{A}|, |\mathcal{A}|, |\mathcal{A}|, |\mathcal{A}|)$ pair-HMM tensor). The Felsenstein passes add $O(|E| \cdot L \cdot |\mathcal{A}|^2)$, which is subdominant.

B.5.4 Stochastic Variational VarAnc (svi-VarAnc)

The VarAnc algorithm of Section B.5.3 computes a variational posterior over ancestral residues for a *fixed* TKF92 model θ on a *single* fixed-MSA family. We now extend it to a training algorithm that updates θ on a corpus of tree-structured MSAs. The construction parallels SVI-BW (Section B.1.13) but uses an MSA-conditioned VarAnc E-step in place of the per-pair Pair-HMM Forward–Backward.

Outer EM. The outer iteration is a stochastic-EM loop over MSAs. At each iteration:

1. Sample a minibatch of MSAs (with their phylogenetic trees, obtained e.g. by FastTree).
2. For each MSA, run a few sweeps of the VarAnc coordinate-ascent algorithm of Section B.5.3 to refine q . This is the *family-level E-step*.
3. Read out, from the converged q and the pairwise edge marginals $q_c^{(u,v)}$, the expected sufficient statistics $(B, D, L, M, S, T, F, E, W, U, V)$ for the TKF92 model:
 - Indel/fragment statistics (B, D, L, M, S, T, F, E) are read column-by-column from the per-edge expected event types (match/insert/delete) implied by the gap structure, weighted by the pairwise edge marginals. These are the same TKF92 sufficient statistics defined in Section A.2.3.
 - Substitution statistics (W, U, V) are accumulated by contracting the pairwise edge marginal $q_c^{(u,v)}(a_u, a_v)$ against the bridge expectations $\mathcal{C}_i^W(a_u, a_v, t_e)$ and $\mathcal{C}_{ij}^U(a_u, a_v, t_e)$ at each match column.
4. Apply the SVI exponential moving average update (equation (B.52)) to the running sufficient statistics, mixing the minibatch contribution with the prior pseudocounts and the previous EMA estimate.
5. Apply the closed-form M-steps of Section A.2.3 (quadratic for λ, μ , ratio for r , GTR coordinate ascent for Q, π).

Comparison to SVI-BW. The two algorithms differ in their E-step and converge to different fixed points:

- SVI-BW (Section B.1.13) treats each pair as conditionally independent given the population indel/substitution parameters, and sums over alignments with a 2D Pair-HMM Forward–Backward. It uses no tree information.
- svi-VarAnc takes a tree-structured MSA as input (alignment fixed, gap structure fixed, ancestral residues marginalised) and shares information across all branches in the tree at each MSA column.

For pairwise data (a tree of two leaves), the two algorithms coincide. For deep trees with many siblings, svi-VarAnc tightens the indel-rate posterior considerably by coupling sibling pairs through the tree. ELBO monitoring, mini-batch construction, and warm-start initialisation from cherry-trained TKF92 parameters follow the SVI-BW recipe of Section B.1.13 verbatim, replacing the Pair-HMM Forward–Backward E-step with the family-level VarAnc sweep of Section B.5.3.

B.6 MixFrag Training from Alignment-Summary Counts

The MixFrag Baum-Welch algorithm of Section A.3.2 marginalises the latent fragtype inside a per-pair Forward-Backward, so its per-fragtype counts F_f, E_f are posterior expectations that must be recomputed for every cherry on every iteration. It would seem that the cherry-count summarisation used for TKF92 (Section B.4), which precomputes a fixed tensor of alignment statistics and then fits it by autograd independently of corpus size, cannot apply, precisely because the fragtypes are latent. In fact it can: the fragtype-augmented MixFrag transfer matrices have a rank-one off-diagonal structure that lets each pairwise alignment be summarised, per discretised time, by a small set of count tensors on which exact EM runs in time independent of the number of cherries.

B.6.1 The fragtype-marginalised transfer decomposition

Fix a pairwise cherry alignment at (discretised) time T , written as a column sequence $c_1 \dots c_L$ with each $c_t \in \{\mathbf{M}, \mathbf{I}, \mathbf{D}\}$. Under MixFrag (Section A.3.1) the indel likelihood of this column sequence is a sum over the latent fragtype $f_t \in \{1, \dots, \mathcal{F}\}$ carried by each column,

$$P_{\text{indel}}(c_{1:L}) = \sum_{f_1 \dots f_L} \tau_{\mathbf{S}, c_1 f_1}^{\mathcal{F}} \left[\prod_{t=1}^{L-1} \tau_{c_t f_t, c_{t+1} f_{t+1}}^{\mathcal{F}} \right] \tau_{c_L f_L, \mathbf{E}}^{\mathcal{F}}.$$

Collecting the fragtype index into $\mathcal{F}$ -vectors and writing $\rho_f = 1 - r_f$, the column-to-column transfers are the $\mathcal{F} \times \mathcal{F}$ matrices

$$(\mathbf{A}_a)_{fg} = r_f \delta_{fg} + \tau_{aa} \rho_f w_g \quad (\text{within a run, } a \rightarrow a), \quad (\mathbf{B}_{ab})_{fg} = \tau_{ab} \rho_f w_g \quad (a \neq b),$$

with start row $(\tau_{\mathbf{S}a} w_g)_g$ and end column $(\rho_f \tau_{a\mathbf{E}})_f$, where $\tau_{ab} \equiv \tau_{ab}(\lambda, \mu, T)$ is the TKF91 Pair HMM matrix of Section A.1.6.

The off-diagonal transfer is *rank one*: $\mathbf{B}_{ab} = \tau_{ab} \rho_{\bullet} w_{\bullet}^{\top}$, the outer product of the column vector $\rho_{\bullet} = (\rho_f)_f$ and the row vector $w_{\bullet} = (w_f)_f$ – the fragtype is redrawn afresh from w at every new link. A rank-one factor cuts the transfer-matrix product, so at every change of column type the chain separates and the likelihood factorises over the alignment’s maximal single-type runs. Writing the runs as $a_1^{k_1} a_2^{k_2} \dots a_m^{k_m}$ (with $a_i \neq a_{i+1}$ and lengths $k_i \geq 1$),

$$P_{\text{indel}}(c_{1:L}) = \tau_{\mathbf{S}a_1} \left[\prod_{i=1}^m \phi_{a_i}(k_i) \right] \left[\prod_{i=1}^{m-1} \tau_{a_i a_{i+1}} \right] \tau_{a_m \mathbf{E}}, \quad \phi_a(k) = w_{\bullet}^{\top} \mathbf{A}_a^{k-1} \rho_{\bullet}, \quad (\text{B.96})$$

where the *run factor* $\phi_a(k)$ depends only on the run type a and length k .

B.6.2 Match-run and gap factors

Splitting $A_a = \text{diag}(r_\bullet) + \tau_{aa} \rho_\bullet w_\bullet^\top$ into extension ($\text{diag} r$) and new-link ($\tau_{aa} \rho_\bullet w_\bullet^\top$) steps shows that a run of length k is a concatenation of $p \geq 1$ same-type fragments of lengths $\ell_1 + \dots + \ell_p = k$, each an independent geometric draw, separated by new links:

$$\phi_a(k) = \sum_{p \geq 1} \tau_{aa}^{p-1} \sum_{\ell_1 + \dots + \ell_p = k} \prod_{u=1}^p m(\ell_u), \quad m(\ell) = \sum_f w_f r_f^{\ell-1} \rho_f, \quad (\text{B.97})$$

with $m(\ell)$ the MixFrag fragment-length marginal (a mixture of geometrics). In generating-function form, with $\Phi_a(z) = \sum_{k \geq 1} \phi_a(k) z^k$ and $\mathcal{M}(z) = \sum_{\ell \geq 1} m(\ell) z^\ell = \sum_f w_f \rho_f z / (1 - r_f z)$,

$$\Phi_a(z) = \frac{\mathcal{M}(z)}{1 - \tau_{aa} \mathcal{M}(z)}, \quad (\text{B.98})$$

so each $\phi_a(k)$ is a rational-generating-function coefficient, computable by an $O(k)$ recursion. A *match-run* of length k contributes exactly $\phi_M(k)$.

A *gap* – a maximal block of I and D columns – factorises by the same rank-one cuts into its alternating I- and D-runs. A pairwise alignment does not identify the relative order of insertions and deletions inside a gap; summing over those orderings (as for the TKF92 gap-block probabilities of Section A.2.2) yields a factor that depends only on the gap’s deletion/insertion totals (i, j) and its exit state. Tracking deletions by z_D and insertions by z_I , let P_I, P_D be the generating functions of gap prefixes (entered from a M state) whose last run is an I- resp. D-run; they solve the 2×2 linear system

$$P_I = (\tau_{MI} + P_D \tau_{DI}) \Phi_I(z_I), \quad P_D = (\tau_{MD} + P_I \tau_{ID}) \Phi_D(z_D), \quad (\text{B.99})$$

and the gap generating function with exit state $Y \in \{M, E\}$ is

$$\mathcal{G}_Y(z_D, z_I) = P_I \tau_{IY} + P_D \tau_{DY} = \sum_{i,j} G_Y(i, j) z_D^i z_I^j. \quad (\text{B.100})$$

The ordering-summed gap-block factor $G_Y(i, j)$ is recovered by the bivariate recursion obtained on clearing the denominators of (B.99). The entry state is fixed to M because the TKF91 rows S, M, I coincide (Section A.1.6): a gap entered from the start S has the same factor as one entered from a match.

B.6.3 Alignment-summary sufficient statistics

Combining (B.96) with the gap factorisation, every cherry alignment enters the indel likelihood, per time bin T , only through the following counts:

- $N_{\text{match}}(k)$ – the number of maximal M-runs of length k (factor $\phi_M(k)$);
- $N_{\text{gap}}^M(i, j)$, $(i, j) \neq (0, 0)$ – the number of gaps exiting into a match (factor $G_M(i, j)$); by the row coincidence above this also covers *leading* gaps (entered from S);
- $N_{\text{gap}}^E(i, j)$, $(i, j) \neq (0, 0)$ – the number of gaps exiting at the alignment end (factor $G_E(i, j)$); this covers a *trailing* gap and, for an alignment with no matches at all, the entire match-free $S \rightarrow (I/D)^+ \rightarrow E$ path;
- $N_{\text{start}}, N_{\text{end}}$ – indicators that the alignment begins, resp. ends, with a match (scalars $\tau_{SM} = \tau_{MM}$ and τ_{ME}), and $N_\emptyset$ – the number of empty cherries (no columns at all), each the bare $S \rightarrow E$ path of weight τ_{SE} (degenerate, and absent from real corpora);

- the substitution counts $N_{\text{sub}}(a, b)$ – aligned residue pairs at $\mathbf{M}$ columns – exactly as in Maraschino (Section B.4), and fragtype-independent because the substitution model is shared.

The case $(i, j) = (0, 0)$ is excluded from the gap counts because two matches with no intervening indel form one longer $\mathbf{M}$ -run, already counted by N_{match} . With $\theta = (\lambda, \mu, \{r_f\}, \{w_f\}, Q, \pi)$ the corpus log-likelihood is

$$\begin{aligned} \log P(\theta) = \sum_T \bigg[& \sum_k N_{\text{match}}(k; T) \log \phi_{\mathbf{M}}(k) + \sum_{i,j} N_{\text{gap}}^{\mathbf{M}}(i, j; T) \log G_{\mathbf{M}}(i, j) \\ & + \sum_{i,j} N_{\text{gap}}^{\mathbf{E}}(i, j; T) \log G_{\mathbf{E}}(i, j) + N_{\text{start}}(T) \log \tau_{\mathbf{MM}} + N_{\text{end}}(T) \log \tau_{\mathbf{ME}} \\ & + N_{\emptyset}(T) \log \tau_{\text{SE}} + \sum_{a,b} N_{\text{sub}}(a, b; T) \log (\pi_a \exp(RT)_{ab}) \bigg], \quad (\text{B.101}) \end{aligned}$$

each factor evaluated at θ and bin time T .

B.6.4 EM on the summary counts

Equation (B.101) is an explicit function of the count tensors, so training is exact EM on those tensors. In the E-step, the expected complete-data sufficient statistics contributed by a segment are read off the gradient of its log-factor by the score-function identity (A.11): the birth, death and time-integral statistics B, D, S , the link count L , and the per-fragtype fragment-extension and -end counts F_f, E_f (the latter also the expected number of type- f fragments) contributed by each $\phi_{\mathbf{M}}(k)$, $G_{\mathbf{M}}(i, j)$, $G_{\mathbf{E}}(i, j)$ and start/end scalar are the partial derivatives of its logarithm with respect to the relevant natural parameters; the trajectory count M and the observation time T (the bin time times the number of cherries in the bin) are tallied directly; and the substitution statistics W, U, V come from N_{sub} as in Maraschino. Weighting these per-segment expectations by the tensor counts and summing over bins gives the corpus-level expectations. The M-step is then the closed form of Section A.3.2: the w_f (categorical) and r_f (geometric) updates from E_f, F_f , together with the TKF91 (λ, μ) quadratic and the (Q, π) substitution update. After one preprocessing pass that builds the tensors, each EM iteration costs only $O(\sum_T (\mathcal{F} k_{\text{max}} + i_{\text{max}} j_{\text{max}}))$ – independent of the number of cherries – so an entire training corpus is summarised once and then fit in time set only by the tensor sizes.

B.7 Mixture-of-trees variational ancestral presence/absence

This appendix derives a variational lower bound on the indel log-likelihood of an MSA under the TKF92 conditional WFST $\mathbb{T}'(\lambda, \mu, T, r)$ of Section A.4 when applied independently on each branch of a phylogenetic tree, combined with a TKF stationary HMM at the root. The bound is parameterised by a tree-structured graphical model on per-column ancestral presence/absence indicators with three irreversible states $\{\text{NotYetInserted}, \text{Present}, \text{Deleted}\}$. It is a proper lower bound on the full TKF92 marginal log-likelihood $\log p(\text{MSA} \mid \mathbf{t}, \theta)$, with gap split into a standard q -dependent variational KL term and a q -independent “restriction” term arising because our latent space cannot represent ancestor residues that get inserted and then fully deleted on every leaf-bound lineage before reaching any leaf (Section B.7.2). For ancestral-presence inference at the observed columns the restriction term is irrelevant, since it cancels from the q -optimal posterior; for parameter learning it does not, and a tighter bound would require explicit modelling of those ghost-column histories. Special-case parameter settings recover an irreversible 3-state analogue of Felsenstein gap-CTMC reconstruction and Fitch parsimony.

B.7.1 Setting and approximation

Fix a rooted phylogenetic tree $\mathbf{t}$ with internal-node set $\mathcal{I}$, leaf set $\mathcal{L}$, and a branch length $d_e > 0$ on each edge e . We are given a multiple-sequence alignment with L columns whose rows are the leaf sequences of $\mathbf{t}$; let $X_n^v \in \{0, 1\}$ denote the observed presence indicator (1 if leaf $v \in \mathcal{L}$ has a residue in column n). The task is to infer per-internal-node presence indicators $\{X_n^v\}_{v \in \mathcal{I}, 1 \leq n \leq L}$.

We replace the full TKF92 generative model on the tree with the following *per-branch conditional approximation*: each branch $v \rightarrow w$ is taken to evolve its row according to the TKF92 conditional WFST $\mathbb{T}'$ acting independently on the whole parent sequence, with no shared latent fragment structure across branches. A residue inserted by the WFST on one branch may therefore be partially deleted, character by character, on a later branch (unlike in the strict TKF92 process, where fragments are indivisible). The WFST $\mathbb{T}'$ is itself a closed-form approximation to the finite-time transition probabilities of the General Geometric Indel (GGI) model — the maximum-entropy CTMC with given expected per-site indel rates and expected inserted-fragment length — and has been shown empirically to fit GGI Gillespie simulations (29) and pairwise-alignment data (38) well. Using it as the per-branch transition law of the tree-level model in this appendix is therefore a small additional approximation on top of the GGI approximation itself.

B.7.2 Restricted generative model

The model joint over leaf and internal-node presence patterns at the L MSA columns is

$$p(\text{MSA}, \{X^v\}_{v \in \mathcal{I}} \mid \mathbf{t}, \theta) = p_{\text{singlet}}(X^{\text{root}} \mid \theta) \prod_{(v \rightarrow w) \in \mathbf{t}} P(X^w \mid X^v, d_{vw}, \theta), \quad (\text{B.102})$$

where p_{singlet} is the TKF stationary distribution (Section B.7.3, below) and each $P(X^w \mid X^v, t)$ is the per-branch presence-pattern conditional defined by the WFST path log-likelihood (Section B.7.4). Marginalising (B.102) over internal patterns supported on the L observed MSA columns gives a quantity we denote $\tilde{p}(\text{MSA} \mid \mathbf{t}, \theta)$. The full TKF92 marginal $p(\text{MSA} \mid \mathbf{t}, \theta)$ also sums over evolutionary histories with ancestral residues that get inserted and then fully deleted on every leaf-bound lineage before reaching any leaf — *ghost columns* that leave no MSA trace and so are not representable as values of X_n^v at observed columns $1, \dots, L$. Since each ghost-column history adds a non-negative probability to p that is absent from $\tilde{p}$,

$$\tilde{p}(\text{MSA} \mid \mathbf{t}, \theta) \leq p(\text{MSA} \mid \mathbf{t}, \theta). \quad (\text{B.103})$$

The gap $\log p - \log \tilde{p} \geq 0$ depends on the model parameters and the tree but not on the variational distribution q introduced below. For the ancestral-presence reconstruction task — fix θ , optimise q to recover internal-node marginals at the observed columns — this gap is a q -independent constant and the restricted-model posterior at observed columns coincides with the full-model posterior conditioned on the same support, so all inferences below proceed unchanged.

B.7.3 Singlet HMM at the root

The TKF92 stationary HMM (Section A.2.2) emits a sequence of length L_{root} with

$$p_{\text{singlet}}(L_{\text{root}}) = \begin{cases} 1 - \kappa & L_{\text{root}} = 0, \\ \kappa p^{L_{\text{root}}-1} (1 - r)(1 - \kappa) & L_{\text{root}} \geq 1, \end{cases} \quad (\text{B.104})$$

where $\kappa = \lambda/\mu$ and $p = r + (1 - r)\kappa$. Identifying L_{root} with the number of MSA columns at which the root is in state xP ($L_{\text{root}} = \sum_n \delta(X_n^{\text{root}} = 1)$) gives the root prior in the joint (B.102). The

two-cases form arises because the TKF92 singlet HMM has distinct “from $\mathbf{S}$ ” (probability κ to emit, $1 - \kappa$ to end) and “from emitting state” (probability p to extend, $(1 - r)(1 - \kappa)$ to end) transition contexts.

For TKF91 ($r = 0$): $p = \kappa$ and the two cases collapse to the single form $p_{\text{singlet}}(L_{\text{root}}) = (1 - \kappa)\kappa^{L_{\text{root}}}$.

Note on a tempting shortcut. One might hope to avoid the singlet by interpreting the root as the output of a pseudo-branch with $T \rightarrow \infty$ and an all-absent pseudo-parent. For TKF91 this works — the $t \rightarrow \infty$ limit of $\mathbb{T}$ with an empty input does recover the singlet sequence-length distribution. For TKF92 it does *not*: the WFST’s $(1 - r)$ factors are tied to the fragment-extension structure of the Pair HMM and the limiting output mass for length $L \geq 1$ is $\kappa p^{L-1}(1 - \kappa)$ rather than the correct $\kappa p^{L-1}(1 - r)(1 - \kappa)$ — i.e. the WFST does not normalise on an empty input under TKF92. We therefore include the singlet term explicitly.

B.7.4 Per-branch path log-likelihood

Consider a single branch $v \rightarrow w$ of length d . Reading the parent and child rows of the MSA column by column gives, for each $n \in \{1, \dots, L\}$, an ordered pair $(X_n^v, X_n^w) \in \{0, 1\}^2$. Map each pair to a WFST column-state $S_n \in \{\mathbf{M}, \mathbf{I}, \mathbf{D}, \mathbf{lg}\}$ via

$$S_n = \begin{cases} \mathbf{M} & \text{if } (X_n^v, X_n^w) = (1, 1) \\ \mathbf{I} & \text{if } (X_n^v, X_n^w) = (0, 1) \\ \mathbf{D} & \text{if } (X_n^v, X_n^w) = (1, 0) \\ \mathbf{lg} & \text{if } (X_n^v, X_n^w) = (0, 0), \end{cases} \quad (\text{B.105})$$

where the symbol $\mathbf{lg}$ (“Ignore”) marks columns absent in both parent and child and so contributing no transition to the WFST path. Add boundary sentinels $S_0 = \mathbf{S}$ and $S_{L+1} = \mathbf{E}$. Stripping out columns with $S_n = \mathbf{lg}$ collapses the sequence $(S_0, S_1, \dots, S_L, S_{L+1})$ to a unique path through the WFST whose nontrivial transitions are between consecutive non-Ignore columns. The branch log-likelihood is the sum of the corresponding entries of $\log \mathbb{T}'(\lambda, \mu, d, r)$ from equation (A.29). Equivalently, summing over each column’s incoming transition,

$$\log P(X^w \mid X^v, d, \theta) = \sum_{N=1}^{L+1} \delta(S_N \neq \mathbf{lg}) \sum_{M=0}^{N-1} \delta(S_M \neq \mathbf{lg}) \log \mathbb{T}'_{S_M S_N} \prod_{K=M+1}^{N-1} \delta(S_K = \mathbf{lg}), \quad (\text{B.106})$$

where $\theta = (\lambda, \mu, r)$, $\delta(\cdot)$ is a Kronecker indicator, and the empty product (when $M + 1 > N - 1$) equals 1. For each non-Ignore column N the inner sum picks out the unique $M < N$ that is also non-Ignore with all columns strictly between M and N in state $\mathbf{lg}$; that pair contributes $\log \mathbb{T}'_{S_M S_N}$ and all other M contribute zero.

B.7.5 Tree-structured variational family

We approximate the joint posterior $P(\{X_n^v\}_{v \in \mathcal{I}} \mid \{X_n^v\}_{v \in \mathcal{L}})$ by a product over MSA columns

$$q(\{X_n^v\}_{v \in \mathcal{I}, 1 \leq n \leq L}) = \prod_{n=1}^L q_n(\{X_n^v\}_{v \in \mathcal{I}}), \quad (\text{B.107})$$

where each per-column distribution q_n is a directed graphical model on the tree with node values in

$$\mathcal{Z} = \{\text{NOTYETINSERTED}, \text{PRESENT}, \text{DELETED}\} \equiv \{xN, xP, xD\}. \quad (\text{B.108})$$

The variable at the root is drawn from a free distribution $q_n^{\text{root}} \in \Delta^{\mathcal{Z}}$; along each edge $v \rightarrow w$ the variable $Z_n^w \in \mathcal{Z}$ is drawn conditionally on Z_n^v from a free distribution $q_n^{v \rightarrow w}(\cdot | \cdot)$. The 9-entry matrix $q_n^{v \rightarrow w}(z' | z)$ has 4 entries pinned to zero to enforce irreversibility:

$$q_n^{v \rightarrow w}(xD | xN) = q_n^{v \rightarrow w}(xN | xP) = q_n^{v \rightarrow w}(xN | xD) = q_n^{v \rightarrow w}(xP | xD) = 0. \quad (\text{B.109})$$

The two unconstrained rows (xN and xP parents) each have two surviving outcomes, contributing one free parameter per row; the xD row is deterministic ($q_n^{v \rightarrow w}(xD | xD) = 1$). This leaves two free parameters per (edge, column). Leaf-node states Z_n^v for $v \in \mathcal{L}$ are clamped: if $X_n^v = 1$, then $Z_n^v = xP$; if $X_n^v = 0$, then $Z_n^v \in \{xN, xD\}$ with the variational mass split between the two by q_n .

The presence indicator at internal node v is recovered as

$$X_n^v = \delta(Z_n^v = xP), \quad (\text{B.110})$$

so that any realisation drawn from q_n has the property that the nodes with $X_n^v = 1$ form a connected subtree of $\mathbf{t}$ (or the empty set, which by hypothesis cannot arise once the leaf clamps are respected, since every column of the supplied MSA contains at least one Present leaf).

Why this support. Under the strict TKF92 process, each MSA column corresponds to a single insertion event in the history of the tree; the residue descends, and the set of Present nodes is necessarily a connected subtree containing the insertion point. A factorised mean-field q with independent Bernoulli marginals at each (v, n) would assign positive mass to disconnected Present-sets, which (i) are unreachable under the model, and (ii) would force the substitution likelihood to factor as a product of independent Felsenstein recursions on the two pieces — corresponding to two separate insertions co-located in one MSA column, a different generative process. The 3-state irreversible q rules out both pathologies by construction.

B.7.6 Expected log-likelihood under q

For a parent-child pair (v, w) on the tree, define the per-column state probability under q_n :

$$P_q(S_n = \mathbf{M}) = q_n(Z_n^v = xP, Z_n^w = xP), \quad (\text{B.111})$$

$$P_q(S_n = \mathbf{D}) = q_n(Z_n^v = xP, Z_n^w = xD), \quad (\text{B.112})$$

$$P_q(S_n = \mathbf{I}) = q_n(Z_n^v = xN, Z_n^w = xP), \quad (\text{B.113})$$

$$P_q(S_n = \mathbf{lg}) = q_n(Z_n^v = xN, Z_n^w = xN) + q_n(Z_n^v = xD, Z_n^w = xD). \quad (\text{B.114})$$

The four other (z, z') joint marginals are zero by equation (B.109). Each pairwise marginal $q_n(Z_n^v, Z_n^w)$ is computed by belief propagation (Section B.7.9).

The product factorisation (B.107) makes the per-branch expected log-likelihood factor across columns:

$$\mathbb{E}_q[\log P(X^w | X^v, d, \theta)] = \sum_{s, s'} \log \mathbb{T}_{ss'}' W_{ss'}^{(v \rightarrow w)}, \quad (\text{B.115})$$

where s ranges over $\{\mathbf{S}, \mathbf{M}, \mathbf{I}, \mathbf{D}, \mathbf{E}\} \setminus \{\mathbf{E}\}$ and s' ranges over $\{\mathbf{S}, \mathbf{M}, \mathbf{I}, \mathbf{D}, \mathbf{E}\} \setminus \{\mathbf{S}\}$ (so $\mathbf{S}$ enters only as a source and $\mathbf{E}$ only as a destination, both via the deterministic boundary sentinels), and

$$W_{ss'}^{(v \rightarrow w)} = \sum_{N=1}^{L+1} \sum_{M=0}^{N-1} P_q(S_M = s) P_q(S_N = s') \prod_{K=M+1}^{N-1} P_q(S_K = \mathbf{lg}) \quad (\text{B.116})$$

is the expected number of times the path uses transition $s \rightarrow s'$ on this branch. Boundary terms have $P_q(S_0 = \mathbf{S}) = 1$ and $P_q(S_{L+1} = \mathbf{E}) = 1$ deterministically.

Root-prior contribution. Under the per-column factorisation (B.107), the expected singlet log-likelihood at the root (equation (B.104)) reduces to

$$\mathbb{E}_q[\log p_{\text{singlet}}(X^{\text{root}})] = \log p \cdot \sum_{n=1}^L q_n(X_n^{\text{root}} = 1) + (1 - P_q(L_{\text{root}}=0)) c_1 + P_q(L_{\text{root}}=0) \log(1 - \kappa), \quad (\text{B.117})$$

with $c_1 = \log[\kappa(1-r)(1-\kappa)/p]$ and $P_q(L_{\text{root}}=0) = \prod_{n=1}^L q_n(X_n^{\text{root}} = 0)$ the variational mass on an empty root sequence. For TKF91 ($r = 0$) the second and third terms collapse to a single q -independent constant $\log(1 - \kappa)$ and (B.117) reduces to the textbook form $\log(1 - \kappa) + \log \kappa \cdot \sum_n q_n(X_n^{\text{root}} = 1)$. For TKF92 the $L_{\text{root}}=0$ special case must be retained; note that $P_q(L_{\text{root}}=0)$ is small but generically nonzero, since some columns may have positive variational mass on root being xN (the column was inserted on a strict subtree below the root).

B.7.7 ELBO

For any product-of-trees q that gives joint marginals $q_n(Z_n^v, Z_n^w)$ at every (edge, column),

$$\log p(\text{MSA} \mid \mathbf{t}, \theta) \geq \log \tilde{p}(\text{MSA} \mid \mathbf{t}, \theta) \geq \mathcal{L}[q] = \mathbb{E}_q[\log p_{\text{singlet}}(X^{\text{root}})] + \sum_{(v \rightarrow w) \in \mathbf{t}} \mathbb{E}_q[\log P(X^w \mid X^v, d_{vw}, \theta)] + \dots \quad (\text{B.118})$$

The first two terms are given by (B.117) and (B.115) respectively; the third is the entropy of the leaf-conditioned variational distribution. The leftmost inequality is (B.103); the right inequality is the standard Jensen bound on the restricted joint (B.102).

Entropy decomposition. The variational q is parameterised through a directed graphical model on the full tree (root + internal + leaf nodes) with edge conditionals $q_n^{v \rightarrow w}(\cdot \mid \cdot)$. Writing q_{joint} for this joint and conditioning on the observed leaf states gives $q(\{X^v\}_{v \in \mathcal{I}} \mid \text{MSA}) = q_{\text{joint}}(\text{internals}, \text{MSA}) / Z_q$, where $Z_q = \sum_{\text{internals}} q_{\text{joint}}$ is the per-column belief-propagation partition function summed over columns. The leaf-conditioned entropy then decomposes as

$$H[q(\text{internals} \mid \text{MSA})] = -\mathbb{E}_q[\log q_{\text{joint}}] + \log Z_q, \quad (\text{B.119})$$

where the first term factors over the directed graphical model:

$$-\mathbb{E}_q[\log q_{\text{joint}}] = -\sum_{n=1}^L \sum_{\mathbf{z} \in \mathcal{Z}} q_n^{\text{root}}(\mathbf{z}) \log q_n^{\text{root}}(\mathbf{z}) - \sum_{(v \rightarrow w) \in \mathbf{t}} \sum_{n=1}^L \sum_{\mathbf{z}, \mathbf{z}' \in \mathcal{Z}} q_n(Z_n^v = \mathbf{z}, Z_n^w = \mathbf{z}') \log q_n^{v \rightarrow w}(\mathbf{z}' \mid \mathbf{z}). \quad (\text{B.120})$$

The first term is the per-column root entropy. Each per-edge term is a *cross-entropy* between the BP-derived joint marginal $q_n(Z_n^v, Z_n^w)$ and the prior edge conditional $q_n^{v \rightarrow w}$; it is *not* the parent-marginal-weighted entropy $\mathbb{E}_{q_n(Z_n^v)}[H(q_n^{v \rightarrow w}(\cdot \mid Z_n^v))]$, which only equals (B.120) when the leaves impose no evidence on the edge — generically a different quantity once the BP-conditioning shifts pair marginals away from the prior factorisation. The $\log Z_q$ term in (B.119) comes out of the up-pass partition function and ensures the entropy is on the conditioned distribution rather than the joint.

Bound interpretation. Equation (D.140) is a proper lower bound on the full TKF92 marginal log-likelihood $\log p(\text{MSA})$, with the gap split into two non-negative pieces:

$$\log p(\text{MSA}) - \mathcal{L}[q] = \underbrace{(\log p(\text{MSA}) - \log \tilde{p}(\text{MSA}))}_{\text{restriction gap}} + \underbrace{\text{KL}[q(\text{internals} \mid \text{MSA}) \parallel \tilde{p}(\text{internals} \mid \text{MSA})]}_{\text{variational gap}}. \quad (\text{B.121})$$

The variational gap is the standard Jensen slack on the restricted joint (B.102) and depends on q ; the restriction gap is the contribution of ghost-column histories that our latent space cannot represent (Section B.7.2) and depends on the model parameters θ and the tree but not on q . For ancestral-presence reconstruction (fix θ , optimise q) the restriction gap is a constant additive offset, so maximising (D.140) is equivalent to minimising the KL divergence from q to the true model posterior on the support of observed-column internal-node configurations. For parameter learning over θ this equivalence breaks: the restriction gap shifts with θ , so the bound is not tight up to a parameter-independent constant. A tighter bound would require augmenting q to model ghost columns explicitly — for instance via per-(branch, gap-region) latent counts of inserted-then-fully-deleted residues — at the cost of additional machinery beyond the per-MSA-column factorisation used here. Equivalently and intuitively, one recovers the full marginal in the limit of inserting infinitely many additional “empty” columns into the alignment between (and around) every observed column, with all leaves clamped to absent at those columns and the variational q free over their internal states; the present formulation simply truncates this padding to zero.

B.7.8 Stable computation: cumulant trick

The expected transition count $W_{ss'}^{(v \rightarrow w)}$ in (B.116) appears at first sight to require $O(L^2)$ work per (branch, state-pair). A standard prefix-sum reduces this to $O(L)$.

For brevity drop the branch superscript and write $P_n^{\text{lg}} = P_q(S_n = \text{lg})$, $P_n^s = P_q(S_n = s)$ for $s \in \{\text{S}, \text{M}, \text{D}, \text{I}, \text{E}\}$ (with $P_0^{\text{S}} = 1$, $P_{L+1}^{\text{E}} = 1$, all other boundary values zero). Define the running log-Ignore mass

$$C_N = \sum_{k=1}^N \log P_k^{\text{lg}}, \quad C_0 = 0. \quad (\text{B.122})$$

Then $\prod_{K=M+1}^{N-1} P_K^{\text{lg}} = \exp(C_{N-1} - C_M)$, and (B.116) factorises as

$$W_{ss'} = \sum_{N=1}^{L+1} P_N^{s'} e^{C_{N-1}} \cdot \underbrace{\sum_{M=0}^{N-1} P_M^s e^{-C_M}}_{\text{prefix sum in } M}. \quad (\text{B.123})$$

Both inner factors are non-negative; the difference $C_{N-1} - C_M$ is non-positive (it is a sum of log-probabilities), so the geometric factor lies in $(0, 1]$ and (B.123) is numerically benign in float64 for any practical L . For very long alignments where individual exponents could underflow, one ϵ -floors P_n^{lg} before computing C_n ; the floor is harmless because the product weights paths that traverse many Ignore columns, which contribute negligibly to the expectation.

The total work for the per-branch expected log-likelihood is $O(L \cdot |S|^2)$ with $|S| \leq 5$ states, yielding $O(BL)$ overall for B branches. This is fully vectorisable across columns and branches.

Why the inner sum is in probability space. Equation (B.106) for a fixed deterministic realisation of the indicators is purely a sum of $\log \mathbb{T}^v$ entries: no probabilities, no products, no exponentials.

The probability-space arithmetic in (B.123) arises only when we marginalise over the variational distribution: each summand is a probability mass (product of independent column factors) times a real-valued $\log \mathbb{T}'$ entry, and the sum collapses these weighted log-transitions into the expected transition count.

B.7.9 Belief propagation for pairwise marginals

The pairwise marginals $q_n(Z_n^v, Z_n^w)$ feeding equations (B.111) are computed by the standard Felsenstein up-down algorithm on $\mathbf{t}$. For each column n :

1. *Up pass*: for every node v and every state $z \in \mathcal{Z}$, accumulate $\beta_v^n(z) = \prod_{w:\text{child of } v} \sum_{z'} q_n^{v \rightarrow w}(z' | z) \beta_w^n(z')$, with leaf base case $\beta_v^n(z) = \delta(\text{leaf clamp at } v, n \text{ is consistent with } z)$.
2. *Down pass*: for every internal v and state z , accumulate the root-conditioned posterior $\alpha_v^n(z)$ in standard fashion.
3. *Marginals*: for every edge $v \rightarrow w$ and every state pair, $q_n(Z_n^v = z, Z_n^w = z') = \alpha_v^n(z) q_n^{v \rightarrow w}(z' | z) \beta_w^n(z') / \mathcal{Z}_n$, where $\mathcal{Z}_n$ is the per-column partition function from the up pass at the root.

The up and down passes are $O(|\mathcal{I}| \cdot |\mathcal{Z}|^2)$ per column and trivially vectorise across columns (same tree topology, different leaf clamps and different per-column edge conditionals). For typical protein alignments ($L \approx 360$ amino acids on average in UniProtKB/Swiss-Prot (61), with the long tail extending to a few thousand) this is unproblematic; for DNA MSAs (gene-scale regions in the few-kbp range (68), longer for syntenic blocks) the per-column independence keeps cost linear in L .

B.7.10 Special cases and scalability

Irreversible Felsenstein-3. Tie the per-(edge, column) variational conditionals $q_n^{v \rightarrow w}(\cdot | \cdot)$ across columns to a single per-edge transition matrix $\bar{q}^{v \rightarrow w}(\cdot | \cdot)$ — equivalently, parameterise each $\bar{q}^{v \rightarrow w}$ as the transition matrix of an irreversible 3-state CTMC over (xN, xP, xD) with rates fitted by maximum likelihood on the leaf data. The variational marginals q_n are then exactly the Felsenstein posteriors of this CTMC at column n , and the ELBO reduces to the log-likelihood of the leaf clamps under that CTMC plus the (now column-independent) expected log of $\mathbb{T}'$.

Fitch parsimony. The zero-temperature limit of the irreversible Felsenstein-3 posterior — assigning each internal node deterministically to its most-likely state under the CTMC — recovers the Fitch parsimony labelling of presence/absence under uniform priors and unit transition costs.

Scalability of stochastic ELBO optimisation. The free-parameter count scales as $2 \cdot |\mathcal{E}| \cdot L$ (plus a per-column root parameter) where $|\mathcal{E}|$ is the number of tree edges and L is the number of MSA columns. The variational q factorises over columns, but the ELBO is non-trivially coupled across columns through the prefix-of-Ignore factor in equation (B.123), which weights each (M, N) transition contribution by the chain of intervening Ignore probabilities. Match-rich columns (where most lineages are clamped to xP) have $P_n^{\text{lg}} \approx 0$ and so act as anchors that break the chain: contributions from (M, N) pairs that straddle such an anchor column are damped to zero, decoupling the gradient on either side. For both protein-scale and DNA-scale alignments with non-trivial column coverage, ELBO optimisation should therefore scale linearly with total residue count without running into long-range coupling pathologies.

B.8 Theory: structural bias of the BP cumulant under column-factorised q

We give a precise derivation of the structural bias displayed empirically in Section ?? . The bias is not a finite-data noise artifact — it is a property of the column-factorised variational family used

by SVI-VarAnc, and persists at any q whose per-column marginals match the truth.

Setup. On a single branch e of length t_e , the per-branch WFST is a chain HMM over the 5 states $\{S, M, I, D, E\}$ with chain transition matrix $\mathbf{T}_e[s, s']$ given by the standard TKF91 / TKF92 closed forms (`tkf92_wfst_T`, eq. A.29). The branch's contribution to the data log-likelihood is $\sum_{n=1}^{L+1} \log T_e[\zeta_{n-1}, \zeta_n]$ where $\zeta = (\zeta_0, \dots, \zeta_{L+1})$ is the chain trajectory across the $L + 2$ columns (including boundaries). Conventionally $\zeta_n \in \{S, M, I, D, E, \text{Ig}\}$ where Ig denotes a non-emitting "insertion-gap" column where the branch is in NYI/NYI or D/D joint state and contributes nothing to the chain transitions; chain transitions are indexed by consecutive non-Ig columns.

The variational q factorises across columns: $q(\zeta) = \prod_{n=1}^L q_n(\zeta_n)$ with each q_n a tree-structured distribution over edge presence/absence carrying the leaf clamps of column n . Define the per-column WFST-state probabilities at branch e :

$$P_n^{(e)}(s) = q_n(\text{branch state } e \text{ is } s), \quad s \in \{M, I, D\}, \quad P_n^{(e)}(\text{Ig}) = q_n(\text{NYI}, \text{NYI}) + q_n(D, D). \quad (\text{B.124})$$

The cumulant formula and what it computes exactly. Section A.2.3 introduces the BP cumulant $\mathbf{W}_e \in \mathbb{R}^{5 \times 5}$ as the sufficient statistic the M-step ingests:

$$W_e[s, s'] = \sum_{1 \leq M < N \leq L+1} P_M^{(e)}(s) \cdot P_N^{(e)}(s') \cdot \prod_{k=M+1}^{N-1} P_k^{(e)}(\text{Ig}). \quad (\text{B.125})$$

Under the factorised q , $W_e[s, s']$ is the *exact* expected count of WFST chain transitions $s \rightarrow s'$ on branch e :

$$\begin{aligned} \mathbb{E}_q[\#\{\text{chain transitions } s \rightarrow s'\}] &= \sum_{M < N} q(\zeta_M = s, \zeta_N = s', \zeta_{M+1} = \dots = \zeta_{N-1} = \text{Ig}) \\ &= \sum_{M < N} P_M(s) P_N(s') \prod_{k=M+1}^{N-1} P_k(\text{Ig}) = W_e[s, s'], \end{aligned}$$

where the second equality uses the column-factorisation of q . The sum over (M, N) counts each chain transition once because of the explicit Ig-padding constraint: any single trajectory realisation ζ with non-Ig columns at indices $n_1 < n_2 < \dots < n_K$ contributes weight 1 to exactly the adjacent-pair transitions $(n_0, n_1), (n_1, n_2), \dots, (n_K, n_{K+1})$ (with $n_0 = 0, n_{K+1} = L+1$); for any other (M, N) pair, at least one intervening column is non-Ig and the indicator product is zero. This $O(L)$ prefix-sum implementation agrees with the $O(L^2)$ direct sum to machine precision (6.7×10^{-16}).

The structural bias. The factorisation assumption is *not* satisfied by the truth posterior $p^*(\zeta \mid \text{data})$, even for TKF91 ($r = 0$). Although TKF91's generative model is per-position i.i.d., the per-branch WFST chain has nonzero transition probabilities $T_e[D, D]$, $T_e[I, I]$, $T_e[I, D]$ and so on (closed-form expressions in (A.29) reduce, at $r = 0$, to the standard TKF91 Pair HMM weights). These chain memories propagate from data conditioning into a non-factorised joint posterior:

$$p^*(\zeta_M = s, \zeta_N = s', \zeta_{M+1..N-1} = \text{Ig} \mid \text{data}) \neq \prod_n p^*(\zeta_n \mid \text{data}), \quad (\text{B.126})$$

even though the per-column truth marginal $p^*(\zeta_n \mid \text{data})$ is exactly recoverable by per-column Felsenstein BP. The bias of the cumulant against the truth-posterior chain expectation is the integrated difference between the factorised-q product of marginals and the truth-posterior joint:

$$W_e[s, s'] - \mathbb{E}_{p^*}[\#\{s \rightarrow s'\} \mid \text{data}] = \sum_{M < N} \left[p_M^*(s) p_N^*(s') \prod_{k=M+1}^{N-1} p_k^*(\text{Ig}) - p^*(\zeta_M = s, \zeta_N = s', \zeta_{M+1..N-1} = \text{Ig} \mid \text{data}) \right] \quad (\text{B.127})$$

The bracketed term is the connected $(N - M + 1)$ -way correlation of the truth posterior over the columns $\{M, M + 1, \dots, N\}$; it vanishes only when the truth posterior factorises across these columns. For TKF91, $T_e[D, D] > 0$ and $T_e[I, I] > 0$ at $r = 0$ alone suffice to make the right-hand side of (B.127) nonzero on the $(s, s') = (I, I)$ and (D, D) entries: an opened gap is positively correlated with continued gap by the chain transition kernel itself, even before any explicit fragment-extension parameter is invoked.

Entropy reward exacerbates the bias on rare entries. The ELBO objective decomposes as $\text{ELBO} = \mathbb{E}_q[\log p(z, \text{data}; \theta)] - \mathbb{E}_q[\log q]$ with the entropy term $-\mathbb{E}_q[\log q]$ rewarding diffuse q . At ELBO stationarity, q 's per-column marginals balance the data-fit term against the entropy term; for tied or weakly-fitted logits, the resulting q is slightly more diffuse than the truth posterior on individual rare states (i.e., $P_n^{(e)}(I)$ above the per-column truth marginal $p_n^*(I)$). Through the cumulant (B.125), even a small per-column inflation $P_n(s) = p_n^*(s) + \epsilon$ contributes a leading linear-in- ϵ bias on $W[s, s']$ with positive coefficient $\partial W / \partial P_n(s) = \sum_{N > n} P_N^{(e)}(s') \prod_{k=n+1}^{N-1} P_k^{(e)}(\text{Ig}) > 0$, summed over L source columns; the cumulant amplifies per-column diffuseness via its $O(L^2)$ pair structure.

Why the EM converges to a fixed point. The EM map $\theta \mapsto \text{MLE}(\mathbf{W}(\theta))$ is self-consistent at the biased fixed point θ_∞ when the cumulant bias on θ saturates. Each entry of the cumulant is bounded uniformly in θ : $W_e[s, s'] \leq L + 1$ on a single branch (each (M, N) pair contributes at most 1, and the geometric Ig-product is summable), giving a θ -independent upper bound on the aggregated sufficient-statistic count. The exposure denominator $T_{\text{eff}} = \sum_e t_e$ in the BDI rate-MLE is itself θ -independent — it depends only on the tree branch lengths, not on the inferred rates — and is bounded below by the smallest tree branch. Composing these, the rate-MLE $\hat{\lambda}(\theta) \propto \sum_e W[\cdot, I] / T_{\text{eff}}$ (with the precise BDI quadratic-MLE form of the closed-form indel M-step) is a continuous, bounded self-map on a compact rate interval $[0, \bar{\lambda}]$ with $\bar{\lambda} \leq (L + 1) / \min_e t_e$ (continuity follows because $W(\theta)$ depends continuously on θ via the per-edge TKF92 transition matrix (A.29), and the M-step quadratic root is continuous in its coefficients). Brouwer's fixed-point theorem applies: a fixed point θ_∞ exists. Consistent with this boundedness argument, the EM trajectory of Section ?? reaches a finite biased fixed point and does not diverge.

Implications for parameter inference. The structural bias is a property of any inference scheme that uses the cumulant tensor $\mathbf{W}$ as a sufficient statistic under a column-factorised variational family. It does not affect: (i) the ELBO value itself (which is a coherent $\log p(\text{data})$ lower bound regardless of what the cumulant looks like), nor (ii) ancestral indel-presence reconstruction (VarAnc and Fitch parsimony are statistically indistinguishable on the unified Pfam test splits, $|\Delta F1| < 0.006$ across short / hard / xhard), nor (iii) downstream MSA-conditioned rate estimation via exact-EM (Maraschino). It does affect any rate estimate computed from the variational cumulant, including the SVI-VarAnc refinement of cherry-trained parameters, where parameters

drift toward θ_∞ rather than toward truth. We report both initial cherry-trained and θ_∞ estimates without correction; the structural bias is the principal known limitation of variational training in this paper.

C Recursive TKF

MixDOM adds two levels of nested mixture structure on top of TKF92: a top-level TKF91 process governing domain births and deaths, and per-domain TKF92 processes governing fragments within. Each fragment also carries a substitution-class index. The exact Baum–Welch M-step proceeds via a six-step chain-restoration identity through a fully exploded null-state model. The same nesting pattern — a parent BDI / TKF process emitting child fragments that themselves carry latent class / domain / structure indices — supports a family of recursive TKF models that this appendix collects: MixDOM, its order-1 Maraschino adjacency distillation, the algebraic full-Woodbury distillation, the MixDOM-specific SVI-BW convergence theory, the tree-level VEM and ancestral-reconstruction algorithms, the generalised phylo-HMM, the formal recursive-grammar-elaboration rules, and four worked recursive examples (L-TKF, TKFST, TKFStack, TKF-Genome).

C.1 The TKF-Mixed Domain Model (MixDom)

This model was preliminarily described and empirically evaluated in (38).

C.1.1 The MixDom Model

The Mixture of TKF92 Domains (MixDom) is a multiply-nested hierarchical mixture model. At the top level is a TKF91-like links process where each link is associated with a domain of random type $n \sim \text{Categorical}(v_1, \dots, v_N)$. Each top-level link emits its own domain sequence, with model parameters determined by domain type.

Three nesting levels. MixDom is generated as three nested processes:

1. **Top-level TKF91 over domains:** a sequence of top-level links, each of domain type $n \sim \text{Categorical}(v_1, \dots, v_N)$, evolving under its own per-domain TKF92 indel process.
2. **Per-domain TKF92 over fragments:** within each domain of type n , a TKF92 process generates a sequence of nested links. Each such nested link is called a *fragment*. Different fragments are statistically independent.
3. **Intra-fragment Markov chain on fragment-types:** a single fragment consists of a sequence of fragment-type characters drawn from a per-domain Markov chain with $\mathcal{F} + 2$ states (start, end, and $\mathcal{F}$ emitting states $f \in \{1, \dots, \mathcal{F}\}$). The initial fragment-type is drawn from w_{nf} . From type f , the chain advances within the current fragment to type g with probability $r_{fg}^{(n)}$, or terminates the fragment with probability $\rho_f^{(n)} = 1 - \sum_g r_{fg}^{(n)}$.

The chain restarts at Start for each fresh fragment; different fragments are independent Markov realisations.

Each emitted site within a fragment of domain n and fragment-type f independently draws a site class $c \sim \text{Categorical}(u_{nf1}, \dots, u_{nfC})$, with C the number of site classes. The site is then governed by the substitution process $\text{Subst}(Q^{(c)}, \pi^{(c)})$.

We denote the per-domain process (TKF92 fragments with an intra-fragment fragment-type chain and per-(domain, fragment-type) site-class mixture) by $\text{HMM}(\{Q^{(c)}, \pi^{(c)}\}_c; r^{(n)}, u_n)$. The full MixDom model is then

$$\begin{aligned} M_{\text{dom}} &= \text{Links}(\text{HMM}(\{Q^{(c)}, \pi^{(c)}\}_c; r^{(n)}, u_n); \lambda_n, \mu_n) \\ \text{MixDom} &= \text{Links}(\text{Mix}_{n \sim v_n}(M_{\text{dom}}); \lambda_0, \mu_0) \end{aligned}$$

where $\text{Mix}(\dots)$ denotes a mixture model, with weights p , over parameters θ for model M

$$x \sim \text{Mix}_{k \sim p}(M(\theta_k); p) \Leftrightarrow k \sim \text{Categorical}(p), x \sim M(\theta_k)$$

Remark C.1 (Relationship to TKF92). When $\mathcal{F} = 1$ and $C = 1$, each fragment's intra-fragment chain has a single emitting state, so the fragment length is $\text{Geometric}(r_{11}^{(n)})$ and all sites share the single substitution model $\text{Subst}(Q, \pi)$, recovering TKF92. In general, the $\mathcal{F} \times \mathcal{F}$ transition matrix $r^{(n)}$ allows *intra-fragment correlations* between the fragment-types of adjacent positions within a single fragment (e.g., fragment-type 1 positions tend to follow fragment-type 1 positions when $r_{11}^{(n)}$ is the dominant row entry), and the per-(fragment-type) class distributions u_{nfc} allow the resulting substitution patterns to vary with fragment-type. *Different fragments remain statistically independent under this scheme*; Markov correlations are strictly *within* a fragment, carried by the fragment-type chain.

C.1.2 Singlet HMM for MixDom

The Singlet HMM generates sequences from the stationary distribution. Each domain may be empty with probability $\varphi_n \equiv 1 - \kappa_n$, so the probability that a top-level link generates a zero-length domain sequence is $z_0 = \sum_{n \in \mathcal{N}} v_n \varphi_n$. This leads to null cycles (a link is entered but immediately terminates). Eliminating these—by the Schur complement procedure described in the next section—yields a collapsed Singlet HMM with state space $\Psi^{(\text{eqm})} = \{\mathbf{S}, \mathbf{E}\} \cup \{\mathbf{I}_{nf} : n \in \mathcal{N}, f \in \mathcal{F}\}$ ($\mathcal{NF} + 2$ states). Each emitting state $\mathbf{I}_{nf}$ emits a character from $\sum_c u_{nfc} \pi^{(c)}$. The transition matrix $\mathcal{T}^{(\text{eqm})}$ for this collapsed Singlet HMM has entries

$$\mathcal{T}_{\mathbf{I}_{lf}, \mathbf{I}_{mg}}^{(\text{eqm})} = \frac{\rho_f^{(l)} \varphi_l \cdot \kappa_0 \cdot v_m \kappa_m w_{mg}}{1 - \kappa_0 z_0} + \delta(l=m) \rho_f^{(l)} \kappa_l w_{lg} + \delta(l=m) r_{fg}^{(l)}$$

where $\rho_f^{(n)} = 1 - \sum_g r_{fg}^{(n)}$ is the fragment termination probability, and the three terms represent (respectively) inter-domain transitions via the null-corrected top-level geometric process, same-domain new-fragment transitions, and same-domain intra-fragment Markov transitions on fragment-types. The $\mathbf{S}$ row is $\mathcal{T}_{\mathbf{S}, \mathbf{I}_{mg}}^{(\text{eqm})} = \kappa_0 v_m \kappa_m w_{mg} / (1 - \kappa_0 z_0)$ and $\mathcal{T}_{\mathbf{S}, \mathbf{E}}^{(\text{eqm})} = (1 - \kappa_0) / (1 - \kappa_0 z_0)$; the $\mathbf{E}$ column is $\mathcal{T}_{\mathbf{I}_{lf}, \mathbf{E}}^{(\text{eqm})} = \rho_f^{(l)} \varphi_l (1 - \kappa_0) / (1 - \kappa_0 z_0)$.

C.1.3 Pair HMM for MixDom

In the joint TKF92 Pair HMM, the start→end weight is $\tau(\lambda, \mu, T)_{\text{SE}}$. In MixDom , the probability that a $\mathbf{M}$ state emits no sequence is obtained by summing this TKF92 null output probability over domain types, $z_T = \sum_{n \in \mathcal{N}} v_n \tau_{\text{SE}}^{(n)}$ where $\tau^{(n)} \equiv \tau(\lambda_n, \mu_n, T)$. The Singlet HMM's null probability z_0 also appears in the Pair HMM, governing the $\mathbf{I}$ and $\mathbf{D}$ state null outputs (which involve only one sequence).

Start with the Pair HMM and split the M, I, and D states into non-emitting and emitting states. Let A, B, C denote the separated empty-match, empty-insert, and empty-delete states, respectively. The 8×8 transition matrix for this null-separated joint pair HMM is

$$v = \begin{pmatrix} & \text{S} & \text{M} & \text{I} & \text{D} & \text{E} & \text{A} & \text{B} & \text{C} \\ \text{S} & 0 & (1 - z_T)\tau_{\text{SM}} & (1 - z_0)\tau_{\text{SI}} & (1 - z_0)\tau_{\text{SD}} & \tau_{\text{SE}} & z_T\tau_{\text{SM}} & z_0\tau_{\text{SI}} & z_0\tau_{\text{SD}} \\ \text{M} & 0 & (1 - z_T)\tau_{\text{MM}} & (1 - z_0)\tau_{\text{MI}} & (1 - z_0)\tau_{\text{MD}} & \tau_{\text{ME}} & z_T\tau_{\text{MM}} & z_0\tau_{\text{MI}} & z_0\tau_{\text{MD}} \\ \text{I} & 0 & (1 - z_T)\tau_{\text{IM}} & (1 - z_0)\tau_{\text{II}} & (1 - z_0)\tau_{\text{ID}} & \tau_{\text{IE}} & z_T\tau_{\text{IM}} & z_0\tau_{\text{II}} & z_0\tau_{\text{ID}} \\ \text{D} & 0 & (1 - z_T)\tau_{\text{DM}} & (1 - z_0)\tau_{\text{DI}} & (1 - z_0)\tau_{\text{DD}} & \tau_{\text{DE}} & z_T\tau_{\text{DM}} & z_0\tau_{\text{DI}} & z_0\tau_{\text{DD}} \\ \text{E} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \text{A} & 0 & (1 - z_T)\tau_{\text{MM}} & (1 - z_0)\tau_{\text{MI}} & (1 - z_0)\tau_{\text{MD}} & \tau_{\text{ME}} & z_T\tau_{\text{MM}} & z_0\tau_{\text{MI}} & z_0\tau_{\text{MD}} \\ \text{B} & 0 & (1 - z_T)\tau_{\text{IM}} & (1 - z_0)\tau_{\text{II}} & (1 - z_0)\tau_{\text{ID}} & \tau_{\text{IE}} & z_T\tau_{\text{IM}} & z_0\tau_{\text{II}} & z_0\tau_{\text{ID}} \\ \text{C} & 0 & (1 - z_T)\tau_{\text{DM}} & (1 - z_0)\tau_{\text{DI}} & (1 - z_0)\tau_{\text{DD}} & \tau_{\text{DE}} & z_T\tau_{\text{DM}} & z_0\tau_{\text{DI}} & z_0\tau_{\text{DD}} \end{pmatrix} \quad (\text{C.1})$$

with $\tau \equiv \tau(\lambda_0, \mu_0, T)$; or, elementwise,

$$v_{ij} = \begin{cases} 0 & \text{if } j = \text{S} \\ (1 - z_T)\tau_{i\text{M}} & \text{if } j = \text{M} \\ (1 - z_0)\tau_{ij} & \text{if } j \in \{\text{I}, \text{D}\} \\ \tau_{i\text{E}} & \text{if } j = \text{E} \\ z_T\tau_{i\text{M}} & \text{if } j = \text{A} \\ z_0\tau_{i\text{I}} & \text{if } j = \text{B} \\ z_0\tau_{i\text{D}} & \text{if } j = \text{C} \end{cases} \quad (\text{C.2})$$

Let $\mathcal{U}_{\Phi_1, \Phi_2}$ be the matrix formed by zeroing all rows except $i \in \Phi_1$ and all columns except $j \in \Phi_2$ of v . Consider the matrix of transitions between the empty states A, B, C

$$\mathcal{U}_{\text{ABC}, \text{ABC}} = \begin{pmatrix} & \dots & \text{A} & \text{B} & \text{C} \\ \dots & \dots & \dots & \dots & \dots \\ \text{A} & \dots & z_T\tau_{\text{MM}} & z_0\tau_{\text{MI}} & z_0\tau_{\text{MD}} \\ \text{B} & \dots & z_T\tau_{\text{IM}} & z_0\tau_{\text{II}} & z_0\tau_{\text{ID}} \\ \text{C} & \dots & z_T\tau_{\text{DM}} & z_0\tau_{\text{DI}} & z_0\tau_{\text{DD}} \end{pmatrix}$$

where the unshown rows and columns (corresponding to transitions to, from, or between S, M, I, D, E) have zero entries. Summing over paths of all (including zero) lengths via A, B, C (i.e. performing the Schur complement) yields

$$\sum_{k=0}^{\infty} \mathcal{U}_{\text{ABC}, \text{ABC}}^k = (I - \mathcal{U}_{\text{ABC}, \text{ABC}})^{-1} = \mathcal{Z}$$

which has a relatively simple closed form (reducing to 3×3 matrix inversion). The effective nonempty 5×5 transition matrix (with A, B, C summed out) is

$$\mathcal{T} \equiv \mathcal{T}(\theta, T) = \mathcal{U}_{\text{SMIDE}, \text{SMIDE}} + \mathcal{U}_{\text{SMIDE}, \text{ABC}} \cdot \mathcal{Z} \cdot \mathcal{U}_{\text{ABC}, \text{SMIDE}}$$

where $\theta = (\lambda_0, \mu_0, \{v_n\}, \{\lambda_n, \mu_n\}, \{w_{nff}\}, \{r_{fg}^{(n)}\}, \{u_{nfc}\}, \{Q^{(c)}, \pi^{(c)}\})$.

Following null state elimination, the collapsed Pair HMM has $5\mathcal{NF} + 2$ states, namely

$$\Psi = \{\text{SS}, \text{EE}\} \cup \{\text{UX}_{lf} : \text{UX} \in \{\text{MM}, \text{MI}, \text{MD}, \text{II}, \text{DD}\}, 1 \leq l \leq \mathcal{N}, 1 \leq f \leq \mathcal{F}\}$$

where l is a domain index and f is a fragment index. For notational convenience, we treat the distinguished states SS and EE as carrying sentinel indices $(l, f) = (0, 0)$, which are unused by other

states as true domain and fragment indices are 1-based. Thus expressions written in terms of a generic state $\mathbf{UX}_{lf}$ are understood to include the cases **SS** and **EE** by setting $(l, f) = (0, 0)$, except where formulae explicitly require $1 \leq l \leq \mathcal{N}$ or $1 \leq f \leq \mathcal{F}$.

Loosely speaking, and noting the above caveat, every transition $\mathbf{UX}_{lf} \rightarrow \mathbf{VY}_{mg}$ involves a potential domain exit transition from $\mathbf{X} \rightarrow \mathbf{E}$ in the nested model (weight p_{out}), an inter-domain transition $\mathbf{U} \rightarrow \mathbf{V}$ in the top-level model that factors in paths through empty domains (weight $\mathcal{T}_{\mathbf{UV}}$), and a potential domain re-entry transition from $\mathbf{S} \rightarrow \mathbf{Y}$ in the nested model (weight p_{in}), which must include a factor of $(1 - z_T)^{-1}$ (for top-level Match states) or $(1 - z_0)^{-1}$ (for top-level Insert/Delete states) to account for domain entry being conditional on the domain being nonempty (these factors will precisely cancel out the factors in the **M**, **I**, and **D** columns of v , which have been included here solely to preserve row-normalization of v and χ). If the domain type and top-level state are the same for source and destination state ($\mathbf{U} = \mathbf{V}$ and $l = m$), then an additional intra-domain transition which extends the current domain, starting a new fragment, is folded in (weight p_{SameDom}). If the domain is the same ($l = m$), then an additional intra-fragment fragment-type transition from f to g is folded in (weight p_{SameFrag}), governed by the $\mathcal{F} \times \mathcal{F}$ transition matrix $r_{fg}^{(l)}$ of the Markov chain within the current fragment.

The transition matrix $\chi(\theta, T)$ for this collapsed Pair HMM has entries

Source i ($\mathbf{UX}_{lf}$)	Destination j ($\mathbf{VY}_{mg}$)	$\chi_{ij} = p_{\text{out}} \times \mathcal{T}_{\mathbf{UV}} \times p_{\text{in}} + \delta(\mathbf{U} = \mathbf{V})\delta(l = m)(p_{\text{SameDom}} + \delta(\mathbf{X} = \mathbf{Y})p_{\text{SameFrag}})$
		$p_{\text{out}}(\mathbf{U}, \mathbf{X}, l, f) \quad \mathcal{T}_{\mathbf{UV}} \quad p_{\text{in}}(\mathbf{V}, \mathbf{Y}, m, g) \quad p_{\text{SameDom}}(\mathbf{X}, \mathbf{Y}, l, f, g) \quad p_{\text{SameFrag}}(l, f, g)$
SS	MY _{mg}	1 $\mathcal{T}_{\mathbf{SM}} (1 - z_T)^{-1} v_m \tau_{\mathbf{SY}}^{(m)} w_{mg}$ 0 0
	II _{mg}	1 $\mathcal{T}_{\mathbf{SI}} (1 - z_0)^{-1} v_m \kappa_m w_{mg}$ 0 0
	DD _{mg}	1 $\mathcal{T}_{\mathbf{SD}} (1 - z_0)^{-1} v_m \kappa_m w_{mg}$ 0 0
	EE	1 $\mathcal{T}_{\mathbf{SE}}$ 1 0 0
MX _{lf}	MY _{mg}	$\rho_f^{(l)} \tau_{\mathbf{XE}}^{(l)} \mathcal{T}_{\mathbf{MM}} (1 - z_T)^{-1} v_m \tau_{\mathbf{SY}}^{(m)} w_{mg}$ $\rho_f^{(l)} \tau_{\mathbf{XY}}^{(l)} w_{lg}$ $r_{fg}^{(l)}$
	II _{mg}	$\rho_f^{(l)} \tau_{\mathbf{XE}}^{(l)} \mathcal{T}_{\mathbf{MI}} (1 - z_0)^{-1} v_m \kappa_m w_{mg}$ 0 0
	DD _{mg}	$\rho_f^{(l)} \tau_{\mathbf{XE}}^{(l)} \mathcal{T}_{\mathbf{MD}} (1 - z_0)^{-1} v_m \kappa_m w_{mg}$ 0 0
	EE	$\rho_f^{(l)} \tau_{\mathbf{XE}}^{(l)} \mathcal{T}_{\mathbf{ME}}$ 1 0 0
II _{lf}	MY _{mg}	$\rho_f^{(l)} \varphi_l \mathcal{T}_{\mathbf{IM}} (1 - z_T)^{-1} v_m \tau_{\mathbf{SY}}^{(m)} w_{mg}$ 0 0
	II _{mg}	$\rho_f^{(l)} \varphi_l \mathcal{T}_{\mathbf{II}} (1 - z_0)^{-1} v_m \kappa_m w_{mg}$ $\rho_f^{(l)} \kappa_l w_{lg}$ $r_{fg}^{(l)}$
	DD _{mg}	$\rho_f^{(l)} \varphi_l \mathcal{T}_{\mathbf{ID}} (1 - z_0)^{-1} v_m \kappa_m w_{mg}$ 0 0
	EE	$\rho_f^{(l)} \varphi_l \mathcal{T}_{\mathbf{IE}}$ 1 0 0
DD _{lf}	MY _{mg}	$\rho_f^{(l)} \varphi_l \mathcal{T}_{\mathbf{DM}} (1 - z_T)^{-1} v_m \tau_{\mathbf{SY}}^{(m)} w_{mg}$ 0 0
	II _{mg}	$\rho_f^{(l)} \varphi_l \mathcal{T}_{\mathbf{DI}} (1 - z_0)^{-1} v_m \kappa_m w_{mg}$ 0 0
	DD _{mg}	$\rho_f^{(l)} \varphi_l \mathcal{T}_{\mathbf{DD}} (1 - z_0)^{-1} v_m \kappa_m w_{mg}$ $\rho_f^{(l)} \kappa_l w_{lg}$ $r_{fg}^{(l)}$
	EE	$\rho_f^{(l)} \varphi_l \mathcal{T}_{\mathbf{DE}}$ 1 0 0

(C.3)

where $\tau^{(n)} \equiv \tau(\lambda_n, \mu_n, T)$, $\rho_f^{(n)} \equiv 1 - \sum_g r_{fg}^{(n)}$ is the fragment termination probability for fragment state f in domain n , and (as before) $\varphi_l \equiv 1 - \kappa_l$.

The substitution parameters ($Q^{(c)}, \pi^{(c)}$) do not appear in the transition matrix, but in the emission probabilities of the various states. The probability of emitting token (a, b) from state $\mathbf{MM}_{nf}$ is

$$P(a, b \mid \mathbf{MM}_{nf}, T) = \sum_{c=1}^C u_{nfc} \pi_a^{(c)} \exp(R^{(c)} T)_{a,b} \quad (\text{C.4})$$

where $R^{(c)} = Q^{(c)} \cdot \text{diag}(\pi^{(c)})$ is the rate matrix for site class c , and u_{nfc} is the probability that fragment state f in domain n generates site class c . The probability of emitting ancestral token a from states $\{\text{MD}_{nf}, \text{DD}_{nf}\}$, or descendant token a from states $\{\text{MI}_{nf}, \text{II}_{nf}\}$, is $\sum_c u_{nfc} \pi_a^{(c)}$.

C.1.4 Baum-Welch Algorithm for MixDom Pair HMM

In order to map HMM transition counts back to the BDI sufficient statistics, we need to correct for the null transition elimination performed in the previous section, resolve the transition counts in the nested Pair HMM onto the separate components of the model, and then apply the formulas from earlier sections.

E-step. Run Forward-Backward on the collapsed $(5\mathcal{NF} + 2)$ -state Pair HMM (Section C.1.1). This yields expected transition counts $\hat{n}_{ij}''$ for all state pairs i, j in the collapsed Pair HMM, and expected emission counts $\hat{e}_{(a,b)}''$ for each state i and input-output token pair (a, b) . The sufficient statistics for the mixture component selectors $\{v_n, w_{nf}\}$ can be recovered directly at this stage, and the emission counts accumulated onto the appropriate W, U, V statistics for the per-domain CTMCs.

Conceptually speaking, we next resolve the collapsed-HMM transition counts $\hat{n}_{ij}''$ to TKF91 Pair HMM-like transition counts $\hat{n}_{ij}^{(0)}$ for transitions in the top-level inter-domain model; TKF92 Pair HMM-like transition counts $\hat{n}_{\text{XY}}^{\text{M}(l)'}$ for the nested intra-domain match, insert, and delete states MX_{lf} of each domain-match submodel (for $\text{X}, \text{Y} \in \{\text{M}, \text{I}, \text{D}\}$); and singlet-style transition counts for the within-domain fragment Markov chain $\hat{n}_0^{\text{II}(l)'}$, $\hat{n}_1^{\text{II}(l)'}$, $\hat{n}_0^{\text{DD}(l)'}$, $\hat{n}_1^{\text{DD}(l)'}$ for the domain-insert and domain-delete submodels (where the “continuation” is now governed by the $\mathcal{F} \times \mathcal{F}$ Markov chain $r^{(l)}$ rather than a scalar geometric parameter).

The intra-domain transition counts arise solely from the p_{SameDom} -weighted transitions, and can be isolated proportionally, while the inter-domain counts must correct for the Schur complement null cycle elimination. The zero-adjustment length counts $\hat{n}_\kappa^{\text{II}(l)'}$, $\hat{n}_\varphi^{\text{II}(l)'}$, $\hat{n}_\kappa^{\text{DD}(l)'}$, $\hat{n}_\varphi^{\text{DD}(l)'}$ accounting for the number of times a domain was empty *vs* nonempty must also receive contributions from the Schur complement correction.

The intra-domain counts can then be resolved to TKF91-like counts $\hat{n}_{ab}^{(l)}$ and Intra-fragment fragment-type transition counts $\hat{n}_{fg}^{(n)}$ (the expected number of transitions from fragment-type f to fragment-type g within a single fragment of domain n), together with fragment termination counts $\hat{n}_{\rho,f}^{(n)}$. The singlet-style counts for each domain $\hat{n}_0^{\text{II}(l)'}$, $\hat{n}_1^{\text{II}(l)'}$, $\hat{n}_0^{\text{DD}(l)'}$, $\hat{n}_1^{\text{DD}(l)'}$ can be resolved via a similar procedure to $\hat{n}_\kappa$, $\hat{n}_\varphi$, which are directly link sequence extension/termination counts $L^{(l)}$, $M^{(l)}$, to be accumulated onto the running totals for those counts. Finally, all TKF91-like transition counts for the top-level and nested models are accumulated onto the BDI sufficient statistics S, B, D, L, M as in Section A.1.8.

We now consider the null count restorations in more detail. In practice, we will use a notational shortcut (that is, nevertheless, correct and reliable, and entirely equivalent to the procedure we just outlined) to greatly simplify many of these calculations, bypassing much of this conceptual tower of piecewise count restorations. In doing this we again exploit a form of the score function identity—in this case, that the expected transition usage is the derivative of the log-likelihood with respect to the log-transition weight.

Resolving transition counts. Following the notation of Section C.1.1, let $(\mathcal{U}_{\Phi_1, \Phi_2})_{ij} = v_{ij} \delta(i \in \Phi_1, j \in \Phi_2)$ be the masked transition matrix so that $\mathcal{Z} = (I - \mathcal{U}_{\text{ABC}, \text{ABC}})^{-1}$ represents sums over null

paths and $\mathcal{T} = \mathcal{U}_{\text{SMIDE,SMIDE}} + \mathcal{U}_{\text{SMIDE,ABC}} \cdot \mathcal{Z} \cdot \mathcal{U}_{\text{ABC,SMIDE}}$ the null-eliminated transition matrix.

Considering paths in v that begin and end in $i, j \in \{\text{S, M, I, D, E}\}$, whose intermediate states (if any) are in $\{\text{A, B, C}\}$, the expected transition usages and state occupancies are

$$\begin{aligned} \mathbb{E}[n_{ab}|i \rightarrow j] &= \frac{(I + \mathcal{U}_{\text{SMIDE,ABC}} \cdot \mathcal{Z})_{ia} \mathcal{T}_{ab} (\mathcal{Z} \cdot \mathcal{U}_{\text{ABC,SMIDE}})_{bj}}{\mathcal{T}_{ij}} \\ &= \frac{\partial \log \mathcal{T}_{ij}}{\partial \log v_{ab}} \end{aligned}$$

The expected transition usage is the score function identity, appearing in a new form: the n_{ab} are sufficient statistics for the v path log-likelihood. We can use this identity, with the chain rule, for many of the counts we seek:

$$\hat{n}_{\text{UV}}^{(0)} = \sum_{i,j} \hat{n}_{ij}'' \sum_{k,l} \frac{\partial \log \chi_{ij}}{\partial \log \mathcal{T}_{kl}} \sum_{a,b} \frac{\partial \log \mathcal{T}_{kl}}{\partial \log v_{ab}} \frac{\partial \log v_{ab}}{\partial \log \tau_{\text{UV}}^{(0)}} \quad (\text{C.5})$$

$$\hat{n}_{\text{UV}}^{(l)} = \sum_{i,j} \hat{n}_{ij}'' \frac{\partial \log \chi_{ij}}{\partial \log \tau_{\text{UV}}^{(l)}} \quad (\text{C.6})$$

$$\hat{n}_{\vartheta} = \sum_{i,j} \hat{n}_{ij}'' \left(\frac{\partial \log \chi_{ij}}{\partial \log \vartheta} + \sum_{k,l} \frac{\partial \log \chi_{ij}}{\partial \log \mathcal{T}_{kl}} \sum_{a,b} \frac{\partial \log \mathcal{T}_{kl}}{\partial \log v_{ab}} \frac{\partial \log v_{ab}}{\partial \log \vartheta} \right) \quad (\text{C.7})$$

for $\vartheta \in \{v_n, \kappa_n, \varphi_n, w_{nf}, r_{fg}^{(n)}, \rho_f^{(n)}\}$. A few notes:

1. In these expressions we have written $\tau_{\text{UV}}^{(0)}$ for the inter-domain TKF91 transition probabilities that just appear as τ_{UV} in Equation (C.1).
2. The term $\frac{\partial \log \mathcal{T}_{ij}}{\partial \log v_{ab}}$ is used to highlight use of the chain rule, but the actual calculation of this term can be done via the matrix formula for $\mathbb{E}[n_{ab}|i \rightarrow j]$ above.
3. We must be careful to treat (κ_n, φ_n) as independent free parameters of χ_{ij} for the purpose of the partial derivatives in Equation (C.7), and similarly for $(r_{fg}^{(n)}, \rho_f^{(n)})$; even though they are deterministically related by $\kappa_n + \varphi_n = 1$ and $\rho_f^{(n)} = 1 - \sum_g r_{fg}^{(n)}$, we should not differentiate through those constraints when calculating the derivatives of v_{ab} with respect to $\kappa_n, \varphi_n, r_{fg}^{(n)}$, and $\rho_f^{(n)}$.
4. We are also treating v_{ab} and ϑ as independent free parameters of χ_{ij} for the purpose of Equation (C.7), which is why we expand the derivative into two terms: one term involving $\frac{\partial \log \chi_{ij}}{\partial \log \vartheta}$ which captures the direct dependence of χ_{ij} on ϑ (e.g. via v_m or w_{mg} in the transition formulas above), and another term which captures the indirect dependence of χ_{ij} on ϑ via v_{ab} .
5. Similar points apply to the roles of $\tau_{\text{UV}}^{(l)}$ in Equation (C.6), and of $\tau_{\text{UV}}^{(0)}$ in Equation (C.5): these must be treated as free parameters and not differentiated through.
6. In contrast, however, we must differentiate through the z_0 and z_T terms in v when calculating the derivatives of v_{ab} with respect to v_n, φ_n , and $\tau_{\text{SE}}^{(n)}$.
7. Similarly, we must differentiate through $p_{\text{out}}, p_{\text{in}}, p_{\text{SameDom}}$, and p_{SameFrag} when calculating the derivatives of $\mathcal{T}_{ij}$. These terms do not represent free parameters.

Finally, we use these counts to accumulate onto the BDI sufficient statistics for the top-level inter-domain TKF91 model and nested intra-domain TKF92 models as described in Section A.1.8. The fragment correction for the intra-domain counts has effectively already been performed by the way we resolved the intra-domain counts from the collapsed Pair HMM counts, which account for the probability of fragment continuation *vs* new fragments via the p_{SameDom} -weighted terms which were differentiated through by the score function identity. We also need to accumulate the expected counts for the mixture component selectors $\{v_n, w_{nf}\}$, which are $\hat{n}_\vartheta$ for $\vartheta \in \{v_n, w_{nf}\}$ as calculated above.

Accumulating sufficient statistics. Initialize all sufficient statistics to zero. Then, for each training pair (x, y) with evolutionary time T , accumulate as follows.

Top-level BDI (inter-domain TKF91). From the top-level count matrix $\hat{n}_{ij}^{(0)}$, compute BDI expectations using (A.15)–(A.17) with parameters (λ_0, μ_0, T) , and accumulate:

$$\begin{aligned} B^{(0)} &+= \mathcal{C}^B(\hat{\mathbf{n}}^{(0)}, T) \\ D^{(0)} &+= \mathcal{C}^D(\hat{\mathbf{n}}^{(0)}, T) \\ S^{(0)} &+= \mathcal{C}^S(\hat{\mathbf{n}}^{(0)}, T) \\ L^{(0)} &+= \hat{n}_\kappa^{(0)} \quad (\text{expected ancestral \& inserted domain count}) \\ M^{(0)} &+= 1 \quad (\text{expected top-level terminations}) \\ T^{(0)} &+= T \quad (\text{top-level BDI observation time}) \end{aligned}$$

Note that $L^{(0)}$ counts *domains* (top-level links), not residues. It is not directly observed, but is recovered from the M and D column sums of $\hat{n}_{ij}^{(0)}$: $\hat{n}_\kappa^{(0)} = \sum_i (\hat{n}_{iM}^{(0)} + \hat{n}_{iD}^{(0)})$. The time statistic $T^{(0)}$ is weighted by $\hat{n}_\varphi^{(0)}$, since each independent BDI process contributes one trajectory of duration T .

Per-domain BDI (intra-domain TKF92, domain l). The intra-domain counts $\hat{n}_{ij}^{(l)}$ have already had the TKF92 fragment correction applied (fragment continuation *vs* new fragments was resolved by differentiating through the p_{SameFrag} terms). Thus:

$$\begin{aligned} B^{(l)} &+= \mathcal{C}^B(\hat{\mathbf{n}}^{(l)}, T) \\ D^{(l)} &+= \mathcal{C}^D(\hat{\mathbf{n}}^{(l)}, T) \\ S^{(l)} &+= \mathcal{C}^S(\hat{\mathbf{n}}^{(l)}, T) \\ L^{(l)} &+= \hat{n}_\kappa^{(l)} \quad (\text{expected ancestral \& inserted fragment count}) \\ M^{(l)} &+= \hat{n}_\varphi^{(l)} \quad (\text{expected domain terminations}) \\ T^{(l)} &+= T \cdot \hat{n}_\varphi^{(l)} \quad (\text{expected domain-level BDI time}) \\ F_{fg}^{(n)} &+= \hat{n}_{r,fg}^{(n)} \quad (\text{fragment } f \rightarrow g \text{ transitions}) \\ E_f^{(n)} &+= \hat{n}_{\rho,f}^{(n)} \quad (\text{fragment } f \text{ terminations}) \end{aligned}$$

Here $L^{(l)}$ counts fragments (links within the domain), not residues. Each domain- l entry runs an independent fragment-level BDI process for time T , so $T^{(l)}$ is weighted by the expected number of entries. The fragment-type transition counts $F_{fg}^{(n)}$ form an $\mathcal{F} \times \mathcal{F}$ matrix per domain, recording the expected number of intra-fragment Markov transitions between fragment-types within a fragment of domain n .

Mixture selectors.

$$\begin{aligned} N_{v_l} & += \hat{n}_{v_l} \\ N_{w_{lf}} & += \hat{n}_{w_{lf}} \end{aligned}$$

CTMC substitution (per site class). For each match emission at state $\mathbf{MM}_{nf}$ with observed pair (a, b) , the posterior probability of site class c given the emission is $\gamma_c \propto u_{nfc} \pi_a^{(c)} \exp(R^{(c)}T)_{a,b}$. Using this posterior weight, accumulate endpoint-conditioned CTMC expectations (A.2)–(A.3) on the $|\mathcal{A}|$ -state chain with rate matrix $R^{(c)}$:

$$\begin{aligned} W_a^{(c)} & += \text{(dwell in state } a, \text{ weighted by } \gamma_c \text{ and HMM posterior)} \\ U_{a,a'}^{(c)} & += \text{(transition } a \rightarrow a', \text{ weighted by } \gamma_c) \\ V_a^{(c)} & += \text{(composition count for } a \text{ at match-position ancestors, insertions, and deletions, weighted by } \gamma_c) \end{aligned}$$

Site class assignment counts. For each emission at state $\mathbf{MM}_{nf}$ (or the corresponding insert/delete states), accumulate the posterior site class assignment:

$$N_{u_{nc}} += \gamma_c$$

Remark C.2 (Genealogical correction terms in nested models). As noted in Section A.1.8, the CTMC sufficient statistics W, U, V omit genealogical correction terms from transient and partially-observed lineages. An analogous omission applies at each nesting level of the MixDom model: transient domain insertions and deletions (domains that are born and die between times 0 and T without being directly observed) contribute to the top-level BDI sufficient statistics B, D, S via the null count restoration, but their internal fragment-level processes (intra-fragment fragment-type Markov chain and BDI) are not modeled. Similarly, domains that are inserted after time 0 or deleted before time T do not accumulate fragment-level statistics for the period of their non-existence. This is consistent with the principle that the M-step optimizes only the complete-data log-likelihood for structures whose existence is certified by the HMM state path.

M-step. All M-step updates use MAP estimates whose priors contribute additive pseudocounts to the sufficient statistics, so the maximizer formulas are the same as the MLE formulas applied to prior-augmented statistics. For the multinomial parameter groups (mixture weights, fragment-type transitions, site-class distributions) the priors below are the standard Dirichlet conjugates; for the BDI rates and the reversible GTR submodel the priors below are non-conjugate regularizers and the true conjugate priors are different (see “Comments on conjugacy” below for what the proper conjugate priors look like and why we use the simpler ones here).

Priors.

- Gamma($\alpha_\lambda, \beta_{\lambda\mu}$) on λ and Gamma($\alpha_\mu, \beta_{\lambda\mu}$) on μ , sharing the rate parameter $\beta_{\lambda\mu}$. Augmented statistics: $B \rightarrow B + \alpha_\lambda - 1$, $D \rightarrow D + \alpha_\mu - 1$, $S \rightarrow S + \beta_{\lambda\mu}$.
- Gamma(α_Q, β_Q) on each Q_{ij} (shared β_Q per row): $U_{ij} \rightarrow U_{ij} + \alpha_Q - 1$, $W_i \rightarrow W_i + \beta_Q$.
- Dirichlet(α_π) on π : $V_i \rightarrow V_i + \alpha_\pi - 1$.
- Dirichlet(α_v) on domain weights: $N_{v_n} \rightarrow N_{v_n} + \alpha_v - 1$.
- Dirichlet(α_w) on fragment weights: $N_{w_{nf}} \rightarrow N_{w_{nf}} + \alpha_w - 1$.
- Dirichlet(α_r) on each row of the fragment transition matrix (including the termination probability): $F_{fg}^{(n)} \rightarrow F_{fg}^{(n)} + \alpha_r - 1$, $E_f^{(n)} \rightarrow E_f^{(n)} + \alpha_r - 1$.
- Dirichlet(α_u) on site class distributions: $N_{u_{nc}} \rightarrow N_{u_{nc}} + \alpha_u - 1$.

Comments on conjugacy. The Dirichlet priors on v , w , u , and the fragment-type transition matrix are conjugate to the corresponding multinomial likelihoods in the standard way. The two remaining cases warrant comment:

Reversible CTMC. Treated as an irreversible CTMC (independent off-diagonal rates R_{ij}), the complete-data likelihood (A.1) is a regular exponential family and a product of independent Gammas on R_{ij} together with a Dirichlet on the initial distribution is conjugate. For the *reversible* parameterization $R_{ij} = Q_{ij}\pi_j$ with symmetric exchangeabilities, the conjugate prior is *not* a product of independent Gammas on Q_{ij} and a Dirichlet on π : detailed balance couples the two factors, and the conjugate prior is the cycle-corrected edge-flow density of Diaconis and Rolles (17), which arises as the de Finetti mixing measure for an edge-reinforced random walk on the state graph (15, 53). Reparameterizing in terms of an undirected edge weight $x_e = \pi_i Q_{ij} \pi_j$ on each edge $e = \{i, j\}$ (with loops $x_{ii} = \pi_i^2 R_{ii}^{\text{loop}}$ in the Grassmann-uniformized variant) and normalising so that $\sum_e x_e = 1$, the prior takes the form

$$\phi_{v_0, a}(x) \propto \left(\prod_e x_e^{a_e - 1/2} \right) x_{v_0}^{a_{v_0}/2} \prod_{v \neq v_0} x_v^{-(a_v + 1)/2} \sqrt{\det A(x)},$$

where $x_v = \sum_{e \ni v} x_e$ is the total flow at vertex v , $a_v = \sum_{e \ni v} a_e$ is the corresponding pseudocount, and $A(x)$ is a matrix indexed by a basis of cycles in the state graph whose determinant equals $\sum_T \prod_{e \notin T} x_e^{-1}$ (sum over spanning trees T), by Kirchhoff's matrix-tree theorem. For the complete graph with loops (i.e. GTR), $\binom{|A|-1}{2}$ independent cycles contribute and the joint prior cannot be factored across edges. The Diaconis–Rolles prior conditions on the initial state v_0 ; including a stationary observation of $X(0) = v_0$ multiplies the prior by $\pi(v_0) \propto \sqrt{x_{v_0}}$ in the edge-flow chart, which shifts the v_0 -exponent and stays within the same family with adjusted hyperparameters. The independent Gamma $\times$ Dirichlet priors used here are non-conjugate regularizers in the reversible parameterization and are perfectly valid for MAP, but the closed-form posterior is in the Diaconis–Rolles family rather than back in Gamma $\times$ Dirichlet.

TKF91 BDI rates. The joint complete-data log-likelihood $\ell_1(\lambda, \mu)$ from (A.18), which includes the prior probability of the ancestral sequence length, is a *curved* exponential family in (λ, μ) because the $\log(\mu - \lambda)$ term couples the natural parameters. A product of independent Gammas on λ and μ is therefore not conjugate; it is conjugate to the complete-data likelihood of the underlying linear birth–death process *conditioned on* the initial sequence length, but not to the joint likelihood that includes a stationary L -prior. The proper conjugate prior, derived in the queueing-theory literature by Armero and Bayarri (3) and applied to the linear-growth BDI by Conti (14), is most naturally written in terms of $\kappa = \lambda/\mu \in (0, 1)$ and μ :

$$\phi(\kappa, \mu) \propto \kappa^{a-1} (1 - \kappa)^{b-1} \mu^{c-1} \exp(-\mu(\tau_1 \kappa + \tau_2)),$$

with five hyperparameters $(a, b, c, \tau_1, \tau_2)$ updated by $a \rightarrow a + B + L$, $b \rightarrow b + M$, $c \rightarrow c + B + D$, $\tau_1 \rightarrow \tau_1 + S + T$, $\tau_2 \rightarrow \tau_2 + S$. Marginalising μ yields a Gauss-hypergeometric distribution on κ (in the Johnson–Kotz–Balakrishnan family); the normaliser is a ${}_2F_1$ value and the posterior is tractable by one-dimensional quadrature or Gibbs sampling. Equivalently, the $M \log(\mu - \lambda) + L \log \lambda$ contribution from incorporating the stationary initial-length prior amounts to an extra Beta($L + 1, M + 1$)-like factor in κ on top of the dynamics-only Gamma evidence. As above, our independent Gamma priors on λ and μ are non-conjugate regularizers in this parameterisation. In the long-time stationary regime where $B \approx D$, the $M \log(\mu - \lambda)$ and $L \log \lambda - (L + M) \log \mu$ stationary contributions are doing the work of identifying κ separately from the overall rate scale; ignoring them entirely (i.e. Gamma EM with no L, M counts) silently underuses the data when only one long observation is available.

We use simple $\text{Gamma} \times \text{Dirichlet}$ pseudocounts throughout because they are easy to set, behave well as MAP regularizers, and keep the M-step closed-form (the augmented sufficient statistics still have a unique maximiser via the same quadratic in κ and the same pooled GTR formula). A fully Bayesian treatment would substitute the Diaconis–Rolles and Armero–Bayarri/Conti priors above; we leave that to future work.

Indel rates. For the top-level rates (λ_0, μ_0) and each per-domain rate pair (λ_n, μ_n) , solve the quadratic (A.22) with augmented (B, D, L, M, S, T) and extract κ, μ, λ via (A.23)–(A.25).

Fragment transition matrix. Row-normalize the augmented fragment-type transition counts per domain: $r_{fg}^{(n)} \leftarrow F_{fg}^{(n)} / (E_f^{(n)} + \sum_{g'} F_{fg'}^{(n)})$ (with augmented F, E).

Mixture weights. Normalize augmented counts: $v_n \propto N_{v_n}, w_{nf} \propto N_{w_{nf}}$.

Site class distributions. $u_{nfc} \propto N_{u_{nfc}}$.

CTMC parameters. The sufficient statistics are projected onto parameter groups.

Per-class equilibrium: $\pi_a^{(c)} \propto V_a^{(c)} + \alpha_\pi - 1$. (This is the standard empirical-frequency estimator; the exact EM M-step couples π with Q through dwell-time statistics, but the approximation is standard practice for GTR models.)

Per-class exchangeability: The rate is $Q_{a,a'}^{(c)} \cdot \pi_{a'}^{(c)}$. The exchangeability is estimated from the bridge-expectation transition and dwell counts:

$$Q_{a,a'}^{(c)} = \frac{U_{a,a'}^{(c)} + U_{a',a}^{(c)}}{W_a^{(c)} \cdot \pi_{a'}^{(c)} + W_{a'}^{(c)} \cdot \pi_a^{(c)}}$$

C.1.5 WFSTs for MixDom

As with TKF92 (Section A.2.2), constructing a WFST for MixDom is complicated by latent information—in this case, the domain type and fragment type associated with each position. We here outline two approaches to this issue.

The first approach is to preserve the latent information by promoting it to the transducer’s input/output alphabet: each character is decorated with its domain and fragment labels, yielding a *Labeled-MixDom WFST* whose state space is comparable to the Pair HMM but whose alphabet is enlarged. This approach is exact but produces larger machines.

The second approach integrates out the latent variables and approximates the result using compact *order-1* machines whose transitions depend only on the most recently emitted characters. These are smaller and more efficient for tree-based inference, at the cost of approximating the full latent structure via local context.

C.2 Selected Inference Algorithms for MixDom

C.2.1 Fast Statistical Alignment (FSA)

Given a set of sequences and a phylogenetic tree with branch-specific pair HMMs (TKF92, MixDom, or distilled order-1 transducers), we construct a multiple sequence alignment using the *sequence annealing* approach of (9), to which we refer the reader for a full description of the algorithm.

Briefly, the method proceeds as follows. For each pair of sequences (x, y) in a selected subset (either all $\binom{N}{2}$ pairs or an $O(N \log N)$ Erdős–Rényi sample), we compute pairwise residue alignment posteriors $P(x_i \sim y_j)$ by running the Forward-Backward algorithm on the pair HMM at an optimized evolutionary time $\hat{\tau}$. The time $\hat{\tau}$ is found by Newton–Raphson optimization of the expected log-likelihood (the “NR step”):

$$\hat{\tau} = \underset{\tau}{\operatorname{argmax}} \mathbb{E}_{P(\pi|x,y,\tau_0)} [\log P(x, y, \pi \mid \tau)] \quad (\text{C.8})$$

where π ranges over alignment paths and τ_0 is an initial estimate. This expectation is computed from Forward-Backward expected counts at τ_0 , and typically converges in 3–5 Newton steps. (This time-maximization differs slightly from the approach of (9) which attempts to optimize all model parameters via unregularized EM for every pair, and consequently must terminate the EM recursion early to avoid instability.)

The pairwise posteriors are then assembled into a multiple alignment by the greedy sequence annealing procedure of (9), which iteratively merges alignment columns to maximize a sum-of-pairs posterior objective.

C.2.2 Beam Search Ancestral Sequence Reconstruction (BeamASR)

We now describe an alternative progressive reconstruction method that finds the maximum-likelihood ancestral sequence at each internal node by beam search, without materializing the full composite automaton.

At each internal node v with children l, r and observed descendant sequences c_l, c_r , we seek

$$\hat{a}_v = \operatorname{argmax}_a [\log P(a, c_l \mid B_l) + \log P(a, c_r \mid B_r) - \log P(a \mid R)] \quad (\text{C.9})$$

where $P(a, c_k \mid B_k)$ is the pair HMM forward probability on branch k and $P(a \mid R)$ is the singlet probability under the root generator, subtracted to avoid double-counting the prior on a .

Incremental forward profiles Since the branches are conditionally independent given the ancestor, we can evaluate (C.9) by maintaining *incremental forward profiles*: for each branch k , a 1D forward table $F_k[i, q]$ giving the log-probability that descendant positions $1, \dots, i$ have been emitted and the branch machine is in state q , given ancestor positions $1, \dots, j$ processed so far.

Each ancestor character extends both profiles independently in $O(L_k)$ time per branch, where $L_k = |c_k|$.

Beam search The ancestor sequence $\hat{a}_v$ is built left-to-right by beam search. At each position j , the beam maintains B candidate partial ancestors. For each candidate and each alphabet character σ :

1. Extend both branch profiles by one ancestor character σ , comprising a *match/delete phase* (the ancestor emits σ , descendant positions advance via M or D transitions) and an *insertion phase* (descendant-only insertions following the ancestor emission).
2. Update the singlet forward score for σ .
3. Score the extension: $\Delta(j, \sigma) = \Delta F_l + \Delta F_r - \Delta_{\text{singlet}}$.

The top B extensions (by cumulative score) are retained. Total cost per node is $O(K \cdot B \cdot A \cdot (L_l + L_r))$ where $K = |\hat{a}_v|$ and A is the alphabet size.

Insertion phase via associative scan The insertion recurrence within each branch profile has the form

$$x_{i+1} = \operatorname{logsumexp}(A_{II} x_i, b_i) + e_i \quad (\text{C.10})$$

where A_{II} is the I-to-I log-transition submatrix, b_i collects transitions into insertion states from M and D, and e_i is the emission score. This is a log-semiring affine recurrence, parallelizable via an associative scan with operator

$$(A_1, b_1) \oplus (A_2, b_2) = (A_2 \otimes A_1, \text{logsumexp}(A_2 b_1, b_2))$$

where $\otimes$ denotes log-semiring matrix multiplication. This reduces the insertion phase from $O(L)$ sequential depth to $O(\log L)$.

Supported model types The beam search interface is generic over the pair HMM used on each branch:

1. **TKF92** — order-0, 5-state pair HMM (the standard model).
2. **MixDom** — the full latent-state pair HMM (Section C.1.1) with $2 + 5NK$ states; latent-state correlations are marginalized in the forward pass without distillation.

Potentials from neighboring columns The inter-column coupling enters through the order-1 WFST transitions. For each edge $e = (u, v)$ and MSA column c where e has an event of type τ_c , let c^- denote the predecessor column (the previous column where e had an event) and c^+ the successor column. The *potential* at column c for edge e receives two contributions:

As-child term (from c^-). The transition from column c^- to c on edge e depends on the characters at both columns. Using the pairwise marginal from q_{c^-} :

$$\log \phi_e^{\text{child}}(a_{u,c}, a_{v,c}) = \sum_{a', b'} q_{c^-}^{(u,v)}(a', b') \log w_e(\tau_{c^-}, \tau_c, a', b', a_{u,c}, a_{v,c}) \quad (\text{C.11})$$

where the sum over (a', b') uses the *joint* pairwise marginal $q_{c^-}^{(u,v)}(a', b')$ —not the product of independent marginals. This is the key advantage over mean-field: the within-tree parent-child correlation at the predecessor column is preserved exactly.

As-parent term (from c^+). Symmetrically, column c acts as the predecessor for column c^+ :

$$\log \phi_e^{\text{parent}}(a_{u,c}, a_{v,c}) = \sum_{a'', b''} q_{c^+}^{(u,v)}(a'', b'') \log w_e(\tau_c, \tau_{c^+}, a_{u,c}, a_{v,c}, a'', b'') \quad (\text{C.12})$$

For insert transitions (only the descendant is present at c), the potential reduces to a per-node function $\psi_v(a_{v,c})$. For delete transitions (only the ancestor is present), it becomes $\psi_u(a_{u,c})$. For match transitions, it contributes a per-edge potential $\phi_e(a_{u,c}, a_{v,c})$. The start and end transitions contribute analogous per-node or per-edge terms.

Felsenstein coordinate ascent Each coordinate ascent step updates q_c for a single MSA column c , holding all other columns fixed. We accumulate, for each edge e and node v in the tree at column c :

- Per-edge log-potentials $\log \phi_e(a_u, a_v) = \log \phi_e^{\text{child}} + \log \phi_e^{\text{parent}}$ (for match transitions where both endpoints are present).

- Per-node log-potentials $\log \psi_v(a_v)$ (from insert/delete transitions on incident edges, plus the root prior $\log \pi(a)$ at the root node).

The optimal q_c , given the potentials, is the Gibbs distribution on the tree at column c :

$$q_c(\mathbf{h}_c) \propto \prod_v \psi_v(a_v) \prod_{e=(u,v)} \phi_e(a_u, a_v) \prod_{\ell \in \text{leaves}} \delta(a_\ell = y_\ell) \quad (\text{C.13})$$

Since this is a tree-structured MRF, the normalizing constant and all node and edge marginals can be computed *exactly* by Felsenstein peeling (postorder) and unpeeling (preorder) in $O(|E| \cdot |\mathcal{A}|^2)$ time.

Peeling (postorder). For each node v in postorder, compute the conditional likelihood:

$$\text{CL}_v(a) = \psi_v(a) \prod_{\text{children } c} \left[\sum_{a_c} \phi_{(v,c)}(a, a_c) \text{CL}_c(a_c) \right] \quad (\text{C.14})$$

with $\text{CL}_\ell(a) = \delta(a = y_\ell)$ for observed leaves. The log-partition function is $\log Z_c = \log \sum_a \pi(a) \text{CL}_{\text{root}}(a)$.

Unpeeling (preorder). Propagate top-down to obtain the posterior marginal at each node:

$$q_c^{(v)}(a) \propto \text{CL}_v(a) \cdot \text{msg}_{\text{parent} \rightarrow v}(a) \quad (\text{C.15})$$

and the pairwise marginal on each edge:

$$q_c^{(u,v)}(a_u, a_v) \propto \text{msg}_{\text{above } u}(a_u) \cdot \phi_{(u,v)}(a_u, a_v) \cdot \text{CL}_v(a_v) \quad (\text{C.16})$$

where the “message from above u ” combines the top-down message to u with u ’s conditional likelihood excluding child v .

Sweep. One iteration sweeps through all MSA columns $c = 1, \dots, L$: for each column, recompute the potentials from the current neighbor marginals, run peeling/unpeeling, and store the updated node and edge marginals. The sweep order is left-to-right; the “as-child” potentials use the just-updated predecessor marginals, while the “as-parent” potentials use stale successor marginals from the previous iteration.

Properties The product-of-trees approximation enjoys the same monotonic convergence guarantee as mean-field coordinate ascent (each column update minimizes the free energy in its coordinate), with the additional guarantee that the ELBO is at least as tight as the fully-factored mean-field bound. This follows because the product-of-trees family *contains* the mean-field family as a special case (where each q_c is itself fully factored).

Computational cost. The dominant cost per sweep is the potential computation: $O(|E| \cdot L \cdot |\mathcal{A}|^4)$ for Match→Match transitions (contracting the $(|\mathcal{A}|, |\mathcal{A}|)$ pairwise marginal against the $(|\mathcal{A}|, |\mathcal{A}|, |\mathcal{A}|, |\mathcal{A}|)$ WFST tensor). The Felsenstein passes add $O(|E| \cdot L \cdot |\mathcal{A}|^2)$, which is subdominant.

C.2.3 Phylogenetic Hidden Markov Model (PhyloHMM)

If the top-level indel rates in MixDom are low, and the ancestral presence/absence fully specified by the MSA, the phylogenetic likelihood calculation and ancestral reconstruction problems admit systematic approximation by a generalized Phylo-HMM, yielding $O(L^2)$ -complexity versions of the Forward and Forward-Backward algorithms. This approach is described in Section C.9.

C.2.4 Phylogenetic composition

The order-1 HMM (Section C.4.5) and transducers (Section C.4.6) can be composed on a phylogenetic tree to yield a single composite machine whose state encodes the joint configuration of all branch machines.

Given a rooted binary tree with n leaves:

- Number nodes $0, \dots, 2n - 2$ in *preorder* (root = 0).
- Place the order-1 Singlet HMM on a notional branch above the root (node 0).
- Place an order-1 Pair Transducer on each real branch.
- Each node v carries a *tag* $\in \mathcal{A} \cup \{\varepsilon\}$, initially ε .

Each branch machine is either the root HMM or a branch transducer. All machines are converted to waiting-machine form (they already are, by construction above).

Composition rules

State constraint. A node’s machine may advance (take a transition) only if all higher-numbered nodes’ machines are in waiting states.

Tagging. When an internal node v ’s transducer takes a transition with output symbol $b \neq \varepsilon$, node v is tagged with b . If v ’s tag is non- ε , the next move must feed v ’s tag as input to both child branch transducers (forcing them out of their waiting states). This clears v ’s tag (but the children’s transitions may tag downstream nodes).

Priority. If multiple nodes are tagged simultaneously, the lowest-numbered (closest to root) tagged node is cleared first.

Cascading. This system of tags allows upward propagation from observed leaf emissions (MSA columns) via Felsenstein-style pruning: working from leaves (known emissions) upward, each internal node’s ancestral character is a latent variable marginalized by the DP.

Composite state space A *configuration* of the composed machine is a tuple

$$\sigma = (q_0, q_1, \dots, q_{2n-2}, \tau_0, \tau_1, \dots, \tau_{2n-2})$$

where q_v is node v ’s machine state and $\tau_v \in \mathcal{A} \cup \{\varepsilon\}$ is its tag. The start configuration has all machines in **S** and all tags ε . The end configuration has all machines in **E** and all tags ε .

Practical caveat. While this composition defines a valid single machine whose language is the set of MSAs weighted by the full phylogenetic likelihood, the composite state space is $O(|Q|^{2n-1} \cdot |\mathcal{A}|^{2n-1})$ where $|Q|$ is the number of states per branch machine—i.e. geometric in the number of taxa. Explicitly constructing the composed machine is therefore impractical for all but the smallest trees. The beam algorithms that follow (Sections C.2.5–C.2.6) avoid this by enumerating only the configurations reachable within a pruned beam, so that the effective state space remains manageable. The composition formalism is nonetheless useful as a *specification*: it defines the target distribution from which the beam search samples or whose expected counts the Forward-Backward algorithm estimates.

C.2.5 Beam Backward algorithm (BeamMSA)

Given a multiple sequence alignment (MSA) with L columns and the composite machine from Section C.2.4, we compute the alignment likelihood using a *beam Backward* algorithm, working from the end configuration backward.

Columns and the emission constraint. Each MSA column $\ell = 1, \dots, L$ specifies, for each leaf v , either a character $y_v^\ell \in \mathcal{A}$ or a gap. A configuration σ is *compatible* with column ℓ if the set of leaf emissions implied by σ 's tags matches the column.

Backward recurrence. Let $B(\sigma)$ denote the Backward variable: the total probability of generating MSA columns $\ell, \ell + 1, \dots, L$ and reaching the end state, given that the composite machine is currently in configuration σ just before column ℓ .

$$B(\sigma_E) = 1 \quad (\text{end configuration}) \quad (\text{C.17})$$

$$B(\sigma) = \sum_{\sigma'} T(\sigma, \sigma') B(\sigma') \quad (\text{all other configurations}) \quad (\text{C.18})$$

where $T(\sigma, \sigma')$ is the composite transition weight (product of individual machine transitions, subject to the composition rules above) and the sum is over all successor configurations σ' .

The alignment likelihood is $B(\sigma_S)$.

Beam pruning. Maintain a beam $\mathcal{B}_\ell$ of at most W configurations per column, ranked by $B(\sigma)$. When expanding $\mathcal{B}_\ell$ from $\mathcal{B}_{\ell+1}$, discard any σ whose $B(\sigma)$ falls below $B_{\max}/\Delta$ where $B_{\max}$ is the current maximum and Δ is the beam width ratio. If the beam collapses (no configurations remain), backtrack.

Epsilon closures within a column. A *column-emitting move* is any transition whose output cascades down the tree to produce a new MSA column: specifically, an insertion (a transition that outputs a character without consuming input), which then tags the node and cascades to its descendants. Between column-emitting moves, machines may make silent transitions (tag propagation, waiting-state transitions, etc.). These form an ε -closure that must be computed at each step. For each configuration in the beam, enumerate all reachable configurations via silent transitions (respecting the priority ordering), accumulating weights multiplicatively along each path. In practice, null cycles (if any) can either be ignored (assuming the model's null-state topology is acyclic) or handled by allowing a configurable number of extra exploratory steps in the beam search.

Forward traceback After the Backward pass reaches σ_S , a stochastic Forward traceback samples a path from σ_S to σ_E :

$$P(\sigma'|\sigma) = \frac{T(\sigma, \sigma') B(\sigma')}{B(\sigma)} \quad (\text{C.19})$$

At each step, sample the next configuration proportional to (C.19). This yields a sampled alignment (including ancestral sequences at internal nodes).

Beam Forward-Backward Alternatively, after the Backward beam pass:

1. Prune dead-end configurations from each $\mathcal{B}_\ell$ (those with no predecessor in $\mathcal{B}_{\ell-1}$).
2. Run a Forward pass over the pruned beam:

$$F(\sigma_s) = 1 \quad (\text{C.20})$$

$$F(\sigma') = \sum_{\sigma \in \mathcal{B}} T(\sigma, \sigma') F(\sigma) \quad (\text{C.21})$$

3. Posterior marginals for any feature ϕ :

$$P(\phi|\text{MSA}) = \frac{1}{B(\sigma_s)} \sum_{\sigma \rightarrow \sigma'} F(\sigma) T(\sigma, \sigma') B(\sigma') [\phi(\sigma, \sigma')] \quad (\text{C.22})$$

Beam Viterbi Replace $\sum$ with $\max$ in the Backward recurrence (C.18) and store argmax pointers:

$$B^V(\sigma_E) = 1 \quad (\text{C.23})$$

$$B^V(\sigma) = \max_{\sigma'} T(\sigma, \sigma') B^V(\sigma') \quad (\text{C.24})$$

The optimal alignment is recovered by Forward traceback following the argmax pointers.

C.2.6 Progressive alignment via profile construction (ProgRec)

We now describe a progressive multiple sequence alignment algorithm using the order-1 machines from Sections C.4.5–C.4.6, with model parameters from (38). This follows the transducer-composition approach of (67), adapted here for Mealy machines (I/O on transitions rather than states).

The antecedents of this approach are the full multidimensional alignment algorithm of (24) which computes the Forward algorithm for TKF91 on a binary tree. This may be seen as unifying the tree-based Viterbi multiple alignment approach of (54) with the statistical phylogenetics of (22). The approach described here also maintains a partial order graph of intermediate alignments (41), which essentially is the approach used by (44).

Recognizers and Profiles Let the phylogenetic tree have n leaves with observed sequences $\{y_v : v \in \text{leaves}\}$, nodes numbered in preorder. Let R denote the order-1 Singlet HMM (root generator, Section C.4.5) and B_v the order-1 Pair Transducer on the branch to node v (Section C.4.6).

Recognizers. For each leaf v , the *exact-match recognizer* $\mathcal{R}(y_v)$ is a transducer with empty output alphabet that accepts only y_v : it has $|y_v| + 1$ states (positions $0, \dots, |y_v|$), with a single input-consuming transition $i \rightarrow i + 1$ labeled by $y_v[i + 1]$ at each position. All states are waiting states except the start.

Profiles. A *profile* E_v at node v is a recognizer (empty output alphabet) that accepts a set of plausible ancestral sequences at v , weighted by their approximate posterior probability given the descendants of v . For leaves, $E_v = \mathcal{R}(y_v)$.

Progressive reconstruction Working from the leaves toward the root, for each internal node v with children l, r :

Step 1: Compose branch and profile. For each child $c \in \{l, r\}$, form the composition $B_c \circ E_c$, which is a transducer mapping the sequence at v to the constrained sequences at child c . Since E_c has empty output, $B_c \circ E_c$ is a recognizer (it reads a candidate parent sequence and recognizes it with weight proportional to the probability of generating the descendant data at c).

Step 2: Intersect siblings. Form the intersection

$$H_v = (B_l \circ E_l) \cap (B_r \circ E_r)$$

This recognizer reads a candidate sequence at v and scores it by the joint probability of both children's descendant data, given that parent sequence. In the Mealy-machine intersection, the composite state is (q_l, e_l, q_r, e_r) where $q_c \in B_c$ and $e_c \in E_c$. Both sides must agree on the same input symbol when both are ready (waiting); when one side is not waiting, it advances silently while the other stays put.

Step 3: Compose with root prior. Form the generator

$$M_v = R \circ H_v$$

This has empty input and empty output: it is a weighted automaton over the empty string, whose total weight $Z = \sum_{\pi} w(\pi)$ over all paths π is the marginal likelihood of the descendant data below v (under the stationary prior R at v).

Step 4: Sample paths and construct profile. Sample K paths from $P(\pi|M_v) = w(\pi)/Z$ using a Forward pass followed by stochastic traceback (C.19). For each sampled path π , extract the H_v -component states visited. The profile E_v is the sub-recognizer of H_v containing exactly those states visited by at least $\tau \geq 2$ of the K sampled paths. This bounds $|E_v| = O(KL)$ where $L = \max_v |y_v|$.

Mealy-machine composition and intersection For completeness, we state the composition and intersection rules for Mealy machines in waiting-machine normal form ("ready" states = waiting states with input-consuming transitions only; "unready" states = non-waiting, silent transitions only).

Composition. Given $T = (\Omega_X, \Omega_Y, \dots)$ and $U = (\Omega_Y, \Omega_Z, \dots)$ in Mealy normal form, $T \circ U$ has states $\subseteq \mathcal{S}_T \times \mathcal{S}_U$ with transition weight:

$$w''((t, u), \omega_x, \omega_z, (t', u')) = \begin{cases} \delta_{tt'} \delta_{\omega_x \varepsilon} w'(u, \varepsilon, \omega_z, u') & \text{if } u \text{ unready} \\ \delta_{uu'} \delta_{\omega_z \varepsilon} w(t, \omega_x, \varepsilon, t') + \sum_{\omega_y} w(t, \omega_x, \omega_y, t') w'(u, \omega_y, \omega_z, u') & \text{if } u \text{ ready} \end{cases}$$

Intersection. Given $T = (\Omega_X, \Omega_T, \dots)$ and $U = (\Omega_X, \Omega_U, \dots)$ in Mealy normal form, $T \cap U$ has states $\subseteq \mathcal{S}_T \times \mathcal{S}_U$ with output alphabet $\Omega_T \times \Omega_U$ and transition weight:

$$w''((t, u), \omega_x, (\omega_y, \omega_z), (t', u')) = \begin{cases} \delta_{tt'} \delta_{\omega_x \varepsilon} \delta_{\omega_y \varepsilon} w'(u, \varepsilon, \omega_z, u') & \text{if } u \text{ unready} \\ \delta_{uu'} \delta_{\omega_x \varepsilon} \delta_{\omega_z \varepsilon} w(t, \varepsilon, \omega_y, t') & \text{if } t \text{ unready, } u \text{ ready} \\ w(t, \omega_x, \omega_y, t') w'(u, \omega_x, \omega_z, u') & \text{if both ready} \end{cases}$$

Forward recursion for M_v The generator $M_v = R \circ H_v$ has states $m = (\rho, q_l, e_l, q_r, e_r)$ where $\rho \in R$, $(q_l, e_l) \in B_l \circ E_l$, and $(q_r, e_r) \in B_r \circ E_r$. The Forward variable $Z(m)$ satisfies:

$$Z(\mathbf{S}_M) = 1 \tag{C.25}$$

$$Z(m') = \sum_{m: (m, \varepsilon, \varepsilon, m') \in \mathcal{T}} w(m, \varepsilon, \varepsilon, m') Z(m) \tag{C.26}$$

where $\mathcal{T}$ is the transition set of M_v . The total likelihood is $Z(\mathbf{E}_M)$.

The fill order iterates over e_l and e_r in topological order (corresponding to positions in the child profiles), with an inner loop over $Q_v = R \circ (B_l \cap B_r)$ states (the “comparison kernel” Pair HMM). This has time complexity $O(|B|^2 |E_l| |E_r|)$ per internal node, where $|B|$ is the branch transducer state count.

Profile extraction Given the Forward table Z , sample paths $\pi^{(1)}, \dots, \pi^{(K)}$ from M_v using the stochastic traceback (C.19). For each path $\pi^{(k)}$, let $\mathcal{H}^{(k)} = \{(q_l, e_l, q_r, e_r) : (\rho, q_l, e_l, q_r, e_r) \in \pi^{(k)}\}$ be the H_v -states visited.

The profile E_v is the sub-automaton of H_v induced by the states

$$\mathcal{S}_{E_v} = \{h \in H_v : |\{k : h \in \mathcal{H}^{(k)}\}| \geq \tau\}$$

with the same transition weights as H_v , restricted to $\mathcal{S}_{E_v}$. Adding appropriate wait states places E_v in Mealy normal form.

Bubble merging. Paths through E_v that traverse the same sequence of wait states but differ only in latent-state assignments can be merged by collapsing bubbles (identifying states with identical incoming and outgoing wait-state connectivity). This further compresses the profile without changing the recognized language.

MSA extraction A sampled path through M_1 (the root) determines, at each position, which H_1 -state is visited, and therefore which states of E_l, E_r are aligned. Recursing into the child profiles yields a full column assignment for all leaves.

Specifically, each emitting transition in the sampled path implies:

- a character at the current node (from the root generator or branch match),
- advancement of the left profile, right profile, or both,
- and therefore a column in the MSA (with gaps for non-advancing sides).

When bubble merging has been applied, the canonical path (chosen during merging) is used to resolve any ambiguity in the sub-alignment of the clade below the merged bubble.

Viterbi Progressive Reconstruction An alternative to the sampling-based profile construction of Step 4 above is to use Viterbi decoding at each internal node. Instead of sampling K paths from M_v and building a multi-path profile, the Viterbi variant computes a single maximum-likelihood path through M_v and uses the resulting ancestral sequence directly as the reconstructed sequence at node v . This gives a deterministic progressive reconstruction that avoids the $O(KL)$ profile size but sacrifices the ability to represent uncertainty in the ancestral sequence.

Table C.1: Exploded MixDom Pair HMM state space.

Category	States	Count
Start/End	S, E	2
Domain-level (top-level TKF91 states)	MatDom, InsDom, DelDom, MatDomEnd, InsDomEnd, DelDomEnd	6
Domain type selection (one per domain type k)	MatDomType[k], InsDomType[k], DelDomType[k]	$3\mathcal{N}$
Fragment-level (inner TKF states within MatDomType[k])	MatFrag[k], InsFrag[k], DelFrag[k]	$3\mathcal{N}$
Fragment-level (single looping state within InsDomType[k], DelDomType[k])	IFrag[k], DFrag[k]	$2\mathcal{N}$
Fragment type selection (one per fragment type f)	MatFragType[k, f], InsFragType[k, f], DelFragType[k, f], IFragType[k, f], DFragType[k, f]	$5\mathcal{N}\mathcal{F}$
Emit states (the only emitting states)	MatEmit[k, f], InsEmit[k, f], DelEmit[k, f], IEmit[k, f], DEmit[k, f]	$5\mathcal{N}\mathcal{F}$
Fragment end (fragment termination)	MatFragEnd[k, f], InsFragEnd[k, f], DelFragEnd[k, f], IFragEnd[k, f], DFragEnd[k, f]	$5\mathcal{N}\mathcal{F}$
Total		$8 + 8\mathcal{N} + 15\mathcal{N}\mathcal{F}$

C.3 Exploded MixDom Pair HMM

C.3.1 State Space

The exploded MixDom Pair HMM makes every structural decision explicit as a separate state transition. Let $\mathcal{N}$ denote the number of domain types, $\mathcal{F}$ the number of fragment types per domain. Parameters are indexed by domain type k and fragment type f .

The states are shown in Table C.1 (emitting states marked with $\star$).

The emitting states correspond to the compound states of the collapsed model: $\text{MatEmit}[k, f] = \text{MM}_{kf}$, $\text{InsEmit}[k, f] = \text{MI}_{kf}$, $\text{DelEmit}[k, f] = \text{MD}_{kf}$, $\text{IEmit}[k, f] = \text{II}_{kf}$, $\text{DEmit}[k, f] = \text{DD}_{kf}$.

C.3.2 Transition Weights

All transitions are between non-emitting states, or from non-emitting to emitting, or from emitting to non-emitting (Mealy machine: emissions occur on the transitions into emit states).

We use BDI parameters for two TKF91 processes:

- Top-level (domain sequence): $\alpha_0, \beta_0, \gamma_0, \kappa_0$ from (λ_0, μ_0, T)
- Per-domain k (fragment sequence): $\alpha_k, \beta_k, \gamma_k, \kappa_k$ from (λ_k, μ_k, T)

Top-level transitions These implement the TKF91 Pair HMM structure with MatDom/InsDom/DelDom/E (for incoming connections) and S/MatDomEnd/InsDomEnd/DelDomEnd (for outgoing connections)

playing the roles of S/M/I/D/E:

$$S \rightarrow \text{MatDom} : \tau_{\text{SM}}(\lambda_0, \mu_0, T) \quad (\text{C.27})$$

$$S \rightarrow \text{InsDom} : \tau_{\text{SI}}(\lambda_0, \mu_0, T) \quad (\text{C.28})$$

$$S \rightarrow \text{DelDom} : \tau_{\text{SD}}(\lambda_0, \mu_0, T) \quad (\text{C.29})$$

$$S \rightarrow E : \tau_{\text{SE}}(\lambda_0, \mu_0, T) \quad (\text{C.30})$$

$$\text{MatDomEnd} \rightarrow \text{MatDom} : \tau_{\text{MM}}(\lambda_0, \mu_0, T) \quad (\text{C.31})$$

$$\text{MatDomEnd} \rightarrow \text{InsDom} : \tau_{\text{MI}}(\lambda_0, \mu_0, T) \quad (\text{C.32})$$

$$\text{MatDomEnd} \rightarrow \text{DelDom} : \tau_{\text{MD}}(\lambda_0, \mu_0, T) \quad (\text{C.33})$$

$$\text{MatDomEnd} \rightarrow E : \tau_{\text{ME}}(\lambda_0, \mu_0, T) \quad (\text{C.34})$$

$$\text{InsDomEnd} \rightarrow \text{MatDom} : \tau_{\text{IM}}(\lambda_0, \mu_0, T) \quad (\text{C.35})$$

$$\text{InsDomEnd} \rightarrow \text{InsDom} : \tau_{\text{II}}(\lambda_0, \mu_0, T) \quad (\text{C.36})$$

$$\text{InsDomEnd} \rightarrow \text{DelDom} : \tau_{\text{ID}}(\lambda_0, \mu_0, T) \quad (\text{C.37})$$

$$\text{InsDomEnd} \rightarrow E : \tau_{\text{IE}}(\lambda_0, \mu_0, T) \quad (\text{C.38})$$

$$\text{DelDomEnd} \rightarrow \text{MatDom} : \tau_{\text{DM}}(\lambda_0, \mu_0, T) \quad (\text{C.39})$$

$$\text{DelDomEnd} \rightarrow \text{InsDom} : \tau_{\text{DI}}(\lambda_0, \mu_0, T) \quad (\text{C.40})$$

$$\text{DelDomEnd} \rightarrow \text{DelDom} : \tau_{\text{DD}}(\lambda_0, \mu_0, T) \quad (\text{C.41})$$

$$\text{DelDomEnd} \rightarrow E : \tau_{\text{DE}}(\lambda_0, \mu_0, T) \quad (\text{C.42})$$

Domain type selection

$$\text{MatDom} \rightarrow \text{MatDomType}[k] : v_k \quad (\text{domain weight}) \quad (\text{C.43})$$

$$\text{InsDom} \rightarrow \text{InsDomType}[k] : v_k \quad (\text{C.44})$$

$$\text{DelDom} \rightarrow \text{DelDomType}[k] : v_k \quad (\text{C.45})$$

Domain-to-fragment entry (M-type domains) Within $\text{MatDomType}[k]$, the fragment-level TKF91 begins. Again, this follows the TKF91 Pair HMM structure, now with $\text{MatFrag}[k]/\text{InsFrag}[k]/\text{DelFrag}[k]$ (for incoming connections) and $\text{MatDomType}[k]/\text{MatFragEnd}[k, /]/\text{InsFragEnd}[k, /]/\text{DelFragEnd}[k, /]$ states

(for outgoing connections) playing the roles of S/M/I/D/E:

$$\text{MatDomType}[k] \rightarrow \text{MatFrag}[k] : \tau_{\text{SM}}(\lambda_k, \mu_k, T) \quad (\text{C.46})$$

$$\text{MatDomType}[k] \rightarrow \text{InsFrag}[k] : \tau_{\text{SI}}(\lambda_k, \mu_k, T) \quad (\text{C.47})$$

$$\text{MatDomType}[k] \rightarrow \text{DelFrag}[k] : \tau_{\text{SD}}(\lambda_k, \mu_k, T) \quad (\text{C.48})$$

$$\text{MatDomType}[k] \rightarrow \text{MatDomEnd} : \tau_{\text{SE}}(\lambda_k, \mu_k, T) \quad (\text{C.49})$$

$$\text{MatFragEnd}[k,] \rightarrow \text{MatFrag}[k] : \tau_{\text{MM}}(\lambda_k, \mu_k, T) \quad (\text{C.50})$$

$$\text{MatFragEnd}[k,] \rightarrow \text{InsFrag}[k] : \tau_{\text{MI}}(\lambda_k, \mu_k, T) \quad (\text{C.51})$$

$$\text{MatFragEnd}[k,] \rightarrow \text{DelFrag}[k] : \tau_{\text{MD}}(\lambda_k, \mu_k, T) \quad (\text{C.52})$$

$$\text{MatFragEnd}[k,] \rightarrow \text{MatDomEnd} : \tau_{\text{ME}}(\lambda_k, \mu_k, T) \quad (\text{C.53})$$

$$\text{InsFragEnd}[k,] \rightarrow \text{MatFrag}[k] : \tau_{\text{IM}}(\lambda_k, \mu_k, T) \quad (\text{C.54})$$

$$\text{InsFragEnd}[k,] \rightarrow \text{InsFrag}[k] : \tau_{\text{II}}(\lambda_k, \mu_k, T) \quad (\text{C.55})$$

$$\text{InsFragEnd}[k,] \rightarrow \text{DelFrag}[k] : \tau_{\text{ID}}(\lambda_k, \mu_k, T) \quad (\text{C.56})$$

$$\text{InsFragEnd}[k,] \rightarrow \text{MatDomEnd} : \tau_{\text{IE}}(\lambda_k, \mu_k, T) \quad (\text{C.57})$$

$$\text{DelFragEnd}[k,] \rightarrow \text{MatFrag}[k] : \tau_{\text{DM}}(\lambda_k, \mu_k, T) \quad (\text{C.58})$$

$$\text{DelFragEnd}[k,] \rightarrow \text{InsFrag}[k] : \tau_{\text{DI}}(\lambda_k, \mu_k, T) \quad (\text{C.59})$$

$$\text{DelFragEnd}[k,] \rightarrow \text{DelFrag}[k] : \tau_{\text{DD}}(\lambda_k, \mu_k, T) \quad (\text{C.60})$$

$$\text{DelFragEnd}[k,] \rightarrow \text{MatDomEnd} : \tau_{\text{DE}}(\lambda_k, \mu_k, T) \quad (\text{C.61})$$

The $\text{MatDomType}[k] \rightarrow \text{MatDomEnd}$ transition is the “phantom” null path: the domain is entered but the inner model immediately terminates with no fragments emitted. This leads to null cycles that must be eliminated by Schur complement.

Domain-to-fragment entry (I/D-type domains) $\text{InsDomType}[k]$ and $\text{DelDomType}[k]$ have a single looping fragment state:

$$\text{InsDomType}[k] \rightarrow \text{IFrag}[k] : \kappa_k \quad (\text{C.62})$$

$$\text{InsDomType}[k] \rightarrow \text{InsDomEnd} : 1 - \kappa_k \quad (\text{C.63})$$

$$\text{DelDomType}[k] \rightarrow \text{DFrag}[k] : \kappa_k \quad (\text{C.64})$$

$$\text{DelDomType}[k] \rightarrow \text{DelDomEnd} : 1 - \kappa_k \quad (\text{C.65})$$

Again, the $\rightarrow \text{InsDomEnd}/\text{DelDomEnd}$ transitions are null (empty domain).

Fragment type selection

$$\text{MatFrag}[k] \rightarrow \text{MatFragType}[k, f] : w_{kf} \quad (\text{fragment weight}) \quad (\text{C.66})$$

$$\text{InsFrag}[k] \rightarrow \text{InsFragType}[k, f] : w_{kf} \quad (\text{C.67})$$

$$\text{DelFrag}[k] \rightarrow \text{DelFragType}[k, f] : w_{kf} \quad (\text{C.68})$$

$$\text{IFrag}[k] \rightarrow \text{IFragType}[k, f] : w_{kf} \quad (\text{C.69})$$

$$\text{DFrag}[k] \rightarrow \text{DFragType}[k, f] : w_{kf} \quad (\text{C.70})$$

Fragment emission These are the only transitions with emissions (the exploded HMM is represented as a Mealy machine, so emissions occur on transitions; the collapsed HMM treats emissions

as state-based). The emission probability is summed over site classes $c \in \{1, \dots, C\}$, weighted by the per-fragment class distribution u_{kfc} :

$$\text{MatFragType}[k, f] \xrightarrow{(a,b)} \text{MatEmit}[k, f] : \sum_{c=1}^C u_{kfc} \pi_a^{(c)} \exp(R^{(c)}T)_{ab} \quad (\text{align } (a, b)) \quad (\text{C.71})$$

$$\text{InsFragType}[k, f] \xrightarrow{(\epsilon, b)} \text{InsEmit}[k, f] : \sum_{c=1}^C u_{kfc} \pi_b^{(c)} \quad (\text{insert } b) \quad (\text{C.72})$$

$$\text{DelFragType}[k, f] \xrightarrow{(a, \epsilon)} \text{DelEmit}[k, f] : \sum_{c=1}^C u_{kfc} \pi_a^{(c)} \quad (\text{delete } a) \quad (\text{C.73})$$

$$\text{IFragType}[k, f] \xrightarrow{(\epsilon, b)} \text{IEmit}[k, f] : \sum_{c=1}^C u_{kfc} \pi_b^{(c)} \quad (\text{insert } b) \quad (\text{C.74})$$

$$\text{DFragType}[k, f] \xrightarrow{(a, \epsilon)} \text{DEmit}[k, f] : \sum_{c=1}^C u_{kfc} \pi_a^{(c)} \quad (\text{delete } a) \quad (\text{C.75})$$

Intra-fragment fragment-type transition vs. fragment termination Within each fragment of domain k the fragment-type process is a Markov chain on $\mathcal{F}+2$ states (start, end, and $\mathcal{F}$ fragment-type states): from the current emit state with fragment-type f , the chain either advances within the fragment to fragment-type g with probability $r_{fg}^{(k)}$, or terminates the fragment (transition to the end state) with probability $\rho_f^{(k)} = 1 - \sum_g r_{fg}^{(k)}$. Different fragments are statistically independent realisations of this chain. The transition from each emit state goes to any fragment-type's type-selection state within the current fragment (not just the same type), or to the fragment end:

$$\text{MatEmit}[k, f] \rightarrow \text{MatFragType}[k, g] : r_{fg}^{(k)} \quad (\text{intra-fragment type transition } f \rightarrow g) \quad (\text{C.76})$$

$$\text{MatEmit}[k, f] \rightarrow \text{MatFragEnd}[k, f] : \rho_f^{(k)} \quad (\text{fragment termination}) \quad (\text{C.77})$$

$$\text{InsEmit}[k, f] \rightarrow \text{InsFragType}[k, g] : r_{fg}^{(k)} \quad (\text{C.78})$$

$$\text{InsEmit}[k, f] \rightarrow \text{InsFragEnd}[k, f] : \rho_f^{(k)} \quad (\text{C.79})$$

$$\text{DelEmit}[k, f] \rightarrow \text{DelFragType}[k, g] : r_{fg}^{(k)} \quad (\text{C.80})$$

$$\text{DelEmit}[k, f] \rightarrow \text{DelFragEnd}[k, f] : \rho_f^{(k)} \quad (\text{C.81})$$

$$\text{IEmit}[k, f] \rightarrow \text{IFragType}[k, g] : r_{fg}^{(k)} \quad (\text{C.82})$$

$$\text{IEmit}[k, f] \rightarrow \text{IFragEnd}[k, f] : \rho_f^{(k)} \quad (\text{C.83})$$

$$\text{DEmit}[k, f] \rightarrow \text{DFragType}[k, g] : r_{fg}^{(k)} \quad (\text{C.84})$$

$$\text{DEmit}[k, f] \rightarrow \text{DFragEnd}[k, f] : \rho_f^{(k)} \quad (\text{C.85})$$

where g ranges over all $\mathcal{F}$ fragment types. For $\mathcal{F} = 1$, this reduces to a scalar self-extension with $r_{11}^{(k)} = r_{k1}$ and $\rho_1^{(k)} = 1 - r_{k1}$.

C.3.3 Null State Classification

Every state except **S**, **E**, and the five emit states $\text{MatEmit}[k, f]$, $\text{InsEmit}[k, f]$, $\text{DelEmit}[k, f]$, $\text{IEmit}[k, f]$, $\text{DEmit}[k, f]$ is a **null state** (non-emitting). The collapsed model retains only **S**, **E**, and the $5\mathcal{NF}$ emit states.

C.3.4 Null Elimination

The null states are eliminated by the standard HMM null closure:

$$\chi_{\text{emit,emit}} = T_{\text{emit,emit}} + T_{\text{emit,null}}(I - T_{\text{null,null}})^{-1}T_{\text{null,emit}}$$

Since there are no direct emit $\rightarrow$ emit transitions in the exploded model (every path between emit states passes through at least one null state), $T_{\text{emit,emit}} = 0$ and:

$$\chi = T_{\text{emit,null}}(I - T_{\text{null,null}})^{-1}T_{\text{null,emit}}$$

This gives the collapsed χ matrix with states $\{\mathbf{S}, \mathbf{E}, \mathbf{MM}_{kf}, \mathbf{MI}_{kf}, \mathbf{MD}_{kf}, \mathbf{II}_{kf}, \mathbf{DD}_{kf}\}$, matching the collapsed MixDom Pair HMM in Section C.1.1.

Each state in the $(5\mathcal{NF}+2)$ -state collapsed HMM corresponds to an uneliminated state in the $(15\mathcal{NF}+8\mathcal{N}+8)$ -state exploded HMM: either $\mathbf{S}$ (SS), $\mathbf{E}$ (EE), or one of the emit states $\mathbf{MatEmit}[l, f]$ ($\mathbf{MM}_{lf}$), $\mathbf{InsEmit}[l, f]$ ($\mathbf{MI}_{lf}$), $\mathbf{DelEmit}[l, f]$ ($\mathbf{MD}_{lf}$), $\mathbf{IEmit}[l, f]$ ($\mathbf{II}_{lf}$), $\mathbf{DEmit}[l, f]$ ($\mathbf{DD}_{lf}$). The transition weight from $\mathbf{UX}_{lf}$ to $\mathbf{VY}_{mg}$ in the collapsed Pair HMM has the form $p_{\text{out}}(\mathbf{U}, \mathbf{X}, l, f) \times \mathcal{T}_{\mathbf{UV}}(\mathbf{U}, \mathbf{V}) \times p_{\text{in}}(\mathbf{V}, \mathbf{Y}, m, g) + \delta_{\mathbf{UV}}\delta(l = m)(p_{\text{SameDom}}(\mathbf{U}, \mathbf{X}, l, f, g) + \delta_{\mathbf{XY}}p_{\text{SameFrag}}(l, f, g))$ corresponding to the following path segments

- $p_{\text{out}}(\mathbf{U}, \mathbf{X}, l, f)$ represents transitions from the emit state to the end state of the domain, e.g. $\tau_{\mathbf{ME}}(\mathbf{M}, \mathbf{I}, l, f)$ represents $\mathbf{InsEmit}[l, f] \rightarrow \mathbf{InsFragEnd}[l, f] \rightarrow \mathbf{MatDomEnd}$;
- $\mathcal{T}_{\mathbf{UV}}(\mathbf{U}, \mathbf{V})$ represents the sum over all paths from the domain end state (or $\mathbf{S}$), through zero or more empty domains, to the next (nonempty) domain start (or $\mathbf{E}$), e.g. $\tau_{\mathbf{MD}}(\mathbf{M}, \mathbf{D})$ represents paths like $\mathbf{MatDomEnd} \rightarrow (\dots \mathbf{InsDom} \rightarrow \mathbf{InsDomType}[\mathcal{N}'] \rightarrow \mathbf{InsDomEnd} \dots)^* \rightarrow \mathbf{DelDom}$. This is where null cycle elimination happens;
- $p_{\text{in}}(\mathbf{V}, \mathbf{Y}, m, g)$ represents paths from the domain start state to an emit state inside the domain, e.g. $\tau_{\mathbf{SM}}(\mathbf{M}, \mathbf{I}, m, g)$ represents $\mathbf{MatDom} \rightarrow \mathbf{MatDomType}[m] \rightarrow \mathbf{InsFrag}[m] \rightarrow \mathbf{InsFragType}[m, g]$;
- $p_{\text{SameDom}}(\mathbf{X}, \mathbf{Y}, l, f, g)$ represents paths from the emit state to another emit state of similar profile but (potentially) different fragment type within the same domain, e.g. $p_{\text{SameDom}}(\mathbf{M}, \mathbf{I}, l, f, g)$ represents $\mathbf{InsEmit}[l, f] \rightarrow \mathbf{MatDomEnd} \rightarrow \mathbf{MatDom} \rightarrow \mathbf{MatDomType}[l] \rightarrow \mathbf{InsFrag}[l] \rightarrow \mathbf{InsFragType}[l, g]$;
- $p_{\text{SameFrag}}(l, f, g) = r_{fg}^{(l)}$ represents intra-fragment Markov transitions between fragment-types (extending the current fragment by one position), e.g. $\mathbf{MatEmit}[l, f] \rightarrow \mathbf{MatFragType}[l, g]$ with weight $r_{fg}^{(l)}$. Different fragments are independent; the Markov structure is strictly *within* a single fragment, allowing the fragment-type to change at each position (not just self-loop).

C.3.5 Exact Count Restoration

Given Forward-Backward expected transition counts $\hat{n}_\chi(i, j)$ on the collapsed model, we recover the expected counts on the exploded model using the null closure inverse.

Define:

$$C = (I - T_{\text{null,null}})^{-1} \quad (\text{null closure}) \tag{C.86}$$

$$C_{ab} = \text{expected visits to null state } b, \text{ starting from null state } a \tag{C.87}$$

For each collapsed transition $\hat{n}_\chi(s, s')$ from emit state s to emit state s' , the path in the exploded model is:

$$s \xrightarrow{1} \mathbf{FragEnd}(s) \xrightarrow{\text{null chain}} \mathbf{FragType}(s') \xrightarrow{1} s'$$

The null chain from $\text{FragEnd}(s)$ to $\text{FragType}(s')$ passes through a sequence of null states. The expected count for each null transition ($a \rightarrow b$) along this chain is:

$$\hat{n}_{\text{exploded}}(a, b) = \sum_{s, s'} \hat{n}_{\chi}(s, s') \cdot \frac{T_{\text{emit}, a'} \cdot C_{a', a} \cdot T_{a, b} \cdot C_{b, b'} \cdot T_{b', \text{emit}'}}{\chi(s, s')} \quad (\text{C.88})$$

where a' is the first null state entered from s , and b' is the last null state before reaching s' .

More explicitly, each collapsed transition decomposes into contributions to the following parameter groups.

Intra-fragment type transitions vs. new-fragment transitions For a transition $\hat{n}_{\chi}(s, s')$ where $s = \text{Emit}_{kf}^X$ and $s' = \text{Emit}_{kg}^Y$ with the same domain k , the path splits into:

- **Intra-fragment fragment-type transition:** $s \rightarrow \text{FragType}_{kg} \rightarrow s'$, with weight $r_{fg}^{(k)}$. This applies to all transitions where $X = Y$ (same TKF state type) and allows $f \neq g$. It extends the current fragment by one position without generating a new TKF92 link.
- **New fragment via domain loop:** $s \rightarrow \text{FragEnd}_{kf} \rightarrow \dots \rightarrow \text{Frag}_k \rightarrow \text{FragType}_{kg} \rightarrow s'$, with weight $\rho_f^{(k)} \cdot \tau_k[X, Y] \cdot w_{kg}$ (for M-type) or $\rho_f^{(k)} \cdot \kappa_k \cdot w_{kg}$ (for I/D-type). This terminates the current fragment and initiates an independent fresh fragment via the TKF92 process.

The expected intra-fragment fragment-type transition count from f to g is:

$$\hat{n}_{\text{ext}}(k, f, g) = \hat{n}_{\chi}(s, s') \cdot \frac{r_{fg}^{(k)}}{r_{fg}^{(k)} + \rho_f^{(k)} \cdot p_{\text{new}, g}} \quad (\text{C.89})$$

where $p_{\text{new}, g}$ is the new-fragment-to-type- g probability. These counts form an $\mathcal{F} \times \mathcal{F}$ matrix per domain, and the M-step row-normalizes ($\hat{n}_{\text{ext}}(k, f, g), \hat{n}_{\rho}(k, f)$) to obtain the updated $r_{fg}^{(k)}$.

Intra-domain TKF transitions Each new-fragment transition (after fragment termination) contributes one TKF transition at the domain level: $\tau_k[X, Y]$ for M-type domains, or $\kappa_k / (1 - \kappa_k)$ for I/D-type domains.

These counts go into the domain- k TKF91 count matrix (for M-type) or the $\kappa_k / (1 - \kappa_k)$ accumulators (for I/D-type).

Domain entry/exit and phantom counts Inter-domain transitions pass through $\text{MatDomEnd}/\text{InsDomEnd}/\text{DelDom}$ (exit from source domain) and $\text{MatDom}/\text{InsDom}/\text{DelDom}$ then $\text{MatDomType}[k]$ (entry to destination domain).

Within the entry, the path $\text{MatDomType}[k] \rightarrow \text{MatFrag}[k]$ uses $\tau_k[\mathbf{S}, \cdot]$. The phantom path $\text{MatDomType}[k] \rightarrow \text{MatDomEnd}$ has probability $\tau_k[\mathbf{S}, \mathbf{E}] = (1 - \beta_k)(1 - \kappa_k)$ and contributes a phantom birth-death event to domain k 's BDI statistics.

Similarly for I/D-type entries: $\text{InsDomType}[k] \rightarrow \text{InsDomEnd}$ with probability $(1 - \kappa_k)$ is a phantom I/D-type domain.

Top-level TKF transitions Each inter-domain transition contributes one TKF transition at the top level: $\tau_0[U, V]$ where $U \in \{\mathbf{S}, \mathbf{M}, \mathbf{I}, \mathbf{D}\}$ is the domain-end type and V is the domain-start type.

The null domain paths (empty domains via $\text{MatDomType}[k] \rightarrow \text{MatDomEnd}$) contribute additional phantom top-level transitions via the null closure $(I - T_{\text{null}, \text{null}})^{-1}$.

Domain and fragment weight counts Each domain entry contributes one count to v_k (domain weight). Each fragment-type selection contributes one count to w_{kf} (fragment weight).

C.3.6 Parameter Group Decomposition

Each transition in the exploded model involves exactly one of the following parameter factors:

Parameter	Factor	Where it appears
α_0	$(1 - \beta_0)\kappa_0\alpha_0$	Top-level $\rightarrow$ MatDom
$1 - \alpha_0$	$(1 - \beta_0)\kappa_0(1 - \alpha_0)$	Top-level $\rightarrow$ DelDom
β_0	β_0	Top-level $\rightarrow$ InsDom
$1 - \beta_0$	$(1 - \beta_0)$	Top-level $\rightarrow$ MatDom, DelDom, E
γ_0	γ_0	DelDomEnd $\rightarrow$ InsDom
$1 - \gamma_0$	$(1 - \gamma_0)$	DelDomEnd $\rightarrow$ MatDom, DelDom, E
κ_0	κ_0	Top-level $\rightarrow$ MatDom, DelDom
$1 - \kappa_0$	$(1 - \kappa_0)$	Top-level $\rightarrow$ E
α_k	$(1 - \beta_k)\kappa_k\alpha_k$	Domain- k $\rightarrow$ MatFrag
β_k	β_k	Domain- k $\rightarrow$ InsFrag
γ_k	γ_k	Domain- k DelFragEnd $\rightarrow$ InsFrag
κ_k	κ_k	Domain- k $\rightarrow$ MatFrag/DelFrag, I/D-type continuation
$1 - \kappa_k$	$(1 - \kappa_k)$	Domain- k $\rightarrow$ DomEnd
$r_{fg}^{(k)}$	$r_{fg}^{(k)}$	Intra-fragment fragment-type transition $f \rightarrow g$
$\rho_f^{(k)}$	$1 - \sum_g r_{fg}^{(k)}$	Fragment termination
v_k	v_k	Domain type selection
w_{kf}	w_{kf}	Fragment type selection

Because each exploded transition involves a product of these factors, and each factor’s log depends on at most one natural parameter (λ_k or μ_k), the Q-function on the exploded model decomposes into independent BDI score terms. The null-state count restoration maps collapsed counts exactly onto exploded counts, allowing the M-step to decompose into the same parameter-group updates used in the component TKF91/TKF92 models, together with standard mixture-weight updates.

C.4 Order-1 Maraschino: Distilled Adjacency Frequencies

“Cherries” are pairwise training examples chosen, in place of full phylogenetically-annotated multiple sequence alignments, as a composite likelihood approximation to the full phylogenetic likelihood (50).

“Maraschino Cherries” are order-1 counts tensors that summarize the adjacency statistics of such pairwise alignments. MixDom’s Maraschino Cherries generalize CherryML to include context-dependent substitution and indel patterns (50).

The Maraschino pipeline has two phases. First (Section C.4.1), pairwise alignments are reduced to fixed-shape *cherry-count* tensors that aggregate adjacency statistics binned by divergence time. Second (Section C.4.2), the parameters of the MixDom Pair HMM (Section C.1.1) are estimated by maximizing the cherry-count log-likelihood: a composite likelihood that scores each adjacency in each time bin under the latent-marginalized collapsed Pair HMM transition matrix $\chi(\theta, T)$ derived in Section C.1.1. After fitting, the MixDom model is then distilled to compact order-1 machines (an HMM and a WFST) suitable for use in tree algorithms (Sections C.4.5 and C.4.6).

C.4.1 Cherry-count summary statistics

The input to the Maraschino fitter is a precomputed tensor of pairwise adjacency counts. For each multiple sequence alignment, sibling pairs are extracted, gapped columns dropped, and the resulting pairwise alignment classified into *adjacency contexts*: for every pair of consecutive non-empty alignment columns, we record the source column type (S, M, I, D), the destination column type (M, I, D, E), and the ancestor/descendant characters in each.

The pairwise p -distance of each cherry is converted to an estimated divergence time T , and T is discretized into n_τ geometric bins $\{T_1, \dots, T_{n_\tau}\}$ with representative bin centres $\bar{T}_b$. Counts are accumulated per bin into the following tensors over the amino-acid alphabet $\mathcal{A}$ ($|\mathcal{A}| = 20$), with extended vocabulary $\mathcal{A} \cup \{\mathbf{S}, \mathbf{E}\}$ for boundary positions:

Tensor	Shape	Meaning
B	$n_\tau \times (\mathcal{A} + 2)^2$	Singlet bigrams (incl. S/E)
C^{MM}	$n_\tau \times \mathcal{A} ^4$	Match→Match: (a, b, a', b')
C^{MI}	$n_\tau \times \mathcal{A} ^3$	Match→Insert: (a, b, b')
C^{MD}	$n_\tau \times \mathcal{A} ^3$	Match→Delete: (a, b, a')
C^{IM}	$n_\tau \times \mathcal{A} ^3$	Insert→Match: (b, a', b')
C^{II}	$n_\tau \times \mathcal{A} ^2$	Insert→Insert: (b, b')
C^{ID}	$n_\tau \times \mathcal{A} ^2$	Insert→Delete: (b, a')
C^{DM}	$n_\tau \times \mathcal{A} ^3$	Delete→Match: (a, a', b')
C^{DD}	$n_\tau \times \mathcal{A} ^2$	Delete→Delete: (a, a')
C^{DI}	$n_\tau \times \mathcal{A} ^2$	Delete→Insert: (a, b')
C^{SM}	$n_\tau \times \mathcal{A} ^2$	Start→Match: (a', b')
C^{SI}	$n_\tau \times \mathcal{A} $	Start→Insert: (b')
C^{SD}	$n_\tau \times \mathcal{A} $	Start→Delete: (a')
C^{ME}	$n_\tau \times \mathcal{A} ^2$	Match→End: (a, b)
C^{IE}	$n_\tau \times \mathcal{A} $	Insert→End: (b)
C^{DE}	$n_\tau \times \mathcal{A} $	Delete→End: (a)
C^{SE}	n_τ	Start→End (empty alignment)

The post-Insert and post-Delete tensors carry only the adjacent emitted character (b for inserts, a for deletes) as context, not the previous match's full (a, b) context: the model will marginalise its context-rich frequencies down to this reduced context when computing the likelihood. The Match-to-Match tensor C^{MM} is the largest and dominates the parameter budget. Boundary tensors record alignments that begin or end in a particular adjacency type.

C.4.2 Cherry-count likelihood for the MixDom Pair HMM

The cherry-count tensors of Section C.4.1 are scored against the collapsed MixDom Pair HMM $\chi(\theta, T)$ defined by equation (C.3) of Section C.1.1. The free parameters are

$$\theta = (\lambda_0, \mu_0, \{\lambda_n, \mu_n\}_{n=1}^{\mathcal{N}}, \{v_n\}_{n=1}^{\mathcal{N}}, \{w_{nf}\}_{n,f}, \{r_{fg}^{(n)}\}_{n,f,g}, \{u_{nfc}\}_{n,f,c}, \{\pi^{(c)}, Q^{(c)}\}_{c=1}^C),$$

parameterised in unconstrained space (log-rates, log-Dirichlet weights, log-exchangeabilities) so that gradient methods are unconstrained. Note in particular:

- w_{nf} is the per-domain Dirichlet over fragment entry types (initial state of the intra-fragment Markov chain).

- $r_{fg}^{(n)}$ is a per-domain $\mathcal{F} \times \mathcal{F}$ row-stochastic matrix (with row sums ≤ 1) giving the intra-fragment Markov transition probability from fragment-type f to fragment-type g within the same fragment; the residual mass $\rho_f^{(n)} = 1 - \sum_g r_{fg}^{(n)}$ is the fragment-termination probability. The TKF92 scalar self-extension is the special case $\mathcal{F} = 1$, with $r_{11}^{(n)}$ playing the role of the TKF92 extension probability (the M-step closed forms for both reduce to a single binary count split per row of $r^{(n)}$).
- u_{nfc} is a per-(domain, fragment-type) Dirichlet over C static site classes (drawn independently per emitted site, with no chain-time class switching).
- Each site class c has its own reversible substitution model $\text{Subst}(Q^{(c)}, \pi^{(c)})$ with rate matrix $R^{(c)} = Q^{(c)} \text{diag}(\pi^{(c)})$. All rate variation across sites is captured by these per-class GTR matrices; there is no separate Yang-style discretized-gamma rate-multiplier mechanism.

Pair adjacency frequencies. For each T -bin centre $\bar{T}_b$, the collapsed $(5\mathcal{NF}+2)$ -state Pair HMM transition matrix $\chi^{(b)} \equiv \chi(\theta, \bar{T}_b)$ is constructed via the closed form of equation (C.3). The marginal stationary distribution π_b^{stat} over emitting states is obtained as the left null vector of $I - \chi_{\Omega\Omega}^{(b)}$, where Ω denotes the set of $5\mathcal{NF}$ emitting states.

For source machine state $u \in \{\mathbf{S}, \mathbf{M}, \mathbf{I}, \mathbf{D}\}$, destination machine state $v \in \{\mathbf{M}, \mathbf{I}, \mathbf{D}, \mathbf{E}\}$, and characters (a, b, a', b') in their respective slots, the *model-side adjacency frequency*

$$F_b^{uv}(a, b; a', b') = \sum_{\substack{s \in \Omega^u \\ s' \in \Omega^v}} (\pi_{b,s}^{\text{stat}}) e_s(a, b) \chi_{ss'}^{(b)} e_{s'}(a', b')$$

sums over latent (domain, fragment-type) realisations of source and destination collapsed states s, s' , weighting by their stationary probability and emission probabilities. The emission probability of a Match state $\mathbf{MM}_{nf}$ emitting (a, b) marginalises the static site-class mixture:

$$e_{\mathbf{MM}_{nf}}(a, b) = \sum_{c=1}^C u_{nfc} \pi_a^{(c)} \exp(R^{(c)}T)_{ab}, \quad (\text{C.90})$$

and emission probabilities for $\mathbf{MI}, \mathbf{II}, \mathbf{MD}, \mathbf{DD}$ analogously marginalise the same class mixture but emit only one of (a, b) :

$$e_{\mathbf{II}_{nf}}(b) = e_{\mathbf{MI}_{nf}}(b) = \sum_c u_{nfc} \pi_b^{(c)}, \quad e_{\mathbf{DD}_{nf}}(a) = e_{\mathbf{MD}_{nf}}(a) = \sum_c u_{nfc} \pi_a^{(c)}.$$

Boundary frequencies $F_b^{\mathbf{S}v}, F_b^{u\mathbf{E}}, F_b^{\mathbf{SE}}$ replace the corresponding endpoint factor with the $\mathbf{S}$ row or $\mathbf{E}$ column of $\chi^{(b)}$. The reduced-context frequencies needed by post-Insert and post-Delete counts are obtained by marginalisation:

$$F_b^{\mathbf{I}v}(b; \cdot) = \sum_a F_b^{\mathbf{I}v}(a, b; \cdot), \quad F_b^{\mathbf{D}v}(a; \cdot) = \sum_b F_b^{\mathbf{D}v}(a, b; \cdot).$$

For each context $(u; a, b)$ at bin b , the row normalisation constant is

$$Z_b^u(a, b) = \sum_v \sum_{a', b'} F_b^{uv}(a, b; a', b'),$$

where the inner sum runs over the characters carried by the destination adjacency type, and the post-Insert/post-Delete normalisations use the reduced-context frequencies above.

Cherry-count log-likelihood. The composite log-likelihood that the Maraschino fitter maximises is

$$\mathcal{L}_{\text{cherry}}(\theta) = \mathcal{L}_{\text{singlet}}(\theta) + \sum_{b=1}^{n_\tau} \mathcal{L}_{\text{pair},b}(\theta), \quad (\text{C.91})$$

$$\mathcal{L}_{\text{singlet}}(\theta) = \sum_{X,Y \in \mathcal{A} \cup \{\text{S}, \text{E}\}} \left(\sum_b B_{XY}^{(b)} \right) \log P_{XY}^{\text{singlet}}(\theta), \quad (\text{C.92})$$

$$\mathcal{L}_{\text{pair},b}(\theta) = \sum_{u,v} \sum_{a,b,a',b'} C_{ab a'b'}^{uv,(b)} \log \left(\frac{F_b^{uv}(a,b;a',b')}{Z_b^u(a,b)} \right). \quad (\text{C.93})$$

Here P^{singlet} is the order-1 transition matrix obtained from the MixDom Singlet HMM by row-normalising its adjacency frequencies (Section C.4.5); the singlet term scores the bigram counts B summed over time bins, since the singlet model is time-independent. The pair term scores each per-bin adjacency tensor under its own row-normalised conditional distribution.

Optimisation. $\mathcal{L}_{\text{cherry}}(\theta)$ is differentiable in θ . The maximisation is carried out by gradient methods (Adam, optionally L-BFGS for refinement) on the unconstrained parameterisation, using the same MixDom initialiser as the exact Baum–Welch trainer (including the same flags for the number of site classes, the class-equilibrium initialisation, and the fragment-class assignment), so the two fitters can be started from identical parameters and compared directly. Because the M-step of the exact-EM trainer is closed-form and Maraschino’s gradient optimiser is not, this provides a controlled comparison of cherry-count fitting against full Baum–Welch on the same data and architecture.

The fitted MixDom parameters are written to a checkpoint with the same key layout as a `train_pfam`-produced checkpoint, so that either trainer’s output can be loaded and refined by the other and either can be used as input to the order-1 distillation (Sections C.4.5 and C.4.6).

C.4.3 Distillation From MixDom To Order-1 Machines

The MixDom HMMs defined above have large structured state spaces (domains $\times$ fragments). We now show how to distill these into compact *order-1* machines—an HMM and a transducer—whose transition probabilities depend only on the most recently emitted characters. These machines are approximations of the full MixDom model. The MixDom model parameters are assumed to have been trained, e.g. via Baum–Welch or via Maraschino cherry-count fitting (Section C.4.2); see also (38).

The distillation operates on the alphabet $|\mathcal{A}|$ (with the per-(domain, fragment-type) site-class mixture marginalised implicitly inside the match emission tensors of equation (C.90)). The Woodbury structural weights depend only on indel parameters; the per-(class) emission tensors $\pi_a^{(c)} \exp(R^{(c)}T)_{ab}$, weighted by u_{nfc} , are contracted with those weights to produce the order-1 transition probabilities.

Let $|\mathcal{A}|$ denote the alphabet size and $\circ$ denote a distinguished beginning-of-sequence (BOS) symbol.

C.4.4 Notation for path marginalizations

Both distillations require marginalizing over null (non-emitting) states between consecutive emissions. We introduce a compact notation for the expected frequency of partially observed paths.

In any HMM or transducer with states partitioned into emitting states Ω and non-emitting (null) states $\mathcal{Z}$, define the *null closure*

$$N^* \equiv (I - T_{\mathcal{Z}\mathcal{Z}})^{-1}$$

where $T_{\mathcal{Z}\mathcal{Z}}$ is the submatrix of transitions among null states. The effective transition matrix from emitting-or-start to emitting-or-end, marginalizing all intervening null paths, is

$$\hat{T}_{ij} = T_{ij} + \sum_{p,q \in \mathcal{Z}} T_{ip} N_{pq}^* T_{qj} \quad i \in \Omega \cup \{\mathbf{S}\}, j \in \Omega \cup \{\mathbf{E}\}$$

For paths through the MixDom HMMs, we use the notation

$$\mathcal{E}[\mathbf{S} \rightarrow \underbrace{s_1}_{(a_1, b_1)} \xrightarrow{\mathcal{Z}^*} \underbrace{s_2}_{(a_2, b_2)} \rightarrow \cdots \rightarrow \mathbf{E}]$$

to denote the expected number of times the path visits the indicated sequence of emitting states (with indicated emissions in subscript) separated by zero or more null states (denoted $\mathcal{Z}^*$), summed over all completions of the path to the left ($\mathbf{S} \rightarrow \cdots$) and right ($\cdots \rightarrow \mathbf{E}$). Formally, if π is the stationary distribution over states,

$$\mathcal{E}[\underbrace{s_1}_{(a_1, b_1)} \xrightarrow{\mathcal{Z}^*} \underbrace{s_2}_{(a_2, b_2)}] = \sum_{s_1 \in \Omega} \sum_{s_2 \in \Omega} \left(\sum_i \pi_i \hat{T}_{is_1}^* \right) e_{s_1}(a_1, b_1) \hat{T}_{s_1 s_2} e_{s_2}(a_2, b_2) \left(\sum_j \hat{T}_{s_2 j}^* \right)$$

where $e_s(\cdot)$ is the emission probability at state s and $\hat{T}_{ij}^* = [(I - \hat{T}_{\Omega\Omega})^{-1}]_{ij}$ (with $\hat{T}_{\Omega\Omega}$ the submatrix of $\hat{T}$ restricted to emitting states) sums over all paths through emitting states.

C.4.5 Distillation to Order-1 HMM

The MixDom Singlet HMM generates sequences from the stationary distribution. We distill it into an order-1 HMM with states

$$\{\mathbf{S}, \mathbf{E}\} \cup \{a : a \in \mathcal{A}\}$$

where state a deterministically emits character a . Define the adjacency frequency from the full MixDom Singlet HMM:

$$f(a, b) = \mathcal{E}[\underbrace{s_1}_a \xrightarrow{\mathcal{Z}^*} \underbrace{s_2}_b]$$

where the sum is over all emitting states s_1, s_2 weighted by emission probabilities $e_{s_1}(a)$ and $e_{s_2}(b)$ as above. Each singlet emission probability $e_{\mathbf{I}_{nf}}(a) = \sum_c u_{nfc} \pi_a^{(c)}$ marginalises the per-(domain, fragment-type) class mixture.

Parameterization. The order-1 Singlet HMM transition probabilities are:

$$P(b|\mathbf{S}) = \frac{\sum_{b'} f(b, b')}{\sum_{a', b'} f(a', b')} \quad (\text{start} \rightarrow \text{first emission}) \quad (\text{C.94})$$

$$P(b|a) = \frac{f(a, b)}{\sum_{b'} f(a, b')} \quad (\text{emission} \rightarrow \text{emission}) \quad (\text{C.95})$$

$$P(\mathbf{E}|a) = 1 - \sum_b P(b|a) \quad (\text{emission} \rightarrow \text{end}) \quad (\text{C.96})$$

That is: normalize each row of the adjacency matrix f to obtain transition probabilities, allocating the residual probability mass to the end state. The start distribution (C.94) is proportional to the column marginals of f .

C.4.6 Distillation to Order-1 WFST

The MixDom Pair HMM describes the joint distribution over ancestor-descendant sequence pairs. We distill it into an order-1 transducer (a Mealy machine) whose state depends on the last inputted ancestor character and the last outputted descendant character.

Machine states. The transducer has seven machine states $\{\mathbf{S}, \mathbf{M}, \mathbf{I}, \mathbf{D}, \mathbf{V}, \mathbf{W}, \mathbf{E}\}$ organized as a *waiting machine*:

- **Non-waiting** (all outgoing transitions have ε input): $\mathbf{S}$ (start), $\mathbf{M}$ (just matched), $\mathbf{I}$ (just inserted), $\mathbf{D}$ (just deleted)
- **Waiting** (all outgoing transitions consume input): $\mathbf{V}$ (ready after $\mathbf{M}$ or $\mathbf{I}$), $\mathbf{W}$ (ready after $\mathbf{D}$)
- **Terminal**: $\mathbf{E}$ (end)

The distinction $\mathbf{V} \neq \mathbf{W}$ is needed because outgoing transition weights differ: in the MixDom Pair HMM, the $\mathbf{M}/\mathbf{I}$ rows use β while $\mathbf{D}$ rows use γ .

Transitions. For a state with last-input ancestor X and last-output descendant Y (where $X, Y \in \mathcal{A} \cup \{\circ\}$ and $\circ$ denotes BOS):

Source	Dest	Input	Output	Weight
S	V	ε	ε	p_{SV}
S	I	ε	b	$p_{SI}(Y, b)$
S	E	ε	ε	p_{SE}
V	M	a	b	$p_{VM}(X, Y, a, b)$
V	D	a	ε	$p_{VD}(X, Y, a)$
W	M	a	b	$p_{WM}(X, Y, a, b)$
W	D	a	ε	$p_{WD}(X, Y, a)$
M	V	ε	ε	p_{MV}
M	I	ε	b	$p_{MI}(X, Y, b)$
M	E	ε	ε	p_{ME}
I	V	ε	ε	p_{IV}
I	I	ε	b	$p_{II}(X, Y, b)$
I	E	ε	ε	p_{IE}
D	W	ε	ε	p_{DW}
D	I	ε	b	$p_{DI}(X, Y, b)$
D	E	ε	ε	p_{DE}

Note that V and W are *not* associated with emissions; they serve only to enforce the waiting-machine property. Transitions from V and W always consume an ancestor input symbol; transitions from M, I, D, and S never do.

Parameterization from the MixDom Pair HMM. We need the expected frequency of transitions conditioned on (last ancestor input X , last descendant output Y , transition type, new symbols). The key subtlety: through insert states, the last ancestor symbol X must be *propagated* from the preceding match or delete, since inserts do not consume input.

Using the path notation from above, we enumerate all adjacency types that arise in the MixDom Pair HMM. Write $M[X, Y]$ for a match state that inputs X and outputs Y , $I[Y]$ for an insert that outputs Y , and $D[X]$ for a delete that inputs X . The last-ancestor and last-descendant context is carried implicitly.

Adjacency frequencies. The following table lists all adjacency types and their corresponding path marginalizations. In each case, the frequency is computed as a sum over MixDom Pair HMM states, with null states marginalized via N^* . Write $\circ$ for the boundary (start/end) context.

Context	Adjacency	MixDom path	Frequency
<i>Start → Match:</i>			
	$S \rightarrow M[X', Y']$	$S \xrightarrow{Z^*} M[X', Y']$	$f^{SM}(X', Y')$
<i>Start → Insert:</i>			
	$S \rightarrow I[Y']$	$S \xrightarrow{Z^*} I[Y']$	$f^{SI}(Y')$
<i>Start → End (empty sequence):</i>			
	$S \rightarrow E$	$S \xrightarrow{Z^*} E$	f^{SE}
<i>Match → Match (via null states only):</i>			
	$M[X, Y] \rightarrow M[X', Y']$	$M[X, Y] \xrightarrow{Z^*} M[X', Y']$	$f^{MM}(X, Y, X', Y')$
<i>Match → Insert:</i>			
	$M[X, Y] \rightarrow I[Y']$	$M[X, Y] \xrightarrow{Z^*} I[Y']$	$f^{MI}(X, Y, Y')$
<i>Match → Delete:</i>			
	$M[X, Y] \rightarrow D[X']$	$M[X, Y] \xrightarrow{Z^*} D[X']$	$f^{MD}(X, Y, X')$
<i>Match → End:</i>			
	$M[X, Y] \rightarrow E$	$M[X, Y] \xrightarrow{Z^*} E$	$f^{ME}(X, Y)$
<i>Insert → Insert (ancestor context X propagated):</i>			
	$I[Y] \rightarrow I[Y']$	$I[Y] \xrightarrow{Z^*} I[Y']$	$f^{II}(X, Y, Y')$
<i>Insert → Match (ancestor context X propagated):</i>			
	$I[Y] \rightarrow M[X', Y']$	$I[Y] \xrightarrow{Z^*} M[X', Y']$	$f^{IM}(X, Y, X', Y')$
<i>Insert → Delete (ancestor context X propagated):</i>			
	$I[Y] \rightarrow D[X']$	$I[Y] \xrightarrow{Z^*} D[X']$	$f^{ID}(X, Y, X')$
<i>Insert → End (ancestor context X propagated):</i>			
	$I[Y] \rightarrow E$	$I[Y] \xrightarrow{Z^*} E$	$f^{IE}(X, Y)$
<i>Delete → Match (descendant context Y propagated):</i>			
	$D[X] \rightarrow M[X', Y']$	$D[X] \xrightarrow{Z^*} M[X', Y']$	$f^{DM}(X, Y, X', Y')$
<i>Delete → Delete (descendant context Y propagated):</i>			
	$D[X] \rightarrow D[X']$	$D[X] \xrightarrow{Z^*} D[X']$	$f^{DD}(X, Y, X')$
<i>Delete → Insert (descendant context Y propagated):</i>			
	$D[X] \rightarrow I[Y']$	$D[X] \xrightarrow{Z^*} I[Y']$	$f^{DI}(X, Y, Y')$
<i>Delete → End (descendant context Y propagated):</i>			
	$D[X] \rightarrow E$	$D[X] \xrightarrow{Z^*} E$	$f^{DE}(X, Y)$

In all cases, the ancestor context X is propagated through insert states (which do not consume input), and the descendant context Y is propagated through delete states (which do not emit output). The notation $\xrightarrow{Z^*}$ denotes zero or more transitions through null states, marginalized via the null closure N^* .

Computing the frequencies. Each frequency above is computed from the MixDom Pair HMM as follows. Let π_s denote the stationary probability of state s and $\hat{T}_{s_1 s_2}$ the null-marginalized effective transition. For the simplest case (direct adjacency):

$$f^{MD}(X, Y, X') = \sum_{s_1 \in \Omega^M} \sum_{s_2 \in \Omega^D} \left(\sum_i \pi_i \hat{T}_{is_1}^* \right) e_{s_1}(X, Y) \hat{T}_{s_1 s_2} e_{s_2}(X') \left(\sum_j \hat{T}_{s_2 j}^* \right)$$

Boundary frequencies use the start/end rows of $\hat{T}$:

$$f^{\text{SM}}(X', Y') = \sum_{s \in \Omega^{\mathbb{M}}} \hat{T}_{ss} e_s(X', Y') \left(\sum_j \hat{T}_{sj}^* \right), \quad f^{\text{ME}}(X, Y) = \sum_{s \in \Omega^{\mathbb{M}}} \left(\sum_i \pi_i \hat{T}_{is}^* \right) e_s(X, Y) \hat{T}_{s\text{E}}$$

(and analogously for f^{SI} , f^{SE} , f^{IE} , f^{DE}).

Normalization to transducer parameters. Given the adjacency frequencies, the order-1 transducer weights are obtained by normalization. For each context (X, Y) and source machine state, normalize outgoing weights to sum to 1:

Start transitions (context $\circ$):

$$p_{\text{SV}} = \frac{\sum_{X', Y'} f^{\text{SM}}(X', Y')}{\sum_{X', Y'} f^{\text{SM}}(X', Y') + \sum_{Y'} f^{\text{SI}}(Y') + f^{\text{SE}}}, \quad p_{\text{SI}}(b) = \frac{f^{\text{SI}}(b)}{\sum_{X', Y'} f^{\text{SM}}(X', Y') + \sum_{Y'} f^{\text{SI}}(Y') + f^{\text{SE}}}$$

(and p_{SE} uses the same denominator with numerator f^{SE}).

Wait-after-match/insert transitions (context (X, Y) , consuming input a):

$$p_{\text{VM}}(X, Y, a, b) = \frac{f^{\text{M}}(X, Y, a, b)}{\sum_{a'} [\sum_{b'} f^{\text{M}}(X, Y, a', b') + f^{\text{D}}(X, Y, a')]} \\ p_{\text{VD}}(X, Y, a) = \frac{f^{\text{D}}(X, Y, a)}{\sum_{a'} [\sum_{b'} f^{\text{M}}(X, Y, a', b') + f^{\text{D}}(X, Y, a')]}$$

where $f^{\text{M}}(X, Y, a, b) = f^{\text{MM}}(X, Y, a, b) + f^{\text{IM}}(X, Y, a, b)$ and $f^{\text{D}}(X, Y, a) = f^{\text{MD}}(X, Y, a) + f^{\text{ID}}(X, Y, a)$, aggregating over both match and insert sources that share the V wait state.

Wait-after-delete transitions (context (X, Y) , consuming input a):

$$p_{\text{WM}}(X, Y, a, b) = \frac{f^{\text{DM}}(X, Y, a, b)}{\sum_{a'} [\sum_{b'} f^{\text{DM}}(X, Y, a', b') + f^{\text{DD}}(X, Y, a')]} \\ p_{\text{WD}}(X, Y, a) = \frac{f^{\text{DD}}(X, Y, a)}{\sum_{a'} [\sum_{b'} f^{\text{DM}}(X, Y, a', b') + f^{\text{DD}}(X, Y, a')]}$$

Post-match transitions (context (X, Y) , after matching with a, b ; new context becomes (a, b)):

$$p_{\text{MV}}(X, Y) = \frac{\sum_{a', b'} f^{\text{MM}}(X, Y, a', b') + \sum_{a'} f^{\text{MD}}(X, Y, a')}{Z^{\text{M}}(X, Y)} \\ p_{\text{MI}}(X, Y, b) = \frac{f^{\text{MI}}(X, Y, b)}{Z^{\text{M}}(X, Y)}, \quad p_{\text{ME}}(X, Y) = \frac{f^{\text{ME}}(X, Y)}{Z^{\text{M}}(X, Y)}$$

where $Z^{\text{M}}(X, Y) = \sum_{a', b'} f^{\text{MM}}(X, Y, a', b') + \sum_{a'} f^{\text{MD}}(X, Y, a') + \sum_b f^{\text{MI}}(X, Y, b) + f^{\text{ME}}(X, Y)$.

Post-insert transitions are analogous, using $f^{\text{I}\cdot}$ frequencies with $Z^{\text{I}}(X, Y)$.

Post-delete transitions (context (X, Y) , after deleting a ; new context becomes (a, Y)):

$$p_{\text{DW}}(X, Y) = \frac{\sum_{a'} f^{\text{DD}}(X, Y, a') + \sum_{a', b'} f^{\text{DM}}(X, Y, a', b')}{Z^{\text{D}}(X, Y)} \\ p_{\text{DI}}(X, Y, b) = \frac{f^{\text{DI}}(X, Y, b)}{Z^{\text{D}}(X, Y)}, \quad p_{\text{DE}}(X, Y) = \frac{f^{\text{DE}}(X, Y)}{Z^{\text{D}}(X, Y)}$$

where $Z^{\text{D}}(X, Y) = \sum_{a'} f^{\text{DD}}(X, Y, a') + \sum_{a', b'} f^{\text{DM}}(X, Y, a', b') + \sum_b f^{\text{DI}}(X, Y, b) + f^{\text{DE}}(X, Y)$.

C.5 Algebraic Distillation of MixDom

We investigate whether the distillation of the MixDom model to order-1 machines (Section C.4.5 and C.4.6) can be performed in closed algebraic form, and how the computation scales with the number of domain types $\mathcal{N}$, the number of fragment types per domain $\mathcal{F}$, and the number of site classes C . The same algebraic decomposition powers the cherry-count log-likelihood that the Maraschino fitter (Section C.4.2) maximises under MixDom.

C.5.1 Setup

The MixDom model has the following parameters (matching Section C.1.1):

- Top-level TKF91: λ_0, μ_0 (domain birth/death rates).
- Domain weights: $v_n, \sum_n v_n = 1$.
- Per-domain TKF91 rates governing the per-domain TKF92 fragment process: λ_n, μ_n for $n = 1, \dots, \mathcal{N}$.
- Per-domain fragment-type entry distribution: w_{nf} with $\sum_f w_{nf} = 1$.
- Per-domain intra-fragment Markov ext matrix $r_{fg}^{(n)}, \mathcal{F} \times \mathcal{F}$, with row sums ≤ 1 . The fragment-termination probability is $\rho_f^{(n)} = 1 - \sum_g r_{fg}^{(n)}$.
- Per-(domain, fragment-type) site-class Dirichlet: u_{nfc} with $\sum_c u_{nfc} = 1$.
- Per-class reversible substitution model: $(Q^{(c)}, \pi^{(c)})$ for $c = 1, \dots, C$; rate matrix $R^{(c)} = Q^{(c)} \text{diag}(\pi^{(c)})$.

The total scalar parameter count is $2 + (\mathcal{N} - 1) + 2\mathcal{N} + \mathcal{N}\mathcal{F}^2 + \mathcal{N}(\mathcal{F} - 1) + \mathcal{N}\mathcal{F}(C - 1)$ plus C rate matrices.

Write $\kappa_n = \lambda_n/\mu_n$, $\alpha_n = \alpha(\lambda_n, \mu_n, T)$, $\beta_n = \beta(\lambda_n, \mu_n, T)$, $\gamma_n = \gamma(\lambda_n, \mu_n, T)$, and similarly $\alpha_0, \beta_0, \gamma_0, \kappa_0$ for the top-level parameters.

C.5.2 Class-mixture emissions

In MixDom the per-(domain, fragment-type) site-class mixture appears in every emission probability of the collapsed Pair HMM. Define the per-class match emission as

$$P^{(c)}(a, b | T) = \pi_a^{(c)} \exp(R^{(c)}T)_{ab}.$$

The model's per-(domain, fragment-type) emission factors at evolutionary time T are:

$$\begin{aligned} \phi_{nf}^{\text{M}}(a, b) &\equiv e_{\text{MM}_{nf}}(a, b) = \sum_{c=1}^C u_{nfc} P^{(c)}(a, b | T), \\ \phi_{nf}^{\text{I}}(b) &\equiv e_{\text{MI}_{nf}}(b) = e_{\text{II}_{nf}}(b) = \sum_c u_{nfc} \pi_b^{(c)}, \\ \phi_{nf}^{\text{D}}(a) &\equiv e_{\text{MD}_{nf}}(a) = e_{\text{DD}_{nf}}(a) = \sum_c u_{nfc} \pi_a^{(c)}. \end{aligned}$$

Each of these is a C -term mixture over the per-class GTRs. The match emission ϕ_{nf}^{M} inherits the time dependence from the per-class transition kernels $P^{(c)}(T)$; the singlet emissions $\phi_{nf}^{\text{I}}, \phi_{nf}^{\text{D}}$ are time-independent.

Remark C.3 (Bilinear emission tensor). The match emission factors as

$$\phi_{nf}^M(a, b) = \sum_c u_{nfc} \pi_a^{(c)} \exp(R^{(c)}T)_{ab}$$

which is rank- C in the latent class index, and the resulting $|\mathcal{A}| \times |\mathcal{A}|$ emission tensor at (n, f) has rank at most C . The single-class case $u = \delta_{nc}$ makes this rank-1 per domain (a single GTR per domain). The general parameterisation strictly generalises by sharing the GTR pool across (domain, fragment-type) contexts and allowing any soft mixture.

C.5.3 Single HMM Distillation

State space The MixDom Singlet HMM generates sequences from the stationary distribution. Its collapsed state space (Section C.1.1) is $\{\mathbf{S}, \mathbf{E}\} \cup \{\mathbf{I}_{nf} : n \in \mathcal{N}, f \in \mathcal{F}\}$, so there are $\mathcal{N}\mathcal{F}$ emitting states with emissions $\phi_{nf}^I(a)$.

Within a domain, fragment continuation is governed by the intra-fragment Markov ext matrix $r_{fg}^{(n)}$ and the new-fragment $\kappa_n w_{n,g}$ branch. Adapting the formulae in Section C.1.1 for the singlet, the effective transition matrix between emitting states $\mathbf{I}_{lf} \rightarrow \mathbf{I}_{mg}$ has entries

$$\hat{T}_{\mathbf{I}_{lf}, \mathbf{I}_{mg}} = r_{fg}^{(l)} \delta_{lm} + \rho_f^{(l)} \kappa_l w_{lg} \delta_{lm} + \frac{\rho_f^{(l)} (1 - \kappa_l) \kappa_0 v_m \kappa_m w_{mg}}{1 - \kappa_0 z_0}$$

where $z_0 = \sum_n v_n (1 - \kappa_n)$. The path-sum matrix $(I - \hat{T}_{\Omega, \Omega})^{-1}$ is an $\mathcal{N}\mathcal{F} \times \mathcal{N}\mathcal{F}$ inversion in general.

Adjacency frequencies The Singlet adjacency frequency is

$$f(a, b) = \sum_{(l,f),(m,g)} W_{(l,f),(m,g)} \phi_{lf}^I(a) \phi_{mg}^I(b) \quad (\text{C.97})$$

where the structural weights $W_{(l,f),(m,g)} = L_{(l,f)} \hat{T}_{(l,f),(m,g)} R_{(m,g)}$ are character-independent, with $L_{(n,f)} = \sum_i \pi_i \hat{T}_{i,(n,f)}^*$ and $R_{(n,f)} = \sum_j \hat{T}_{(n,f),j}^*$. Substituting the class-mixture emission yields a bilinear sum over both the (domain, fragment-type) latent and the site-class latent:

$$f(a, b) = \sum_{l,f,m,g,c_1,c_2} W_{(l,f),(m,g)} u_{lf c_1} u_{mg c_2} \pi_a^{(c_1)} \pi_b^{(c_2)}.$$

The order-1 HMM transition is therefore

$$P(b | a) = \frac{f(a, b)}{\sum_{b'} f(a, b')} = \frac{\sum_{c_2} \left(\sum_{l,f,m,g,c_1} W_{l,f,m,g,c_1} u_{lf c_1} u_{mg c_2} \right) \pi_a^{(c_1)} \pi_b^{(c_2)}}{\sum_{l,f,c_1} \pi_a^{(c_1)} \sum_{m,g,c_2} W_{l,f,m,g,c_1} u_{lf c_1} u_{mg c_2}}.$$

Remark C.4 (Non-trivial character correlations). When the per-class equilibria $\pi^{(c)}$ differ, the previous character a carries information about the latent class c_1 of the source state via the Bayesian posterior $P(c_1 | a) \propto \pi_a^{(c_1)} \sum_{l,f,m,g,c_2} W_{l,f,m,g,c_1} u_{lf c_1} u_{mg c_2}$, and consequently about which (domain, fragment-type) is likely to be generating this region. The order-1 HMM therefore has genuinely a -dependent transitions: a mixture of $\pi^{(c_2)}$ tilted by the joint posterior over latent state and source class.

Remark C.5 (Closed form). All quantities in (C.97) are rational functions of the model parameters: the $\mathcal{N}\mathcal{F} \times \mathcal{N}\mathcal{F}$ matrix $(I - \hat{T})^{-1}$ has entries that are ratios of polynomials in $(r_{fg}^{(n)}, \kappa_n, \kappa_0, \beta_0, w_{nf}, v_n)$. The class mixture enters linearly through u_{nfc} , and the per-class GTRs appear only inside $\pi^{(c)}, P^{(c)}(T)$.

C.5.4 Pair HMM Distillation

Emitting state space The collapsed MixDom Pair HMM with $\mathcal{N}$ domain types and $\mathcal{F}$ fragment types per domain has $5\mathcal{N}\mathcal{F} + 2$ states: $\{\mathbf{SS}, \mathbf{EE}\}$ and $\{\mathbf{MM}_{nf}, \mathbf{MI}_{nf}, \mathbf{MD}_{nf}, \mathbf{II}_{nf}, \mathbf{DD}_{nf} : n \in \mathcal{N}, f \in \mathcal{F}\}$.

Group the $5\mathcal{N}\mathcal{F}$ emitting states by emission type: match $\mathcal{M} = \{\mathbf{MM}_{nf}\}$, insert $\mathcal{I} = \{\mathbf{MI}_{nf}, \mathbf{II}_{nf}\}$, delete $\mathcal{D} = \{\mathbf{MD}_{nf}, \mathbf{DD}_{nf}\}$.

Class-mixture emissions in the Pair HMM The per-state emissions at (d, f) are the class-mixture emissions $\phi_{nf}^{\mathbf{M}, \mathbf{I}, \mathbf{D}}$ defined in Section C.5.2. Two states with the same (d, f) but different top-level types share the same equilibrium mixture, but the match state combines this with the time-evolved class-mixture transition kernel.

The per-(domain, fragment-type) emission factors do *not* lift cleanly out of the structural sum; however, they have low rank in the class index c (Section C.5.2). Within a fixed (d, f) , all match states emit from $\phi_{nf}^{\mathbf{M}}$ and all insert/delete singlets from $\phi_{nf}^{\mathbf{I}} / \phi_{nf}^{\mathbf{D}}$.

Per-(domain, fragment)-pair structural weights The pair adjacency frequencies decompose as sums over (domain, fragment-type) pairs. For match-to-match, writing $X = a, Y = b, X' = a', Y' = b'$:

$$f^{\mathcal{MM}}(X, Y, X', Y') = \sum_{(n_1, f_1), (n_2, f_2)} C_{(n_1, f_1), (n_2, f_2)}^{\mathcal{MM}} \phi_{n_1 f_1}^{\mathbf{M}}(X, Y) \phi_{n_2 f_2}^{\mathbf{M}}(X', Y') \quad (\text{C.98})$$

where

$$C_{(n_1, f_1), (n_2, f_2)}^{\mathcal{MM}} = \sum_{s_1 \in \mathcal{M}_{n_1 f_1}} \sum_{s_2 \in \mathcal{M}_{n_2 f_2}} L_{s_1} \hat{T}_{s_1 s_2} R_{s_2}$$

are structural weights indexed by (domain, fragment-type) pairs (not just emission types). Similarly for other adjacency types ($\mathcal{MI}$, $\mathcal{MD}$, etc.), with the appropriate per-(domain, fragment-type) emissions.

Each adjacency frequency is therefore a sum of $(\mathcal{N}\mathcal{F})^2$ structural-weight terms, each multiplied by per-(domain, fragment-type) emission factors that are themselves rank- C class mixtures.

Context dependence In the order-1 WFST, the wait-after-match/insert transition involves

$$\begin{aligned} f^{\mathcal{M}}(X, Y, a', b') = & \sum_{(n_1, f_1), (n_2, f_2)} \left(C_{(n_1, f_1), (n_2, f_2)}^{\mathcal{MM}} \phi_{n_1 f_1}^{\mathbf{M}}(X, Y) \right. \\ & \left. + C_{(n_1, f_1), (n_2, f_2)}^{\mathcal{IM}} \phi_{n_1 f_1}^{\mathbf{I}}(Y) \right) \phi_{n_2 f_2}^{\mathbf{M}}(a', b') \end{aligned}$$

After normalization, $p_{\mathbf{VM}}(X, Y, a', b')$ depends on (X, Y) through the joint posterior over (domain, fragment-type, class) of the previous emission. Specifically, the context enters through the $\mathcal{NF}$ -dimensional vector

$$\rho(X, Y) = \left(\sum_{(n_2, f_2)} C_{(1, 1), (n_2, f_2)}^{\mathcal{MM}} \phi_{1, 1}^{\mathbf{M}}(X, Y) + \sum_{(n_2, f_2)} C_{(1, 1), (n_2, f_2)}^{\mathcal{IM}} \phi_{1, 1}^{\mathbf{I}}(Y), \dots \right)$$

which captures the relative likelihood of each source (domain, fragment-type).

Remark C.6 (Richer context than per-domain shared- π case). With class-mixture emissions, the WFST transition probabilities depend on context (X, Y) through a joint (domain, fragment-type, class) posterior that is sensitive to *both* X and Y independently—not just through a single scalar ratio. The right-side character dependence (a', b') also varies by (domain, fragment-type, class): the next emission is a mixture of $\pi^{(c_2)} P^{(c_2)}(T)$ weighted by the structural weights, the source posterior, and the destination u . Despite this richer structure, all quantities are closed-form rational functions of the model parameters (composed with C rate-matrix exponentials).

C.5.5 Block Structure and Matrix Inversions

Top-level null closure (always 3×3) The null states **A, B, C** yield a 3×3 submatrix with entries depending on

$$z_0 = \sum_n v_n (1 - \kappa_n)$$

$$z_T = \sum_n v_n (1 - \kappa_n) (1 - \beta_n)$$

The null closure is a 3×3 inversion with closed-form determinant, independent of $\mathcal{N}, \mathcal{F}, C$ (the dependence on $\mathcal{N}$ enters only through these two scalar sums; $\mathcal{F}$ and C do not appear at all because empty domains are independent of fragment and class structure inside the domain).

Block-diagonal-plus-low-rank decomposition The $5\mathcal{N}\mathcal{F} \times 5\mathcal{N}\mathcal{F}$ effective transition matrix between emitting states decomposes as

$$\hat{T}_{\Omega, \Omega} = \underbrace{\text{diag}(D_1, \dots, D_{\mathcal{N}})}_{\text{intra-domain}} + \underbrace{E \mathcal{T}_{\bullet} S^{\top}}_{\text{inter-domain (rank } \leq 3)}$$

where each D_n is a $5\mathcal{F} \times 5\mathcal{F}$ within-domain block that combines the intra-fragment Markov ext matrix $r_{fg}^{(n)}$ (acting on the fragment-type axis) with the same-domain new-fragment TKF92 transitions $(\rho_f^{(n)} \tau_{\text{XY}}^{(n)} w_{n,g})$ for M-type entries, $\rho_f^{(n)} \kappa_n w_{n,g}$ for I/D-type entries), $E \in \mathbb{R}^{5\mathcal{N}\mathcal{F} \times 3}$ stacks per-(domain, fragment-type) exit vectors projected to top-level types $\{\text{M, I, D}\}$ via $(\rho_f^{(n)} \varphi_n)$ for the I/D singlet rows and $(\rho_f^{(n)} \tau_{\text{XE}}^{(n)})$ for the matched-domain rows, $S \in \mathbb{R}^{5\mathcal{N}\mathcal{F} \times 3}$ stacks per-(domain, fragment-type) start vectors with $(1 - z_T)^{-1} v_m \tau_{\text{SY}}^{(m)} w_{mg}$ for M-type entries and $(1 - z_0)^{-1} v_m \kappa_m w_{mg}$ for I/D-type entries (these $1/(1 - z)$ factors are the domenter normalisation that conditions on the destination domain being non-empty), and $\mathcal{T}_{\bullet}$ is the relevant 5×5 submatrix of the null-eliminated top-level matrix.

Note that $D_{n_1} \neq D_{n_2}$ in general (different $r_{fg}^{(n)}, w_{nf}, \lambda_n, \mu_n$). For $\mathcal{F} > 1$, D_n has no further emission-type block-diagonal sub-structure within the matched-domain block (the intra-fragment Markov ext couples $\{\text{MM, MI, MD}\}$ across fragment-types); however, it admits a Kronecker-plus-low-rank decomposition that preserves the linear-in- $\mathcal{N}\mathcal{F}$ scaling (Section C.5.6). The diagonal-ext baseline (`-freeze-offdiag-ext` in the fitter) recovers the diagonal-extension special case block structure as a special case.

Remark C.7 (Domenter normalization). The χ matrix (Section C.1.1, Equation C.3) includes factors $(1 - z_T)^{-1}$ for M-type destinations and $(1 - z_0)^{-1}$ for I/D-type destinations, conditioning domain entry on the domain being non-empty. These factors cancel the corresponding $(1 - z_T)$ and $(1 - z_0)$ column scaling in $\mathcal{T}_{\bullet}$ (which arises from the v null-elimination Schur complement). In the distillation, the start vectors $\mathbf{s}_{n,f}$ carry the domenter factors while $\mathcal{T}_{\bullet}$ retains the column factors; the product $\mathcal{T}_{\bullet} \cdot \mathbf{s}_{n,f}$ thus reproduces the correct χ entries.

Woodbury identity Writing $G_n = (I - D_n)^{-1}$ (each $5\mathcal{F} \times 5\mathcal{F}$), $E = (\mathbf{e}_{n,f})$ (exit vectors, $5\mathcal{N}\mathcal{F} \times 3$), $S = (\mathbf{s}_{n,f})$ (start vectors, $5\mathcal{N}\mathcal{F} \times 3$), and $\mathcal{T}_{\text{mid}}$ for the 3×3 subblock of $\mathcal{T}_\bullet$ restricted to $\{\mathbf{M}, \mathbf{I}, \mathbf{D}\}$ (the only top-level types that couple across domains), the path-sum matrix is

$$(I - \hat{T})^{-1} = G + G E \mathcal{T}_{\text{mid}} (I_3 - S^\top G E \mathcal{T}_{\text{mid}})^{-1} S^\top G$$

using the push-through form of the Woodbury identity, where $G = \text{diag}(G_1, \dots, G_N)$. This avoids inverting $\mathcal{T}_{\text{mid}}$, which is singular (the TKF91 M and I rows are identical, so $\mathcal{T}_{\text{mid}}$ has rank 2).

Proposition C.1 (Closed-form distillation for MixDom). *All adjacency frequencies, and hence all order-1 HMM and WFST parameters, are closed-form rational functions of the model parameters (composed with the matrix exponentials $\exp(R^{(c)}T)$), for any finite $N, \mathcal{F}, C$. The computation requires:*

1. A 3×3 inversion for the top-level null closure.
2. $\mathcal{N}$ within-domain inversions $(I - D_n)^{-1}$, each of size $5\mathcal{F} \times 5\mathcal{F}$. By the Kronecker-plus-rank-3 decomposition of D_n (Section C.5.6), this reduces to one $\mathcal{F} \times \mathcal{F}$ inversion of $(I - r^{(n)})$ plus a 3×3 inner Woodbury kernel per domain. For $\mathcal{F} = 1$ the whole within-domain block collapses to a 3×3 adjugate plus two scalar inversions (Section C.5.7).
3. One 3×3 outer Woodbury correction for inter-domain coupling.
4. Summation of $(\mathcal{N}\mathcal{F})^2$ (domain, fragment-type)-pair terms per adjacency entry, with each term involving a class-mixture emission factor of length C .

Every step is a rational function of the model parameters: no numerical iteration is required to evaluate the closed form.

Remark C.8 (Contrast with shared-emission case). If all (d, f) shared a single class mixture (e.g. $u_{nfc} = u_c$ uniform), the adjacency frequencies would still factor as in (C.98), but the emission tensor would have rank-1 structure across (d, f) , collapsing the WFST context dependence. With per-(domain, fragment-type) class mixtures, the structural constants are full $(\mathcal{N}\mathcal{F}) \times (\mathcal{N}\mathcal{F})$ matrices indexed by (domain, fragment-type) pairs, the emission tensors are class-rank- C , and the context dependence is genuinely multi-dimensional.

C.5.6 Within-Domain Inversion: closed form

For general $\mathcal{F}$, the within-domain block D_n has a Kronecker-plus-rank-3 structure that yields a closed-form inverse without ever inverting a $5\mathcal{F} \times 5\mathcal{F}$ matrix directly.

Kronecker-plus-rank-3 decomposition of the matched-domain block The matched-domain inner block (acting on $\{\mathbf{MM}, \mathbf{MI}, \mathbf{MD}\}$ at each fragment-type, total $3\mathcal{F}$ states) combines intra-fragment Markov continuation with fragment termination plus a TKF92 same-domain new-fragment branch:

$$D_n^{\text{match}}[(X, f), (Y, g)] = \delta_{XY} r_{fg}^{(n)} + \rho_f^{(n)} \cdot \tau_{XY}^{(n)} \cdot w_{n,g}. \quad (\text{C.99})$$

The first term is $I_3 \otimes r^{(n)}$ (the intra-fragment Markov chain, identical at every emission type); the second is $\tau_{3 \times 3}^{(n)} \otimes (\rho^{(n)} w_n^\top)$ where $\rho^{(n)} w_n^\top$ is the rank-1 $\mathcal{F} \times \mathcal{F}$ outer product of fragment-termination probabilities and fragment-entry weights. The whole rank-3 correction is therefore of rank ≤ 3 in

the joint $(\mathbf{X}, f)$ space (one factor of 3 from $\tau_{3 \times 3}^{(n)}$, and rank 1 in the fragment slot from the outer product). Writing

$$D_n^{\text{match}} = I_3 \otimes r^{(n)} + U_n V_n^\top, \quad U_n = \tau_{3 \times 3}^{(n)} \otimes \rho^{(n)}, \quad V_n = I_3 \otimes w_n,$$

U_n and V_n are $3\mathcal{F} \times 3$ matrices.

Inverse via inner Woodbury Since $I_3 \otimes r^{(n)}$ commutes with itself, $I - I_3 \otimes r^{(n)} = I_3 \otimes (I_{\mathcal{F}} - r^{(n)})$, which inverts blockwise: $(I_3 \otimes (I_{\mathcal{F}} - r^{(n)}))^{-1} = I_3 \otimes (I_{\mathcal{F}} - r^{(n)})^{-1}$. Note $(I_{\mathcal{F}} - r^{(n)})^{-1}$ is a single $\mathcal{F} \times \mathcal{F}$ inversion per domain. Applying the Sherman–Morrison–Woodbury identity to the rank-3 correction gives

$$\begin{aligned} (I - D_n^{\text{match}})^{-1} &= G_0 + G_0 U_n K_n^{\text{inner}} V_n^\top G_0, \\ G_0 &\equiv I_3 \otimes (I_{\mathcal{F}} - r^{(n)})^{-1}, \\ K_n^{\text{inner}} &\equiv (I_3 - V_n^\top G_0 U_n)^{-1}. \end{aligned} \tag{C.100}$$

K_n^{inner} is a 3×3 matrix whose entries are rational functions of w_n , $r^{(n)}$, $\tau^{(n)}$ via a single $\mathcal{F}$ -fold inner product $w_n^\top (I - r^{(n)})^{-1} \rho^{(n)}$ times $\tau_{3 \times 3}^{(n)}$.

The two singlet-domain blocks act on $\mathcal{F}$ states each ($\{\text{II}_{nf} : f\}$ and $\{\text{DD}_{nf} : f\}$):

$$d_n^{\text{ins}}[f, g] = d_n^{\text{del}}[f, g] = r_{fg}^{(n)} + \rho_f^{(n)} \kappa_n w_{n,g}.$$

Their inverse follows the same pattern: rank-1 correction to $(I - r^{(n)})$, giving

$$(I - d_n^{\text{ins}})^{-1} = (I - r^{(n)})^{-1} + (I - r^{(n)})^{-1} \rho^{(n)} k_n^{\text{ins}} w_n^\top (I - r^{(n)})^{-1},$$

with scalar inner kernel $k_n^{\text{ins}} = \kappa_n [1 - \kappa_n w_n^\top (I - r^{(n)})^{-1} \rho^{(n)}]^{-1}$, and identically for delete.

Every entry of $(I - D_n)^{-1}$ is therefore a rational function of $(r^{(n)}, w_n, \alpha_n, \beta_n, \gamma_n, \kappa_n)$, computed at cost $O(\mathcal{F}^3)$ per domain (the inversion of $I_{\mathcal{F}} - r^{(n)}$) plus $O(\mathcal{F}^2)$ for the singlet blocks and $O(1)$ for the inner 3×3 Woodbury kernel. No numerical iteration is required; the inversion can be carried out analytically (for small $\mathcal{F}$, by Cramer’s rule on $I_{\mathcal{F}} - r^{(n)}$) or directly evaluated for larger $\mathcal{F}$.

Remark C.9 (Compactness). For small $\mathcal{F}$ the inverse $(I_{\mathcal{F}} - r^{(n)})^{-1}$ has a compact symbolic form (e.g. at $\mathcal{F} = 2$ it is a 2×2 adjugate over a scalar determinant). The further reductions of the matched-domain block via (C.100) and the singlet blocks above involve only 3×3 and scalar inner inversions, so the overall within-domain inverse can be written down symbolically without ever exceeding a $\mathcal{F} \times \mathcal{F}$ inversion. The $\mathcal{F} = 1$ scalar-extension special case collapses $(I_{\mathcal{F}} - r^{(n)})^{-1}$ to the scalar $1/(1 - r_n)$ and recovers the more compact 3×3 adjugate of Section C.5.7 below.

C.5.7 Within-Domain Inversion: $\mathcal{F} = 1$ closed form

When $\mathcal{F} = 1$ (scalar self-extension $r_n \equiv r_{11}^{(n)}$, no off-diagonal Markov coupling), D_n collapses to 5×5 with the additional block structure described below; we record this special case because it admits a particularly compact algebraic form.

Block decomposition: $3 \times 3 + 1 + 1$ In the $\mathcal{F} = 1$ limit, the five emitting state types $\{\text{MM}, \text{MI}, \text{MD}, \text{II}, \text{DD}\}$ cannot transition between top-level types within a domain: if the domain is inserted (top-level I), all emissions stay in II until the domain ends; similarly for deleted domains (DD). Therefore D_n is block-diagonal:

$$D_n = \begin{pmatrix} D_n^{\text{match}} & 0 & 0 \\ 0 & d_n^{\text{ins}} & 0 \\ 0 & 0 & d_n^{\text{del}} \end{pmatrix}$$

where D_n^{match} is 3×3 (for matched-domain states **MM**, **MI**, **MD**) and the singlet-domain states have scalar self-loops:

$$d_n^{\text{ins}} = d_n^{\text{del}} = r_n + (1 - r_n)\kappa_n$$

The scalar inversions are trivial: $(1 - d_n^{\text{ins}})^{-1} = [(1 - r_n)(1 - \kappa_n)]^{-1}$.

The 3×3 matched-domain block The matched-domain block combines fragment extension (self-loop at rate r_n) with intra-domain TKF transitions:

$$D_n^{\text{match}} = r_n I_3 + (1 - r_n) \tau_{\text{MID}, \text{MID}}^{(n)}$$

where $\tau_{\text{MID}, \text{MID}}^{(n)}$ is the 3×3 submatrix of $\tau^{(n)}$ restricted to rows and columns **M**, **I**, **D**.

A key property: in the TKF transition matrix, rows **M** and **I** are identical. Defining shorthand (suppressing domain subscript n):

$$\begin{aligned} \mathfrak{a} &= (1 - \beta)\kappa\alpha, & \mathfrak{b} &= \beta, & \mathfrak{c} &= (1 - \beta)\kappa(1 - \alpha) \\ \mathfrak{d} &= (1 - \gamma)\kappa\alpha, & \mathfrak{g} &= \gamma, & \mathfrak{h} &= (1 - \gamma)\kappa(1 - \alpha) \end{aligned}$$

the 3×3 TKF submatrix is

$$\tau_{\text{MID}, \text{MID}}^{(n)} = \begin{pmatrix} \mathfrak{a} & \mathfrak{b} & \mathfrak{c} \\ \mathfrak{a} & \mathfrak{b} & \mathfrak{c} \\ \mathfrak{d} & \mathfrak{g} & \mathfrak{h} \end{pmatrix}$$

with row sums $\mathfrak{a} + \mathfrak{b} + \mathfrak{c} = (1 - \beta)\kappa + \beta$ for the first two rows and $\mathfrak{d} + \mathfrak{g} + \mathfrak{h} = (1 - \gamma)\kappa + \gamma$ for the third.

Factoring out the extension rate Since $I - D^{\text{match}} = I - rI - (1 - r)\tau_{3 \times 3} = (1 - r)(I - \tau_{3 \times 3})$, the extension rate factors out as a scalar:

$$(I - D_n^{\text{match}})^{-1} = \frac{1}{1 - r_n} P_n^{-1}$$

where

$$P_n \equiv I - \tau_{\text{MID}, \text{MID}}^{(n)} = \begin{pmatrix} 1 - \mathfrak{a} & -\mathfrak{b} & -\mathfrak{c} \\ -\mathfrak{a} & 1 - \mathfrak{b} & -\mathfrak{c} \\ -\mathfrak{d} & -\mathfrak{g} & 1 - \mathfrak{h} \end{pmatrix}$$

with row sums $(1 - \beta)(1 - \kappa)$, $(1 - \beta)(1 - \kappa)$, $(1 - \gamma)(1 - \kappa)$ (the domain-exit probabilities from each inner state).

Determinant of P_n Since rows 1 and 2 of $\tau_{3 \times 3}$ are identical, subtracting row 2 from row 1 in P yields $(1, -1, 0)$. Expanding the determinant along this simplified row gives

$$\begin{aligned} \det(P_n) &= (1 - \mathfrak{b})(1 - \mathfrak{h}) - \mathfrak{c}\mathfrak{g} - \mathfrak{a}(1 - \mathfrak{h}) - \mathfrak{c}\mathfrak{d} \\ &= (1 - \beta_n)(1 - \kappa_n) \end{aligned}$$

This factors cleanly: since $\mathfrak{a} + \mathfrak{b} + \mathfrak{c}$ and $\mathfrak{d} + \mathfrak{g} + \mathfrak{h}$ are the row sums of $\tau_{3 \times 3}$, we have

$$\boxed{\det(P_n) = (1 - \beta_n)(1 - \kappa_n)}$$

The determinant is the product of the M/I-row exit probability $(1 - \beta)$ and the stationary emptiness probability $(1 - \kappa)$.

Inverse of P_n The adjugate $\text{adj}(P_n)$ has entries (again suppressing domain subscripts):

$$\text{adj}(P) = \begin{pmatrix} (1-\beta)(1-\kappa(1-\alpha)) & \beta + \kappa(1-\alpha)(\gamma-\beta) & (1-\beta)\kappa(1-\alpha) \\ (1-\beta)\kappa\alpha & 1 - (1-\beta)\kappa\alpha - (1-\gamma)\kappa(1-\alpha) & (1-\beta)\kappa(1-\alpha) \\ (1-\beta)\kappa\alpha & \gamma + \kappa\alpha(\beta-\gamma) & (1-\beta)(1-\kappa\alpha) \end{pmatrix}$$

Note the symmetries: $\text{adj}(P)_{21} = \text{adj}(P)_{31}$ and $\text{adj}(P)_{13} = \text{adj}(P)_{23}$ (reflecting the identical rows of $\tau_{3 \times 3}$), while $\text{adj}(P)_{11} - \text{adj}(P)_{33} = (1-\beta)\kappa(2\alpha-1)$, which vanishes only when $\alpha = \frac{1}{2}$.

The full inverse is

$$P_n^{-1} = \frac{\text{adj}(P_n)}{(1-\beta_n)(1-\kappa_n)}$$

and therefore

$$(I - D_n^{\text{match}})^{-1} = \frac{\text{adj}(P_n)}{(1-r_n)(1-\beta_n)(1-\kappa_n)}$$

Remark C.10 (Compactness in the $\mathcal{F} = 1$ limit). Each entry of $(I - D_n^{\text{match}})^{-1}$ is a ratio of a polynomial with 1–3 terms (numerator, from the adjugate) over a polynomial with 2–3 terms (denominator $(1-r)(1-\beta)(1-\kappa)$). These are genuinely compact closed-form expressions in the TKF parameters $(\alpha_n, \beta_n, \gamma_n, \kappa_n, r_n)$. For $\mathcal{F} > 1$ the inversion is still closed-form (a rational function in $r^{(n)}, w_n, \alpha_n, \beta_n, \gamma_n, \kappa_n$), via the Kronecker-plus-rank-3 Woodbury decomposition of Section C.5.6: the only inversion of dimension exceeding 3×3 is a single $\mathcal{F} \times \mathcal{F}$ inverse $(I_{\mathcal{F}} - r^{(n)})^{-1}$ per domain, which can itself be written symbolically via the matrix adjugate.

C.5.8 Bilinear Factored Form of Adjacency Frequencies

The Woodbury identity gives the full path-sum matrix in a form that makes the adjacency computation practical for any $\mathcal{N}, \mathcal{F}$.

Structure of the path-sum matrix Writing $G_n = (I - D_n)^{-1}$ ($5\mathcal{F} \times 5\mathcal{F}$ per domain), the Woodbury expansion gives, for states s_1 in (n_1, f_1) and s_2 in (n_2, f_2) :

$$[(I - \hat{T})^{-1}]_{s_1 s_2} = \delta_{n_1 n_2} [G_{n_1}]_{s_1 s_2} + [G_{n_1} \mathbf{e}_{n_1}]_{s_1}^{\top} \mathcal{T}_{\text{mid}} K [\mathbf{s}_{n_2}^{\top} G_{n_2}]_{s_2}$$

where $K = (I_3 - \sum_n \mathbf{s}_n^{\top} G_n \mathbf{e}_n \mathcal{T}_{\text{mid}})^{-1}$ is the 3×3 Woodbury kernel (push-through form), computed once.

Bilinear form of structural weights The structural weight $C_{(n_1, f_1), (n_2, f_2)}^{\alpha\beta}$ decomposes as a within-domain diagonal plus a cross-domain bilinear form:

$$C_{(n_1, f_1), (n_2, f_2)}^{\alpha\beta} = \delta_{n_1 n_2} c_{n_1 f_1 f_2}^{\alpha} + \mathbf{a}_{n_1 f_1}^{\alpha\top} M^{-1} \mathbf{b}_{n_2 f_2}^{\beta}$$

where $\mathbf{a}_{nf}^{\alpha}, \mathbf{b}_{nf}^{\beta} \in \mathbb{R}^5$ are per-(domain, fragment-type) vectors derived from G_n , $\mathbf{e}_n$, $\mathbf{s}_n$, and the emission-type projections, and $c_{nf_1 f_2}^{\alpha}$ is the within-domain entry of G_n between fragment-types f_1 and f_2 at emission type α (a $\mathcal{F} \times \mathcal{F}$ matrix per domain rather than a scalar).

Closed-form adjacency formula Substituting into the adjacency frequency (C.98) and using the per-(domain, fragment-type) class-mixture emissions ϕ_{nf}^α from Section C.5.2, the full adjacency table takes the form:

$$f^{\alpha\beta}(\text{chars}_L, \text{chars}_R) = \underbrace{\left[\sum_{n,f} \mathbf{a}_{nf}^\alpha \phi_{nf}^\alpha(\text{chars}_L) \right]^\top M^{-1} \left[\sum_{n,f} \mathbf{b}_{nf}^\beta \phi_{nf}^\beta(\text{chars}_R) \right]}_{\text{cross-domain: bilinear in two 5-vectors}} + \underbrace{\sum_n \sum_{f_1, f_2} c_{nf_1 f_2}^\alpha \phi_{nf_1}^\alpha(\text{chars}_L) \phi_{nf_2}^\beta(\text{chars}_R)}_{\text{within-domain: } \mathcal{F} \times \mathcal{F} \text{ inner sum}} \quad (\text{C.101})$$

This is algebraic closed form: every quantity appearing— G_n entries, $\mathbf{a}_{nf}^\alpha$, $\mathbf{b}_{nf}^\beta$, $c_{nf_1 f_2}^\alpha$, M^{-1} , and ϕ_{nf}^α —is a rational function of the model parameters (composed with the matrix exponentials $\exp(R^{(c)}T)$). All inversions reduce to closed-form rational expressions: the only matrix inversion of dimension exceeding 3×3 in the entire distillation pipeline is $(I_{\mathcal{F}} - r^{(n)})^{-1}$ per domain (Section C.5.6), which is a single $\mathcal{F} \times \mathcal{F}$ adjugate-over-determinant. The Woodbury correction M^{-1} is a single 3×3 inversion regardless of $\mathcal{N}, \mathcal{F}, C$.

Remark C.11 (Practical computation for $\mathcal{N} = \mathcal{F} = C = 3$). For $\mathcal{N} = 3, \mathcal{F} = 3, C = 3$:

1. Precompute $C = 3$ per-class transition kernels $\exp(R^{(c)}T)$.
2. Precompute 3 within-domain inverses G_n (each 15×15 , i.e. $5\mathcal{F} = 15$).
3. Precompute 9 pairs $(\mathbf{a}_{nf}^\alpha, \mathbf{b}_{nf}^\beta)$ for each (α, β) pair, plus the within-domain $\mathcal{F} \times \mathcal{F}$ matrices $c_{nf_1 f_2}^\alpha$.
4. Accumulate $\Sigma = \sum_n \mathbf{s}_n^\top G_n \mathbf{e}_n$ and compute the push-through kernel K (one 3×3 inversion).
5. For each character entry: evaluate (C.101) by summing 9 scaled 3-vectors for the left and right factors of the cross-domain term, then a 3×3 bilinear product, plus a within-domain double sum over (f_1, f_2) .

The per-entry cost is $O(\mathcal{N}\mathcal{F} + C)$ multiplications.

C.5.9 Full-Context Distillation: Passthrough Context for Insert and Delete

In a two-tape transducer, the *state* at any point should encode the most recent character on *each* tape:

- After Match(X, Y): context is (X, Y) — both tapes updated.
- After Insert emitting Y' : context is (X, Y') — ancestor context X unchanged (passthrough from prior Match or Delete), descendant updated to Y' .
- After Delete consuming X' : context is (X', Y) — descendant context Y unchanged (passthrough from prior Match or Insert), ancestor updated to X' .

The adjacency frequencies in Section C.5 track only partial context for Insert and Delete states: $\phi_{nf}^I(Y) = \sum_c u_{nfc} \pi_Y^{(c)}$ depends only on the descendant character, and $\phi_{nf}^D(X) = \sum_c u_{nfc} \pi_X^{(c)}$ depends only on the ancestor character. This loses information about which (domain, fragment-type, class) generated the passthrough context.

We now show that the full-context adjacency frequencies—with both (X, Y) tracked in all states—can be computed in closed algebraic form, preserving the Woodbury structure.

Insert chain Green's function Define the *Insert chain* as the sub-process restricted to Insert states $\mathcal{I} = \{\text{MI}_{nf}, \text{II}_{nf} : n \in \mathcal{N}, f \in \mathcal{F}\}$. The restricted transition matrix $T_{\text{eff}}^{\mathcal{II}}$ (Insert→Insert transitions within $\hat{T}$) has the same block-diagonal-plus-low-rank structure as the full $\hat{T}$:

$$T_{\text{eff}}^{\mathcal{II}} = D^{II} + E^{II} \mathcal{T}_{\text{mid}}^{II} S^{II\top}$$

where $D^{II} = \text{diag}(D_1^{II}, \dots, D_N^{II})$ with each D_n^{II} being the within-domain Insert self-loop block ($2\mathcal{F} \times 2\mathcal{F}$ for $\{\text{MI}_{nf}, \text{II}_{nf}\}$ over fragment-types), and the cross-domain coupling has rank ≤ 2 through the Insert rows of $\mathcal{T}_\bullet$.

The Insert chain Green's function is therefore

$$G^{II} = (I - T_{\text{eff}}^{\mathcal{II}})^{-1}$$

computable via Woodbury with a kernel of dimension ≤ 2 . Each within-domain inverse $(I - D_n^{II})^{-1}$ is a $2\mathcal{F} \times 2\mathcal{F}$ inversion — simpler than the $5\mathcal{F} \times 5\mathcal{F}$ within-domain block of the full system.

An analogous *Delete chain Green's function* $G^{DD} = (I - T_{\text{eff}}^{\mathcal{DD}})^{-1}$ handles the descendant passthrough through Delete states, with identical structure.

Ancestor-conditioned structural weights To enter the Insert chain with ancestor context X , the process must have come from a prior Match or Delete state in some (n_0, f_0) that emitted ancestor X . The match emission satisfies $\sum_Y \phi_{nf}^{\text{M}}(X, Y) = \phi_{nf}^{\text{D}}(X)$ (row-stochastic per ancestor), so the entry weight from (n_0, f_0) with ancestor X is proportional to $\phi_{n_0 f_0}^{\text{D}}(X)$ for both Match and Delete source states.

Define the ancestor-conditioned left vector:

$$\tilde{L}_{n_1 f_1}^{\mathcal{I}}(X) = \sum_{n_0, f_0} \underbrace{(L_{\text{MM}n_0 f_0} + L_{\text{MD}n_0 f_0} + L_{\text{DD}n_0 f_0})}_{\lambda_{n_0 f_0}} \cdot \phi_{n_0 f_0}^{\text{D}}(X) \cdot [\hat{T}_{\text{entry}} G^{II}]_{(n_0, f_0), (n_1, f_1)}$$

where L_s are the standard left boundary weights from the full path-sum computation, $\lambda_{n_0 f_0}$ collects all ancestor-emitting (Match and Delete) boundary contributions for (n_0, f_0) , $\hat{T}_{\text{entry}}$ is the entry transition from ancestor-emitting states $\{\text{MM}, \text{MD}, \text{DD}\}$ into the Insert chain (i.e. the $\{M, D\} \rightarrow \{I\}$ block of $\hat{T}$), and $G_{(n_0, f_0), (n_1, f_1)}^{II}$ sums over all Insert-chain continuations.

Key observation: $\phi_{n_0 f_0}^{\text{D}}(X)$ factors out multiplicatively from the structural sum, preserving the bilinear structure.

Full-context adjacency for Insert-sourced transitions The full-context adjacency for Insert→Match is:

$$f_{\text{full}}^{\mathcal{IM}}(X, Y, X', Y') = \sum_{(n_1, f_1), (n_2, f_2)} \tilde{L}_{n_1 f_1}^{\mathcal{I}}(X) \cdot \eta_{(n_1, f_1), (n_2, f_2)}^{\mathcal{IM}} \cdot \phi_{n_1 f_1}^{\text{I}}(Y) \cdot \phi_{n_2 f_2}^{\text{M}}(X', Y') \quad (\text{C.102})$$

where $\eta_{(n_1, f_1), (n_2, f_2)}^{\mathcal{IM}} = T_{I_{n_1 f_1}, M_{n_2 f_2}} \cdot R_{M_{n_2 f_2}}$ and Y is the descendant character at the Insert, X the passthrough ancestor.

The bilinear factored form (cf. (C.101)) generalizes to:

$$f_{\text{full}}^{\mathcal{IM}}(X, Y, X', Y') = \left[\sum_{n, f} \tilde{\mathbf{a}}_{nf}^{\mathcal{I}}(X) \phi_{nf}^{\text{I}}(Y) \right]^\top K_I \left[\sum_{n, f} \mathbf{b}_{nf}^{\mathcal{M}} \phi_{nf}^{\text{M}}(X', Y') \right] + \sum_{n, f_1, f_2} (\text{diagonal terms}) \quad (\text{C.103})$$

where $\tilde{\mathbf{a}}_{nf}^{\mathcal{I}}(X)$ are modified per-(domain, fragment-type) left vectors incorporating the ancestor context through $\tilde{L}_{nf}^{\mathcal{I}}(X)$, and K_I is the Woodbury kernel for the Insert-chain correction.

Similarly, for Insert→Insert with ancestor passthrough:

$$f_{\text{full}}^{\mathcal{II}}(X, Y, X, Y') = \sum_{(n_1, f_1), (n_2, f_2)} \tilde{L}_{n_1 f_1}^{\mathcal{I}}(X) \cdot \eta_{(n_1, f_1), (n_2, f_2)}^{\mathcal{II}} \cdot \phi_{n_1 f_1}^{\mathcal{I}}(Y) \cdot \phi_{n_2 f_2}^{\mathcal{I}}(Y')$$

where the ancestor X is preserved on both sides (3 effective character dimensions, not 4).

By symmetry, the Delete-sourced adjacencies with descendant passthrough are:

$$f_{\text{full}}^{\mathcal{DM}}(X, Y, X', Y') = \sum_{(n_1, f_1), (n_2, f_2)} \tilde{L}_{n_1 f_1}^{\mathcal{D}}(Y) \cdot \eta_{(n_1, f_1), (n_2, f_2)}^{\mathcal{DM}} \cdot \phi_{n_1 f_1}^{\mathcal{D}}(X) \cdot \phi_{n_2 f_2}^{\mathcal{M}}(X', Y') \quad (\text{C.104})$$

where $\tilde{L}_{n_1 f_1}^{\mathcal{D}}(Y)$ is the descendant-conditioned left vector for the Delete chain, defined analogously with $G^{\mathcal{DD}}$.

Computational cost The affected adjacency tables gain one character dimension:

Tensor	Original	Full-context	Per-entry cost
$f^{\mathcal{MM}}$	$ \mathcal{A} ^4$	$ \mathcal{A} ^4$ (unchanged)	$O(\mathcal{NF} + C)$
$f^{\mathcal{MI}}$	$ \mathcal{A} ^3$	$ \mathcal{A} ^3$ (unchanged)	$O(\mathcal{NF} + C)$
$f^{\mathcal{MD}}$	$ \mathcal{A} ^3$	$ \mathcal{A} ^3$ (unchanged)	$O(\mathcal{NF} + C)$
$f^{\mathcal{IM}}$	$ \mathcal{A} ^3$	$ \mathcal{A} ^4$	$O(\mathcal{NF} + C)$
$f^{\mathcal{II}}$	$ \mathcal{A} ^2$	$ \mathcal{A} ^3$ (X preserved)	$O(\mathcal{NF} + C)$
$f^{\mathcal{ID}}$	$ \mathcal{A} ^2$	$ \mathcal{A} ^3$ (X preserved, $Y \rightarrow X'$)	$O(\mathcal{NF} + C)$
$f^{\mathcal{DM}}$	$ \mathcal{A} ^3$	$ \mathcal{A} ^4$	$O(\mathcal{NF} + C)$
$f^{\mathcal{DD}}$	$ \mathcal{A} ^2$	$ \mathcal{A} ^3$ (Y preserved)	$O(\mathcal{NF} + C)$
$f^{\mathcal{DI}}$	$ \mathcal{A} ^2$	$ \mathcal{A} ^3$ (Y preserved, $X \rightarrow Y'$)	$O(\mathcal{NF} + C)$

The per-entry cost is $O(\mathcal{NF} + C)$ after Woodbury factoring (cf. the same per-entry cost for $f^{\mathcal{MM}}$ noted in Remark on p. 139), since the Insert and Delete chain Green’s functions have the same bilinear decomposition. The total cost for the full adjacency table is $O((\mathcal{NF} + C) \cdot |\mathcal{A}|^4)$ — the same asymptotic scaling as the existing match-to-match computation. The Woodbury kernel remains 3×3 for the full system; the Insert and Delete chain kernels are $\leq 2 \times 2$. All quantities remain closed-form rational functions of the model parameters (composed with the C rate-matrix exponentials).

Remark C.12 (Why the original formulation lost context). The emission functions $\phi_{nf}^{\mathcal{I}}(Y)$ and $\phi_{nf}^{\mathcal{D}}(X)$ are genuinely single-character: the Insert emission does not depend on the ancestor, and vice versa. The lost context is not an emission effect but a *transition routing* effect: the joint posterior $P((n, f, c) \mid X, Y)$ depends on both characters, so the passthrough character informs which (domain, fragment-type, class) we are in, and hence which transition probabilities apply. The correction above recovers this information by conditioning the structural weights on the passthrough character.

C.5.10 Domains versus Fragments versus Classes for Adjacency Capture

The order-1 WFST has $O(|\mathcal{A}|^4)$ free parameters (from match-to-match transition weights parameterized by context (X, Y) and next pair (a', b')), while the MixDom model has far fewer. We analyse which type of model complexity most efficiently captures adjacency structure.

Adjacency tensor rank The match-to-match adjacency table $f^{\mathcal{M}\mathcal{M}}(X, Y, X', Y')$ is a sum of $(\mathcal{NF})^2$ structural-weight terms (in the (domain, fragment-type)-pair sense), each weighted by a rank- C emission factor on each side. To approach saturating the WFST capacity ($|\mathcal{A}|^4$ entries), the tensor rank must approach $|\mathcal{A}|^2$; this is achievable through a combination of $(\mathcal{N}, \mathcal{F}, C)$.

Domains: independent TKF rates and weights Each domain type n brings its own $(\alpha_n, \beta_n, \gamma_n, \kappa_n)$, creating genuinely different:

- M $\rightarrow$ D vs. M $\rightarrow$ I ratios (different $\kappa_n(1 - \alpha_n)$ vs. β_n),
- V vs. W transition behaviour (different β_n vs. γ_n),
- overall indel/match balance.

These features drive cross-type adjacency diversity in the WFST and are not reproducible by fragments or classes alone.

Fragments: vary intra-fragment transition pattern as well as boundary frequency The intra-fragment Markov ext matrix $r_{fg}^{(n)}$ allows transitions *between fragment-types within a single fragment*, without invoking the TKF92 new-fragment branch. These intra-fragment $f \rightarrow g$ transitions at each emitted site can carry persistent emission-class context: a chain that prefers one fragment-type tends to stay in that fragment-type’s class mixture, producing richer M/I/D adjacency patterns than fragments without intra-fragment correlation. The off-diagonal entries of $r^{(n)}$ thus contribute genuine new M/I/D adjacency content; setting them to zero (**-freeze-offdiag-ext** in the fitter) recovers the diagonal-extension special case behaviour where fragments only modulate boundary frequency, and the $\mathcal{F} = 1$ case collapses fragments to a scalar self-extension per domain.

Classes: emission-mixture diversity, decoupled from structure Each site class c has its own GTR $(Q^{(c)}, \pi^{(c)})$. Classes contribute *emission* diversity: the rank- C class-mixture emission $\phi_{nf}^{\mathbf{M}, \mathbf{I}, \mathbf{D}}$ allows the joint match emission tensor to deviate from a single GTR’s product structure, even within a single (d, f) . Classes do not contribute to the M/I/D transition pattern (they do not appear in $\hat{T}$ or in any of the structural weights $C^{\alpha\beta}$); they affect only the character-side of each adjacency frequency. Sharing the class pool across (d, f) via u gives a low-parameter way to enrich the emission tensor without growing the latent-state space.

Remark C.13 (Parameter allocation). For adjacency capture in MixDom:

- Domains are essential to enrich the M/I/D transition pattern via independent TKF rates.
- Fragments enrich the M/I/D structure (via the intra-fragment Markov ext) and the boundary frequency (via $\rho_f^{(n)}$); they also support an additional layer of emission diversity through u_{nfc} .
- Classes enrich emissions only, but cheaply (decoupled from $\mathcal{N}, \mathcal{F}$): with C classes, every (d, f) has access to the full pool via its Dirichlet u_{nfc} .

The total adjacency-tensor rank scales as $\mathcal{NF}$ (structural-weight pairs) $\times C^2$ (left/right class mixture); in practice, allocating the parameter budget across all three axes gives the most efficient match to the WFST capacity.

C.5.11 Identifiability

Are the MixDom parameters recoverable from the distilled order-1 models?

Generic identifiability (up to standard ambiguities). The distilled WFST provides $O(|\mathcal{A}|^4)$ constraints on the MixDom parameters. With $\mathcal{N}$ domains, $\mathcal{F}$ fragment types per domain, and C site classes, there are $2 + (\mathcal{N} - 1) + 2\mathcal{N} + \mathcal{N}\mathcal{F}^2 + \mathcal{N}\mathcal{F}(C - 1) + C(|\mathcal{A}|^2 + |\mathcal{A}| - 2)$ free parameters (the last term accounts for C symmetric exchangeability matrices and equilibrium distributions). The system is heavily over-determined for moderate $(\mathcal{N}, \mathcal{F}, C)$.

The per-domain TKF parameters $(\lambda_n T, \mu_n T, r_{fg}^{(n)})$ are recoverable from the within-(domain, fragment-type) M/I/D transition structure: α_n and κ_n determine $\mu_n T$ and λ_n/μ_n ; β_n provides a redundant check; $r^{(n)}$ separates from κ_n because intra-fragment Markov transitions decompose differently from new-fragment transitions (see Section C.3.5 of the Maraschino fitter and Baum–Welch derivation). The top-level parameters $(\lambda_0 T, \mu_0 T)$ are identifiable from the inter-domain transition pattern. The per-class GTRs $(Q^{(c)}, \pi^{(c)})$ and per-(domain, fragment-type) u_{nfc} are jointly identifiable up to label permutation of classes, because the WFST context dependence reveals the mixture components as the joint posterior shifts across different (X, Y) contexts.

Unavoidable ambiguities.

1. **Label permutation:** permuting domain labels gives the same model ($\mathcal{N}!$ -fold), permuting fragment-type labels at fixed domain gives the same model ($\mathcal{F}!$ -fold), and permuting class labels gives the same model ($C!$ -fold).
2. **Rate-time confounding:** only products $\lambda T, \mu T$ are identifiable from pairwise data (a single evolutionary time).
3. **Class-domain mixing:** when $C \geq \mathcal{N}\mathcal{F}$, the class structure can absorb domain-level emission differences, creating identifiability issues unless the structural weights $C^{\alpha\beta}$ resolve them.

Lossy in distribution, injective in parameters. The distilled model captures only pairwise-adjacent correlations; the full MixDom has higher-order structure (e.g., runs of characters from the same fragment, intra-fragment Markov correlations between fragment-types). The distillation map $\text{MixDom} \rightarrow \text{order-1}$ is therefore *lossy* for the sequence distribution but generically *injective* for the parameters: one can recover the MixDom parameters from the distilled model, even though the distilled model cannot reproduce all statistics of the MixDom.

C.5.12 Scaling to $\mathcal{N}, \mathcal{F}, C$

The top-level null closure does not grow with $\mathcal{N}, \mathcal{F}, C$ The null states A, B, C are always exactly three, regardless of the latent-state cardinalities. Their transition submatrix depends on $\mathcal{N}$ only through the scalars $z_0 = \sum_n v_n(1 - \kappa_n)$ and $z_T = \sum_n v_n(1 - \kappa_n)(1 - \beta_n)$; $\mathcal{F}$ and C do not appear at this top level. The null closure and the effective 5×5 top-level matrix $\mathcal{T}$ are $O(\mathcal{N})$ to compute but $O(1)$ in matrix dimension.

Domain types are drawn i.i.d., not as a Markov chain When a new domain begins, its type is drawn independently from v , regardless of the previous domain’s type. This i.i.d. structure means the cross-domain transitions factor as $\hat{T}(s, s') = e_l(s) \times \mathcal{T}_\bullet \times s_m(s')$ (exit $\times$ top-level $\times$ start), and the cross-domain contribution to the full $5\mathcal{N}\mathcal{F} \times 5\mathcal{N}\mathcal{F}$ transition matrix has rank at most 3. Note that this i.i.d. property holds at the *domain* level only; within a fragment, the fragment-type process is a Markov chain on $\mathcal{F}$ states (see Section C.1.1, Equation C.3), which is encoded in the within-domain block D_n and does not break the cross-domain factorisation.

Woodbury for general $\mathcal{N}, \mathcal{F}$ The emitting state space has $5\mathcal{N}\mathcal{F}$ states. The effective transition matrix decomposes as

$$\hat{T} = D + E \mathcal{T}_\bullet S^\top$$

where $D = \text{diag}(D_1, \dots, D_{\mathcal{N}})$ with each D_n a $5\mathcal{F} \times 5\mathcal{F}$ within-domain block (now domain-specific and, for $\mathcal{F} > 1$, with intra-fragment Markov coupling), $E \in \mathbb{R}^{5\mathcal{N}\mathcal{F} \times 3}$ stacks exit vectors projected to $\{\mathbf{M}, \mathbf{I}, \mathbf{D}\}$, and $S \in \mathbb{R}^{5\mathcal{N}\mathcal{F} \times 3}$ stacks start vectors projected to $\{\mathbf{M}, \mathbf{I}, \mathbf{D}\}$.

The Woodbury identity gives $(I - \hat{T})^{-1}$ via:

1. $\mathcal{N}$ independent $5\mathcal{F} \times 5\mathcal{F}$ inversions $(I - D_n)^{-1}$, each evaluated in closed form via the Kronecker-plus-rank-3 inner Woodbury of Section C.5.6: one $\mathcal{F} \times \mathcal{F}$ adjugate $(I_{\mathcal{F}} - r^{(n)})^{-1}$ plus a 3×3 inner kernel per domain. For $\mathcal{F} = 1$ this further collapses to a 3×3 adjugate (closed-form determinant $(1 - r)(1 - \beta)(1 - \kappa)$) plus two scalar inversions (Section C.5.7). The total cost is $O(\mathcal{N}\mathcal{F}^3)$ per evaluation.
2. Computing $\sum_n \mathbf{s}_n^\top G_n \mathbf{e}_n$, a 3×3 matrix built by summing $\mathcal{N}$ contributions: $O(\mathcal{N}\mathcal{F}^2)$ work.
3. One 3×3 inversion for the Woodbury correction: $O(1)$ work.

Proposition C.2 (Linear-in- $\mathcal{N}$, cubic-in- $\mathcal{F}$, linear-in- C scaling). *The order-1 distillation computation scales as $O(\mathcal{N}\mathcal{F}^3)$ in within-domain inversions and $O(C)$ in per-class GTR exponentials:*

- $O(\mathcal{N})$ to compute z_0, z_T and the 5×5 top-level matrix.
- $O(\mathcal{N}\mathcal{F}^3)$ for within-domain path sums.
- $O(\mathcal{N}\mathcal{F}^2)$ to accumulate the 3×3 Woodbury correction.
- $O((\mathcal{N}\mathcal{F})^2)$ (domain, fragment-type)-pair terms per adjacency entry (or $O((\mathcal{N}\mathcal{F})^2 |\mathcal{A}|^d \cdot C)$ for the full adjacency table, with $d \in \{2, 3, 4\}$ character dimensions depending on state type; see Section C.5.9).
- $O(C|\mathcal{A}|^3)$ for per-class transition kernels $\exp(R^{(c)}T)$.

The Woodbury kernel is always 3×3 , regardless of the latent state cardinalities.

C.5.13 Summary

Component	Matrix size	Scaling
Top-level null closure	3×3	$O(1)$
Top-level eff. matrix	5×5	$O(\mathcal{N})$ to compute
Within-domain inversions	$5\mathcal{F} \times 5\mathcal{F}$ each	$O(\mathcal{N}\mathcal{F}^3)$
Woodbury correction	3×3	$O(\mathcal{N}\mathcal{F}^2)$ to accumulate
Per-class GTR exponentials	$ \mathcal{A} \times \mathcal{A} $	$O(C \mathcal{A} ^3)$
(Domain, fragment-type)-pair adjacency terms	—	$O((\mathcal{N}\mathcal{F})^2 C)$ per entry

The distillation is closed-form for any finite $\mathcal{N}, \mathcal{F}, C$: all quantities are rational functions of the model parameters (composed with C exponentials of rate-times-time products). The key structural features enabling this are:

1. **Top-level null closure is $O(1)$ in dimension.** The MixDom null states $\mathbf{A}, \mathbf{B}, \mathbf{C}$ are always exactly three, regardless of $\mathcal{N}, \mathcal{F}, C$. All dependence on $\mathcal{N}$ enters through the scalars z_0, z_T .

2. **Cross-domain transitions factor.** The exit-vector $\times$ top-level $\times$ start-vector structure caps the rank of the inter-domain coupling at 3 (the M, I, D top-level states that couple across domains), enabling Woodbury reduction to a fixed 3×3 inversion.
3. **Domain types are i.i.d., not Markov.** If domain type depended on the previous domain (a Markov chain on domain types), the cross-domain block would lose its factored structure and the Woodbury reduction would fail—the full $5\mathcal{NF} \times 5\mathcal{NF}$ inversion would be required, scaling as $O((\mathcal{NF})^3)$. The domain-level i.i.d. mixture is what keeps the cross-domain part linear in $\mathcal{N}$. The intra-fragment Markov chain on fragment-types is encoded inside D_n and does not break this factorisation.
4. **Class mixture is bilinear in emissions.** The class index c enters every emission as a linear sum weighted by u_{nfc} , never inside the structural weights $C^{\alpha\beta}$. This makes the class mixture a low-rank emission factor that does not interact with the block-diagonal-plus-low-rank structure of the transition matrix.

C.6 MixDom-Specific SVI-BW Convergence Considerations

This appendix specialises the model-agnostic convergence analysis of Appendix B.2 and the BDI expected-statistic formulae of Appendix B.3 to the hierarchical MixDom model. Each subsection corresponds to a question that arises specifically because MixDom carries multiple interacting parameter groups (top-level vs. per-domain BDI, intra-fragment fragment-type chains, and per-class substitution models).

C.6.1 Parameter groups and Fisher information

The MixDom model has several parameter groups, each with different Fisher information characteristics:

Parameter group	# params	Info per pair	Bottleneck
Top-level (λ_0, μ_0)	2	$O(1)$	few domains per sequence
Per-domain (λ_d, μ_d)	$2N_{\text{dom}}$	$O(w_d)$	domain frequency w_d
Intra-fragment fragment-type transitions $r_{fg}^{(d)}$	$N_{\text{dom}}F^2$	$O(w_d\bar{L}_d)$	fragment count
Domain weights w_d	$N_{\text{dom}} - 1$	$O(1)$	multinomial
Substitution Q_c	$O(\mathcal{A} ^2C)$	$O(\bar{L} \cdot w_c)$	alignment length

The bottleneck for SVB convergence is the **indel parameters of rare domains**. A domain with frequency $w_d = 0.05$ contributes useful BDI statistics to only $\sim 5\%$ of pairs, so its effective sample size is $\sim 0.05B$ per minibatch.

C.6.2 Substitution vs. indel information

Substitution parameters benefit from $O(\bar{L})$ information per pair (one observation per aligned column), while indel parameters get $O(1)$ information per pair. For a protein with $\bar{L} \approx 200$ residues, substitution parameters converge $\sim 200\times$ faster than indel parameters. This motivates **decoupled update frequencies**: update substitution parameters every minibatch, but average indel parameter estimates over multiple minibatches.

C.6.3 MixDom expected statistics

We now aggregate the per-process expectations of Appendix B.3 to the full MixDom model with N_{dom} domain types.

Top-level (domain birth-death) The top-level BDI process has rates (λ_0, μ_0) with $\kappa_0 = \lambda_0/\mu_0$ and creates/destroys domains. At stationarity:

$$L_0 = \kappa_0/(1 - \kappa_0) \quad (\text{expected } \# \text{ domains per sequence}) \quad (\text{C.105})$$

$$\mathbb{E}[B_0] = \lambda_0 T/(1 - \kappa_0) \quad (\text{domain births per pair}) \quad (\text{C.106})$$

$$\mathbb{E}[D_0] = \mathbb{E}[B_0] \quad (\text{domain deaths per pair}) \quad (\text{C.107})$$

$$\mathbb{E}[S_0] = L_0 T \quad (\text{time-integrated domain count}) \quad (\text{C.108})$$

$$M_0 = 1, \quad T_0 = T \quad (\text{one endpoint, one process}) \quad (\text{C.109})$$

Per-domain type d (fragment birth-death within a domain) Each surviving domain link of type d (with probability w_d) contains a TKF92 fragment process with rates (λ_d, μ_d) , $\kappa_d = \lambda_d/\mu_d$.

The number of domain links of type d is approximately $L_d = w_d L_0$ at stationarity. Each such link contributes one independent BDI fragment process, so the *aggregated* fragment-level statistics for domain type d are:

$$L_d = w_d L_0 \cdot \frac{\kappa_d}{1 - \kappa_d} \quad (\text{total fragment links, type } d) \quad (\text{C.110})$$

$$\mathbb{E}[B_d] = w_d L_0 \cdot \frac{\lambda_d T}{1 - \kappa_d} \quad (\text{fragment births, type } d, \text{ per pair}) \quad (\text{C.111})$$

$$\mathbb{E}[D_d] = \mathbb{E}[B_d] \quad (\text{fragment deaths, type } d, \text{ per pair}) \quad (\text{C.112})$$

$$\mathbb{E}[S_d] = w_d L_0 \cdot \frac{\kappa_d T}{1 - \kappa_d} \quad (\text{time-integrated fragment count, type } d) \quad (\text{C.113})$$

$$M_d = w_d L_0 \quad (\# \text{ independent processes}) \quad (\text{C.114})$$

$$T_d = M_d T = w_d L_0 T \quad (\text{total observation time}) \quad (\text{C.115})$$

Note that M_d and T_d are themselves random (they depend on how many domains of type d survive), but at stationarity their expectations are as above.

Intra-fragment fragment-type transitions and sequence length Within each fragment of domain d the fragment-type chain is governed by an $F \times F$ Markov transition matrix $r_{fg}^{(d)}$ (intra-fragment; different fragments are independent realisations). The termination probability from fragment-type f is $\rho_f^{(d)} = 1 - \sum_g r_{fg}^{(d)}$. The expected sojourn length starting from fragment state f is determined by the fundamental matrix $(I - r^{(d)})^{-1}$: specifically, the expected number of sites emitted starting from a fragment initiated in state f is $\sum_g [(I - r^{(d)})^{-1}]_{fg}$. Averaging over the initial fragment distribution $w_{d,f}$, the expected number of residues per fragment is $\bar{K}_d = \sum_f w_{d,f} \sum_g [(I - r^{(d)})^{-1}]_{fg}$.

The expected number of residues per domain of type d is:

$$\bar{C}_d = \frac{\kappa_d}{1 - \kappa_d} \bar{K}_d, \quad (\text{C.116})$$

and the total expected sequence length is:

$$\bar{L}_{\text{seq}} = L_0 \sum_d w_d \bar{C}_d = \frac{\kappa_0}{1 - \kappa_0} \sum_d w_d \frac{\kappa_d}{1 - \kappa_d} \bar{K}_d. \quad (\text{C.117})$$

Remark C.14 (Convergence of Markov chain transition counts). When fragments are IID (the $\mathcal{F} = 1$ scalar-extension special case), the sufficient statistics for the fragment extension parameter r_f are Bernoulli counts (extend vs. terminate), and the convergence analysis reduces to a Beta-distributed posterior. With the general fragment-type Markov, the sufficient statistics are rows of a Markov chain transition count matrix: for each domain d and source fragment state f , we observe counts $\hat{n}_{fg}^{(d)}$ of transitions to each target state g plus termination counts $\hat{n}_{f,\text{end}}^{(d)}$. The M-step row-normalizes these counts (a Dirichlet posterior), and the per-pair Fisher information for each row scales with the expected number of visits to state f per pair. For fragment states that are rarely visited (low stationary probability under $r^{(d)}$), the effective sample size is correspondingly small, mirroring the rare-domain bottleneck at the top level.

Numerical example: d3 checkpoint We evaluate these formulas using the MixDom d3 checkpoint parameters:

$$\begin{aligned}\lambda_0 &= 0.01328, & \mu_0 &= 0.01412, & \kappa_0 &= 0.9405 \\ w &= (0.662, 0.075, 0.264) \\ \lambda_d &= (0.00302, 0.01033, 0.12370) \\ \mu_d &= (0.00372, 0.04686, 0.17775) \\ \kappa_d &= (0.812, 0.220, 0.696) \\ T &= 0.5\end{aligned}$$

	Top-level ($d=0$)	Dom 1 ($d=1$)	Dom 2 ($d=2$)	Dom 3 ($d=3$)
κ	0.941	0.812	0.220	0.696
$L = \kappa/(1-\kappa)$	15.81	4.31	0.283	2.29
$w_d \cdot L_0$	—	10.47	1.19	4.17
$\mathbb{E}[B]$ per pair	0.112	0.084	0.008	0.849
$\mathbb{E}[D]$ per pair	0.112	0.084	0.008	0.849
$\mathbb{E}[S]$	7.91	22.58	0.168	4.78
M	1	10.47	1.19	4.17
T	0.5	5.23	0.59	2.09
v_θ (approx.)	~ 18	~ 24	~ 255	~ 2.4

Computation of the table entries. *Top level:* $L_0 = 0.9405/0.0595 = 15.81$. $\mathbb{E}[B_0] = 0.01328 \times 0.5/0.0595 = 0.112$. $\mathbb{E}[S_0] = 15.81 \times 0.5 = 7.91$.

Domain type 1 ($w_1 = 0.662$): $M_1 = 0.662 \times 15.81 = 10.47$ independent fragment processes. $L_1 = 10.47 \times 0.812/0.188 = 10.47 \times 4.31 = 45.1$ total fragment links. $\mathbb{E}[B_1] = 10.47 \times 0.00302 \times 0.5/0.188 = 0.084$. (The table shows L per process and $\mathbb{E}[B]$ aggregated over M_d processes.)

Domain type 3 ($w_3 = 0.264$): $M_3 = 0.264 \times 15.81 = 4.17$. $\mathbb{E}[B_3] = 4.17 \times 0.12370 \times 0.5/0.304 = 0.849$. This domain has high λ_3 , so fragment births are frequent and its indel parameters are easy to estimate ($v \approx 2.4$).

v_{λ_0} : Using (B.77) with $\rho \approx 0.5$: $v_{\lambda_0} \approx 2/\mathbb{E}[B_0] = 2/0.112 = 17.9$.

v_{λ_2} : $\mathbb{E}[B_2] = 1.19 \times 0.01033 \times 0.5/0.780 = 0.0079$. $v_{\lambda_2} \approx 2/0.008 = 255$.

C.6.4 Convergence rate estimates

From (B.62) and the per-pair relative variance v_θ , the number of pairs $N = BK$ needed for target relative error ε is:

$$N \geq \frac{v_\theta}{\varepsilon^2}. \quad (\text{C.118})$$

Parameter	v_θ	N for $\varepsilon=10\%$	N for $\varepsilon=5\%$	N for $\varepsilon=1\%$
λ_0 (top-level ins)	18	1 800	7 200	180 000
μ_0 (top-level del)	18	1 800	7 200	180 000
λ_1 (dom 1 ins)	24	2 400	9 600	240 000
λ_2 (dom 2 ins)	255	25 500	102 000	2 550 000
λ_3 (dom 3 ins)	2.4	240	960	24 000
w_d (domain weights)	$\sim 1/L_0 \approx 0.06$	6	25	630
$r_{fg}^{(d)}$ (fragment trans.)	$\sim 1/\bar{C}_d$	depends on domain		
Substitution (Q)	$\sim 1/\bar{L}_{\text{seq}}$	$\ll 100$		

C.6.5 Discussion: why top-level indel rates are hardest

The table reveals a clear hierarchy of estimation difficulty:

1. Substitution parameters are easiest. Each aligned residue pair contributes one independent observation to the substitution sufficient statistics. With $\bar{L}_{\text{seq}} \approx 200\text{--}400$ residues per sequence, the per-pair Fisher information for substitution parameters is $O(\bar{L}_{\text{seq}})$, so the per-pair relative variance is $O(1/\bar{L}_{\text{seq}})$. A few dozen pairs suffice for $\varepsilon = 5\%$ accuracy.

2. Domain weights converge fast. Each domain in the ancestor contributes one multinomial observation of the domain type. With $L_0 \approx 16$ domains per sequence, the per-pair Fisher information is $O(L_0)$, giving $v_{w_d} \sim 1/(w_d L_0)$. For the most common domain type ($w_1 = 0.662$), $v_{w_1} \approx 0.095$, and even for the rarest ($w_2 = 0.075$), $v_{w_2} \approx 0.84$. Domain weights are precisely estimated with $N \sim 100$ pairs.

3. Fragment-type transition parameters are moderate. Each intra-fragment fragment-type transition contributes one observation of the $F \times F$ Markov chain row $r_{f,:}^{(d)}$. With $\sim L_0 w_d \kappa_d / (1 - \kappa_d)$ fragments of type d per pair, the per-pair information scales with the total fragment count. For domain 1 (which dominates), there are ~ 45 fragment links, so the per-row variance scales as $\sim F/(45 \cdot \pi_f)$ where π_f is the stationary probability of fragment state f . Domain 2 fragments are rare (~ 0.3 per pair) and harder to estimate.

4. Indel parameters for active domains are easy. Domain type 3 has high insertion rate ($\lambda_3 = 0.124$) and $\mathbb{E}[B_3] \approx 0.85$ births per pair, giving $v_{\lambda_3} \approx 2.4$. A few hundred pairs suffice for 5% accuracy.

5. Indel parameters for common but slow domains are moderate. Domain type 1 has $\mathbb{E}[B_1] \approx 0.084$ births per pair, giving $v_{\lambda_1} \approx 24$. This requires $N \approx 10,000$ pairs for 5% accuracy—feasible but not trivial.

6. Indel parameters for rare domains are the bottleneck. Domain type 2 has weight $w_2 = 0.075$, low $\kappa_2 = 0.22$ (short domains), and low $\lambda_2 = 0.01$. The expected fragment births per pair are only $\mathbb{E}[B_2] \approx 0.008$. This means on average, only 1 in ~ 125 pairs shows even a single fragment birth event in a domain of type 2. The per-pair relative variance $v_{\lambda_2} \approx 255$ requires $N > 100,000$ pairs for 5% accuracy.

7. Top-level indel rates are intrinsically hard. The top-level BDI controls domain creation and destruction. With $\lambda_0 = 0.013$, $T = 0.5$, and $L_0 \approx 16$ domains, the expected number of domain births per pair is only $\mathbb{E}[B_0] \approx 0.11$. The per-pair relative variance $v_{\lambda_0} \approx 18$ requires $N \approx 7,200$ pairs for 5% accuracy.

The fundamental reason is that domain-level indel events are rare compared to residue-level observations. A sequence of 300 residues organized into 16 domains gives ~ 300 substitution observations but only ~ 0.1 domain birth observations per pair at $T = 0.5$. The information ratio is roughly $300/0.1 = 3,000\times$ in favor of substitution parameters.

Implications for training. These estimates motivate the decoupled update strategy recommended in Section 5: substitution parameters can be accurately estimated from small minibatches ($B \sim 10$), while indel parameters (especially for rare domain types) require accumulation over $N \sim 10^4\text{--}10^5$ pairs. The Maraschino pipeline (Section 5.5) achieves this by processing all training pairs in a single count tensor, at the cost of the composite-likelihood efficiency gap. The hybrid Maraschino $\rightarrow$ SVB pipeline is optimal: Maraschino provides accurate initial estimates using all data, and SVB refines the indel parameters using the full model.

C.7 Variational EM training of MixDom from tree-structured data

The variational ELBO of Appendix C.8 treats the model parameters $\theta = (\lambda_0, \mu_0, \{\lambda_n\}, \{\mu_n\}, v, w, r^{(n)}, u, \{\pi^{(c)}\}, \{S_{\text{ex}}^{(c)}\})$ as fixed inputs and optimises only the variational distribution q over per-(node, column) latent states. In this appendix we extend the framework to a Variational Bayesian EM (VBEM) training algorithm that learns θ by alternating between a per-family E-step (Adam ascent on the variational q) and a global M-step (closed-form θ update from aggregated sufficient statistics).

This is structurally analogous to the SVI Baum-Welch pipeline of Section C.1.4 (which trains MixDom from sequence *pairs* via the labelled Pair HMM), but operates on whole MSAs with their tree topology — each MSA family becomes a single training datum, with the variational distribution capturing the posterior over internal-node states and column-wide tuples.

C.7.1 Outer EM loop

Given the corpus $\{\mathcal{D}_i\}_{i=1}^N$ of MSA-with-tree training data and an initial parameter estimate $\theta^{(0)}$, the outer loop is

- **E-step** (per family i): fit the variational $q_i = (q_i^{(\tau)}, q_i^{(\pi|\tau)})$ by Adam ascent on the ELBO $\mathcal{L}_i(q_i; \theta^{(t)})$ at the current parameter estimate, returning per-family sufficient statistics $\Phi_i = \{W_i^{(v \rightarrow w)}, q_i^{(\tau)}, L_{i,n,c}^{(\text{sub})}\}$.
- **M-step** (global): aggregate sufficient statistics across families and apply closed-form parameter updates for each parameter group, yielding $\theta^{(t+1)}$.

The corpus may be subsampled minibatch-style each iteration (SVI-style), in which case the M-step uses an exponential-moving-average of sufficient statistics across iterations (Section B.1.13).

C.7.2 Per-family E-step

For family i with binary tree $\mathbf{t}_i$, branch lengths d_i , and observed leaf data X_i , the E-step maximises

$$\mathcal{L}_i(q_i; \theta) = \sum_{(v \rightarrow w) \in \mathbf{t}_i} \mathbb{E}_{q_i}[\log P^{\text{WFST}}(Z^w \mid Z^v)] + \sum_n \mathbb{E}_{q_i^{(f)}}[\log L_n^{(\text{sub}), \text{tot}}] + \log p_{\text{singlet}}^{\text{red}}(Z^{\text{root}}) + H[q_i^{(\tau)}] + H[q_i^{(\pi|\tau)}] + \log Z_q \quad (\text{C.119})$$

over the variational logits (edge_logits, root_logit, tuple_logits) via JIT-compiled Adam, identical to the inference-time ELBO of Appendix C.8 but treating θ as fixed input.

After convergence (typically 100–300 Adam iterations with $\text{lr} \approx 0.05$, Fitch-seeded init), the E-step extracts the following sufficient statistics from q_i :

Per-branch reduced expected counts. For each branch $(v \rightarrow w)$, the cumulant-trick prefix sum (Section C.8.6) yields

$$W_{ss', \tau\tau'}^{(v \rightarrow w)} = \sum_{N=1}^{L+1} \sum_{M=0}^{N-1} q_{n(M)}^{(\tau)}(\tau) q_{n(N)}^{(\tau')}(\tau') P_{q,M}^{v \rightarrow w}(s) P_{q,N}^{v \rightarrow w}(s') \prod_{K=M+1}^{N-1} P_{q,K}^{v \rightarrow w}(\mathbf{lg}), \quad (\text{C.120})$$

a $5 \times T \times 5 \times T$ tensor that summarises the expected labelled WFST transitions on this branch.

Per-column class posteriors. For each column n and class c , compute

$$q_n^{(c|f)}(c) = \frac{u_{f,c} L_n^{(\text{sub})}(c; \mathcal{F}_n)}{\sum_{c'} u_{f,c'} L_n^{(\text{sub})}(c'; \mathcal{F}_n)}, \quad q_n^{(c)}(c) = \sum_f q_n^{(f)}(f) q_n^{(c|f)}(c), \quad (\text{C.121})$$

where $\mathcal{F}_n$ is the Fitch subtree of column n and $L_n^{(\text{sub})}(c)$ is the Felsenstein up-pass likelihood under class- c rate matrix.

Per-class substitution counts. For each class c and column n , weighted Felsenstein expected substitution counts on $\mathcal{F}_n$:

$$\hat{M}_{i,n}^{(c)}[a, b] = q_n^{(c)}(c) \mathbb{E}[N_{ab}(\text{branch}; \mathcal{F}_n) \mid \text{leaf data}, c, \theta], \quad \hat{T}_{i,n}^{(c)}[a] = q_n^{(c)}(c) \mathbb{E}[\text{dwell time in } a \mid \text{leaf data}, c, \theta]. \quad (\text{C.122})$$

These are computed via the standard bridge-expectation posterior formulae for substitution-only CTMCs on trees, weighted by the per-class responsibility $q_n^{(c)}(c)$.

C.7.3 M-step from aggregated sufficient statistics

Let $\Phi = \{\Phi_i\}_i$ denote the per-family sufficient statistics aggregated across the (mini)batch. The M-step decomposes parameter group by parameter group, exploiting the route decomposition (C.144) and the responsibilities derived above.

Route attribution. For each per-character labelled WFST transition the route posterior

$$\rho_{ss', \tau\tau'}^{(r)}(\theta) = \frac{\omega_{\tau\tau'}^{(r)} \tilde{T}_{ss'}^{\text{lab}, (r)}}{\tilde{T}_{ss'}(\tau, \tau'; \theta)} \quad (\text{C.123})$$

partitions the per-branch counts $W_{ss', \tau\tau'}^{(v \rightarrow w)}$ into route-attributable soft counts

$$\tilde{W}_{ss', \tau\tau'}^{(r), (v \rightarrow w)} = W_{ss', \tau\tau'}^{(v \rightarrow w)} \cdot \rho_{ss', \tau\tau'}^{(r)}(\theta^{(t)}). \quad (\text{C.124})$$

Each route $r \in \{R1, R2, R3\}$ has a fixed BDI / fragment / domain factor signature (Section C.8.4): R1 contributes only to within-fragment extension counts; R2 contributes to fragment-level BDI counts in domain d ; R3 contributes to both top-level BDI counts and to fragment-level BDI counts in the *destination* domain d' .

Indel rate M-step. For each domain d , the per-route soft counts in $\tilde{W}^{(R2)}$ (restricted to $d = d'$ in source/dest tuples) and $\tilde{W}^{(R3)}$ (with destination domain d') accumulate to a per-domain 5×5 WFST transition count matrix $\hat{n}_{ss'}^{(d)}$ analogous to the SVI-BW pair-counts. The standard transition-count groups (Section C.1.4, `transition_count_groups`) and quadratic-in- κ closed form (`m_step_indel_quadratic`) then deliver $(\hat{\lambda}_d, \hat{\mu}_d)$:

$$(\hat{\lambda}_d, \hat{\mu}_d) = \text{m_step_indel_quadratic}(\hat{B}_d, \hat{D}_d, \hat{S}_d, \hat{L}_d, \hat{M}_d, \hat{T}_d; \text{prior}). \quad (\text{C.125})$$

The top-level rates $(\hat{\lambda}_0, \hat{\mu}_0)$ are recovered analogously from $\tilde{W}^{(R3)}$ contributions to the top-level WFST transition counts (with R3 supplying both a top-level BDI event and a destination-domain BDI event per transition).

Within-fragment Markov M-step. For each domain d , the per-fragchar transition matrix $r_{f,f'}^{(d)}$ has Dirichlet-conjugate update from the R1 soft counts:

$$\hat{r}_{f,f'}^{(d)} = \frac{\hat{E}_{d,f,f'} + \alpha_r - 1}{\sum_{f''} \hat{E}_{d,f,f''} + \hat{N}_{d,f} + (N_{\mathcal{F}} + 1)(\alpha_r - 1)}, \quad \hat{\rho}_f^{(d)} = 1 - \sum_{f'} \hat{r}_{f,f'}^{(d)}, \quad (\text{C.126})$$

where $\hat{E}_{d,f,f'} = \sum_{(v \rightarrow w), s, s'} \tilde{W}_{ss', (d,f), (d,f')}^{(R1), (v \rightarrow w)}$ and $\hat{N}_{d,f} = \sum_{(v \rightarrow w), s, s'} [\tilde{W}_{ss', (d,f), (d,\cdot)}^{(R2)} + \tilde{W}_{ss', (d,f), (\cdot, \cdot)}^{(R3)}]$ are the within-fragment and fragment-end soft counts respectively.

Domain and fragment weight M-step. The tuple priors update via Dirichlet-conjugate from per-route soft-counts:

$$\hat{v}_{d'} \propto \sum_{(v \rightarrow w)} \sum_{(d,f,f'), s, s'} \tilde{W}_{ss', (d,f), (d',f')}^{(R3), (v \rightarrow w)} + \alpha_n - 1, \quad (\text{C.127})$$

$$\hat{w}_{d',f'} \propto \sum_{(v \rightarrow w)} \sum_{(d,f), s, s'} \left[\delta_{d,d'} \tilde{W}_{ss', (d,f), (d',f')}^{(R2)} + \tilde{W}_{ss', (d,f), (d',f')}^{(R3)} \right] + \alpha_w - 1, \quad (\text{C.128})$$

both normalised over the appropriate index.

Substitution M-step. Per-class rate-matrix and stationary updates use the standard class-weighted bridge-expectation closed forms (Section B.1.10 and the `substitution_mstep.tex` appendix), with per-class soft-counts $\hat{M}^{(c)} = \sum_i \sum_n \hat{M}_{i,n}^{(c)}$ and $\hat{T}^{(c)} = \sum_i \sum_n \hat{T}_{i,n}^{(c)}$. The class distribution itself updates via Dirichlet-conjugate:

$$\hat{u}_{d,f,c} \propto \sum_i \sum_n q_{i,n}^{(\tau)}(d, f) \cdot q_{i,n}^{(c|f)}(c) + \alpha_u - 1. \quad (\text{C.129})$$

C.7.4 Stochastic VBEM (SVI-VBEM)

For corpora with $N \gg 100$ families a full-batch outer iteration is infeasible (Appendix C.7.7). We adopt the SVI-BW machinery of Appendix B.2.4 verbatim, with one substitution: the per-batch sufficient-statistic vector s_{batch_k} in (B.56) is now the per-family Tree-VBEM aggregate

$$s_{\text{batch}_k} = \sum_{i \in \mathcal{B}_k} \left\{ \tilde{W}_i^{(R1)}, \tilde{W}_i^{(R2)}, \tilde{W}_i^{(R3)}, q_i^{(\tau)}, \hat{M}_i^{(c)}, \hat{T}_i^{(c)} \right\}, \quad (\text{C.130})$$

each component being aggregated linearly over the families $i \in \mathcal{B}_k$ in minibatch k . (The route-attributed soft counts $\tilde{W}_i^{(r)}$ are themselves linear in the per-branch $W_i^{(v \rightarrow w)}$ tensors of (C.120) and the route posteriors (C.123), so the per-family contributions sum directly without bias.) The pseudocount EMA carries one such state $\tilde{\alpha}_k^{(g)}$ per parameter group g and is updated each iteration as

$$\tilde{\alpha}_k^{(g)} = (1 - \eta_k) \tilde{\alpha}_{k-1}^{(g)} + \eta_k (\alpha^{(g)} + (N/|\mathcal{B}_k|) s_{\text{batch}_k}^{(g)}), \quad (\text{C.131})$$

with the closed-form M-step applied on demand to derive $\theta_g^{(t+1)} = f^{(g)}(\tilde{\alpha}_k^{(g)})$ for each group: the BDI rates per domain via (C.125); the Dirichlet groups $v, w, u, r_{f,\cdot}^{(d)}$ direct from (C.126)–(C.128) and (C.129); and the per-class GTR substitution parameters via the closed forms of Appendix B.1.10.

The Polyak–Ruppert / ESS / Fisher analyses of Appendix B.2.4 (eqs. (B.58) onward) carry over verbatim, with s_{batch_k} replaced by the family-aggregate (C.130).

Step-size schedule. The standard Robbins–Monro choice $\eta_k = (k + \tau)^{-\kappa}$ with $\tau \in [10, 100]$, $\kappa \in [0.5, 1]$ guarantees almost-sure convergence as in SVI-BW. For tree-VBEM we suggest defaults $\tau = 10$, $\kappa = 0.7$.

Breadth-first minibatch sampling. I.i.d. uniform minibatch sampling concentrates updates on the $\sim |\mathcal{B}|/N$ most-recently-visited families and under-represents the long tail; per-group ESS for a parameter that is informative in only a fraction ε of families collapses to εESS_K (the “rare-parameter bottleneck” of Appendix B.2.4). We instead maintain a per-family visit count $c_i^{(0)} = 0$ and select each minibatch $\mathcal{B}_k$ as the $|\mathcal{B}|$ families with the smallest $c_i^{(k-1)}$, breaking ties at random; visit counts then update $c_i^{(k)} = c_i^{(k-1)} + 1$ for $i \in \mathcal{B}_k$. This deterministic round-robin guarantees every family is visited at least once every $\lceil N/|\mathcal{B}| \rceil$ iterations (“one epoch”), so the per-family contribution arrives in $\Theta(K|\mathcal{B}|/N)$ of the first K batches with no starvation.

Convergence diagnostics. Per outer iteration k , log:

- Per-batch ELBO sum, scaled by $N/|\mathcal{B}_k|$ for cross-iteration comparison.
- Per-group $\tilde{\alpha}_k^{(g)}$ effective sample size, $\text{ESS}_k = (\sum_j w_{j,k})^2 / \sum_j w_{j,k}^2$ with $w_{j,k} = \eta_j \prod_{i>j}^k (1 - \eta_i)$, to monitor warm-up.
- Visit-count distribution $\min_i c_i^{(k)}, \max_i c_i^{(k)}$, to confirm the breadth schedule is achieving uniform coverage.

Validation ELBO on a held-out family subset is computed once per “epoch” (every $\lceil N/|\mathcal{B}| \rceil$ iterations).

C.7.5 Convergence and ELBO monitoring

Each E-step monotonically increases the per-family ELBO at fixed θ ; each M-step monotonically increases the corpus-aggregate ELBO at the new q (this is the standard EM monotonicity result, modulo the variational E-step’s sub-optimality).

For monitoring purposes, log:

- Per-iteration corpus ELBO (sum across families).
- Per-domain BDI sufficient statistics aggregates.
- Per-class soft-count totals and effective sample size.
- Validation-set ELBO on a held-out family subset (for early stopping and overfitting monitoring; same role as val LL/pair in SVI-BW).

Convergence in practice: corpus ELBO increase plateaus below a relative threshold ($\sim 10^{-4}$ per outer iteration), or validation ELBO begins to decrease.

C.7.6 Initialisation and warm-start

For warm-start from an existing SVI-BW checkpoint: $\theta^{(0)}$ is loaded directly from the checkpoint (the variational EM operates on the same parameter space as SVI-BW provided the checkpoint includes `class_pis` and `class_S_exch`; checkpoints without a class layer are auto-promoted to a 1-class-per-domain structure).

For the variational $q^{(0)}$: per-family Fitch-seeded init for the inner 3-state q (as in the inference-only benchmark of Appendix C.8); per-family tuple init biased toward the substitution-likelihood-maximising fragchar (via `class_marginalised_sub_LL_per_column` applied to the initial θ).

The first outer iteration will see a large ELBO improvement as the variational distributions are fit to the warm-start parameters. Subsequent iterations refine θ toward the tree-aggregated likelihood maximum, which differs in general from the pair-aggregated SVI-BW maximum.

C.7.7 Computational scaling and minibatching

Per-family E-step cost is dominated by Adam-ELBO evaluation: $O(|\mathcal{E}| \cdot L \cdot T^2)$ per Adam step where $T = N_n N_{\mathcal{F}}$ is the reduced tuple count, plus $O(L \cdot N_{\text{cl}} \cdot \text{Felsenstein cost})$ for the substitution likelihoods (computed once per family at the start of the E-step). Typical numbers for unified-short ($T = 6$, $N_{\text{cl}} = 3$, $|\mathcal{E}| \approx 40$, $L \approx 100$, 100 Adam iters) give ~ 30 s/family on a single GPU.

For the full Pfam corpus ($\sim 17,000$ families) a full E-step pass is ~ 140 GPU-hours. Stochastic VBEM (Appendix C.7.4) replaces the full pass with a breadth-first minibatch of ~ 10 –200 families per outer iteration and an EMA accumulation of sufficient statistics across iterations, reducing per-iteration cost to a few GPU-minutes with convergence in $\sim 10 N/|\mathcal{B}|$ outer iterations (i.e. ~ 10 epochs).

C.7.8 Comparison to SVI-BW

The SVI-BW pipeline trains θ from sequence pairs sampled from the corpus, using the labelled Pair HMM forward-backward as its inference primitive. Tree-VBEM trains from whole MSAs with their tree topology, using the variational TreeVarAnc-MixDom inference of Appendix C.8.

Differences:

- **Information per training datum.** A pair contributes two-leaf data; a family contributes $|\text{leaves}|$ -leaf joint data with phylogenetic structure. Tree-VBEM extracts more information per datum but requires more compute per datum.

- **Bias.** SVI-BW assumes pairs are i.i.d. samples from the model; real pairs share evolutionary history (within-family pairs are not independent under the tree). Tree-VBEM correctly handles this correlation. The cost: SVI-BW’s pair-likelihood is the exact data likelihood under its assumption; tree-VBEM’s ELBO is a lower bound on the family likelihood (with a non-trivial variational gap).
- **Application alignment.** Parameters trained by tree-VBEM are likely better-suited to tree-based downstream tasks (ancestral reconstruction, progressive alignment) since the training objective matches the inference task. Parameters trained by SVI-BW may be better for pairwise tasks (pairwise alignment scoring).

Empirically, the two regimes can be combined: SVI-BW for fast warm-up followed by tree-VBEM for task-specific fine-tuning, exploiting both algorithms’ strengths.

C.8 Mixture-of-trees variational MixDom ancestral inference

This appendix generalises the variational ancestral-presence reconstruction of Section B.7 from the TKF92-conditional-WFST-as-MaxEnt-GGI approximation to the labelled MixDom model, with two further simplifications relative to a fully labelled treatment:

- the per-character substitution class c is integrated out at the model level via the per-column prior $u_{f,c}$ (rather than carried as a variational latent);
- the fragchar-boundary indicator g and domain-end indicator e are absorbed into a *reduced* per-character WFST kernel that sums over the implicit fragment/domain bookkeeping at each step (rather than tracked explicitly in the variational state).

The variational state per (internal node, MSA column) is therefore $\{N, D\} \cup \{(d, f)\}$, $|\mathcal{Z}| = N_{\text{dom}}N_{\text{fr}} + 2$ — a $(4N_{\text{cl}}) \times$ reduction over the fully-labelled state space. Three concrete payoffs: (i) the variational bound is *strictly tighter* than the fully-labelled ELBO at any non-Bayes-optimal class posterior, because the analytic marginalisation of c and (g, e) replaces two Jensen inequalities with equalities; (ii) the labelled-variant cross-column hard-zero structural constraint disappears, so per-column factorised $q^{(\tau)}$ is adequate (the column-Markov-chain upgrade is no longer required); (iii) per-branch evaluation is roughly $|\mathcal{T}^{\text{lab}}|^2/|\mathcal{T}|^2$ -times cheaper.

The bound is $\log p(\text{MSA} \mid \mathbf{t}, \theta) \geq \log \tilde{p}(\text{MSA} \mid \mathbf{t}, \theta) \geq \mathcal{L}[q]$, in the same restricted-model sense as appendix B.7; the restriction-gap (q -independent) and variational-gap (KL) decomposition carries over.

C.8.1 Setting and reduced state space

Fix a rooted phylogeny $\mathbf{t}$ with internal-node set $\mathcal{I}$, leaf set $\mathcal{L}$, and branch lengths d_e on each edge e . The MSA has L columns; each leaf $v \in \mathcal{L}$ has an observed presence indicator $X_n^v \in \{0, 1\}$ and (when present) a residue $a_n^v \in \mathcal{A}$.

The per-(node, column) variational state space is

$$\mathcal{Z} = \{N, D\} \cup \mathcal{T}, \quad \mathcal{T} = [N_{\text{dom}}] \times [N_{\text{fr}}]. \quad (\text{C.132})$$

A tuple-valued state $\tau = (d, f) \in \mathcal{T}$ encodes only the column’s domain d and fragchar f ; the per-character class c is marginalised at the model level (Section C.8.7) and the fragchar-boundary / domain-end indicators (g, e) are absorbed into the reduced WFST kernel (Section C.8.4).

The presence indicator at internal node v , column n is $X_n^v = \mathbb{I}\{Z_n^v \in \mathcal{T}\}$. Leaf states are partially clamped: $X_n^v = 0$ pins $Z_n^v \in \{N, D\}$; $X_n^v = 1$ pins $Z_n^v \in \mathcal{T}$ (consistent with the column-wide tuple τ_n — see below).

C.8.2 Restricted generative model

Following Section B.7.2, the model joint over MSA columns and internal-node states is

$$p(\text{MSA}, \{Z^v\}_{v \in \mathcal{I}} \mid \mathbf{t}, \theta) = p_{\text{singlet}}^{\text{red}}(Z^{\text{root}}) \prod_{(v \rightarrow w) \in \mathbf{t}} \hat{P}^{\text{WFST}}(Z^w \mid Z^v, d_{vw}, \theta) \prod_{n: \mathcal{F}_n \neq \emptyset} L_n^{(\text{sub}), \text{tot}}(f_n; \mathcal{F}_n), \quad (\text{C.133})$$

where $\mathcal{F}_n$ is the Fitch-parsimony subtree of column n , $L_n^{(\text{sub}), \text{tot}}(f; \mathcal{F}_n)$ is the class-marginalised Felsenstein column-substitution likelihood under fragchar f (Section C.8.7), and $p_{\text{singlet}}^{\text{red}}, \hat{P}^{\text{WFST}}$ are the *reduced* singlet HMM and WFST defined on (d, f) alone (Section C.8.4). As in the simple case we marginalise (C.133) over internal patterns supported on the L observed columns to obtain $\tilde{p}(\text{MSA} \mid \mathbf{t}, \theta) \leq p(\text{MSA} \mid \mathbf{t}, \theta)$ (with the same ghost-column caveat), and bound the latter from below.

C.8.3 Variational family

The variational q factorises over MSA columns, $q(\{Z_n^v\}) = \prod_{n=1}^L q_n$, and within each column,

$$q_n = q_n^{(\tau)}(\tau_n) \cdot q_n^{(\pi|\tau)}(\{Z_n^v\}_{v \in \mathcal{I}} \mid \tau_n), \quad (\text{C.134})$$

where $q_n^{(\tau)}$ is a free categorical over $\mathcal{T}$ on the column-wide tuple, and $q_n^{(\pi|\tau)}$ is the same 3-state irreversible tree-structured graphical model of Section B.7.5 (with $xP \equiv \tau_n$). By construction q assigns zero mass to within-column configurations in which Present nodes carry differing tuples — the column has one (d, f) , full stop. The cross-column structural-label constraints that forced a column-Markov chain in the labelled formulation *vanish* under reduction (Section C.8.9).

Free parameter count. Per column: $|\mathcal{T}| - 1 = N_{\text{dom}}N_{\text{fr}} - 1$ free parameters in $q_n^{(\tau)}$, plus $2|\mathcal{E}|$ in the inner 3-state graphical model (as in Section B.7.10), plus 2 for the inner root distribution. For typical $(N_{\text{dom}}, N_{\text{fr}}) = (3, 2)$ that is 5 tuple parameters per column, vs $4N_{\text{cl}}N_{\text{fr}}N_{\text{dom}} - 1$ in the labelled formulation (e.g. 239 for $N_{\text{cl}} = 10$) and $\sim |\mathcal{T}|^2$ in the labelled column-Markov upgrade ($\sim 6 \times 10^4$).

C.8.4 Reduced WFST: marginalising (g, e) and the class c

The reduced per-character WFST kernel is

$$\hat{T}_{ss'}((d, f), (d', f'); d, \theta), \quad s, s' \in \{\mathbf{S}, \mathbf{M}, \mathbf{I}, \mathbf{D}, \mathbf{E}\}, \quad (\text{C.135})$$

obtained by marginalising the labelled-MixDom WFST $T_{ss'}^{\text{lab}}((c, f, g, d, e), (c', f', g', d', e'); d, \theta)$ over (c, c', g, e, g', e') at each end. The reduced state space is $\{\mathbf{S}, \mathbf{E}\} \cup \{(s, d, f) : s \in \{\mathbf{M}, \mathbf{I}, \mathbf{D}\}\}$ — i.e. $3N_{\text{dom}}N_{\text{fr}} + 2$ states.

Routes from (d, f) to (d', f') . Under the labelled MixDom model, the per-character labelled transition $(d, f) \rightarrow (d', f')$ admits up to three latent routes, indexed by the value of (g, e) at the source position:

- R1. *Intra-fragment fragchar transition* ($g = 0$): the current character is not the last of its fragment, and the next character stays in the same fragment with fragchar transitioning $f \rightarrow f'$ via $r_{f, f'}^{(d)}$. Restricted to $d' = d$.

- R2. *New fragment, same domain* ($g = 1, e = 0$): the current fragment terminates ($\rho_f^{(d)}$) and the same domain starts a new fragment (κ_d) with first fragchar drawn from $w_{d,f'}$. Restricted to $d' = d$.
- R3. *New domain* ($g = 1, e = 1$): the current fragment terminates, the current domain ends ($1 - \kappa_d$), and (after possibly skipping geometrically-many empty domains) a new domain d' begins with its first fragment's first fragchar drawn from $w_{d',f'}$. Available for any d' , including the self-recurrence $d' = d$.

For $d' \neq d$ only R3 is enabled, so the source's $(g, e) = (1, 1)$ is uniquely determined by the transition. For $d' = d$, multiple routes contribute and the latent (g, e) at the source carries a non-trivial posterior over routes given the observed transition. The diagonal-extension special case ($r_{f,f'}^{(d)} = r_f \delta_{f,f'}$) suppresses R1 whenever $f' \neq f$, but R2 and R3 still mix for any $(d, f) \rightarrow (d, f')$, so the route-sum machinery introduced below is required even there. With a general off-diagonal $r^{(d)}$ matrix, R1 admits $f' \neq f$ transitions, so the route-sum acquires a third contributing term.

The per-route singlet contribution, accounting for the joint prior on (g, e) at the source and the singlet emission of the next character, is:

$$\omega_{(d,f) \rightarrow (d',f')}^{(R1)} = \delta_{d,d'} r_{f,f'}^{(d)}, \quad (\text{C.136})$$

$$\omega_{(d,f) \rightarrow (d',f')}^{(R2)} = \delta_{d,d'} \rho_f^{(d)} \kappa_d w_{d,f'}, \quad (\text{C.137})$$

$$\omega_{(d,f) \rightarrow (d',f')}^{(R3)} = \frac{\rho_f^{(d)} (1 - \kappa_d) \kappa_0 v_{d'} \kappa_{d'} w_{d',f'}}{1 - \zeta}, \quad (\text{C.138})$$

where $\zeta = \kappa_0 \sum_{d''} v_{d''} (1 - \kappa_{d''})$ is the empty-domain renormalisation. Each $\omega^{(r)}$ factorises as $P((g, e) = (g_r, e_r) \mid (d, f)) \cdot P_{\text{singlet}}((d', f') \mid (d, f), g_r, e_r)$, so summing over routes recovers the marginal singlet emission probability:

$$\omega^{(d,f,d',f')} = \sum_r \omega_{(d,f) \rightarrow (d',f')}^{(r)} = \delta_{d,d'} [r_{f,f'}^{(d)} + \rho_f^{(d)} \kappa_d w_{d,f'}] + \frac{\rho_f^{(d)} (1 - \kappa_d) \kappa_0 v_{d'} \kappa_{d'} w_{d',f'}}{1 - \zeta}. \quad (\text{C.139})$$

Marginalisation of c . The class c at column n is generated once per column by $u_{f_n, \cdot}$ at column birth and governs the substitution likelihood across all branches. The labelled WFST's indel block is class-independent, so $\sum_c u_{f,c} = 1$ trivially. Class-marginalisation of $\hat{T}$ has no residue at the indel level; it is handled separately in the substitution term (Section C.8.7).

Step-by-step derivation of the reduced kernel The route-sum (C.144) is not stipulated; it follows from two facts about the labelled MixDom model. We state and prove both.

Notation. Write $Z_n = (d_n, f_n, g_n, e_n)$ for the labelled chain state at position n (dropping c_n , which is class-independent for the indel block) and $\tau_n = (d_n, f_n)$ for the reduced state. Let σ denote the labelled singlet HMM transition kernel, and let $T_{ss'}^{\text{lab}}$ denote the labelled conditional WFST kernel of the full labelled-MixDom model (the per-character transducer on the labelled state (c, f, g, d, e) , before the (c, g, e) -marginalisation performed here). The labelled per-character Pair HMM joint factorises as

$$P((s', Z') \mid (s, Z)) = T_{ss'}^{\text{lab}}(Z, Z') \sigma(Z' \mid Z), \quad (\text{C.140})$$

i.e. the WFST conditional weight times the singlet emission weight — the *Singlet* $\circ$ *WFST* = *Pair HMM* factorisation of the labelled-MixDom model.

Singlet-table convention. Throughout the proof we use the labelled singlet HMM table with the following normalisation. Two structural facts about the labelled MixDom chain underpin this:

- (a) *e is fragment-level.* The domain-end indicator e is constant for all characters within a single fragment (it records whether the current fragment is the last in its domain, which is set when the fragment begins and does not change as the fragment extends). Hence within-fragment transitions ($g=0$ rows) have $e' = e$ implicitly, and a single “ $g=0$ row” covers all source e values without depending on e .
- (b) *Destination indicators (g', e') are summed implicitly.* The destination (g', e') in each row is treated as a free index ranging over $\{0, 1\}^2$, with the implicit prior $\pi(g', e' \mid d', f')$ to be applied separately if a finer-grained joint over destination indicators is needed.

Under these conventions, the table entry $\tilde{\sigma}$ for a given source-row (d, f, g, e) and destination structural state (d', f') is the joint

$$\tilde{\sigma}((d, f, g, e) \rightarrow (d', f')) = \pi(g, e \mid d, f) \cdot \sigma_{\text{cond}}((d', f') \mid (d, f, g, e)),$$

where σ_{cond} is the source-conditional next-character emission probability and $\pi(g, e \mid d, f)$ is the source-row marginal. As a sanity check: summing each row of the table over (d', f') recovers exactly $\pi(g, e \mid d, f)$, which sums in turn to 1 over the three rows (with termination column included). For the $g=0$ row: $\sum_{f'} r_{f, f'}^{(d)} = 1 - \rho_f^{(d)} = \pi(g=0 \mid d, f)$. For the $g=1, e=0$ row: $\sum_{f'} \rho_f^{(d)} \kappa_d w_{d, f'} = \rho_f^{(d)} \kappa_d = \pi(g=1, e=0 \mid d, f)$. For the $g=1, e=1$ row, the row sum (over destinations plus termination) equals $\rho_f^{(d)} (1 - \kappa_d) = \pi(g=1, e=1 \mid d, f)$.

Lemma 1 (Conditional independence of (g, e)). *Under the labelled MixDom singlet construction, $(g_n, e_n) \perp \{Z_k : k < n\} \mid (d_n, f_n)$ for every $n \geq 1$, with*

$$\pi(g=0 \mid d, f) = 1 - \rho_f^{(d)}, \quad (\text{C.141})$$

$$\pi(g=1, e=0 \mid d, f) = \rho_f^{(d)} \kappa_d, \quad (\text{C.142})$$

$$\pi(g=1, e=1 \mid d, f) = \rho_f^{(d)} (1 - \kappa_d). \quad (\text{C.143})$$

Proof. By the singlet-table convention above, the labelled-chain joint factorises position-by-position as

$$P(Z_n \mid Z_{n-1}) = P(d_n, f_n \mid Z_{n-1}) \cdot \pi(g_n, e_n \mid d_n, f_n),$$

because the table entry at position n has the destination prior $\pi(g_n, e_n \mid d_n, f_n)$ as an explicit multiplicative factor, independent of Z_{n-1} beyond what flows through (d_n, f_n) . Marginalising the chain joint over $Z_{1:n-1}$ at fixed (d_n, f_n) leaves the $\pi(g_n, e_n \mid d_n, f_n)$ factor untouched, establishing $P(g_n, e_n \mid d_n, f_n, Z_{1:n-1}) = \pi(g_n, e_n \mid d_n, f_n)$. Substituting the explicit Bernoulli forms ($g_n \sim \text{Bern}(\rho_{f_n}^{(d_n)})$) and, given $g_n = 1$, $e_n \sim \text{Bern}(1 - \kappa_{d_n})$, with e_n at $g_n = 0$ folded into the joint as the residual $\rho_f^{(d)}$ row) gives (C.141).

Lemma 2 (Singlet route-decomposition). *The marginal singlet emission $\omega^{(d, f, d', f')} := \sum_{g, e, g', e'} \tilde{\sigma}((d, f, g, e) \rightarrow (d', f', g', e'))$ decomposes as $\omega = \omega^{(R1)} + \omega^{(R2)} + \omega^{(R3)}$, with the per-route weights of eq. (C.136).*

Proof. The labelled singlet table partitions the source (g, e) values into three rows:

- Row $g=0$ (combining all source e values, since the $g=0$ row weight does not depend on source e): table entry $\tilde{\sigma}_{R1} = r_{f, f'}^{(d)}$ for destination (d, f', g', e') summed over destination indicators.
- Row $g=1, e=0$: $\tilde{\sigma}_{R2} = \rho_f^{(d)} \kappa_d w_{d, f'}$.

- Row $g=1, e=1$: $\tilde{\sigma}_{R3} = \rho_f^{(d)}(1-\kappa_d)\kappa_0 v_{d'}\kappa_{d'} w_{d',f'}/(1-\zeta)$.

By the singlet-table convention, each row is the *joint* contribution including the source- (g, e) prior; in particular $\tilde{\sigma}_{R1} = \pi(g=0 \mid d, f) \cdot \sigma_{\text{cond}}((d, f', g'=\cdot, e'=\cdot) \mid (d, f, g=0, e))$ summed over destination (g', e') . Thus $\omega^{(r)} = \tilde{\sigma}_r$ directly: the row weight already equals the route contribution to the marginal singlet emission, so no separate “prior cancellation” step is needed — the source prior was baked in from the start. Summing the three rows gives $\omega^{(d,f,d',f')}$ in (C.139). The fact that each row has support only on destinations consistent with its route is by inspection of the singlet table.

Proposition (Reduced kernel). *The reduced per-character marginal Pair HMM kernel, $\hat{T}_{ss'}((d, f), (d', f')) := \sum_{g,e,g',e'} \pi(g, e \mid d, f) \cdot P((s', d', f', g', e') \mid (s, d, f, g, e))$, satisfies the route-sum (C.144).*

Proof. Substitute the Pair HMM factorisation (C.140), and replace $\pi(g, e \mid d, f) \cdot \sigma_{\text{cond}}(\dots) = \tilde{\sigma}(\dots)$ via the singlet-table convention:

$$\begin{aligned} \hat{T}_{ss'}((d, f), (d', f')) &= \sum_{g,e,g',e'} \pi(g, e \mid d, f) T_{ss'}^{\text{lab}}((d, f, g, e), (d', f', g', e')) \sigma_{\text{cond}}((d', f', g', e') \mid (d, f, g, e)) \\ &= \sum_{g,e,g',e'} T_{ss'}^{\text{lab}}((d, f, g, e), (d', f', g', e')) \tilde{\sigma}((d, f, g, e) \rightarrow (d', f', g', e')). \end{aligned}$$

By Lemma 2, $\tilde{\sigma}$ has support partitioned into the three routes $R1, R2, R3$, each with a unique source (g_r, e_r) . The labelled WFST entry for fixed source (g_r, e_r) and structural transition (s, s') depends on the destination (g', e') in only two ways: (i) the label-preservation constraint on $\mathbf{M}$ (which forces $g' = g_{r,\text{dest}}$ and $e' = e_{r,\text{dest}}$ where the destination-side (g_r, e_r) are determined by the route and the destination structural label, trivialising the destination sum); and (ii) for $\mathbf{I}$, the destination prior on (g', e') at the new (d', f') is folded into the WFST as a $\pi(g', e' \mid d', f')$ factor (the WFST, defined as the labelled Pair HMM joint divided by the labelled singlet emission, makes this division explicit). In both cases the (g', e') summation cleanly factors out:

$$\sum_{g',e'} T_{ss'}^{\text{lab}}((d, f, g_r, e_r), (d', f', g', e')) \pi(g', e' \mid d', f')^{-1} \pi(g', e' \mid d', f') = \tilde{T}_{ss'}^{\text{lab},(r)}((d, f), (d', f'))$$

(the inverse-prior in the WFST cancels the prior in $\tilde{\sigma}$ ’s destination factor; for $\mathbf{M}$ the inverse prior is trivially 1 because of label preservation). Combining with the source weights $\tilde{\sigma}_r$ summed over destinations giving $\omega^{(r)}$ from Lemma 2, we obtain eq. (C.144).

Corollary (Markov property of reduced chain). *The marginal chain in (d, f) obtained by summing (g, e) out of the labelled singlet chain is exactly Markov, with per-step kernel $\hat{T}_{\mathbf{S}\mathbf{S}}$ at the reduced state space.*

Proof. We show $P(\tau_n \mid \tau_{1:n-1}) = P(\tau_n \mid \tau_{n-1})$ for all $n \geq 2$, which is the defining property of a Markov chain. By the singlet’s chain factorisation, $P(\tau_n \mid Z_{1:n-1}) = \sigma_{\text{cond, marg}}(\tau_n \mid Z_{n-1})$, which depends on $Z_{n-1} = (\tau_{n-1}, g_{n-1}, e_{n-1})$. Marginalising (g_{n-1}, e_{n-1}) against its conditional distribution given $\tau_{1:n-1}$:

$$P(\tau_n \mid \tau_{1:n-1}) = \sum_{g_{n-1}, e_{n-1}} P(g_{n-1}, e_{n-1} \mid \tau_{1:n-1}) \sigma_{\text{cond, marg}}(\tau_n \mid \tau_{n-1}, g_{n-1}, e_{n-1}).$$

By Lemma 1 (Markov property of (g, e)), $P(g_{n-1}, e_{n-1} \mid \tau_{1:n-1}) = \pi(g_{n-1}, e_{n-1} \mid \tau_{n-1})$, depending only on τ_{n-1} . Substituting:

$$P(\tau_n \mid \tau_{1:n-1}) = \sum_{g_{n-1}, e_{n-1}} \pi(g_{n-1}, e_{n-1} \mid \tau_{n-1}) \sigma_{\text{cond, marg}}(\tau_n \mid \tau_{n-1}, g_{n-1}, e_{n-1})$$

which is $\hat{T}_{ss}$ (eq. (C.144) for the $s = s' = \text{trivial-transition}$ case, i.e. depending only on (τ_{n-1}, τ_n)). This depends only on τ_{n-1} , establishing the Markov property. The product factorisation $P(\tau_1, \dots, \tau_L) = \prod_n \hat{T}(\tau_{n-1}, \tau_n)$ follows by chain rule.

Caveats. Two technical conditions buried in the proof deserve explicit flagging:

1. *Independence of WFST weights from destination (g', e') :* the labelled WFST entries $T_{ss'}^{\text{lab}}((d, f, g, e), (d', f', g', e))$ depend on the structural transition type (M/I/D at both ends) and on the source (g, e) via the case split into mid-fragment / fragment-boundary / domain-boundary, but they do not depend on the destination (g', e') except through the trivial label-preservation (M requires matching labels) constraint already implicit in T^{lab} . This is by inspection of the labelled-MixDom WFST tables.
2. *Boundary entries:* the proof above covers non-boundary positions $1 \leq n \leq L$ with $(s, s') \in \{\text{M, I, D}\}$. The **S** and **E** row/column entries of $\hat{T}$ require separate derivation because the **S** source has no preceding (g, e) to enumerate routes over, and the **E** destination requires the singlet's termination weight rather than a next-character emission. These are flagged as Section C.8.11, item I.2.

Markov property of the reduced chain (consequence). The latent (g, e) at position n depends only on (d_n, f_n) and the model parameters (Lemma 1). Consequently, summing out (g, e) at each position yields a marginal chain in (d, f) that remains *exactly Markov*, with per-step transition kernel $\hat{T}$ defined below. This justifies the path-LL formulation in Section C.8.5 as exact (not a mean-field approximation) under the reduced state space.

Reduced kernel as a route-sum. Combining the route enumeration with the labelled WFST entries gives the reduced per-character kernel as a sum of route contributions:

$$\hat{T}_{ss'}((d, f), (d', f'); d, \theta) = \sum_{r \in \mathcal{R}((d, f), (d', f'))} \omega_{(d, f) \rightarrow (d', f')}^{(r)} \tilde{T}_{ss'}^{\text{lab}, (r)}((d, f), (d', f'); d, \theta), \quad (\text{C.144})$$

with route set $\mathcal{R}((d, f), (d', f')) = \{R3\}$ for $d' \neq d$ and $\{R1, R2, R3\}$ for $d' = d$, and per-route labelled WFST entry

$$\tilde{T}_{ss'}^{\text{lab}, (r)}((d, f), (d', f'); d, \theta) = \sum_{g', e'} T_{ss'}^{\text{lab}}((d, f, g_r, e_r), (d', f', g', e'); d, \theta), \quad (\text{C.145})$$

where T^{lab} is the *conditional* labelled WFST, defined as the labelled Pair HMM joint divided by the labelled singlet emission, and (g_r, e_r) is the source (g, e) for route r : $(g_{R1}, e_{R1}) = (0, e)$ for any e (since within-fragment transitions do not constrain e ; the labelled-singlet table treats e as a free variable in the $g=0$ row, so we adopt $(0, 0)$ as the canonical representative); $(g_{R2}, e_{R2}) = (1, 0)$; and $(g_{R3}, e_{R3}) = (1, 1)$. The product $\omega^{(r)} \cdot \tilde{T}_{ss'}^{\text{lab}, (r)}$ then equals the per-route contribution to the labelled Pair HMM joint $(d, f) \rightarrow (d', f')$ summed over destination indicators (g', e') , so summing over routes recovers the marginal Pair HMM joint at the reduced state space; division of T^{lab} by the singlet was applied per-route inside $\omega^{(r)}$ to avoid double-counting.

When the route-sum collapses. The previously-claimed factorisation $\hat{T} = \omega \cdot \tilde{T}^{\text{lab}}$ holds only if, for the given (s, s') , the labelled WFST entry $\tilde{T}_{ss'}^{\text{lab}, (r)}$ is independent of r across the enabled

routes. This essentially never happens for the indel block. For $\mathbf{M} \rightarrow \mathbf{M}$, the per-route conditional WFST weights (Pair HMM joint divided by the route’s singlet emission) are

$$W_{\mathbf{MM}}^{(R1)} = 1, \quad W_{\mathbf{MM}}^{(R2)} = (1-\beta_d)\alpha_d, \quad W_{\mathbf{MM}}^{(R3)} \propto \tau_{\mathbf{M},\mathbf{E}}^{(d)} \mathcal{T}_{\mathbf{M},\mathbf{M}}^{[d \rightarrow d']} \tau_{\mathbf{S},\mathbf{M}}^{(d')},$$

where $W_{\mathbf{MM}}^{(R3)}$ involves the destination-domain-specific top-level effective transition (in the implementation, `T_exit_k`; structurally, $\mathcal{T}_{\mathbf{M},\mathbf{M}}^{[d \rightarrow d']}$ is the destination-typed version of the empty-domain-summed top-level $\mathbf{M} \rightarrow \mathbf{M}$ kernel, with the $v_{d'}$ and other normalisation factors absorbed). The three weights are distinct in general: R1 carries no separate BDI factor because in-fragment extension is generated by the singlet alone (the descendant character matches trivially via fragment continuation); R2 carries the standard new-fragment match factor $(1-\beta_d)\alpha_d$ (the singlet’s κ_d cancels into $\tau_{\mathbf{M},\mathbf{M}}^{(d)} = (1-\beta_d)\alpha_d\kappa_d$); R3 carries the cross-domain entry weight, involving both the source domain’s exit $\tau_{\mathbf{M},\mathbf{E}}^{(d)} = (1-\beta_d)(1-\kappa_d)$ and the destination domain’s TKF92 entry $\tau_{\mathbf{S},\mathbf{M}}^{(d')} = (1-\beta_{d'})\alpha_{d'}\kappa_{d'}$. Hence the route-sum genuinely cannot be collapsed to a single labelled-WFST entry for any $(d, f) \rightarrow (d', f')$ transition that admits more than one route, which includes every same-domain transition $(d, f) \rightarrow (d, f')$.

The reduced kernel is no more row-stochastic than the labelled one was — both are conditional transducers — but the input-conditional normalisation of the transducer is preserved by the route-sum marginalisation.

Numerical verification. The route-sum (C.144) has been verified at $t = 0.1$ on a $(N_{\text{dom}}, N_{\text{fr}}) = (2, 2)$ instance (random parameters, fixed seed) by reconstructing every $\mathbf{M} \rightarrow \mathbf{M}$ entry of the Pair HMM as $\sum_r \omega_r^{(r)} W_{\mathbf{MM}}^{(r)}$ and comparing against the corresponding entry of `build_nested_trans`; the maximum absolute discrepancy across the 16-entry $\mathbf{M} \rightarrow \mathbf{M}$ block is 1.1×10^{-16} , i.e. floating-point noise. The script is at `python/verify_reduced_wfst_routes.py`. The previous $t = 0$ check did not test the inter-route discrepancy in $W_{ss'}^{(r)}$ because at $t = 0$ the WFST collapses to (near-)identity and all routes coincide trivially.

C.8.5 Per-branch path log-likelihood

Read parent and child labelled states column-by-column on branch $v \rightarrow w$ to obtain the joint sequence $(Z_n^v, Z_n^w)_{n=1}^L$. By the same-tuple invariant of q_n , only five joint configurations carry positive variational mass, mapping to WFST states as in equation (B.105) (with τ now in the reduced $\mathcal{T}$):

$$S_n = \begin{cases} \mathbf{M} & (Z_n^v, Z_n^w) = (\tau, \tau), \\ \mathbf{D} & (Z_n^v, Z_n^w) = (\tau, xD), \\ \mathbf{I} & (Z_n^v, Z_n^w) = (xN, \tau), \\ \mathbf{lg} & (Z_n^v, Z_n^w) \in \{(xN, xN), (xD, xD)\}. \end{cases} \quad (\text{C.146})$$

With sentinels $S_0 = \mathbf{S}$, $S_{L+1} = \mathbf{E}$, the strip-Ignore reduction of equation (B.106) carries over:

$$\log P(X^w \mid X^v, d, \theta) = \sum_{N=1}^{L+1} \delta(S_N \neq \mathbf{lg}) \sum_{M=0}^{N-1} \delta(S_M \neq \mathbf{lg}) \log \hat{T}_{S_M S_N}(\tau_{n(M)}, \tau_{n(N)}; d, \theta) \prod_{K=M+1}^{N-1} \delta(S_K = \mathbf{lg}), \quad (\text{C.147})$$

with the reduced WFST $\hat{T}$ from equation (C.144) replacing the labelled T^{lab} . Tuples are now in $\mathcal{T} = [N_{\text{dom}}] \times [N_{\text{fr}}]$.

C.8.6 Per-column expected indel log-likelihood under q

The variational column-state probabilities $P_q^{v \rightarrow w}(S_n = s)$ depend only on the presence/absence pattern through the inner $q_n^{(\pi|\tau)}$ (which is τ -independent in our factorisation), so the same belief propagation machinery from the simple case (equation (B.111)) computes them.

The per-branch expected indel log-likelihood factorises as

$$\mathbb{E}_q[\log P(X^w | X^v, d, \theta)] = \sum_{s, s'} \sum_{\tau, \tau' \in \mathcal{T}} W_{ss', \tau\tau'}^{(v \rightarrow w)} \log \hat{T}_{ss'}(\tau, \tau'; d, \theta), \quad (\text{C.148})$$

with the reduced expected transition counts

$$W_{ss', \tau\tau'}^{(v \rightarrow w)} = \sum_{N=1}^{L+1} \sum_{M=0}^{N-1} q_{n(M)}^{(\tau)}(\tau) q_{n(N)}^{(\tau)}(\tau') P_{q, M}^{v \rightarrow w}(s) P_{q, N}^{v \rightarrow w}(s') \prod_{K=M+1}^{N-1} P_{q, K}^{v \rightarrow w}(\text{lg}), \quad (\text{C.149})$$

where the per-column factorisation of $q^{(\tau)}$ has reduced the pairwise tuple weight to a product of per-column marginals (no chain forward-backward needed). The cumulant trick of Section B.7.8 lifts unchanged to the inner K -product, with stable computation in $O(L \cdot |s|^2 \cdot |\mathcal{T}|^2)$ per branch via the running-logaddexp prefix on $\log P_q^{v \rightarrow w}(\text{lg})$. For $(N_{\text{dom}}, N_{\text{fr}}) = (3, 2)$ this gives $|\mathcal{T}|^2 = 36$ — versus $|\mathcal{T}^{\text{lab}}|^2 = 57600$ in the labelled variant ($\sim 1600\times$ cheaper per branch).

C.8.7 Per-column expected substitution log-likelihood

For each column n , define the *Fitch subtree* $\mathcal{F}_n \subseteq \mathbf{t}$ as the smallest connected subtree containing all leaves v with $X_n^v = 1$. By stationarity and reversibility of the substitution model, only nodes inside $\mathcal{F}_n$ contribute to the substitution likelihood; nodes outside are treated as missing data and absorbed into normalisation. The Fitch subtree is determined by the leaf data alone, independently of q and τ .

Because c is a per-column model latent (one class per column, governing all branches at that column), the column-substitution likelihood under fragchar f marginalises c at the model level:

$$L_n^{(\text{sub}), \text{tot}}(f; \mathcal{F}_n) = \sum_{c=1}^{N_{\text{cl}}} u_{f, c} L_n^{(\text{sub})}(c; \mathcal{F}_n), \quad (\text{C.150})$$

with

$$L_n^{(\text{sub})}(c; \mathcal{F}_n) = \sum_{a \in \mathcal{A}} \pi_a^{(c)} \beta_n^{r_n}(a; c) \quad (\text{C.151})$$

the standard Felsenstein up-pass likelihood at column n on $\mathcal{F}_n$ under class- c 's rate matrix $R^{(c)}$ and stationary $\pi^{(c)}$ (r_n is the Fitch-determined root of $\mathcal{F}_n$, $\beta_n^v(a; c)$ the standard Felsenstein up-message). The expected substitution log-likelihood under q involves only the fragchar marginal $q_n^{(f)}(f) = \sum_d q_n^{(\tau)}(d, f)$:

$$\mathbb{E}_{q_n^{(\tau)}}[\log L_n^{(\text{sub}), \text{tot}}(f_n; \mathcal{F}_n)] = \sum_{f=1}^{N_{\text{fr}}} q_n^{(f)}(f) \log \left[\sum_c u_{f, c} L_n^{(\text{sub})}(c; \mathcal{F}_n) \right]. \quad (\text{C.152})$$

The presence factor $q_n^{(\pi|\tau)}$ does not enter — the substitution likelihood is fully determined by the (data-determined) Fitch subtree and the (variational-determined) fragchar marginal.

Why $\log \sum_c u L$, not $\sum_c u \log L$. This is the proper integration over the per-column class prior. A labelled-variant counterpart that carried c as a variational latent would give $\sum_c q_n^{(c)}(c) \log L_n^{(\text{sub})}(c)$; by Jensen’s inequality $\log \sum_c p_c L_c \geq \sum_c p_c \log L_c$ with $p_c = u_{f,c}$, equality holding only when $q_n^{(c)}$ is at the Bayes-optimal class posterior $q^*(c) \propto u_{f,c} L_n^{(\text{sub})}(c)$. The reduction analytically marginalises c at the place where the labelled formulation left an inequality, so the reduced ELBO is at least as tight as the labelled ELBO and strictly tighter for any $q^{(c)}$ off the Bayes-optimal class posterior.

Numerical implementation. $\log L_n^{(\text{sub}),\text{tot}}(f)$ is computed in log-space:

$$\log L_n^{(\text{sub}),\text{tot}}(f) = \text{logsumexp}_c(\log u_{f,c} + \log L_n^{(\text{sub})}(c; \mathcal{F}_n)), \quad (\text{C.153})$$

with the expected log under $q_n^{(f)}$ a simple weighted sum.

C.8.8 ELBO

Combining the four contributions:

$$\boxed{\log p(\text{MSA} \mid \mathbf{t}, \theta) \geq \log \tilde{p}(\text{MSA} \mid \mathbf{t}, \theta) \geq \mathcal{L}[q]}, \quad (\text{C.154})$$

with

$$\mathcal{L}[q] = \mathbb{E}_q[\log p_{\text{singlet}}^{\text{red}}(Z^{\text{root}})] + \sum_{(v \rightarrow w) \in \mathbf{t}} \mathbb{E}_q[\log \hat{P}^{\text{WFST}}(Z^w \mid Z^v, d_{vw}, \theta)] + \sum_{n=1}^L \sum_f q_n^{(f)}(f) \log L_n^{(\text{sub}),\text{tot}}(f; \mathcal{F}_n) + H[q]. \quad (\text{C.155})$$

The first two terms use the reduced singlet HMM (defined on $\mathcal{T} = [N_{\text{dom}}] \times [N_{\text{fr}}]$ with the singlet kernel ω from equation (C.139)) and the reduced WFST $\hat{T}$ from equation (C.144). The third is the marginalised substitution term (C.152).

Entropy decomposition.

$$H[q] = \sum_{n=1}^L H[q_n^{(\tau)}] + \sum_{n=1}^L H[q_n^{(\pi|\tau)} \mid \text{MSA}], \quad (\text{C.156})$$

with $H[q_n^{(\tau)}]$ the simple categorical entropy (closed-form, $|\mathcal{T}| - 1$ free parameters per column) and $H[q_n^{(\pi|\tau)} \mid \text{MSA}]$ the leaf-conditioned entropy of the simple-case 3-state graphical model (equations (B.119) and (B.120) of Section B.7.7, with the same $\log Z_q$ correction).

Bound interpretation. Identical to equation (B.121): the bound is on $\log p$ with $\text{gap} = q$ -independent restriction-gap (ghost-column histories) + variational KL. The reduction (c, g, e) -collapse introduces no additional restriction gap; the analytic marginalisations are exact at the model level, replacing two labelled-variant Jensen inequalities with equalities and yielding a strictly tighter bound at the same variational optimum.

C.8.9 Cross-column constraint vanishes

The labelled-variant cross-column constraint (Section M.8 of the labelled draft) forced $q^{(\tau)}$ to be a column-Markov chain to avoid hard-zero violations of structural rules ($d_{n+1} = d_n$ when $e_n = 0$,

etc.). Under the (g, e) -marginalisation those rules disappear: every entry of $\omega^{(d, f, d', f')}$ is positive (assuming irreducibility of $r^{(d)}$ and positivity of $v, w, \rho^{(d)}, \kappa_0, \kappa_n$), so no (τ_n, τ_{n+1}) pair is hard-zero in $\hat{T}$ either. Per-column factorised $q^{(\tau)}(\tau_1, \dots, \tau_L) = \prod_n q_n^{(\tau)}(\tau_n)$ puts positive mass everywhere on the model’s support and $\mathbb{E}_q[\log p]$ is finite for any positive q .

A column-Markov $q^{(\tau)}$ remains an *optional refinement* — the true posterior on column-tuples is correlated across columns even after (g, e) -marginalisation (the singlet kernel is genuinely Markov on (d, f) via ω) — but the per-column factorised $q^{(\tau)}$ is sufficient for $\mathcal{L}[q]$ to be a finite proper bound, eliminating the $|\mathcal{T}|^2$ tuple-Markov parameters per column-pair that the labelled variant required.

C.8.10 Special cases and recovery

Recovery of simple TreeVarAnc. Setting $|\mathcal{T}| = 1$ (single domain, single fragchar) collapses $\hat{T}$ to the GGI-approximation WFST $\mathbb{T}'$ of Section A.4 and renders the substitution term constant, recovering Section B.7 verbatim.

Single-class limit. Setting $u_{f,c} = \delta_{c,c_0}$ for a fixed class c_0 gives $L_n^{(\text{sub}),\text{tot}}(f) = L_n^{(\text{sub})}(c_0; \mathcal{F}_n)$, and the substitution term reduces to the standard Felsenstein log-likelihood under class c_0 .

Single-fragchar-per-domain limit. $N_{\text{fr}} = 1$ collapses $r^{(d)}$ to a scalar self-loop, the within-domain fragchar Markov reduces to a geometric extension, and the reduced model recovers a simpler scalar-extension form.

Fragment-boundary inference caveat. Were the labelled g indicator a deterministic function of consecutive fragchar transitions ($g_n = \delta(f_n \neq f_{n+1})$ within the same domain — the special case where intra-fragment fragchar transitions are forbidden), the variational marginal would identify fragment boundaries directly. Under the reduced formulation the variational marginal $q_n^{(\tau)}(d, f)$ at each column gives the ancestral (d, f) directly, but does *not* uniquely identify “fragment boundaries” in that restrictive sense: a fragchar transition $(d, f) \rightarrow (d, f')$ with $f' \neq f$ in the inferred ancestral trajectory may be either a within-fragment Markov move or a new-fragment start, depending on the latent route. For ancestral (d, f) inference at observed columns this distinction is irrelevant. For downstream tasks that need fragment boundaries explicitly, one option is to augment $q^{(\tau)}$ post-hoc with a fragment-boundary indicator inferred from the route posterior over each $(d, f) \rightarrow (d, f')$ transition — but this is a derived quantity rather than an independent variational latent.

C.8.11 Open issues

Two points warrant explicit verification before relying on the construction quantitatively (a third was resolved by numerical verification at $t = 0$, see Section C.8.4).

(I.1) $\hat{T}$ row-sum identity. The reduced WFST $\hat{T}$ inherits the labelled WFST’s input-conditional normalisation by construction (Section C.8.4), but we have not explicitly verified the row-sum identity. A small numerical check is recommended: for $(N_{\text{dom}}, N_{\text{fr}}) \in \{(2, 2), (3, 2)\}$, build $\hat{T}$ from the labelled T^{lab} and verify that composition with the reduced singlet HMM on (d, f) yields the same column-marginal Pair HMM probabilities as the labelled construction.

(I.2) Start/end-row boundary effects. Boundary entries $\hat{T}_{\mathbf{S}\mathbf{S}'}(\mathbf{S}, \tau')$ and $\hat{T}_{\mathbf{S}\mathbf{E}}(\tau, \mathbf{E})$ involve the labelled WFST's start and end rows, which carry domain-level boundary indicators ($\mathcal{T}_{\mathbf{S}}, \mathcal{T}_{\mathbf{E}}$) that are not column-internal. These rows may need explicit derivation rather than mechanical application of equation (C.144); the construction in Section C.8.4 is correct for interior $(s, s') \in \{\mathbf{M}, \mathbf{I}, \mathbf{D}\}^2$ but should be verified at the boundaries.

These open issues are flagged here for transparency. Their resolution is a precondition for production use of the reduced ELBO in parameter learning; for ancestral-tuple inference at fixed θ the resulting bias (if any) is q -independent and so cancels from the variational optimum.

C.9 Generalized Phylo-HMM for MixDom

This appendix presents a polynomial-time algorithm for marginal ancestral reconstruction in a restricted MixDom model. The restriction is: the top-level TKF91 process has vanishing indel rates $\lambda_0, \mu_0 \rightarrow 0$ at fixed ratio $\kappa_0 = \lambda_0/\mu_0$. As input we assume an MSA with a tree and a per-node gap/residue annotation (in practice obtained by Fitch parsimony on gaps).

C.9.1 The Vanishing-Top-Level-Indel Limit

Setting $\lambda_0 = \mu_0 = 0$ literally produces 0/0 indeterminate forms in the TKF91 transition probabilities (eq. A.5). We instead take the limit $\lambda_0, \mu_0 \rightarrow 0^+$ with κ_0 fixed. In this limit the TKF91 α, β, γ coefficients evaluated at a branch length $T > 0$ satisfy $\alpha = e^{-\mu_0 T} \rightarrow 1$, $\beta = \lambda_0(1 - e^{-(\mu_0 - \lambda_0)T})/(\mu_0 - \lambda_0 e^{-(\mu_0 - \lambda_0)T}) \rightarrow 0$, and $\gamma \rightarrow 0$. Inspecting the entries of the 5×5 top-level matrix τ_0 :

$$\begin{aligned}\tau_0(\mathbf{S}, \mathbf{M}) &= (1 - \beta)\kappa_0\alpha \rightarrow \kappa_0, \\ \tau_0(\mathbf{S}, \mathbf{I}) &= \beta \rightarrow 0, \\ \tau_0(\mathbf{S}, \mathbf{D}) &= (1 - \beta)\kappa_0(1 - \alpha) \rightarrow 0, \\ \tau_0(\mathbf{S}, \mathbf{E}) &= (1 - \beta)(1 - \kappa_0) \rightarrow 1 - \kappa_0, \\ \tau_0(\mathbf{M}, \mathbf{M}) &\rightarrow \kappa_0, \quad \tau_0(\mathbf{M}, \mathbf{I}), \tau_0(\mathbf{M}, \mathbf{D}) \rightarrow 0, \\ \tau_0(\mathbf{M}, \mathbf{E}) &\rightarrow 1 - \kappa_0.\end{aligned}$$

The **I** and **D** columns become structurally unreachable on every branch, while the $\mathbf{S} \rightarrow \mathbf{M} \rightarrow \dots \rightarrow \mathbf{M} \rightarrow \mathbf{E}$ chain survives with probability $\kappa_0^{\mathcal{N}}(1 - \kappa_0)$ for a chain of $\mathcal{N}$ top-level match states. In other words, the number of top-level domains $\mathcal{N}$ is distributed Geometric(κ_0) at the root, and every descendant preserves the same ordered list of domains (no top-level births or deaths). The ratio κ_0 thus becomes the single top-level parameter that survives the limit; all branch-length dependence at the top level vanishes.

C.9.2 Partition Decomposition

Let the MSA have L columns. Conditional on the number of top-level domains $\mathcal{N}_*$ and their classes $d_1, \dots, d_{\mathcal{N}_*}$, the full model decomposes into $\mathcal{N}_*$ independent nested processes, each responsible for some contiguous subset of columns. Since there are no top-level births or deaths on any branch, each domain is either entirely absent or entirely present on each branch, and the correspondence between columns and domains is a single partition shared across the whole tree. Let

$$P = (b_1, b_2, \dots, b_B), \quad b_i = [l_{i-1} + 1, l_i],$$

with $0 = l_0 < l_1 < \dots < l_B = L$, be a partition of $\{1, \dots, L\}$ into B contiguous blocks, and let $d_{b_i} \in \{1, \dots, \mathcal{N}\}$ be the class label of block i , where $\mathcal{N}$ is the number of top-level domain classes in

the model. Then

$$P(\text{MSA} \mid \text{tree}, \text{model}) = \sum_P P(P \mid \text{model}) \prod_{i=1}^B G(l_{i-1} + 1, l_i, d_{b_i}), \quad (\text{C.157})$$

where the block likelihood $G(k, l, n)$ is the standalone phylogenetic likelihood of the sub-MSA on columns $k..l$ under the within-domain model for domain class n , now including the Markovian fragment process with transition matrix $r^{(n)}$ and per-fragment site class distributions u_{nfc} . The partition prior is

$$P(P \mid \text{model}) = (1 - \kappa_0) \kappa_0^B \prod_{i=1}^B v_{db_i}, \quad (\text{C.158})$$

where v is the top-level class (stationary) distribution. When marginalising over P the number of summands is exponential in L , so direct evaluation of (C.157) is intractable. It is, however, a generalised hidden Markov model in which each “emission” is an arbitrarily long contiguous block of columns.

C.9.3 Why the State Space Cannot Be Collapsed

A standard reduction of a generalised HMM to an ordinary HMM would introduce a hidden state at every column encoding “which within-domain Pair-HMM state each phylogenetic lineage is in”. Under a single TKF92, each lineage has three relevant states (**M**, **I**, **D**), but the per-column state vector across T tree nodes has 3^T values. Furthermore, once a new top-level domain begins the within-domain Pair HMMs on every lineage all reset to **S**, so the memory of the within-domain state on any previously gapped lineage is lost at every block boundary: a newly present lineage can only be reasoned about as starting from **S** at the first column of the new block. Conditioning on the partition P , on the other hand, makes each block a self-contained problem on a sub-MSA, and eliminates the combinatorial explosion at the price of a quadratic sum over start columns.

C.9.4 Setup and Definitions

We pretend there is an “infinitely long” branch above the root node, so every fragment on the root row is modeled as an insertion. We partition the alignment into blocks of domain type n .

Column presence profile. Let $A(j)$ denote the column presence/absence profile for column j —a binary vector with one entry per tree node, where 1 indicates the node has a residue at column j and 0 indicates a gap (as determined by Fitch parsimony).

Fragment continuity. Define

$$k_{\min}(i, j) = \min\{k : i \leq k \leq j, A(k') = A(j) \text{ for all } k \leq k' \leq j\}$$

This is the first column in $i..j$ that can be part of the same fragment as column j (i.e., the earliest column from which an unbroken run of identical presence profiles extends to j).

Per-branch TKF state. Let $S_{\text{tkf}}(r, i, j)$ be the TKF Pair HMM emit state for row (branch) r at column j , given block start i . It is a deterministic function of the presence/absence pattern, defined by the following cases. Write $(\text{anc}_r(j), \text{desc}_r(j)) = (\text{ancestor_present}_r(j), \text{descendant_present}_r(j))$ for the parent/child presence of branch r at column j . We say column j *touches* branch r when

at least one of $\text{anc}_r(j), \text{desc}_r(j)$ is 1; a column with $(\text{anc}_r(j), \text{desc}_r(j)) = (0, 0)$ leaves branch r *untouched*. Then

$$S_{\text{tkf}}(r, i, j) = \begin{cases} \text{M} & (\text{anc}_r(j), \text{desc}_r(j)) = (1, 1), \\ \text{D} & (\text{anc}_r(j), \text{desc}_r(j)) = (1, 0), \\ \text{I} & (\text{anc}_r(j), \text{desc}_r(j)) = (0, 1), \\ \text{S} & (\text{anc}_r(j), \text{desc}_r(j)) = (0, 0) \text{ and branch } r \text{ is untouched for all of } i..j, \\ S_{\text{tkf}}(r, i, j-1) & (\text{anc}_r(j), \text{desc}_r(j)) = (0, 0) \text{ otherwise (last-touched carry).} \end{cases}$$

The last case is the key subtlety: a column in which branch r is gapped at *both* its parent and child node is not part of that branch’s pairwise (parent, child) alignment at all, so its TKF Pair HMM neither emits nor moves — the branch simply retains the state from the most recent column that touched it (still **S** if no earlier column in the block did). The fourth case (all-untouched-so-far) is the special case of this carry in which the carried state has never advanced past **S**.

Per-branch TKF transitions. Let $B_{\text{tkf}}(n, r, s, s')$ be the TKF91 branch transition matrix entry for domain n , branch r , from state s to state s' . Crucially, each per-branch product below runs over the rows *touched* at the column in question, i.e. those r for which $(\text{anc}_r(j), \text{desc}_r(j)) \neq (0, 0)$; an untouched branch contributes the identity factor 1 (equivalently, B_{tkf} is taken to be 1 for it). Writing $\text{Touched}(j) = \{r : \text{column } j \text{ touches branch } r\}$, define

$$T_{\text{tkf}}(n, i, j) = \prod_{r \in \text{Touched}(j)} B_{\text{tkf}}(n, r, S_{\text{tkf}}(r, i, j-1), S_{\text{tkf}}(r, i, j)).$$

The source state $S_{\text{tkf}}(r, i, j-1)$ is the branch’s *last-touched* state (which, by the carry rule above, may pre-date column $j-1$ if intervening columns left r untouched), so a branch’s TKF transition is charged exactly once, at the next column that touches it, and never at the both-gapped columns it skips over. Omitting the untouched rows is essential: charging them a self-transition $B_{\text{tkf}}(s, s)$ would double-count fragment extensions that never occurred, while forcing them to **S** would inject invalid transitions such as $B_{\text{tkf}}(\text{M}, \text{S})$. Similarly define

$$T_{\text{tkf}, \text{start}}(n, j) = \prod_{r \in \text{Touched}(j)} B_{\text{tkf}}(n, r, \text{S}, S_{\text{tkf}}(r, j, j))$$

for block-start transitions and

$$T_{\text{tkf}, \text{end}}(n, i, j) = \prod_r B_{\text{tkf}}(n, r, S_{\text{tkf}}(r, i, j), \text{E})$$

for block-closing transitions, where the block-end product runs over *all* rows using each branch’s last-touched state $S_{\text{tkf}}(r, i, j)$ (every branch must close, whether or not the final column j touched it).

Felsenstein emission likelihood. Let $U(j, c)$ be the Felsenstein pruning likelihood for column j under site class c , computed over the present subtree at column j with the substitution model $(Q^{(c)}, \pi^{(c)})$.

C.9.5 Intra-Block Forward Recurrence

Within a block assigned to domain n spanning columns $i..j$, the Markovian fragment process induces a forward recurrence over fragment states.

Transition probabilities. Fragment-to-fragment transition (same fragment continues):

$$T_{\text{ext}}(n, j, f, g) = r_{fg}^{(n)}$$

This transition is only available when $A(j-1) = A(j)$ (the presence profile has not changed, so the same fragment can continue).

Fragment termination followed by new fragment start (the presence profile changes, or the Markov chain starts a new fragment):

$$T_{\text{notext}}(n, i, j, f, g) = \rho_f^{(n)} \cdot T_{\text{tkf}}(n, i, j) \cdot w_{ng}$$

Emission weight. Each column j in fragment state g contributes the class-averaged Felsenstein likelihood:

$$E(n, j, g) = \sum_{c=1}^C u_{ngc} U(j, c)$$

Forward recurrence. Define $F_{i,j,n,g}$ as the probability of columns $i..j$ within a block starting at i in domain n , with column j in fragment state g .

Base case:

$$F_{i,i,n,g} = T_{\text{tkf,start}}(n, i) \cdot w_{ng} \cdot E(n, i, g) \quad (\text{C.159})$$

Recursion for $j > i$:

$$F_{i,j,n,g} = \sum_{f=1}^{\mathcal{F}} F_{i,j-1,n,f} \cdot (\delta(A(j-1)=A(j)) T_{\text{ext}}(n, j, f, g) + T_{\text{notext}}(n, i, j, f, g)) \cdot E(n, j, g) \quad (\text{C.160})$$

where $\delta(A(j-1)=A(j))$ restricts the fragment extension term to columns with identical presence profiles.

Block likelihood.

$$G(i, j, n) = \sum_{f=1}^{\mathcal{F}} F_{i,j,n,f} \cdot \rho_f^{(n)} \cdot T_{\text{tkf,end}}(n, i, j) \quad (\text{C.161})$$

C.9.6 The Forward Recursion

Define the Forward quantity

$$F(l, n) = P(\text{columns } 1..l \text{ of MSA, with last block ending at } l \text{ in domain class } n \mid \text{tree, model}). \quad (\text{C.162})$$

Encoding the partition prior (C.158) into the recursion, we initialise with a virtual start marker $F(0, \emptyset) = 1$ and advance by

$$F(l, n) = \kappa_0 v_n \sum_{k=0}^{l-1} \bar{F}(k) G(k+1, l, n), \quad (\text{C.163})$$

where $\bar{F}(0) := 1$ and $\bar{F}(k) := \sum_m F(k, m)$ for $k > 0$. The total data log-likelihood is obtained by closing off the final block with the top-level termination factor:

$$P(\text{MSA} \mid \text{tree, model}) = (1 - \kappa_0) \sum_n F(L, n). \quad (\text{C.164})$$

Full MSA probability. Given a specific partition into blocks $(i_1, j_1, n_1), \dots, (i_B, j_B, n_B)$, the probability of the partitioned MSA given the tree is

$$P(\text{partitioned MSA} \mid \text{tree}) = (1 - \kappa_0) \prod_{b=1}^B \kappa_0 v_{n_b} G(i_b, j_b, n_b)$$

C.9.7 The Backward Recursion

Since the domain class of the *next* block does not depend on the class of the current block (the partition prior factorises over blocks), the backward variable can be collapsed to a scalar. Define

$$\bar{\beta}(l) = P(\text{columns } l+1..L \text{ of MSA} \mid \text{last block ended at column } l, \text{ tree, model}). \quad (\text{C.165})$$

The boundary condition and recursion are

$$\bar{\beta}(L) = 1 - \kappa_0, \quad (\text{C.166})$$

$$\bar{\beta}(k) = \sum_{l=k+1}^L \left[\kappa_0 \sum_n v_n G(k+1, l, n) \right] \bar{\beta}(l), \quad k < L. \quad (\text{C.167})$$

The total likelihood is recoverable as $\bar{\beta}(0)$ —equivalent to (C.164) (since the $k=0$ case expands to $(1-\kappa_0) \sum_n \kappa_0 v_n \sum_{l=1}^L G(1, l, n) [\dots]$, matching the forward expression).

C.9.8 Intra-Block Backward Recurrence

To compute posterior fragment state and site class probabilities, we need an intra-block backward recurrence. Define $B_{i,k,j,n,f}$ as the probability of columns $k..j$ given a block $i..j$ in domain n , with column $k-1$ in fragment state f .

Boundary condition:

$$B_{i,j+1,j,n,f} = \rho_f^{(n)} \cdot T_{\text{tkf,end}}(n, i, j) \quad (\text{C.168})$$

(Note: this depends on f through $\rho_f^{(n)}$, the probability that fragment f terminates.)

Recursion for $k \leq j$:

$$B_{i,k,j,n,f} = \sum_{g=1}^{\mathcal{F}} (\delta(A(k-1)=A(k)) T_{\text{ext}}(n, k, f, g) + T_{\text{notext}}(n, i, k, f, g)) \cdot E(n, k, g) \cdot B_{i,k+1,j,n,g} \quad (\text{C.169})$$

C.9.9 Posterior Domain and Fragment State Assignment

From the inter-block forward F and backward β , and the intra-block forward $F_{i,k,n,f}$ and backward $B_{i,k,j,n,f}$, we recover per-column posteriors.

Posterior domain assignment. Writing $Z = P(\text{MSA} \mid \text{tree, model})$,

$$P(c \text{ in block of class } n \mid \text{MSA}) = \frac{1}{Z} \kappa_0 v_n \sum_{k=0}^{c-1} \sum_{l=c}^L \bar{F}(k) G(k+1, l, n) \bar{\beta}(l). \quad (\text{C.170})$$

Posterior fragment state. The posterior probability that column k is in fragment state f , given a block $i..j$ of domain n , is:

$$P(\text{col } k \text{ is frag } f \mid \text{block } i..j, n) = \frac{F_{i,k,n,f} \cdot B_{i,k+1,j,n,f}}{G(i, j, n)} \quad (\text{C.171})$$

where $B_{i,k+1,j,n,f}$ uses $k+1$ because B gives the probability of the *remaining* columns $k+1..j$ given fragment state f at column k .

Full unconditional fragment state posterior. The full posterior for fragment state at column c , marginalised over all block placements, is

$$P(\text{col } c \text{ is fragtype } f \mid \text{MSA}) = \sum_n \sum_{i \leq c} \sum_{j \geq c} P(\text{block } i..j, n \mid \text{MSA}) \frac{F_{i,c,n,f} B_{i,c+1,j,n,f}}{G(i, j, n)}, \quad (\text{C.172})$$

where the block–domain posterior is

$$P(\text{block } i..j, n \mid \text{MSA}) \propto \bar{F}(i-1) \kappa_0 v_n G(i, j, n) \bar{\beta}(j). \quad (\text{C.173})$$

Posterior site class. The posterior probability of site class c at column k is obtained by mixing over the fragment state posterior:

$$P(\text{class } c \text{ at col } k \mid \text{block } i..j, n) = \sum_f P(\text{col } k \text{ is frag } f \mid \text{block } i..j, n) \cdot \frac{u_{nfc} U(k, c)}{\sum_{c'} u_{nfc'} U(k, c')} \quad (\text{C.174})$$

The full (unconditional) posterior over site class at column k is obtained by further marginalizing over blocks and domain types using the inter-block posterior (C.170).

C.9.10 Root Residue Reconstruction

Given the posterior in (C.170) and (C.174), we obtain a posterior over root residues at column c by mixing over the class assignment and re-using the Felsenstein pruning posterior under each class:

$$P(\text{root}_c = a \mid \text{MSA}) = \sum_n P(c \in n \mid \text{MSA}) \cdot \sum_c P(\text{class } c \text{ at } c \mid n) \cdot P(\text{root}_c = a \mid \text{MSA}, c \in c), \quad (\text{C.175})$$

where the per-class conditional on the right uses Felsenstein pruning at column c with the substitution model $(Q^{(c)}, \pi^{(c)})$. The MAP root sequence is obtained by taking argmax_a of this posterior column-by-column; gaps at the root are determined by the presence annotation obtained from Fitch parsimony.

C.9.11 Why the Trick Fails with Top-Level Indels

If we retained nonzero top-level insertion and deletion rates we would also need to decide on which branch each top-level domain was born or died. Given only the presence pattern at each column, a block whose residues appear on only a single clade is consistent with both (i) an ancestrally present domain that was deleted on every other branch and (ii) a recent insertion. Conditioning on the partition no longer factorises the likelihood across blocks because the per-branch top-level state sequence is correlated across blocks. The algorithm above relies crucially on the absence of top-level births and deaths so that every block is a fully independent problem with a fresh $\mathbf{S}$ at every branch.

Remark C.15 (Relaxing the top-level constraint). The zero top-level indel-rate limit ($n_{\text{top,indel}} = 0$) can be partially relaxed by conditioning on domain presence/absence profiles. At $n_{\text{top,indel}} = 1$, we allow a single top-level domain insertion or deletion event on the tree, yielding $O(R)$ possible domain subtrees (where R is the number of tree rows/edges). At $n_{\text{top,indel}} = 2$, there are $O(R^2)$ configurations, and so on. The probability of each truncation level is roughly exponential in $n_{\text{top,indel}}$ (controlled by $\lambda_0 + \mu_0$), so this provides a systematic expansion controlling the state space explosion. The leading term ($n_{\text{top,indel}} = 0$) is the algorithm presented here.

C.9.12 Complexity

Let $\mathcal{N}$ be the number of top-level domain classes, $\mathcal{F}$ the number of fragment types per domain, C the number of site classes, T the number of tree nodes, and L the number of MSA columns.

- Computing all $G(k+1, l, n)$ by incremental forward updates requires, for each starting column $k+1$, a left-to-right pass over $l = k+1, \dots, L$. Each column update involves $O(\mathcal{F}^2)$ work for the fragment Markov chain transitions and $O(\mathcal{F} \cdot C)$ for the emission weights, plus $O(T)$ for the TKF transition products. Total: $O(L^2 (\mathcal{F}^2 + \mathcal{F} \cdot C + T) \mathcal{N})$.
- Computing F and β given G takes $O(L^2 \mathcal{N})$.
- Computing the per-column posteriors including fragment state requires the intra-block backward, making the total $O(L^3 \mathcal{F}^2 \mathcal{N})$ in the worst case (since $B_{i,k,j}$ depends on both i and j for each k).
- Each column likelihood $U(j, c)$ is computed once and cached as an $O(L \cdot C \cdot T \cdot |\mathcal{A}|)$ precomputation.

Total for the G pass and inter-block DP: $O(L^2 (\mathcal{F}^2 + T) \mathcal{N})$. The intra-block backward for posteriors adds $O(L^3 \mathcal{F}^2 \mathcal{N})$ due to the dependence of $B_{i,k,j,n,f}$ on both i (block start) and j (block end) for each interior column k .

The algorithm is naturally vectorisable: all G updates for a fixed starting column can be computed in parallel across $\mathcal{N}$ domain classes and across tree branches.

C.9.13 Simulation from MixDom

To generate data from the MixDom model on a phylogenetic tree:

1. **Sample domain sequence.** Draw the number of top-level domains $\mathcal{N}_* \sim \text{Geometric}(\kappa_0)$. For each domain, draw its type $n_i \sim \text{Categorical}(v_1, \dots, v_{\mathcal{N}})$.
2. **Sample fragment trajectory.** For each domain of type n , sample a fragment trajectory through the $\mathcal{F}$ -state Markov chain. The initial fragment state is $f_1 \sim \text{Categorical}(w_{n1}, \dots, w_{n\mathcal{F}})$. At each step, transition to fragment state f_{k+1} with probability $r_{f_k, f_{k+1}}^{(n)}$, or terminate the domain with probability $\rho_{f_k}^{(n)} = 1 - \sum_g r_{f_k, g}^{(n)}$.
3. **Sample site class and root residue.** For each emitted site from fragment state f , draw site class $c \sim \text{Categorical}(u_{nf1}, \dots, u_{nfC})$. Draw the root residue $a \sim \pi^{(c)}$.
4. **Evolve down the tree.** For each site, evolve the root residue down the phylogenetic tree using the substitution model $(Q^{(c)}, \pi^{(c)})$ at evolutionary time T_r on each branch r : $b \sim P^{(c)}(T_r) \cdot e_a$, where $P^{(c)}(T_r) = \exp(R^{(c)} T_r)$. Apply the per-branch TKF91 indel process (with domain-specific parameters λ_n, μ_n) to create insertions and deletions.

C.10 Formal Grammar Elaboration Rules

The TKF family of evolutionary models—TKF91, TKF92, MixDom, the TKF Structure Tree, and the TKF Genome—describes the joint evolution of biological sequences subject to insertions, deletions, and substitutions. Despite their apparent diversity, all these models share a common constructive pattern: they begin with a simple weighted context-free grammar (WCFG) for a geometrically distributed number of links, and then systematically elaborate that grammar through a series of formal transformations.

In the TKF91 model (62), each link carries a single character evolving by a continuous-time Markov chain (CTMC). TKF92 (63) extends this by replacing each character with a geometrically distributed fragment of characters. The MixDom model nests a TKF92 process inside a TKF91 process, decorating links with mixtures of domain types, fragment types, and substitution classes. The TKF Structure Tree (28) uses stochastic context-free grammar (SCFG) recursion to model RNA secondary structure with stems (emitting paired characters left and right) and loops. The TKF Genome extends these ideas to entire genomes with coding sequences, introns, RNA structures, and conserved elements.

Making the elaboration steps explicit and composable has several benefits:

- (i) **Correctness:** each transformation can be verified independently, rather than checking a large monolithic grammar.
- (ii) **Modularity:** new models (e.g., RNA models with basepair stacking, codon models with reading-frame-aware indels) can be constructed by composing well-understood building blocks.
- (iii) **Automation:** the transformations are sufficiently formal to be implemented as software operations on grammar objects, enabling automatic derivation of dynamic programming algorithms from high-level model specifications.

This appendix defines seven elaboration rules and the associated null-state management procedures, then shows how each known TKF-family model arises as a specific sequence of elaborations.

C.10.1 Base Grammar

Weighted Context-Free Grammars

Definition C.1 (Weighted Context-Free Grammar). A weighted context-free grammar (WCFG) is a tuple $\mathcal{G} = (\mathcal{N}, \Sigma, \mathcal{P}, S, w)$ where:

- $\mathcal{N}$ is a finite set of nonterminal symbols.
- Σ is a finite set of terminal symbols, disjoint from $\mathcal{N}$.
- $S \in \mathcal{N}$ is the start symbol.
- $\mathcal{P}$ is a finite set of production rules of the form $X \rightarrow \alpha$ where $X \in \mathcal{N}$ and $\alpha \in (\mathcal{N} \cup \Sigma)^*$.
- $w : \mathcal{P} \rightarrow \mathbb{R}_{\geq 0}$ assigns a nonneg weight to each production.

The grammar is proper if for every nonterminal X , the weights of all productions with left-hand side X sum to 1: $\sum_{(X \rightarrow \alpha) \in \mathcal{P}} w(X \rightarrow \alpha) = 1$. In a proper WCFG, weights are probabilities and the grammar defines a stochastic context-free grammar (SCFG).

Definition C.2 (Elaboration Rule). *An elaboration rule (or grammar transformation) is a map $\mathcal{E} : \mathcal{G} \rightarrow \mathcal{G}'$ that takes a WCFG and a set of elaboration parameters, and produces a new WCFG. An elaboration is validity-preserving if it maps proper grammars to proper grammars.*

We distinguish between the *single-sequence grammar* (describing the stationary distribution over sequences) and the *pair grammar* (describing the joint distribution over ancestor–descendant sequence pairs). Most elaborations operate on the single-sequence grammar; the Evolution elaboration (Section C.10.2) converts a single-sequence grammar into a pair grammar.

The Link Grammar The fundamental building block of all TKF-family models is a grammar generating a geometrically distributed number of “links.” This grammar arises from the stationary distribution of the linear birth-death-immigration (BDI) process with per-capita birth rate λ , per-capita death rate $\mu > \lambda$, and immigration rate $\nu = \lambda$.

Definition C.3 (Link Grammar). *The link grammar $\mathcal{G}_{\text{link}}(\kappa)$ with parameter $\kappa = \lambda/\mu \in [0, 1)$ is the WCFG with nonterminals $\{\text{IMM}, \text{MOR}\}$, start symbol IMM , and productions:*

$$\text{IMM} \rightarrow \text{MOR IMM} \quad \text{weight } \kappa \quad (\text{C.176})$$

$$\text{IMM} \rightarrow \epsilon \quad \text{weight } 1 - \kappa \quad (\text{C.177})$$

$$\text{MOR} \rightarrow \text{MOR MOR} \quad \text{weight } \kappa \quad (\text{C.178})$$

$$\text{MOR} \rightarrow \epsilon \quad \text{weight } 1 - \kappa \quad (\text{C.179})$$

Remark C.16. In this grammar, IMM (the “immortal link”) generates a sequence of $n \sim \text{Geometric}(\kappa)$ mortal links. Each MOR can recursively generate further mortal links; the recursive self-loop in (C.178) reflects the offspring-generating property of the BDI process. The distinction between IMM and MOR captures the different roles: the immortal link corresponds to the BDI regime $\nu = \lambda, X(0) = 0$ (immigration from nothing), while mortal links correspond to $\nu = 0, X(0) = 1$ (a single founder that can die).

Remark C.17. The grammar in Definition C.3 generates only the empty string ϵ , since MOR has no terminal-producing rules. The elaboration rules below will add terminal emissions (characters, character pairs, etc.) to the mortal links.

Proposition C.3. *The link grammar $\mathcal{G}_{\text{link}}(\kappa)$ is proper for any $\kappa \in [0, 1)$. Under the start symbol IMM , the number of MOR expansions before reaching ϵ is distributed as $\text{Geometric}(\kappa)$ (with support $\{0, 1, 2, \dots\}$ and mean $\kappa/(1 - \kappa)$).*

Proof. For IMM : the productions (C.176) and (C.177) have weights κ and $1 - \kappa$, summing to 1. Similarly for MOR . The number of MOR nonterminals generated by IMM before choosing ϵ is geometric with parameter κ by the standard geometric series argument. $\square$

C.10.2 Elaboration Rules

We now define each elaboration rule as a formal grammar transformation. For each rule, we specify:

- The *input* grammar fragment (which productions are targeted).
- The *output* grammar fragment (the replacement productions).
- The *parameters* introduced by the elaboration.
- The *validity conditions* under which the transformation preserves properness.

CTMC Expansion The most basic elaboration decorates each link with a character (or tuple of characters) that evolves according to a finite-state CTMC. This is the step that takes the bare link grammar to a model with observable sequences.

Definition C.4 (Emission Type). *An emission type specifies where terminal symbols appear relative to the recursive expansion of a nonterminal:*

- **Left-emission:** *terminal appears to the left of the recursive part. Production form: $X \rightarrow c \alpha$ where $c \in \Sigma$.*
- **Right-emission:** *terminal appears to the right. Production form: $X \rightarrow \alpha c$.*
- **LR-emission:** *terminals appear on both sides. Production form: $X \rightarrow c_L \alpha c_R$ where $c_L, c_R \in \Sigma$. This is the form needed for RNA basepair models (stems).*

Definition C.5 (CTMC Expansion). *Let $\mathcal{G}$ be a WCFG containing a nonterminal X with an ϵ -generating production used for link termination. Let $\mathcal{A}$ be a finite alphabet, π a probability distribution over $\mathcal{A}$, and Q a rate matrix with stationary distribution π . The left CTMC expansion of X with parameters $(\mathcal{A}, \pi, Q)$ replaces every production $X \rightarrow \alpha$ (where $\alpha \neq \epsilon$) with the family of productions:*

$$X \rightarrow c \alpha \quad \text{weight } w(X \rightarrow \alpha) \cdot \pi_c \quad \text{for each } c \in \mathcal{A} \quad (\text{C.180})$$

The ϵ -production $X \rightarrow \epsilon$ is left unchanged.

The right CTMC expansion replaces $X \rightarrow \alpha$ ($\alpha \neq \epsilon$) with:

$$X \rightarrow \alpha c \quad \text{weight } w(X \rightarrow \alpha) \cdot \pi_c \quad \text{for each } c \in \mathcal{A} \quad (\text{C.181})$$

The LR CTMC expansion with alphabet $\mathcal{A} \times \mathcal{A}$ and equilibrium distribution $\pi(c_L, c_R)$ replaces $X \rightarrow \alpha$ ($\alpha \neq \epsilon$) with:

$$X \rightarrow c_L \alpha c_R \quad \text{weight } w(X \rightarrow \alpha) \cdot \pi(c_L, c_R) \quad \text{for each } (c_L, c_R) \in \mathcal{A} \times \mathcal{A} \quad (\text{C.182})$$

Proposition C.4. *Left, right, and LR CTMC expansion are validity-preserving: if $\mathcal{G}$ is proper, then so is the elaborated grammar.*

Proof. For left CTMC expansion: the total weight of productions with LHS X becomes $\sum_{c \in \mathcal{A}} \sum_{\alpha \neq \epsilon} w(X \rightarrow \alpha) \cdot \pi_c + w(X \rightarrow \epsilon) = \sum_{\alpha \neq \epsilon} w(X \rightarrow \alpha) \cdot 1 + w(X \rightarrow \epsilon) = \sum_{\alpha} w(X \rightarrow \alpha) = 1$. The right and LR cases are analogous. $\square$

Example C.1 (TKF91). Applying left CTMC expansion to MOR in the link grammar $\mathcal{G}_{\text{link}}(\kappa)$ with alphabet $\mathcal{A}$ and equilibrium distribution π yields:

IMM $\rightarrow$ MOR IMM	κ
IMM $\rightarrow \epsilon$	$1 - \kappa$
MOR $\rightarrow c$ MOR MOR	$\kappa \cdot \pi_c \quad (c \in \mathcal{A})$
MOR $\rightarrow c$	$(1 - \kappa) \cdot \pi_c \quad (c \in \mathcal{A})$

This is the stationary (single-sequence) grammar for TKF91: it generates a geometric number of links, each carrying a character drawn i.i.d. from π .

Example C.2 (TKF Structure Tree stems). In the TKF Structure Tree, stems use LR CTMC expansion. The nonterminal S (“stem”) generates basepairs:

$$\begin{array}{ll} S \rightarrow c_L S c_R & (1 - \kappa_S) \cdot \pi_S(c_L, c_R) \\ S \rightarrow L & \kappa_S \end{array}$$

where L is the loop nonterminal (which uses left CTMC expansion), and $\pi_S(c_L, c_R)$ is the joint equilibrium distribution over basepairs. The LR emission enables the grammar to generate palindromic structures characteristic of RNA secondary structure.

Fragment Expansion Fragment expansion takes TKF91 to TKF92 by replacing each single-character link with a geometrically distributed sequence of characters.

Definition C.6 (Fragment Expansion). *Let $\mathcal{G}$ be a WCFG with a nonterminal MOR representing a mortal link. The fragment expansion with parameter $r \in [0, 1)$ (the extension probability) replaces MOR with three nonterminals MOR_S (fragment start), MOR_X (fragment extend), and MOR_E (fragment end), defined as follows.*

Every production in $\mathcal{G}$ that references MOR on its right-hand side is updated to reference MOR_S instead. Then:

Before (link with single terminal slot):

$$MOR \rightarrow [terminal] \alpha \quad w_{orig}$$

After (link with geometric fragment):

$$\begin{array}{ll} MOR_S \rightarrow MOR_X & 1 \\ MOR_X \rightarrow [terminal] MOR_X & r \\ MOR_X \rightarrow [terminal] MOR_E & 1 - r \\ MOR_E \rightarrow \alpha & w_{orig} \end{array}$$

Here $[terminal]$ denotes whatever terminal emission was associated with MOR (a single character from a CTMC expansion, or a character pair, etc.), and α denotes the rest of the original production’s right-hand side (the recursive continuation).

More precisely, if MOR had productions $MOR \rightarrow c MOR MOR$ (weight $\kappa \cdot \pi_c$) and $MOR \rightarrow c$ (weight $(1 - \kappa) \cdot \pi_c$) after CTMC expansion, the fragment expansion produces:

$$\begin{array}{ll} MOR_S \rightarrow MOR_X & 1 \\ MOR_X \rightarrow c MOR_X & r \cdot \pi_c \quad (c \in \mathcal{A}) \\ MOR_X \rightarrow c MOR_E & (1 - r) \cdot \pi_c \quad (c \in \mathcal{A}) \\ MOR_E \rightarrow MOR_S MOR_S & \kappa \\ MOR_E \rightarrow \epsilon & 1 - \kappa \end{array}$$

where MOR_E takes over the inter-link continuation logic from the original MOR .

Proposition C.5. *Fragment expansion is validity-preserving. The expected number of terminals per fragment is $1/(1 - r)$.*

Remark C.18. Fragment expansion must be applied *after* CTMC expansion (or simultaneously), because it assumes the link already has terminal emissions. If applied to a bare link grammar (no terminals), the result would have fragments of ϵ ’s, which is degenerate.

Example C.3 (TKF92). TKF92 = link grammar $\rightarrow$ (fragment expansion with parameter r) $\rightarrow$ (CTMC expansion with $\mathcal{A}, \pi, Q$). Equivalently, CTMC expansion first, then fragment expansion. Both orderings yield the same grammar, because fragment expansion simply wraps the terminal emission in a geometric self-loop. The resulting grammar generates sequences where each link produces a fragment of $K \sim \text{Geometric}(r)$ characters from π .

Mixture Expansion Mixture expansion decorates a link with a latent categorical variable whose value determines subsequent model parameters.

Definition C.7 (Mixture Expansion). Let $\mathcal{G}$ be a WCFG containing a nonterminal X . Let $\{1, \dots, K\}$ be a finite set of mixture components with weights $p_1, \dots, p_K > 0$ satisfying $\sum_k p_k = 1$. The mixture expansion of X with components K and weights (p_k) replaces X with K new nonterminals $X_1, \dots, X_K$ and modifies all productions referencing X as follows.

Every production $Y \rightarrow \alpha X \beta$ in $\mathcal{G}$ (where $Y \neq X$ and $\alpha, \beta \in (\mathcal{N} \cup \Sigma)^*$) is replaced by K productions:

$$Y \rightarrow \alpha X_k \beta \quad \text{weight } w(Y \rightarrow \alpha X \beta) \cdot p_k \quad \text{for } k = 1, \dots, K \quad (\text{C.183})$$

The productions of X itself are copied to each X_k : for each production $X \rightarrow \gamma$, create

$$X_k \rightarrow \gamma_k \quad \text{weight } w(X \rightarrow \gamma) \quad (\text{C.184})$$

where γ_k is γ with any self-references to X replaced by X_k (maintaining the component assignment within a link).

Each X_k may then undergo different subsequent elaborations (e.g., different CTMC parameters, different fragment extension rates).

Proposition C.6. Mixture expansion is validity-preserving.

Proof. For Y : the total weight of productions referencing X_k is $\sum_k w(\cdot) \cdot p_k = w(\cdot) \cdot 1$. For each X_k : the production weights are copied from X , hence sum to 1. $\square$

Remark C.19. In principle, the mixture component selector could itself evolve via a CTMC across evolutionary time, provided that nested expansions are resampled from equilibrium upon a change of component. In this appendix, we restrict attention to the simpler case where the component assignment is fixed throughout the lifetime of a link (i.e., it is sampled once at birth and does not change).

Example C.4 (MixDom: three levels of mixture). In the MixDom model, mixture expansion is applied at three levels:

1. **Domain mixture:** each top-level link is assigned a domain type $n \sim \text{Categorical}(v_1, \dots, v_N)$, determining (λ_n, μ_n) for the nested TKF92 process.
2. **Fragment process:** within domain n , the initial fragment type is $f \sim \text{Categorical}(w_{n1}, \dots, w_{n\mathcal{F}})$. Subsequent fragments are drawn from the $\mathcal{F} \times \mathcal{F}$ transition matrix $r_{fg}^{(n)}$ (the $\mathcal{F} = 1$ case reduces to IID geometric extension).
3. **Site class mixture:** within fragment state f of domain n , each character position is assigned a class $c \sim \text{Categorical}(u_{nf1}, \dots, u_{nfC})$, determining the substitution parameters $(Q^{(c)}, \pi^{(c)})$.

Link Sequence Concatenation Concatenation decorates a single link with two or more consecutive sub-links.

Definition C.8 (Link Sequence Concatenation). *Let $\mathcal{G}$ be a WCFG with nonterminal X representing a link. The binary concatenation of X replaces it with two new nonterminals X_A, X_B and the production:*

$$X \rightarrow X_A X_B \quad \text{weight } 1 \quad (\text{C.185})$$

X_A and X_B may then be independently elaborated. For n -ary concatenation, X is replaced by $X \rightarrow X_1 X_2 \cdots X_n$.

Remark C.20 (Concatenation combined with mixture and fragments). A powerful pattern combines mixture, fragment, and concatenation to decorate a link with a variable-length sequence of categorically typed sub-links:

1. Apply fragment expansion to the link, creating a geometric number of sub-link slots.
2. Apply mixture expansion to each sub-link slot, assigning it a categorical type.
3. If correlations between adjacent types are desired, replace the i.i.d. mixture with an HMM output distribution: the type sequence is generated by a hidden Markov model whose transition matrix captures adjacency preferences.

This yields a variable number of concatenated sub-links of varying types with Markovian correlations. In the TKF Genome, this pattern is used for genomic regions (coding, noncoding, structural) within a top-level link sequence.

Example C.5 (TKF Genome: region concatenation). The TKF Genome's top-level grammar has:

GENOME $\rightarrow$ REGION GENOME	$1 - \kappa_R$
GENOME $\rightarrow \epsilon$	κ_R
REGION $\rightarrow$ INTER	p_N
FWDCDS	$p_G/2$
REVCDS	$p_G/2$
STRUCT	p_S
CONS	p_C

Here **REGION** is both a mixture expansion (over region types) and a concatenation point (each region type expands into its own sub-grammar).

Non-Recursive Nesting Non-recursive nesting splices a complete sub-grammar into the transitions of a mortal link, without introducing bifurcation or self-reference. This is how nesting works in MixDom.

Definition C.9 (Non-Recursive Nesting). *Let $\mathcal{G}_{\text{outer}}$ be a link grammar with nonterminal **MOR** (mortal link), and let $\mathcal{G}_{\text{inner}}$ be an independent link grammar with its own start symbol **IMM**_{inner} and parameters. The non-recursive nesting of $\mathcal{G}_{\text{inner}}$ into **MOR** of $\mathcal{G}_{\text{outer}}$ replaces each terminal-emitting production of **MOR** with:*

$$\text{MOR} \rightarrow \text{IMM}_{\text{inner}} \alpha \quad \text{weight } w_{\text{orig}} \quad (\text{C.186})$$

where α is the original continuation. The inner grammar generates a complete sequence of inner links, each with its own parameters, for every outer mortal link.

More precisely, wherever *MOR* previously emitted a terminal symbol c , it now expands into the entire inner grammar $\mathcal{G}_{\text{inner}}$, which itself may generate zero or more characters. The inner grammar's ϵ -productions (zero-length inner sequences) give rise to null states in the combined grammar.

Remark C.21. The key difference from recursive nesting (Section C.10.2) is that $\mathcal{G}_{\text{inner}}$ does *not* reference any nonterminals of $\mathcal{G}_{\text{outer}}$. There is no possibility of re-entering the outer grammar from within the inner grammar. This ensures that the combined grammar generates strings from a regular language (at each level), rather than a context-free language.

Example C.6 (MixDom as non-recursive nesting). The MixDom model nests a Markovian fragment process (inner grammar) into each mortal link of a TKF91 process (outer grammar). The outer link grammar has parameters (λ_0, μ_0) and the inner grammar, for domain type n , has parameters $(\lambda_n, \mu_n, r_{fg}^{(n)}, u_{nfc}, Q^{(c)}, \pi^{(c)})$. Each outer mortal link, instead of emitting a single character, expands into a domain sequence governed by the Markovian fragment process.

Since the inner grammar can generate the empty string (the inner link sequence may have length zero), this nesting creates null states in the combined Pair HMM. These must be eliminated by the procedures of Section C.10.3.

Recursive Nesting Recursive nesting, used in the TKF Structure Tree, allows transitions into a mortal link to spawn a bifurcation: a new nonterminal whose sub-grammar may reference the original grammar's nonterminals.

Definition C.10 (Recursive Nesting). Let $\mathcal{G}$ be a link grammar with nonterminals including *MOR* (mortal link). Let *BIF* be a new nonterminal with its own sub-grammar $\mathcal{G}_{\text{bif}}$ that may reference nonterminals of $\mathcal{G}$ (including *IMM* and *MOR*).

The right recursive nesting of *BIF* at *MOR* replaces each terminal-emitting production of *MOR* with a mixture of terminal emission and bifurcation:

Before:

$$MOR \rightarrow c \alpha \qquad w_{\text{orig}} \cdot \pi_c$$

After:

$$\begin{array}{lll} MOR \rightarrow c \alpha & w_{\text{orig}} \cdot s \cdot \pi_c & (\text{terminal link, probability } s) \\ MOR \rightarrow BIF \alpha & w_{\text{orig}} \cdot (1 - s) \cdot v_{\text{bif}} & (\text{bifurcation, probability } 1 - s) \end{array}$$

where $s \in (0, 1]$ is the probability that a link is a terminal (character-emitting) link rather than a nesting point, and v_{bif} is a distribution over bifurcation types if there are multiple *BIF* variants.

For left recursive nesting:

$$MOR \rightarrow \alpha BIF \qquad w_{\text{orig}} \cdot (1 - s) \cdot v_{\text{bif}}$$

The sub-grammar $\mathcal{G}_{\text{bif}}$ for *BIF* defines how the bifurcation expands. Since it may reference the start symbol of $\mathcal{G}$ (e.g., *IMM*), the combined grammar is genuinely recursive: link sequences can contain nested link sequences of arbitrary depth.

Remark C.22. The recursive nesting creates null cycles whenever the nested sub-grammar can generate the empty string. These must be handled by the null-state management procedures (Section C.10.3). In the TKF Structure Tree, the nullability fixed-point iteration solves for the probability that each nonterminal generates ϵ .

Example C.7 (TKF Structure Tree: stems and loops). The TKF Structure Tree has two types of link sequences:

- *Loop sequences* (L): left-emitting links generating single nucleotides. Nonterminal rule: $L \rightarrow c_L L$ with CTMC expansion (left emission).
- *Stem sequences* (S): LR-emitting links generating basepairs. Nonterminal rule: $S \rightarrow c_L S c_R$ with CTMC expansion (LR emission).

The recursive nesting works as follows. Within a loop sequence, a mortal link may either emit a character (probability s_L) or spawn a stem (probability $1 - s_L$):

$$L \rightarrow c L \quad | \quad S L$$

At the base of a stem (when the self-loop terminates), the grammar transitions to a loop:

$$S \rightarrow c_L S c_R \quad | \quad L$$

This creates the alternating stem-loop structure characteristic of RNA secondary structure. The recursion arises because a loop can spawn a stem, which eventually returns to a loop, which can spawn another stem, *ad infinitum*.

Example C.8 (TKF92 with Recursive Domains). The recursive domain model has nonterminals L_n (link sequence for domain n), A_n (aligned component), I_n (inserted component), D_n (deleted component), etc. The aligned component rule is:

$$\begin{aligned} A_n &\rightarrow c & s_n \cdot \pi_{n,c} & \text{(terminal: character)} \\ A_n &\rightarrow L_{n'} & (1 - s_n) \cdot v_{nn'} & \text{(bifurcation: nested link sequence)} \end{aligned}$$

Since $L_{n'}$ can reference L_n (if $n' = n$ or through a chain of domain transitions), this creates genuine recursion: link sequences contain domains that contain further link sequences.

Evolution The Evolution elaboration converts a single-sequence grammar (describing the stationary distribution) into a pair grammar (describing the joint ancestor-descendant distribution at evolutionary time T).

Definition C.11 (Evolution Elaboration). *Let $\mathcal{G}_1$ be a single-sequence grammar with link nonterminal X generating terminals from alphabet $\mathcal{A}$ with equilibrium distribution π . Let Q be the CTMC rate matrix and $T > 0$ the evolutionary time.*

Define the TKF parameters:

$$\alpha = e^{-\mu T}, \quad \beta = \frac{\lambda(e^{-\lambda T} - e^{-\mu T})}{\mu e^{-\lambda T} - \lambda e^{-\mu T}}, \quad \gamma = 1 - \frac{\mu\beta}{\lambda(1 - \alpha)}, \quad \kappa = \frac{\lambda}{\mu}$$

The evolution elaboration replaces each link nonterminal X in $\mathcal{G}_1$ with three nonterminals X_M , X_I , X_D in the pair grammar $\mathcal{G}_2$:

Before (single-sequence grammar, left-emitting):

$$\begin{aligned} X &\rightarrow c X X & \kappa \cdot \pi_c & \text{(link with offspring)} \\ X &\rightarrow c & (1 - \kappa) \cdot \pi_c & \text{(terminal link)} \end{aligned}$$

After (pair grammar):

For the immortal link continuation, each nonterminal Y that previously generated the link sequence $Y \rightarrow X \ Y \mid \epsilon$ transforms as follows:

$$\begin{array}{ll}
Y_M \rightarrow X_M Y_M & (1 - \beta_Y)\kappa\alpha \\
Y_M \rightarrow X_I Y_M & \beta_Y \\
Y_M \rightarrow X_D Y_D & (1 - \beta_Y)\kappa(1 - \alpha) \\
Y_M \rightarrow \epsilon & (1 - \beta_Y)(1 - \kappa) \\
\\
Y_D \rightarrow X_M Y_M & (1 - \gamma_Y)\kappa\alpha \\
Y_D \rightarrow X_I Y_M & \gamma_Y \\
Y_D \rightarrow X_D Y_D & (1 - \gamma_Y)\kappa(1 - \alpha) \\
Y_D \rightarrow \epsilon & (1 - \gamma_Y)(1 - \kappa)
\end{array}$$

where β_Y and γ_Y use the appropriate (λ, μ) for the link sequence that Y belongs to.

The subscript indicates the alignment type:

- X_M (**match**): ancestral link survived; emits aligned pair (c_a, c_d) with probability $\pi_{c_a} \cdot \exp(RT)_{c_a c_d}$.
- X_I (**insert**): new link born in descendant; emits descendant-only character c_d with probability π_{c_d} .
- X_D (**delete**): ancestral link died; emits ancestor-only character c_a with probability π_{c_a} .

The transition weights are precisely the entries of the TKF91 Pair HMM transition matrix $\tau(\lambda, \mu, T)$.

Proposition C.7. The evolution elaboration roughly triples the number of nonterminals (each single-sequence nonterminal becomes three pair nonterminals). For LR-emitting nonterminals (as in stem sequences), match states emit paired tuples $(c_L^a, c_R^a, c_L^d, c_R^d)$, insert states emit (c_L^d, c_R^d) , and delete states emit (c_L^a, c_R^a) .

Remark C.23. When fragment expansion has been applied before evolution, the fragment self-loop interleaves with the alignment states. In TKF92, the Pair HMM transition matrix becomes τ' where the M, I, and D self-loops gain a fragment-extension component:

$$\tau'_{aa} = r + (1 - r)\tau_{aa} \quad \text{for } a \in \{M, I, D\}$$

and off-diagonal transitions are scaled by $(1 - r)$:

$$\tau'_{ab} = (1 - r)\tau_{ab} \quad \text{for } a \neq b$$

Remark C.24. For LR-emitting grammars (such as stem sequences in the TKF Structure Tree), the evolution elaboration creates nonterminals X_M, X_I, X_D whose left and right emissions are correlated:

- $X_M \rightarrow c_L^x c_L^y X_M c_R^y c_R^x$: ancestral basepair (c_L^x, c_R^x) evolved to descendant basepair (c_L^y, c_R^y) (match).
- X_I : emits only descendant basepair $c_L^y \cdots c_R^y$ (insertion).
- X_D : emits only ancestral basepair $c_L^x \cdots c_R^x$ (deletion).

The ancestor terminals go on the outside, the descendant terminals on the inside (or vice versa), preserving the palindromic nesting.

Example C.9 (TKF91 Pair HMM). Applying evolution to the TKF91 single-sequence grammar (Example C.1) yields the standard 5-state Pair HMM ($\mathbf{S}, \mathbf{M}, \mathbf{I}, \mathbf{D}, \mathbf{E}$) with transition matrix $\tau(\lambda, \mu, T)$.

C.10.3 Null State Management

Null State Identification

Definition C.12 (Null state). *A nonterminal X in a WCFG is nullable if there exists a derivation $X \Rightarrow^* \epsilon$. The nullability $\eta(X) = P(X \Rightarrow^* \epsilon)$ is the probability that a parse tree rooted at X yields the empty string.*

Elaborations that create null states include:

1. **Non-recursive nesting**: the inner grammar may generate the empty string (e.g., a domain in MixDom may contain zero fragments).
2. **Recursive nesting**: nested link sequences can be empty.
3. **Mixture expansion** combined with nesting: a mixture component that expands into a nullable sub-grammar.

In the MixDom Pair HMM, null states arise because each domain's inner TKF92 process can generate an empty sequence. The probability of an empty domain at evolutionary time T is $z_T = \sum_n v_n(1 - \kappa_n)(1 - \beta_n)$.

Null State Removal: The $(I - T_{NN})^{-1}$ Closure

Definition C.13 (Null Closure). *Let $\mathcal{G}$ be a WCFG (or equivalently an HMM/transducer) with states partitioned into emitting states Ω and non-emitting (null) states $\mathcal{Z}$. Let $T_{\mathcal{Z}\mathcal{Z}}$ be the submatrix of transition weights among null states. The null closure is:*

$$N^* = (I - T_{\mathcal{Z}\mathcal{Z}})^{-1} = \sum_{k=0}^{\infty} T_{\mathcal{Z}\mathcal{Z}}^k \quad (\text{C.187})$$

This converges provided the spectral radius $\rho(T_{\mathcal{Z}\mathcal{Z}}) < 1$.

Proposition C.8 (Effective Transition Matrix). *The effective transition matrix between emitting (and start/end) states, with all null-state paths summed out, is:*

$$\hat{T} = T_{\Omega\Omega} + T_{\Omega\mathcal{Z}} \cdot (I - T_{\mathcal{Z}\mathcal{Z}})^{-1} \cdot T_{\mathcal{Z}\Omega} \quad (\text{C.188})$$

where subscripts denote submatrices restricted to the indicated state sets.

Null Cycle Detection and Removal

Definition C.14 (Null Cycle). *A null cycle in a WCFG is a chain of unit productions (productions whose right-hand side is a single nonterminal) that returns to the starting nonterminal: $X \rightarrow Y_1 \rightarrow Y_2 \rightarrow \dots \rightarrow X$. In the context of HMMs/transducers, this corresponds to a cycle among non-emitting states.*

Null cycles arise in two main situations:

1. **Non-recursive nesting with empty inner grammars:** when the inner grammar can produce ϵ , a path $S \rightarrow \text{MOR}_{\text{outer}} \rightarrow \text{IMM}_{\text{inner}} \rightarrow \epsilon \rightarrow \text{MOR}_{\text{outer}}$ creates a cycle through null states.
2. **Recursive nesting:** the chain $L'_n \rightarrow M'_n \rightarrow S'_n \rightarrow A'_n \rightarrow L'_{n'}$ creates a null cycle when $n' = n$ or when the chain of domain transitions eventually returns to n .

Definition C.15 (Null Cycle Removal). *To remove null cycles from a WCFG:*

1. **Compute nullabilities:** for each nonterminal X , compute $\eta(X) = P(X \Rightarrow^* \epsilon)$. In the non-recursive case, this can be done in closed form. In the recursive case, iterate the fixed-point equations:

$$\eta(X) = \sum_{(X \rightarrow \alpha) \in \mathcal{P}} w(X \rightarrow \alpha) \cdot \prod_{Y \in \alpha} \eta(Y) \quad (\text{C.189})$$

where the product is over nonterminals in α , with $\eta(\text{terminal}) = 0$ and $\eta(\epsilon) = 1$.

Initialize $\eta^{(0)}(X) = 0$ for all X and iterate until convergence.

2. **Create non-nullable copies:** for each nullable nonterminal X , create X' whose productions never generate ϵ . For a bifurcation rule $X \rightarrow Y Z$, the non-nullable version adds:

$$\begin{array}{ll} X' \rightarrow Y' Z' & w(X \rightarrow YZ) \\ X' \rightarrow Y' & w(X \rightarrow YZ) \cdot \eta(Z) \\ X' \rightarrow Z' & w(X \rightarrow YZ) \cdot \eta(Y) \end{array}$$

This accounts for the two ways one child can be null.

3. **Remove unit-production cycles:** identify cycles $X' \rightarrow Y'_1 \rightarrow \dots \rightarrow X'$ among the non-nullable nonterminals. For each such cycle, compute the transition matrix $\mathcal{A}$ among the cycle's nonterminals and replace the cycle with its closure $(I - \mathcal{A})^{-1}$, distributing the accumulated weight to the non-cyclic continuations.

Example C.10 (Recursive domains: nullability fixed point). In the recursive domain model, $\eta(\mathbf{C}_n)$ (nullability of the child link sequence nonterminal) satisfies:

$$\eta(\mathbf{C}_n) = \frac{1 - \kappa_n}{1 - \kappa_n(1 - s_n) \sum_{n'} v_{nn'} \eta(\mathbf{C}_{n'})}$$

This is solved by initializing $x_n^{(0)} = 0$ and iterating:

$$x_n^{(k+1)} = \frac{1 - \kappa_n}{1 - \kappa_n(1 - s_n) \sum_{n'} v_{nn'} x_{n'}^{(k)}}$$

After convergence, the null cycles $L'_n \rightarrow L'_{n'}$ and $\mathbf{C}'_n \rightarrow \mathbf{C}'_{n'}$ are removed using the $(I - \mathcal{A})^{-1}$ and $(I - \mathcal{B})^{-1}$ closures respectively, where $\mathcal{A}$ and $\mathcal{B}$ are the $\mathcal{N} \times \mathcal{N}$ transition matrices among domain types.

C.10.4 Composition Properties

Commutativity and Order The elaboration rules do not, in general, commute. The following table summarizes the ordering constraints.

Constraint	Reason
CTMC expansion commutes with mixture expansion	Mixture selects which CTMC parameters to use; the order of these two operations does not affect the final grammar. Both orderings produce the same set of productions.
Fragment expansion must follow (or be simultaneous with) CTMC expansion	Fragment expansion wraps terminal emissions in a geometric self-loop. Without terminals, fragment expansion produces fragments of ϵ 's.
Non-recursive nesting must follow both CTMC and fragment expansion of the inner grammar	The inner grammar must be fully specified before it can be spliced into the outer grammar.
Evolution must be applied last (after all structural elaborations)	Evolution triples the nonterminals and introduces alignment-dependent transition weights (α, β, γ) . Applying structural elaborations after evolution would require modifying all three copies independently.
Mixture expansion commutes with concatenation	Both are structural operations on different aspects of a link.
Null state removal must follow all nullable elaborations but precede distillation	All null states must be identified before they can be summed out. The distilled order-1 machines assume null-free grammars.

Validity Conditions

Definition C.16 (Well-Formed Elaborated Grammar). *An elaborated grammar $\mathcal{G}'$ is well-formed if:*

1. **Properness:** *for every nonterminal X , the production weights sum to 1.*
2. **No unresolved null cycles:** *after null state removal, no cycles among non-emitting states remain. Equivalently, $\rho(T_{\mathbb{Z}\mathbb{Z}}) < 1$ for the null-state transition matrix.*
3. **Convergent nullability:** *for recursive grammars, the fixed-point iteration for nullabilities converges. A sufficient condition is that $\kappa_n < 1$ and $s_n > 0$ for all domain types (every link has a positive probability of being terminal rather than a nesting point).*
4. **Finite expected derivation length:** *the expected total number of terminals generated is finite. For the link grammar, this requires $\kappa < 1$. For recursive nesting, additional conditions on the nesting probabilities are needed.*

Proposition C.9. *Each elaboration rule defined above is validity-preserving under its stated conditions. Composition of validity-preserving elaborations is validity-preserving, provided the ordering constraints above are respected.*

Derivation of Existing Models We now show explicitly how each known TKF-family model arises as a sequence of elaborations applied to the base link grammar.

TKF91.

$$\boxed{\text{TKF91}} = \mathcal{G}_{\text{link}}(\kappa) \xrightarrow{\text{CTMC}(\mathcal{A}, \pi, Q)} \mathcal{G}_{\text{TKF91}}$$

Steps:

1. Start with the link grammar $\mathcal{G}_{\text{link}}(\kappa)$ (Definition C.3).
2. Apply left CTMC expansion (Definition C.5) to MOR with alphabet $\mathcal{A}$, equilibrium π , rate matrix Q .

Result: each mortal link emits a single character from π . The pair grammar is obtained by applying evolution (Definition C.11), yielding the standard 5-state Pair HMM with transition matrix $\tau(\lambda, \mu, T)$.

TKF92.

$$\boxed{\text{TKF92}} = \mathcal{G}_{\text{link}}(\kappa) \xrightarrow{\text{Frag}(r)} \xrightarrow{\text{CTMC}(\mathcal{A}, \pi, Q)} \mathcal{G}_{\text{TKF92}}$$

Steps:

1. Start with $\mathcal{G}_{\text{link}}(\kappa)$.
2. Apply fragment expansion (Definition C.6) with extension probability r .
3. Apply left CTMC expansion to each fragment position.

Result: each mortal link emits a fragment of $K \sim \text{Geometric}(r)$ characters. The pair grammar has self-looping match/insert/delete states with fragment-extension probability r .

MixDom: Markovian fragments.

$$\boxed{\text{MixDom}} = \mathcal{G}_{\text{link}}(\kappa_0) \xrightarrow{\text{Mix}(v_n)} \xrightarrow{\text{NonRecNest}\left(\mathcal{G}_{\text{link}}(\kappa_n) \xrightarrow{\text{HMM}(r^{(n)})} \xrightarrow{\text{Mix}(w_{nf})} \xrightarrow{\text{Mix}(u_{nfc})} \xrightarrow{\text{CTMC}(\mathcal{A}, \pi^{(c)}, Q^{(c)})} \right)}_{\text{for each } n} \mathcal{G}_{\text{MixDom}}$$

Steps:

1. Start with $\mathcal{G}_{\text{link}}(\kappa_0)$ (outer/top-level link grammar).
2. Apply mixture expansion (Definition C.7) to MOR with domain types $n \sim \text{Categorical}(v_1, \dots, v_N)$.
3. For each domain type n , construct an inner grammar:
 - (a) Start with $\mathcal{G}_{\text{link}}(\kappa_n)$ (inner link grammar).
 - (b) Replace geometric fragment extension with the Markovian fragment HMM governed by the $\mathcal{F} \times \mathcal{F}$ transition matrix $r_{fg}^{(n)}$.
 - (c) Apply mixture expansion for initial fragment types $f \sim \text{Categorical}(w_{n1}, \dots, w_{nF})$.

- (d) Apply mixture expansion for site classes $c \sim \text{Categorical}(u_{nf1}, \dots, u_{nfC})$.
 - (e) Apply CTMC expansion with $(\mathcal{A}, \pi^{(c)}, Q^{(c)})$.
4. Apply non-recursive nesting (Definition C.9): splice each domain's inner grammar into the corresponding outer mortal link.
 5. Apply null state removal (Section C.10.3): the inner grammar can generate empty sequences (null domains), creating the null states **A**, **B**, **C** in the null-separated Pair HMM. The $(I - T_{ZZ})^{-1}$ closure reduces the 8-state null-separated Pair HMM to the effective 5-state matrix $\mathcal{T}$.

TKF Structure Tree.

$$\boxed{\text{TKF Structure Tree}} = \mathcal{G}_{\text{link}}(\kappa_L) \xrightarrow{\text{CTMC}_L(\mathcal{A}, \pi_L, Q_L)} \xrightarrow{\text{RecNest}(\mathcal{G}_{\text{link}}(\kappa_S) \xrightarrow{\text{CTMC}_{LR}(\mathcal{A}^2, \pi_S, Q_S)})} \mathcal{G}_{\text{ST}}$$

Steps:

1. Start with $\mathcal{G}_{\text{link}}(\kappa_L)$ (loop link grammar).
2. Apply left CTMC expansion for loops with $(\mathcal{A}, \pi_L, Q_L)$.
3. Apply recursive nesting (Definition C.10): within the loop, a mortal link may spawn a stem. The stem sub-grammar is:
 - (a) Start with $\mathcal{G}_{\text{link}}(\kappa_S)$ (stem link grammar).
 - (b) Apply LR CTMC expansion with basepair alphabet $(\mathcal{A} \times \mathcal{A}, \pi_S, Q_S)$.
 - (c) At stem termination, return to the loop grammar (creating the recursion $S \rightarrow L$).
4. The loop grammar now has two types of mortal links:
 - Character-emitting links (probability s_L): emit a single nucleotide c with $\pi_L(c)$.
 - Stem-spawning links (probability $1 - s_L$): expand into a stem S nonterminal.

The LR emission in the stem grammar is critical: the rule $S \rightarrow c_L S c_R$ emits characters on *both* sides of the recursive expansion, generating the palindromic base-pairing structure of RNA stems.

TKF Genome.

$$\boxed{\text{TKF Genome}} = \mathcal{G}_{\text{link}}(\kappa_R) \xrightarrow{\text{Concat+Mix}(\text{region types})} \xrightarrow{\text{various nested elaborations per region type}} \mathcal{G}_{\text{Genome}}$$

Steps:

1. Start with $\mathcal{G}_{\text{link}}(\kappa_R)$ (top-level genomic region grammar).
2. Apply concatenation + mixture: each link is a “region” selected from types $\{\text{INTER}, \text{FWD CDS}, \text{REV CDS}, \text{STRUCT}, \text{CO}\}$ with probabilities $(p_N, p_G/2, p_G/2, p_S, p_C)$.
3. Each region type undergoes its own elaboration chain:
 - **INTER**: link grammar + left CTMC expansion (single nucleotides, neutral evolution).

- FWDGDS/REVGDS: link grammar + concatenation into codons ($c_1c_2c_3$) + CTMC expansion with codon substitution model + recursive nesting for introns (introns contain a nested GENOME nonterminal, flanked by splice donor/acceptor sites).
- STRUCT: link grammar with LR CTMC expansion (stems) + recursive nesting into loops (left CTMC expansion).
- CONS: link grammar + left CTMC expansion (conserved elements).

4. The intron nesting creates recursion: GENOME $\rightarrow$ CDS $\rightarrow$ CODON $\rightarrow$ INTRON $\rightarrow$ GENOME.

C.10.5 Toward Implementation

Grammar Objects with Transformation Methods The elaboration rules defined above can be implemented as methods on a grammar object:

```
class WCFG:
    nonterminals: Set[str]
    terminals: Set[str]
    productions: List[Production]
    start: str

    def ctmc_expand(self, nonterminal, alphabet,
                    equilibrium, emission_type='left'):
        """CTMC Expansion"""

    def fragment_expand(self, nonterminal, extension_prob):
        """Fragment Expansion"""

    def mixture_expand(self, nonterminal, components, weights):
        """Mixture Expansion"""

    def concatenate(self, nonterminal, parts):
        """Link Sequence Concatenation"""

    def nest_nonrecursive(self, nonterminal, inner_grammar):
        """Non-Recursive Nesting"""

    def nest_recursive(self, nonterminal, bifurcation_grammar,
                       terminal_prob, side='right'):
        """Recursive Nesting"""

    def evolve(self, time, rates):
        """Evolution"""

    def remove_null_states(self):
        """Null State Removal"""
```

Each method validates the preconditions (e.g., that fragment expansion targets a nonterminal with terminal emissions), performs the transformation, and returns the modified grammar.

Automatic Derivation of DP Algorithms Given an elaborated grammar, the dynamic programming algorithm (Forward, Backward, Inside, Viterbi) can be derived automatically by:

1. **State space identification:** each nonterminal in the elaborated grammar corresponds to a state in the DP. Emitting states correspond to observable positions (sequence characters); non-emitting states are handled by null closure.
2. **Transition structure:** the productions define the recurrence relations. Linear (non-branching) productions yield HMM-style recurrences; branching (bifurcation) productions yield CYK-style recurrences.
3. **Emission probabilities:** determined by the CTMC parameters and the emission type (left, right, LR, match/insert/delete).
4. **Fill order:** determined by the topological sort of nonterminals (after null cycle removal). For recursive grammars, the fill order follows the CYK pattern (by span length for context-free rules).

Connection to Existing Frameworks The elaboration rules defined here are compatible with existing grammar and transducer frameworks:

- **Transducer composition:** the Evolution elaboration produces a transducer (Pair HMM / WFST). These can be composed on phylogenetic trees using the standard composition and intersection operations for Mealy machines in waiting-machine normal form.
- **SCFG parsers:** the elaborated grammars (especially those with recursive nesting and LR emission) are SCFGs amenable to Inside/Outside parsing algorithms.
- **Distillation:** the elaborated pair grammars can be distilled to order-1 machines (HMMs and WFSTs) by computing adjacency frequencies and normalizing, as described in Section C.4.5. This step loses the hierarchical structure but produces compact machines suitable for phylogenetic composition.

C.11 Recursive TKF Models

Another way to nest TKF models is to borrow from grammar theory and allow recursion. This was previously used to develop RNA evolutionary models (28), and is developed here in a general way for proteins and genomic DNA.

We illustrate how the TKF-mixed domain model can be interpreted as a stochastic grammar, developing four examples—recursive protein domains (i.e. arbitrary nesting of motifs), a basic model of RNA foldback structure, a second more sophisticated model of RNA structure, and a basic model of a genome—that highlight the structure of such models as a series of stepwise grammar elaborations that constitute tree-adjointing moves on the space of grammars.

C.11.1 Example One: Left-Recursive TKF (L-TKF)

We can imagine a links model where, in a domain of type l , each mortal link is either (with probability s_l) associated with a character, or (with probability $(1 - s_l)v_{lm}$ associated with its own

independently-evolving links model of domain type m . Links models can thus be nested *ad infinitum*

$$\begin{aligned} M_l &= \text{Links}(\text{Mix}_{k \sim s}(L_l^{(k)}); \lambda_l, \mu_l) \\ L_l^{(1)} &= \text{HMM}(\{Q^{(c)}, \pi^{(c)}\}_c; r^{(l)}, u_l) \\ L_l^{(0)} &= \text{Mix}_{m \sim v_m}(M_m) \end{aligned}$$

where HMM denotes the Markovian fragment process (Section C.1.1), and $\text{Mix}_{k \sim s}$ is defined for the Bernoulli index variable k and probability s as it was for categorical index variables

$$x \sim \text{Mix}_{k \sim p}(M(\theta_k); p) \Leftrightarrow k \sim \text{Bernoulli}(p), x \sim M(\theta_k)$$

We postpone the TKF92-like augmentations of $L_l^{(1)}$ (fragment types f and site classes c) for now, and start with a simplified TKF91-like version that allows full recursively-nested domains but allows only single-character fragments with one site class per domain

$$\begin{aligned} M'_l &= \text{Links}(\text{Mix}_{k \sim s}(L_l'^{(k)}); \lambda_l, \mu_l) \\ L_l'^{(1)} &= \text{Subst}(Q_l, \pi_l) \\ L_l'^{(0)} &= \text{Mix}_{m \sim v_l}(M'_m) \end{aligned}$$

As with the nested TKF HMM, we have to account for the probability of zero-length components, leading to null cycles during likelihood and inference computations.

The joint distribution over ancestor-descendant alignments under the recursive TKF model is described by the following stochastic context-free grammar (SCFG)

Symbol interpretation	LHS	RHS	Probability
<u>L</u> ink sequence, domain l	$L_l \rightarrow$	$L_l \quad M_l$	κ_l
	$ $	K_l	$1 - \kappa_l$
<u>I</u> mmortal link	$K_l \rightarrow$	N_l	β_l
	$ $	ϵ	$1 - \beta_l$
<u>M</u> ortal link	$M_l \rightarrow$	S_l	α_l
	$ $	E_l	$1 - \alpha_l$
<u>S</u> urviving mortal link	$S_l \rightarrow$	$A_l \quad N_l$	β_l
	$ $	A_l	$1 - \beta_l$
<u>E</u> xpired mortal link	$E_l \rightarrow$	$D_l \quad N_l$	γ_l
	$ $	D_l	$1 - \gamma_l$
<u>N</u> ewborn mortal link(s)	$N_l \rightarrow$	$I_l \quad N_l$	β_l
	$ $	I_l	$1 - \beta_l$
<u>A</u> ligned component	$A_l \rightarrow$	T_{ab}	$s_l \pi_{la} \exp(Q_l T)_{ab}$
	$ $	L_m	$(1 - s_l) v_{lm}$
<u>I</u> nserted component	$I_l \rightarrow$	$T_{\bullet b}$	$s_l \pi_{lb}$
	$ $	C_m	$(1 - s_l) v_{lm}$
<u>D</u> eleted component	$D_l \rightarrow$	$T_{a\bullet}$	$s_l \pi_{la}$
	$ $	P_m	$(1 - s_l) v_{lm}$
<u>C</u> hild link sequence (inserted)	$C_l \rightarrow$	$C_l \quad I_l$	κ_l
	$ $	ϵ	$1 - \kappa_l$
<u>P</u> arent link sequence (deleted)	$P_l \rightarrow$	$P_l \quad D_l$	κ_l
	$ $	ϵ	$1 - \kappa_l$

The standard path from here is to transform the grammar to Chomsky Normal Form (55). We don't need to go all the way down that path; we only need to remove ϵ -productions and null cycles.

In order to train by EM, every time we remove ϵ -productions and null cycles, we need to be able to convert a rule count in the ϵ -eliminated grammar to a set of rule counts in the original grammar. For each non-nullable nonterminal $\mathbf{X}'$, when a rule $\mathbf{X}' \rightarrow \mathbf{Y}'$ appears (arising from an original bifurcation $\mathbf{X} \rightarrow \mathbf{Y} \mathbf{Z}$ where $\mathbf{Z}$ was nullable), the expected count $c(\mathbf{X}' \rightarrow \mathbf{Y}')$ should contribute $c(\mathbf{X}' \rightarrow \mathbf{Y}') \cdot \eta(\mathbf{Z})$ to the count of the original rule $\mathbf{X} \rightarrow \mathbf{Y} \mathbf{Z}$, and symmetrically when the first child was nullable.

More precisely, for the nullability fixed-point iteration, this means tracking an additional set of *expected counts* alongside the nullabilities. For each nonterminal $\mathbf{X}$ with nullability $\eta(\mathbf{X})$, define $\bar{c}(\mathbf{X} \rightarrow \alpha)$ as the expected number of times rule $\mathbf{X} \rightarrow \alpha$ would have been used in the original grammar, conditioned on $\mathbf{X}$ generating the empty string. These satisfy analogous fixed-point equations:

$$\bar{c}(\mathbf{X} \rightarrow \mathbf{Y}_1 \cdots \mathbf{Y}_k) = w(\mathbf{X} \rightarrow \mathbf{Y}_1 \cdots \mathbf{Y}_k) \prod_{i=1}^k \eta(\mathbf{Y}_i) + \sum_{i=1}^k \frac{w(\mathbf{X} \rightarrow \mathbf{Y}_1 \cdots \mathbf{Y}_k) \prod_{j=1}^k \eta(\mathbf{Y}_j)}{\eta(\mathbf{X})} \sum_{\alpha'} \bar{c}(\mathbf{Y}_i \rightarrow \alpha')$$

and can be iterated to convergence alongside the nullabilities. Given posterior counts from the Inside-Outside algorithm on the ϵ -eliminated grammar, the original-grammar rule counts are recovered by:

1. For each non-null production $\mathbf{X}' \rightarrow \alpha'$ in the ϵ -eliminated grammar, its posterior count $c'(\mathbf{X}' \rightarrow \alpha')$ contributes directly to the corresponding original rule.
2. For each “nullability shortcut” production $\mathbf{X}' \rightarrow \mathbf{Y}'$ (arising from $\mathbf{X} \rightarrow \mathbf{Y} \mathbf{Z}$ with $\mathbf{Z}$ nullable), the count $c'(\mathbf{X}' \rightarrow \mathbf{Y}')$ contributes $c'(\mathbf{X}' \rightarrow \mathbf{Y}')$ to the original rule $\mathbf{X} \rightarrow \mathbf{Y} \mathbf{Z}$, plus $c'(\mathbf{X}' \rightarrow \mathbf{Y}') \cdot \bar{c}(\mathbf{Z} \rightarrow \alpha)$ to each rule within $\mathbf{Z}$'s null derivation subtree.

The general theory of ϵ -elimination with EM count recovery, including the null closure $(I - T_{ZZ})^{-1}$ and null cycle removal, is developed in Appendix C.10 (Section C.10.3)

First we find the nullability $\eta(\mathbf{X})$ of each nonterminal $\mathbf{X}$. The nullability is the probability that a parse tree rooted in that nonterminal yields the empty string. We can't solve for these in closed form (at least not in the general recursive model, where a link sequence can contain another link sequence of the same type). Instead we can solve approximately by iterating towards a fixed point.

We first observe that the nullabilities collectively satisfy the following

$$\begin{aligned}
\eta(\mathbf{L}_l) &= \eta(\mathbf{K}_l) \frac{1 - \kappa_l}{1 - \kappa_l \eta(\mathbf{M}_l)} \\
\eta(\mathbf{K}_l) &= 1 - \beta_l + \beta_l \eta(\mathbf{N}_l) \\
\eta(\mathbf{M}_l) &= \alpha_l \eta(\mathbf{S}_l) + (1 - \alpha_l) \eta(\mathbf{E}_l) \\
\eta(\mathbf{S}_l) &= \eta(\mathbf{A}_l) (\beta_l \eta(\mathbf{N}_l) + (1 - \beta_l)) \\
\eta(\mathbf{E}_l) &= \eta(\mathbf{D}_l) (\gamma_l \eta(\mathbf{N}_l) + (1 - \gamma_l)) \\
\eta(\mathbf{N}_l) &= \eta(\mathbf{I}_l) \frac{1 - \beta_l}{1 - \beta_l \eta(\mathbf{I}_l)} \\
\eta(\mathbf{A}_l) &= (1 - s_l) \sum_m v_{lm} \eta(\mathbf{L}_m) \\
\eta(\mathbf{I}_l) &= (1 - s_l) \sum_m v_{lm} \eta(\mathbf{C}_m) \\
\eta(\mathbf{D}_l) &= \eta(\mathbf{I}_l) \\
\eta(\mathbf{C}_l) &= \frac{1 - \kappa_l}{1 - \kappa_l \eta(\mathbf{I}_l)} \\
\eta(\mathbf{P}_l) &= \eta(\mathbf{C}_l)
\end{aligned}$$

A general procedure is (i) iterate to solve for the $\eta(\mathbf{C}_l)$ (substituting in the downstream definition of $\eta(\mathbf{I}_l)$ so the formula becomes self-referential); (ii) this directly yields $\eta(\mathbf{P}_l), \eta(\mathbf{I}_l), \eta(\mathbf{D}_l), \eta(\mathbf{N}_l), \eta(\mathbf{E}_l), \eta(\mathbf{K}_l)$; (iii) iterate to solve for $\eta(\mathbf{L}_l)$ (again, first substituting to make it self-referential); (iv) this directly yields the remaining $\eta(\mathbf{M}_l), \eta(\mathbf{S}_l), \eta(\mathbf{A}_l)$.

In detail: initialize $x_l^{(0)} \leftarrow 0$ for $l \in \mathcal{N}$. Iterate to convergence

$$x_l^{(k+1)} \leftarrow \frac{1 - \kappa_l}{1 - \kappa_l (1 - s_l) \sum_m v_{lm} x_m^{(k)}}$$

We then set $\eta(\mathbf{C}_l) \leftarrow \lim_{k \rightarrow \infty} x_l^{(k)}$ and set $\eta(\mathbf{P}_l), \eta(\mathbf{I}_l), \eta(\mathbf{D}_l), \eta(\mathbf{N}_l), \eta(\mathbf{E}_l), \eta(\mathbf{K}_l)$ using the above equations. Now set $y_l^{(0)} \leftarrow 0$ for $l \in \mathcal{N}$ and again iterate

$$y_l^{(k+1)} \leftarrow \frac{(1 - \kappa_l) \eta(\mathbf{K}_l)}{1 - \kappa_l (1 - \alpha_l) \eta(\mathbf{E}_l) - \kappa_l \alpha_l (1 - \beta_l (1 - \eta(\mathbf{N}_l))) (1 - s_l) \sum_m v_{lm} y_m^{(k)}}$$

Then set $\eta(\mathbf{L}_l) \leftarrow \lim_{k \rightarrow \infty} y_l^{(k)}$ and set the remaining $\eta(\mathbf{M}_l), \eta(\mathbf{S}_l), \eta(\mathbf{A}_l)$ using the above equations.

We next develop a “non-nullable” version of the grammar that yields the same Inside probabilities, but explicitly separates out ϵ -generations. For every nonterminal $\mathbf{X}_l$ we create a new nonterminal $\mathbf{X}'_l$ with rules that (by construction) never generate empty parse trees, but are otherwise identical to those transforming $\mathbf{X}_l$. In cases where the original grammars has bifurcation rules $\mathbf{X}_i \rightarrow \mathbf{X}_j \mathbf{X}_k$, we need to introduce transitions $\mathbf{X}'_i \rightarrow \mathbf{X}'_j$ and $\mathbf{X}'_i \rightarrow \mathbf{X}'_k$, to account for the missing nullability of $\mathbf{X}_j$ and $\mathbf{X}_k$. (We can also eliminate $\mathbf{K}_l$, which only has one outgoing rule when its ϵ -production is removed, and fold $\mathbf{L}_l \rightarrow \mathbf{K}_l \rightarrow \mathbf{M}_l$ into $\mathbf{L}_l \rightarrow \mathbf{M}_l$.)

Symbol interpretation	LHS	RHS	Probability
<u>L</u> ink sequence, domain l (nonempty)	$L'_l \rightarrow$ 	$L'_l \quad M'_l$ L'_l M'_l N'_l	κ_l $\kappa_l \eta(M_l)$ $\kappa_l \eta(L_l)$ $(1 - \kappa_l) \beta_l$
<u>M</u> ortal link (nonempty)	$M'_l \rightarrow$ 	S'_l E'_l	α_l $1 - \alpha_l$
<u>S</u> urviving mortal link (etc.; all X'_l are nonempty)	$S'_l \rightarrow$ 	$A'_l \quad N'_l$ N'_l A'_l	β_l $\beta_l \eta(A_l)$ $1 - \beta_l(1 - \eta(N_l))$
<u>E</u> xpired mortal link	$E'_l \rightarrow$ 	$D'_l \quad N'_l$ N'_l D'_l	γ_l $\gamma_l \eta(D_l)$ $1 - \gamma_l(1 - \eta(N_l))$
<u>N</u> ewborn mortal link(s)	$N'_l \rightarrow$ 	$I'_l \quad N'_l$ N'_l I'_l	β_l $\beta_l \eta(I_l)$ $1 - \beta_l(1 - \eta(N_l))$
<u>A</u> ligned component	$A'_l \rightarrow$ 	T_{ab} L'_m	$s_l \pi_{la} \exp(Q_l T)_{ab}$ $(1 - s_l) v_{lm}$
<u>I</u> nserted component	$I'_l \rightarrow$ 	$T_{\bullet b}$ C'_m	$s_l \pi_{lb}$ $(1 - s_l) v_{lm}$
<u>D</u> eleted component	$D'_l \rightarrow$ 	$T_{a\bullet}$ P'_m	$s_l \pi_{la}$ $(1 - s_l) v_{lm}$
<u>C</u> hild link sequence (inserted)	$C'_l \rightarrow$ 	$C'_l \quad I'_l$ C'_l I'_l	κ_l $\kappa_l \eta(I_l)$ $\kappa_l \eta(C_l)$
<u>P</u> arent link sequence (deleted)	$P'_l \rightarrow$ 	$P'_l \quad D'_l$ P'_l D'_l	κ_l $\kappa_l \eta(D_l)$ $\kappa_l \eta(P_l)$

The final step is to remove null cycles: chains of unit productions resulting in the same nonterminal. Specifically we need to remove $L'_l \rightarrow M'_l \rightarrow S'_l \rightarrow A'_l \rightarrow L'_m$, $C'_l \rightarrow I'_l \rightarrow C'_m$ and $P'_l \rightarrow D'_l \rightarrow P'_m$. We do this by deleting the $A'_l \rightarrow L'_m$, $I'_l \rightarrow C'_m$ and $D'_l \rightarrow P'_m$ transitions, adding compensatory self-loops and self-bifurcations to L'_l , P'_l , and C'_l to account for the now-broken paths through A'_l , I'_l and D'_l . The modified grammar is

Symbol	LHS		RHS		Probability
<u>L</u> ink	L_l''	$\rightarrow$	L_l''	M_l''	κ_l
		$ $	L_l''	L_m''	$\kappa_l \alpha_l (1 - \beta_l (1 - \eta(N_l))) (1 - s_l) v_{lm}$
		$ $	L_m''		$\mathcal{A}_{lm}$
		$ $	M_l''		$\kappa_l \eta(L_l)$
		$ $	N_l''		$(1 - \kappa_l) \beta_l$
<u>M</u> ortal	M_l''	$\rightarrow$	S_l''		α_l
		$ $	E_l''		$1 - \alpha_l$
<u>S</u> urviving	S_l''	$\rightarrow$	A_l''	N_l''	β_l
		$ $	N_l''		$\beta_l \eta(A_l)$
		$ $	A_l''		$1 - \beta_l (1 - \eta(N_l))$
<u>E</u> xpired	E_l''	$\rightarrow$	D_l''	N_l''	γ_l
		$ $	P_m''	N_l''	$\gamma_l (1 - s_l) v_{lm}$
		$ $	N_l''		$\gamma_l \eta(D_l)$
		$ $	D_l''		$1 - \gamma_l (1 - \eta(N_l))$
		$ $	P_m''		$(1 - \gamma_l (1 - \eta(N_l))) (1 - s_l) v_{lm}$
<u>N</u> ewborns	N_l''	$\rightarrow$	I_l''	N_l''	β_l
		$ $	C_m''	N_l''	$\beta_l (1 - s_l) v_{lm}$
		$ $	N_l''		$\beta_l \eta(I_l)$
		$ $	I_l''		$1 - \beta_l (1 - \eta(N_l))$
		$ $	C_m''		$(1 - \beta_l (1 - \eta(N_l))) (1 - s_l) v_{lm}$
<u>A</u> ligned	A_l''	$\rightarrow$	T_{ab}		$s_l \pi_{la} \exp(Q_l T)_{ab}$
<u>I</u> nserted	I_l''	$\rightarrow$	$T_{\bullet b}$		$s_l \pi_{lb}$
<u>D</u> eleted	D_l''	$\rightarrow$	$T_{a\bullet}$		$s_l \pi_{la}$
<u>C</u> hild	C_l''	$\rightarrow$	C_l''	I_l''	κ_l
		$ $	C_l''	C_m''	$\kappa_l (1 - s_l) v_{lm}$
		$ $	C_m''		$\mathcal{B}_{lm}$
		$ $	I_l''		$\kappa_l \eta(C_l)$
<u>P</u> arent	P_l''	$\rightarrow$	P_l''	D_l''	κ_l
		$ $	P_l''	P_m''	$\kappa_l (1 - s_l) v_{lm}$
		$ $	P_m''		$\mathcal{B}_{lm}$
		$ $	D_l''		$\kappa_l \eta(C_l)$

where $\mathcal{A}, \mathcal{B}$ are the $\mathcal{N} \times \mathcal{N}$ transition matrices

$$\begin{aligned} \mathcal{A}_{lm} &= \kappa_l (\eta(M_l) \delta_{lm} + \eta(L_l) \alpha_l (1 - \beta_l (1 - \eta(N_l))) (1 - s_l) v_{lm}) \\ \mathcal{B}_{lm} &= \kappa_l (\eta(I_l) \delta_{lm} + \eta(C_l) (1 - s_l) v_{lm}) \end{aligned}$$

Letting $\mathcal{G} = (I - \mathcal{A})^{-1}$ and $\mathcal{H} = (I - \mathcal{B})^{-1}$ be the geometric series sums of these matrices, update

the rules to sum over all $L_l'' \rightarrow L_m'', C_l'' \rightarrow C_m'',$ and $P_l'' \rightarrow P_m''$ transition chains explicitly

Symbol	LHS	RHS	Probability
<u>Link</u>	$L_l''' \rightarrow$	$L_m''' M_m'''$	$\mathcal{G}_{lm}\kappa_m$
		$L_m''' L_n'''$	$\mathcal{G}_{lm}\kappa_m\alpha_m(1 - \beta_m(1 - \eta(N_m))) (1 - s_m)v_{mn}$
		M_m'''	$\mathcal{G}_{lm}\kappa_m\eta(L_m)$
		N_m'''	$\mathcal{G}_{lm}(1 - \kappa_m)\beta_m$
<u>Child</u>	$C_l''' \rightarrow$	$C_m''' I_m'''$	$\mathcal{H}_{lm}\kappa_m$
		$C_m''' C_n'''$	$\mathcal{H}_{lm}(1 - s_m)v_{mn}$
		I_m'''	$\mathcal{H}_{lm}\kappa_m\eta(C_m)$
<u>Parent</u>	$P_l''' \rightarrow$	$P_m''' D_l'''$	$\mathcal{H}_{lm}\kappa_m$
		$P_m''' P_n'''$	$\mathcal{H}_{lm}(1 - s_m)v_{mn}$
		D_m'''	$\mathcal{H}_{lm}\kappa_m\eta(C_m)$

Rules for $X_l''' \dots$ where $X \in \{M, S, E, N\}$ are just copied over from the corresponding $X_l'' \rightarrow \dots$ rules, changing X'' to X''' on the right-hand side as well.

Symbol	LHS	RHS	Probability
<u>Mortal</u>	$M_l''' \rightarrow$	S_l'''	α_l
		E_l'''	$1 - \alpha_l$
<u>Surviving</u>	$S_l''' \rightarrow$	$A_l''' N_l'''$	β_l
		N_l'''	$\beta_l\eta(A_l)$
		A_l'''	$1 - \beta_l(1 - \eta(N_l))$
<u>Expired</u>	$E_l''' \rightarrow$	$D_l''' N_l'''$	γ_l
		$P_m''' N_l'''$	$\gamma_l(1 - s_l)v_{lm}$
		N_l'''	$\gamma_l\eta(D_l)$
		D_l'''	$1 - \gamma_l(1 - \eta(N_l))$
		P_m'''	$(1 - \gamma_l(1 - \eta(N_l))) (1 - s_l)v_{lm}$
		N_l'''	$\gamma_l\eta(D_l)$
<u>Newborns</u>	$N_l''' \rightarrow$	$I_l''' N_l'''$	β_l
		$C_m''' N_l'''$	$\beta_l(1 - s_l)v_{lm}$
		N_l'''	$\beta_l\eta(I_l)$
		I_l'''	$1 - \beta_l(1 - \eta(N_l))$
		C_m'''	$(1 - \beta_l(1 - \eta(N_l))) (1 - s_l)v_{lm}$

We use this last opportunity to reintroduce mixtures of site and fragment classes

Symbol	LHS	RHS	Probability
<u>Aligned</u>	$A_l''' \rightarrow$	F_f	$s_l w_{lf}$
<u>Inserted</u>	$I_l''' \rightarrow$	G_f	$s_l w_{lf}$
<u>Deleted</u>	$D_l''' \rightarrow$	R_f	$s_l w_{lf}$
<u>Fragment</u>	$F_f \rightarrow$	$T_{ab} F_g$	$r_{fg}^{(l)} \sum_c u_{lfc} \pi_a^{(c)} \exp(R^{(c)}T)_{ab}$
		T_{ab}	$\rho_f^{(l)} \sum_c u_{lfc} \pi_a^{(c)} \exp(R^{(c)}T)_{ab}$
<u>Generated</u>	$G_f \rightarrow$	$T_{\bullet b} G_g$	$r_{fg}^{(l)} \sum_c u_{lfc} \pi_b^{(c)}$
		$T_{\bullet b}$	$\rho_f^{(l)} \sum_c u_{lfc} \pi_b^{(c)}$
<u>Removed</u>	$R_f \rightarrow$	$T_{a\bullet} R_g$	$r_{fg}^{(l)} \sum_c u_{lfc} \pi_a^{(c)}$
		$T_{a\bullet}$	$\rho_f^{(l)} \sum_c u_{lfc} \pi_a^{(c)}$

TKF Structure Tree singlet rules (a)
Sequence a terminals: {u, v}.

lhs	→	rhs	$P(a)$
L_a	→	$u L_a$	$\kappa_l \pi_l(u)$
		$S_a L_a$	$\kappa_l \pi_l(S)$
		ϵ	$1 - \kappa_l$
S_a	→	$u S_a v$	$\kappa_s \pi_s(uv)$
		L_a	$1 - \kappa_s$

Table C.2: TKF Structure Tree. Singlet rule-set for a .

and add the top-level rule:

Symbol	LHS	RHS	Probability
<u>B</u> egin	B →	L_1'''	1
		ϵ	$\eta(L_1)$

With no ϵ -productions and a strictly acyclic topological transition sort order on nonterminals of $B \rightarrow L \rightarrow M \rightarrow (S \rightarrow (A, N \rightarrow I \rightarrow C), E \rightarrow D \rightarrow P)$, this grammar is ready for an Inside parser.

At this point our model is the desired

$$\begin{aligned}
M_l &= \text{Links}(\text{Mix}_{k \sim s}(L_l^{(k)}); \lambda_l, \mu_l) \\
L_l^{(1)} &= \text{HMM}(\{Q^{(c)}, \pi^{(c)}\}_c; r^{(l)}, u_l) \\
L_l^{(0)} &= \text{Mix}_{m \sim v_m}(M_m)
\end{aligned}$$

C.11.2 Example Two: The TKF Structure Tree (TKFST)

Consider now the RNA evolutionary model derived in (28).

That model has stem (S) and loop (L) sequences, each of which is a TKF91 sequence with its own rates. Stems are sequences of base pairs, loops of individual bases.

The model was developed as a proof of concept, but suffers similar deficiencies to TKF91 concerning the absence of an affine gap penalty, as well as a lack of basepair stacking effects or other empirically observed features of biological RNA structures.

Parameters Let $\overline{xyz}$ denote reverse complement e.g. $\overline{AAG} = CTT$.

Parameters: insertion and deletion rates $\lambda_F < \mu_F$, fragment extension probability r_F , substitution rate matrix Q_F , equilibrium probability vector q_F (so $q_x Q_F = 0$) for $F \in \{R, N, G, S, L, C\}$. Splice donor/acceptor site distribution $q_{D1}, q_{D2}, q_{A1}, q_{A2}$. Region-type probabilities $p_G + p_N + p_S + p_C = 1$. Intron probability p_I .

The Q_N, Q_S, Q_L and Q_C models should be strand-invariant, so e.g. $Q_N(x_1, x_2) = Q_N(\overline{x_1}, \overline{x_2})$.

Functions For $F \in \{R, N, G, S, L, C\}$:

$$\kappa_F = \left(1 - \frac{\lambda_F}{\mu_F}\right) (1 - r_F)$$

TKF Structure Tree singlet rules (b)

Sequence b terminals: $\{w, x\}$.

lhs	$\rightarrow$	rhs	$P(b)$
L_b	$\rightarrow$	$w L_b$	$\kappa_l \pi_l(w)$
	$ $	$S_b L_b$	$\kappa_l \pi_l(S)$
	$ $	ϵ	$1 - \kappa_l$
S_b	$\rightarrow$	$w S_b x$	$\kappa_s \pi_s(wx)$
	$ $	L_b	$1 - \kappa_s$

Table C.3: TKF Structure Tree. Singlet rule-set for b .

The Stationary Grammar

LHS	$\rightarrow$	RHS	Transition	Emission
GENOME	$\rightarrow$	REGION GENOME	$1 - \kappa_R$	
	$ $	ϵ	κ_R	
REGION	$\rightarrow$	INTER	p_N	
	$ $	FWDCDS	$p_G/2$	
	$ $	REVCDS	$p_G/2$	
	$ $	STRUCT	p_S	
	$ $	CONS	p_C	
INTER	$\rightarrow$	x_1 INTER	$1 - \kappa_N$	$q_N(x_1)$
	$ $	ϵ	κ_N	
FWDCDS	$\rightarrow$	FWDCOD FWDCDS	$1 - \kappa_G$	
	$ $	ϵ	κ_G	
FWDCOD	$\rightarrow$	$x_1 x_2 x_3$	$1 - p_I$	$q_G(xyz)$
	$ $	$x_1 x_2 x_3$ FWDINT	$p_I/3$	$q_G(xyz)$
	$ $	$x_1 x_2$ FWDINT x_3	$p_I/3$	$q_G(xyz)$
	$ $	x_1 FWDINT $x_2 x_3$	$p_I/3$	$q_G(xyz)$
FWDINT	$\rightarrow$	$x_1 x_2$ GENOME $x_3 x_4$	1	$q_{D1}(x_1)q_{D2}(x_2)q_{A1}(x_3)q_{A2}(x_4)$
REVCDS	$\rightarrow$	REVCD S REVCOD	$1 - \kappa_G$	
	$ $	ϵ	κ_G	
REVCOD	$\rightarrow$	$x_1 x_2 x_3$	$1 - p_I$	$q_G(\overline{xyz})$
	$ $	$x_1 x_2 x_3$ REVINT	$p_I/3$	$q_G(\overline{xyz})$
	$ $	$x_1 x_2$ REVINT x_3	$p_I/3$	$q_G(\overline{xyz})$
	$ $	x_1 REVINT $x_2 x_3$	$p_I/3$	$q_G(\overline{xyz})$
REVINT	$\rightarrow$	$x_1 x_2$ GENOME $x_3 x_4$	1	$q_{D1}(\overline{x_4})q_{D2}(\overline{x_3})q_{A1}(\overline{x_2})q_{A2}(\overline{x_1})$
STRUCT	$\rightarrow$	x_1 STRUCT x_2	$1 - \kappa_S$	$q_S(xy)$
	$ $	LOOP	κ_S	
LOOP	$\rightarrow$	x_1 LOOP	$1 - \kappa_L$	$q_L(x_1)$
	$ $	ϵ	κ_L	
CONS	$\rightarrow$	x_1 CONS	$1 - \kappa_C$	$q_C(x_1)$
	$ $	ϵ	κ_C	

<i>TKF Structure Tree pair rules ($a \xrightarrow{t} b$)</i>				
<i>Sequence a terminals: $\{u, v\}$. Sequence b terminals: $\{w, x\}$.</i>				
lhs	$\rightarrow$	rhs	$P(a)$	$P(b a)$
L_{ab}	$\rightarrow$	$u \ w \ L_{ab}$	$\kappa_l \pi_l(u)$	$(1 - \beta_l) \alpha_l M_l(u, w)$
		$w \ L_{ab}$	1	$\beta_l \pi_l(w)$
		$u \ L_{a \bullet b}$	$\kappa_l \pi_l(u)$	$(1 - \beta_l)(1 - \alpha_l)$
	$ $	$S_{ab} \ L_{ab}$	$\kappa_l \pi_l(S)$	$(1 - \beta_l) \alpha_l$
		$S_b \ L_{ab}$	1	$\beta_l \pi_l(S)$
		$S_a \ L_{a \bullet b}$	$\kappa_l \pi_l(S)$	$(1 - \beta_l)(1 - \alpha_l)$
		ϵ	$1 - \kappa_l$	$1 - \beta_l$
	$\rightarrow$	$u \ w \ S_{ab} \ x \ v$	$\kappa_s \pi_s(uv)$	$(1 - \beta_s) \alpha_s M_s(uv, wx)$
		$w \ S_{ab} \ x$	1	$\beta_s \pi_s(wx)$
		$u \ S_{a \bullet b} \ v$	$\kappa_s \pi_s(uv)$	$(1 - \beta_s)(1 - \alpha_s)$
		L_{ab}	$1 - \kappa_s$	$1 - \beta_s$
$L_{a \bullet b}$	$\rightarrow$	$u \ w \ L_{ab}$	$\kappa_l \pi_l(u)$	$(1 - \gamma_l) \alpha_l M_l(u, w)$
		$w \ L_{ab}$	1	$\gamma_l \pi_l(w)$
		$u \ L_{a \bullet b}$	$\kappa_l \pi_l(u)$	$(1 - \gamma_l)(1 - \alpha_l)$
	$ $	$S_{ab} \ L_{ab}$	$\kappa_l \pi_l(S)$	$(1 - \gamma_l) \alpha_l$
		$S_b \ L_{ab}$	1	$\gamma_l \pi_l(S)$
		$S_a \ L_{a \bullet b}$	$\kappa_l \pi_l(S)$	$(1 - \gamma_l)(1 - \alpha_l)$
		ϵ	$1 - \kappa_l$	$1 - \gamma_l$
	$\rightarrow$	$u \ w \ S_{ab} \ x \ v$	$\kappa_s \pi_s(uv)$	$(1 - \gamma_s) \alpha_s M_s(uv, wx)$
		$w \ S_{ab} \ x$	1	$\gamma_s \pi_s(wx)$
		$u \ S_{a \bullet b} \ v$	$\kappa_s \pi_s(uv)$	$(1 - \gamma_s)(1 - \alpha_s)$
		L_{ab}	$1 - \kappa_s$	$1 - \gamma_s$

Table C.4: TKF Structure Tree. Pair rule-set for $a \xrightarrow{t} b$ branch. Requires singlet rule-sets for a and b .

The Joint Finite-Time Grammar The general rules for forming the joint grammar from the conditional grammar are as follows. For every nonterminal of the following form (here L, R, and/or E are allowed to be ϵ)

LHS	$\rightarrow$	RHS	Transition	Emission
F	$\rightarrow$	L F R	$1 - \kappa_F$	$q_F(LR)$
		E	κ_F	

...that is, for GENOME, INTER, FWDCDS, REVCDS, STRUCT, LOOP, and CONS (with $F \in \{R, N, G, S, L, C\}$), replace these rules with

LHS	$\rightarrow$	RHS	Transition	Emission
F_M	$\rightarrow$	$L_M F_M R_M$	$(1 - \beta_{M,F})(1 - \kappa_F)\alpha_F$	$q_F(LR_x) \exp(Q_F t)(LR_x, LR_y)$
		$L_Y F_M R_Y$	$\beta_{M,F}$	$q_F(LR_y)$
		$L_X F_D R_X$	$(1 - \beta_{M,F})(1 - \kappa_F)(1 - \alpha_F)$	$q_F(LR_x)$
		E_M	$(1 - \beta_{M,F})\kappa_F$	
F_D	$\rightarrow$	$L_M F_M R_M$	$(1 - \beta_{D,F})(1 - \kappa_F)\alpha_F$	$q_F(LR_x) \exp(Q_F t)(LR_x, LR_y)$
		$L_Y F_M R_Y$	$\beta_{D,F}$	$q_F(LR_y)$
		$L_X F_D R_X$	$(1 - \beta_{D,F})(1 - \kappa_F)(1 - \alpha_F)$	$q_F(LR_x)$
		E_M	$(1 - \beta_{D,F})\kappa_F$	

...that is, two versions F_M and F_D , with different β 's for each type.

For every other nonterminal N, there need to be three versions N_M , N_X and N_Y , with outgoing rules for each type going to other nonterminals of the same type.

Top level.

LHS	$\rightarrow$	RHS	Transition	Emission
$GENOME_k$	$\rightarrow$	$REGION_M GENOME_M$	$(1 - \beta_{k,R})(1 - \kappa_R)\alpha_R$	
		$REGION_Y GENOME_M$	$\beta_{k,R}$	
		$REGION_X GENOME_D$	$(1 - \beta_{k,R})(1 - \kappa_R)(1 - \alpha_R)$	
		ϵ	$(1 - \beta_{k,R})\kappa_R$	
$REGION_j$	$\rightarrow$	$INTER_j$	p_N	
		$FWDCDS_j$	$p_G/2$	
		$REVCDS_j$	$p_G/2$	
		$STRUCT_j$	p_S	
		$CONS_j$	p_C	
$INTER$	$\rightarrow$	$x_1 INTER$	$1 - \kappa_N$	$q_N(x_1)$
		ϵ	κ_N	

Coding sequences.

LHS	$\rightarrow$	RHS	Transition	Emission
FWDCDS	$\rightarrow$	FWDCOD FWDCDS	$1 - \kappa_G$	
		ϵ	κ_G	
FWDCOD	$\rightarrow$	$x_1 x_2 x_3$	$1 - p_I$	$q_G(xyz)$
		$x_1 x_2 x_3 FWDINT$	$p_I/3$	$q_G(xyz)$
		$x_1 x_2 FWDINT x_3$	$p_I/3$	$q_G(xyz)$
		$x_1 FWDINT x_2 x_3$	$p_I/3$	$q_G(xyz)$
FWDINT	$\rightarrow$	$x_1 x_2 GENOME x_3 x_4$	1	$q_{D1}(x_1)q_{D2}(x_2)q_{A1}(x_3)q_{A2}(x_4)$

LHS	→	RHS	Transition	Emission
REVCDS	→	REVCDS REVCOD	$1 - \kappa_G$	
		ϵ	κ_G	
REVCOD	→	$x_1 x_2 x_3$	$1 - p_I$	$q_G(\overline{xyz})$
		$x_1 x_2 x_3 \text{ REVINT}$	$p_I/3$	$q_G(\overline{xyz})$
		$x_1 x_2 \text{ REVINT } x_3$	$p_I/3$	$q_G(\overline{xyz})$
		$x_1 \text{ REVINT } x_2 x_3$	$p_I/3$	$q_G(\overline{xyz})$
REVINT	→	$x_1 x_2 \text{ GENOME } x_3 x_4$	1	$q_{D1}(\overline{x_4})q_{D2}(\overline{x_3})q_{A1}(\overline{x_2})q_{A2}(\overline{x_1})$

RNA structures.

LHS	→	RHS	Transition	Emission
STRUCT	→	$x_1 \text{ STRUCT } x_2$	$1 - \kappa_S$	$q_S(xy)$
		LOOP	κ_S	
LOOP	→	$x_1 \text{ LOOP}$	$1 - \kappa_L$	$q_L(x_1)$
		ϵ	κ_L	
CONS	→	$x_1 \text{ CONS}$	$1 - \kappa_C$	$q_C(x_1)$
		ϵ	κ_C	

C.11.3 Example Three: The TKF Basepair Stack (TKFStack)

The simple TKF Structure Tree models RNA secondary structure with alternating stems (basepair sequences) and loops (single-nucleotide sequences). While a useful proof of concept, it lacks basepair stacking, multiloop junctions, bulges, and internal loops—features critical for realistic RNA structure modeling.

We now define an enhanced stem-loop grammar that incorporates these features while remaining within the TKF evolutionary framework. We then show how evolution elaboration, profile SCFG construction, and the triplet model for progressive reconstruction all apply to this grammar.

Parameters The grammar has the following parameters:

- Stem link TKF91 rates $\lambda_S < \mu_S$, giving $\kappa_S = \lambda_S/\mu_S$.
- Loop link TKF91 rates $\lambda_L < \mu_L$, giving $\kappa_L = \lambda_L/\mu_L$.
- Bulge extension probability r_B .
- Stacked-pair fragment extension probability r_K .
- Stem link type probabilities $p_{\text{bp}} + p_{\text{st}} + p_{\text{bu}} = 1$.
- Loop link type probabilities $p_{\text{lf}} + p_{\text{rf}} + p_{\text{sl}} = 1$.
- Nucleotide equilibrium $\pi(c)$ over $\mathcal{A} = \{A, C, G, U\}$, with rate matrix Q .
- Single basepair equilibrium $\pi_{\text{bp}}(c_L, c_R)$ over $|\mathcal{A}|^2 = 16$ states, rate matrix Q_{bp} .
- Closing basepair equilibrium $\pi_{\text{cl}}(c_L, c_R)$ over 16 states, rate matrix Q_{cl} .
- Stacked-pair equilibrium $\pi_K(c_L^1, c_L^2, c_R^2, c_R^1)$ over the $6^2 = 36$ canonical stacked pairs, rate matrix Q_K . A stacked pair consists of two consecutive canonical basepairs (c_L^1, c_R^1) and (c_L^2, c_R^2) ; the state space is restricted to the six Watson-Crick and wobble pairs (AU, CG, GC, UA, GU, UG) for each position, giving 36 states rather than the full $16^2 = 256$.

Nonterminals and Productions A *stem-loop* consists of a stem (nested basepairs with decorations), a closing basepair, and a loop (with possible multiloop branches). The start symbol is **SL** (stem-loop).

Stem-loop:

$$\text{SL} \rightarrow \text{STEM} \quad \text{weight } 1 \quad (\text{C.190})$$

Stem (TKF91 sequence of stem-links, from outer to inner):

$$\text{STEM} \rightarrow c_L \text{STEM} c_R \quad \kappa_S p_{\text{bp}} \pi_{\text{bp}}(c_L, c_R) \quad [\text{single basepair, LR}] \quad (\text{C.191})$$

$$\rightarrow c_L^1 c_L^2 \text{STEM} c_R^2 c_R^1 \quad \kappa_S p_{\text{st}} (1 - r_K) \pi_K(\cdot) \quad [\text{terminal stacked pair, LLRR}] \quad (\text{C.192})$$

$$\rightarrow c_L^1 c_L^2 \text{STACK} c_R^2 c_R^1 \quad \kappa_S p_{\text{st}} r_K \pi_K(\cdot) \quad [\text{extended stacked pair, LLRR}] \quad (\text{C.193})$$

$$\rightarrow \text{LDECO} \text{STEM} \text{RDECO} \quad \kappa_S p_{\text{bu}} \quad [\text{bulge}] \quad (\text{C.194})$$

$$\rightarrow \text{CLOSE LOOP} \quad 1 - \kappa_S \quad [\text{end stem}] \quad (\text{C.195})$$

A bulge link (C.194) represents an internal loop, a single-sided bulge, or a multihelix junction branch between basepairs. All non-basepair content between consecutive basepairs is consolidated into a single TKF link type whose internal structure is governed by LDECO and RDECO.

Bulge decorations (L-side and R-side content):

$$\text{LDECO} \rightarrow \text{LFRAG} \text{LDECO} \quad [\text{L-fragment, then more L-content}] \quad (\text{C.196})$$

$$\rightarrow \text{SL} \text{LDECO} \quad [\text{left branch: nested stem-loop}] \quad (\text{C.197})$$

$$\rightarrow \epsilon \quad [\text{end L-decorations}] \quad (\text{C.198})$$

$$\text{RDECO} \rightarrow \text{RDECO} \text{RFRAG} \quad [\text{R-fragment, more R-content}] \quad (\text{C.199})$$

$$\rightarrow \text{RDECO} \text{SL} \quad [\text{right branch: nested stem-loop}] \quad (\text{C.200})$$

$$\rightarrow \epsilon \quad [\text{end R-decorations}] \quad (\text{C.201})$$

LDECO generates all content on the 5' side (between the outer basepair's L-half and the continuation), while RDECO generates all content on the 3' side (between the continuation and the outer basepair's R-half). An internal loop has both $\text{LDECO} \neq \epsilon$ and $\text{RDECO} \neq \epsilon$; a single-sided bulge has content on only one side; a multihelix junction branch has **SL** nested within LDECO or RDECO.

Stacked-pair fragment continuation (LLRR emission):

$$\text{STACK} \rightarrow c_L^1 c_L^2 \text{STEM} c_R^2 c_R^1 \quad (1 - r_K) \pi_K(\cdot) \quad [\text{terminal}] \quad (\text{C.202})$$

$$\rightarrow c_L^1 c_L^2 \text{STACK} c_R^2 c_R^1 \quad r_K \pi_K(\cdot) \quad [\text{extend}] \quad (\text{C.203})$$

Closing basepair (LR emission):

$$\text{CLOSE} \rightarrow c_L \text{CLOSE}' c_R \quad \pi_{\text{cl}}(c_L, c_R) \quad (\text{C.204})$$

where **CLOSE'** is a unit nonterminal ($\text{CLOSE}' \rightarrow \epsilon$, weight 1) that serves as a placeholder for the inside of the closing basepair.

Loop (TKF91 sequence of loop-links):

$$\text{LOOP} \rightarrow \text{LOOPLINK} \text{LOOP} \quad \kappa_L \quad [\text{add loop link}] \quad (\text{C.205})$$

$$\rightarrow \epsilon \quad 1 - \kappa_L \quad [\text{end loop}] \quad (\text{C.206})$$

Loop link types:

$$\text{LOOPLINK} \rightarrow \text{LFRAG} \quad p_{\text{lf}} \quad [\text{L-fragment}] \quad (\text{C.207})$$

$$\rightarrow \text{RFRAG} \quad p_{\text{rf}} \quad [\text{R-fragment}] \quad (\text{C.208})$$

$$\rightarrow \text{SL} \quad p_{\text{sl}} \quad [\text{nested stem-loop (multiloop)}] \quad (\text{C.209})$$

Unpaired nucleotide fragments in loops (geometric length ≥ 1):

$$\text{LFRAG} \rightarrow c \text{ LFRAG} \quad r_L \pi(c) \quad [\text{extend, L-emission}] \quad (\text{C.210})$$

$$\rightarrow c \quad (1 - r_L) \pi(c) \quad [\text{terminal, L-emission}] \quad (\text{C.211})$$

$$\text{RFRAG} \rightarrow \text{RFRAG } c \quad r_R \pi(c) \quad [\text{extend, R-emission}] \quad (\text{C.212})$$

$$\rightarrow c \quad (1 - r_R) \pi(c) \quad [\text{terminal, R-emission}] \quad (\text{C.213})$$

Emission Types and Span Tracking The grammar has four emission patterns, each determining how terminals consume positions from the span $[i, j]$ of the input sequence:

1. **L-emission** (c): consumes from the left end, advancing $i \rightarrow i + 1$. Used by LFRAG, LDECO content.
2. **R-emission** (c): consumes from the right end, retreating $j \rightarrow j - 1$. Used by RFRAG, RDECO content.
3. **LR-emission** (c_L, c_R): consumes from both ends simultaneously, $i \rightarrow i + 1, j \rightarrow j - 1$. Used by STEM (single basepair), CLOSE.
4. **LLRR-emission** ($c_L^1, c_L^2, c_R^2, c_R^1$): consumes two from each end, $i \rightarrow i + 2, j \rightarrow j - 2$. Used by STEM (stacked pair), STACK. Each LLRR unit represents a *stacked dinucleotide pair*: two consecutive canonical basepairs (c_L^1, c_R^1) and (c_L^2, c_R^2) treated as a single evolutionary unit with nearest-neighbor stacking energy. The state space is restricted to the $6 \times 6 = 36$ combinations of canonical (Watson-Crick and wobble) pairs.

Remark C.25 (L/R distinction in terminals vs. productions). The terminal alphabet is $\mathcal{A} = \{A, C, G, U\}$ without L/R copies: both L-emission and R-emission produce characters from the same alphabet, differing only in which end of the span $[i, j]$ is consumed. At leaf nodes, the observed sequence carries no L/R annotation.

However, the L/R distinction is essential at the *production* level and therefore in the *profile SCFG*. Each nonterminal instance in a profile has a fixed emission direction (L, R, LR, or LLRR), determined by the parse tree from which it was extracted—analogous to the MATL, MATR, and MATP node types in Infernal covariance models (48). The branch PTT must preserve emission-direction compatibility when mapping parent productions to child productions. Consequently, a profile position that is LR-emitting (basepaired) at one node of the phylogeny may correspond to separate L-emitting and R-emitting positions (unpaired) at another node, reflecting structural change along the evolutionary lineage.

Remark C.26 (Structural interpretation). Consider a stem with links (from outer to inner): basepair (c_L^1, c_R^1), bulge (left-side $b_1 b_2$, right-side branch SL'), basepair (c_L^2, c_R^2). The yield is:

$$c_L^1 \ b_1 \ b_2 \ c_L^2 \ [\text{loop}] \ c_R^2 \ [\text{yield}(\text{SL}')] \ c_R^1$$

The bulge nucleotides b_1, b_2 appear on the 5' side (via LDECO), while the branch structure appears on the 3' side (via RDECO). This represents an internal loop with unpaired nucleotides on the 5' strand and a branching sub-structure on the 3' strand—a configuration that the consolidated bulge link type (C.194) captures as a single TKF event.

The Pair Grammar via Evolution Elaboration Applying the evolution elaboration rules (Appendix C.10) to the singlet grammar produces the pair grammar for an ancestor–descendant pair separated by evolutionary time T . Each nonterminal participating in a TKF91 link sequence (STEM, LOOP) gains M/I/D versions with the standard TKF91 transition weights (α, β, γ) derived from (λ, μ, T) .

Stem Link Elaboration. The stem is a TKF91 sequence of links with rates (λ_S, μ_S) . Each STEM nonterminal becomes STEM_M (post-match/insert) and STEM_D (post-delete):

$$\text{STEM}_M \rightarrow c_L^x c_L^y \text{STEM}_M c_R^y c_R^x \quad (1 - \beta_S) \kappa_S p_{bp} \alpha_S \pi_{bp}(c_L^x, c_R^x) P_{bp}(c_L^y, c_R^y | c_L^x, c_R^x) \quad (\text{C.214})$$

$$\rightarrow c_L^y \text{STEM}_M c_R^y \quad \beta_S \kappa_S p_{bp} \pi_{bp}(c_L^y, c_R^y) \quad [\text{BP insert}] \quad (\text{C.215})$$

$$\rightarrow c_L^x \text{STEM}_D c_R^x \quad (1 - \beta_S) \kappa_S p_{bp} (1 - \alpha_S) \pi_{bp}(c_L^x, c_R^x) \quad [\text{BP delete}] \quad (\text{C.216})$$

with analogous rules for STEM_D using γ_S in place of β_S .

For matched single basepairs (C.214), the nesting is $c_L^x c_L^y \cdots c_R^y c_R^x$: ancestor terminals on the outside, descendant terminals on the inside, preserving palindromic structure. For inserted basepairs (C.215), only descendant terminals appear (LR emission). For deleted basepairs (C.216), only ancestor terminals appear (LR emission).

Stacked-Pair Elaboration. Matched stacked pairs have LLRR emission on both ancestor and descendant, yielding an $L^4 R^4$ nesting pattern in the pair grammar:

$$\text{STEM}_M \rightarrow c_{L1}^x c_{L2}^x c_{L1}^y c_{L2}^y \text{STEM}_M c_{R2}^y c_{R1}^y c_{R2}^x c_{R1}^x \quad (1 - \beta_S) \kappa_S p_{st} (1 - r_K) \alpha_S \pi_K(\cdot) P_K(\cdot | \cdot) \quad (\text{C.217})$$

with inserted stacked pairs emitting only descendant LLRR, and deleted stacked pairs emitting only ancestor LLRR.

Bulge Elaboration. A bulge link (C.194) gains M/I/D versions like any other stem link. A matched bulge has the form LDECO_M STEM_M RDECO_M, with each side elaborated independently.

Within LDECO, each sub-element (LFRAG, nested SL) gains M/I/D versions with L-emission direction preserved. For example, a matched left-branch within a bulge produces:

$$\text{LDECO}_M \rightarrow \text{SL}_M \text{LDECO}_M \quad \alpha_{\text{deco}} P(\text{branch}) \quad [\text{matched left branch}] \quad (\text{C.218})$$

$$\rightarrow \text{SL}_I \text{LDECO}_M \quad \beta_{\text{deco}} \quad [\text{inserted left branch}] \quad (\text{C.219})$$

$$\rightarrow \text{SL}_D \text{LDECO}_D \quad (1 - \alpha_{\text{deco}}) P(\text{branch}) \quad [\text{deleted left branch}] \quad (\text{C.220})$$

Here SL_M generates aligned ancestor–descendant sub-structures, SL_I generates descendant-only sub-structures, and SL_D generates ancestor-only sub-structures.

RDECO is elaborated symmetrically, with R-emission direction preserved for RFRAG elements and right-side branches.

Loop Elaboration. The loop is a TKF91 sequence of loop-links with rates (λ_L, μ_L) . Elaboration proceeds identically to the stem: each **LOOP** becomes **LOOP_M**/**LOOP_D**, and each loop-link type (**LFRAG**, **RFRAG**, nested **SL**) gains **M/I/D** versions.

Nested stem-loops within the loop (multiloop junctions) are handled recursively: **SL_M** aligns both ancestor and descendant sub-structures, allowing multiloop branches to be independently inserted, deleted, or matched.

C.11.4 Example Four: The TKF Genome

Parameters Let $\overline{xyz}$ denote reverse complement e.g. $\overline{AAG} = CTT$.

Parameters: insertion and deletion rates $\lambda_F < \mu_F$, fragment extension probability r_F , substitution rate matrix Q_F , equilibrium probability vector q_F (so $q_x Q_F = 0$) for $F \in \{R, N, G, S, L, C\}$. Splice donor/acceptor site distribution $q_{D1}, q_{D2}, q_{A1}, q_{A2}$. Region-type probabilities $p_G + p_N + p_S + p_C = 1$. Intron probability p_I .

The Q_N, Q_S, Q_L and Q_C models should be strand-invariant, so e.g. $Q_N(x_1, x_2) = Q_N(\overline{x_1}, \overline{x_2})$.

Functions For $F \in \{R, N, G, S, L, C\}$:

$$\kappa_F = \left(1 - \frac{\lambda_F}{\mu_F}\right) (1 - r_F)$$

The Stationary Grammar

LHS	→	RHS	Transition	Emission
GENOME	→	REGION GENOME	$1 - \kappa_R$	
		ϵ	κ_R	
REGION	→	INTER	p_N	
		FWDCDS	$p_G/2$	
		REVCDS	$p_G/2$	
		STRUCT	p_S	
		CONS	p_C	
INTER	→	x_1 INTER	$1 - \kappa_N$	$q_N(x_1)$
		ϵ	κ_N	
FWDCDS	→	FWDCOD FWDCDS	$1 - \kappa_G$	
		ϵ	κ_G	
FWDCOD	→	$x_1 x_2 x_3$	$1 - p_I$	$q_G(xyz)$
		$x_1 x_2 x_3$ FWDINT	$p_I/3$	$q_G(xyz)$
		$x_1 x_2$ FWDINT x_3	$p_I/3$	$q_G(xyz)$
		x_1 FWDINT $x_2 x_3$	$p_I/3$	$q_G(xyz)$
FWDINT	→	$x_1 x_2$ GENOME $x_3 x_4$	1	$q_{D1}(x_1)q_{D2}(x_2)q_{A1}(x_3)q_{A2}(x_4)$
REVCDS	→	REVCD S REVCOD	$1 - \kappa_G$	
		ϵ	κ_G	
REVCOD	→	$x_1 x_2 x_3$	$1 - p_I$	$q_G(\overline{xyz})$
		$x_1 x_2 x_3$ REVINT	$p_I/3$	$q_G(\overline{xyz})$
		$x_1 x_2$ REVINT x_3	$p_I/3$	$q_G(\overline{xyz})$
		x_1 REVINT $x_2 x_3$	$p_I/3$	$q_G(\overline{xyz})$
REVINT	→	$x_1 x_2$ GENOME $x_3 x_4$	1	$q_{D1}(\overline{x_4})q_{D2}(\overline{x_3})q_{A1}(\overline{x_2})q_{A2}(\overline{x_1})$
STRUCT	→	x_1 STRUCT x_2	$1 - \kappa_S$	$q_S(xy)$
		LOOP	κ_S	
LOOP	→	x_1 LOOP	$1 - \kappa_L$	$q_L(x_1)$
		ϵ	κ_L	
CONS	→	x_1 CONS	$1 - \kappa_C$	$q_C(x_1)$
		ϵ	κ_C	

The Joint Finite-Time Grammar

How to form the joint grammar The general rules for forming the joint grammar from the conditional grammar are as follows. For every nonterminal of the following form (here L, R, and/or E are allowed to be ϵ)

LHS	→	RHS	Transition	Emission
F	→	L F R	$1 - \kappa_F$	$q_F(LR)$
		E	κ_F	

...that is, for GENOME, INTER, FWDCDS, REVCDS, STRUCT, LOOP, and CONS (with $F \in \{R, N, G, S, L, C\}$), replace these rules with

LHS	→	RHS	Transition	Emission
F_M	→	$L_M F_M R_M$	$(1 - \beta_{M,F})(1 - \kappa_F)\alpha_F$	$q_F(LR_x) \exp(Q_F t)(LR_x, LR_y)$
		$L_Y F_M R_Y$	$\beta_{M,F}$	$q_F(LR_y)$
		$L_X F_D R_X$	$(1 - \beta_{M,F})(1 - \kappa_F)(1 - \alpha_F)$	$q_F(LR_x)$
		E_M	$(1 - \beta_{M,F})\kappa_F$	
F_D	→	$L_M F_M R_M$	$(1 - \beta_{D,F})(1 - \kappa_F)\alpha_F$	$q_F(LR_x) \exp(Q_F t)(LR_x, LR_y)$
		$L_Y F_M R_Y$	$\beta_{D,F}$	$q_F(LR_y)$
		$L_X F_D R_X$	$(1 - \beta_{D,F})(1 - \kappa_F)(1 - \alpha_F)$	$q_F(LR_x)$
		E_M	$(1 - \beta_{D,F})\kappa_F$	

...that is, two versions F_M and F_D , with different β 's for each type.

For every other nonterminal N , there need to be three versions N_M , N_X and N_Y , with outgoing rules for each type going to other nonterminals of the same type.

Top level

LHS	→	RHS	Transition	Emission
$GENOME_k$	→	$REGION_M GENOME_M$	$(1 - \beta_{k,R})(1 - \kappa_R)\alpha_R$	
		$REGION_Y GENOME_M$	$\beta_{k,R}$	
		$REGION_X GENOME_D$	$(1 - \beta_{k,R})(1 - \kappa_R)(1 - \alpha_R)$	
		ϵ	$(1 - \beta_{k,R})\kappa_R$	
$REGION_j$	→	$INTER_j$	p_N	
		$FWDCDS_j$	$p_G/2$	
		$REVCDS_j$	$p_G/2$	
		$STRUCT_j$	p_S	
		$CONS_j$	p_C	
$INTER$	→	$x_1 INTER$	$1 - \kappa_N$	$q_N(x_1)$
		ϵ	κ_N	

Coding sequences

LHS	→	RHS	Transition	Emission
$FWDCDS$	→	$FWDCOD FWDCDS$	$1 - \kappa_G$	
		ϵ	κ_G	
$FWDCOD$	→	$x_1 x_2 x_3$	$1 - p_I$	$q_G(xyz)$
		$x_1 x_2 x_3 FWDINT$	$p_I/3$	$q_G(xyz)$
		$x_1 x_2 FWDINT x_3$	$p_I/3$	$q_G(xyz)$
		$x_1 FWDINT x_2 x_3$	$p_I/3$	$q_G(xyz)$
$FWDINT$	→	$x_1 x_2 GENOME x_3 x_4$	1	$q_{D1}(x_1)q_{D2}(x_2)q_{A1}(x_3)q_{A2}(x_4)$
LHS	→	RHS	Transition	Emission
$REVCDS$	→	$REVCDS REVCOD$	$1 - \kappa_G$	
		ϵ	κ_G	
$REVCOD$	→	$x_1 x_2 x_3$	$1 - p_I$	$q_G(\overline{xyz})$
		$x_1 x_2 x_3 REVINT$	$p_I/3$	$q_G(\overline{xyz})$
		$x_1 x_2 REVINT x_3$	$p_I/3$	$q_G(\overline{xyz})$
		$x_1 REVINT x_2 x_3$	$p_I/3$	$q_G(\overline{xyz})$
$REVINT$	→	$x_1 x_2 GENOME x_3 x_4$	1	$q_{D1}(\overline{x_4})q_{D2}(\overline{x_3})q_{A1}(\overline{x_2})q_{A2}(\overline{x_1})$

RNA structures

LHS	→	RHS	Transition	Emission
STRUCT	→	x_1 STRUCT x_2	$1 - \kappa_S$	$q_S(xy)$
		LOOP	κ_S	
LOOP	→	x_1 LOOP	$1 - \kappa_L$	$q_L(x_1)$
		ϵ	κ_L	
CONS	→	x_1 CONS	$1 - \kappa_C$	$q_C(x_1)$
		ϵ	κ_C	

D TKF-DP: Dirichlet-process Potts coupling

The TKF-DP model decorates TKF92 with three independent Dirichlet processes that govern (i) the per-site key partition into Potts coevolutionary cliques; (ii) the per-site assignment to a substitution class; and (iii) the per-class-pair assignment to a Potts coupling atom. The class-level substitution dynamics is a multi-site reversible CTMC; the alignment likelihood remains plain TKF92. Inference proceeds by a variational E-step on the continuous-time class clique, Gibbs sweeps and Jain–Neal split–merge proposals on the three Dirichlet partitions, recursive traceback over a time-indexed gravestone-augmented pair SCFG for the augmented branch history, and a stochastic M-step on the corpus-level hyperparameters with closed-form Dirichlet–multinomial / Gamma–Poisson / Laplace conjugacies unlocking the new-class / new-atom base-measure integrals.

D.1 The TKF-DP generative model

The TKF92 model (63) of insertion, deletion, and substitution has the convenient property that the alignment path and the substitution history factorize: the marginal distribution over alignments is independent of the amino acid dynamics, and inference proceeds via a pair HMM on the alignment with substitution likelihoods plugged in pointwise. This factorization is exactly what is lost when site–site coevolutionary couplings are introduced naively, since the substitution process at any site then depends on the joint state of its coupled partners, and any coupling rule that consults the existing alignment to decide partners destroys the marginal pair HMM.

We describe a minimal extension of TKF92 that re-introduces site–site Potts couplings while preserving the alignment–substitution factorization. Each newborn site is assigned a discrete key drawn i.i.d. from a Dirichlet process, and sites that share a key co-evolve under a Potts-modulated joint CTMC. Because the keys are drawn independently of the alignment, the alignment marginal is unchanged.

The alignment–substitution factorization survives at the parameter level but, once couplings are present, breaks at the latent-variable level: the substitution likelihood conditional on the alignment depends on the full timed indel history, including transient (alignment-invisible) lineages whose presence transiently modifies the rates of co-class members. Inference therefore requires a tractable representation of timed indel histories. We obtain one via a gravestone-augmented latent state and a time-indexed pair SCFG whose inside–outside is closed form (§D.10).

Definition D.1 (TKF-DP). *Fix TKF92 indel parameters (λ, μ, r) , a key concentration parameter $\alpha_z > 0$, and a symmetric 20×20 Potts coupling matrix H . The process jointly generates a TKF92 alignment path, a key assignment for every site, and an amino acid trajectory at every site, as follows.*

1. Sample stick-breaking weights once, globally:

$$\pi_k = \beta_k \prod_{l < k} (1 - \beta_l), \quad \beta_k \stackrel{iid}{\sim} \text{Beta}(1, \alpha_z).$$

2. Run TKF92 indel dynamics with parameters (λ, μ, r) , giving a stochastic set of alive sites $S(t)$ over time.
3. At the birth of each new site s , draw a key

$$z_s \mid \pi \sim \text{Cat}(\pi),$$

independently of all other sites, of the alignment path, and of the substitution history. The key is fixed for the site's lifetime.

4. At any time t , the alive sites $S(t)$ are partitioned into equivalence classes $\{C_k(t)\}$ by the relation $z_s = z_{s'}$. The amino acid states $\{x_s\}_{s \in S(t)}$ evolve as independent CTMCs across classes; the CTMC on class C is a $|C|$ -site reversible chain with stationary distribution

$$\pi_C(x_C) \propto \exp\left(-\sum_{\{i,j\} \subset C} H(x_i, x_j)\right).$$

Class composition is piecewise constant between birth/death events; on each event, the affected class is rebuilt and the surviving sites re-equilibrate under the new class generator.

Remark D.1 (Factorization). Conditional on π , the key draws are i.i.d. across sites, so the joint distribution factorizes as

$$P(\text{alignment}, \pi, \{z_s\}, \text{substitution}) = P(\text{alignment}) \cdot P(\pi) \cdot \prod_s P(z_s \mid \pi) \cdot P(\text{substitution} \mid \text{alignment}, \{z_s\}).$$

In particular, the marginal alignment likelihood is exactly that of TKF92, and the pair HMM one would use under $\alpha_z = \infty$ (no coupling) is unchanged.

Remark D.2 (Partition statistics). Marginalizing π , the partition induced on any fixed finite set of sites is the Ewens partition with parameter α_z , equivalent to a Chinese restaurant process draw (2). The expected number of co-evolving classes among L sites is $\mathcal{O}(\alpha_z \log L)$, and tuning α_z controls the sparsity of couplings: $\alpha_z \gg L$ yields $\mathcal{O}(L^2/\alpha_z)$ co-evolving pairs with higher-order classes suppressed by additional factors of L/α_z . Smaller α_z gives Ewens-distributed class sizes with heavier tails, including occasional larger cliques.

D.2 IBP variant

To allow each site to participate in multiple independent couplings, replace the Dirichlet process with an Indian Buffet Process (23).

Definition D.2 (TKF92-IBP). Sample feature frequencies $\{\pi_k\}_{k \geq 1}$ from an IBP with concentration α_z . At the birth of each site s , draw a binary feature vector $b_s \in \{0, 1\}^{\mathbb{N}}$ with

$$b_{s,k} \mid \pi_k \stackrel{ind}{\sim} \text{Bern}(\pi_k),$$

independently across sites. Two alive sites s, s' are coupled iff there exists k with $b_{s,k} = b_{s',k} = 1$. The amino acid dynamics is the reversible CTMC on the alive sites with stationary distribution

$$\pi(x_S) \propto \exp\left(-\sum_{\{s,s'\} : \exists k, b_{s,k} = b_{s',k} = 1} H(x_s, x_{s'})\right).$$

The same factorization observation applies: feature draws are i.i.d. given $\{\pi_k\}$, so the marginal alignment likelihood remains plain TKF92. The IBP variant naturally accommodates hub structure (residues that interact with many partners) via heavy-tailed feature frequencies.

D.3 Site classes and a GTR-parameterized generator

Real proteins exhibit substantial heterogeneity in equilibrium amino acid usage and in coupling profile: positions in α -helices, β -sheets, buried cores, and disordered regions have distinct frequencies and distinct interaction patterns with their structural neighbours. We extend the model with a per-site latent class variable that determines both the uncoupled (point) substitution model and the per-pair Potts interaction tensor. The variable is non-evolving, drawn once at site birth, and we put a second Dirichlet process on it — independent of the key DP — so that the number of site classes is itself learned from data, in line with the original infinite-mixture CAT model of Lartillot and Philippe (39).

Definition D.3 (Site-class TKF-DP). *Fix the alphabet $\mathcal{A}$ with $|\mathcal{A}| = A = 20$, a symmetric exchangeability matrix $S \in \mathbb{R}_{\geq 0}^{A \times A}$ ($S_{xy} = S_{yx}$), per-site rate-multiplier hyperparameters (a_η, b_η) , a site-class concentration $\alpha_c > 0$, a Potts-atom concentration $\alpha_H > 0$, and base measures*

$$G_0 : \quad \pi^{(c)} \stackrel{iid}{\sim} \text{Dirichlet}(\kappa_\pi \bar{\pi}), \quad G_0^H : \quad H_h(i, j) \stackrel{iid}{\sim} \mathcal{N}(\mu_{kl}, \tau_{kl}^{-1}) \text{ for } i \leq j, \quad H_h(j, i) := H_h(i, j),$$

where $(k, l) = (\min(i, j), \max(i, j))$ index the $A(A+1)/2$ unordered amino-acid pairs, and each Potts atom $H_h \in \mathbb{R}^{A \times A}$ is symmetric in (i, j) by construction. Site classes are drawn from a Dirichlet process $D_c \sim \text{DP}(\alpha_c, G_0)$ via stick-breaking; Potts atoms are drawn from a separate Dirichlet process $D_H \sim \text{DP}(\alpha_H, G_0^H)$. At the birth of each site s , draw an independent per-site rate multiplier and a site class:

$$\eta_s \stackrel{iid}{\sim} \text{Gamma}(a_\eta, b_\eta), \quad c_s \mid D_c \stackrel{iid}{\sim} D_c,$$

in addition to the DP key z_s (§D.1); η_s , c_s , and z_s are mutually independent and independent of the alignment path, and are fixed for the site’s lifetime. The Potts coupling tensor is decomposed by class-pair: for each unordered pair of classes $\{c, c'\}$ that ever co-occur within a key DP class, draw a Potts-atom assignment

$$h_{cc'} \mid D_H \stackrel{iid}{\sim} D_H,$$

at first co-occurrence and fixed thereafter; the materialized slice for the pair is $H_{cc'}(i, j) := h_{cc'}(i, j)$. The amino-acid trajectory on a DP key class C is the single-site reversible CTMC with stationary distribution

$$\pi_C(x_C) \propto \prod_{s \in C} \pi^{(c_s)}(x_s) \cdot \exp\left(-\sum_{\{s, s'\} \subset C} H_{c_s c_{s'}}(x_s, x_{s'})\right),$$

and (taking the F81 instance reversible with respect to this stationary)

$$Q^s(x \rightarrow x'; x_{C \setminus s}) = \eta_s S_{xx'} \pi^{(c_s)}(x') \exp\left(-\frac{1}{2} \Delta H_s\right), \quad \Delta H_s = \sum_{s' \in C \setminus s} [H_{c_s c_{s'}}(x', x_{s'}) - H_{c_s c_{s'}}(x, x_{s'})].$$

Remark D.3 (GTR limit and singletons). For a singleton key DP class, $\Delta H_s \equiv 0$ and the rate reduces to $Q_{xy}^s = \eta_s S_{xy} \pi^{(c_s)}(y)$, the F81 instance of GTR with class-specific equilibrium and a per-site rate multiplier. Marginalized over c_s this is exactly the CAT infinite-mixture profile model of Lartillot and Philippe (39), augmented with Yang’s gamma per-site rate variation (69). Coupling enters only via ΔH_s when the key DP class size exceeds one, so the model strictly extends the CAT–gamma combination by adding a coevolutionary tail.

Remark D.4 (F81 vs. symmetric-Metropolis form). The F81 form $Q_{xy} \propto S_{xy}\pi(y)$ chosen above is one of two natural reversible instances of the GTR family with stationary π and exchangeability S ; the other is the symmetric-Metropolis instance $Q_{xy} \propto S_{xy}\sqrt{\pi(y)/\pi(x)}$. Both satisfy detailed balance with respect to the joint Potts stationary above. The symmetric-Metropolis form has the property that the eigendecomposition of its symmetrized similarity transform is independent of π , allowing one corpus-wide A^3 decomposition to be reused across all classes. We adopt the F81 form here because, combined with the secret-destination augmentation of §D.12, it yields strict Dirichlet–multinomial conjugacy on $\pi^{(c)}$, which the symmetric-Metropolis form does not. The eigendecomposition then becomes per-class (one A^3 per class per outer step), which is small at $A = 20$ and is dominated by the Potts cluster work in any case.

Remark D.5 (Three independent Dirichlet processes). The construction has *three* Dirichlet processes acting on the substitution side: the key DP with concentration α_z partitions sites into co-evolving cliques (§D.1); the site-class DP with concentration α_c assigns each site a profile $\pi^{(c)}$; and a Potts-atom DP with concentration α_H and base measure G_0^H assigns each unordered class-pair $\{c, c'\}$ to a coupling atom $H_{h_{cc'}}$. With α_H small the $K(K+1)/2$ class-pair indices collapse onto a small number of canonical Potts atoms shared across class-pairs; with $\alpha_H \rightarrow \infty$ each class-pair has its own atom. The Potts-atom DP is what makes the model identifiable for $K_c \gg 1$: without it, the number of free Potts parameters scales as $K_c^2 \cdot A^2$; with it the count scales as $K_H \cdot A^2 + K_c^2$ class-pair indices, where K_H is data-driven and typically much smaller than K_c^2 .

Remark D.6 (Per-class-pair side potentials). A useful refinement of the per-class profile $\pi^{(c)}$ is to give each unordered class-pair $\{c, c'\}$ its own pair of *side potentials* $h_a^{(c,c')}, h_b^{(c,c')} \in \mathbb{R}^A$, with independent Gaussian priors centered at zero,

$$h_a^{(c,c')}(x), h_b^{(c,c')}(y) \sim \mathcal{N}(0, \tau_h^{-1}). \quad (\text{D.1})$$

The per-pair joint stationary then becomes

$$\log \pi_{\text{joint}}^{(c,c')}(x, y) = \log \pi^{(c)}(x) + \log \pi^{(c')}(y) - h_a^{(c,c')}(x) - h_b^{(c,c')}(y) - H_{cc'}(x, y) - \log Z_{cc'}, \quad (\text{D.2})$$

i.e. h is the per-pair deviation of the effective per-site stationary $\pi_{cc'}^{\text{eff},a}(x) \propto \pi^{(c)}(x)e^{-h_a^{(c,c')}(x)}$ from the per-class background $\pi^{(c)}$. For self-pairs (c, c) the two sites are exchangeable, so the joint distribution must be symmetric under swap $(x, y) \leftrightarrow (y, x)$. Combined with H_{cc} symmetric in its arguments, this forces $h_a^{(c,c)} = h_b^{(c,c)}$; we parameterise self-pairs with a single shared vector $h^{(c,c)} \in \mathbb{R}^A$ (and prior $h^{(c,c)} \sim \mathcal{N}(0, \tau_h^{-1})$, contributing A free parameters per self-pair rather than $2A$). Off-diagonal pairs $\{c, c'\}$ with $c \neq c'$ retain two independent vectors $h_a^{(c,c')}, h_b^{(c,c')}$ ($2A$ free parameters each). Reversibility of the F81-form joint generator is preserved by replacing the destination factor $\pi^{(c)}(x')$ in the site-1 flip rate with $\pi_{cc'}^{\text{eff},a}(x')$ (and analogously for site 2).

The motivation is that $\pi^{(c)}$ captures the population-mean amino-acid usage of class c *averaged over all class-pair contexts*, while the actual usage of class c when paired with class c' may deviate (e.g. a Cys-bearing class paired with itself is more Cys-rich than the per-class average, because we are conditioning on disulfide-forming sites). The h vectors absorb that deviation, leaving the Potts atom $H_{cc'}$ to capture the coupling *above the marginal compositional shift*. The Gaussian prior pulls h to zero so that any pair-specific deviation must be data-supported.

The free- h Gaussian-prior formulation of h presented here is the original “A3” parameterisation. [subsection D.4](#) shows that A3 is non-reversible across partial-presence pair histories; [subsubsection D.4.2](#) replaces it with the unique reversible (“A1”) alternative in which h is deterministically

Sinkhorn-determined by $(\pi^{(c)}, \pi^{(c')}, H_{cc'})$ rather than sampled. Under A1 the lone single-site stationary $\pi^{(c)}$ used on indel branches agrees with the joint pair-marginal exactly, restoring detailed balance at every indel seam. The remainder of this appendix adopts A1.

Remark D.7 (Hierarchical structure of the H base measure). The amino-acid-pair-indexed Gaussian base measure $\mathcal{N}(\mu_{kl}, \tau_{kl}^{-1})$ embeds an empirical-Bayes hierarchy: the corpus-level (μ_{kl}, τ_{kl}) encode the population-mean coupling propensity for each amino-acid pair (typically attractive for hydrophobic–hydrophobic, charge-complementary, etc., and repulsive for charge-like-like), and atom-specific deviations $H_h(i, j) - \mu_{kl}$ are learned around that mean with atom-specific magnitude controlled by the data. Symmetry of each H_h in (i, j) encodes the unordered-edge convention of Potts couplings; this is standard practice in the DCA literature (18) and is biologically natural in the absence of any a priori labelling distinguishing the two ends of an inter-class edge. The Gaussian prior on each H_h is conjugate to the Gaussian (Hessian-based) Laplace approximation of the path-DCA likelihood, giving a closed-form Gaussian posterior on each materialized atom under the local quadratic expansion of §D.12.

D.4 Reversibility under partial presence

The generative model of §D.1, with the side potentials of D.6 treated as free parameters with Gaussian priors, is the construction we will refer to as the *A3 free-h* formulation. Under A3 the per-pair joint stationary

$$\pi_{\text{joint}}^{(c,c')}(a, b) \propto \pi^{(c)}(a) \pi^{(c')}(b) e^{-h_a^{(c,c')}(a) - h_b^{(c,c')}(b) - H_{cc'}(a, b)} \quad (\text{D.3})$$

need not have row-marginal $\pi^{(c)}$ or column-marginal $\pi^{(c')}$; the freely-chosen h shifts both. The lone single-site stationary used on indel branches where one column is gapped is $\pi^{(c)}$, but the pair-marginalised stationary used on both-present branches is the marginal of (D.3). These two stationaries disagree at any nonzero H , with the disagreement carried by h . We show below that this disagreement costs the model reversibility on *partial-presence* pair histories (column pairs that are present at some leaves and gapped at others), and we develop the unique fix.

For a pair of classes (c, c') the bare joint

$$\pi_{\text{joint}}^{(c,c')}(a, b) \propto \pi^{(c)}(a) \pi^{(c')}(b) e^{-H_{cc'}(a, b)}, \quad H_{cc'} = H_{cc'}^\top \in \mathbb{R}^{k \times k}, \quad (\text{D.4})$$

with $k = |\mathcal{A}|$, is the $h \equiv 0$ specialisation of (D.3). On a tree branch where both columns are present the pair evolves under a reversible CTMC $\mathbf{Q}_{\text{pair}}$ with stationary (D.4); on a branch where one column is absent (inserted or deleted) the surviving column evolves under its lone CTMC with stationary $\pi^{(c)}$.

The defect. The per-pair Felsenstein likelihood is *root-dependent* whenever the pair has partial presence and $H_{cc'} \neq 0$. On a diagnostic fixture (binary alphabet, tree $((A, B)P, C)R$, one column present at $\{A, P, R\}$ and gapped at B, C), the pair log-likelihood evaluated at every admissible root is -3.0452 at the three roots present in both columns (A, P, R) but -2.4401 and -2.5831 at the two gapped-leaf roots (B, C) : the same physical (alignment, partition) state receives two different log-densities (~ 1.25 nats apart) on different visits, silently breaking detailed balance of any column-changing MCMC move.

D.4.1 Reversibility is marginal-consistency

Model the augmented state as (C, x_C) : the alive set C and residues $x_C \in \mathcal{A}^C$. The generator has three parts: (S) substitution within a fixed alive set, a CTMC reversible w.r.t. π_C ; (D) deletion of a site at a residue-independent TKF rate μ (drop the coordinate); (I) insertion of a site at rate λ , drawing the new residue from an insertion kernel $g_C(y | x_C)$.

Theorem D.1 (Reversibility criterion). *Suppose the within-level CTMC is reversible w.r.t. π_C and the candidate stationary factors as $P(C, x_C) = w_{|C|} \pi_C(x_C)$. Then the augmented process is reversible—equivalently the phylogenetic likelihood that peels with per-level equilibrium π_C is root-independent—if and only if the family $\{\pi_C\}$ is marginal-consistent:*

$$\sum_{x_s} \pi_C(x_C) = \pi_{C \setminus s}(x_{C \setminus s}) \quad \text{for every } C \text{ and every } s \in C, \quad (\text{D.5})$$

in which case the insertion kernel is forced to the true conditional $g_C(y | x_C) = \pi_{C \cup s}(y | x_C)$ (the conditional-Boltzmann insertion) and deletion is exact marginalisation; the level weights satisfy $w_k = (\lambda/\mu)^k w_0$.

Proof. Within a fixed C , detailed balance (DB) holds by the assumed substitution reversibility ($w_{|C|}$ is a common factor). The binding constraints are the cross-level (birth/death) edges. For $s \notin C$ and every x_C, y ,

$$w_{|C|} \pi_C(x_C) \lambda g_C(y | x_C) = w_{|C \cup s|} \pi_{C \cup s}(x_C, y) \mu. \quad (\text{D.6})$$

Summing over y (with $\sum_y g_C = 1$) gives $w_{|C|} \pi_C(x_C) \lambda = w_{|C \cup s|} \mu (\pi_{C \cup s} \downarrow C)(x_C)$. Both π_C and the marginal $\pi_{C \cup s} \downarrow C$ are normalised, so the proportionality constant is 1, forcing $w_{|C \cup s|} \mu = w_{|C|} \lambda$ and (D.5). Substituting back into (D.6) yields $g_C(y | x_C) = \pi_{C \cup s}(y | x_C)$, the true conditional. Conversely, given a marginal-consistent family these equations are satisfied for all edges, so the full generator is reversible. Reversibility of a CTMC is equivalent to root-independence of its Felsenstein likelihood (pulley principle). $\square$

Remark D.8 (The base $h \equiv 0$ model is not reversible). The bare joint (D.4) violates (D.5): its marginal $\sum_b \pi_{\text{joint}}^{(c, c')}(a, b) \propto \pi^{(c)}(a) \sum_b \pi^{(c')}(b) e^{-H_{cc'}(a, b)}$ equals $\pi^{(c)}(a)$ only when the bracket is constant in a , i.e. when $H_{cc'} \equiv 0$. For any nonzero coupling the pair marginal differs from the lone stationary $\pi^{(c)}$ used on indel branches, and the process is non-reversible / root-dependent on composition-changing partial-presence pairs.

Equivalently (the time-reversal view): a death is residue-independent, so for a birth to be its exact time-reverse the surviving residues must already sit at the reduced-level law; that is precisely (D.5), and it forces the conditional-Boltzmann insertion uniquely. The seam-DB cross ratio is 1.000000 for a marginal-consistent build and 2.927 for a marginal-inconsistent one; the conditional-Boltzmann insertion satisfies DB to 1.8×10^{-16} while bare- π insertion breaks it (5.08×10^{-3}). A randomised sweep (2000 trials at $k = 2, 3$, random symmetric H) gives $\max |\Delta_{\text{root}}| \leq 1.8 \times 10^{-15}$ when the lone stationary equals the joint marginal, versus up to 1.52 nats with bare π .

D.4.2 Imposing reversibility determines the side potentials

Returning to the side-potential joint (D.3), imposing marginal-consistency (D.5) promotes h from a free parameter (the A3 formulation, where h was given a Gaussian prior and sampled) to a function of (π, H) determined by Sinkhorn / Iterative Proportional Fitting.

Proposition D.1 (Sinkhorn determination). *Imposing marginal-consistency (D.5) on (D.3)—i.e. requiring row-marginals $\pi^{(c)}$ and column-marginals $\pi^{(c')}$ —is exactly the Sinkhorn / IPF matrix-scaling problem for the strictly positive kernel $K(a, b) = \pi^{(c)}(a)\pi^{(c')}(b)e^{-H_{cc'}(a, b)}$ with prescribed marginals. By Sinkhorn’s theorem the scaled joint exists and is unique, and the potentials are unique up to the gauge $h_a^{(c, c')} \mapsto h_a^{(c, c')} + \kappa_1$, $h_b^{(c, c')} \mapsto h_b^{(c, c')} + \kappa_2$ (absorbed by the normaliser). The coupling is preserved exactly: every log-odds-ratio $\log \frac{\pi_{\text{joint}}(a, b)\pi_{\text{joint}}(a', b')}{\pi_{\text{joint}}(a, b')\pi_{\text{joint}}(a', b)} = -(H(a, b) + H(a', b') - H(a, b') - H(a', b))$ is invariant under the scaling, so H carries correlation only and the marginals are corrected without touching the interaction.*

At $k = 4$ with non-uniform π and non-trivial symmetric H : base ($h = 0$) marginal error 6.2×10^{-2} ; Sinkhorn restores marginals to 1.1×10^{-16} in 18 iterations; six random initialisations give an identical joint ($\leq 5.6 \times 10^{-17}$) with potentials differing only by additive constants (the gauge); the interaction log-odds-ratios match base versus Sinkhorn to 1.6×10^{-15} . The side potentials collapse from free prior-sampled parameters (A3) to a deterministic function of (π, H) , removing both the sampling cost and the non-reversibility of A3 in one move.

D.4.3 The impossibility triangle

Write $\mu_L^{(c, c')}(x) = \sum_y \pi_{\text{joint}}^{(c, c')}(x, y)$ for the partner-marginal at the first site of a pair, and $g(c)$ for the lone stationary used when a paired site’s partner is absent. Three properties:

- (R) *Reversible / root-independent* $\Leftrightarrow$ marginal-consistency: $g(c) = \mu_L^{(c, c')}$ and $g(c') = \mu_R^{(c, c')}$ for every co-occurring pair (Theorem D.1).
- (I) *Table-independent*: g depends only on a site’s own class, $g = g(c)$, with no partner index. A site “chooses a table”, not “to be paired”; the lone law of a paired-but-currently-alone site equals the singleton law of its class. This is the defining factorisation of the TKF-DP construction (keys i.i.d. and independent of the alignment, so the alignment marginal stays exactly TKF92).
- (F) *Free per-pair marginal shift*: $\exists c, c_1 \neq c_2$ with $\mu_L^{(c, c_1)} \neq \mu_L^{(c, c_2)}$ (the marginal of a class genuinely depends on its partner; the biological motivation for h).

Theorem D.2 (Exactly two of three). *No cap-2 model satisfies (R), (I), (F) simultaneously; each pair of the three is realisable.*

Proof. (R) pins $\mu_L^{(c, c')} = g(c)$ for every partner c' ; (I) strips the partner index from g ; equating, $\mu_L^{(c, c_1)} = g(c) = \mu_L^{(c, c_2)}$ for all partners, so the partner-marginal is partner-independent and (F) fails. The argument references only the marginal of the joint, so it is invariant to how the coupling is parameterised. $\square$

The three corners are all attainable: **A1** = (R, I) is the Sinkhorn fix of [subsubsection D.4.2](#) with $g = \pi^{(c)}$ (coupling alive, $|H|_{\text{max}} = 2.63$; F dead to 10^{-13}); **A2** = (R, F) is reversible only with a partner-indexed $g(c, c')$ (no class-only g exists, so (I) fails); **A3** = (I, F) is the original free-sampled- h formulation of D.6, with lone $= \pi^{(c)}$ but root-violation 0.30 (non-reversible). The model adopted in the remainder of this appendix is A1.

Remark D.9 (What A1 keeps). A1 does not make the two members of a pair symmetric. Because the two sites draw their site-classes independently (the site-class DP, distinct from the key/coupling DP), generically $c_i \neq c_j$, so the pair marginalises to different per-class backgrounds $\pi^{(c_i)} \neq \pi^{(c_j)}$. Marginal asymmetry between paired sites, to the full extent their classes differ, is retained; only the

partner-context-dependence of a class’s marginal is lost. The two Dirichlet processes divide labour cleanly: the site-class DP carries marginal composition, the key/Potts DP carries correlation.

Remark D.10 (Why site classes are required under A1). With a single site class every site shares one background π , so a reversible cap-2 coupling can express only correlation in the marginal sense—never per-site compositional bias. All positional amino-acid structure must then come from the site-class DP; reversibility makes site classes a prerequisite for any composition signal beyond pure pair correlation, and H alone buys none of it (it cancels in every log-odds-ratio, Proposition D.1).

D.5 The order rule: compensatory potentials of order $m-1$

Proposition D.2 (No single-site repair beyond pairs). *For clusters of size ≥ 3 with fixed pairwise Potts atoms, single-site side potentials cannot enforce marginal-consistency. Marginalising one site of a size-3 cluster induces an effective two-site factor $T(x_1, x_2) = \sum_{x_3} \pi^{(c_3)}(x_3) e^{-H_{13}(x_1, x_3) - H_{23}(x_2, x_3)}$ whose log is generically non-separable in (x_1, x_2) , i.e. it injects interaction into the $(k-1)^2$ -dimensional interaction block. Single-site potentials move only the $2(k-1)$ main effects (zero interaction), so they cannot reproduce the exact $3 \rightarrow 2$ marginal on the same atom.*

Across 100 random clusters at each of $k = 3, 4, 5, 20$, the induced-interaction Frobenius norm exceeds 10^{-9} in all 100 cases (up to 1.85); the best IPF size-2 fit to the exact $3 \rightarrow 2$ marginal leaves residual KL = 1.1×10^{-2} nats. **Cap-2 is therefore the maximal cluster reversible by single-site repair**, beyond its computational motivation.

Proposition D.3 (Order rule). *Order- r compensatory potentials restore marginal-consistency (hence reversibility, Theorem D.1) for clusters up to size $r+1$, and are generically necessary beyond it. Equivalently: a size- m cluster carrying a full m -way Potts tensor stays reversible iff it acquires compensatory potentials of every order $\leq m-1$, fitted Sinkhorn / IPF-style so that each lower-order marginal equals the corresponding reduced-cluster stationary. Single-site repair ($r = 1$) thus reaches size 2 (Sinkhorn, Appendix D.4.2); pair repair ($r = 2$) reaches size 3; and so on.*

Mechanism. Eliminating one site from a size- m cluster whose interaction graph is complete leaves an $(m-1)$ -clique among the survivors (graph fill-in): the induced log-marginal carries an effective interaction of order exactly $m-1$ (the connected cumulant, scaling as $(\text{coupling})^{m-1}$). A cluster whose potentials are capped at order r parameterises exactly the log-densities of interaction order $\leq r$, so it can represent that $(m-1)$ -site induced marginal iff $m-1 \leq r$. By exponential-family duality, fixing all $\leq (m-1)$ -subset marginals determines a unique max-entropy Gibbs distribution whose sufficient statistics are exactly the $\leq (m-1)$ -subset indicators; the order- $(m-1)$ IPF/Sinkhorn fit therefore reproduces any realisable target lower-marginals exactly, with the order- m term set to zero, while any cap below $m-1$ is missing the required sufficient statistics. Size 3 is the special case where the $m \rightarrow (m-1)$ marginal lands on 2 sites, whose most general interaction is already pairwise. $\square$

The induced log-marginal’s top nonzero ANOVA order is exactly $m-1$ for $m = 3, 4, 5, 6$, with the leading term scaling cubically as coupling $\rightarrow 0$ at $m = 4$ (connected-cumulant signature); IPF reaches machine-zero full-table residual precisely when the fitted order reaches $m-1$ and only then. Pairwise repair succeeds 12/12 at $m = 3$ and fails generically at $m \geq 4$; order- $(m-1)$ is exact in every cell.

Why dynamic latent fields buy out of the hierarchy. Carrying the full order- $(m-1)$ potential hierarchy is the price of a *direct* (potential-based) reversible cluster (Proposition D.3). A latent

field (Appendix D.8) is the same projective family written through a hidden mediator: a categorical latent of cardinality $\leq k$ that decouples the pair (Proposition D.7) replaces the entire compensatory-potential tower, at the cost of an extra sum over the latent variable in the per-cluster likelihood.

D.6 Class-level variational substitution likelihood

Inference under the model requires, for each branch of a phylogeny and each co-evolving DP class C , the per-branch transition probability $P(\mathbf{x}_C(t) \mid \mathbf{x}_C(0), \{c_s\}_{s \in C})$. Direct evaluation requires exponentiating a generator on $\mathcal{A}^{|C|}$ and is intractable beyond $|C| \approx 5$. The bulk of the key-DP partition is small-class for α_z in the practical regime, so exact computation handles most of the work; what remains is to control the tail of large classes.

We derive a variational lower bound from an exact path-measure factorization on the class clique. The factorization is *exact* (Proposition below)—unlike the static pseudolikelihood of DCA inference (18), each factor still conditions site s on the *full* neighbour trajectory $\sigma_{C \setminus s}$ —and the intractable joint stationary is absorbed into the conditioned-on initial state. The genuine approximation, and with it the loss of direct-versus-indirect coupling resolution, enters only through the mean-field variational family below; see D.14. Two variational families on the resulting Girsanov ELBO are natural. The first is the simpler scheme we develop here, parameterized by per-site distributions at internal tree nodes plus a single constant rate matrix per site per branch, with a closed-form ELBO via eigendecomposition and a Felsenstein-style pruning recursion for optimization. The second is the fully inhomogeneous mean-field scheme of Cohn et al. (13), with time-varying rates and forward-backward ODE sweeps, which we describe as an upgrade for branches where the constant-rate scheme is too loose.

Proposition D.4 (Path-measure factorization). *Let $\sigma_C \in D([0, t]; \mathcal{A}^{|C|})$ be the trajectory of a single-site reversible CTMC on class C with rate $Q^s(x_s \rightarrow x'_s; x_{C \setminus s})$ and stationary distribution π_C . The path measure satisfies*

$$\mathbb{P}(\sigma_C) = \pi_C(\sigma_C(0)) \prod_{s \in C} \mathbb{P}_s(\sigma_s \mid \sigma_{C \setminus s}),$$

where $\mathbb{P}_s(\cdot \mid \sigma_{C \setminus s})$ is the law of an inhomogeneous CTMC on site s with rate $Q^s(\cdot; x_{C \setminus s}(u))$ at time u .

The factorization is exact: simultaneous jumps at distinct sites have zero Lebesgue measure, every jump is therefore attributable to a unique site, and the path log-density splits site-wise. The only intractable factor is $\pi_C(\sigma_C(0))$, which is conditioned upon for transition probabilities and so cancels.

Constant-rate variational family

Remark D.11 (Variational family). The variational family is parameterized at two levels. At each internal tree node v and site s , a discrete distribution $\nu_v^s \in \Delta^{|\mathcal{A}|-1}$ over residue states; node distributions factorize over sites, $\nu_v(\mathbf{x}_v) = \prod_s \nu_v^s(x_v^s)$. At each branch b between nodes p and c of length t_b and each site $s \in C$, a constant rate matrix $\hat{Q}_b^s$ on $[0, t_b]$. Conditional on sampled endpoint states $(x_p^s, x_c^s) \sim \nu_p^s \otimes \nu_c^s$, the variational law on the branch is the CTMC bridge of $\hat{Q}_b^s$ from x_p^s to x_c^s , with sites independent given endpoints. The joint variational law on the tree is

$$\mathbb{Q} = \prod_v \nu_v \cdot \prod_b \prod_{s \in C} \mathbb{Q}_b^s(\sigma_b^s \mid x_p^s, x_c^s).$$

Proposition D.5 (Closed-form per-branch ELBO). *The path-measure KL contribution from branch b and site s has Girsanov form*

$$\text{KL}_{b,s} = \mathbb{E}_{\mathbb{Q}} \int_0^{t_b} \sum_{x,x'} q_b^s(x; u) \left[\mathbb{E}_u [Q^s(x \rightarrow x'; x_{C \setminus s}(u))] - \hat{Q}_b^s(x \rightarrow x') + \hat{Q}_b^s(x \rightarrow x') \log \frac{\hat{Q}_b^s(x \rightarrow x')}{\exp \mathbb{E}_u [\log Q^s(\cdot)]} \right] du,$$

where $q_b^s(x; u) = \mathbb{E}_{x_p, x_c} [q_b^s(x; u \mid x_p, x_c)]$ is the variational bridge marginal of site s at time u averaged over endpoint distributions, and the inner expectations $\mathbb{E}_u[\cdot]$ are over the product of instantaneous neighbour bridge marginals at u , $\prod_{s' \neq s} q_b^{s'}(\cdot; u)$. The integrand is therefore a product of two time-varying bridge marginals at the same instant u , and the integral over $[0, t_b]$ does not factorise into time-averages: the cross-temporal correlation between $q_b^s(\cdot; u)$ and $q_b^{s'}(\cdot; u)$ contributes to $\text{KL}_{b,s}$ and is what makes the formula a strict lower bound. The bridge-expectation closed form below evaluates the integral correctly.

In the constant-rate family, the stationary point of $\text{KL}_{b,s}$ in $\hat{Q}_b^s$ admits no closed form in general because of the cross-temporal correlation. The geometric-mean rate

$$\log \hat{Q}_b^{s,*}(x \rightarrow x') = \mathbb{E}_{\bar{q}_b^{s,*}} [\log Q^s(x \rightarrow x'; x_{C \setminus s})],$$

where $\bar{q}_b^{s'}(x) = (1/t_b) \int_0^{t_b} q_b^{s'}(x; u) du$ is the time-averaged neighbour distribution for each $s' \neq s$, is the stationary point under the additional “mean-field-of-bridges” assumption that $q_b^s(\cdot; u)$ and $q_b^{s'}(\cdot; u)$ are approximately uncorrelated in u on $[0, t_b]$. This assumption is exact at $H = 0$ (sites independent under the variational law as well as under the truth), close to exact for small coupling, and only mildly violated at typical phylogenetic time scales; we take $\hat{Q}_b^{s,*}$ above as the chosen variational rate and evaluate the strict $\text{KL}_{b,s}$ at this rate via the bridge-expectation formula in the Remark below. Each term of the resulting bound is ≥ 0 by Jensen, so $\text{KL}_{b,s} \geq 0$ and the ELBO is a strict lower bound on $\log P$, as required.

Remark D.12 (Closed-form ELBO via pairwise bridge expectations). Each rate $\hat{Q}_b^s$ admits an eigen-decomposition $\hat{Q}_b^s = U^s D^s (V^s)$ with $V^s = (U^s)^{-1}$ and eigenvalues $\{\lambda_k^s\}$ (we drop the branch subscript b for legibility). The forward and backward propagators have the same expansion:

$$[\exp(\hat{Q}_b^s u)]_{x_p, x} = \sum_k U_{x_p, k}^s V_{k, x}^s e^{\lambda_k^s u}, \quad [\exp(\hat{Q}_b^s (t - u))]_{x, x_c} = \sum_l U_{x, l}^s V_{l, x_c}^s e^{\lambda_l^s (t - u)},$$

so the per-site bridge marginal is

$$q_b^s(x; u \mid x_p, x_c) = \frac{1}{P_s} \sum_{k, l} U_{x_p, k}^s V_{k, x}^s U_{x, l}^s V_{l, x_c}^s e^{\lambda_k^s u} e^{\lambda_l^s (t - u)},$$

with $P_s = [\exp(\hat{Q}_b^s t)]_{x_p, x_c}$ the bridge normalising constant. The branch integrand of $\text{KL}_{b,s}$ becomes a sum over (k, l) and over (k', l') from the neighbour expansion, with pair-time integrals

$$\text{HR}(\alpha, \beta, t) = \begin{cases} (e^{\alpha t} - e^{\beta t})/(\alpha - \beta) & \alpha \neq \beta, \\ t e^{\alpha t} & \alpha = \beta, \end{cases}$$

which is the standard bridge-expectation (31) expression. The full integral over $[0, t_b]$ is a finite sum of HR evaluations and closed-form coefficients in U, V, λ .

Optimization: Felsenstein-like pruning over node distributions

The node distributions ν_v^s optimise the ELBO under a Felsenstein-style upwards-downwards pass: each ν_v^s is the geometric mean of (i) the upwards-propagated likelihood from ν_v^s 's subtree, (ii) the downwards-propagated prior from the rest of the tree, and (iii) the per-branch forward and backward propagators of $\hat{Q}^s$ on each adjacent branch. The pass is $O(\text{edges} \cdot A^2)$ per outer ELBO step, dominated by the per-branch matrix exponential. The rate $\hat{Q}_b^{s,*}$ is then re-computed under the fresh node distributions and the loop iterates to convergence; the ELBO is monotone increasing in each step.

Jensen gap and the Cohn upgrade as endpoints of an adaptive refinement

The constant-rate scheme of the previous subsection and the inhomogeneous mean-field scheme of Cohn et al. (13) are not separate algorithms but the $N = 1$ and $N \rightarrow \infty$ ends of a single adaptive refinement. Subdivide a branch of length t_b into N sub-intervals at times $0 = u_0 < u_1 < \dots < u_N = t_b$ with constant rate $\hat{Q}_{b,k}^s$ on each $[u_{k-1}, u_k]$, and a variational node distribution ν_k^s at each intermediate midpoint. The resulting variational law is a piecewise-constant CTMC bridge: a piecewise-constant approximation to whatever time-varying rate Cohn's scheme would use at the same fidelity. As $N \rightarrow \infty$ with mesh refining, piecewise-constant rates become dense in continuous time-varying rates, so the variational family becomes Cohn's and the ELBO converges monotonically up. At the optimum, the ν_k^s are pinned by stationarity to the bridge marginals of the surrounding piecewise-constant CTMC — the explicit-midpoint parameterisation is slack at the optimum.

The Jensen-gap diagnostic per segment is the integrated AM-GM discrepancy of Q^s against neighbour bridge marginals at the segment's resolution; it is closed form via the same eigendecomposition machinery and tells the algorithm where to refine. For typical phylogenetic branches ($t_b \in [0.1, 1]$) most segments need $N = 1$; high-coupling segments may benefit from $N = 2$ or 3; only pathological branches need $N \rightarrow \infty$. Forced segmentation by composition-change events from the augmented indel history (§D.10) composes cleanly with this Jensen-driven refinement: subdivide first at every composition-change event, then optionally N -refine within each composition-constant segment.

Remark D.13 (Cohn's inhomogeneous fixed point as the $N \rightarrow \infty$ limit). For a composition-constant segment requiring $N \rightarrow \infty$ refinement, the limit is Cohn et al.'s full inhomogeneous fixed point: a time-varying rate $\tilde{Q}^s(\cdot; u)$ obtained by alternating forward sweeps $\dot{q}_s = q_s \tilde{Q}^s$ from $\delta_{x_p^s}$, backward sweeps $\dot{\rho}_s = -\tilde{Q}^s \rho_s$ from $\delta_{x_e^s}$, and rate updates $\log \tilde{Q}^{s,*}(\cdot; u) = \mathbb{E}[\log Q^s(\cdot)]$ pointwise in u , with bridge marginals $q_s^{\text{bridge}}(x; u) \propto q_s(x; u) \rho_s(x; u)$. The midpoint construction above is the discretisation of this fixed point at mesh size $u_k - u_{k-1}$; the Cohn limit is recovered as the mesh shrinks to zero. Cohn et al. (13, §7) also give the internal-node rule used by Felsenstein pruning under the time-varying rate.

This is the lower-bound construction referenced in the Discussion for Potts components of size > 2 : independent-coupling pairwise factorisation gives the $N=1$ lower bound, midpoint augmentation extends it to a formal ELBO, and $N \rightarrow \infty$ recovers the Cohn fixed point. Classes whose Jensen gap remains large after refinement are candidates for the k -clique cluster-variational upgrade of Linzner & Koepl (43), or for fallback to exact uniformization-based bridge sampling via Rao & Teh (51).

Fixed product-bridge path ELBO

The scheme above optimises an inhomogeneous variational rate. A cheaper bound, used for the coupled substitution models of the companion paper, fixes the variational law outright: it is the

product of endpoint-conditioned bridges of a single *time-homogeneous* generator. Everything below is stated for one branch of length t with endpoints $x(0) = a$, $x(t) = b$.

Let the cluster C carry $L = |C|$ sites with Potts target $\Pi(x) \propto e^{-E(x)}$ on $\mathcal{A}^L$, $E(x) = -\sum_s h_s(x_s) - \sum_{s < s'} H_{ss'}(x_s, x_{s'})$, let $R_{ic} = S_{ic}\pi_c$ be a single-site GTR proposal acting site-wise, and let W be the square-root Metropolis target rate for the move $x \rightarrow x^{s:c}$ that replaces residue $i = x_s$ by c ,

$$W_{x, x^{s:c}} = R_{ic} \sqrt{\frac{\Pi(x^{s:c}) R_{ci}}{\Pi(x) R_{ic}}} = S_{ic} \sqrt{\pi_i \pi_c} e^{-\frac{1}{2} \Delta_s^c E(x)}, \quad \Delta_s^c E(x) = E(x^{s:c}) - E(x),$$

which is reversible with respect to Π for any S, h, H . Among acceptance functions $A(z) \propto z^\theta$, reversibility ($A(z) = zA(1/z)$) forces $\theta = \frac{1}{2}$, so the square root is the unique log-linear reversible choice; it is also the unique one for which the partition function of Π cancels from every rate ratio.

The bound. Girsanov's formula for jump processes writes the log-density ratio of the two path measures as a difference of complete-data log-likelihoods: for a trajectory $\mathbb{X}$ with transition usages U and dwell times T ,

$$\log \frac{d\mathbb{P}_W}{d\mathbb{P}_R}(\mathbb{X}) = \ell_W(\mathbb{X}) - \ell_R(\mathbb{X}), \quad \ell_Q(\mathbb{X}) = \sum_{x \neq y} U_{xy} \log Q_{xy} - \sum_x T_x q(x),$$

with $q(x) = \sum_{y \neq x} Q_{xy}$ the exit rate. The reference measure (counting measure on jump counts $\times$ Lebesgue measure on jump times) carries no dependence on the generator, so this is a genuine Radon–Nikodym derivative and ℓ_Q has no additive Q -independent term. Averaging over the R -bridge and applying Jensen's inequality gives

$$\mathcal{L}_{ab} = \sum_s \log[e^{tR}]_{a_s b_s} + \sum_{x, s, c \neq x_s} \bar{N}_{x, s: c}^{ab} \log \frac{W_{x, x^{s:c}}}{R_{x_s c}} - \sum_x \bar{T}_x^{ab} [w(x) - r(x)] \leq \log[e^{tW}]_{ab}, \quad (\text{D.7})$$

where w, r are the exit rates of W, R and $\bar{T}, \bar{N}$ are the endpoint-conditioned dwell and usage statistics of the A^L -state product chain. Equation (D.7) is the mean-field free energy of Cohn et al. (13) evaluated at one member of their factored family, so their Theorem 3 certifies it; it is not the optimal member, and is therefore looser than the fixed point of the previous subsection. In exchange the bridge rates never appear: the factors $\log[e^{(t-u)R}]_{yb}$, which diverge as $u \rightarrow t$ for $y \neq b$ and are the source of the endpoint stiffness in the inhomogeneous scheme, telescope analytically into the normaliser $\sum_s \log[e^{tR}]_{a_s b_s}$.

Log-linearity, and the two contractions. Because the acceptance is log-linear, the log-rate ratio is a *potential difference*,

$$\log \frac{W_{x, x^{s:c}}}{R_{x_s c}} = \psi(x^{s:c}) - \psi(x), \quad \psi(x) = -\frac{1}{2} \sum_s \log \pi_{x_s} - \frac{1}{2} E(x). \quad (\text{D.8})$$

Both terms of (D.7) then reduce to per-site quantities, for two separate consequences of (D.8).

The usage term is a boundary term. Every jump of the product chain moves exactly one site, so a path is a chain $a = x_0 \rightarrow \dots \rightarrow x_K = b$ with $x_k = x_{k-1}^{s_k: c_k}$, and by (D.8) the increments $\psi(x_k) - \psi(x_{k-1})$ telescope to $\psi(b) - \psi(a)$ — the same value for every path, whatever its length or the order of its jumps. (Equivalently: the energy telescopes globally, and the equilibrium term telescopes site by site, each site's intermediate states cancelling.) Since $\sum_{x, s, c} \bar{N}_{x, s: c}^{ab} f$ is by definition

the bridge expectation of $\sum_{\text{jumps}} f$, and the expectation of a path-independent quantity is that quantity,

$$\sum_{x,s,c \neq x_s} \bar{N}_{x,s;c}^{ab} \log \frac{W_{x,x^{s:c}}}{R_{xc}} = \frac{1}{2} \sum_s \log \frac{\pi_{a_s}}{\pi_{b_s}} + \frac{1}{2} [E(a) - E(b)]. \quad (\text{D.9})$$

No bridge statistic and no integral is required, and the couplings enter through the endpoints alone.

The dwell term contracts to a one-dimensional integral. The product chain has $M_{ab}(t) = \prod_s M_{a_s b_s}^s(t)$ with $M^s(u) = e^{uR}$, so $\bar{T}_x^{ab} = \int_0^t \prod_s p_u^s(x_s) du$ and hence, for any ϕ ,

$$\sum_x \bar{T}_x^{ab} \phi(x) = \int_0^t \mathbb{E}_u[\phi] du, \quad p_u^s(y) = \frac{M_{a_s y}^s(u) M_{y b_s}^s(t-u)}{M_{a_s b_s}^s(t)}, \quad (\text{D.10})$$

with $\mathbb{E}_u$ the expectation under the product measure $\prod_s p_u^s$: the sum over $\mathcal{A}^L$ has become a time integral of an expectation over *independent* sites. Now $-\frac{1}{2} \Delta_s^c E(x)$ is a sum over the neighbours of s , so W is a *product* over them, and the expectation of a product of functions of distinct independent sites is the product of the expectations,

$$\mathbb{E}_u[W_{x,x^{s:c}}] = \sum_i p_u^s(i) S_{ic} \sqrt{\pi_i \pi_c} e^{\frac{1}{2}[h_s(c) - h_s(i)]} \prod_{s' \neq s} \left(\sum_z p_u^{s'}(z) e^{\frac{1}{2}[H_{ss'}(c,z) - H_{ss'}(i,z)]} \right). \quad (\text{D.11})$$

Every factor is a single sum over one site's alphabet, so the A^L configuration sum never forms, even though the exit rate depends on all neighbours jointly and nonlinearly: what is required is *separability* across neighbours, not linearity. A genuinely k -body term in E would cost A^k . The proposal exit rate r is one-body and contracts likewise. At each quadrature node, evaluating (D.11) costs $O(A^3)$ per coupled pair and $O(A^2)$ per site, so the bound costs $O(m(LA^2 + PA^3))$ for m quadrature nodes and P coupled pairs — polynomial in cluster size rather than exponential.

When the time integral is exact. Factorisation across sites is not factorisation across time: $\int_0^t \prod_s \neq \prod_s \int_0^t$, so (D.10) leaves a genuine integral. Each p_u^s is a sum of A^2 exponentials in u , with exponents $\xi_k - \xi_l$ drawn from the spectrum of R ; the integrand of (D.11) is a product of one such sum per coupled site, L in all. For $L = 2$ that product is of two, the integral is the exponential divided-difference kernel of Holmes & Rubin (31), and $\mathcal{L}_{ab}$ is exactly closed-form. For $L > 2$ the expansion carries A^{2L} terms; grouping the product eigenvalues $\sum_s \xi_{k_s}$ into their $\binom{L+A-1}{A-1}$ multiplicity classes reduces this substantially but not enough to be practical at $A = 20$, and we integrate numerically instead. Because h, H, S and π enter (D.11) only as u -independent weights, the exponents are set by R alone and are bounded by $L\rho(R)t$ with ρ the spectral radius; the integrand is therefore entire and Gauss–Legendre converges spectrally. With R set to LG08 and normalised to one expected substitution per unit time, $m \leq 14$ nodes reach machine precision for $L \leq 5$, $t \leq 3$ and couplings up to $|H| \sim 4$, the required m growing roughly as $6 + L\rho(R)t/3$. It is this conditioning, rather than any closed form, that distinguishes the bound from the inhomogeneous mean field, whose endpoint-singular rate integral converges at first order.

Remark D.14 (What the mean-field bound recovers: direct vs. indirect coupling). Because the variational family factorises over sites ($\mathbb{Q} = \prod_s \mathbb{Q}^s$), the coupling enters the ELBO only through the rank-1 outer product of per-site bridge marginals,

$$\sum_{s < s'} \int_{\mathcal{T}} \sum_{a,b} H_{ss'}(a,b) \gamma_s(a;u) \gamma_{s'}(b;u) du,$$

and the connected correlation $\text{Cov}_{\mathbb{Q}}(\mathbb{1}_{x_s=a}, \mathbb{1}_{x_{s'}=b})$ vanishes identically. Consequently $\partial \text{ELBO} / \partial H_{ss'}$ matches, per pair, that pair's own endpoint-conditioned marginals and nothing else. This is *naive*

mean-field learning: the bound never forms the connected covariance C nor inverts it, so it performs neither the fluctuation-response deconvolution $J \approx -C^{-1}$ of mean-field DCA (47) nor the all-sites conditioning of pseudolikelihood DCA (18). The self-consistency of the E-step couples the γ_s through the *forward* mean-field map $H \mapsto \gamma$ (cavity fields $\sum_{s'} H_{ss'} \gamma_{s'}$); linear response is the Jacobian $d\gamma/dh = \chi = C$, and disentangling requires forming *and inverting* χ —coordinate-ascent variational EM uses only the fixed-point values γ , never $d\gamma/dh$, so no implicit inversion is taken (35, 57). Therefore, whenever a site has ≥ 2 coupled partners (a genuine $m \geq 3$ contact triangle), the fitted $H_{ss'}$ absorbs the indirect s – k – s' correlation, and the bound recovers indirect-inclusive (mutual-information-like) couplings, not direct ones. The exact path factorisation is not to blame—it conditions on the full correlated neighbour trajectory and is plmDCA-like—the loss is incurred by the mean-field product family. At genuine cap-2 (no triangles) the distinction is moot and the bound is exact.

D.7 Composite-likelihood phylogenetic ELBO for multi-site Potts coupling under a Metropolis–Hastings substitution form

The class-level bound of subsection D.6 controls the coupled CTMC on a single clique of arbitrary size m . Two costs remain for a general phylogeny. First, the *joint* clique likelihood on a tree carries an $\mathcal{A}^m$ -state conditional-likelihood vector at every node, super-exponential in tree depth. Second, reversibility of the joint stationary $\pi_C(x_C) \propto \prod_s \pi^{(c_s)}(x_s) \exp(-\sum_{\{s,s'\} \subset C} H_{c_s c_{s'}}(x_s, x_{s'}))$ across the insertion/deletion seam forces the order- $(m-1)$ potential hierarchy of D.3 and the Sinkhorn side-potential fix of D.1. This subsection shows that a *composite* (pairwise) likelihood dissolves both costs at once: the corpus objective becomes a *product over coupled pairs of independent two-site phylogenetic trees*, the Potts coupling enters *only* through the two-site stationary, and—under a Metropolis–Hastings substitution form—the per-pair generator, its exact reversibility, and its Holmes–Rubin bridge statistics all collapse to a single shared exchangeability acting on single-residue moves.

Composite decomposition. Let E be the set of coupled site-pairs: under the key DP this is the set of two-element blocks of the partition (the cap-2 TKF-DP training regime, where every co-evolving class has $|C| \leq 2$); under structure supervision it is a fixed contact map. Replace the joint clique likelihood on the phylogeny $\mathcal{T}$ by the composite (pseudo-)likelihood

$$\mathcal{L}_{\text{comp}} = \sum_{(i,j) \in E} \log P(\mathbf{x}_i, \mathbf{x}_j \mid \mathcal{T}, \pi^{(c_i)}, \pi^{(c_j)}, H_{c_i c_j}, S), \quad (\text{D.12})$$

a composite *marginal* likelihood (42) (see 65 for the general theory). This is the product-of-*marginals* form, and is to be distinguished from Besag’s product-of-*conditionals* pseudolikelihood (6) (the object underlying plmDCA (18)): the two agree only in the absence of coupling triangles. Each summand is the phylogenetic likelihood of a single ordered residue pair (x_i, x_j) evolving down $\mathcal{T}$ under a two-site reversible CTMC. The Potts energy is *genuinely pairwise*, so the composite reproduces the model’s energy *form*; but each H_{ij} is fitted from the pair *marginal* π_{ij} , which is exact for the direct coupling only when i, j share no common partner (D.14).

The two-site chain and its stationary. Index the pair state $x = (a, b) \in \mathcal{A}^2$ by a (site i), b (site j), and write $H := H_{c_i c_j}$. The composite factor treats both sites as present throughout the branch (no partner insertion/deletion *inside* the factor), so the only requirement on its generator is

reversibility with respect to the pair stationary

$$\pi_{ij}(a, b) = \frac{1}{Z_{ij}} \pi^{(c_i)}(a) \pi^{(c_j)}(b) e^{-H(a, b)}, \quad Z_{ij} = \sum_{a, b} \pi^{(c_i)}(a) \pi^{(c_j)}(b) e^{-H(a, b)}. \quad (\text{D.13})$$

This is exactly the pair-marginalised joint stationary (D.4). The marginal-consistency condition of Theorem D.1— $\sum_b \pi_{ij}(a, b) = \pi^{(c_i)}(a)$ —is *not* imposed here, because it is an indel-seam requirement (needed only when a partner is born or dies mid-branch, coupling a singleton stationary to a pair stationary). Fixing both sites present within the composite factor removes it, and with it the entire order- $(m-1)$ hierarchy and the Sinkhorn correction: the pair chain needs only detailed balance for (D.13), which any GTR or Metropolis instance provides.

Metropolis–Hastings generator. Take the single-transition (one residue changes at a time) Metropolis–Hastings generator with a single shared symmetric exchangeability S and Hastings acceptance $g(u, v) = \min(1, v/u)$,

$$Q_{ij}[(a, b), (a', b)] = S(a, a') \min\left(1, \frac{\pi_{ij}(a', b)}{\pi_{ij}(a, b)}\right), \quad a' \neq a \quad (\text{site-}i \text{ move}), \quad (\text{D.14})$$

$$Q_{ij}[(a, b), (a, b')] = S(b, b') \min\left(1, \frac{\pi_{ij}(a, b')}{\pi_{ij}(a, b)}\right), \quad b' \neq b \quad (\text{site-}j \text{ move}), \quad (\text{D.15})$$

$$Q_{ij}[(a, b), (a', b')] = 0, \quad a' \neq a, b' \neq b, \quad (\text{D.16})$$

with $Q_{ij}[x, x] = -\sum_{y \neq x} Q_{ij}[x, y]$; simultaneous double jumps have zero rate, as in any single-transition (CTBN-style) pair chain. Three properties of the Hastings kernel do the work.

Proposition D.6 (Exact reversibility, spectator cancellation, symmetric min-flux). *For any symmetric S and any H , the generator (D.14)–(D.16) satisfies detailed balance with respect to (D.13), with the symmetric flux*

$$F_{xy} := \pi_{ij}(x) Q_{ij}[x, y] = S_{xy} \min(\pi_{ij}(x), \pi_{ij}(y)) = F_{yx}. \quad (\text{D.17})$$

Moreover the acceptance ratio of a single-site move depends on the Potts coupling only through the local increment $\Delta H_i(a \rightarrow a'; b) = H(a', b) - H(a, b)$: the spectator stationary cancels,

$$\frac{\pi_{ij}(a', b)}{\pi_{ij}(a, b)} = \frac{\pi^{(c_i)}(a')}{\pi^{(c_i)}(a)} e^{-\Delta H_i(a \rightarrow a'; b)}, \quad (\text{D.18})$$

so the site- i block of F carries the spectator as an overall factor, $F[(a, b), (a', b)] = Z_{ij}^{-1} S(a, a') \pi^{(c_j)}(b) \min(\pi^{(c_i)}(a) e^{-H(a, b)}, \pi^{(c_i)}(a') e^{-H(a', b)})$ and symmetrically for the site- j block.

Proof. Detailed balance: $\pi_{ij}(x) Q_{ij}[x, y] = S_{xy} \pi_{ij}(x) \min(1, \pi_{ij}(y)/\pi_{ij}(x)) = S_{xy} \min(\pi_{ij}(x), \pi_{ij}(y))$, manifestly symmetric in x, y (and $= 0$ off single-transition edges, where $S_{xy} = 0$). (D.18) is immediate from (D.13) since $\pi^{(c_j)}(b)$ appears in numerator and denominator; the flux form follows by multiplying through by $\pi_{ij}(a, b) = \pi^{(c_j)}(b) \pi^{(c_i)}(a) e^{-H(a, b)} / Z_{ij}$. $\square$

What M–H buys. Compared with the F81 pair $Q_{xy} \propto S_{xy} \pi_{ij}(y)$ —which is *also* reversible with respect to (D.13), so reversibility alone does not distinguish them—the Hastings form gives (i) the symmetric min-flux (D.17), whose exchangeability is a *single* shared S (the 190-parameter symmetric object of the pair-model family, versus $\mathcal{O}(K_c^2 A^2)$ free Potts entries), with a closed-form usage/exposure M-step below; (ii) the spectator cancellation (D.18), which localises H to the increment ΔH_i so the pair generator is assembled from a per-site exchangeability and a Potts-modulated

acceptance rather than a dense $\mathcal{A}^2 \times \mathcal{A}^2$ rate table; and (iii) support of F (hence of the Holmes–Rubin usage N) *only* on the $2\mathcal{A}(\mathcal{A}-1)$ single-transition edges, so the bridge statistics never populate the $\mathcal{A}^2$ dense block. (The symmetric-Metropolis $\sqrt{\cdot}$ variant of the remark in subsection D.3 instead buys a π -independent symmetrised eigenbasis; the min variant here buys the exact acceptance and the sparsest flux.)

The composite phylo-ELBO objective. Each summand of (D.12) is the reversible-CTMC Felsenstein forward on the $\mathcal{A}^2$ -state pair chain Q_{ij} over $\mathcal{T}$: a post-order recursion of pair conditional-likelihood vectors with per-branch propagators $P_{ij}(\tau) = \exp(Q_{ij}\tau)$ from the reversible eigendecomposition of D.12. The corpus objective is thus a *product of pair-trees*,

$$\mathcal{L}_{\text{comp}} = \sum_{(i,j) \in E} \log \left(\sum_{x_r} \pi_{ij}(x_r) F_{ij}^T(x_r) \right), \quad F_{ij}^v(x_v) = \prod_{u \in \text{ch}(v)} [P_{ij}(\tau_u) F_{ij}^u](x_v), \quad (\text{D.19})$$

with leaf CLVs $F_{ij}^\ell(a, b) = \mathbb{I}[a=x_{i,\ell}^{\text{obs}}] \mathbb{I}[b=x_{j,\ell}^{\text{obs}}]$. Each tree runs on $\mathcal{A}^2$ states independently of every other pair; no $\mathcal{A}^m$ state and no $|\Theta|$ product ever appears. Where a co-evolving clique genuinely has $m > 2$ sites, each pair term may be further mean-fielded by the product-of-single-site-trees family of subsection D.6 (constant-rate Cohn/Girsanov variational generators per site, coupled only through the pairwise bridge expectations $\mathbb{E}[\Delta H]$), yielding a strict lower bound $\mathcal{L}_{ij} \leq \log P(\text{pair}_{ij})$; at $m = 2$ that gap is zero because (D.19) keeps the exact two-site chain. Hence at cap-2, $\mathcal{L}_{\text{comp}} = \log P_{\text{comp}}(\text{data})$ exactly—the ELBO is tight and equals the composite log-likelihood.

E-step: Holmes–Rubin sufficient statistics. For each pair and each branch $u \rightarrow v$ of length τ , one reversible forward–backward pass over Q_{ij} gives the endpoint-conditioned edge posterior $w_{ij}^{uv}(x, y)$ (the two-slice variational marginal of the pair state at the branch endpoints). The Holmes–Rubin bridge (31), evaluated by the reversible eigendecomposition and the divided-difference kernel $\text{HR}(\alpha, \beta, \tau)$ of D.12, returns the expected transition usage and dwell for that branch, aggregated over pairs and branches into

$$N_{xy} = \sum_{(i,j) \in E} \sum_{u \rightarrow v} \mathbb{E}[\#\{x \rightarrow y \text{ jumps}\} \mid w_{ij}^{uv}], \quad T_x = \sum_{(i,j) \in E} \sum_{u \rightarrow v} \mathbb{E}[\text{time in } x \mid w_{ij}^{uv}]. \quad (\text{D.20})$$

By D.6, N_{xy} is supported only on single-transition edges. These are the complete-data sufficient statistics of the pair chain.

M-step for S , π/H . The expected complete-data log-likelihood is $\sum_{x \neq y} [N_{xy} \log Q_{ij}[x, y] - T_x Q_{ij}[x, y]]$ up to π -only terms. Two blocks alternate. (i) *Exchangeability.* All single-transition edges sharing an unordered residue pair $\{a, a'\}$ (across both coordinates, pooled by the component-exchange symmetry $\pi_{ij}(a, b) = \pi_{ji}(b, a)$) carry the same $S(a, a')$; stationarity in $S(a, a')$ gives the closed form

$$S(a, a') \leftarrow \frac{\sum_{\text{edges } \{a, a'\}} N_{xy}}{\sum_{\text{edges } \{a, a'\}} T_x \min(1, \pi_{ij}(y)/\pi_{ij}(x))}, \quad (\text{D.21})$$

usage over exposure—the exact maximiser, hence non-decreasing in the expected complete-data objective. (ii) *Stationary/coupling.* With the per-class profiles $\pi^{(c)}$ frozen (LG08 or the archetype vocabulary (D.23)), the free coupling is H ; equivalently one may fit the symmetric pair stationary π_{ij} directly and read $H(a, b) = -\log \pi_{ij}(a, b) + \log \pi^{(c_i)}(a) + \log \pi^{(c_j)}(b) + \text{const}$, identified up to the usual Potts gauge (additive functions of a alone or b alone, absorbable into $\pi^{(c_i)}, \pi^{(c_j)}$). Because the min acceptance makes π_{ij} enter Q_{ij} nonlinearly, there is no closed form; a damped gradient

ascent on $\log \pi_{ij}$ of the same expected complete-data objective (with residual $N_{xy} - T_x Q_{ij}[x, y]$) increases it. *Monotonicity.* Since (D.14)–(D.16) is reversible with respect to π_{ij} for *every* iterate (S, π_{ij}) (D.6), each pair term of (D.19) remains a valid observed-data likelihood of a reversible CTMC, and the standard EM inequality applies per pair: the E-step computes exact expected sufficient statistics (D.20) under the current Q_{ij} and the M-step raises the expected complete-data log-likelihood (exactly in S , ascent in π_{ij}), so $\mathcal{L}_{\text{comp}}$ is non-decreasing under coordinate ascent.

What the composite approximates, and where it is exact. At cap-2 (every co-evolving class has $|C| \leq 2$, the TKF-DP training geometry) the set E is the class partition itself, the composite objective *is* the joint likelihood, and each pair-tree forward (D.19) is exact, so $\mathcal{L}_{\text{comp}} = \log P(\text{data})$ with no gap. For genuine $m > 2$ cliques the product over the $\binom{m}{2}$ pairs is a composite *marginal* likelihood (42, 65): it retains the pairwise energy *form* (the model has no higher-body term to lose), but each H_{ij} is fitted from the pair *marginal* π_{ij} . When i and j share a common partner k , that marginal absorbs the indirect i – k – j correlation, so the estimator converges to the indirect-inclusive (mutual-information-like) coupling, *not* the direct one—it is *inconsistent* for the direct H_{ij} , and the failure is genuine, not merely a loss of Godambe efficiency. This is the marginal-composite counterpart of the mean-field limitation of D.14; contrast Besag’s conditional pseudolikelihood (the plmDCA object (6, 18)), which conditions each site on all others and *is* consistent for the direct couplings, and mean-field DCA (47), which recovers direct couplings by inverting the connected covariance. The two-site tree forwards themselves remain exact; the bias is entirely the per-pair marginalisation. Thus the composite phylo-ELBO is exact where TKF-DP trains (cap-2, no triangles), and above it a controlled but *direct-indirect-conflating* surrogate that reduces the multi-site Potts objective to a product of independent two-site trees with the coupling entering only through the pair stationary (D.13).

D.8 Dynamic latent-field variant

The Potts construction of subsection D.3 routes coupling through an explicit pairwise residue-space tensor $H_{cc'}$ and pays for arbitrary-size reversibility with the order- $(m - 1)$ potential hierarchy of D.3. An alternative construction routes the coupling instead through a shared categorical *latent field* per coupled cluster, evolving on the tree as a small reversible CTMC. The two sites of a coupled pair are then conditionally independent given the field at every time; the cost of arbitrary cluster size is one extra sum (per cluster) over the small latent alphabet rather than a hierarchy of compensatory potentials. The construction is reversible by induction, exactly tractable at the cap-2 cluster geometry used in TKF-DP training, and extends to general phylogenies via a structured mean-field variational bound (subsubsection D.8.6). Its closest phylogenetic antecedents are the multi-level covarion / nested-iHMM families (4, 59, 64) and the F81/parent-independent-mutation foundations of population genetics (21, 22, 37).

D.8.1 Categorical mediator

Proposition D.7 (Categorical mediator). *Any pair joint $\pi_{\text{joint}}(x, y)$ on $\mathcal{A} \times \mathcal{A}$ can be written $\pi_{\text{joint}}(x, y) = \sum_h w_h a_h(x) b_h(y)$ with $\{a_h\}$, $\{b_h\}$ probability vectors and $\{w_h\}$ a probability—i.e. there is a latent h rendering the two sites conditionally independent, $x \perp y \mid h$. The minimal cardinality $|h|$ is the nonnegative rank $\text{nn-rank}(\pi_{\text{joint}}) \leq k$: it is 1 iff the coupling is separable (independent sites), and up to k for full-rank coupling. Computing nn-rank is NP-hard in general (66); for $k \leq 3$ Cohen–Rothblum gives $\text{nn-rank} = \text{rank}$ (12), and the first gap appears at $k = 4$ (a strictly-positive symmetric size-4 joint of ordinary rank 3 can have nonnegative rank 4).*

A Potts edge “is” a hidden common cause: a categorical field of size nn -rank that decouples the pair. In cap-2 every edge is its own small ($\leq k$ -state) mediator.

Gaussian copula is a strict subfamily. A scalar latent Gaussian per site (one correlation ρ) induces only *monotone* association. For binary alphabets that is the whole story; for $k \geq 3$ a Potts has $\binom{k}{2}$ coupling degrees of freedom and can be non-monotone or cyclic, which no single- ρ copula reproduces. Matching arbitrary H needs a *vector* latent Gaussian (rank up to $k - 1$) plus general links, i.e. a latent-factor model.

D.8.2 Instantly-re-equilibrating dynamic latent-field selector

A categorical latent field $\theta(t)$ evolves on the tree as a small reversible CTMC with state space the number of fields $L \ll q^m$; a coupled site evolves under the field-conditioned single-site generator $\mathbf{Q}(\theta)$ between field jumps and *instantly re-equilibrates*—resamples from $f(\cdot \mid \theta')$ —at each jump. This is not a continuous-time Bayesian network (the residue process has instantaneous jumps slaved to θ), but it is exactly solvable at the cap-2 cluster geometry used in TKF-DP training; on general phylogenies the joint multi-component Felsenstein admits the variational rescue of [subsubsection D.8.6](#).

Proposition D.8 (Hierarchical Felsenstein). *Resampling at every field jump makes residues conditionally independent across sites and memoryless across jumps given the θ -trajectory. Hence conditional on a θ -trajectory $\boldsymbol{\theta}$ on the tree, the joint likelihood factorises over sites,*

$$L(\text{obs} \mid \boldsymbol{\theta}) = \prod_s L_s(\text{obs at site } s \mid \boldsymbol{\theta}). \quad (\text{D.22})$$

At the cap-2 cluster geometry (one internal parent, two leaves; the cherry geometry of TKF-DP training and IPHMM pairwise postprocessing), integrating $\boldsymbol{\theta}$ via a single Felsenstein step over the field alphabet gives the exact joint cluster emission at cost $O(\text{edges} \cdot L^2 + \text{edges} \cdot L \cdot m \cdot k^2)$, linear in cluster size m . For phylogenies of depth > 1 , the per-node conditional-likelihood representation under exact marginalisation expands non-linearly with tree depth; a structured mean-field ELBO ([subsubsection D.8.6](#)) restores tractable linear-in- m cost on general trees and is tight at cap-2 and at the $\rho_{\text{chain}} \rightarrow \{0, \infty\}$ rate limits.

The θ -CTMC is reversible by construction; residues are conditionally independent given θ , so masking is trivial; and sites read the field but never a partner directly, so the construction is *table-independent* in the sense of (I) in [Theorem D.2](#). The expressiveness knob is L (the field truncation cap) and the breadth of $f(\cdot \mid \theta)$ (the per-field residue stationary spread).

D.8.3 F81-on-DP: a Dirichlet-process field selector

Let the field $\theta(t)$ evolve by F81, $Q(\theta \rightarrow \theta') = \rho_{\text{chain}} \pi_{\theta'}$ for $\theta \neq \theta'$, with π the stick-breaking weights of a Dirichlet process and ρ_{chain} a scalar rate multiplier. This is reversible w.r.t. π conditional on the DP draw; F81’s defining feature—jump to the destination type independently of the source—is exactly parent-independent mutation (PIM), whose stationary allelic partition in the infinite-alleles model is the Ewens sampling formula equivalent to the DP / CRP ([20, 21, 37](#)). It is the continuous-time cousin of the infinite HMM and HDP-HMM families ([4, 59](#)). Tractability is exact: a finite tree occupies finitely many atoms, and F81 is *strongly lumpable* onto $\{\text{occupied atoms}\} \cup \{\text{one tail block}\}$, so the occupied-atoms-plus-tail field Felsenstein is an exact collapse of the infinite-state recursion.

Per-state rate: the reversible form. A naive per-state exit-rate multiplier $Q(\theta \rightarrow \theta') = \eta_\theta \pi_{\theta'}$ is *not* reversible (detailed-balance residual $\pi_\theta \pi_{\theta'} (\eta_\theta - \eta_{\theta'})$; the chain stays π -stationary but is irreversible). The reversible way to vary rate by field-type is a rank-1 symmetric exchangeability $S_{\theta\theta'} = a_\theta a_{\theta'}$, $Q(\theta \rightarrow \theta') = a_\theta a_{\theta'} \pi_{\theta'}$: the exit rate is $\propto a_\theta$ (a genuine per-state multiplier) but jumps land in the a -tilted π ($\propto a_{\theta'} \pi_{\theta'}$). The clean dichotomy: constant rate $\Leftrightarrow$ bare- π jump $\Leftrightarrow$ exact Ewens marginal; per-state rate $\Leftrightarrow$ tilted jump $\Leftrightarrow$ non-Ewens marginal.

Carry vs. resample under a modulator jump. When any modulator θ jumps and resets the modulated layer’s generator, the modulated state x is either carried (kept) or resampled. Detailed balance across the jump leaves the residual factor $\pi(x | \theta)$ vs. $\pi(x | \theta')$: carrying is reversible iff the modulator preserves the modulated stationary (changes only *rate*, not *distribution*—the Tuffley–Steel covarion regime, 64). If the modulator changes the equilibrium (the expressive case here), carrying breaks detailed balance and one must resample x from $\pi(\cdot | \theta')$. The dynamic-field variant therefore mandates resampling at each field jump.

D.8.4 Dynamic site classes: a nested hierarchy

The same construction layers. A slower per-site reversible DP-CTMC (an *evolving profile*) sits above the field selector: class $\rightarrow$ field $\rightarrow$ residue, each an F81-on-DP modulating the next—a multi-level covarion / nested-iHMM. By induction the composition is reversible, table-independent, and de-Finetti-safe under indels; tractability nests, with the per-node CLV a tensor over (occupied classes $\times$ occupied fields). The natural timescale ordering residue $\gg$ field $\gg$ class (point substitution fastest, coevolutionary regime shifts slower, site identity slowest) aids identifiability but is not required for reversibility.

The two dynamic DP layers are independent options: a dynamic *field* selector alone gives an evolving coevolutionary coupling over background-composed sites; a dynamic *site-class* selector alone gives evolving per-site composition with no coupling; the general dynamic latent-field variant carries both.

D.8.5 Archetype vocabulary for per-(class, field) stationaries

We parameterise the per-(class, field) residue stationary $\pi^{(c,\theta)}$ through a shared vocabulary of simplex points $\{\pi^{(k)}\}_{k=1}^{K_a}$ (“archetypes”): each (c, θ) is mapped to one archetype by a discrete assignment,

$$\pi^{(c,\theta)} = \pi^{(\text{arch}(c,\theta))}, \quad \text{arch} : \{1, \dots, K_c\} \times \{1, \dots, L_{\max}\} \rightarrow \{1, \dots, K_a\}. \quad (\text{D.23})$$

This shared-vocabulary constraint is what makes the field selector θ act as a *coupling mediator*. If instead each $\pi^{(c,\theta)}$ were an unconstrained free simplex point, the model could pick each independently and a field flip would often produce a near-identical emission — no salt-bridge-like covariation would emerge. A site class is thus a *pattern* of archetype selections across θ . Some classes ($\text{arch}(c, \cdot) \equiv k$) are specializers: the field flip is a no-op. Others (e.g. (k_+, k_-, k_+, k_-) with k_+ positive-charge and k_- negative-charge archetypes) are mediators: a field flip is a discrete archetype swap that swings the emission across biophysically distinct simplex points. Because each archetype retains a full 20-simplex mass, standard GTR substitution physics within an archetype is untouched — no expressiveness is sacrificed. Priors: $\pi^{(k)} \sim \text{Dirichlet}(\alpha_{\text{prior}} \mathbf{1})$; $\text{arch}(c, \theta) \sim \text{Cat}(\rho_a)$ with ρ_a a truncated stick-breaking draw with concentration α_{arch} .

Compound CTMC state and dynamics. Consider a cluster of N sites of classes $(c_1, \dots, c_N)$ sharing a single field selector θ . The compound state $(\theta, x_1, \dots, x_N)$ evolves according to the following generator:

$$Q_\theta[(\theta, \mathbf{x}), (\theta', \mathbf{x}')] = \rho_{\text{chain}} \rho[\theta'] \prod_{n=1}^N \pi^{(k(c_n, \theta'))}[x'_n], \quad \theta' \neq \theta, \quad (\text{D.24})$$

$$Q^{(n)}[(\theta, \mathbf{x}), (\theta, \mathbf{x} \oplus_n y)] = Q[x_n, y] \pi^{(k(c_n, \theta))}[y], \quad y \neq x_n, \quad (\text{D.25})$$

where $\mathbf{x} \oplus_n y$ denotes the residue tuple $\mathbf{x}$ with position n set to y and other positions unchanged. Equation (D.24) is the field-jump generator: when θ jumps to θ' , *every* residue is simultaneously resampled from the new archetype stationary $\pi^{(k(c_n, \theta'))}$. This joint resampling is what preserves the joint stationary $\rho[\theta] \prod_n \pi^{(k(c_n, \theta))}[x_n]$ and enforces detailed balance of the compound chain (cf. 64 on carrying vs. resampling under covarion dynamics). Equation (D.25) is the standard GTR residue substitution at fixed field: rate $Q[x_n, y] \pi^{(k(c_n, \theta))}[y]$ with the LG08 exchangeabilities Q carried unchanged from Section D.1.

Reversibility. Both marginals are reversible: ρ is stationary for the F81-on-DP field chain (Section D.8); $\pi^{(k)}$ is stationary for GTR with Q symmetric. The joint $\rho[\theta] \prod_n \pi^{(k(c_n, \theta))}[x_n]$ satisfies detailed balance for the compound generator by direct verification: the field-jump rate $\rho_{\text{chain}} \rho[\theta']$ times the pre-jump weight $\rho[\theta] \prod_n \pi^{(k(c_n, \theta))}[x_n]$ equals the reverse rate $\rho_{\text{chain}} \rho[\theta]$ times the post-jump weight $\rho[\theta'] \prod_n \pi^{(k(c_n, \theta'))}[x'_n]$ only when each per-site resampled distribution $\pi^{(k(c_n, \theta'))}[x'_n]$ matches the post-jump stationary — which is exactly the resampling rule (D.24). The residue-only generator (D.25) is reversible w.r.t. $\pi^{(k(c_n, \theta))}$ site-by-site, so is reversible w.r.t. the compound stationary as well.

Rooting via the pulley principle. The compound chain is reversible, so Felsenstein’s pulley principle lets us re-root a cherry anywhere without changing the likelihood. Rooting at leaf X replaces the cherry $((X, X_1, \dots, X_N), (\text{LCA}), (Y, Y_1, \dots, Y_N))$ with a single branch of length t from the observed X -endpoint to the observed Y -endpoint. The θ_X at the root is a latent draw from ρ ; site residues at the root are the observed X_n . This eliminates the LCA state marginalisation used in the cherry-doublet form of Section D.8 (the four-case decomposition indexed by no-jump vs. ≥ 1 -jump on each half-edge) and reduces it to a two-case (no-jump vs. ≥ 1 -jump) decomposition on the whole branch, refined below to the three-case $\{0, 1, \geq 2\}$ decomposition needed to isolate stationary middle segments.

Archetype orbits: a permutation-structured field selector. The free assignment $\text{arch}(c, \theta)$ of (D.23) is more expressive than the covariation it is meant to capture: an unstructured $(c, \theta) \rightarrow k$ table lets a class drift to an arbitrary archetype at each field value, so a “flip” need not be a *charge-complementary swap*, and empirically the learned table shows no acid $\leftrightarrow$ base structure above a label-permutation null. We therefore restrict the field selector to act by *permutation of archetypes within orbits*, which makes “flip = swap” hold by construction and replaces the $K_c \times L_{\text{max}}$ assignment table by a single partition of the archetype vocabulary. Take the frozen diagonal base $K_c = K_a = K$, $\text{arch}_0(c) = c$ (class $\equiv$ archetype), and draw a partition $\mathcal{B} = \{B_1, \dots, B_m\}$ of $\{1, \dots, K\}$ into *archetype orbits* from a Dirichlet process with concentration α_{orb} ; large α_{orb} keeps $\mathcal{B}$ near all-singletons, with occasional small blocks.

Permutation field. The field alphabet is the product of symmetric groups on the orbits,

$$\Theta = \prod_{b=1}^m \text{Sym}(B_b), \quad |\Theta| = \prod_{b=1}^m |B_b|!, \quad (\text{D.26})$$

so a field state is a tuple $\theta = (\sigma_1, \dots, \sigma_m)$ of per-orbit permutations, and the archetype seen by class c is its base archetype permuted within its own orbit,

$$k(c, \theta) = \sigma_{b(c)}(c), \quad b(c) = \{b : c \in B_b\}. \quad (\text{D.27})$$

This is the constrained form of (D.23): as θ ranges over Θ , class c visits only the archetypes of its orbit, as a group action shared across classes. A singleton orbit contributes the trivial group ($|B_b|! = 1$): its classes are *specialisers* with no reachable flip. A size-2 orbit $B_b = \{c, c'\}$ contributes $\text{Sym} = \{e, (cc')\}$, whose two states leave the pair fixed or transpose it.

Field dynamics. Each orbit B_b carries its *own* F81-on-DP field chain (D.24) on $\text{Sym}(B_b)$, evolving *independently* (asynchronously) of the other orbits: at rate $\rho_{\text{chain}} r$ its permutation resamples from $\text{Unif}(\text{Sym}(B_b))$, on its own jump times. The *only* coupling is therefore between columns that *share* an orbit — a same-orbit pair is a 2-cycle ($\{AB, BA\}$, or $\{AA, BB\}$ if both columns sit at the same pole), the compensatory flip. Columns in *different* orbits are conditionally independent: the $\{AC, AD, BC, BD\}$ configuration collapses to two unsynchronised singletons (one oscillating $A \leftrightarrow B$, the other $C \leftrightarrow D$). This keeps a cluster’s field *local to its orbits* — independent of how the rest of the archetype vocabulary is partitioned. The single-orbit F81 chain and its reversibility are exactly (D.24)–(D.25); a per-cluster rate r with an invariant bin $r = 0$ gives the $+\Gamma + \text{I}$ field-rate (singleton-orbit or invariant-bin cluster = static).

Per-cluster locality and the state-space cap. The cluster likelihood *factorises* over the distinct orbits its columns touch:

$$\log p(\text{data}_C) = \sum_{b \in \text{touched}(C)} \log p(\text{cols}_b \mid \sigma_b),$$

each term a Felsenstein forward over $\text{Sym}(B_b)$ alone. So every forward runs on $L = |B_b|! \leq s_{\text{max}}! = 2$ states — the *maximum orbit size*, never a product — and neither $|\Theta_C|$ nor the global $|\Theta| = \prod_b |B_b|!$ enters. For cap-2 clusters a forward is a coupled pair (two columns sharing an orbit, $L \leq 2$) or a singleton/covarian single ($L \leq 2$); a different-orbit pair is two independent single forwards. We cap s_{max} (e.g. $s_{\text{max}} = 2$, transpositions only).

Compensatory flip and its posterior. A cap-2 cluster $C = \{i, j\}$ exhibits a compensatory flip iff its columns’ classes share a pair-orbit, $c_i, c_j \in B_b$ with $|B_b| = 2$: the transposition $\sigma_b = (c_i c_j)$ swaps their archetypes, $\pi^{(k(c_i, \theta))} \leftrightarrow \pi^{(k(c_j, \theta))}$ — a charge-complementary swap when B_b pairs a positive- and a negative-charge archetype. If the two classes are singletons or lie in distinct orbits, no field state couples them and C is static. The flip posterior is then a structural object,

$$\varphi(C) = \Pr[\sigma_b \text{ non-identity on some tree edge} \mid \text{data}], \quad (\text{D.28})$$

identically zero for singleton-orbit clusters rather than an emergent near-zero — removing the “flip vs. chance” ambiguity of the free table.

Inference: split–merge on orbits. The learned field structure is now the partition $\mathcal{B}$ of the K_a archetypes, sampled by Jain–Neal split–merge moves — as in [section D.8.5](#), but on archetype *orbits* rather than vocabulary slots: a merge places two archetypes in a common orbit (creating a candidate flip), a split separates them. This replaces the $O(K_c L_{\max})$ resampling of the free arch(c, θ) table — step (3) of the SVI loop below — with moves on a partition of K_a elements; the site-class Gibbs (c_s) and the $+\Gamma + I$ rate updates are unchanged. Acceptance uses the same class-marginal / field-rate-marginal cluster evidence as the rest of the loop, summed over the small Θ_C .

Remark D.15 (Exchangeability reading). Acting on the archetype vocabulary by the subgroup of S_{K_a} that preserves $\mathcal{B}$ is an exchangeability symmetry: the model is invariant to relabelling archetypes *within* an orbit, and the field selects the coset. This is the finite-group analogue of the de-Finetti coupling of the companion paper, giving the field selector a representation-theoretic rather than tabular description.

Inference: soft-EM archetype update with sampled discrete latents. The archetype variant is the sole learned inference target in the dynamic-latent-field SVI loop ([Section D.12](#)). Per outer iter:

1. CRP-Gibbs on cluster partitions π (unchanged).
2. Class-Gibbs on per-column labels c_s (unchanged).
3. Sample $\text{arch}(c, \theta) \sim \text{Cat}(\rho_a[k] \cdot \prod_a \pi^{(k)}[a]^{N[c, \theta, a]})$ from the archetype posterior given the current per- (c, θ) substitution counts $N[c, \theta, \cdot]$.
4. Accumulate Holmes–Rubin sufficient statistics ([Section D.8.5](#)) by soft attribution over the field trajectory, given the currently-sampled discrete latents.
5. M-steps: closed-form Gamma-posterior update on ρ_{chain} ; GTR fixed-point update on $\pi^{(k)}$; TSB Beta-posterior update on ρ_a .

The pattern matches the recommendation of Beal et al. ([5](#), [Section 4](#)) for hybrid Gibbs / variational schemes with discrete-continuous latent decompositions.

Field-chain spectrum: rank-1 collapse. The F81-on-DP field-chain generator $Q_\theta = \rho_{\text{chain}}(\mathbf{1}\rho^\top - I)$ has two distinct eigenvalues: $\lambda_0 = 0$ (right eigenvector $\mathbf{1}$, left eigenvector ρ) and $\lambda_1 = -\rho_{\text{chain}}$ (multiplicity $L_{\max} - 1$, eigenspace $\{v : \rho^\top v = 0\}$). Spectral projectors: $P_0 = \mathbf{1}\rho^\top$ (rank 1, element-wise $P_0[i, j] = \rho[j]$) and $P_1 = I - P_0$. This gives the closed-form transition probability

$$P_\theta(i \rightarrow j; \tau) = \rho[j] + g_\tau (\delta_{ij} - \rho[j]) = g_\tau \delta_{ij} + (1 - g_\tau) \rho[j], \quad (\text{D.29})$$

with $g_\tau := e^{-\rho_{\text{chain}}\tau}$. The state-dependent “no jump on interval τ ” survival probability is a distinct quantity:

$$\beta_\tau(i) := e^{-\rho_{\text{chain}}(1-\rho[i])\tau} = g_\tau^{1-\rho[i]}, \quad (\text{D.30})$$

smaller than $P_\theta(i \rightarrow i; \tau)$ because the latter also credits trajectories that jump away and return.

Number-of-jumps decomposition. With state-dependent exit rate $r'(\theta) = \rho_{\text{chain}}(1 - \rho[\theta])$ for the exit from field state θ , decompose the branch $[0, t]$ by the number of real field jumps $M \in \{0, 1, 2+\}$:

$$P(M = 0 \mid \theta_X, t) = \beta_t(\theta_X), \quad (\text{D.31})$$

$$P(M = 1, \theta_1 = j, \tau_1 \mid \theta_X, t) d\tau_1 = \rho_{\text{chain}} \rho[j] \beta_{\tau_1}(\theta_X) \beta_{t-\tau_1}(j) d\tau_1, \quad j \neq \theta_X, \quad (\text{D.32})$$

$$P(M \geq 2 \mid \theta_X, t) = 1 - P(M = 0 \mid \theta_X, t) - P(M = 1 \mid \theta_X, t). \quad (\text{D.33})$$

Marginalising (D.32) over θ_1 and τ_1 yields $P(M = 1 \mid \theta_X, t)$ as a sum over $j \neq \theta_X$ of

$$\rho[j] \cdot \int_0^t \rho_{\text{chain}} \beta_{\tau_1}(\theta_X) \beta_{t-\tau_1}(j) d\tau_1 = \rho[j] \cdot \begin{cases} \frac{\beta_t(j) - \beta_t(\theta_X)}{\rho[j] - \rho[\theta_X]}, & \rho[j] \neq \rho[\theta_X], \\ \rho_{\text{chain}} t \beta_t(\theta_X), & \rho[j] = \rho[\theta_X]. \end{cases} \quad (\text{D.34})$$

The $\rho[j] = \rho[\theta_X]$ degenerate case is the L'Hôpital limit of the generic form and appears when two field states have identical stick-breaking weight.

Sufficient statistics per site. For each (cluster, cherry q of length t_q , site n) we accumulate the following expectations, weighted by the posterior over (θ_X, θ_Y, M) implied by the site-level cluster likelihood factorisation:

$$V[k, a] := \text{expected \# of arch-}k \text{ segments starting at residue } a, \quad (\text{D.35})$$

$$U[k, i, j] := \text{expected \# of } i \rightarrow j \text{ substitutions within arch-}k \text{ intervals}, \quad (\text{D.36})$$

$$W[k, i] := \text{expected dwell time in state } i \text{ within arch-}k \text{ intervals}, \quad (\text{D.37})$$

$$N_\theta[q] := \text{expected \# of field jumps on cherry } q, \quad (\text{D.38})$$

$$T_q := t_q \text{ (deterministic branch length)}. \quad (\text{D.39})$$

These aggregate over the corpus by direct summation. Per-case contributions:

Case $M = 0$: $k_X = \text{arch}(c_n, \theta_X)$.

$$V[k_X, X_n] += P(M = 0 \mid \theta_X) \cdot \rho[\theta_X], \quad (\text{D.40})$$

$$W[k_X, i] += P(M = 0 \mid \theta_X) \cdot \rho[\theta_X] \cdot W_{k_X}^{\text{br}}(X_n \rightarrow Y_n; i; t), \quad (\text{D.41})$$

$$U[k_X, i, j] += P(M = 0 \mid \theta_X) \cdot \rho[\theta_X] \cdot U_{k_X, ij}^{\text{br}}(X_n \rightarrow Y_n; t), \quad (\text{D.42})$$

where $W_k^{\text{br}}(X \rightarrow Y; i; t)$ is the standard GTR bridge expected dwell in state i given endpoints X, Y and branch length t under stationary $\pi^{(k)}$, computed by the standard HR spectral formula (31). The symmetrisation $D_k^{1/2} Q_k D_k^{-1/2}$ (with $D_k = \text{diag}(\pi^{(k)})$; write $d_i^k := \sqrt{\pi^{(k)}[i]}$ and let ξ_α^k, U^k be the eigenvalues/eigenvectors of the symmetrised generator) yields a real-symmetric matrix whose eigendecomposition serves both the expected-dwell integral

$$W_i^{\text{br}}(X \rightarrow Y; t) = \frac{d_Y^k}{d_X^k P_{XY}^k(t)} \sum_{\alpha, \beta} J^{\alpha\beta}(t) U_{X\alpha}^k U_{i\alpha}^k U_{i\beta}^k U_{Y\beta}^k$$

and the expected-transition integral

$$U_{ij}^{\text{br}}(X \rightarrow Y; t) = \frac{Q[i, j] d_i^k d_j^k d_Y^k}{d_X^k P_{XY}^k(t)} \sum_{\alpha, \beta} J^{\alpha\beta}(t) U_{X\alpha}^k U_{i\alpha}^k U_{j\beta}^k U_{Y\beta}^k, \quad i \neq j, \quad (\text{D.43})$$

by the well-known $J^{\alpha\beta}$ formula (with $J^{\alpha\beta}(t) = (e^{\xi_\alpha^k t} - e^{\xi_\beta^k t})/(\xi_\alpha^k - \xi_\beta^k)$ for distinct eigenvalues and $t e^{\xi_\alpha^k t}$ in the degenerate limit). Note that $U_{ij}^{\text{br}} \neq Q_{ij}^k W_i^{\text{br}}$: the identity $E[N_{ij}] = Q_{ij} E[W_i]$ holds for the *unconditional* CTMC (Dynkin), but the Doob h -transform for the bridge makes the generator time-inhomogeneous, so the sum-and-integrate in (D.43) must be carried out with distinct α (before the jump) and β (after the jump) eigenvalue indices rather than collapsing on the site index.

Case $M \geq 1$: the trajectory decomposes into a first segment $[0, \tau_1]$ under $k_X = \text{arch}(c_n, \theta_X)$, a (possibly empty) sequence of middle segments under $\text{arch}(c_n, \theta_m)$ for intermediate field states θ_m , and a last segment $[\tau_M, t]$ under $k_Y = \text{arch}(c_n, \theta_Y)$. The three segment kinds have qualitatively different HR structure and are handled separately below. Contributions integrate over τ_1, θ_1 (and, for $M \geq 2$, over the intermediate segment counts) using (D.32) for the $M=1$ slice and the residual construction described below for the $M \geq 2$ slice; the $M=1$ and $M \geq 2$ contributions are combined into a unified $M \geq 1$ attribution because the first/last-segment formulas take the same form (with $M \geq 2$ contributing additional middle-segment terms only).

First segment (start X_n observed, end resampled from $\pi^{(k_1)}$ at τ_1^+). Because the residue at τ_1^+ is resampled independently under Interp 1, the first segment is an *unconditional* CTMC of duration τ_1 starting at X_n — its end residue is not conditioned. Dynkin's identity therefore holds: $E[N_{ij}] = Q_{ij}^{k_X} E[T_i]$, and specifically the Doob h -transform correction used in the $M=0$ bridge formula (D.43) is *not* needed.

Exact $M \geq 1$ density for τ_1 . We compute the marginal density of τ_1 conditional on $M \geq 1, \theta_X, \theta_Y$ in closed form: $p(\tau_1) \propto r'_X e^{-r'_X \tau_1} \sum_{\theta_1 \neq \theta_X} [\rho(\theta_1)/(1 - \rho(\theta_X))] P_\theta(\theta_1 \rightarrow \theta_Y; t - \tau_1)$. Using stationarity of ρ under the field CTMC ($\sum_{\theta_1} \rho(\theta_1) P_\theta(\theta_1 \rightarrow \theta_Y; s) = \rho(\theta_Y)$) and the F81-on-DP form $P_\theta(u \rightarrow v; s) = g_s \delta_{u,v} + (1 - g_s) \rho(v)$ with $g_s = e^{-\rho_{\text{chain}} s}$, the density collapses to a *two-exponential mixture*:

$$p(\tau_1 \mid M \geq 1) \propto \beta_1 e^{\delta_1 \tau_1} + \beta_2 e^{\delta_2 \tau_1} \quad (\text{D.44})$$

with

$$\delta_1 = -r'_X, \quad \delta_2 = \rho_{\text{chain}} \rho(\theta_X), \quad (\text{D.45})$$

$$\beta_1 = r'_X \rho(\theta_Y), \quad \beta_2 = r'_X \frac{\rho(\theta_X)}{1 - \rho(\theta_X)} [\rho(\theta_Y) - \delta_{\theta_X, \theta_Y}] g_t. \quad (\text{D.46})$$

(For $\theta_X \neq \theta_Y$ both coefficients are positive; for the $\theta_X = \theta_Y$ self-return case $\beta_2 < 0$, reflecting that $M = 1$ is impossible and the density is a difference of two exponentials.) The last-segment density on $t - \tau_M$ is the same mixture with θ_X and θ_Y exchanged (time reversal of the field bridge).

Elementary truncated-exp integrals on $[0, t]$. For any $\delta \in \mathbb{R}$ and $t > 0$, define the truncated-exp

normalizer, mean, dwell-moment, and transition-average primitives:

$$Z(\delta, t) := \int_0^t e^{\delta s} ds = \begin{cases} \frac{e^{\delta t} - 1}{\delta}, & \delta \neq 0, \\ t, & \delta = 0, \end{cases} \quad (\text{D.47})$$

$$\bar{\tau}(\delta, t) := \frac{1}{Z(\delta, t)} \int_0^t \tau e^{\delta \tau} d\tau = \begin{cases} \frac{t e^{\delta t}}{e^{\delta t} - 1} - \frac{1}{\delta}, & \delta \neq 0, \\ t/2, & \delta = 0, \end{cases} \quad (\text{D.48})$$

$$\begin{aligned} H(\xi; \delta, t) &:= \frac{1}{Z(\delta, t)} \int_0^t \frac{e^{\xi \tau} - 1}{\xi} e^{\delta \tau} d\tau \\ &= \begin{cases} \frac{Z(\xi + \delta, t) - Z(\delta, t)}{\xi Z(\delta, t)}, & \xi \neq 0, \xi + \delta \neq 0, \\ \frac{1}{\delta} - \frac{t}{e^{\delta t} - 1}, & \xi + \delta = 0, \delta \neq 0, \\ \bar{\tau}(\delta, t), & \xi = 0, \end{cases} \end{aligned} \quad (\text{D.49})$$

$$T(\xi; \delta, t) := \frac{1}{Z(\delta, t)} \int_0^t e^{\xi \tau} e^{\delta \tau} d\tau = \frac{Z(\xi + \delta, t)}{Z(\delta, t)}, \quad T(-\delta; \delta, t) = \frac{t}{Z(\delta, t)}. \quad (\text{D.50})$$

H and T are the two spectral-index primitives appearing in the free-end dwell and last-jump target-distribution formulas below.

Combination under the two-exponential mixture. For $p(\tau) \propto \beta_1 e^{\delta_1 \tau} + \beta_2 e^{\delta_2 \tau}$ on $[0, t]$, the mixture normalizer, weights, and expectation of any linear functional f of τ are

$$Z_{\text{mix}} = \beta_1 Z(\delta_1, t) + \beta_2 Z(\delta_2, t), \quad a_i = \frac{\beta_i Z(\delta_i, t)}{Z_{\text{mix}}}, \quad a_1 + a_2 = 1, \quad (\text{D.51})$$

$$E_{\text{mix}}[f(\tau)] = a_1 E_{p_{\delta_1}}[f(\tau)] + a_2 E_{p_{\delta_2}}[f(\tau)]. \quad (\text{D.52})$$

This linearity lets us evaluate every SS attribution by calling each single- δ primitive twice — once at δ_1 , once at δ_2 — and taking the a_1, a_2 -weighted sum. No mixture-specific closed forms are needed. The construction is verified against the joint (field $\times$ residue) matrix-exponential sufficient statistics to relative error $\lesssim 10^{-14}$ across the full parameter grid.

First-segment W, U in terms of H . The free-end dwell (start at X_n, τ_1 drawn from case-1 density with rate δ) is

$$W_i^{\text{free}}(X_n; t, \delta) = \frac{d_i^{k_X}}{d_{X_n}^{k_X}} \sum_{\alpha} U_{X_n \alpha}^{k_X} U_{i \alpha}^{k_X} H(\xi_{\alpha}^{k_X}; \delta, t), \quad (\text{D.53})$$

and the free-end U (Dynkin identity holds because the endpoint is resampled independently under Interp 1) is

$$U_{ij}^{\text{free}}(X_n; t, \delta) = Q_{ij}^{k_X} W_i^{\text{free}}(X_n; t, \delta), \quad i \neq j. \quad (\text{D.54})$$

Under the exact $M \geq 1$ density (Eq. D.44), the first-segment contributions to $\{W, U\}$ at $(c_n, \theta_X, \cdot)$ are the mixture-averaged versions:

$$W_i^{\text{first}} = a_1^{\text{first}} W_i^{\text{free}}(X_n; t, \delta_1^{\text{first}}) + a_2^{\text{first}} W_i^{\text{free}}(X_n; t, \delta_2^{\text{first}}), \quad (\text{D.55})$$

$$\begin{aligned} U_{ij}^{\text{first}} &= a_1^{\text{first}} U_{ij}^{\text{free}}(X_n; t, \delta_1^{\text{first}}) + a_2^{\text{first}} U_{ij}^{\text{free}}(X_n; t, \delta_2^{\text{first}}) \\ &= Q_{ij}^{k_X} W_i^{\text{first}}, \end{aligned} \quad (\text{D.56})$$

with $(\delta_i^{\text{first}}, \beta_i^{\text{first}})$ from Eq. D.44 and a_i^{first} from Eq. D.51.

Last-jump target-distribution via T. The V attribution at $(c_n, \theta_Y, \cdot)$ receives the expected residue distribution at the moment of the last field jump. Under Interp 1 with a resample from $\pi^{(k_Y)}$, the observed Y_n at time t gives (by reversibility) a bridge-conditional expected resample distribution which by time-reversal equals $E_s[P^{k_Y}(Y_n \rightarrow a; s)]$ averaged over $s = t - \tau_M$:

$$P_a^{\text{res},k}(X; t, \delta) := E_s[P^k(X \rightarrow a; s)] = \frac{d_a^k}{d_X^k} \sum_{\alpha} U_{X\alpha}^k U_{a\alpha}^k T(\xi_{\alpha}^k; \delta, t). \quad (\text{D.57})$$

Under the exact $M \geq 1$ last-segment density (mixture with θ_X, θ_Y exchanged in Eq. D.44), the V attribution is

$$V_a^{\text{last}} = a_1^{\text{last}} P_a^{\text{res},k_Y}(Y_n; t, \delta_1^{\text{last}}) + a_2^{\text{last}} P_a^{\text{res},k_Y}(Y_n; t, \delta_2^{\text{last}}), \quad (\text{D.58})$$

with $\delta_1^{\text{last}} = -r_Y'$, $\delta_2^{\text{last}} = \rho_{\text{chain}}\rho(\theta_Y)$, $\beta_1^{\text{last}} = r_Y'\rho(\theta_X)$, $\beta_2^{\text{last}} = r_Y'(\rho(\theta_Y)/(1 - \rho(\theta_Y)))(\rho(\theta_X) - \delta_{\theta_X, \theta_Y})gt$, and a_i^{last} from Eq. D.51.

Middle segments (both endpoints at stationarity), $M \geq 2$ only. When $M \geq 2$, each middle segment has both endpoints i.i.d. from $\pi^{(k)}$ (the entering jump resamples the residue from stationarity; the exiting jump does the same on the next segment). The segment is a stationary CTMC of expected duration τ_{mid} , and Dynkin on the stationary chain gives $E[T_i] = \tau_{\text{mid}}\pi^{(k)}[i]$ and $E[N_{ij}] = \tau_{\text{mid}}\pi^{(k)}[i] Q[i, j] \pi^{(k)}[j]$ (using $Q_{ij}^k = Q[i, j] \pi^{(k)}[j]$). The per-site middle-segment contributions to (V, W, U) are therefore

$$V[k, a] += E_{M \geq 1}[\# \text{ middle arrivals at arch } k] \pi^{(k)}[a], \quad (\text{D.59})$$

$$W[k, a] += E_{M \geq 1}[\text{middle time in arch } k] \pi^{(k)}[a], \quad (\text{D.60})$$

$$U[k, i, j] += E_{M \geq 1}[\text{middle time in arch } k] \pi^{(k)}[i] Q[i, j] \pi^{(k)}[j]. \quad (\text{D.61})$$

The middle-arrival count and middle-time expectations are the residuals of the whole-branch bridge-conditional quantities after subtracting the first- and last-segment contributions. Under Interp 1 the first segment sits entirely at θ_X and the last entirely at θ_Y ; the expected first- and last-segment durations under the exact $M \geq 1$ mixture densities are

$$\overline{\tau_1} = a_1^{\text{first}} \bar{\tau}(\delta_1^{\text{first}}, t) + a_2^{\text{first}} \bar{\tau}(\delta_2^{\text{first}}, t), \quad (\text{D.62})$$

$$\overline{t - \tau_M} = a_1^{\text{last}} \bar{\tau}(\delta_1^{\text{last}}, t) + a_2^{\text{last}} \bar{\tau}(\delta_2^{\text{last}}, t). \quad (\text{D.63})$$

(Both use Eq. D.48 at the corresponding δ and the mixture weights from Eq. D.51.) The middle-time at intermediate field state θ_{int} is then

$$\tau_{\text{mid}}(\theta_{\text{int}}) = \max(0, E[T_{\theta_{\text{int}}} | \theta_X, \theta_Y, t, M \geq 1] - \overline{\tau_1} \mathbb{1}[\theta_{\text{int}} = \theta_X] - \overline{t - \tau_M} \mathbb{1}[\theta_{\text{int}} = \theta_Y]), \quad (\text{D.64})$$

and the middle-arrival count (excluding the last-arrival at θ_Y which initiates the last segment, not a middle segment) is

$$N_{\text{mid}}(\theta_{\text{int}}) = \max(0, E[N_{\theta_{\text{int}}} | \theta_X, \theta_Y, t, M \geq 1] - \mathbb{1}[\theta_{\text{int}} = \theta_Y]). \quad (\text{D.65})$$

where $E[T_{\theta} | \cdot]$ and $E[N_{\theta} | \cdot]$ are the whole-branch bridge-conditional expectations from Eq. D.71 and Eq. D.77 below. The middle-segment SS attribution (D.59)–(D.61) then uses τ_{mid} for the “middle time in arch k ” factor and N_{mid} for the “middle arrivals at arch k ” factor. At $M = 1$ the residuals (D.64)–(D.65) vanish identically ($\overline{\tau_1} + \overline{t - \tau_M} = t$ when $M = 1$; the last-arrival subtraction cancels the single arrival at θ_Y), so a single $M \geq 1$ attribution correctly covers both the $M = 1$ and $M \geq 2$ slices.

Last segment (marginal start from $\pi^{(k_Y)}$, end Y_n observed). The last segment is a “reversed bridge”: the forward start z is drawn from $\pi^{(k_Y)}$ (the last jump’s resample) and the forward end is observed as Y_n . Under GTR reversibility $\pi^{(k_Y)}[z] P^{k_Y}(z \rightarrow Y_n; \tau) = \pi^{(k_Y)}[Y_n] P^{k_Y}(Y_n \rightarrow z; \tau)$, so integrating over $z \sim \pi^{(k_Y)}$ the conditional dwell in state i collapses to a free-end from Y_n in *reversed* time:

$$W_i^{\text{last}}(Y_n; \tau) = E_{z \sim \pi^{(k_Y)}} [W_i^{\text{br}}(z \rightarrow Y_n; \tau)] = \int_0^\tau P^{k_Y}(Y_n \rightarrow i; s) ds = W_i^{\text{free}}(Y_n; \tau). \quad (\text{D.66})$$

For expected transitions the same reversibility argument, applied to the Doob- h -transformed bridge integrand, gives

$$U_{ij}^{\text{last}}(Y_n; \tau) = \pi^{(k_Y)}[i] Q[i, j] W_j^{\text{free}}(Y_n; \tau), \quad i \neq j, \quad (\text{D.67})$$

which is *not* the Dynkin form $Q_{ij}^{k_Y} W_i^{\text{last}} = Q[i, j] \pi^{(k_Y)}[j] W_i^{\text{free}}(Y_n; \tau)$ (differences: the $i \leftrightarrow j$ swap in both the stationary factor and the free-end-dwell argument). This is the same phenomenon as in the $M=0$ bridge — endpoint conditioning breaks the unconditional-CTMC Dynkin identity — but here the closed form is simpler because one endpoint is stationary rather than observed, and one 4-index spectral summation collapses to a 2-index (single- α) one. Under the exact $M \geq 1$ last-segment density (Eq. D.44 with θ_X, θ_Y exchanged), the mixture-averaged last-segment attributions to (W, U) at $(c_n, \theta_Y, \cdot)$ are

$$W_j^{\text{last}} = a_1^{\text{last}} W_j^{\text{free}}(Y_n; t, \delta_1^{\text{last}}) + a_2^{\text{last}} W_j^{\text{free}}(Y_n; t, \delta_2^{\text{last}}), \quad (\text{D.68})$$

$$U_{ij}^{\text{last}} = \pi^{(k_Y)}[i] Q[i, j] W_j^{\text{last}}, \quad i \neq j. \quad (\text{D.69})$$

(Note the $i \leftrightarrow j$ swap between W^{last} indexing and Dynkin’s would-be $Q_{ij}^{k_Y} W_i^{\text{last}}$ formula, forced by the reversibility argument.) The V attribution at $(c_n, \theta_Y, \cdot)$ is V_a^{last} from Eq. D.58.

The $M=1$ attribution follows immediately from Eqs. D.53, D.54, D.66, D.67 (first + last only; no middle contribution). For $M \geq 2$, the first- and last-segment formulas are unchanged and the middle-segment attribution (D.59)–(D.61) accounts for the intermediate field time. Together with the whole-branch bridge $E[N_\theta | \theta_X, \theta_Y, t]$ (Eq. D.71) and the per-destination arrivals (Eq. D.77), these formulas express (V, U, W, N_θ) as closed-form functions of $\{\beta_t(i), g_t, t\}$ and the GTR bridge/free-end primitives, without any integral remaining to be evaluated numerically.

Whole-branch bridge $E[N_\theta | \theta_X, \theta_Y, t]$. The total field-jump expectation entering the middle-segment residual construction of the previous paragraph is the bridge-conditional

$$E[N_\theta | \theta_X, \theta_Y, t] \cdot P_\theta(\theta_X \rightarrow \theta_Y; t) = \sum_{i \neq j} Q_\theta[i, j] \int_0^t P_\theta(\theta_X \rightarrow i; s) P_\theta(j \rightarrow \theta_Y; t-s) ds, \quad (\text{D.70})$$

which is *not* equal to $\rho_{\text{chain}} \sum_i (1 - \rho[i]) E_T[i | \theta_X, \theta_Y, t]$. That shortcut multiplies the unconditional exit rate $-Q_\theta[i, i] = \rho_{\text{chain}}(1 - \rho[i])$ against the bridge-conditional dwell, and only recovers the correct expectation when the trajectory is at field-stationarity throughout (endpoints i.i.d. from ρ , as inside a case-(2+) middle segment). Under the rank-1 spectral form (D.29) substitution into (D.70) collapses to a closed form in $\{\rho, \rho_{\text{chain}}, t, g_t, \sigma\}$ with $\sigma := \sum_i \rho[i]^2$:

$$E[N_\theta | \theta_X, \theta_Y, t] \cdot P_\theta(\theta_X \rightarrow \theta_Y; t) = \Sigma_1(\theta_Y) + \Sigma_{23}(\theta_X, \theta_Y) + \Sigma_4(\theta_X, \theta_Y), \quad (\text{D.71})$$

with

$$\Sigma_1(b) = \rho_{\text{chain}} t \rho[b] (1 - \sigma), \quad (\text{D.72})$$

$$\Sigma_{23}(a, b) = (1 - g_t) \rho[b] (2\sigma - \rho[a] - \rho[b]), \quad (\text{D.73})$$

$$\Sigma_4(a, b) = -\rho_{\text{chain}} g_t t (\delta_{a,b} \rho[a] - \rho[a] \rho[b] - \rho[b]^2 + \rho[b] \sigma). \quad (\text{D.74})$$

Deriving this: rewrite $P_\theta(u \rightarrow v; \tau) = \rho[v] + g_\tau(\delta_{u,v} - \rho[v])$, integrate the four cross-products of the $\{\rho, g_\tau\}$ terms against $Q_\theta[i, j] = \rho_{\text{chain}}\rho[j]$ ($i \neq j$), and use $\int_0^t 1 ds = t$, $\int_0^t g_s ds = (1 - g_t)/\rho_{\text{chain}}$, $\int_0^t g_s g_{t-s} ds = t g_t$. This is verified against Gillespie simulation.

Per-destination arrival counts. The bridge-conditional expected count of field jumps that *arrive* at a specific destination θ_{int} is needed for the V attribution (each arrival triggers a residue resample from $\pi^{(\text{arch}(c_n, \theta_{\text{int}}))}$). Applying the same argument as (D.70) but marginalising over jump sources (not destinations):

$$E[N_{\theta_{\text{int}}} \mid \theta_X, \theta_Y, t] \cdot P_\theta(\theta_X \rightarrow \theta_Y; t) = \sum_{j \neq \theta_{\text{int}}} Q_\theta[j, \theta_{\text{int}}] \int_0^t P_\theta(\theta_X \rightarrow j; s) P_\theta(\theta_{\text{int}} \rightarrow \theta_Y; t-s) ds. \quad (\text{D.75})$$

With $Q_\theta[j, \theta_{\text{int}}] = \rho_{\text{chain}}\rho[\theta_{\text{int}}]$ for $j \neq \theta_{\text{int}}$, factor the destination-independent part and use $\sum_{j \neq i} P_\theta(a \rightarrow j; s) = 1 - P_\theta(a \rightarrow i; s)$ to get

$$E[N_i \mid a, b, t] \cdot P_\theta(a \rightarrow b; t) = \rho_{\text{chain}} \rho[i] \int_0^t (1 - P_\theta(a \rightarrow i; s)) P_\theta(i \rightarrow b; t-s) ds \quad (\text{D.76})$$

$$= \rho_{\text{chain}} \rho[i] \left[(1 - \rho[i]) \rho[b] t + ((1 - \rho[i]) d_{ib} - d_{ai} \rho[b]) I_1 - d_{ai} d_{ib} t g_t \right], \quad (\text{D.77})$$

with $I_1 := (1 - g_t)/\rho_{\text{chain}}$, $d_{ai} := \delta_{a,i} - \rho[i]$, and $d_{ib} := \delta_{i,b} - \rho[b]$. Summing (D.77) over i recovers (D.71) (an analytic identity, verified numerically).

Case- $M=1$ second-segment V attribution via time reversal. The case- $M=1$ second segment $[\tau_1, t]$ has resampled start $z \sim \pi^{(k_1)}$ at τ_1^+ and observed end Y_n in arch k_1 . Under residue-endpoint conditioning, z is biased away from $\pi^{(k_1)}$ by the Y-endpoint. GTR reversibility gives the bridge-conditional distribution $\pi^{(k_1)}[z] P^{k_1}(z \rightarrow Y_n; t-\tau_1)/\pi^{(k_1)}[Y_n] = P^{k_1}(Y_n \rightarrow z; t-\tau_1)$, so the second segment's forward-time start z is distributed exactly as the reversed segment's target from Y_n . Averaging over the case- $M=1$ posterior $\tau_1 \sim \exp(\delta\tau_1)/Z_1$ on $[0, t]$ (with $\delta := r'(\theta_1) - r'(\theta_X)$; the residue-endpoint factor cancels stationarily, cf. appendix D.8.5), substitute $\tau' := t - \tau_1$ and apply the GTR spectral form $P^k(X \rightarrow a; \tau') = (D_{1/2}^k[a]/D_{1/2}^k[X]) \sum_\alpha U^k[X, \alpha] U^k[a, \alpha] e^{\xi_\alpha^k \tau'}$:

$$E[P^{k_1}(Y_n \rightarrow a; t-\tau_1) \mid M=1, \theta_X, \theta_1] = \frac{D_{1/2}^{k_1}[a]}{D_{1/2}^{k_1}[Y_n]} \sum_\alpha U^{k_1}[Y_n, \alpha] U^{k_1}[a, \alpha] E_\tau[e^{\xi_\alpha^{k_1} \tau'}], \quad (\text{D.78})$$

with $E_\tau[e^{\xi \tau'}] = (e^{\delta t}/Z_1) \cdot (e^{(\xi-\delta)t} - 1)/(\xi - \delta)$ for $\xi \neq \delta$ (and the L'Hôpital limit $(e^{\delta t}/Z_1) \cdot t$ otherwise). The V second-segment contribution is then $V[k_1, a] += P(\text{case-}M=1, \theta_X, \theta_1 \mid X_{\text{obs}}, Y_{\text{obs}}, t) \cdot E[P^{k_1}(Y_n \rightarrow a; t-\tau_1)]$, and *not* $\pi^{(k_1)}[a]$ times the case-1 mass – that shortcut ignores the residue-endpoint bias captured by (D.78) and can be off by $P^{k_1}(Y_n \rightarrow a; \cdot)/\pi^{(k_1)}[a]$, which is $\gg 1$ at $a \approx Y_n$ and short remaining branch length. Verified against Gillespie simulation.

HR under soft leaves (PSWM observations). The default HR pass of Section D.8.5 conditions on hard leaf residues $(X_n, Y_n) \in \{1, \dots, A\}^2$ at every cherry site. In the PSWM training path each cherry endpoint carries instead a distribution $\hat{p}_n^A, \hat{p}_n^B \in \Delta^{A-1}$ derived from LG08 Felsenstein peeling at the parent / child node of a tree branch. The bridge posterior at that branch integrates against $\hat{p}_n^A(X_n) \hat{p}_n^B(Y_n)$:

$$P_{\theta_X \theta_Y}^{\text{soft}}(t) = \sum_{X_n, Y_n} \hat{p}_n^A(X_n) \hat{p}_n^B(Y_n) P_{X_n Y_n}^{k_n}(t),$$

and every HR statistic becomes the ratio $\mathcal{N}/P^{\text{soft}}$ where the numerator $\mathcal{N}$ is the same integrand bilinearly weighted by $\hat{p}_n^A, \hat{p}_n^B$. Under the F81-on-DP field the site observations are conditionally independent given (θ_X, θ_Y, M) , so the cluster-level ratio factorises over sites $n = 1, \dots, N$ by a leave-one-out product

$$\mathcal{N}_{\text{cluster}} = \sum_n \mathcal{N}_n \cdot \prod_{m \neq n} P_m^{\text{soft}},$$

and closed forms for every $\mathcal{N}_n$ follow directly from the spectral primitives of Section D.8.5.

Bridge primitive. Define the effective end-state vectors

$$v_\alpha^X := \sum_X \frac{\hat{p}_n^A(X) U_{X\alpha}^k}{d_X^k}, \quad v_\alpha^Y := \sum_Y \hat{p}_n^B(Y) d_Y^k U_{Y\alpha}^k, \quad (\text{D.79})$$

with $d_a^k := \sqrt{\pi^{(k)}[a]}$ as in (D.43). When $\hat{p}_n^A, \hat{p}_n^B$ are delta functions on X, Y these reduce to $v_\alpha^X = U_{X\alpha}^k/d_X^k$ and $v_\alpha^Y = d_Y^k U_{Y\alpha}^k$, matching the hard-observation combination that appears in every X, Y -conditional expectation. The soft bridge probability collapses to a single spectral sum,

$$P_n^{\text{soft}}(t) = \sum_\alpha e^{\xi_\alpha^k t} v_\alpha^X v_\alpha^Y, \quad (\text{D.80})$$

and the soft dwell / transition numerators inherit the same rank-1 structure that the hard forms have in $U_{X\alpha}^k, U_{Y\beta}^k$:

$$\mathcal{N}_n^{W,i} = \sum_{\alpha, \beta} J^{\alpha\beta}(t) v_\alpha^X v_\beta^Y U_{i\alpha}^k U_{i\beta}^k, \quad (\text{D.81})$$

$$\mathcal{N}_n^{U,ij} = Q_{ij} d_i^k d_j^k \sum_{\alpha, \beta} J^{\alpha\beta}(t) v_\alpha^X v_\beta^Y U_{i\alpha}^k U_{j\beta}^k, \quad i \neq j. \quad (\text{D.82})$$

The conditional HR statistics are $W_i^{\text{soft}} = \mathcal{N}_n^{W,i}/P_n^{\text{soft}}$ and $U_{ij}^{\text{soft}} = \mathcal{N}_n^{U,ij}/P_n^{\text{soft}}$; both reduce to the hard closed forms (D.43) when $\hat{p}_n^A, \hat{p}_n^B$ are delta. The per-site compute cost is $O(A^2)$ (one $A \times A$ spectral sum) – identical to the hard version.

Case- $M \geq 1$ free-end and second-segment primitives. The first-segment substitution factor $E_{\tau_1}[\text{HR at } X_n \mid \delta]$ from Section D.8.5 is linear in $U_{X_n, \cdot}^k$, so under a soft start distribution $\hat{p}_n^A$ we replace the single row $U_{X_n, \cdot}^k/d_{X_n}^k$ with the weighted average

$$\tilde{v}_\alpha^X := \sum_X \frac{\hat{p}_n^A(X) U_{X\alpha}^k}{d_X^k},$$

(same construction as v^X above, evaluated at the class-index k_X at θ_X). The free-end dwell and its U_{ij} attribution become

$$W_i^{\text{soft}}(\delta, t) = d_i^k \sum_\alpha U_{i\alpha}^k \tilde{v}_\alpha^X E_h(\xi_\alpha^k, \delta, t), \quad U_{ij}^{\text{soft}}(\delta, t) = Q_{\text{off}}^k[i, j] W_i^{\text{soft}}(\delta, t), \quad (\text{D.83})$$

with E_h the truncated-exponential moment already used by the hard case-1 primitive. The soft second-segment transition average (D.78) is likewise reweighted:

$$P_{\text{avg}}^{\text{soft}}(a; \delta, t) = d_a^k \sum_\alpha U_{a\alpha}^k \tilde{v}_\alpha^Y E_\tau[e^{\xi_\alpha^k \tau'}],$$

which is linear in $\hat{p}_n^B$ (so an unnormalised weight $\hat{p}_n^B \odot \pi^{(k_Y)}$, needed by the middle-segment attribution for the joint case-jump likelihood $L_{\text{res},j} = \prod_n \langle \hat{p}_n^B, \pi^{(k_Y(n))} \rangle$, plugs in unchanged).

Cluster-level aggregation. Fix (θ_X, θ_Y) . Under the factorisation over sites the leave-one-out identity gives

$$V_{\text{cluster}}^{\text{soft}} = \rho[\theta_X] P_{\theta_X \theta_Y}^{M=0}(t) \left(\prod_m P_m^{\text{soft}} \right) \sum_n \frac{V_n^{\text{soft}}}{P_n^{\text{soft}}}, \quad (\text{D.84})$$

and analogously for $W_{\text{cluster}}, U_{\text{cluster}}$ in case $M = 0$; case $M \geq 1$ uses $L_{\text{res},j}$ as the leave-one-out product with the same structural form. In the delta limit $\hat{p}_n^A = \eta_{X_n}, \hat{p}_n^B = \eta_{Y_n}$ every P_n^{soft} collapses to $P_{X_n Y_n}^{k_n}(t)$ and the aggregation matches the hard Section D.8.5 formulae exactly.

Cost. Building v^X, v^Y (and $\tilde{v}^X, \tilde{v}^Y$) takes $O(A^2)$ per site; the spectral sums are $O(A^2)$ per site as in the hard path. Total additional cost is a small constant factor over the hard HR pass – *not* the naive $O(A^4)$ that arises from enumerating $(X_n, Y_n) \in A^2$ before summing. When soft leaves are available at every branch of every family the closed forms (D.79)-(D.84) replace both the E-step (hard-conditional bridge posteriors on X_n, Y_n) and the M-step SS accumulation of Section D.8.5 with the marginalised soft-observation analogues; the pi-arch and ρ_{chain} M-steps below apply unchanged.

M-steps. The M-steps follow the standard GTR / F81-on-DP forms from Bruno (10), Holmes and Rubin (31).

Archetype stationaries $\pi^{(k)}$. With $V'[k, a] = V[k, a] + \sum_i U[k, i, a]$ (start-plus-entry count for state a) and $r_k[a] = \sum_{b \neq a} Q[a, b] W[k, b]$ (exchangeability-weighted opportunity), iterate

$$\pi^{(k)}[a] \leftarrow \frac{V'[k, a] + \alpha_{\text{prior}} - 1}{V_{\bullet}[k] + \sigma_k \cdot r_k[a]}, \quad \sigma_k \text{ chosen so that } \sum_a \pi^{(k)}[a] = 1, \quad (\text{D.85})$$

where $V_{\bullet}[k] = \sum_a V[k, a]$ and σ_k is the unique positive root of a monotone one-dimensional equation solved by a few Newton iterations. This is the standard GTR π update with Dirichlet(α_{prior}) pseudocounts on $\pi^{(k)}$. The MAP numerator $V'[k, a] + \alpha_{\text{prior}} - 1$ is negative for under-observed states when $\alpha_{\text{prior}} < 1$; we clamp both the numerator and the resulting $\pi^{(k)}[a]$ elementwise to 10^{-30} before renormalising, which preserves sparsity (unused residues land at $\sim 10^{-30}$ not $\sim 1/A$) while keeping $\sqrt{\pi^{(k)}[a]} \geq 10^{-15}$ well above the subnormal float64 threshold needed by the spectral decomposition (D.43). The default $\alpha_{\text{prior}} = 0.5$ is a sparse simplex-corner prior encouraging peaky archetype distributions; setting $\alpha_{\text{prior}} \geq 1$ recovers a proper MAP with Laplace-style smoothing.

Field-chain rate ρ_{chain} under Gamma(α_ρ, β_ρ) prior:

$$\rho_{\text{chain}} \leftarrow \frac{\alpha_\rho - 1 + \sum_q N_\theta[q]}{\beta_\rho + \sum_q T_q}. \quad (\text{D.86})$$

Closed form.

Archetype selector ρ_a : unchanged TSB Beta posterior mean update on the (soft) archetype assignment counts, mirroring the field selector ρ update of Section D.8.

Rate heterogeneity (+ Γ + I): per-cluster field-rate and per-site substitution rate. A single global ρ_{chain} forces every cluster to share one field-jump rate. Empirically most structural contacts are orientation-conserved (static) and only a minority are genuinely compensatory, so a global rate is dragged toward the static regime and no cluster can flip decisively. We give each cluster its own field-rate and each site its own substitution rate, each a discretised Gamma-plus-invariant mixture in the sense of Yang (70, 71).

Per-cluster field-rate. Cluster C draws a field-rate multiplier r_C : with probability p_I the *invariant* value $r_C = 0$, else one of G equiprobable $\text{Gamma}(a_f, a_f)$ quantile midpoints $\{\gamma_1, \dots, \gamma_G\}$ (mean 1). Its field chain uses $\rho_{\text{chain}} r_C$ in place of ρ_{chain} throughout (D.29)–(D.33), i.e. $g_\tau^{(C)} = e^{-\rho_{\text{chain}} r_C \tau}$. The invariant bin is the crux: $r_C = 0 \Rightarrow g_\tau^{(C)} = 1 \Rightarrow \beta_\tau \equiv 1 \Rightarrow P(M \geq 1) = 0$, so θ is constant on the whole tree and the cluster collapses to a single field draw $\theta \sim \rho$ with independent per-column emissions under $\pi^{(\text{arch}(c_n, \theta))}$ — a static class-mixture carrying *no* field flip. Marginalising r_C ,

$$L(C) = p_I L_0(C) + \frac{1-p_I}{G} \sum_{g=1}^G L(C \mid r_C = \gamma_g), \quad L_0(C) = \sum_{\theta} \rho[\theta] \prod_n L_n^{\text{stat}}(\theta), \quad (\text{D.87})$$

with L_0 the no-jump (static) evidence. The natural per-cluster flip statistic is the posterior responsibility of the non-invariant bins,

$$\varphi(C) := P(r_C \neq 0 \mid C) = \frac{\frac{1-p_I}{G} \sum_g L(C \mid \gamma_g)}{L(C)}, \quad (\text{D.88})$$

replacing hard flip-detection: a cluster flips to the extent its data prefer a nonzero field rate over the static bin, and the corpus flip rate is $\mathbb{E}_C[\varphi(C)]$. Conserved contacts have $\varphi \approx 0$; compensatory (salt-bridge-flip) contacts $\varphi \rightarrow 1$.

Per-site substitution rate. Each site n carries a multiplier m_n on the residue generator (D.25), $Q[x_n, y] \mapsto m_n Q[x_n, y]$ (with $m_n = 0$ the invariant bin, a frozen site), drawn from a discretised $\text{Gamma}(a_s, a_s) + I$ mixture. We fix $\{m_n\}$ *once at initialisation* by empirical Bayes — m_n set to its posterior-mean bin under the fixed LG08 per-site likelihoods (per-column conservation) — and hold it constant. This absorbs across-site rate variation into the substitution layer so it is not misattributed to field dynamics (a fast column is explained by large m_n , not by a spurious flip); being fixed, it is not a free latent during training. In the covarion dichotomy of Section D.8.3 the site-rate is a Tuffley–Steel *rate-only* modulator (distribution-preserving, residue carried), whereas the field-rate multiplies the distribution-changing field chain (residue resampled at each jump); the two heterogeneities act on the two reversible layers without interfering.

Evidence unit and cached primitives. The unit of the class marginalisation is the per-(cluster, class-labeling) evidence: K_c labelings for a singleton, K_c^2 for a doublet. It *cannot* be partitioned by archetype. A class is a *sequence* $\text{arch}(c, \cdot)$ of archetypes across the field states, and as the shared field θ evolves on the tree a site of class c is carried through whichever archetype $\text{arch}(c, \theta(t))$ is live, resampled at each jump (D.24); the substitution evidence therefore integrates this archetype-flip trajectory (coupled across the two columns of a doublet by the *shared* θ). That integration is exactly the field-HMM forward of D.8, and a class’s evidence is *not* the sum of its archetypes’ evidences. What does cache are the *substitution primitives* entering the number-of-jumps decomposition (D.31)–(D.34): the per-(site n , archetype k) no-jump branch evidence $\varepsilon_n^{(k)}$ and the per-(site n , ordered archetype pair $k \rightarrow k'$) single-jump branch evidence $\varepsilon_n^{(k, k')}$ (endpoint transitions under Q , π , m_n with a stationary middle segment). With frozen archetypes and fixed site rates these primitives are *invariant* under every discrete update; only the K_a archetype rate-matrix eigendecompositions are ever recomputed (never, here). The per-(cluster, class-labeling) forward assembles them via the field transition (D.29), at cost $O(L_{\text{max}}^2 \cdot \text{edges})$ per labeling — reusing the cached ε , not re-peeling. Hence: (i) an arch move on (c, θ) leaves the primitives untouched, re-indexes which primitive each field state selects, and reassembles only the labelings whose class set contains c (1 of K_c per singleton — the class- c labeling; $2K_c - 1$ of K_c^2 per doublet — the (c, d) and (d, c) labelings for all d); every other labeling’s cached forward is reused, so one arch move triggers only $N_{\text{sing}} + (2K_c - 1)N_{\text{pair}}$ field-HMM reassemblies across the corpus; (ii) field-rate marginalisation (D.87)

reassembles from the same ε with the scalar jump-count weights $\beta_\tau^{(C)}$, $P(M \mid r_C)$ rescaled at $\rho_{\text{chain}} r_C$ ($G+1$ assemblies per labeling). The substitution eigen-cost is thus paid once per archetype and the per-branch primitive once per (n, k) and (n, k, k') ; the residual per-move work is the $O(L_{\text{max}}^2 \cdot \text{edges})$ field-HMM reassembly of the affected labelings. The factorisation is exact — a finite discrete mixture over field-rate bins and class labelings — the sole approximation being the empirical-Bayes point estimate of $\{m_n\}$, standard in $+\Gamma + I$ phylogenetics.

Remark D.16 (Exact cap-2 Felsenstein — future refinement). Every TKF-DP cluster is size ≤ 2 , so the moment-matching / structured-mean-field forward (subsubsection D.8.6) is not strictly necessary: the cap-2 cluster evidence admits an *exact* Felsenstein that is also *cheaper* than the naive A^2 -state (400-state) coupled chain. The compound state is (θ, x_1, x_2) , but conditional on θ the two columns evolve independently, so a no-jump branch never forms the dense $A^2 \times A^2$ transition: it applies two per-column $A \times A$ GTRs, $P^{(\text{arch}(c_1, \theta))} \otimes P^{(\text{arch}(c_2, \theta))}$, and the field is integrated in closed form via the rank-1 collapse (D.29) (jumps resample both residues from the new archetype stationaries, a rank-1 outer product). Tracking the (θ, x_1, x_2) message thus costs $O(L_{\text{max}} A^2)$ storage and $O(L_{\text{max}} A^2)$ per branch — factored $A \times A$ work, not $A^2 \times A^2$ — giving the exact cap-2 evidence at $O(\text{edges} \cdot L_{\text{max}} A^2)$ per (cluster, class-labeling) with no variational gap. Singletons ($m = 1$) are already exact. This exact pair scorer drops into the same batched (cluster, class-labeling) $\times$ rate-bin enumeration and arch-move invalidation described above; we note it as a future refinement and use the moment-matching forward in the current implementation.

M-step. ρ_{chain} keeps its Gamma-posterior update (D.86), with $\sum_q N_\theta[q]$ and $\sum_q T_q$ now accumulated under the per-cluster responsibilities over r_C . The invariant weight has a Beta(a_p, b_p)-conjugate update

$$p_I \leftarrow \frac{a_p - 1 + \sum_C q_I(C)}{a_p + b_p - 2 + |\mathcal{C}|}, \quad q_I(C) = P(r_C = 0 \mid C) = 1 - \varphi(C), \quad (\text{D.89})$$

over the clusters $\mathcal{C}$, the standard $+I$ update. The Gamma shapes a_f, a_s are fixed hyperparameters (or profiled on a grid).

Full training algorithm. Each outer iteration of the archetype trainer applies the following sub-steps in order (Algorithm 3). We split them into the alignment/partition-side Gibbs updates (which sample from the CRP posterior on cluster partitions and column class labels) and the archetype/substitution-side updates (which follow an EM-style split with a deliberate loose coupling between the E-step for the discrete archetype assignment arch and the M-step for the continuous parameters $\pi^{(k)}, \rho_{\text{chain}}, \rho_a$).

Algorithm 3 One outer iteration of the archetype-variant trainer

- 1: **CRP-Gibbs** on cluster partitions (Section D.8), per MSA.
- 2: **Class-Gibbs** on per-column class labels c_s , per MSA (if $K_c > 1$).
- 3: α_z -**MH**: random-walk MH on $\log \alpha_z$ under a $\text{Gamma}(a_\alpha, b_\alpha)$ prior.
- 4: Extract cluster observations (classes, $X_{\text{obs}}, Y_{\text{obs}}, t$) from all MSAs.
- 5: **E-step (arch assignment)**: accumulate the per- (c, θ, a) soft-attribution counts $N[c, \theta, a]$ (residue counts attributed to (c, θ) buckets, marginalised over field- trajectory histories under a per-column Categorical emission model). Compute the Categorical posterior on arch

$$q(\text{arch}(c, \theta) = k \mid N) \propto \rho_a[k] \prod_a \pi^{(k)}[a]^{N[c, \theta, a]}, \quad (\text{D.90})$$

and hard-assign $\hat{\text{arch}}(c, \theta) = \arg \max_k q$.

- 6: **E-step (HR pass)**: with the fresh $\hat{\text{arch}}$, accumulate per- (c, θ) HR sufficient statistics $\{V_{c, \theta, a}, U_{c, \theta, i, j}, W_{c, \theta, i}\}$ (Section D.8.5) via Algorithm 4 below (composition of the closed-form primitives (D.71), (D.77), (D.78)). Also accumulate the branch-level scalars $\sum_q N_\theta[q]$ and $\sum_q T_q$.
 - 7: **M-step (aggregate)**: aggregate per-archetype sums $V[k, a] = \sum_{c, \theta: \hat{\text{arch}}(c, \theta) = k} V_{c, \theta, a}$ and similarly $U[k, i, j], W[k, i]$.
 - 8: **M-step** ($\pi^{(k)}$): Newton fixed-point update (D.85) on each $\pi^{(k)}$.
 - 9: **M-step** (ρ_{chain}): closed-form Gamma posterior mode (D.86).
 - 10: **M-step** (ρ_a): TSB posterior mean from soft archetype counts $n_k = \sum_{c, \theta} q(\text{arch}(c, \theta) = k \mid N)$.
 - 11: Materialise $\pi^{(c, \theta)} = \pi^{(\hat{\text{arch}}(c, \theta))}$.
-

Loose coupling between the E-step and M-step. Algorithm 3 runs the E-step for arch on the Categorical posterior (D.90), whose sufficient statistic is the residue-count tensor $N[c, \theta, a]$; the M-step for $\pi^{(k)}$ uses the trajectory-based HR sufficient statistics $\{V, U, W\}$ that carry dwell and transition information that N alone does not. Formally these two updates target related but non-identical variational bounds — the Categorical posterior is correct for the per-column emission model, and the GTR Newton update is correct for the compound-CTMC trajectory model — and their composition is *not* a monotone EM step on any single objective. The overall procedure has a fixed point at the true maximum-likelihood solution in the F81 limit (where the residue-count tensor N captures all the trajectory information the HR sufficient statistics do); away from F81 the two objectives share stationary points but reach them along different paths. Empirically the composition converges within the K=4 top-1000 Pfam configuration used throughout this paper.

Cluster-level HR pass algorithm. For completeness, Algorithm 4 states the per-cluster HR pass invoked in step 6 above; it composes the closed forms established in the preceding paragraphs.

Algorithm 4 Cluster HR pass: composes the closed-form primitives

Require: Cluster observations (classes, $X_{\text{obs}}, Y_{\text{obs}}$), branch length t , current $\{\rho, \rho_{\text{chain}}, \pi^{(k)}, \text{arch}\}$, exchangeabilities Q .

- 1: Precompute per-archetype eigendecompositions of $Q^{(k)}$ (Section D.8.5).
 - 2: For each $(\theta_X, \theta_Y) \in \{1, \dots, L_{\text{max}}\}^2$:
 - 3: Compute the field-side quantities $P_\theta(\theta_X \rightarrow \theta_Y; t)$, $E[T_\theta \mid \theta_X, \theta_Y, t]$ (Section D.8), $E[N_\theta \mid \theta_X, \theta_Y, t]$ (Eq. D.71), $E[N_i \mid \theta_X, \theta_Y, t]$ (Eq. D.77).
 - 4: **Case** $M = 0$: for each site n , add the GTR bridge HR contribution under $\text{arch}(c_n, \theta_X)$ to $\{V_{c_n, \theta_X, \cdot}, W_{c_n, \theta_X, \cdot}, U_{c_n, \theta_X, \cdot}\}$, weighted by $\rho[\theta_X] \cdot \beta_t(\theta_X) \cdot \prod_n P^{(\text{arch}(c_n, \theta_X))}(X_n \rightarrow Y_n; t)$.
 - 5: **Case** $M \geq 1$: first-segment free-end HR from X_n (Eq. D.70), last-segment time-reversed free-end HR from Y_n , and last-jump V attribution via $E[P^{(k)}(Y_n \rightarrow a; t - \tau_1)]$ (Eq. D.78). Middle-segment stationary HR accounts for the intermediate θ time.
 - 6: Divide by $P(\text{obs})$ to convert joint sufficient statistics to conditional expectations required by the M-step.
 - 7: **return** $\{V_{c, \theta, \cdot}, U_{c, \theta, \cdot}, W_{c, \theta, \cdot}\}, N_\theta[q], T_q, \log P(\text{obs}_q)$.
-

Split-merge moves on the archetype block. The archetype budget K_a is fixed at construction; under truncated stick-breaking (TSB) the training loop cannot promote a dead slot back into use on its own, so the effective vocabulary can collapse below K_a . Symmetrically, an overloaded archetype with heavy $\rho_a[k]$ and diffuse $\pi^{(k)}$ may be absorbing what would otherwise be two distinct atomic types. We periodically apply hill-climbing *split-merge* moves to re-allocate archetype slots, following the Jain-Neal split-merge motif for Dirichlet mixture models (33) but with the full reversible MH acceptance ratio replaced by a deterministic multinomial-log-likelihood improvement threshold, consistent with the surrounding stochastic-EM regime.

Let

$$\mathcal{L}_{\text{mult}}(\pi, \text{arch}; N) = \sum_{c, \theta, a} N[c, \theta, a] \log \pi^{(\text{arch}[c, \theta])}[a],$$

be the archetype-emission log-likelihood evaluated on the current residue-count tensor $N[c, \theta, a]$ (the same tensor used in (D.90)). A split move is accepted iff $\mathcal{L}_{\text{mult}}(\pi', \text{arch}'; N) > \mathcal{L}_{\text{mult}}(\pi, \text{arch}; N) + \tau_{\text{split}}$ with $\tau_{\text{split}} = 1$ nat (a minimum improvement threshold to prevent split-merge from chasing numerical noise). A merge move is accepted iff $\mathcal{L}_{\text{mult}}(\pi', \text{arch}'; N) \geq \mathcal{L}_{\text{mult}}(\pi, \text{arch}; N) - \tau_{\text{merge}}^{\text{rel}} \cdot |\mathcal{L}_{\text{mult}}(\pi, \text{arch}; N)|$ with $\tau_{\text{merge}}^{\text{rel}} = 10^{-4}$; the tolerance is *relative* to the current LL magnitude because merging two near-identical archetypes (TVD $< \varepsilon_{\text{tvd}}$) is LL-neutral by construction but may nudge down by numerical $O(10^{-6})$ due to the ρ_a -weighted averaging, and an absolute tolerance calibrated to one corpus size fails on others. This asymmetry — strict for splits, tolerant for merges — reflects an implicit MDL/Ockham preference for lower archetype capacity when the data cannot distinguish between two atomic types.

Dead slots. A slot k_{dead} counts as *dead* and eligible for revival if either

- (a) $\rho_a[k_{\text{dead}}] < \varepsilon_{\text{dead}}$ (default 10^{-2}): the slot has essentially no data pulling it anywhere, or
- (b) $H(\pi^{(k_{\text{dead}})}) < H_{\text{coll}}$ (default 0.5 bits): the slot’s emission has collapsed onto one or two residues under sub-1 Dirichlet prior + SS starvation, so observations of any other residue at (c, θ) positions pointing at k_{dead} contribute $\log \pi^{(k_{\text{dead}})}[a]$ near $\log 10^{-30} \approx -69$, a per-residue penalty large enough that even a modest number of such observations dominates the joint LL.

The second criterion is essential: rho-only detection misses the failure mode where a slot has non-trivial usage ($\rho_a > 10^{-2}$) but has collapsed to a delta-like emission, which is what actually produces

the catastrophic ΔLL drops observed in practice.

Split proposal. Given a dead slot k_{dead} , choose an overloaded slot k_{over} maximising $\rho_a[k] \cdot H(\pi^{(k)})$ (heavy and diffuse) subject to $k_{\text{over}} \notin \{\text{dead slots}\}$. Collect the per- (c, θ) residue-count rows $\{N[c, \theta, \cdot] : \text{arch}[c, \theta] = k_{\text{over}}\}$, normalise each to the amino-acid simplex, and run weighted $k = 2$ -means with L^1 (TVD) distance and row-mass weights. The two centroids $\mathbf{m}_0, \mathbf{m}_1$ become the proposed $\pi'^{(k_{\text{over}})}$ and $\pi'^{(k_{\text{dead}})}$; the (c, θ) assignments split accordingly, with $\text{arch}'[c, \theta]$ set to whichever centroid is nearest under TVD.

Merge proposal. Enumerate archetype pairs (k_1, k_2) with $\rho_a[k_i] > \varepsilon_{\text{dead}}$ for both, and choose the pair minimising the total-variation distance $\frac{1}{2} \|\pi^{(k_1)} - \pi^{(k_2)}\|_1$. If the minimum is below a threshold ε_{tvd} (default 0.05), merge by setting

$$\pi'^{(k_1)} = \frac{\rho_a[k_1] \pi^{(k_1)} + \rho_a[k_2] \pi^{(k_2)}}{\rho_a[k_1] + \rho_a[k_2]},$$

redirecting all $\text{arch}[c, \theta] = k_2$ to k_1 , and freeing slot k_2 to a uniform distribution (available for future revival by a split).

After each accepted move the TSB weights ρ_a and their underlying Beta sticks $\beta_{a,i}$ are recomputed from the updated arch by the same posterior-mean rule used elsewhere in the archetype block (the archetype-selector line of the M-step in Section D.8.5) applied to the new occupancy counts $E[n_k]$.

On the acceptance rule. A full Jain-Neal MH move would multiply the $\mathcal{L}_{\text{mult}}$ ratio by (i) the ratio of Dirichlet priors on $\pi^{(k)}$ and (ii) the restricted-scan proposal-density ratio $q(\text{old}|\text{new})/q(\text{new}|\text{old})$ governing the 2-means initialisation and Gibbs mini-sweep. Both are close to unity for the $K_a=8$ configurations of interest here, and retaining them would only slow the walk in cases where the raw LL ratio is already tightly bimodal. In the stochastic-EM setting where the surrounding updates are not themselves a valid MH kernel, the combined chain is not sampling from a well-defined posterior anyway, so the simpler LL-improvement rule is preferred and does not lose guarantees that are not already forfeit.

Where in the algorithm. The split-merge step is invoked after the atom / archetype M-steps, every J_{sm} outer iterations. The choice of J_{sm} has qualitative consequences. $J_{\text{sm}}=1$ (every iter) is the safe default when using the sparse Dirichlet prior $\alpha_{\text{prior}} < 1$: on any single iter the atom M-step can drive an under-observed archetype into an entropy-collapsed configuration (all mass on one residue), and split-merge fires at end-of-iter to revive it before the next E-step propagates the damage. Sparser J_{sm} (e.g. $J_{\text{sm}}=5$) is only safe when $\alpha_{\text{prior}} \geq 1$ so that no single M-step can collapse an archetype past the entropy revival threshold in the $J_{\text{sm}}-1$ intervening iterations. In practice we use $J_{\text{sm}}=1$ and pay the modest overhead of an extra multinomial-LL scan per iter, which is $O(K_c \cdot L_{\text{max}} \cdot A)$ per proposal and does not depend on corpus size.

Because $\mathcal{L}_{\text{mult}}$ omits the substitution-side dwell term $-\sum_a r_k[a] \pi^{(k)}[a]$ from the full HR objective (cf. (D.85)), an accepted split-merge move is not guaranteed to improve the joint HR log-likelihood; empirically the correlation between the two objectives is strong enough that accepted moves do improve the joint LL after the next atom step re-refines $\pi^{(k)}$ from HR sufficient statistics under the new arch.

Alternative training: phylogenetic ELBO on general trees. The composite-likelihood training regime factorises over independent cherries: each cherry’s parent field state θ_P is marginalised independently against the field prior $\rho[\theta_P]$, so different cherries at the same MSA are free to select different θ_P values. This freedom is empirically visible: a diagnostic run on a trained checkpoint shows a large fraction of size- ≥ 2 clusters where the argmax posterior θ_P varies across cherries (Shannon entropy ≥ 1 bit) while the per-cherry probability of any within-cherry field jump is essentially

zero (mean $\approx 10^{-2}$ under the trained ρ_{chain}). The cluster “co-variation” the model is reporting is thus explained by cross-cherry θ_P selection — a genuine pathway that the CRP prior does not resist — rather than by within-cherry coordinated substitution. We call this the *composite-likelihood freedom* and it is inherent to any pair-wise cherry training over an underlying multi-sequence MSA.

The remedy is to train against a joint likelihood that shares field trajectories across sibling branches emanating from the same internal node. On a tree $\mathcal{T}$ with branch lengths, the correct joint Felsenstein-on-DP recursion carries state $(\theta_v, r_1, \dots, r_m)$ at every node v and is exponential in tree depth, because the per-node conditional-likelihood-vector (CLV) factorisation (D.93) holds only conditional on θ_v : at an internal node with two or more children, the field-marginal combine re-couples the columns through the shared parent field.

We therefore introduce a moment-matching variational family whose form is preserved by per-edge propagation and closed under projective combining at internal nodes. At every node v restrict the joint (θ_v, x_v) marginal to

$$q_v(x_v, \theta_v) = q_v(\theta_v) \left[(1 - \lambda_v(\theta_v)) \prod_{n \in \text{block}} p_{n,v}(x_{v,n} | \theta_v) + \lambda_v(\theta_v) \prod_{n \in \text{block}} \pi^{(\text{arch}[c_n, \theta_v])}(x_{v,n}) \right], \quad (\text{D.91})$$

with per-node parameters $q_v(\theta_v) \in \Delta^{L-1}$ (distribution over field states), $p_{n,v}(\cdot | \theta_v) \in \Delta^{A-1}$ (per-site, per-field residue marginal on the “tracking” component), and $\lambda_v(\theta_v) \in [0, 1]$ (mixture weight of the “regenerated” component: probability that at least one field event has occurred on the path from v back to the last ancestor whose residues are still evidence-informed).

The family (D.91) is chosen so that it contains the exact posterior in three regimes: (i) depth-1 cherries under the 2×2 jump-case decomposition of Section D.8.5; (ii) $\rho_{\text{chain}} \rightarrow 0$ (no field events; $\lambda \rightarrow 0$ everywhere; $p_{n,v}$ carries the per-site per-class $\mathbf{Q}$ -evolution exactly); (iii) $\rho_{\text{chain}} \rightarrow \infty$ (every edge regenerates; $\lambda \rightarrow 1$ everywhere; $q_v \rightarrow \rho$ and $p_{n,v}$ is irrelevant). At intermediate ρ_{chain} on deeper trees the ELBO is a controlled approximation.

The ELBO factorises over the tree:

$$\text{ELBO}(q) = \sum_{v \in \text{leaves}} \mathbb{E}_{q_v} [\log P(\mathbf{X}_v^{\text{obs}} | x_v, \theta_v)] + \sum_{v \rightarrow u \in \text{edges}} \mathbb{E}_q [\log K_{\text{edge}}(x_u, \theta_u | x_v, \theta_v; \tau_{vu})] - \sum_v \mathbb{E}_{q_v} [\log q_v], \quad (\text{D.92})$$

with K_{edge} the Interp-2 per-edge kernel with state-dependent no-jump probability $\beta(\theta_v, \tau)$ and jump weight $W(\theta_v, \theta_u; \tau)$ constructed in the 2×2 -case decomposition. Each term is closed form under the family (D.91): the leaf-emission term is $O(L \cdot m \cdot A)$, the per-edge cross-entropy is $O(L^2 \cdot m \cdot A^2)$, and the per-node entropy is $O(L \cdot m \cdot A)$.

Per-edge propagation preserves the form. The variational family (D.91) is stated in posterior form; the corresponding CLV form $F_v(x_v, \theta_v)$ (with $q_v \propto \rho \prod \pi \cdot F_v$) is

$$F_v(x_v, \theta_v) = r_v(\theta_v) \prod_n A_v(x_{v,n} | \theta_v, n) + s_v(\theta_v) \cdot 1, \quad (\text{D.93})$$

with the invariant $\langle \pi^{(\text{arch}[c_n, \theta])}, A_v(\cdot | \theta, n) \rangle = 1$ and, correspondingly, $\langle 1, \pi \rangle = 1$. The scalar component of the CLV is *constant* in x_v (with weight s_v), reflecting that a regenerated ancestor contributes an x_v -independent factor $\prod_n \pi^{(\text{arch}[c_n, \theta_p])}(x_v)$ to the observed-evidence integral (already absorbed by the prior in the CLV normalisation). Substituting (D.93) into the upward message $\tilde{U}_{u \rightarrow p}(x_p, \theta_p) = \sum_{x_u, \theta_u} K(x_u, \theta_u | x_p, \theta_p; \tau) F_u(x_u, \theta_u)$ and expanding the two-component K (D.112) produces four cross-terms, each of which is already in the target family form: no-jump $\times$ rank-1 $\rightarrow$ rank-1 with kernel $\tilde{A}(a | \theta_p, n) := \sum_y P^{(c_n, \theta_p)}(a, y; \tau) A_u(y | \theta_p, n)$ (invariant preserved by left-stationarity $\pi^\top P = \pi^\top$); no-jump $\times$ scalar $\rightarrow$ scalar with weight $\beta(\theta_p, \tau) s_u(\theta_p)$ (using

$\sum_y P^{(c_n, \theta_p)}(x_p, y; \tau) = 1$); jump $\times$ rank-1 $\rightarrow$ scalar with weight $\sum_{\theta_u} W(\theta_p, \theta_u; \tau) r_u(\theta_u)$ (using invariant $\langle \pi, A_u \rangle = 1$); jump $\times$ scalar $\rightarrow$ scalar with weight $\sum_{\theta_u} W(\theta_p, \theta_u; \tau) s_u(\theta_u)$ (using $\langle \pi, 1 \rangle = 1$). The result is exactly in the family with:

$$r_v(\theta_p) = \beta(\theta_p, \tau) r_u(\theta_p), \quad (\text{D.94})$$

$$A_v(a | \theta_p, n) = \tilde{A}(a | \theta_p, n), \quad (\text{D.95})$$

$$s_v(\theta_p) = \beta(\theta_p, \tau) s_u(\theta_p) + \sum_{\theta_u} W(\theta_p, \theta_u; \tau) (r_u(\theta_u) + s_u(\theta_u)). \quad (\text{D.96})$$

No projection is needed: the (rank-1 + constant) CLV form is closed under exact per-edge propagation. The internal-node combine below does require projection because the exact product of two family members is not itself in the family.

Internal-node combining via moment-matching projection. At a binary internal node v combining children u_1, u_2 , the exact product of two (rank-1 + scalar) CLVs decomposes into three rank-1 terms and one scalar; the variational projection matches, for each θ_v : (i) the total mass into $q_v(\theta_v)$; (ii) the per-site per- θ_v marginal of $x_{v,n}$ into $p_{n,v}(\cdot | \theta_v)$; (iii) the fraction of the four-term sum that comes from scalar-involving terms into $\lambda_v(\theta_v)$. Each projection is closed form, cost $O(m \cdot L \cdot A)$ per combine. Non-binary internal nodes are handled by sequential binary combines with intermediate projection.

Coordinate ascent alternates post-order updates (leaves up to root via the combined-then-projected forward messages) and pre-order updates (root down to leaves via backward messages). The ELBO is monotone non-decreasing under each half-sweep and typically converges in 10–50 full sweeps at cost $O(\text{edges} \cdot L^2 \cdot m \cdot A^2)$ per sweep.

Internal-node combine: moment-matching projection. Per-edge propagation is exact under the CLV form (D.93) (derived above). The only place a moment-matching projection is required is at internal nodes with two or more children, where the exact product of two family CLVs $F_{u_1} \cdot F_{u_2}$ is not itself in the family.

Exact product. Write both children in family form (D.93): $F_{u_i}(x, \theta) = r_i \prod A_i + s_i$ for $i \in \{1, 2\}$ (dropping θ -dependence for brevity). The pointwise product decomposes into four cross-terms:

$$F_{u_1}(x, \theta) F_{u_2}(x, \theta) = \underbrace{r_1 r_2 \prod_n A_1 A_2}_{\text{rank-1, kernel } A_1 A_2} + \underbrace{r_1 s_2 \prod_n A_1}_{\text{rank-1, kernel } A_1} + \underbrace{s_1 r_2 \prod_n A_2}_{\text{rank-1, kernel } A_2} + \underbrace{s_1 s_2}_{\text{constant}}. \quad (\text{D.97})$$

The three rank-1 terms have distinct per-site kernels ($A_1 A_2$, A_1 , A_2) and cannot be exactly represented by a single-rank-1 family member. A projection is required.

Projection. We project onto the target family $F_p = r_p \prod A_p + s_p$ by matching (i) the total equilibrium-weighted mass $\langle \pi_{\text{all}}, F_p(\cdot, \theta) \rangle = r_p + s_p$ per θ , and (ii) the per-site π -weighted marginal at each site n under $\langle \pi, \cdot \rangle$ -weighting at other sites. We assign the scalar component s_p to the constant term $s_1 s_2$ (which is already in the family) and put the three rank-1 terms into $r_p A_p$:

$$s_p(\theta) = s_1(\theta) s_2(\theta), \quad (\text{D.98})$$

$$r_p(\theta) = r_1 r_2 \prod_n \langle \pi, A_1 A_2 \rangle_n + r_1 s_2 + s_1 r_2, \quad (\text{D.99})$$

$$A_p(a | \theta, n) = \frac{r_1 r_2 (\prod_{n' \neq n} \langle \pi, A_1 A_2 \rangle_{n'}) A_1(a) A_2(a) + r_1 s_2 A_1(a) + s_1 r_2 A_2(a)}{r_p(\theta)}. \quad (\text{D.100})$$

Each cross-term contributes to the per-site marginal at site n via its $\prod_{n' \neq n} \langle \pi, \cdot \rangle$ -weighted rest factor (all equal to 1 except for the $A_1 A_2$ term, which picks up $\prod_{n' \neq n} \langle \pi, A_1 A_2 \rangle_{n'}$). The $s_1 s_2$ constant term contributes only to s_p ; it drops out of A_p 's numerator because the family's scalar-per-site marginal $s_p \cdot \langle \pi, 1 \rangle_{n'} = s_p$ absorbs it directly. The family invariant $\langle \pi, A_p \rangle_n = 1$ follows by direct computation using $\langle \pi, A_i \rangle = 1$.

Cost. Building per-site inner products $\langle \pi, A_1 A_2 \rangle_n$ is $O(A)$ per (site, θ); combining is $O(A)$ per site. Total $O(mLA)$ per combine. Non-binary internal nodes reduce to sequential binary combines with intermediate projection.

Approximation quality. The projection is exact when $\rho_{\text{chain}} \rightarrow 0$ (no field jumps, so $s_i = 0$ throughout and only the $r_1 r_2 A_1 A_2$ term survives, which is already rank-1). It is also exact at depth-1 cherries. At depth-2 and deeper, per-branch $s > 0$ activates the projection and induces the first-order moment-matching bias. Empirically the residual is $< 1\%$ in log-likelihood on random depth-2 trees at $\rho_{\text{chain}} \in [0.1, 1.5]$.

The rest of this paragraph is discarded — it derived a projection under an incorrect CLV form that took the scalar as $\prod_n \pi^{(\text{arch})}(x_n)$ rather than the constant 1. Under the correct form (D.93), per-edge propagation is exact and only the internal-node combine needs the projection above.

Alternative training: tied- θ MCMC on whole-tree histories. A third route that side-steps both the variational family of Section D.8.5 and the soft closed forms of Section D.8.5 is to sample the tree history end to end and compute HR sufficient statistics on the fully-observed trajectory. The core insight is that composite training treats each branch as an independent “cherry” with its own marginalised parent field state θ_P , yet all branches emanating from an internal node V physically share the same θ_V : composite likelihood is deliberately breaking a constraint the tree imposes. Tying θ at every internal node undoes the break directly, and does so *without* importance-weight variance, because with tied θ the sample is drawn from (an approximation to) the tree-joint TKF-DP posterior and every HR estimator is just a Monte-Carlo average.

The scheme is a two-block Gibbs step nested in the outer SVI iter: $X \sim p(X | y, \mathcal{T}, \Theta)$ followed by $\theta \sim p(\theta | X, y, \mathcal{T}, \Theta)$ per cluster, plus a binary branch-jump indicator M_v for ρ_{chain} identifiability. Each block is a Felsenstein pass over a discrete state (residue or field), so nothing expensive appears.

Notation. Fix a family with guide tree $\mathcal{T}$, leaf set $\mathcal{L}$ with observed residues y_l^s for leaf l and site s , and branch lengths $\{\tau_v\}_{v \neq \text{root}}$. Let $\pi^{(c, \theta)}$ be the per-(class, field) stationary, and $Q_{ij}^{(c, \theta)} = Q_{ij} \pi^{(c, \theta)}(j)$ for $i \neq j$ the associated GTR generator. Write $X = (x_v^s)$ for the residues at every internal-node/site pair. Cluster assignments $\mathcal{C}$ partition the L columns of the alignment; each cluster carries its own field trajectory θ_v on $\mathcal{T}$.

Block 1: sample X (residues). Use the standard bottom-up / top-down Felsenstein sampler on the tree with the class-conditional field-marginal transition $P_{\text{marg}}^{(cs)}(\tau) = \sum_{\theta} \rho[\theta] \exp(Q^{(cs, \theta)} \tau)$ as the per-site kernel. This is a proper row-stochastic matrix (a mixture of row-stochastic matrices), even though it is not the exponential of a single generator, so peeling and stochastic mapping are unmodified. The draw preserves branch-length correlations exactly and delivers a whole-tree residue sample per site in $O(|\mathcal{T}| \cdot A^2)$ per class per branch-length bin.

Block 2: sample θ (field, per cluster). Given X , the per-branch cluster emission under Interp 2 factorises in a form convenient for Felsenstein-on- θ . Define $\beta(\theta) := \exp(-\rho_{\text{chain}}(1 - \rho[\theta])\tau_v)$ (per-

branch survival probability of θ),

$$\begin{aligned}\Sigma_{\text{prod}}(\theta) &:= \prod_{s \in \mathcal{C}} [\exp(Q^{(c_s, \theta)} \tau_v)]_{x_p^s, x_c^s}, \\ \pi_{\text{prod}}(\theta) &:= \prod_{s \in \mathcal{C}} \pi^{(c_s, \theta)}(x_c^s).\end{aligned}$$

Then the per-branch cluster emission conditional on both endpoints' field values is

$$E_{\mathcal{C}, v}(\theta_p, \theta_c) = \underbrace{\beta(\theta_p) \Sigma_{\text{prod}}(\theta_p) \delta_{\theta_c = \theta_p}}_{M=0 \text{ term}} + \underbrace{(1 - \beta(\theta_p)) \rho[\theta_c] \pi_{\text{prod}}(\theta_c)}_{M \geq 1 \text{ term, Interp 2 resample}}. \quad (\text{D.101})$$

The factorisation $E = A(\theta_p) \delta + B(\theta_p) C(\theta_c)$ with $A = \beta \cdot \Sigma_{\text{prod}}$, $B = 1 - \beta$, $C = \rho \cdot \pi_{\text{prod}}$ makes the Felsenstein-on- θ child update at a node v

$$\phi_{v \leftarrow c}(\theta_v) = A_{v \rightarrow c}(\theta_v) \text{CLV}_c(\theta_v) + B_{v \rightarrow c}(\theta_v) \langle C_{v \rightarrow c}, \text{CLV}_c \rangle, \quad (\text{D.102})$$

which costs $O(L)$ (not $O(L^2)$) per branch per cluster. The full CLV pass is $O(m \cdot L \cdot |\mathcal{T}|)$ per cluster for building the A, B, C caches plus $O(L \cdot |\mathcal{T}|)$ for combining messages. Top-down sampling of θ_v uses (D.101) directly.

Block 3: sample M_v per branch. Once (θ_p, θ_c) is drawn at branch v 's endpoints,

$$P(M_v = 0 \mid \theta_p, \theta_c, X, \tau_v) = \begin{cases} 0 & \theta_p \neq \theta_c, \\ \frac{A_v(\theta_p)}{A_v(\theta_p) + B_v(\theta_p) C_v(\theta_p)} & \theta_p = \theta_c, \end{cases} \quad (\text{D.103})$$

gives a Bernoulli sample of $M_v \in \{0, 1\}$. The Rao-Blackwellised ρ_{chain} SS increment on branch v is

$$\Delta N_\theta^v = M_v \cdot E[N_\theta \mid \theta_p, \theta_c, \tau_v, M \geq 1], \quad (\text{D.104})$$

where the case- $M \geq 1$ expectation

$$E[N_\theta \mid \theta_p, \theta_c, \tau_v, M \geq 1] = \frac{E[N_\theta \mid \theta_p, \theta_c, \tau_v] P_\theta(\theta_p \rightarrow \theta_c; \tau_v)}{P_\theta(\theta_p \rightarrow \theta_c; \tau_v) - \beta(\theta_p) \delta_{\theta_p, \theta_c}} \quad (\text{D.105})$$

divides the whole-branch bridge form (D.71) by the $M \geq 1$ mass $P_\theta - \beta\delta$. Summing ΔN_θ^v over branches gives N_θ^{sum} for the ρ_{chain} Gamma-posterior M-step. A bare $\sum_v M_v$ (as in an early implementation) undercounts jumps on long branches where the $M \geq 1$ posterior concentrates mass above 1, and self-reinforcingly drives $\rho_{\text{chain}} \rightarrow 0$ (fewer $M = 1$ samples, lower estimate, fewer samples, ...); the closed-form expectation avoids this collapse. A finer sample of the exact jump count under the truncated-Poisson posterior would refine the tail but is unnecessary once (D.105) is used.

HR sufficient statistics. For each branch v in each cluster $\mathcal{C}$ with the sampled $(\theta_p, \theta_c, M_v)$ and the observed residues $(x_p^s, x_c^s)_{s \in \mathcal{C}}$, we accumulate:

$M_v = 0$ (so $\theta_p = \theta_c = \theta$): per site s , standard GTR bridge HR at archetype $k = \text{arch}(c_s, \theta)$: increment $V[c_s, \theta, x_p^s]$ by 1 and add the closed-form dwell and transition contributions $W_i^{\text{br}}, U_{ij}^{\text{br}}$ from (D.43) at endpoints (x_p^s, x_c^s) to $W[c_s, \theta, \cdot]$ and $U[c_s, \theta, \cdot, \cdot]$.

$M_v = 1$ (means $M \geq 1$ field jumps on the branch; intermediate trajectory unobserved, but (θ_p, θ_c) are fixed by Block 2 and (x_p^s, x_c^s) by Block 1): per site s apply the case- $M \geq 1$ closed forms of Section D.8.5 with $(\theta_X, \theta_Y) = (\theta_p, \theta_c)$ fixed, i.e. skipping the $L \times L$ marginalisation over (θ_X, θ_Y) that appears in the composite path. The three-segment attribution is:

- First segment, archetype $k_p := \text{arch}(c_s, \theta_p)$, forward end x_p^s , duration τ_1 under the exact mixture density (D.44). Add the start-boundary $V^{\text{first}}[a] = \delta_{a, x_p^s}$ to $V[c_s, \theta_p, \cdot]$, W_i^{first} from (D.55) to $W[c_s, \theta_p, \cdot]$, and U_{ij}^{first} from (D.56) to $U[c_s, \theta_p, \cdot, \cdot]$.
- Middle segments (residual field states $\theta_{\text{int}} \in \{1, \dots, L\}$) at their respective archetypes $k_{\text{int}} := \text{arch}(c_s, \theta_{\text{int}})$. Compute the bridge-conditional-on- $M \geq 1$ quantities

$$E[T_{\theta_{\text{int}}} \mid \theta_p, \theta_c, \tau_v, M \geq 1] = \frac{E[T_{\theta_{\text{int}}} \mid \theta_p, \theta_c, \tau_v] P_{\theta}(\theta_p \rightarrow \theta_c; \tau_v) - \beta(\theta_p) \tau_v \delta_{\theta_p, \theta_c} \delta_{\theta_{\text{int}}, \theta_p}}{P_{\theta}(\theta_p \rightarrow \theta_c; \tau_v) - \beta(\theta_p) \delta_{\theta_p, \theta_c}},$$

$$E[N_{\theta_{\text{int}}} \mid \theta_p, \theta_c, \tau_v, M \geq 1] = \frac{E[N_{\theta_{\text{int}}} \mid \theta_p, \theta_c, \tau_v] P_{\theta}(\theta_p \rightarrow \theta_c; \tau_v)}{P_{\theta}(\theta_p \rightarrow \theta_c; \tau_v) - \beta(\theta_p) \delta_{\theta_p, \theta_c}},$$

from (D.71) and (D.77). Subtract first- and last-segment residuals (D.64)–(D.65) to get $\tau_{\text{mid}}(\theta_{\text{int}})$ and $N_{\text{mid}}(\theta_{\text{int}})$, then add (D.59)–(D.61) to $V[c_s, \theta_{\text{int}}, \cdot]$, $W[c_s, \theta_{\text{int}}, \cdot]$, $U[c_s, \theta_{\text{int}}, \cdot, \cdot]$.

- Last segment, archetype $k_c := \text{arch}(c_s, \theta_c)$, observed end x_c^s , duration $\tau_v - \tau_M$ under the mixture density with $\theta_X \leftrightarrow \theta_Y$ exchanged in (D.44). Add V_a^{last} from (D.58), W_j^{last} from (D.68), and U_{ij}^{last} from (D.69) to $V[c_s, \theta_c, \cdot]$, $W[c_s, \theta_c, \cdot]$, $U[c_s, \theta_c, \cdot, \cdot]$.

Also increment N_{θ}^{sum} by ΔN_{θ}^v from (D.104)–(D.105).

Sum over branches, clusters, and families gives the same per- (c_s, θ) -indexed (V, U, W) tensors and scalar $(N_{\theta}^{\text{sum}}, T^{\text{sum}})$ statistics the composite pass yields; downstream aggregation to per-archetype (V_k, U_k, W_k) via $\text{arch}(c_s, \theta) = k$ and the $\pi^{(k)} / \rho_{\text{chain}}$ M-steps of Section D.8.5 apply unchanged.

Why no importance weights. We are sampling directly from an approximation of the tree-joint TKF-DP posterior (Block 1 uses the class-marginal transition; Block 2 is exact given X under Interp 2 with Interp 1 resample). The M-step estimator is

$$\hat{E}_{\Theta}[\text{SSc}] = \text{SSc}(X, \theta, M, y), \quad (\text{D.106})$$

a plain Monte-Carlo evaluation on the sampled trajectory. There is no target/proposal ratio, no ESS collapse, and no per-cluster or per-cluster-size weight variance. Cross-cluster θ -independence is respected by construction (each cluster gets its own draw); within-cluster θ is tied at every internal node by construction too.

Failure modes of the intermediate designs, for the record. An earlier version of this section proposed *importance sampling* against the composite target with a class-marginal Felsenstein proposal. Three attempts, in order, each diagnosed and abandoned:

- Store per-node marginals $p_v(a) \propto \pi(a) L_v(a)$ and draw each branch endpoint independently. Discards the parent-child correlation carried by $P(\tau_v)$; empirically the sampled tuples are 15–25% less conserved than LG08 predicts and ρ_{chain} diverges to > 1 within an iteration.
- Store CLVs and draw whole-tree histories under a single-stationary LG08 proposal. Fixes (i)’s branch-correlation problem but the per-branch log-ratio between LG08 and the class-marginal target accumulates to ± 20 –50 nats over the $\sim 2N - 1$ nodes of a deep Pfam tree, collapsing ESS/K to $\sim 6 \times 10^{-3}$ at $K_c = 4$.
- Switch the proposal to class-conditional GTR and hope the residual is $O(\tau^2)$. It is, but the numerator of the correction weight ratio was still the composite target $\Sigma_{\text{cherry}}(X, Y; \tau) = \pi(X) \cdot P(X \rightarrow Y; \tau)$ while the denominator was the sampler marginal $q_{\text{tree}}(X_p) \cdot P(X_p \rightarrow X_c)$. Under GTR reversibility this leaves an unavoidable $\log \pi^{(c)}(X_p) - \log q_{\text{tree}}(X_p)$ per site, which

at leaves (where q_{tree} collapses to a delta on the observed residue) is $\approx -\log(1/A) \approx 3$ nats per site and accumulates linearly in cluster size m . That means larger clusters get exponentially down-weighted in the M-step for the wrong reason (tree-conditional vs stationarity mismatch, not field-coupling signal). Measured $\log w/m \approx -2.85$ constant across $m = 1, \dots, 8$ at iter 1, confirming the $\log-\pi$ baseline is dominant.

Tied- θ MCMC dodges all three failure modes because it does not need to reweight anything: the SS is computed on the trajectory that was drawn, whose target the sampler already approximates.

Preprocessing footprint. Preprocessing stores only the tree topology (a $2N-1$ int parent map), branch lengths (a $2N-1$ float64 vector) and the observed leaf residues (an $N \times L$ int8 matrix). Storage is $O(NL)$ per family, dominated by the leaf-residue matrix and totalling only ~ 10 MB for the top-1000 corpus. The CLVs for Block 1 are recomputed per iter from the current $\pi^{(c)}$; the A, B, C tables for Block 2 are recomputed per iter per cluster from the sampled X and the current $\pi^{(c, \theta)}$. Both fit comfortably in RAM at Pfam scale.

Backward recursion for the moment-matching family. We now derive the second half of the message-passing structure needed to extract per-branch pair posteriors: the *downward* conditional partials $D_v(x_v, \theta_v)$ and their propagation through the variational family (D.91). We give the exact recursion first (linear-algebra form), then show it preserves the (rank-1 + scalar) CLV form up to a moment-matching projection at sibling-merge points, and finally state the concrete update rule.

Setup. On a rooted tree $\mathcal{T}$ with the compound-CTMC generative model (root state drawn from $\rho_\theta \prod_n \pi^{(\text{arch}[c_n, \theta])}(x_n)$; each branch of length τ transports (x, θ) under the joint kernel above), define the standard Felsenstein partials

$$F_v(x_v, \theta_v) \equiv P(\text{obs at leaves under } v \mid x_v, \theta_v) \quad (\text{upward}) \quad (\text{D.107})$$

$$D_v(x_v, \theta_v) \equiv P(\text{obs at leaves outside subtree at } v \mid x_v, \theta_v) \quad (\text{downward}). \quad (\text{D.108})$$

Posterior at v :

$$q_v(x_v, \theta_v) = \frac{P_v^{\text{prior}}(x_v, \theta_v) \cdot F_v(x_v, \theta_v) \cdot D_v(x_v, \theta_v)}{P(\text{observations})}, \quad P_v^{\text{prior}} = \rho_{\theta_v} \prod_n \pi^{(\text{arch}[c_n, \theta_v])}(x_{v,n}), \quad (\text{D.109})$$

since the compound CTMC's joint stationary factorises as $\rho \cdot \prod_n \pi^{(\text{arch})}$ at every node (a consequence of the F81-on-DP field being at stationary throughout the tree).

Exact downward recursion. At the root r , no observations lie outside the subtree, so $D_r \equiv 1$. For a non-root v with parent p and siblings $\{s_i\}$ (children of p other than v), condition on parent state and sum:

$$D_v(x_v, \theta_v) = \sum_{x_p, \theta_p} K(x_v, \theta_v \mid x_p, \theta_p; \tau_v) D_p(x_p, \theta_p) \prod_i U_{s_i \rightarrow p}(x_p, \theta_p), \quad (\text{D.110})$$

where $U_{s_i \rightarrow p}(x_p, \theta_p) := \sum_{x_{s_i}, \theta_{s_i}} K(x_{s_i}, \theta_{s_i} \mid x_p, \theta_p; \tau_{s_i}) F_{s_i}$ is precisely the upward message the forward sweep already computes at p . Write

$$M_v(x_p, \theta_p) \equiv D_p(x_p, \theta_p) \prod_i U_{s_i \rightarrow p}(x_p, \theta_p) \quad (\text{D.111})$$

for the aggregated “outside” partial at v 's parent; then (D.110) is a single edge propagation of M_v through the branch $p \rightarrow v$, summed on the source side (parent) rather than the destination side (as in the forward sweep).

Kernel decomposition. Under Interp 2, the joint kernel factors into no-jump and ≥ 1 -jump components:

$$K(x_v, \theta_v \mid x_p, \theta_p; \tau) = \underbrace{\beta(\theta_p, \tau) \delta_{\theta_v=\theta_p} \prod_n P^{(c_n, \theta_p)}(x_{p,n}, x_{v,n}; \tau)}_{\text{no-jump}} + \underbrace{W(\theta_p, \theta_v; \tau) \prod_n \pi^{(\text{arch}[c_n, \theta_v])}(x_{v,n})}_{\text{jump}}, \quad (\text{D.112})$$

with $\beta(\theta_p, \tau) = e^{-\rho_{\text{chain}}(1-\rho_{\theta_p})\tau}$ and $W(\theta_p, \theta_v; \tau) = K_{\text{field}}(\theta_v \mid \theta_p; \tau) - \delta_{\theta_v=\theta_p} \beta(\theta_p, \tau)$.

Substituting into (D.110) and evaluating the $\delta_{\theta_v=\theta_p}$ constraint in the no-jump branch:

$$\begin{aligned} D_v(x_v, \theta_v) &= \beta(\theta_v, \tau) \sum_{x_p} \left[\prod_n P^{(c_n, \theta_v)}(x_{p,n}, x_{v,n}; \tau) \right] M_v(x_p, \theta_v) \\ &\quad + \left[\prod_n \pi^{(\text{arch}[c_n, \theta_v])}(x_{v,n}) \right] \sum_{\theta_p} W(\theta_p, \theta_v; \tau) \sum_{x_p} M_v(x_p, \theta_p). \end{aligned} \quad (\text{D.113})$$

The first term (no-jump) is a per-site product of transposed substitution kernels applied to M_v 's rank-1 content at the diagonal $\theta_p = \theta_v$; the second term (jump) is a scalar in x_v times the per-site stationary product at θ_v . This is the same (rank-1 + scalar) template as the forward message, but with two critical differences:

- (a) Kernel direction: β is evaluated at θ_v (destination in the downward direction) via the δ -collapse, whereas in the forward pass β is evaluated at the parent (also the “destination” from the upward perspective). Under reversibility of the field CTMC these two conventions differ by no factor; $\beta(\theta_v)$ is the correct multiplier in both directions once labels are matched to the resulting message's source axis.
- (b) Jump-weight orientation: $W(\theta_p, \theta_v)$, not $W(\theta_v, \theta_p)$ — the summation variable is the parent's field, i.e., the *source* of the field's forward transition. Using detailed balance $\rho_{\theta_p} W(\theta_p, \theta_v) = \rho_{\theta_v} W(\theta_v, \theta_p)$ (for $\theta_p \neq \theta_v$; equal on the diagonal), we may rewrite

$$\sum_{\theta_p} W(\theta_p, \theta_v; \tau) \bar{M}_v(\theta_p) = \frac{1}{\rho_{\theta_v}} \sum_{\theta_p} W(\theta_v, \theta_p; \tau) [\rho_{\theta_p} \bar{M}_v(\theta_p)], \quad (\text{D.114})$$

where $\bar{M}_v(\theta_p) := \sum_{x_p} M_v(x_p, \theta_p)$ is the theta-slice mass of M_v . Identity (D.114) reduces the backward jump-weight sum to the same $W(\theta_v, \theta_p)$ -structure as the forward mm-edge, provided we replace $\bar{M}_v(\theta_p) \mapsto \rho_{\theta_p} \bar{M}_v(\theta_p)$ on the input and divide by ρ_{θ_v} on the output.

Backward propagation in the moment-matching family. Suppose M_v is in the (rank-1 + scalar) form

$$M_v(x_p, \theta_p) = r_M(\theta_p) \prod_n A_M(x_{p,n} \mid \theta_p) + s_M(\theta_p) \prod_n \pi^{(\text{arch}[c_n, \theta_p])}(x_{p,n}), \quad (\text{D.115})$$

with the normalisation $\langle \pi^{(c_n, \theta)}, A_M(\cdot \mid \theta) \rangle = 1$ inherited from the forward sweep. Substituting into (D.113) and using the per-site GTR reversibility identity $\sum_y P^{(c_n, \theta)}(y, x; \tau) \pi^{(c_n, \theta)}(y) = \pi^{(c_n, \theta)}(x)$ (direct consequence of detailed balance on the substitution chain), the two terms of (D.113) become:

$$\begin{aligned} D_v^{\text{no-jump}}(x_v, \theta_v) &= \beta(\theta_v, \tau) \left[r_M(\theta_v) \prod_n \sum_y P^{(c_n, \theta_v)}(y, x_{v,n}; \tau) A_M(y \mid \theta_v) \right. \\ &\quad \left. + s_M(\theta_v) \prod_n \pi^{(\text{arch}[c_n, \theta_v])}(x_{v,n}) \right], \end{aligned} \quad (\text{D.116})$$

$$D_v^{\text{jump}}(x_v, \theta_v) = \left[\prod_n \pi^{(\text{arch}[c_n, \theta_v])}(x_{v,n}) \right] \sum_{\theta_p} W(\theta_p, \theta_v; \tau) \bar{M}_v(\theta_p). \quad (\text{D.117})$$

Combining, D_v has the (rank-1 + scalar) form:

$$D_v(x_v, \theta_v) = r_D(\theta_v) \prod_n A_D(x_{v,n} \mid \theta_v) + s_D(\theta_v) \prod_n \pi^{(\text{arch}[c_n, \theta_v])}(x_{v,n}), \quad (\text{D.118})$$

with

$$r_D(\theta_v) = \beta(\theta_v, \tau) r_M(\theta_v), \quad (\text{D.119})$$

$$A_D(x \mid \theta_v)_n = \sum_y P^{(c_n, \theta_v)}(y, x; \tau) A_M(y \mid \theta_v) \quad (\text{per site } n; \text{ note transposed convention}), \quad (\text{D.120})$$

$$s_D(\theta_v) = \beta(\theta_v, \tau) s_M(\theta_v) + \sum_{\theta_p} W(\theta_p, \theta_v; \tau) \bar{M}_v(\theta_p). \quad (\text{D.121})$$

The three closed-form updates constitute the backward analog of the forward edge propagation. Compared to it:

- The rank-1 magnitude $r_M(\theta_v) \rightarrow r_D(\theta_v)$ transforms identically (β -scaling at the diagonal $\theta_p = \theta_v$).
- The per-site rank-1 marginal $A_M \rightarrow A_D$ is propagated by the *transposed* substitution kernel $y \rightarrow x$ rather than $x \rightarrow y$; algebraically, $A_D = P^{(c_n, \theta_v)} \cdot A_M$ if we treat A as a column vector, whereas the forward update was $A_v = (P^{(c_n, \theta_v)})^\top \cdot A_c$. Under GTR reversibility these two operations are related by $A_D = D_\pi^{-1} A_v^{\text{fwd}} D_\pi$ where $D_\pi = \text{diag}(\pi)$, but the two are numerically distinct in general.
- The scalar $s_M(\theta_v) \rightarrow s_D(\theta_v)$ picks up a ρ -scaled jump-weight sum with source θ_p (D.114), differing from the forward $\sum_{\theta_u} W(\theta_v, \theta_u) \bar{F}(\theta_u)$ by the ρ -swap of source and destination in W .

The normalisation invariant $\langle \pi, A_D \rangle = 1$ is preserved by the transposed propagation, by the same argument as the forward: $\langle \pi, P^{(c_n, \theta_v)} A_M \rangle_x = \langle A_M, P^{(c_n, \theta_v)^\top} \pi \rangle_y = \langle A_M, \pi \rangle = 1$ using $(P^{(c_n, \theta_v)})^\top \pi = \pi$ (right-invariance of the stationary under reversibility).

Sibling merge (moment-matching at the aggregator M_v). The aggregate $M_v = D_p \cdot \prod_i U_{s_i \rightarrow p}$ is a product of several (rank-1 + scalar) forms and must be projected back onto the family before the edge propagation above can be applied. Under the moment-matching combine rule, each pairwise product of terms $[r_1 R_1 + s_1] \cdot [r_2 R_2 + s_2]$ decomposes into $r_1 r_2 R_1 R_2 + r_1 s_2 R_1 + s_1 r_2 R_2 + s_1 s_2$; the scalar component $s_1 s_2$ is kept as the projected scalar, and the three rank-1 residuals are merged into a single per-site marginal by matching the per-site $\langle \pi, \cdot \rangle$ -moments. This is the same projection used in the forward sweep at internal-node combines; it is exact when there is only one contributing sibling (so $M_v = D_p \cdot U_{s \rightarrow p}$ is a single pairwise product) and introduces a controlled bias for higher-arity nodes, matching the forward’s cost/quality tradeoff.

Root and leaf boundary. At the root, $M_v = D_p \cdot U_{s \rightarrow p}$ reduces to $\prod_{i \neq v} U_{s_i \rightarrow \text{root}}$ (no D_p factor); the root’s “prior” contribution $\rho \cdot \prod \pi$ enters only when computing the posterior via (D.109) so that $D_r \equiv 1$ needs no representation. At a leaf ℓ , $F_\ell(x, \theta) = \delta(x = x_\ell^{\text{obs}})$ and $D_\ell(x, \theta)$ is computed by exactly one backward mm-edge from its parent; the leaf posterior over θ_ℓ then reduces to $q_\ell(\theta) \propto \rho_\theta \pi^{(\text{arch}[\dots, \theta])}(x_\ell^{\text{obs}}) \cdot D_\ell(x_\ell^{\text{obs}}, \theta)$.

Coordinate-ascent convergence. A full backward sweep after each forward sweep gives one ELBO coordinate-ascent iteration; the ELBO (D.92) is non-decreasing per half-sweep and converges in 10–50 full sweeps in practice. At convergence, the forward and backward CLVs at every node are jointly consistent with the marginal posterior (D.109), and (D.110) is satisfied to floating-point precision on cherries and to the moment-matching tolerance on deeper trees.

Bridge posterior on a branch. For extraction of HR sufficient statistics we need not the marginal posterior at a node but the *joint posterior on a branch* — the pair distribution over $((x_v, \theta_v), (x_u, \theta_u))$ at the two ends of the branch $v \rightarrow u$ of length τ_{vu} :

$$q_{vu}(x_v, \theta_v, x_u, \theta_u) \propto P_v^{\text{prior}}(x_v, \theta_v) \cdot D_v(x_v, \theta_v) \cdot K(x_u, \theta_u \mid x_v, \theta_v; \tau_{vu}) \cdot F_u(x_u, \theta_u), \quad (\text{D.122})$$

where F_u and D_v are the converged CLVs and the normalising constant is the tree log-likelihood $P(\text{obs})$.

Substituting the two-term K (D.112) and the (rank-1 + scalar) forms of F_u and D_v into (D.122), and marginalising over positions (x_v, x_u) at all sites but one, each site’s pair distribution $q_{vu,n}(x_{v,n}, x_{u,n} \mid \theta_v, \theta_u)$ is a mixture over $2 \times 2 = 4$ terms (no-jump vs. jump, tracking vs. regenerated sources at each end), each of which factorises across sites given (θ_v, θ_u) . The concrete formulas appear below.

Field-jump-count decomposition of a branch. The branch kernel $K(x_u, \theta_u \mid x_v, \theta_v; \tau)$ marginalises over the entire compound trajectory on the branch, but the HR sufficient statistics require attribution of dwell time and substitution counts to specific archetypes, and archetypes change with θ . We therefore decompose the branch by the number M of field jumps that occurred on the interval $(0, \tau)$:

$$K(x_u, \theta_u \mid x_v, \theta_v; \tau) = \underbrace{K^{(M=0)}(x_u \mid x_v, \theta_v; \tau) \delta_{\theta_u=\theta_v}}_{\text{no jump}} + \underbrace{\sum_{M \geq 1} K^{(M)}(x_u, \theta_u \mid x_v, \theta_v; \tau)}_{M \geq 1 \text{ jumps}}. \quad (\text{D.123})$$

$M = 0$: *no field jump*. The path is a pure GTR evolution under the fixed archetype $\text{arch}[c_n, \theta_v]$ at each site over the whole branch of length τ . Probability of the joint (no-jump event and observed residues):

$$K^{(M=0)}(x_u \mid x_v, \theta_v; \tau) = \beta(\theta_v, \tau) \prod_n P^{(c_n, \theta_v)}(x_{v,n}, x_{u,n}; \tau), \quad (\text{D.124})$$

with $\beta(\theta_v, \tau) = e^{-\rho_{\text{chain}}(1-\rho_{\theta_v})\tau}$.

$M \geq 1$: *at least one field jump*. Under Interp 2 the field-CTMC generator has state-dependent exit rate $\rho_{\text{chain}}(1-\rho_\theta)$; conditional on any jump occurring, the destination distribution is $\rho_{\theta_u}/(1-\rho_{\theta_v})$ for $\theta_u \neq \theta_v$. Aggregating over $M \geq 1$ we may write:

$$\sum_{M \geq 1} K^{(M)}(x_u, \theta_u \mid x_v, \theta_v; \tau) = W(\theta_v, \theta_u; \tau) \prod_n \pi^{(\text{arch}[c_n, \theta_u])}(x_{u,n}), \quad (\text{D.125})$$

i.e., the “regenerated” component of the branch kernel. The per-site product $\prod_n \pi^{(\text{arch}[c_n, \theta_u])}(x_{u,n})$ expresses the collapse assumption that after any field jump the substitution CTMC is at stationary conditional on the current archetype (the “carry” vs. “resample” choice of Section D.8.3), which under Interp 2 corresponds to a full re-equilibration per jump.

Case-conditional HR primitives. On the branch, for each pair (θ_v, θ_u) , the case- $M=0$ contribution to HR statistics attributes all dwell time and transitions to archetype $\text{arch}[c_n, \theta_v]$ for site n . This is exactly the standard *whole-branch GTR bridge* HR pass at site n : given endpoint residues $(x_{v,n}, x_{u,n})$ and rate $Q^{(c_n, \theta_v)}$, the per-site sufficient stats $(W_n^{(0)}, U_n^{(0)})$ are computed by the reversible-bridge closed forms of Section D.8.5, with the “segment start” count $V_n^{(0)}$ receiving a $\delta_{a=x_{v,n}}$ contribution (since the segment starts at the observed root-side residue).

The case- $M \geq 1$ contribution is more intricate: the branch is partitioned into a first segment $[0, \tau_1]$ under archetype $\text{arch}[c_n, \theta_v]$, a middle chain of jumps at random times with free intermediate

archetypes drawn from $\text{arch}[c_n, \theta_{\text{int}}]$ with $\theta_{\text{int}} \sim \rho$, and a last segment $[\tau_M, \tau]$ under archetype $\text{arch}[c_n, \theta_u]$. The whole-branch HR contributions decompose as (dropping the site index n for readability):

$$V_a^{(M \geq 1)} = \delta_{a=x_v} + \left(\sum_{\theta_{\text{int}}} \text{Arr}(\theta_{\text{int}}; \tau) \right) \cdot \pi^{(\text{arch}[c_n, \theta_{\text{int}}])}(a) + P_{\text{avg}}^{\text{last}}(x_u; a; \tau), \quad (\text{D.126})$$

$$W_i^{(M \geq 1)} = W_i^{\text{first}}(x_v; \theta_v; \tau_1) + \mathbb{E}[\tau_{\text{mid}}] \cdot \pi^{(\text{arch}[c_n, \theta_{\text{int}}])}(i) + W_i^{\text{last}}(x_u; \theta_u; \tau - \tau_M), \quad (\text{D.127})$$

$$U_{i,j}^{(M \geq 1)} = U_{i,j}^{\text{first}}(x_v; \theta_v; \tau_1) + \mathbb{E}[\tau_{\text{mid}}] \cdot \pi^{(\text{arch}[c_n, \theta_{\text{int}}])}(i) \cdot S_{i,j} \cdot \pi^{(\text{arch}[c_n, \theta_{\text{int}}])}(j) + U_{i,j}^{\text{last}}(x_u; \theta_u; \tau - \tau_M), \quad (\text{D.128})$$

with Arr , $P_{\text{avg}}^{\text{last}}$, and the first/last-segment HR primitives all computed by the case-1 closed forms of Sections D.8.5–D.8.5 (specifically (D.55)–(D.69) for $W^{\text{first, last}}$, $U^{\text{first, last}}$, (D.62)–(D.63) for $E[\tau_1]$, $E[\tau - \tau_M]$, and (D.65) for $\mathbb{E}[\tau_{\text{mid}}]$). The middle-segment attribution uses the stationary of the intermediate archetype for both dwell and transitions, weighted by the expected number of arrivals at that θ_{int} .

The V-attribution term $P_{\text{avg}}^{\text{last}}(x_u; a; \tau)$ is the expected starting-residue distribution of the last segment, computed by the “transition-avg” closed form (D.57): it is not $\delta_{a=x_u}$ because the last segment starts *internally* on the branch (not at the observed x_u), so the segment-start residue is averaged over the last-jump-time distribution.

Spectral acceleration. The per- (c, k) HR primitives W^{first} , W^{last} , U^{first} , U^{last} have closed forms in the eigenspace of the GTR generator: given the eigendecomposition $Q^{(c_n, \text{arch}[c_n, \theta])} = D_{\pi^{(c_n, \theta)}}^{1/2} U_{c_n, \theta} \Xi_{c_n, \theta} U_{c_n, \theta}^\top D_{\pi^{(c_n, \theta)}}^{-1/2}$ (Section D.8.5), the per-segment integrals in (D.127)–(D.128) factor into a sum over eigenpair-index pairs (a, b) of the elementary H and T integrals (D.49)–(D.50). The per-branch cost is thus $O(L^2 \cdot m \cdot A^2)$ (assembling the per-site pair distribution) plus $O(L^2 \cdot m \cdot A)$ (spectral integration), rather than the naive $O(A^{2m})$ or even $O(A^4)$ per-site cost that a direct enumeration of the (x_v, x_u) pairs would require. This mirrors the “accumulate sums in eigenspace” trick of the composite HR pass (Algorithm 4).

Obtaining HR sufficient statistics from the ELBO’d tree. The M-steps for the archetype parameters (D.85) and ρ_{chain} (D.86) consume aggregated HR sufficient statistics $\{V_{c, \theta, \cdot}, U_{c, \theta, \cdot}, W_{c, \theta, \cdot}\}$, $\sum_q N_\theta[q]$, $\sum_q T_q$. Under composite training these are accumulated over the cluster HR pass of Algorithm 4, one HR pass per cherry. Under tree training the same statistics are accumulated one HR pass per *branch* of the tree, with the branch endpoints’ $(\theta, r_1, \dots, r_m)$ distributions read from the converged variational marginal q .

For each branch $v \rightarrow u$ of length τ_{vu} , given the converged forward CLVs F_v, F_u and backward CLVs D_v, D_u (Section D.8.5):

1. Compute the joint pair posterior $q_{vu}(x_v, \theta_v, x_u, \theta_u)$ via (D.122). Under the moment-matching family it factorises across sites n conditional on (θ_v, θ_u) ; extract the four-term per-site pair distribution $q_{vu, n}(x_{v, n}, x_{u, n} \mid \theta_v, \theta_u)$ as a mixture over $\{\text{no-jump/jump}\} \times \{\text{tracking}_v/\text{regenerated}_v\} \times \{\text{tracking}_u/\text{regenerated}_u\}$ (some cells vanish under $\delta_{\theta_v=\theta_u}$ on the no-jump branch), giving the field-pair weight

$$w^{vu}(\theta_v, \theta_u) = \sum_{x_v, x_u} q_{vu}(x_v, \theta_v, x_u, \theta_u). \quad (\text{D.129})$$

The residue pair distribution conditional on (θ_v, θ_u) factorises across sites.

2. For each pair (θ_v, θ_u) and each site n , compute the per-site HR contributions using the case-conditional primitives of Section D.8.5:

- $\theta_v = \theta_u$ (no jump possible along with the δ -collapse, weight $\beta \cdot q_{vu,n}$): apply the whole-branch GTR bridge HR primitive of Section D.8.5 at archetype $\text{arch}[c_n, \theta_v]$, endpoints $(x_{v,n}, x_{u,n})$ integrated against the per-site pair distribution.
 - $M \geq 1$ -jump contribution (any θ_v, θ_u): apply (D.126)–(D.128), integrated against the same per-site pair distribution but with the residue-marginal at x_u taken from $\pi^{(\text{arch}[c_n, \theta_u])}$ (regenerated component).
3. Weight the per- (θ_v, θ_u) contributions by (D.129) and sum across all field-pair contexts, all sites n , and all branches, indexing into the archetype bins by $k = \text{arch}[c_n, \theta]$ (with $\theta = \theta_v$ for first-segment attribution, θ_u for last-segment, and θ_{int} weighted by $\text{Arr}(\theta_{\text{int}})$ for middle-segment):

$$V[k, a] += \sum_{vu, n, \theta_v, \theta_u} w^{vu}(\theta_v, \theta_u) [\text{arch}[c_n, \theta_v] = k] V_{n, k, a}^{(vu, \theta_v, \theta_u)}, \quad (\text{D.130})$$

$$U[k, i, j], W[k, i] \text{ analogously.} \quad (\text{D.131})$$

The field-jump counts and time sums receive

$$\sum_q N_\theta[q] += \sum_{vu, \theta_v, \theta_u} w^{vu}(\theta_v, \theta_u) \mathbb{E}[N_\theta \mid \theta_v, \theta_u, \tau_{vu}], \quad \sum_q T_q += \sum_{vu} \tau_{vu}, \quad (\text{D.132})$$

with $\mathbb{E}[N_\theta \mid \theta_v, \theta_u, \tau_{vu}]$ from the bridge-conditional field expectation (D.70).

Under the moment-matching family the per-branch cost is dominated by the eigenspace expansion of the per- (θ_v, θ_u) HR primitives at each of m sites: $O(L^2 \cdot m \cdot A^2)$ per branch, matching the forward and backward mm-edge cost. The overall per-iter cost per cluster is thus $O(\text{edges} \cdot L^2 \cdot m \cdot A^2)$ times a small coordinate-ascent factor.

The archetype $\pi^{(k)}$ Newton MAP step (D.85) and the ρ_{chain} closed-form step (D.86) are then unchanged: both consume the aggregated (V, U, W, N_θ, T) , whose *source* has moved from independent cherries to tree-branch expectations under the variational posterior. The M-step targets an ELBO-consistent lower bound on the true tree likelihood (rather than the composite likelihood), so the fixed points of the resulting SEM loop are stationary points of the tree-ELBO objective.

Reduction to composite training. If the tree is a forest of independent depth-1 cherries with no shared internal ancestors, the variational family (D.91) is exact (each cherry runs its own two-leaf projection), and the branch-summed HR statistics above reduce identically to the cherry HR statistics of Algorithm 4. Composite training is therefore the depth-1 special case of tree-ELBO training. Any Pfam MSA represented as a star tree (a single internal ancestor with Q leaf cherries) already forbids the composite-likelihood freedom, since all Q cherries must share the same θ posterior at the root node.

Class assignments. The per-site class c_n is fixed within a tree-ELBO sweep, following the composite training regime (Section D.8.5, sampled by the outer Gibbs sweep). A mean-field per-site class marginal $q_{n,v}(c_n) \in \Delta^{K_c-1}$ could be added to the per-node variational state at cost $O(K_c \cdot L \cdot m \cdot A^2)$ per edge, but at training corpus scale the sampled hard assignment avoids the diffuse-marginal failure mode observed in strong-signal regimes with $(c_i^*, c_j^*) \leftrightarrow (c_j^*, c_i^*)$ symmetry, and the class Gibbs is more cost-effective.

Trigger for tree-ELBO training. The composite (cherry) regime is adequate when (a) the corpus is presented as a set of aligned pairs without tree structure, or (b) the composite-likelihood artifact diagnostics (cross-cherry θ_P entropy vs. mean within-cherry jump probability) show no artifact. When either condition fails — as in the training runs reported in the main text, where the diagnostic reports $\geq 65\%$ of clusters in the artifact quadrant — tree-ELBO training becomes the preferred regime.

Rate heterogeneity: discretised- Γ plus invariant mixture on the field-flip rate. The base model of the previous paragraphs assumes a single global ρ_{chain} governs the field-flip rate for every cluster. Adapting the classical Yang (70, 71) $+\Gamma + I$ rate-heterogeneity framework to the archetype trainer, we introduce a per-cluster latent $b_q \in \{\text{inv}\} \cup \{1, 2, \dots, K_r\}$ that scales the flip rate:

$$\rho_{\text{chain}}^{(q)} = \begin{cases} 0, & b_q = \text{inv}, \\ \rho_{\text{chain}} \cdot m_{b_q}, & b_q \in \{1, \dots, K_r\}, \end{cases} \quad (\text{D.133})$$

where $\{m_1, \dots, m_{K_r}\}$ are the K_r equal-mass quantile means of a $\Gamma(\alpha_\gamma, \alpha_\gamma)$ distribution (so $E[m] = 1$; α_γ controls the variance across bins). The mixing weights are $\Pr[b_q = \text{inv}] = p_{\text{inv}}$ and $\Pr[b_q = k] = (1 - p_{\text{inv}})/K_r$ for the Γ bins, mirroring Yang’s $+\Gamma + I$ default.

Interpretation. The invariant bin ($b_q = \text{inv}$) models clusters whose field-selector trajectory is frozen for the entire branch — structurally invariant sites in Yang’s original substitution-rate application transfer directly to structurally non-switching cluster contexts here. The K_r Γ bins give the model flexibility to have both slow-mixing clusters (low m , mostly one archetype for the branch) and fast-mixing clusters (high m , walking through multiple archetypes) share a single ρ_{chain} base rate, with the shape parameter α_γ learned from data.

Standard choice. $K_r = 4$, following Yang (71); this and the single $p_{\text{inv}} / \alpha_\gamma$ scalar pair adds just three global parameters to the model, without altering the archetype vocabulary $\{\pi^{(k)}\}$ or field-state marginals ρ . The version implemented in this work uses $K_r = 4$ Gamma bins plus one invariant bin ($4\Gamma + I$).

E-step. For each cluster q , evaluate the marginal likelihood under each of the $K_r + 1$ candidate rates (invariant + Gamma bins), either by running the HR pass $K_r + 1$ times with different $\rho_{\text{chain}}^{(q)}$ or by conditioning ρ_{chain} inside a vectorised HR call. The soft posterior is

$$q_q(b) \propto \Pr[b_q = b] \cdot P(\text{obs}_q \mid b_q = b, \pi, \text{arch}, \rho, \rho_{\text{chain}}), \quad (\text{D.134})$$

computable in closed form from the HR pass at each rate.

M-step. The HR sufficient statistics $\{V_{c,\theta}, U_{c,\theta}, W_{c,\theta}\}$ aggregate over clusters with weights $q_q(b)$ (equivalently: each cluster contributes $K_r + 1$ weighted copies of its stats). The Newton update (D.85) on $\pi^{(k)}$ proceeds unchanged. The Gamma-posterior ρ_{chain} update (D.86) becomes

$$\rho_{\text{chain}} \leftarrow \frac{\alpha_\rho - 1 + \sum_q \sum_k q_q(k) N_\theta[q, k]}{\beta_\rho + \sum_q \sum_k q_q(k) m_k T_q}, \quad (\text{D.135})$$

where the sum over k omits the invariant bin (which contributes zero to both numerator and denominator). The p_{inv} update is the Beta-conjugate closed form $p_{\text{inv}} \leftarrow (\sum_q q_q(\text{inv}) + a_p - 1) / (Q + a_p + b_p - 2)$ under a Beta(a_p, b_p) prior; α_γ is updated by Newton on the marginal log-likelihood, given the fixed bin weights and $\{N_\theta[q, k], T_q\}$ sufficient statistics.

Expected impact. Structural specialisers under the base model (classes whose $\text{arch}(c, \theta)$ is constant across θ) currently occupy the low- ρ_{chain} tail; under (D.133) they will preferentially assign to the invariant or lowest- m bin. Dynamic mediators (classes that visit multiple archetypes across θ) will migrate to the higher- m bins, sharpening both the archetype-usage separation and the compensatory-flip signal (Section D.8.5).

Per-site $\Gamma + I$ rate heterogeneity on the substitution rate. Alongside the per-cluster $\Gamma + I$ mixture on the field-flip rate, we add a second, independent $\Gamma + I$ mixture at the per-site level that scales the archetype substitution rate. This mirrors Yang’s original (70) $+\Gamma + I$ on the GTR substitution matrix and is the standard treatment of positional rate heterogeneity in phylogenetics.

For each column s in the corpus, introduce a latent $b_s \in \{\text{inv}\} \cup \{1, \dots, K_r^{\text{site}}\}$ that scales the site's substitution rate:

$$Q^{(k,s)} = \begin{cases} \mathbf{0}, & b_s = \text{inv}, \\ m_{b_s}^{\text{site}} \cdot Q^{(k)}, & b_s \in \{1, \dots, K_r^{\text{site}}\}, \end{cases} \quad (\text{D.136})$$

where $k = \text{arch}(c_s, \theta_s)$ is the archetype visible at column s when the field selector is θ_s , and $m_1^{\text{site}}, \dots, m_{K_r^{\text{site}}}^{\text{site}}$ are the quantile means of a $\Gamma(\alpha_\gamma^{\text{site}}, \alpha_\gamma^{\text{site}})$ distribution (mean 1). Bin priors are $\Pr[b_s = \text{inv}] = p_{\text{inv}}^{\text{site}}$ and $\Pr[b_s = k] = (1 - p_{\text{inv}}^{\text{site}})/K_r^{\text{site}}$. *Composition.* The per-site and per-cluster mixtures act on different rate constants (site-substitution versus field-flip) and are prior-independent; the effective per-(cluster, site) rate is $m_{b_s}^{\text{site}} \cdot Q^{(k)}$ for the residue CTMC and $\rho_{\text{chain}}^{(q)}$ for the field CTMC. *Invariant behaviour (witness-site convention).* When $b_s = \text{inv}$, the site has no continuous substitution ($Q^{(k,s)} = \mathbf{0}$), so between field jumps the residue is preserved: the Case 0 (no-jump) emission collapses to $\llbracket Y_s = X_s \rrbracket$ via the $m_{b_s}^{\text{site}} \xi^{(k)} = 0$ eigenvalue limit. At each field jump, however, the residue resamples from the stationary at the new archetype $\pi^{(k(c_s, \theta_Y))}$, exactly like a variant-rate site under Interpretation 1. The $M \geq 1$ (jump) emission is therefore the same for all bins: $\pi_{Y_s}^{(k(c_s, \theta_Y))}$. This is a deliberate departure from Yang's (1994) original "+I = no evolution at all" semantics. Two motivations: (i) *covariation-signal clarity* — for an invariant-rate site whose observed residue nonetheless changes between X and Y , the only remaining explanation is a field jump; if multiple invariant sites in the same cluster co-change, that is unambiguous evidence of a coordinated field transition (they act as "witness sites" for the field). Under the old "never changes" convention this signal was folded into the likelihood-zero regime and lost; and (ii) *mechanism separation* — 100%-conserved sites in the witness-site convention are captured through a distinct mechanism: a singleton cluster whose per-cluster rate bin (Eq. D.133) is invariant, so its field never jumps. This factorises "site rate" and "cluster field-flip rate" instead of conflating them. Reversibility is preserved: an invariant bin is trivially reversible w.r.t. any stationary, and the Case ≥ 1 formula is the same as for variant bins.

Efficient inference by rate-eigenvalue scaling. Since $Q^{(k,s)}$ shares the eigenvectors $U^{(k)}, D^{(k)}$ of $Q^{(k)}$ and differs only by scaling the eigenvalues by $m_{b_s}^{\text{site}}$, every closed-form HR primitive (bridge, free-end, and case-1 transition average) can be evaluated at any bin by substituting $\xi^{(k)} \mapsto m_{b_s}^{\text{site}} \cdot \xi^{(k)}$ inside the per-site call. No re-diagonalisation is required; the K_a archetype eigendecompositions computed once per outer iter (cached until the next arch M-step) are reused across all $(K_r^{\text{site}} + 1)$ bins and all sites. Numerically this makes per-site bin evaluation $O(A^2)$ per (bin, site), a factor of A faster than a naive per-bin re-diagonalisation.

Per-site update strategy. The per-site bin b_s is treated as persistent state; the HR pass applies each site's rate multiplier $m_{b_s}^{\text{site}}$ to $\xi^{(k)}$ at every iteration without additional per-bin evaluation cost. Two update regimes are available, controlled by a site-resampling period R :

1. Γ -only init, no resample ($R = 0$). Bins are drawn uniformly over the Γ quantile bins $(1 \dots K_r^{\text{site}})$ at training start and held fixed. The invariant bin ($b_s = \text{inv}$) is *not* used at initialisation: assigning bin 0 to a column whose observed (X, Y) is variant ($X \neq Y$ at any cherry) zeroes the Case 0 factor at that cluster on the very first HR pass; the resulting drop in Case-0 log-likelihood is severe enough (though not $-\infty$: Case ≥ 1 still contributes $\pi_Y^{(k(c, \theta_Y))} > 0$) to make the atom step at iter 1 diverge in practice. A safe uniform Γ init sidesteps this.
2. Γ -init + infrequent Gibbs ($R > 0$). Same Γ -only init; then every R outer iters, bin b_s is refreshed by a partial Gibbs step using the Rao-Blackwellised conditional (Eq. D.137). The Gibbs conditional evaluates data-conditional per-site log-likelihoods over all $K_r^{\text{site}} + 1$ bins and

downweights bin 0 on variant columns (where Case 0 pays a large penalty) while upweighting bin 0 on invariant-observation columns in slow clusters (where Case 0 = 1 dominates). Bin 0 is thus introduced only where the data supports it, avoiding the initialisation blow-up of a naïve $+\Gamma + I$ prior draw.

Historical note on the invariant-bin encoding. An earlier version of this section used the strict Yang '94 semantics for invariance (residue preserved in *both* no-jump and jump cases). That model class made a per-cherry Gibbs step unsafe: a column whose observed (X, Y) was $X = Y$ in one cherry and $X \neq Y$ in another could not be assigned $b_s = \text{inv}$ without zeroing the cluster likelihood on the second cherry. The witness-site convention above removes this obstruction (invariant bin now emits $\pi_Y^{(k(c, \theta_Y))} > 0$ in the jump case) and enables $+\Gamma + I$ under the per-cherry Gibbs directly. The pre-witness-site Γ -only sub-model is recovered as an A/B baseline by disallowing the invariant bin. The Gibbs conditional is Rao-Blackwellised over the field trajectory:

$$\tilde{q}_s(b) \propto \Pr[b_s = b] \cdot \sum_{\theta_X, \theta_Y} w_{\theta_X, \theta_Y}^{(q)} \cdot L_{\text{site}}^{(b)}(s; \theta_X, \theta_Y), \quad (\text{D.137})$$

where q is the cluster containing s , $w_{\theta_X, \theta_Y}^{(q)}$ is the pair weight $\rho(\theta_X) P_{\theta_X \rightarrow \theta_Y}(\tau_q)$ produced by the HR pass, and

$$L_{\text{site}}^{(b)}(s; \theta_X, \theta_Y) = \begin{cases} \mathbb{K}[X_s = Y_s], & b = \text{inv}, \\ P_{X_s Y_s}^{k_s, m_b^{\text{site}}}(\tau_q), & b = 1..K_r^{\text{site}} \text{ and } \theta_X = \theta_Y, \\ \pi_{Y_s}^{(k_s(\theta_Y))}, & b = 1..K_r^{\text{site}} \text{ and } \theta_X \neq \theta_Y, \end{cases}$$

with $k_s(\theta) = \text{arch}(c_s, \theta)$. This approximation ignores between-site coupling under the field posterior (which is weak under Interpretation 1: sites are conditionally independent given the field trajectory, and between-site coupling only arises through the pair-averaged posterior over (θ_X, θ_Y)).

M-step for $p_{\text{inv}}^{\text{site}}, \alpha_\gamma^{\text{site}}$. Under a Beta(a_p, b_p) prior: $p_{\text{inv}}^{\text{site}} \leftarrow (\sum_s \mathbb{K}[b_s = \text{inv}] + a_p - 1) / (S_{\text{total}} + a_p + b_p - 2)$, where S_{total} is the total number of columns across the corpus. $\alpha_\gamma^{\text{site}}$ is held fixed at its initialisation (Yang's (1996) original recommendation; can be swept externally).

Summary of cost. Marginalising per-site bins at every outer iter would multiply HR-pass work by $(K_r^{\text{site}} + 1)$; the persistent-plus-infrequent-Gibbs scheme reduces this to $1 + K_r^{\text{site}}/R \approx 1.8 \times$ amortised for $K_r^{\text{site}} = 4$, $R = 5$, giving essentially the runtime of the base model plus a constant every R iters for the resample.

D.8.6 Variational ELBO for general phylogenies

Proposition D.8 is tight at the cap-2 cluster geometry that TKF-DP training uses: the per-node conditional likelihood at a cherry parent has the closed-form decomposition

$$F^v(x_v, \theta_v) = \beta_v(\theta_v) \prod_{n=1}^m \alpha_n^v(x_{v,n}, \theta_v) + \gamma_v(\theta_v) \quad (\text{D.138})$$

— a per-site rank-1 term (the cumulative no-event branch) plus a residue-constant scalar (the regenerated branch); per-edge propagation preserves this form. On phylogenies of depth > 1 , the combine at an internal node v with children u_1, u_2 multiplies the rank-1 count, $K_v = K_{u_1} K_{u_2} + K_{u_1} + K_{u_2}$, giving cost super-exponential in tree depth. A structured mean-field variational restriction recovers linear cost while remaining exact at cap-2.

Variational family. At every tree node v restrict the joint (x_v, θ_v) marginal to the per- θ mixture of a per-site factored marginal and the dynfield stationary:

$$q_v(x_v, \theta_v) = q_v(\theta_v) \left[(1 - \lambda_v(\theta_v)) \prod_{n=1}^m p_n^v(x_{v,n} | \theta_v) + \lambda_v(\theta_v) \prod_{n=1}^m \pi^{(c_n, \theta_v)}(x_{v,n}) \right]. \quad (\text{D.139})$$

Per-node variational parameters: $q_v(\theta_v) \in \Delta^{L-1}$, $\lambda_v(\theta_v) \in [0, 1]^L$, and $p_n^v(\cdot | \theta_v) \in \Delta^{k-1}$ for $n = 1, \dots, m$, total $O(L \cdot m \cdot k + 2L)$ per node. The interpretation of $\lambda_v(\theta_v)$ is the posterior probability that at least one field event has occurred on the path from v back to the last “tracking” ancestor; in the regenerated component residues sit at the model’s joint stationary.

ELBO.

$$\begin{aligned} \mathcal{L}(q) = & \sum_{\text{leaves } v} \mathbb{E}_{q_v} [\log p(\text{obs}_v | x_v, \theta_v)] \\ & + \sum_{\text{edges } u \rightarrow v} \mathbb{E}_{q_v, q_u} [\log K(\phi_v, \phi_u; \tau)] - \sum_{\text{nodes } v} \mathbb{E}_{q_v} [\log q_v(\phi_v)] + \log Z_{\text{prior}}. \end{aligned} \quad (\text{D.140})$$

Each term has a closed form under (D.139): leaf-emission and node-entropy at $O(L \cdot m \cdot k)$ (the mixture entropy uses the standard component-wise upper bound, tight when $\lambda \in \{0, 1\}$); per-edge cross-entropy at $O(L^2 \cdot m \cdot k^2)$ with the per-site $\mathbf{Q}$ -evolution and field-transition kernel both factored over sites given the endpoint fields (θ_v, θ_u) .

Per-edge propagation. For each parent-child pair $u \rightarrow v$ of length τ , the exact propagation (D.138) of one (rank-1 + scalar) form to another is closed form; under the variational family the propagation simplifies to per-site $\mathbf{Q}$ -matrix actions on $p_n^u(\cdot | \theta)$ followed by a renormalisation. Cost per edge: $O(m \cdot L \cdot k^2 + L^2)$.

Internal-node projection (M-projection: moment matching). At an internal node v with children u_1, u_2 , the exact multiplicative combine of two (rank-1 + scalar) child contributions yields four components (3 rank-1 + 1 scalar). The variational M-projection restores the form (D.139) as the $\text{KL}(p_v \| q_v)$ minimiser:

- $q_v(\theta_v)$ to the θ_v -marginal of the exact four-term product;
- $p_n^v(x_{v,n} | \theta_v)$ to the per-site, per- θ_v residue marginal of the exact product;
- $\lambda_v(\theta_v)$ to the proportion of total mass in terms involving at least one child’s scalar component.

All updates are closed form; $O(m \cdot L \cdot k)$ per binary combine. k -ary internal nodes are handled as $k - 1$ sequential binary combines with projection after each.

Internal-node projection (I-projection: KL minimisation). M-projection produces unbiased marginals but can systematically overestimate $\lambda_v(\theta_v)$ when the exact p_v has strong residue-level structure (matching scalar mass via mass proportion alone ignores the residue-level information that distinguishes the rank-1 mode from the regenerated mode), loosening the ELBO and risking pathological over-clustering bias downstream. The tighter alternative is I-projection: minimise $\text{KL}(q_v \| p_v)$ directly. Writing $\mathcal{T}(x_v | \theta_v) := (1 - \lambda_v(\theta_v)) \prod_n p_n^v(x_{v,n} | \theta_v) + \lambda_v(\theta_v) \prod_n \pi^{(c_n, \theta_v)}(x_{v,n})$ (so $q_v(x_v, \theta_v) = q_v(\theta_v) \mathcal{T}(x_v | \theta_v)$), the ELBO at node v decomposes as

$$\text{ELBO}_v = \sum_{\theta_v} q_v(\theta_v) \left[\mathbb{E}_{\mathcal{T}(\cdot | \theta_v)} [\log p_v(\cdot, \theta_v)] - \log q_v(\theta_v) + \text{H}[\mathcal{T}(\cdot | \theta_v)] \right]. \quad (\text{D.141})$$

The per-parameter coordinate-ascent gradients (with simplex multipliers μ for $\sum q_v = \sum p_n^v = 1$ suppressed):

$$\frac{\partial \text{ELBO}_v}{\partial q_v(\theta_v)} = \mathbb{E}_{\mathcal{T}(\cdot|\theta_v)}[\log p_v(\cdot, \theta_v)] - \log q_v(\theta_v) - 1 + H[\mathcal{T}(\cdot|\theta_v)], \quad (\text{D.142})$$

$$\frac{\partial \text{ELBO}_v}{\partial \lambda_v(\theta_v)} = q_v(\theta_v) [\mathbb{E}_{\pi^{\theta_v}}[\log p_v/\mathcal{T}](\cdot, \theta_v) - \mathbb{E}_{\mathbf{p}^v}[\log p_v/\mathcal{T}](\cdot, \theta_v)], \quad (\text{D.143})$$

$$\frac{\partial \text{ELBO}_v}{\partial p_n^v(a|\theta_v)} = q_v(\theta_v) (1 - \lambda_v(\theta_v)) \mathbb{E}_{\mathbf{p}_{-n}^v(\cdot|\theta_v)}[\log \frac{p_v(a, x_{-n}, \theta_v)}{\mathcal{T}(a, x_{-n}|\theta_v)}], \quad (\text{D.144})$$

where $\pi^{\theta_v}(x) := \prod_n \pi^{(c_n, \theta_v)}(x_n)$, $\mathbf{p}^v(x|\theta_v) := \prod_n p_n^v(x_n|\theta_v)$, and $\mathbf{p}_{-n}^v$ omits site n . The $q_v(\theta_v)$ stationarity (D.142) gives the closed-form update

$$q_v(\theta_v) \propto \exp(\mathbb{E}_{\mathcal{T}(\cdot|\theta_v)}[\log p_v(\cdot, \theta_v)] + H[\mathcal{T}(\cdot|\theta_v)]). \quad (\text{D.145})$$

The λ_v stationarity (D.143) is the “equal expected log-evidence ratio under both mixture components” condition: at the stationary λ , the regenerated and tracking components contribute the same average $\log p_v/\mathcal{T}$. No closed form in general; solve by bisection on $\lambda_v(\theta_v) \in [0, 1]$ (the expression is monotone in λ once $\mathcal{T}$ is updated coherently). The p_n^v stationarity (D.144) is an exponential-family fixed point: $\log p_n^v(a|\theta_v)$ equals the $\mathbf{p}_{-n}^v$ -averaged log of the $(p_v/\mathcal{T})$ ratio at site n , up to a constant.

The mixture entropy $H[\mathcal{T}]$ in (D.141) has no closed form in general but admits the Jensen lower bound $H[\mathcal{T}(\cdot|\theta_v)] \geq (1 - \lambda_v)H[\mathbf{p}^v(\cdot|\theta_v)] + \lambda_v H[\pi^{\theta_v}]$, tight at $\lambda_v \in \{0, 1\}$. Substituting the bound makes (D.141) and the gradients fully closed form at the cost of a slightly looser ELBO. In practice M-projection (D.139) provides a fast warm start; I-projection fixed-point updates tighten the bound at additional per-sweep cost and can be invoked when downstream behaviour indicates pathological over-clustering bias. The two projections agree exactly at the cap-2 cherry parent: the exact F^v is already in the variational family, so all four gradient expressions vanish at the exact solution under both projections.

Coordinate ascent. Standard message-passing: post-order sweep (combine + project at internal nodes), pre-order sweep (downward updates), iterate. Per sweep $O(N \cdot m \cdot L \cdot k^2)$. The ELBO is non-decreasing; typical convergence in 10–50 sweeps.

Certified bound from the moment-matching marginals, without iteration. Because $\mathcal{L}(q)$ in (D.140) is a valid lower bound on $\log P(\text{data})$ for *any* q in the family (D.139), the M-projected marginals $\{q_v(\theta), p_n^v(\cdot|\theta), \lambda_v(\theta)\}$ from a single upward moment-matching pass already define such a q : plugging them into (D.140) yields a certified bound with *no* coordinate-ascent iteration. The essential caveat is that this is the ELBO functional *evaluated at* the marginals — one extra tree pass accumulating the leaf-emission terms, the per-edge cross-entropies $\mathbb{E}_{q_u, q_v}[\log K]$, the per-node entropies, and $\log Z_{\text{prior}}$ — and is *not* the log-normaliser $\log Z_{\text{MM}}$ that the moment-matching forward itself accumulates while combining and propagating messages. The two are distinct quantities:

$$\mathcal{L}(q_{\text{MM}}) \leq \log P(\text{data}) \quad \text{always, whereas} \quad \log Z_{\text{MM}} \leq \log P(\text{data}), \quad (\text{D.146})$$

since $\log Z_{\text{MM}}$ is an expectation-propagation-style marginal-likelihood *estimate* that can fall on either side of the truth. Only the functional $\mathcal{L}(q_{\text{MM}})$ is a bound. The evaluation pass costs the same $O(N \cdot m \cdot L \cdot k^2)$ as one message-passing sweep, and bound validity is preserved under the Jensen entropy lower bound of (D.141) (lower-bounding H only lowers $\mathcal{L}$). At the cap-2 cherry

geometry $\mathcal{L}(q_{\text{MM}}) = \log P(\text{data})$ exactly (no projection fires and $q_{\text{MM}} = q^*$); at depth > 1 the M-projection’s tendency to overestimate λ_v merely loosens the bound without invalidating it, so the certified moment-matching bound is the recommended default and the I-projection sweeps (D.145)–(D.144) are reserved for tightening only when downstream over-clustering bias indicates the λ slack matters.

Tree-structured refinement: parent–child correlation at mean-field cost. The family (D.139) is mean-field *over nodes*: the edge term $\mathbb{E}_{q_u, q_v}[\log K]$ in (D.140) averages over *independent* node marginals, discarding the parent–child correlation. One would expect a tree-structured q that instead carries branch conditionals $q(x_v, \theta_v | x_u, \theta_u)$ to reinstate the super-exponential rank growth $K_v = K_{u_1} K_{u_2} + K_{u_1} + K_{u_2}$ that the mean-field restriction was introduced to tame. It does not, because the field couples the m sites *only* through a single per-branch event. Decompose each branch by its field-jump indicator $\Delta \in \{0, 1\}$ ((D.31)–(D.33), with $\Delta = 1$ collecting $M \geq 1$): conditioned on $(\theta_u, \theta_v, \Delta)$ the sites evolve independently, so the cheapest correlation-preserving branch conditional is the two-component mixture

$$q(x_v, \theta_v | x_u, \theta_u) = \sum_{\Delta \in \{0, 1\}} q(\Delta, \theta_v | \theta_u) \prod_{n=1}^m q_{\Delta}(x_{v,n} | x_{u,n}, \theta_u, \theta_v), \quad (\text{D.147})$$

with a per-site tracking kernel $q_0(\cdot | x_{u,n})$ for the no-jump component ($\Delta = 0$, forcing $\theta_v = \theta_u$) and the field-renewed stationary $q_1(x_{v,n}) = \pi^{(c_n, \theta_v)}(x_{v,n})$, independent of $x_{u,n}$, for the regenerated component ($\Delta = 1$). This is exactly the rank-1-plus-scalar form (D.138) read as a variational *branch conditional*: the $\Delta = 0$ term is the per-site rank-1 carry, the $\Delta = 1$ term the residue-constant shared renewal. Its cost is $O(mk^2)$ per branch — the same order as the mean-field edge term — because Δ refactorises the branch over sites; the parent–child correlation is carried entirely by the shared Δ (all sites track together, or all renew together, the “simultaneous across-sites renewal” of the dynamic-field jump). Internal-node combines project the four-component product back onto the two- Δ -component form (D.147) — an M- or I-projection of the *conditional* rather than of the marginal — so the rank stays bounded at 2 instead of growing with depth. Because (D.147) recovers the node mean-field (D.139) when the tracking kernel q_0 is frozen to the product marginal, its optimised ELBO is never looser and generically strictly tighter, at the same asymptotic cost, and remains exact on cap-2 cherries (where no projection fires). The rank-1-plus-scalar propagation is thus not merely a cap-2 computational shortcut but the exact functional form of the cheapest correlation-preserving variational branch conditional for the dynamic field.

Field-exact structured mean-field: the tight, tree-correlated bound. The node mean-field (D.140) and the tree-structured branch conditional above still *approximate the field*; but the field carries only L states, so it is cheap to keep *exact* while mean-fielding only the $\mathcal{A}^m$ residues. Take

$$q(\boldsymbol{\theta}, \mathbf{x}_{\text{int}}) = q_{\Theta}(\boldsymbol{\theta}) \prod_{v \text{ internal}} \prod_{n=1}^m q_{v,n}(x_{v,n} | \theta_v), \quad (\text{D.148})$$

where q_{Θ} is an *unrestricted tree-structured* (Markov-on-the-tree) distribution over the whole field configuration — carried by its node marginals $q_v(\theta_v)$ and edge marginals $q_{uv}(\theta_u, \theta_v)$ — and $q_{v,n}(\cdot | \theta_v)$ is the mean-field residue marginal at internal node v , site n , conditioned on the *local* field state (leaves fix x to the observation). Crucially q_{Θ} is *not* factorised across nodes: the parent–child field correlation that the node mean-field discards is retained in full.

Writing $\mu_{v,n}(\cdot | \theta) := q_{v,n}(\cdot | \theta)$ at internal nodes and $\mu_{v,n} := \delta_{x_{v,n}^{\text{obs}}}$ at leaves, the ELBO is

$$\begin{aligned} \mathcal{L} = & \sum_{\theta} q_r(\theta) \left[\log \rho(\theta) + \sum_n \sum_a \mu_{r,n}(a | \theta) \log \pi^{(c_n, \theta)}(a) \right] \\ & + \sum_{u \rightarrow v} \sum_{\theta_u, \theta_v} q_{uv}(\theta_u, \theta_v) \left[\log P_{\theta}(\theta_u \rightarrow \theta_v; \tau) + \sum_{\Delta} r(\Delta | \theta_u, \theta_v) \sum_n \mathbb{E}_{\mu_{u,n}, \mu_{v,n}} [\log P_{\text{sub},n}^{(\Delta)}] \right] \\ & + H[q_{\Theta}] + \sum_{v \text{ internal}} \sum_n \sum_{\theta} q_v(\theta) H[q_{v,n}(\cdot | \theta)], \end{aligned} \quad (\text{D.149})$$

with the field-only responsibility $r(\Delta | \theta_u, \theta_v) = P_{\text{field}}(\theta_v, \Delta | \theta_u) / P_{\theta}(\theta_u \rightarrow \theta_v)$ (so the Δ -sum inside $\log K$ is Jensen-bounded but its field factor collapses to the exact $\log P_{\theta}$, cf. the certified-bound remark), and the tree-Markov entropy in closed form,

$$H[q_{\Theta}] = \sum_{u \rightarrow v} H[q_{uv}] - \sum_v (\deg v - 1) H[q_v], \quad (\text{D.150})$$

which is *exact* for a tree-structured q_{Θ} — on a tree the junction-tree entropy is the true entropy, no Bethe approximation.

Coordinate ascent. Two blocks. (i) Fixing q_{Θ} , each residue marginal has the closed-form exponential-family update $\log q_{v,n}(a | \theta) \propto [\text{incident-edge and emission expected log-potentials at } (v, n, \theta, a)]$, a per- (v, n, θ) softmax over the messages from v 's parent and children — the residue mean-field of the earlier paragraphs, now conditioned on θ . (ii) Fixing the residues, (D.149) is, in q_{Θ} , the free energy of an L -state tree Markov random field with node potentials $\varphi_v(\theta) = \exp(\text{root-prior} + \text{residue-emission expectations})$ and edge potentials $\psi_{uv}(\theta_u, \theta_v) = P_{\theta}(\theta_u \rightarrow \theta_v) \exp(\sum_{\Delta} r \sum_n \mathbb{E}[\log P_{\text{sub},n}^{(\Delta)}])$; because the graph is a tree, a single exact Felsenstein/belief-propagation pass returns the *exact* maximising node and edge marginals q_v, q_{uv} — the field is never factorised. Cost per sweep $O(N(L^2 + mL\mathcal{A}^2))$: the field block is L^2 per edge, the residues mean-field.

Exactness (corrected). At a single-branch tree and at a cap-2 cherry the residue mean-field is exact (the rank-1+scalar form is closed under a lone edge and under a single two-leaf combine) and the field BP is exact (a two- or three-node MRF), so the maximised (D.149) equals $\log P(\text{data})$: its maximiser reproduces the moment-matching forward wherever that forward is itself exact. This is the precise sense in which the cap-2 ELBO is the exact log-likelihood — it holds for the field-*correlated* family (D.148). The node-factorised family (D.139) does *not* share it: factorising the field across nodes leaves a strictly positive gap even at a cherry (it drops the endpoint field correlation), so its ELBO is a valid but loose floor. The field-exact family removes exactly that gap while keeping the $\mathcal{A}^m$ residues mean-fielded.

Where this mean-field is exact. Like the residue-correlated family analysed below, (D.149) keeps the field factor tree-Markov, so it is exact only when the residue-integrated field posterior is tree-Markov: at a single branch (two field states, no separator to violate) it is exact for any m , and on every tree with three or more field states — cherries included — it is a strict bound, because the shared or chained residues couple field states across their tree separators (the paths made explicit under (D.151)) and a tree-Markov field cannot hold them. Node-factorising the residues on top of this drops the along-branch residue correlation as well, so (D.149) is strictly looser than the residue-correlated bound (D.151) at the same $O(N(L^2 + mL\mathcal{A}^2))$ cost; it is the floor. For $m = 1$ the compound (θ, x) chain has only $L\mathcal{A}$ states, so one keeps that joint and is exact at any depth rather than mean-fielding at all — the $m = 1$ instance of the joint form to which cherry- and singleton-exactness properly belong (below).

Residue-correlated tree-structured family: the headline bound. The residue mean-field’s two remaining gaps — singletons at depth ≥ 2 and the $m \geq 2$ interior — both come from dropping the *along-branch* residue correlation. Restore it without leaving the tractable class by making the residues, too, Markov on the tree *conditioned on the field and its jumps*. Promote the per-branch jump indicator $\Delta \in \{0, 1\}$ to a variational latent and take the branch conditional

$$q(\mathbf{x}_v \mid \mathbf{x}_u, \theta_u, \theta_v, \Delta) = \mathbb{I}[\Delta=0] \mathbb{I}[\theta_v=\theta_u] \prod_{n=1}^m q_n^{uv}(x_{v,n} \mid x_{u,n}) + \mathbb{I}[\Delta=1] \prod_{n=1}^m r_n^v(x_{v,n} \mid \theta_v), \quad (\text{D.151})$$

so the full q is Markov-on-the-tree over $(\boldsymbol{\theta}, \boldsymbol{\Delta}, \mathbf{x})$: a tracking component ($\Delta=0$, no field jump) that carries a free per-site parent $\rightarrow$ child kernel q_n^{uv} — the along-branch correlation the mean-field dropped — and a renewal component ($\Delta=1$) in which the sites are drawn from a free per-site node marginal r_n^v , not the prior stationary, so it absorbs downward evidence. The mean-field (D.148) is the special case $q_n^{uv}(x' \mid x) = q_{v,n}(x')$. Equation (D.151) *specifies* q in the rooted conditional parametrisation (a product of branch conditionals $q(x_v \mid x_u)$ off a root factor); the entropy below is written in the equivalent *moment* parametrisation, whose edge marginal is $q_{uv}(x_u, x_v) = q_u(x_u) q(x_v \mid x_u)$ (node marginal $\times$ branch conditional). The two describe the same tree-Markov family — conditionals are convenient for specifying and sampling q , marginals for the junction-tree entropy — and give the same $H[q]$ ($H[q_{uv}] = H[q_u] + H[x_v \mid x_u]$).

Tractability is unchanged. The decisive property survives: conditioned on $(\theta_u, \theta_v, \Delta)$ the residues *factorise over sites*, so the edge pairwise belief is a product of m blocks of size $\mathcal{A} \times \mathcal{A}$, never the $\mathcal{A}^{2m}$ joint. With Δ in q the energy is a plain sum (no $\log \sum_{\Delta}$), and the junction-tree entropy

$$H[q] = \sum_{u \rightarrow v} H[q_{uv}] - \sum_v (\deg v - 1) H[q_v], \quad H[q_{uv}] = H[q_{uv}^{\Theta\Delta}] + \sum_{\theta_u, \theta_v, \Delta} q_{uv}(\theta_u, \theta_v, \Delta) \sum_n H[q_n^{uv}(\cdot, \cdot \mid \cdot)], \quad (\text{D.152})$$

is computed per site; cost stays $O(N(L^2 + m\mathcal{A}^2))$ per sweep — the *same class* as the residue mean-field. Coordinate ascent alternates the per-site kernel updates (a closed-form $\log q_n^{uv} \propto$ incident-edge + emission expected log-potentials) with an exact field-and-jumps belief propagation over the induced $(\boldsymbol{\theta}, \boldsymbol{\Delta})$ tree MRF; for $m \geq 2$ the two-child combine, which produces three distinct rank-1 residue vectors whose sum does not factorise over sites, is projected back onto (D.151) (the sole approximation).

Where the bound is strict, and where exactness lives instead. Equation (D.151) keeps $q_{\Theta\Delta}$ tree-Markov and couples the residues to it only through the ELBO’s *expectations* — the field-jump law $q(\theta_v, \Delta \mid \theta_u)$ does not depend on the residue *values* x_u . So the maximised ELBO equals $\log P(\text{data})$ exactly when the true field-jump posterior $P(\boldsymbol{\theta}, \boldsymbol{\Delta} \mid \text{data}) \propto P_{\text{field}} \prod_n L_n(\boldsymbol{\theta}, \boldsymbol{\Delta})$ is tree-Markov. Each per-site evidence $L_n = \sum_{\mathbf{x}_n} \prod_e T_{\theta}^{\Delta_e}$ is a whole-tree Felsenstein contraction, so integrating the residues out couples field states through paths that run *through the residues*: on a chain $\theta_1 - x_1 - \theta_2 - x_2 - \theta_3 - x_3$ the route $\theta_1 - x_1 - x_2 - x_3 - \theta_3$ bypasses the separator θ_2 , and at a cherry the shared root residue couples the two leaf field states past θ_r . A tree-Markov $q_{\Theta\Delta}$ cannot carry these. Hence:

- ≤ 2 field nodes (a single branch), any m : *exact*. Two field states have no separator to violate, so $q_{\Theta\Delta}$ is unconstrained and the bound is tight.
- ≥ 3 field nodes (cherries and every deeper tree): *a strict bound*, for $m \geq 2$ and $m = 1$ alike. The residue-induced field couplings above are dropped. The gap is the residue $\rightarrow$ field feedback — second-order and small when the field drives the residues more than the residues identify the field ($\sim 0.06\text{--}0.5$ nat on cherries and depth-2 trees, vanishing as $\rho_{\text{chain}} \rightarrow 0, \infty$). Variationally this is the forward’s combine blow-up ($K_v = K_{u_1} K_{u_2} + K_{u_1} + K_{u_2}$).

The exactness that *is* available for cherries at any m , for singletons at any depth, and at $\rho_{\text{chain}} = 0$ belongs instead to the *joint* form that keeps (θ, x) in one message — the rank-1+scalar CLV (D.138), in which the field jump is coupled to the actual residue values. Its *forward* is exact through $|C| = 2$ and at cherries; but its ELBO *functional* needs $\mathbb{E}_q[\log F]$ over that joint message, and log of an $\mathcal{A}^m$ -site product does not factorise, so the certified joint bound costs $O(\mathcal{A}^m)$ — cheap only at $m = 1$ ($L\mathcal{A}$) or $m \leq 2$. The residue-correlated family (D.151) recovers most of that tightness at $O(N(L^2 + m\mathcal{A}^2))$: it strictly *dominates* the residue mean-field (D.148) (restoring the along-branch residue correlation) at the same asymptotic cost, and is the recommended scalable bound; (D.148) is retained only as the cheaper floor. Its M-projected single forward pass is a valid certified bound provided it is evaluated through (D.152), not its own log-normaliser.

Tight regimes. The maximised ELBO of the node-factorised family (D.139) equals $\log P(\text{data})$ only as $\rho_{\text{chain}} \rightarrow \infty$: every edge regenerates ($\lambda_v \equiv 1$, $q_v(\theta_v) \rightarrow \pi_\theta$), the field states become independent across nodes, and factorising q_Θ across nodes then costs nothing. At the cap-2 cherry and as $\rho_{\text{chain}} \rightarrow 0$ the exact object is instead the field-correlated family (D.151) (equivalently the moment-matching forward’s $\log Z_{\text{MM}}$, exact wherever no combine-projection fires): the cherry couples the root and leaf field states, and at $\rho_{\text{chain}} \rightarrow 0$ the branch factor forces $\theta_v = \theta_u$ — neither of which the node-factorised q_Θ can represent, so (D.139) keeps a strictly positive gap there. The TKF-DP training loop scores cap-2 cherries by that forward; the residue-correlated bound (D.151) carries the exactness into depth- > 1 inference (ancestor reconstruction, tree-level held-out validation) while remaining a certified lower bound at intermediate ρ_{chain} on deeper trees.

Per-site class assignments under the variational family. The form (D.139) treats the per-site class c_n as a fixed attribute of site n , not a variable to be integrated. At the cap-2 cluster geometry the class assignments are marginalised exactly by the K_c^2 -term sum $\sum_{c_1, c_2} \pi_c(c_1) \pi_c(c_2) P^{(c_1, c_2, \theta_P)}$, which is bounded (at $K_c = 8$, $K_c^2 = 64$). At general cluster size m the analogous marginalisation costs K_c^m , destroying the linear-in- m scaling that the dynamic field was designed to deliver. Two paths preserve linearity:

1. *Sampling.* Per-site class is a sampled hard assignment under a separate Gibbs sweep on c_s given the cluster’s coupling-side observations. This is the path taken by the TKF-DP training loop, where cap-2 cherries are the unit and per-cherry signal is too weak to drive a mean-field variational marginal to a hard fixed point reliably.
2. *Mean-field per-site marginals.* $q_n^v(c_n) \in \Delta^{K_c-1}$ is added to the per-node variational state and marginalised inside the cluster emission, $\sum_{c_1, \dots, c_m} \prod_n q_n^v(c_n) P^{(c_1, \dots, c_m, \theta_v)}$. Cost stays $O(L \cdot m \cdot k^2)$ per cluster.

Path 2 captures anti-correlation between coupled sites *only at strong-signal coordinate-ascent fixed points* where each q_n^v collapses to a Dirac on c_n^* , recovering the hard-assignment emission $P^{(c_1^*, \dots, c_m^*, \theta_v)}$ that exposes the per-class compositional bias responsible for anti-correlation. The hard-assignment limit is contained in the mean-field family as a degenerate special case, and at KL-strong evidence the coordinate-ascent fixed point is provably the Dirac. At weak signal (few observations per cluster, K_c large, or two symmetric modes such as $(c_1^*, c_2^*) \leftrightarrow (c_2^*, c_1^*)$ contributing equal likelihood), the mean-field marginals stay diffuse, the emission averages over K_c^m class assignments—including the K_c^{m-1} same-class assignments that are typically neutral on anti-correlation—and the variational emission posterior loses the anti-correlation structure that motivates the per-site class DP in the first place.

Mitigations of escalating richness, applicable when downstream metrics show the diffuse-marginal failure mode: *structured-mean-field per cluster pair*, replacing $\prod_n q_n^v$ with $\prod_{(i,j) \in \text{pairs}(C)} q_{ij}^v(c_i, c_j)$ (adds K_c^2 per pair, $K_c^2 \cdot \binom{m}{2}$ per cluster, recovers $(c_1^*, c_2^*) \leftrightarrow (c_2^*, c_1^*)$ symmetry exactly); *warm-start* the coordinate ascent from a sampled hard c_s so the descent begins inside the basin of the relevant Dirac; *entropy tempering* on $q_n^v(c_n)$ that pushes toward hard assignments early in the optimisation and relaxes once the rest of the variational state has settled. The TKF-DP training loop sidesteps the issue entirely via path 1 (sampling); the mean-field class extension is needed only when extending the ELBO to general phylogenies, where aggregated per-tree signal at each site is stronger than the per-cherry signal that defeats it at training scale.

Parity with the Potts variant. Proposition D.8 together with this ELBO extension puts the dynamic-field variant on equal structural footing with the Potts variant’s Cohn/Girsanov bound (subsection D.6): both are exact at the cap-2 cluster training geometry and both admit a principled variational extension to deeper trees. Combined with the no-Sinkhorn-needed property of the field-marginal joint stationary under the dynamic-field variant—the joint $J^{(c_1, c_2)}(a, b) = \sum_{\theta} \pi_{\theta} \pi^{(c_1, \theta)}(a) \pi^{(c_2, \theta)}(b)$ is marginal-consistent by construction, so Theorem D.1 is automatically satisfied—the dynamic-field variant is a drop-in substitute for the Potts variant throughout the inference stack.

D.9 Survival of the approximate evaluators under A1 and under the dynamic field

The TKF-DP inference loop evaluates the substitution likelihood by three techniques; each carries over under A1 and under the dynamic-field variant of subsection D.8.

Cohn/Girsanov class-level variational ELBO. The bound on the coupled-CTMC path measure on matched (both-present) branches via a constant-rate variational family with closed-form pairwise bridge expectations (subsection D.6) survives unchanged in form: a matched branch is still a coupled reversible CTMC, now with the Sinkhorn-corrected $\mathbf{Q}_{\text{pair}}$; the marginal-consistent stationary is plugged in directly, and the $H = 0$ exactness is preserved. A1 changes only the indel-seam scoring (the lone-branch stationary and the conditional-Boltzmann insertion), which is outside the coupled-interior bound. Under the dynamic-field variant the bound becomes moot at the cap-2 cluster geometry: sites are conditionally independent given the field, the re-equilibrating dynamic field model is not a CTBN, and both are handled by Proposition D.8 at depth 1. For inference on general phylogenies under the dynamic-field variant (ancestor-residue reconstruction at internal nodes, tree-level held-out validation), the structured mean-field ELBO of subsubsection D.8.6 restores tractable linear-in-cluster-size cost while remaining exact at cap-2 and in the rate-limit regimes; it plays the same structural role for the dynamic field as Cohn/Girsanov does for the Potts coupling.

Complete-data conjugacy. The secret-destination Dirichlet conjugacy on $\pi^{(c)}$ and the Gamma–Poisson on rates survive, and A1 makes them cleaner. Under A1 the pair-marginal of class c is exactly $\pi^{(c)}$ regardless of partner, so the per-class sufficient-statistic counts aggregate without partner-dependent reweighting (under the free- h A3 model they would not, since c ’s effective composition varies per partner). The side potentials h , now deterministically Sinkhorn-determined by (π, H) , are demoted from sampled parameters to a deterministic readout, so no prior or update is needed for them. Gamma–Poisson for the indel and rate constants is untouched. Under the

dynamic-field variant the per-(class, field) stationary $\pi^{(c,\theta)}$ admits the same Dirichlet conjugacy on sufficient statistics gathered per (c, θ) bin; the field selector itself is updated by a separate truncated-stick-breaking Beta posterior on the per-cluster field assignment counts.

Laplace for the Potts atoms H . H remains the free, real-valued, non-conjugate coupling parameter under A1, so the Gaussian (Hessian-based) Laplace approximation still applies. The only change is that because $h = \text{Sinkhorn}(\pi, H)$ is now an inner deterministic step, the Laplace objective is $H \mapsto h(H) \mapsto \pi_{\text{joint}} \mapsto \text{likelihood}$, and the Hessian acquires the Sinkhorn-map Jacobian $\partial h / \partial H$ (well-defined and differentiable by the implicit function theorem). The multi-seed Laplace mixture for multimodal H posteriors carries over verbatim. Under the dynamic-field variant the H -parameter is replaced by the per-(class, field) stationary $\pi^{(c,\theta)}$, which is conjugate to its Dirichlet base measure; no Laplace approximation is required on the substitution-side parameters of the dynamic field.

Summary. A1 preserves all three evaluators: the ELBO is unchanged on matched branches; conjugacy is cleaner; Laplace acquires an extra Jacobian. The dynamic-field variant replaces the coupled-CTMC ELBO with the exact hierarchical recursion of Proposition D.8 at cap-2 and its variational extension of subsection D.8.6 on deeper trees, retains conjugacy on the per-(class, field) stationary, and obviates Laplace on the substitution side entirely.

D.10 Augmented indel histories via a time-indexed pair SCFG

The class-level ELBO of §D.6 was derived under fixed class composition over a constant-composition segment of a branch. To compose with the tree we need the timed indel history on each branch, including *transient* insertions: under coupling, an insertion that is born and then deleted on the same branch is not visible in the alignment but does change the rate of every other site in its DP class during its lifetime. The naive scheme of sampling alignment-visible indel timings from their TKF92 prior is inadequate for two distinct reasons: first, transient lineages are missed; second, even for visible events, the conditional density given the alignment depends on substitution evidence and is not what the alignment marginal alone provides.

We resolve both issues by augmenting the latent state with *gravestones*. The augmentation does not change the model. It changes the latent representation: every fragment that ever existed during the branch — alive at start, alive at end, deleted on the branch, or transiently inserted then deleted — is represented as a labeled position in the augmented branch history. With gravestones the alive count $N_a(s)$ is bounded by the visible count $N_v(s)$, so the indel CTMC bridge has a finite state space and admits a closed-form recursive sampler. The gravestone counts at internal nodes of the tree become latent variables resampled jointly with the rest of the augmented history.

The branching process and its grammar

We first present the TKF91 case (single residue per fragment) where the branching structure is cleanest, then read off the TKF92 expansion. Let $L(t)$ denote a single link observed t time units after its birth. Under TKF91 dynamics, $L(t)$ has lifetime $\tau \sim \text{Exp}(\mu)$, and during its lifetime spawns child links at rate λ . Each child is itself an L of the appropriate residual age. The branching process is a continuous-time Galton–Watson process expressible as a stochastic context-free grammar with

productions indexed by continuous time:

$$\begin{aligned}
L(t) &\rightarrow A \cdot D(t, t) && \text{at weight } e^{-\mu t}, \\
L(t) &\rightarrow G_\tau \cdot D(t, \tau) && \text{at density } \mu e^{-\mu\tau} d\tau, \quad \tau \in (0, t), \\
D(t, T) &\rightarrow L(t - u) \cdot D(t, T) && \text{at density } \lambda du, \quad u \in (0, T), \\
D(t, T) &\rightarrow \varepsilon && \text{at weight } e^{-\lambda T}.
\end{aligned}$$

The terminals A and G_τ stand for an alive residue and a gravestone residue that died at relative time τ , respectively, each carrying a DP key, a site class, and an amino-acid trajectory over its lifetime as governed by §D.6. The non-terminal $D(t, T)$ generates a list of child links born during a window of size T inside an observation interval of size t . The reading is: a link either survives to observation (probability $e^{-\mu t}$) and can have spawned children at any time in $(0, t)$, or it died at some $\tau \in (0, t)$ (density $\mu e^{-\mu\tau}$) and can only have spawned children during $(0, \tau)$. The empty-list weight $e^{-\lambda T}$ is the probability of zero events from a Poisson process of rate λ over a window of T .

TKF92 expansion. The generalization to TKF92 replaces each single-residue terminal with a fragment of geometrically-distributed length, leaving the fragment-level branching structure entirely unchanged:

$$\begin{aligned}
A &\rightarrow \mathcal{R}^A \cdot A^*, & A^* &\rightarrow \mathcal{R}^A \cdot A^* \text{ at weight } r, & A^* &\rightarrow \varepsilon \text{ at weight } 1 - r, \\
G_\tau &\rightarrow \mathcal{R}_\tau^G \cdot G_\tau^*, & G_\tau^* &\rightarrow \mathcal{R}_\tau^G \cdot G_\tau^* \text{ at weight } r, & G_\tau^* &\rightarrow \varepsilon \text{ at weight } 1 - r,
\end{aligned}$$

where $\mathcal{R}^A$ is a single alive residue and $\mathcal{R}_\tau^G$ a single gravestone residue with shared death time τ . Fragment death is collective: all residues of a fragment share its lifetime $[0, \tau]$ and die together at τ . Key-DP labels z_s and class-DP labels c_s are drawn independently per residue at fragment birth from their respective Chinese restaurant predictives, even within a single fragment, so a fragment can carry residues of several key classes and several site classes. The expansion is invisible to the indel skeleton — the bivariate Riccati of the next subsection lives at the fragment level — but enters the substitution likelihood and the DP-class composition through the per-residue payload.

Pairing this with an ancestor sequence yields the pair version: each ancestor fragment present at branch start is an independent root of the recursion above, with $t = t_b$ the branch length; insertions appearing in the children list have no ancestor counterpart and carry only a descendant payload. The pair grammar generates an aligned pair of fragment sequences (with alive/gravestone labels on the descendant side) jointly with the parentage tree.

Inside generating function and the bivariate Riccati

Let $f_t(z, w) = \mathbb{E}[z^{N_a(t)} w^{N_g(t)}]$ where N_a, N_g count alive and gravestone descendants of one root fragment of age t . The inside generating function of the grammar satisfies the Volterra fixed point

$$f_t(z, w) = z e^{-\mu t} \exp\left(\lambda \int_0^t (f_s(z, w) - 1) ds\right) + w \int_0^t \mu e^{-\mu\tau} \exp\left(\lambda \int_{t-\tau}^t (f_s(z, w) - 1) ds\right) d\tau,$$

which is equivalent to the forward equation

$$\partial_t f_t(z, w) = [\lambda z^2 - (\lambda + \mu)z + \mu w] \partial_z f_t(z, w), \quad f_0(z, w) = z.$$

The right-hand side is quadratic in z , so the characteristic equation $\dot{z} = -[\lambda z^2 - (\lambda + \mu)z + \mu w]$ is Riccati. Define

$$\Delta(w) = \sqrt{(\lambda - \mu)^2 + 4\lambda\mu(1 - w)}, \quad z_\pm(w) = \frac{(\lambda + \mu) \pm \Delta(w)}{2\lambda}.$$

The Riccati flow $(z(t) - z_+)/ (z(t) - z_-) = (z(0) - z_+)/ (z(0) - z_-) \cdot e^{-\Delta(w)t}$ inverts to give $f_t(z^*, w)$ in closed form, hence closed-form pointwise transition probabilities $P((N_a, N_g)(t) | (N_a, N_g)(0))$ by coefficient extraction. Setting $w = 1$ recovers the standard linear birth-death PGF for N_a alone.

Recursive traceback sampler

Given an augmented history with endpoint counts $(a_0, g_0) = (N_a(0), N_g(0))$ and $(a^*, g^*) = (N_a(t_b), N_g(t_b))$, the within-branch indel history is sampled by recursive midpoint traceback on the bivariate process (N_a, N_g) . At each level of the recursion, sample a midpoint state from

$$P((a_m, g_m) | (a_0, g_0), (a^*, g^*)) \propto P((a_0, g_0) \rightarrow (a_m, g_m); t_b/2) \cdot P((a_m, g_m) \rightarrow (a^*, g^*); t_b/2),$$

both factors closed form via the Riccati. The midpoint state space is finite since $a_m, g_m \leq g^* + a^*$. Recurse on each half. The base case is reached when the residual subproblem has at most a small number of events (births and deaths), at which point the closed-form list of valid event interleavings is enumerable directly and event times are sampled from the corresponding hypoexponential simplex density. Total cost is $O((b + d) \log(b + d))$ per branch where b, d are the branch totals of births and deaths.

After event times are placed, parentage is sampled forward in time: at each birth, a uniform alive fragment is chosen as the parent; at each death, a uniform alive fragment is killed. Parentage uniformity is the standard exchangeability of the linear Galton–Watson tree.

Inserted fragment and residue distributions

When a child fragment is added at time u to a DP class C that already contains alive fragments, the inserted residues are drawn from the conditional Boltzmann

$$P(x_s(u^+) | x_{C \setminus s}(u^-)) \propto \pi^{(c_s)}(x_s(u^+)) \exp\left(- \sum_{s' \in C \setminus s} H_{c_s c_{s'}}(x_s(u^+), x_{s'}(u^-))\right),$$

which is exactly normalizable on the alphabet $\mathcal{A}$ even when the joint stationary on C is intractable. This is the proper coupled generalization of the $\text{Cat}(\pi)$ inserted-residue rule of standard TKF92.

Composition with substitution and with the rest of the tree

The traceback gives a piecewise-constant DP-class composition on each branch, with segment boundaries at every (alive or gravestone) indel event. The variational ELBO of §D.6 applies to each segment as written, with the segment’s class composition and duration. Bridge expectations accumulate over segments within each branch and across branches.

Internal-node augmented states (N_a, N_g) -valued are sampled jointly across the tree. The standard upwards-downwards (Felsenstein-style) recursion applies because the closed-form per-branch transition kernel from the Riccati composes across branches: the marginal at each internal node, conditional on the rest of the tree, is a closed-form pointwise distribution computable in $O(1)$ given the per-branch kernel evaluations. The augmentation thus extends to a full tree-level closed-form sampler.

Marginalizing the augmentation

Gravestones are an augmentation of the latent state, not of the data. The marginal likelihood of the observed alignment under the model is recovered by summing over gravestone configurations

at every internal node and along every branch. The recursive traceback samples the augmented state, and the variational ELBO of §D.6 applied segment-wise on the augmented branches gives the conditional substitution likelihood. No other approximation enters at the indel level: the SCFG is the exact representation of TKF92’s fragment-level dynamics, and the gravestone augmentation is what makes its inside-outside computable.

Remark D.17 (Coalescent priors; ARG augmentation as a parallel construction). The per-branch SCFG depends on branch length only through t_b , so any prior on the tree — including a coalescent prior on branch times — composes with the gravestone-augmented sampler without modification. This is a free generalization that supports joint indel-aware coalescent inference on protein-coding regions, which existing coalescent frameworks typically handle by treating indels as missing data. Separately, the structural mechanism — augmenting an unbounded latent process with non-observable entities so that an observed count provides a state-space bound — carries over to the ancestral recombination graph: non-ancestral lineages in the ARG play the role of gravestones, bounding the per-genome-position bridge state space and admitting (we conjecture) a parallel closed-form recursive sampler.

D.11 Cluster class-composition prior: a mixture of Dirichlet-multinomials

The generative model of §D.1 draws the site classes c_s *iid* from a single global D_c , so it says nothing about which classes prefer to co-occur *within a cluster*. Yet with the partition z and the labels c jointly sampled, each cluster C furnishes a direct multinomial observation of its own class composition. We exploit this to give clusters a learned, *exchangeable*, cap-agnostic preference over class *combinations* — the coevolutionary analogue of “which types like to sit together” — with no pairwise ($O(K_c^2)$, cap-2-bound) Potts term.

Generative model. With H truncated mixture components (e.g. $H = 20$), draw stick-breaking weights $\pi \sim \text{TSB}(\alpha_\pi)$ and, per component, a Dirichlet hyperparameter $\alpha_h \in \mathbb{R}_{>0}^{K_c}$. For each cluster C of any size $m = |C|$,

$$h_C \sim \text{Cat}(\pi), \quad \theta_C \mid h_C \sim \text{Dir}(\alpha_{h_C}), \quad c_s \mid \theta_C \stackrel{\text{iid}}{\sim} \text{Cat}(\theta_C) \quad (s \in C). \quad (\text{D.153})$$

Only the class *labels* are drawn this way; the residues still evolve under the same per-cluster field likelihood $\mathcal{L}(C)$ of §D.8 (the prior multiplies, it does not replace, the emission model). With class counts $n_C = (n_{C,1}, \dots, n_{C,K_c})$, $\sum_k n_{C,k} = m$, and $A_h = \sum_k \alpha_{h,k}$, integrating θ_C out (Dirichlet–multinomial conjugacy) and marginalising the component gives the cluster class-composition prior

$$\Pr(n_C \mid \alpha, \pi) = \sum_{h=1}^H \pi_h \text{DM}(n_C \mid \alpha_h), \quad \text{DM}(n_C \mid \alpha_h) = \frac{\Gamma(A_h)}{\Gamma(A_h + m)} \prod_{k=1}^{K_c} \frac{\Gamma(\alpha_{h,k} + n_{C,k})}{\Gamma(\alpha_{h,k})}. \quad (\text{D.154})$$

Since (D.154) depends only on the count vector n_C it is exchangeable in the columns and defined for *any* cluster size (the cap- > 2 -safe property); the full cluster score is $\mathcal{L}(C) \cdot \Pr(n_C \mid \alpha, \pi)$ and the joint prior is $\prod_C \Pr(n_C \mid \alpha, \pi)$.

Co-occurrence, not arrangement. A component α_h encodes which classes *co-occur* in a type- h cluster; being exchangeable it does not by itself prefer any *arrangement* across columns. For the enumerated-class field model (a class is an ordered pair (k_0, k_1) of archetypes at the two field states) a compensatory flip is a cluster carrying classes (A, B) and (B, A) : the mixture makes such class *sets* likely, while the field likelihood $\mathcal{L}(C)$ — through the shared field trajectory — selects

the compensatory arrangement over the co-flip $(A, B), (A, B)$. The Dirichlet–multinomial’s positive same-class correlation in fact *mildly disfavours* the distinct-class (compensatory) pairing, so the arrangement is carried entirely by $\mathcal{L}$; the concentration A_h should be large enough that this clumping does not fight the likelihood. We add *no* flip affinity to the prior (a per-cluster transposition kernel $c_s \mapsto \text{swap}(c_s)$ w.p. p preserves exchangeability and is available, but is omitted) so that flip structure must *emerge*: a fitted component whose mass concentrates on a charge-complementary class set is then an unsupervised, directly interpretable “salt-bridge type.”

Sampling. All four updates are standard.

1. *Component* h_C : Gibbs, $\Pr(h_C = h \mid n_C) \propto \pi_h \text{DM}(n_C \mid \alpha_h)$ (length- H categorical; or marginalise h via (D.154)).
2. *Site class* c_s : the existing class-Gibbs gains the Dirichlet–multinomial predictive log-prior. Given $h_C = h$, moving s to class k' from k contributes

$$\Delta \log \Pr = \log \frac{\alpha_{h,k'} + n_{C,k'}^{-s}}{\alpha_{h,k} + n_{C,k}^{-s}}, \quad (\text{D.155})$$

with n_C^{-s} the counts excluding s ($\Gamma(A_h + m)$ cancels) — the “rich-get-richer within the cluster’s type” term, additive to the field-likelihood evidence at $O(1)$ (no forward). With h marginalised it is the mixture ratio $\sum_h \pi_h \text{DM}(n_C^{k'} \mid \alpha_h) / \sum_h \pi_h \text{DM}(n_C \mid \alpha_h)$.

3. *Partition* z_s : a column moving $C \rightarrow C'$ updates both $n_C, n_{C'}$; the move already rescores the affected clusters’ likelihood, and (D.154) for the two clusters is added.
4. *Hyperparameters* α_h : a Dirichlet–multinomial MLE (Minka fixed-point) or Metropolis step from the *pooled* counts of clusters currently in component h — each cluster gives only m draws, so the components are identified by pooling, not from any single cluster.
5. *Weights* π : truncated-stick-breaking posterior (Beta conjugacy / Escobar–West auxiliary Gibbs on α_π), as for the other DP concentrations.

Remark D.18 (Interpretability). The fitted $\{\alpha_h\}$ are a taxonomy of coevolutionary cluster types read straight off the model: rank each active component’s classes by $\alpha_{h,k}$ and inspect their archetypes’ charge composition. A component concentrating on charge-complementary flip-classes is a salt-bridge type; one on diagonal (static $k_0 = k_1$) or hydrophobic classes is a specialiser/packing type — with no supervision or built-in flip bias used to obtain them.

D.12 Posterior sampling and parameter learning

We turn to the inverse problem: given a corpus of MSAs $\{X_n\}_{n=1}^N$ with associated trees $\{T_n\}$ and fixed alignment paths, learn the corpus-level substitution hyperparameters

$$\theta = (S, \alpha_z, \alpha_c, \alpha_H, a_\eta, b_\eta, \kappa_\pi, \bar{\pi}, \{\mu_{kl}, \tau_{kl}\}_{k \leq l}),$$

namely the shared exchangeability S , the three DP concentrations $(\alpha_z, \alpha_c, \alpha_H)$, the per-site rate prior (a_η, b_η) , and the base-measure parameters for G_0 on $\pi^{(c)}$ and for G_0^H on the entries of each Potts atom H_h . The per-site rate multipliers $\{\eta_s\}$, the per-class profiles $\{\pi^{(c)}\}$, the materialized Potts atoms $\{H_h\}$, and the class-pair Potts assignments $\{h_{cc'}\}$ are no longer parameters but *latent* draws

from the base measures; their posteriors are inferred per site, per class, per atom, and per class-pair respectively, with the corpus-level parameters governing the priors. Indel parameters (λ, μ, r) decouple cleanly at the parameter level via the alignment–substitution factorization of §D.1 and are learned from the alignment marginal alone by standard TKF92 EM; we focus on the substitution side. The latent variables on each MSA are the key-DP labels $\{z_s\}$, the site classes $\{c_s\}$, the per-site rates $\{\eta_s\}$, the within-class amino-acid trajectories $\{\sigma_C\}$, the unobserved internal-node states, the Potts-atom assignments $\{h_{cc'}\}$, and (per §D.10) the gravestone-augmented indel histories on every branch — comprising the alive and gravestone counts (N_a, N_g) at every internal node, plus the parentage tree and event times within each branch.

Variational EM with bridge-expectation sufficient statistics

For singleton key-DP classes, Q^s reduces to F81 with class-specific equilibrium, shared exchangeability, and per-site rate. The bridge expectations (31) — expected transition counts $N_{xy}^{(b,s)}$ and dwell times $T_x^{(b,s)}$ along each branch b , conditioned on the endpoint amino acids — are computable in closed form by integrating $\int_0^{t_b} e^{Qu} A e^{Q(t_b-u)} du$ in the eigenspace of the rate matrix. Aggregated appropriately they give closed-form posterior updates on every continuous latent on the substitution side:

- $\pi^{(c)}$ has a Dirichlet posterior under the Dirichlet($\kappa_\pi \bar{\pi}$) prior, via the secret-destination augmentation of §D.12 below.
- η_s has a Gamma posterior under the Gamma(a_η, b_η) prior; both shape and rate updates are immediate from the bridge expectations. The marginal site likelihood $\int \text{Gamma}(a_\eta, b_\eta) \cdot \text{Poisson}(N_s^{\text{acc}} \mid \eta_s \tilde{T}_s) d\eta_s$ is closed-form Negative-Binomial.
- Each materialized Potts atom H_h has a Gaussian posterior under the $\mathcal{N}(\mu_{kl}, \tau_{kl}^{-1})$ prior, via the Gaussian (Hessian-based) Laplace approximation of the path-DCA likelihood at the current iterate.

The shared exchangeability S aggregates over all classes as in standard GTR EM. Site-class assignments c_s and Potts-atom assignments $h_{cc'}$ are latent in the CRP sense, integrated over via the Gibbs sweeps below.

For multi-site DP classes the joint generator on $\mathcal{A}^{|C|}$ has no usable eigendecomposition, and the bridge expectations cannot be applied to the coupled bridge directly. The variational construction of §D.6 is the substitute: at the variational fixed point each $s \in C$ has an inhomogeneous single-site bridge with rate $\tilde{Q}^{s,*}$ on each constant-composition segment of the branch (§D.10), and the bridge expectations apply to this bridge in its eigenspace. The integral identity goes through with Q replaced by $\tilde{Q}^{s,*}(u)$ (piecewise-constant on a time grid within each segment), yielding per-site sufficient statistics $\tilde{N}_{xy}^{(b,s)}$, $\tilde{T}_x^{(b,s)}$ as well as per-pair co-occupancy times

$$\tilde{T}_{xy}^{(b,s,s')} = \int_0^{t_b} q_s(x; u) q_{s'}(y; u) du,$$

the latter immediate from the variational independence of single-site bridges given endpoints.

The envelope theorem ensures that, evaluated at the variational fixed point, the gradient of the ELBO with respect to θ has no contribution from the implicit θ -dependence of $\tilde{Q}^{s,*}$:

$$\nabla_\theta \mathcal{L} = \mathbb{E}_\mathbb{Q}[\nabla_\theta \log \frac{d\mathbb{P}}{d\mathbb{Q}}] = \mathbb{E}_\mathbb{Q}[\nabla_\theta \log d\mathbb{P}].$$

For the GTR-Potts generator this gradient has three structurally distinct pieces. The exchangeability gradient $\nabla_S \mathcal{L}$ is a linear combination of the single-site $\tilde{N}_{xy}^{(b,s)}$ counts as in standard GTR EM. The frequency gradient $\nabla_{\pi(c)} \mathcal{L}$ aggregates over sites with $c_s = c$ and is again single-site. The coupling-tensor gradient is a path-space DCA M-step,

$$\nabla_{H_{cc'}(x,x')} \mathcal{L} \propto \sum_{n,b} \sum_{\substack{s,s' \in C(n) \\ c_s=c, c_{s'}=c'}} \left[(\text{expected } x_s = x, x_{s'} = x' \text{ co-occupancy under } \mathbb{P}) - (\text{same under } \mathbb{Q}) \right],$$

expressible in the same bridge-expectation format using the products $q_s(x;u)q_{s'}(x';u)$ for the variational dwell-time terms and contributions from each variational jump intensity for the transition terms.

MCMC over the discrete latents

The combinatorial latents $\{z_s\}$, $\{c_s\}$, and $\{h_{cc'}\}$ admit straightforward Gibbs sampling. All three partitions are integrated over their respective stick-breaking weights via Chinese restaurant predictives. For the key DP,

$$P(z_s = k \mid z_{-s}, \alpha_z) \propto \begin{cases} |C_k^{(-s)}| \cdot \mathcal{L}_{\text{class}}(C_k \cup \{s\}) / \mathcal{L}_{\text{class}}(C_k) & (\text{existing key class}), \\ \alpha_z \cdot \mathcal{L}_{\text{class}}(\{s\}) & (\text{new singleton}), \end{cases}$$

where $\mathcal{L}_{\text{class}}$ is the per-branch-aggregated class-level variational ELBO from §D.6 plus the contribution of the rest of the tree via Felsenstein pruning. Site-class updates use the class DP predictive,

$$P(c_s = c \mid c_{-s}, \alpha_c, \theta) \propto \begin{cases} n_c^{(-s)} \cdot \mathcal{L}_{\text{site}}(s; \pi^{(c)}) & (\text{existing class, profile } \pi^{(c)}), \\ \alpha_c \cdot \int \mathcal{L}_{\text{site}}(s; \pi') G_0(d\pi') & (\text{new class}), \end{cases}$$

with the new-class integral closed form by the Dirichlet–multinomial conjugacy of §D.12. The Potts-atom assignment $h_{cc'}$ for an unordered class-pair $\{c, c'\}$ is itself drawn by CRP-Gibbs,

$$P(h_{cc'} = h \mid \text{rest}, \alpha_H) \propto \begin{cases} m_h^{(-cc')} \cdot \mathcal{L}_{\text{pair}}(\{c, c'\}; H_h) & (\text{existing atom}), \\ \alpha_H \cdot \int \mathcal{L}_{\text{pair}}(\{c, c'\}; H') G_0^H(dH') & (\text{new atom}), \end{cases}$$

where $\mathcal{L}_{\text{pair}}$ is the path-DCA contribution from co-occurrences of classes c and c' across the corpus, and the new-atom integral is approximated by the Laplace scheme of §D.12.

Split–merge proposals of Jain–Neal type (33) are essential for mixing on all three partitions, with acceptance ratios in terms of class-level (for z), site-aggregated (for c), or class-pair-aggregated (for h) ELBO ratios. All three concentrations $\alpha_z, \alpha_c, \alpha_H$ admit closed-form Gamma-prior posteriors given the respective partitions via the auxiliary-variable scheme of Escobar & West (19).

Truncated stick-breaking on Potts atoms

The Potts-atom CRP admits a truncated stick-breaking (TSB) alternative that is cheaper per sweep, requires no new-atom Laplace branch inside the Gibbs step, and removes the empty-atom bookkeeping of the CRP. The construction mirrors Ishwaran & James (32) and Blei & Jordan (7) as applied to the site-class DP, transposed to the Potts-atom DP.

Fix a truncation level $K_H^{\max}$. The natural choice is $K_H^{\max} = K_c(K_c + 1)/2$, the total number of unordered class-pairs — with this bound the truncation is exactly tight (every class-pair could in principle have its own atom). Replace the CRP-Gibbs sweep over $\{h_{cc'}\}$ with the following two-step alternation:

1. **Beta CAVI / Gibbs on stick proportions.** Maintain stick variables $\beta_h \sim \text{Beta}(1, \alpha_H)$ for $h < K_H^{\max}$, with $\beta_{K_H^{\max}} = 1$ and $\rho_h = \beta_h \prod_{h' < h} (1 - \beta_{h'})$. Given the current per-atom counts $m_h = \#\{(c, c') : h_{cc'} = h\}$, the conditional posterior on β_h is the closed-form Beta

$$\beta_h \mid m, \alpha_H \sim \text{Beta}\left(1 + m_h, \alpha_H + \sum_{h' > h} m_{h'}\right).$$

Use either the posterior mean (CAVI) or a Beta sample (Gibbs).

2. **Categorical resample on $\{h_{cc'}\}$.** For each unordered class-pair $\{c, c'\}$, draw the atom assignment from the closed-form Categorical

$$P(h_{cc'} = h \mid \text{rest}) \propto \rho_h \cdot \mathcal{L}_{\text{pair}}(\{c, c'\}; H_h),$$

where $\mathcal{L}_{\text{pair}}$ is the path-DCA contribution from co-occurrences of classes c and c' across the corpus, evaluated at the current materialized atom H_h .

Remark D.19 (Bijective initialization). Initialization at $K_H^{\max} = K_c(K_c + 1)/2$ should set each unordered class-pair to a *distinct* atom drawn from G_0^H , not collapse all class-pairs onto a single shared atom. The latter is a chicken-and-egg failure mode: only the populated atom receives data through the Laplace M-step, so the others stay at the prior; the next TSB resample therefore prefers the populated atom for all pairs (its likelihood dominates the others' prior-only score), and the system collapses. The bijective init guarantees every atom carries at least one class-pair's data through the first Laplace M-step.

Locality of DP updates under the SCFG augmentation

When the augmented indel history is resampled (§D.10), new gravestone fragments appear with new residues that need DP keys, and old gravestones disappear. Naively this would suggest a full DP-key resample after every augmentation step, which would be prohibitive. Exchangeability of the CRP and the segment-wise structure of the variational ELBO together remove almost all of this cost.

The decomposition is two-step within a single sweep. The *indel-skeleton step* resamples the gravestone-augmented history on every branch by the SCFG traceback of §D.10 without touching DP keys; cost $O((b + d) \log(b + d))$ per branch, independent of DP state. The *gravestone-key step* then assigns DP keys to each new residue s on a gravestone fragment with lifetime $[u, v]$ on branch b , using the local CRP predictive over existing classes plus a singleton, evaluated against $\mathcal{L}_{\text{seg}}$ on segments of b within the gravestone's lifetime. Adding s to class C_k changes that class's variational fixed point only on those segments, costing $O(|C_k| \cdot A^3)$ per fixed-point iteration per touched segment. Per gravestone, the work is one Categorical draw with support equal to the existing classes plus a singleton, plus one local fixed-point re-solve on the affected segments.

Closed-form base-measure integrals via complete-data augmentation

The base-measure integrals

$$\int \mathcal{L}_{\text{site}}(s; \pi') G_0(d\pi'), \quad \int \mathcal{L}_{\text{pair}}(\{c, c'\}; H') G_0^H(dH'),$$

that appear in the new-class and new-atom branches of the CRP predictives, and the analogous corpus-level marginals over η_s , all admit closed-form approximations under complete-data augmentation.

Secret-destination augmentation for $\pi^{(c)}$. The F81-form rate factors as $Q_{xy}^s = \eta_s \cdot S_{xy} \cdot \pi^{(c_s)}(y) \cdot \exp(-\frac{1}{2}\Delta H_s)$, which decomposes the dynamics into an S -driven Poisson proposal clock at rate $\eta_s S_{xy}$ from state x to destination y , modulated by a Bernoulli filter that accepts the proposal with probability $\pi^{(c_s)}(y) \exp(-\frac{1}{2}\Delta H_s)$. To recover full multinomial structure on $\pi^{(c)}$ we augment each silent failure of a proposal of y with a categorical *secret destination* $j \neq y$, drawn from $\pi^{(c)}(j)/(1 - \pi^{(c)}(y))$ on the simplex. Each proposal then casts exactly one categorical vote: the destination is y with probability $\pi^{(c)}(y)$ on acceptance, and a secret j with probability $\pi^{(c)}(j)$ on rejection. Marginally over the proposal, the vote is $\text{Categorical}(\pi^{(c)})$, so the augmented count vector $N^{(c)} \in \mathbb{Z}_{\geq 0}^A$ satisfies

$$N^{(c)} \mid \pi^{(c)}, \text{ augmentation} \sim \text{Multinomial}(N_{\text{total}}^{(c)}, \pi^{(c)}),$$

and the posterior is the strict closed-form Dirichlet

$$\pi^{(c)} \mid N^{(c)} \sim \text{Dirichlet}(\kappa_\pi \bar{\pi} + N^{(c)}).$$

In the EM/VBEM setting the augmentation is replaced by its conditional expectation: each proposal contributes $\pi^{(c)}(y)$ to $E[N_y^{(c)}]$ regardless of which destination was proposed. The Dirichlet update is then a fixed-point iteration whose limit is the conjugate posterior under the conditional augmentation. The new-class integral $\int \mathcal{L}_{\text{site}}(s; \pi') G_0(d\pi')$ is the analytic ratio of multivariate Beta functions $B(\kappa_\pi \bar{\pi} + N^{(c)})/B(\kappa_\pi \bar{\pi})$.

Gamma-Poisson conjugacy for η_s . The F81 substitution-count likelihood at site s is itself Gamma-conjugate in η_s without any need for the secret-destination augmentation. The expected substitution count along the tree is Poisson with rate $\eta_s \cdot \tilde{T}_s$, where

$$\tilde{T}_s = \sum_b \sum_x T_x^{(b,s)} \cdot \sum_{y \neq x} S_{xy} \pi^{(c_s)}(y)$$

is the π -weighted dwell-time integral. Under the $\text{Gamma}(a_\eta, b_\eta)$ prior the conjugate posterior is closed form

$$\eta_s \mid \text{path}, \pi^{(c_s)} \sim \text{Gamma}(a_\eta + N_s^{\text{acc}}, b_\eta + \tilde{T}_s),$$

with $N_s^{\text{acc}} = \sum_b \sum_{x \neq y} N_{xy}^{(b,s)}$ the total observed substitution count, and the per-site marginal likelihood is closed-form Negative-Binomial.

Laplace for H_h . Each Potts atom is real-valued and the path-DCA likelihood is locally Gaussian in H_h to second order. We approximate

$$\log p_{\text{path-DCA}}(\text{data} \mid H_h) \approx \log p_{\text{path-DCA}}(\text{data} \mid \hat{H}_h) - \frac{1}{2}(H_h - \hat{H}_h)^\top \Lambda_h (H_h - \hat{H}_h),$$

where $\hat{H}_h$ is the MAP under the current iterate, found by a few Newton or natural-gradient steps starting from a chosen seed, and $\Lambda_h = -\nabla_{H_h}^2 \log p_{\text{path-DCA}}$ is the Hessian at $\hat{H}_h$. Combined with the $\mathcal{N}(\mu_{kl}, \tau_{kl}^{-1})$ prior on each entry, the posterior is the closed-form Gaussian

$$H_h \mid \text{data} \sim \mathcal{N}(\hat{H}_h^{\text{post}}, (\text{diag}(\tau_{kl}) + \Lambda_h)^{-1}).$$

The Laplace estimate of the new-atom integral is the standard $\log p_{\text{path-DCA}}(\hat{H}_h) + \log G_0^H(\hat{H}_h) + \frac{d}{2} \log(2\pi) - \frac{1}{2} \log \det(\text{diag}(\tau_{kl}) + \Lambda_h)$ with $d = A(A+1)/2$ the symmetric-slice dimension.

Multi-seed Laplace mixture for multimodal posteriors. The Laplace approximation collapses around a single mode and underestimates posterior mass elsewhere; for atoms that admit multiple plausible coupling patterns (e.g., charge-complementary vs. hydrophobic-hydrophobic minima) this is a genuine approximation error. We mitigate by running the few-step optimizer from a small number K_{seed} of distinct seeds, obtaining K_{seed} Laplace components, and combining them as a weighted Gaussian mixture.

The recommended hybrid

The natural hybrid is:

1. **Inner variational E-step.** For each MSA, branch and DP class, segment the branch by the current sampled augmented indel history and solve the geometric-mean fixed point of §D.6 on each segment to obtain $\{\tilde{Q}^{s,*}\}$ and the per-class ELBO. No parameter or structure updates here.
2. **MCMC over discrete and augmentation latents.** Resample augmented indel histories on every branch by recursive traceback on the time-indexed pair SCFG of §D.10, with closed-form midpoint marginals from the bivariate Riccati and Felsenstein-style upwards-downwards composition for internal-node augmentation counts. Gibbs sweep over $\{z_s\}$, $\{c_s\}$, and $\{h_{cc'}\}$ using the class-level ELBOs from step 1 as evidence; periodic Jain–Neal split-merge proposals on each partition.
3. **Stochastic M-step.** Compute bridge-expectation sufficient statistics $\tilde{N}$, $\tilde{T}$, $\tilde{T}_{\text{pair}}$ from the variational single-site bridges, summed over augmented constant-composition segments within each branch, over a minibatch of MSAs. Take a stochastic gradient step on the corpus-level hyperparameters $\theta = (S, \alpha_z, \alpha_c, \alpha_H, a_\eta, b_\eta, \kappa_\pi, \bar{\pi}, \{\mu_{kl}, \tau_{kl}\})$. Per-class $\pi^{(c)}$ updates by its closed-form Dirichlet posterior under the secret-destination augmentation; per-site η_s marginalizes in closed form via Gamma–Poisson conjugacy; per-atom H_h updates by Laplace with multi-seed mixture against the prior G_0^H . The three DP concentrations update by the Escobar–West auxiliary-variable scheme.
4. **Iterate.** The inner variational fixed point in step 1 is re-solved cheaply after each step because the previous solution is a warm start.

This factors the overall problem into the three regimes where each tool is strongest: variational for the continuous-time substitution dynamics where it admits closed-form rates and bridge-expectation sufficient statistics; MCMC for the discrete combinatorial latents where it mixes well and avoids mean-field pathologies; stochastic gradient for the continuous parameters θ where it scales to corpus-level training. In SVI form the M-step is over a minibatch of MSAs at each iteration and amortizes naturally over the corpus.

Counts-tensor representation of the training corpus

Per family we precompute a counts tensor (cherry $\times$ column $\times$ AA-pair $\times$ branch length), packed as int8, that decouples inference from raw MSA parsing and lets one outer SVI iteration scale to 10^3 – 10^4 PFAM families. Branch lengths are quantized to $n_t \approx 50$ geometrically spaced bins between $\tau_{\min}$ and $\tau_{\max}$; geometric spacing exploits that $\exp(Q\tau)$ varies fastest at small τ , giving a $\sim 5\times$ cache reduction over linear quantization at no measurable accuracy cost. Per-class single-site log-transition matrices (K_c, n_t, A^2) are precomputed once per outer iteration and shared across all families.

D.13 Pairwise alignment postprocessing

Multiple-sequence alignment tools of the FSA family (8) (and other consistency-based schemes) operate on pairwise residue posterior matrices $Q_{ij}^{(X,Y)} = P(X_i \sim Y_j \mid X, Y)$ produced by per-pair forward-backward passes through a Pair HMM. The TKF-DP postprocessing target is the joint-marginal alignment-and-partition likelihood described next, in which an alignment A together with a size- $\{1, 2\}$ partition E of its Match cells contributes a product of per-block substitution factors with no class or rate-multiplier latent left explicit.

Remark D.20 (Historical note: deprecated first-order correction). An early formulation of TKF-DP postprocessing expressed the coupling as a first-order mean-field correction $\log Q'_{ij} - \log Q_{ij} \approx \varepsilon \sum_{(i',j')} Q_{i'j'} [M(i, j; i', j'; t) - 1]$ to the baseline single-site posterior, with ε a per-pair partner prior under the size- $\{1, 2\}$ Ewens distribution and M a column-class-marginalized Potts boost. That correction is a first-order expansion of the joint marginal in the partner-prior strength and is now considered a heuristic stepping-stone; the modern infinite Pair HMM (§E.3 of Appendix E) treats the same coupling exactly via the joint-marginal target below, with class assignments and rate multipliers integrated analytically rather than re-marginalized at every column.

The unified joint-marginal target

Let $\theta_{\text{indel}} = (\lambda, \mu, r)$ be the TKF92 indel parameters and α_z the Ewens partition concentration. Write $\text{Match}(A)$ for the Match cells of an alignment A , and let E be a set of unordered Match-cell edges (a partition of $\text{Match}(A)$ into singletons of size 1 and doublets of size 2; we let $V(E) \subset \text{Match}(A)$ denote the doublet endpoints). The TKF-DP joint marginal at branch length t is

$$P(X, Y; t) = \sum_A \pi_{\text{TKF92}}(A \mid t, \theta_{\text{indel}}) \sum_E \pi_{\text{Ewens}}(E \mid \alpha_z, |\text{Match}(A)|) \prod_{(i,j) \in \text{Match}(A) \setminus V(E)} P_{\text{singlet}}(x_i, y_j; t) \prod_{e=((i,j),(k,l)) \in E} P_{\text{doublet}}(x_i, y_j, x_k, y_l; t) \quad (\text{D.156})$$

with the per-block primitives

$$P_{\text{singlet}}(a, b; t) = \sum_c \pi_c \pi^{(c)}(a) P_c(a \rightarrow b; t \cdot \eta), \quad (\text{D.157})$$

$$P_{\text{doublet}}((a, c), (b, d); t) = \sum_{c_1, c_2} \pi_c(c_1) \pi_c(c_2) \pi_{\text{joint}}^{c_1, c_2}(a, c) P_{\text{joint}}^{c_1, c_2}((a, c) \rightarrow (b, d); t, \eta_1, \eta_2). \quad (\text{D.158})$$

The doublet endpoint convention (matching the boost API of the released code) is that (a, b) are the amino acids at the left endpoint (i, j) and (c, d) at the right endpoint (k, l) : $a = x_i$, $b = y_j$, $c = x_k$, $d = y_l$. Indices c_1, c_2 run over site classes and should not be confused with the endpoint-amino-acid index c .

The empirical class prior π_c is the per-class column count from training (*not* uniform $1/K_c$); the per-class profile $\pi^{(c)}(\cdot)$ is the trained per-class profile; the class-conditional transition is

$$P_c(a \rightarrow b; t) = [\exp(Q_c \cdot t)]_{a,b}, \quad Q_c = (S_{\text{LG08}} - \text{diag } S_{\text{LG08}}) \cdot \pi^{(c)}[\cdot]^\top, \quad (\text{D.159})$$

the standard F81-form GTR generator with exchangeability S_{LG08} (40) and class-specific stationary $\pi^{(c)}$. The joint stationary $\pi_{\text{joint}}^{c_1, c_2}(a, c)$ at a coupled cell-pair is the joint stationary of the Potts atom $H_{c_1 c_2}$ assigned to that class pair by the atom DP, and $P_{\text{joint}}^{c_1, c_2}$ is its time-evolved transition $\exp(Q_{\text{joint}}^{c_1, c_2}(H_{c_1 c_2}) \cdot t)$, with $Q_{\text{joint}}^{c_1, c_2}$ built from $(H_{c_1 c_2}, \pi_a, \pi_b, S, \eta_{\text{pair}})$ as described in §D.3.

Pair-background convention for the joint generator. Two natural choices for the pair background (π_a, π_b) arise. The first is to use the LG08 single-site equilibrium $\pi_a = \pi_b = \pi^{\text{LG08}}$ for every class-pair (c_1, c_2) . This is the *canonical* convention for the released $K=4$ EM-warmup checkpoint: the trained Potts atoms were fit as deviations of the joint stationary from the LG08-pair-background reference, so inference under that checkpoint must use the same convention for the joint stationary and joint generator to match training. The second choice, $\pi_a = \pi^{(c_1)}$, $\pi_b = \pi^{(c_2)}$, uses the per-class profiles as the pair background; this is the structurally “cleaner” choice for future model variants in which the Potts atoms are retrained against the per-class background, but it is *not* consistent with the released checkpoint.

Discrete-gamma rate-multiplier integration. The per-site rate multiplier $\eta_s \sim \text{Gamma}(a_\eta, b_\eta)$ of §D.3 is integrated analytically by the discrete-gamma quadrature of Yang (69): K_r equiprobable bin medians $\eta_1, \dots, \eta_{K_r}$ are taken from the $\text{Gamma}(a_\eta, b_\eta)$ prior with uniform weight $1/K_r$, and the per-block primitive of equation D.157 (resp. D.158) is averaged over rate-multiplier draws via $P_{\text{singlet}}(a, b; t) = K_r^{-1} \sum_{r=1}^{K_r} P_{\text{singlet}}(a, b; t \cdot \eta_r)$ (resp. a double sum over (η_1, η_2) for the doublet). The Gamma hyperparameters (a_η, b_η) at inference time must match those used at training time; the released $K=4$ EM-warmup checkpoint uses $(a_\eta, b_\eta) = (2, 2)$. The reduction $K_r = 1$ with $\eta = 1$ is the canonical evaluation of that checkpoint, since the stored per-site η posteriors all collapsed to 1.0 during EM warmup; for retrained checkpoints with a non-trivial η posterior, $K_r > 1$ is required.

$t = 0$ reduction identities. At $t = 0$ both per-block primitives reduce to the appropriate marginal stationary:

$$P_{\text{singlet}}(a, a; 0) = \sum_c \pi_c \pi^{(c)}(a) \equiv \pi_{\text{marg}}(a), \quad (\text{D.160})$$

$$P_{\text{doublet}}((a, c), (a, c); 0) = \sum_{c_1, c_2} \pi_c(c_1) \pi_c(c_2) \pi_{\text{joint}}^{c_1, c_2}(a, c) \equiv \pi_{\text{marg, pair}}(a, c). \quad (\text{D.161})$$

These follow from $P_c(a \rightarrow b; 0) = \delta_{ab}$ and from the joint stationary $\pi_{\text{joint}}^{c_1, c_2}$ being a fixed point of $P_{\text{joint}}^{c_1, c_2}(\cdot; 0)$.

The boost tensor. The four-residue Potts *boost* entering equation D.156 through the doublet factor is the multiplicative deviation of the doublet emission from the product of two singlet emissions:

$$M((a, c), (b, d); t) \equiv \frac{P_{\text{doublet}}((a, c), (b, d); t)}{P_{\text{singlet}}(a, b; t) \cdot P_{\text{singlet}}(c, d; t)}. \quad (\text{D.162})$$

Indexing convention as above: (a, b) is the left endpoint, (c, d) the right endpoint. By construction $M \equiv 1$ when every Potts atom is zero (so $\pi_{\text{joint}}^{c_1, c_2}$ factorizes into $\pi^{(c_1)} \otimes \pi^{(c_2)}$ and $P_{\text{joint}}^{c_1, c_2}$ factorizes into the two single-site P_{c_i} factors). The boost tensor M is the object that the bounded-edge augmented Pair HMM methods of Appendix E fold multiplicatively into the Match emissions to score doublets relative to singletons. The MCMC sampler operates on the joint (A, E) directly via equation D.156, with class assignments and rate multipliers integrated analytically as above (*not* sampled).

Sequence annealing with coevolutionary scoring

The joint-marginal of equation D.156 implies a per-Match-cell posterior Q'_{ij} that the augmented Pair HMM machinery of Appendix E computes either exactly under bounded-edge truncation or stochastically under the infinite-Pair-HMM MCMC. An alternative routing of the same coupling

signal applies the boost *during* assembly: the greedy column-merging step inside the sequence annealer is extended to score pairs of merges jointly. With pair-HMM input, the canonical edge score is

$$w(e) = \frac{2Q_{ij}^{(X,Y)}}{g_i^{(X,Y)} + g_j^{(Y,X)}}, \quad e = (\text{column of } X_i, \text{ column of } Y_j), \quad (\text{D.163})$$

where g are the per-residue gap posteriors. The coupled-pair extension admits joint candidates consisting of pairs of merges scored as

$$W(e_1, e_2) = \log Q_{ij}^{(X,Y)} + \log Q_{i'j'}^{(X,Y)} + \log M((X_i, X_{i'}), (Y_j, Y_{j'}); t), \quad (\text{D.164})$$

with M the boost tensor of equation D.162 evaluated at the observed amino acids. The first two terms are the baseline log-posteriors, and the third is the four-residue Potts contribution; the latter is identically zero when every Potts atom is zero.

Greedy schedule and conflict resolution. When a coupled candidate (e_1, e_2) is popped from the queue, conflicts are resolved as follows. If both component merges are still admissible (no same-sequence conflict, no DAG cycle), both are committed in a single step and the boost-driven joint commit is recorded. If e_1 is admissible but e_2 has been invalidated by an intervening merge, the candidate degrades to a single-edge merge with score $\log Q_{ij}$ and is re-inserted into the queue at that priority. If neither component is admissible, the candidate is dropped.

Cost analysis. Under two practical prunings (threshold $q_{\min} = 0.1$ on the single-edge posterior and threshold $\mu_{\min} = 0.1$ nat on $|\log M|$), the per-step cost is $O(L^2)$ amortized and the total cost is $O(L^3)$ in the worst case, comparable to the $O(L^2 \log L)$ baseline assembler.

Augmented-PHMM versus greedy-pairwise tradeoff. The augmented-Pair-HMM and infinite-Pair-HMM methods of Appendix E produce a single corrected posterior matrix Q' that any downstream assembler can consume. The coupled-pair extension above scores strongly-coupled pairs jointly on their native four-residue domain inside the assembler, but commits greedily and is therefore vulnerable to local optima where an early commit on a coupled pair locks out a globally better arrangement. The two routes consume the same boost tensor M of equation D.162 but apply it at different points in the pipeline; the augmented-PHMM route is the principled default for TKF-DP postprocessing in §E.

E The infinite Pair HMM and its MCMC sampler

The Pair HMM postprocessing methods of §D.13 are progressively faithful approximations to a single underlying object: the alignment-and-partition posterior on $\pi(A, \pi_M \mid X, Y)$, where A is the TKF92 alignment and π_M is a Chinese-restaurant-process partition of its Match cells into singletons (block of size 1, contributing the standard per-site CTMC factor) and pairs (block of size 2, contributing the joint Potts CTMC factor). This appendix develops the three-factor factorisation $\pi_{\text{TKF92}} \cdot \pi_{\text{CRP}} \cdot \prod_b P_{\text{block}}$ in full; describes the exact F_2 -SCFG inside-outside formulation that achieves it at $O(L^4)$; re-encodes the same content as a memory-augmented Pair HMM at $O(L^2 A^2)$; and develops the Gibbs+MH+replica-exchange MCMC sampler that draws from the unbounded-edge limit at amortised $O(L^3)$ per sweep. The conceptual hierarchy at the end of the appendix lifts this construction to the full multi-branch *phylogenetic* SCFG, the natural fixed point of the entire ladder of approximations.

E.1 Exact 0-or-1-edge marginal posteriors via Pair-SCFG inside-outside

The deprecated mean-field correction (remark D.20 of Appendix D) and the coupled-pair greedy annealer of §D.13 are two routes to the same intractable posterior of equation D.156; the present subsection derives an exact dynamic-programming alternative under the hard restriction that the coupling partition E contains zero or one edge. The result is a per-residue-pair posterior Q'_{ij} that consumes the same downstream FSA assembler as the baseline, with the coupling evidence absorbed exactly into the posterior rather than fed in via a per-pair correction or via a custom annealer move set.

Three pair-HMM marginal moments

Under the no-coevolution baseline TKF92 Pair HMM with the class-mixture single-site emissions of Appendix D, let $\pi(X, Y, A)$ denote the joint probability of an alignment A between sequences X and Y . Define

$$F_0(X, Y) \equiv \sum_A \pi(X, Y, A), \quad (\text{E.1})$$

$$F_1(X, Y; i, j) \equiv \sum_{A: X_i \sim_A Y_j} \pi(X, Y, A), \quad (\text{E.2})$$

$$F_2(X, Y; i, j; k, l) \equiv \sum_{A: X_i \sim_A Y_j \wedge X_k \sim_A Y_l} \pi(X, Y, A), \quad (\text{E.3})$$

where $X_i \sim_A Y_j$ denotes that the alignment A pairs residue X_i with Y_j in a Match emission. Computationally, F_0 is a Forward in $O(L^2)$, F_1 is a Forward–Backward in $O(L^2)$, and F_2 is the Inside–Outside table of a non-bifurcating Pair SCFG (or equivalently a doubled Forward–Backward) in $O(L^4)$ time and memory. The conventional Pair-HMM match-state posterior is $Q_{ij} \equiv F_1(i, j)/F_0$.

The objects F_1 and F_2 stand in the relation that gradients and Hessians do to a partition function: $F_1(i, j)/F_0$ is the marginal probability that (X_i, Y_j) is paired and $F_2(i, j; k, l)/F_0$ is the marginal joint probability that both (X_i, Y_j) and (X_k, Y_l) are paired, with the no-coevolution model serving as the unperturbed reference point.

The 0-or-1-edge Pair SCFG

Let $M(i, j; k, l; t) \equiv M((X_i, X_k), (Y_j, Y_l); t)$ denote the four-residue Potts coupling boost of equation D.162 of Appendix D, evaluated at the observed residues (X_i, Y_j, X_k, Y_l) and the inferred branch length t . Define the (unnormalised) Pair-SCFG generating function

$$S \rightarrow S_0 \mid S_1, \quad \text{with multiplicative weights 1 and } \varepsilon = 1/\alpha_z, \quad (\text{E.4})$$

where the two non-terminals expand as

$$\begin{aligned} S_0 &\rightarrow \text{TKF92 Pair HMM with zero coupled column-pairs (the baseline),} \\ S_1 &\rightarrow \text{TKF92 Pair HMM with exactly one coupled column-pair.} \end{aligned}$$

The 1-edge nonterminal’s parse picks one ordered column-pair $(i, j; k, l)$ with $i < k$ as the coupled pair and replaces the product of independent single-site Match emissions at columns (i, j) and (k, l) with the joint pair-emission $P_{\text{joint}}((X_i, X_k) \rightarrow (Y_j, Y_l); t, H)$, equivalently the product of independent emissions multiplied by $M(i, j; k, l; t)$. The partition function of the grammar is

$$L_{\text{exact}}(X, Y) = F_0(X, Y) + \varepsilon \sum_{(i, j; k, l), i < k} F_2(X, Y; i, j; k, l) \cdot M(i, j; k, l; t). \quad (\text{E.5})$$

The first term is the no-coupling marginal likelihood; the second is the marginal contribution of all alignments that contain exactly one coupled column-pair, summed over all candidate locations $(i, j; k, l)$ for that pair. The ε prefactor is the per-pair partner prior under the size- $\{1, 2\}$ Ewens partition with concentration α_z ; the assumption restricting to at most one edge per alignment is a hard-truncation of the size- $\{1, 2\}$ partition that retains the leading-order correction in ε .

The exact match posterior

Inside–Outside on the grammar (E.4) yields the marginal probability that residue X_i is aligned to Y_j in the data:

$$Q'_{ij} = \frac{F_1(X, Y; i, j) + \varepsilon \sum_{(k, l): k \neq i} F_2(X, Y; i, j; k, l) \cdot M(i, j; k, l; t)}{L_{\text{exact}}(X, Y)}. \quad (\text{E.6})$$

The numerator collects all alignment-paths through the SCFG that touch (i, j) as a Match: the first term covers S_0 paths (no coupled pair touches (i, j)), the second term covers S_1 paths in which (i, j) is part of the (unique) coupled pair, summed over the choice of partner column (k, l) . Both summands include the boost factor M at exactly the right place. The denominator is the partition function (E.5).

In the limit $\varepsilon \rightarrow 0$ (or $H = 0$ so $M \equiv 1$ everywhere), Q'_{ij} collapses to $F_1(i, j)/F_0 = Q_{ij}$, the no-coupling baseline, as required.

Cost and practical considerations

The bottleneck of the explicit F_2 tensor encoding is the $O(L^4)$ time and memory required to materialise F_2 itself: F_2 has $\binom{L}{2}^2 \approx L^4/4$ free entries, which at $L = 200$ occupies a few GB in single precision. Longer sequences require either (a) prefix/suffix chunking on the second-anchor axis (k, l) to keep memory bounded, or (b) candidate pruning that drops (i, j) or (k, l) pairs with $F_1(\cdot) < q_{\min}$, reducing the candidate set from $O(L^4)$ to $O(L^2)$ in practice. The reference implementation lives in `src/tkfdp/f2_scfg.py`; in the current codebase it is retained primarily as a cross-validation reference for the more efficient memory-augmented Pair HMM described in the next subsection (which agrees with F_2 -SCFG to machine precision and runs at $O(L^2 A^2)$).

The available pathways to a coupled-FSA-consumable per-residue posterior, in order of fidelity:

Pathway	Cost	Approximation
Mean-field pre-correction (deprecated; rem. D.20)	$O(L^2 K_c^2 A^2)$	First-order in ε
Coupled-pair greedy annealer (§D.13)	$O(L^3)$	Greedy commit ordering
F_2 -SCFG explicit enumeration	$O(L^4)$	0-or-1-edge truncation, otherwise e
Aug-PHMM 1-edge (§E.2)	$O(L^2 A^2)$	0-or-1-edge truncation, otherwise e
Aug-PHMM 2-edge (§E.2)	$O(L^2 A^4)$	0-or-1-or-2-edge truncation, otherw
Infinite Pair HMM MCMC (§E.3)	$O(L^4) + O(L^3) \cdot N_{\text{sweep}}$	Stochastic; converges to exact

The bounded-edge approximations capture progressively more of the coupling correction at progressively higher cost; the infinite Pair HMM MCMC is the principled limit, exact in expectation up to MCMC sampling error. For families dominated by a single strong coevolutionary edge per column, the 1-edge aug-PHMM at $O(L^2 A^2)$ is empirically adequate; for families with multiple competing partners per column, the 2-edge aug-PHMM (or the MCMC sampler) is necessary.

E.2 Memory-augmented Pair HMM: the same content at $O(L^2 A^2)$

The F_2 -SCFG of equations E.4–E.6 can be re-encoded as a Pair HMM whose latent state is augmented with a small *tag memory* of in-progress coupled-edge endpoints. For the 0-or-1-edge case the tag set has $|\mathcal{T}| = 1 + A^2 + 1$ values: a sentinel `no_edge` state, A^2 states (a, b) recording that a left endpoint with observed amino acids (a, b) is awaiting its right partner, and a terminal `done` state recording that a coupled pair has been consumed. At any Match cell (i, j) the HMM may either (a) carry the tag forward unchanged, (b) transition `no_edge` $\rightarrow (X_i, Y_j)$ with weight ε (spawn a left endpoint), or (c) transition $(a, b) \rightarrow \text{done}$ with weight $M(\cdot, \cdot; X_i, Y_j; t)$ summed over the recorded (a, b) (close the pair). At all non-Match states the tag is preserved verbatim. The augmented Forward partition function equals L_{exact} of equation E.5, and the marginal posterior at Match-state cell (i, j) , summed over tag values, equals Q'_{ij} of equation E.6. Cost is $O(L^2 \cdot 5 \cdot |\mathcal{T}|) = O(L^2 A^2)$ time and memory: a factor $L^2/A^2 \approx 100$ speedup over the explicit F_2 tensor at typical sequence lengths, with no loss of fidelity. Implementation lives in `src/tkfdp/aug_phmm.py`; cross-validation against the $O(L^4)$ F_2 -SCFG reference confirms agreement to machine precision ($\max |Q'_{ij} - Q_{ij}^{\text{SCFG}}| \approx 10^{-14}$ on tested pairs).

The same construction generalises to allow up to k coupled column-pairs per alignment by enlarging the tag set to record (i) the multiset of in-flight left endpoints (size up to k), (ii) the number of resolved pairs (in $\{0, 1, \dots, k\}$), and (iii) optionally the in-flight endpoints alongside the resolved-pair counter. For $k = 2$ the tag set has $|\mathcal{T}_2| = 1 + A^2 + A^2(A^2 + 1)/2 + A^2 + 1 \approx 81,000$ values for $A = 20$: a sentinel `no_edge`, A^2 singleton states (one in-flight, none resolved), an unordered-multiset pair state of size $A^2(A^2 + 1)/2$ (two in-flight, none resolved; same-element multisets are valid distinct states), A^2 closed-singleton states (one resolved, one in-flight), and a terminal `closed_done`. The match-cell transitions add a spawn from singleton $\rightarrow$ pair with weight ε (multiset construction), and three closure transitions: pair $\rightarrow$ closed-singleton with weight $M(\cdot; X_i, Y_j; t)$ over the closed multiset element, singleton $\rightarrow$ closed-done with weight M , and closed-singleton $\rightarrow$ closed-done with weight M . Cost is $O(L^2 \cdot 5 \cdot |\mathcal{T}_2|) = O(L^2 A^4)$ time and memory.

E.3 The principled formulation: three-factor model and MCMC

The bounded-edge truncations (0-or-1-edge in §E.1; 0-or-1-or-2-edge as a natural extension) are dynamic-programming approximations to a more principled underlying model whose joint posterior factorises cleanly into three independent ingredients: the alignment path A under the TKF92 Pair HMM, a partition π_M of its Match cells under a Chinese Restaurant Process (CRP) prior, and the per-block substitution likelihood under the substitution CTMC:

$$\pi(A, \pi_M \mid X, Y) \propto \pi_{\text{TKF92}}(A \mid X, Y) \cdot \pi_{\text{CRP}}(\pi_M \mid \alpha_z, |\text{Match}(A)|) \cdot \prod_{b \in \pi_M} P_{\text{block}}(b \mid t, H). \quad (\text{E.7})$$

Blocks of size 1 (uninvolved Match cells) contribute the standard single-site CTMC factor P_{singlet} of equation D.157 of Appendix D; blocks of size 2 (coupled-edge endpoints) contribute the joint Potts CTMC factor P_{doublet} of equation D.158. The CRP prior with concentration α_z on a partition of $N = |\text{Match}(A)|$ Match cells into K blocks of sizes $\{n_b\}_{b=1}^K$ is the canonical Ewens partition prior (21, 49)

$$\pi_{\text{CRP}}(\pi_M \mid \alpha_z, N) = \frac{\alpha_z^K \prod_{b=1}^K (n_b - 1)!}{\alpha_z(\alpha_z + 1) \cdots (\alpha_z + N - 1)}; \quad (\text{E.8})$$

large α_z favours many small blocks (mostly singletons, few coupled pairs). In practice we restrict to blocks of size at most 2: blocks of size ≥ 3 would correspond to triangular cliques whose marginal mass is suppressed by factors of $1/\alpha_z$ at high concentration. We call (E.7) the *infinite Pair HMM* in

the sense that the number of coupled edges per alignment is not capped by the prior — in contrast to the bounded-edge dynamic-programming methods of §E.2.

The original *infinite Hidden Markov Model* of Beal, Ghahramani & Rasmussen (5) and the HDP-HMM of Teh, Jordan, Beal & Blei (60) provide the broader Bayesian-nonparametric framework in which equation E.7 sits.

MCMC sampler from the infinite Pair HMM

The factorisation (E.7) admits an efficient Markov-chain Monte Carlo sampler that scales as $O(L^4)$ for one-time setup followed by $O(L^3)$ amortised cost across many sweeps. We give the setup phase first, then the three move types that compose the sweep.

Target distribution. The sampler operates on the joint state (A, E) where A is a TKF92 Pair HMM alignment and E is a set of unordered Match-cell-pair edges. The unnormalised target distribution is

$$\pi(A, E \mid X, Y) \propto P_{\text{baseline}}(A \mid X, Y) \cdot \frac{\alpha_z^{|E|}}{\prod_{m=1}^{N_M(A)} (m - 1 + \alpha_z)} \cdot \prod_{e \in E} M(e \mid t), \quad (\text{E.9})$$

where $N_M(A)$ is the number of Match cells in A and $M(e \mid t)$ is the four-residue Potts boost at edge e . The combinatorial prior is the canonical Ewens distribution restricted to blocks of size ≤ 2 .

Setup phase. The setup phase precomputes (a) the partial-Forward tensor $F^{\text{partial}}[i, j; k, l]$ giving the baseline TKF92 Pair HMM probability of any alignment passing through Match at both anchor cells (i, j) and (k, l) (the same $O(L^4)$ object enumerated by the F_2 inside–outside table of §E.1), and (b) the four-residue Potts boost tensor M at the observed amino acids over all anchor quadruples. Both are stored once per (X, Y) pair and reused across all sweeps and chains.

The three MCMC moves. The MCMC kernel composes three move types:

1. **Segment-resample move (MH).** Pick a contiguous fragment of the current alignment A bounded by two anchor Match cells $a_L < a_R$ (or by the alignment endpoints). Propose A^{new} by sampling the fragment from immediately after a_L to immediately before a_R by stochastic traceback on the segment-conditional Forward partition function $F_{\text{segment}}^{\text{partial}}[a_L; a_R]$ (Gibbs sample of the path conditional on the bounding anchors and the unchanged flanking alignment). Existing edges with endpoints in the resampled region are discarded; their orphaned partners outside the region become singletons in the partition. The proposal is exact under π_{TKF92} , so the alignment-likelihood factor of (E.9) cancels in the Hastings ratio, leaving only the CRP and boost factors.
2. **CRP add-edge move (MH).** Pick a Match cell of A that is currently a singleton in π_M . Propose a partner Match cell from a proposal distribution $q_{\text{add}}(\cdot \mid \text{singleton position})$ (the natural choice weights candidate partners by their boost magnitude $|M - 1|$). Insert the new edge into E . The Hastings ratio combines the CRP prior ratio $\alpha_z / (m - 1 + \alpha_z)$, the boost factor $M(e \mid t)$, and the reverse-proposal ratio (which is symmetric for a uniform remove-edge proposal of the same edge cardinality).
3. **CRP remove-edge move (MH).** Pick an existing edge in E uniformly. Remove it; both endpoints become singletons. The Hastings ratio is the reciprocal of the add-edge ratio.

Replica exchange. Parallel-tempering replica exchange over a ladder of α_z values (corresponding to varying coupling strengths) ensures global mixing across the multimodal posterior. Adjacent replicas swap states with the usual Metropolis acceptance ratio for the temperature-pair exchange.

Cost. The setup phase is $O(L^4)$ one-time per sequence pair. Each MCMC sweep does $O(1)$ inter-edge-anchor segment resamples; each segment resample is an $O(L_{\text{seg}})$ stochastic traceback through the cached partial-Forward tensor, giving an amortised $O(L)$ per-sweep cost. Aggregated per-cell match indicators across post-burn-in samples yield a stochastic estimate of the per-residue posterior Q'_{ij} that the FSA assembler consumes in place of the bounded-edge analytic estimate. The block move reduces to a pure Gibbs “segment between adjacent edge anchors” resample whenever a_L and a_R are adjacent in the current edge-anchor list.

Algorithmic detail

Algorithm 5 One MCMC sweep on the infinite Pair HMM

Require: Current state (A, E) , partial-Forward tensor F^{partial} , boost tensor M , concentration α_z

Ensure: New state (A', E')

- 1: Pick K_{seg} segment-resample positions (e.g. $K_{\text{seg}} = 8$, distributed by alignment length)
 - 2: **for** each segment $[a_L, a_R]$ **do**
 - 3: $A_{\text{seg}}^{\text{new}} \leftarrow$ stochastic traceback of $F_{\text{segment}}^{\text{partial}}[a_L; a_R]$
 - 4: Discard edges in E with at least one endpoint in $[a_L, a_R]$ (orphaned partners become singletons)
 - 5: $\rho \leftarrow$ MH ratio combining CRP and boost factors (alignment factor cancels by construction of the proposal)
 - 6: **if** $u \sim \text{Uniform}(0, 1) < \rho$ **then**
 - 7: $A \leftarrow A$ with $[a_L, a_R]$ replaced by $A_{\text{seg}}^{\text{new}}$
 - 8: $E \leftarrow E$ with orphaned edges removed
 - 9: **end if**
 - 10: **end for**
 - 11: Pick K_{add} add-edge candidates and K_{rem} remove-edge candidates (e.g. $K_{\text{add}} = 4|E|$, $K_{\text{rem}} = |E|$)
 - 12: **for** each candidate **do**
 - 13: Propose under the appropriate move kernel
 - 14: Compute MH ratio (CRP $\times$ boost for add; reciprocal for remove)
 - 15: Accept / reject
 - 16: **end for**
 - 17: Periodically: attempt replica-exchange swap with adjacent α_z rung
 - 18: **return** (A, E)
-

The exact form of the segment-resample MH ratio depends on the edge-handling convention. Under the simplest convention (“Strategy S-1” in the implementation spec), edges whose endpoints lie wholly within the resampled region are discarded unconditionally and their boost contributions enter only the forward MH ratio; edges with one endpoint outside the region are discarded and their orphaned partner becomes a singleton.

Verification protocol

The MCMC sampler is verified against four reference points:

- E.1 Cross-validate against `aug_phmm` (1-edge, large α_z).** At large α_z the infinite Pair HMM truncates to a 0-or-1-edge model, and the per-cell match-marginal Q'_{ij} produced by the MCMC sampler should agree with the exact $O(L^2 A^2)$ aug-PHMM 1-edge inside–outside.
- E.2 Cross-validate against `aug_phmm_2edge` (2-edge, moderate α_z).** At moderate α_z the truncation extends to 0-or-1-or-2-edge; agreement with the exact $O(L^2 A^4)$ aug-PHMM 2-edge inside–outside is the next consistency check.
- E.3 Cross-validate against brute-force enumeration.** On small sequence pairs $(L_x, L_y) \leq (5, 4)$ where all alignments and all ≤ 5 -edge placements can be enumerated directly, the MCMC sample mean of Q'_{ij} must converge to the brute-force value at the canonical $1/\sqrt{n}$ rate.
- E.4 Detailed-balance sanity check (no-data case).** At $H = 0$ (so $M \equiv 1$) the boost factor is identically one, the per-block factor reduces to the baseline emission, and the sampler should recover the no-edge TKF92 Pair HMM marginal exactly — with edge-set sample sizes distributed as the prior Ewens distribution.

The first three checks pin the sampler to the exact reference at each of three orders of edge truncation; the fourth checks detailed balance independently of the data-conditioned target.

E.4 The conceptual hierarchy: infinite phylogenetic SCFG

The infinite Pair HMM is itself an approximation. The infinite Pair HMM of equation E.7 is the principled limit of the bounded-edge dynamic-programming approximations of §E.2, but it is itself an approximation to a still-more-principled object: the infinite-edge generalisation of the gravestone-augmented time-indexed pair SCFG of §D.10 of Appendix D. That fuller model samples not only the visible alignment A and the edge set E but also the timed gravestone-augmented branch history G comprising every fragment that ever existed during the branch (whether visible at the alignment endpoints, deleted on the branch, or transiently inserted then deleted), with the same canonical CRP rule of equation E.7 governing edge spawn at every Match-class member of either an alive or a gravestone fragment. On the substitution side, transient gravestones inserted during the branch can be assigned the same z_s key-DP class as their alive class-mates and therefore contribute coupling terms to those class-mates’ rates during the gravestone’s lifetime — so the substitution likelihood at a Match cell (i, j) depends on G , not just on A . The infinite Pair HMM of this appendix marginalises G implicitly via the standard TKF92 Pair HMM substitution emissions, which is correct in expectation when transient lineages are rare or when their coupling contribution is small, but loses the within-branch dynamics that the gravestone-augmented SCFG captures.

The resulting hierarchy is nested: the bounded-edge dynamic-programming methods of §E.2 are exact in the limit of the infinite Pair HMM of this section; the infinite Pair HMM is exact in the limit of the infinite gravestone-augmented pair SCFG; and that fuller object is the natural fixed point at which the indel-side and the substitution-side principled formulations meet. The MCMC sampler described above samples from the middle level of this hierarchy; lifting it to the full infinite gravestone-augmented pair SCFG — by interleaving the infinite-HMM CRP edge moves with the gravestone-augmented branch-history moves of §D.10 of Appendix D — is a natural extension.

Further: the infinite gravestone-augmented *phylogenetic* SCFG. The pairwise hierarchy above lifts to the full multi-branch phylogenetic setting at no further modelling cost, by composition with the Felsenstein-style upwards–downwards machinery of §D.10 that already glues per-branch gravestone-augmented histories into a tree-level posterior. At every internal node and along every

branch of the tree, the gravestone-augmented latent state is sampled jointly with the alignment, while the same canonical CRP rule of equation E.7 governs edge spawn at every Match-class member of every fragment (alive or gravestone) at every node. The MCMC sampler above lifts by interleaving (i) per-branch gravestone-augmented history moves of §D.10, (ii) per-class-pair CRP edge moves of this appendix, and (iii) the per-internal-node Felsenstein-style updates already present in the SVI loop of §D.10. Each ingredient is already load-bearing in the existing pipeline; no new modelling assumptions are required to assemble them.

The full conceptual ladder is therefore:

mean-field pre-correction (deprecated; rem. D.20) $\rightarrow$ bounded-edge aug-PHMM (§E.2)
 $\rightarrow$ infinite Pair HMM (§E.3) $\rightarrow$ infinite gravestone-augmented pair SCFG (§E.3 + §D.10)
 $\rightarrow$ infinite gravestone-augmented *phylogenetic* SCFG (§E.4),

each rung exact in the limit of the next, the entire ladder falling out of the simple stick-breaking-TKF92-with-Potts construction of Appendices D–E with no further mathematical machinery.

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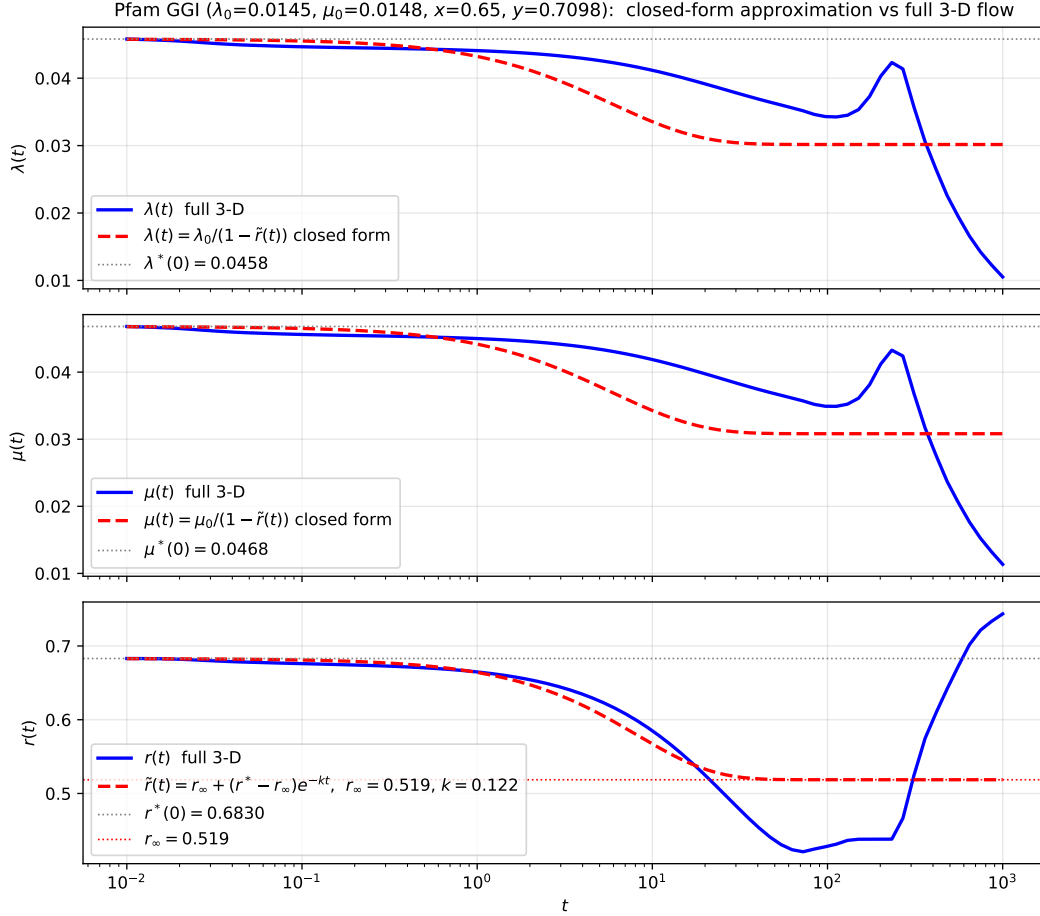


Figure A.1: Pfam-scale ($\lambda_0 \approx 0.0145$, $\mu_0 \approx 0.0148$, $x = 0.65$, $y = 0.71$) flow out to $T = 10^3$. Full three-parameter flow (A.79) (solid blue) vs the leading-order closed form (A.86) with the slaved rates $\lambda(T) = \lambda_0/(1 - r(T))$, $\mu(T) = \mu_0/(1 - r(T))$ (dashed red). Agreement is tight up to $T \sim 5$. The closed form asymptotes to $r_\infty = r^*(0)/(2 - r^*(0)) = 0.52$; the full flow dips below this near $T \sim 70$ and then rises back, a higher-order kink the linearised closed form does not capture. For typical phylogenies the relevant branch-length range ($T \lesssim 10$) lies well inside the regime where the closed form is accurate.

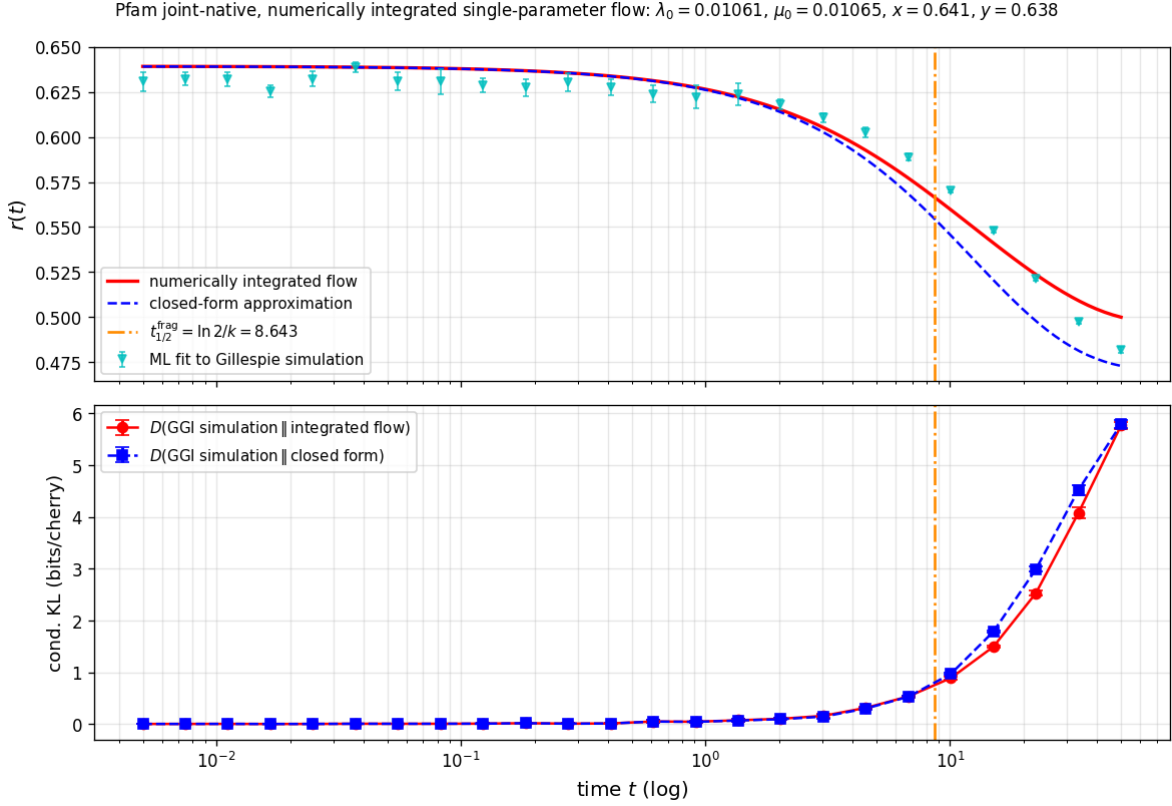


Figure A.2: Pfam joint-native ($\lambda_0 = 0.01061$, $\mu_0 = 0.01065$, $x = 0.641$, $y = 0.638$), $T \in [0.005, 50]$. Top panel: $r(T)$ trajectories under (i) the numerically-integrated single-parameter (fragment-length) flow (A.89) (red solid), (ii) the closed-form approximation (A.87) (blue dashed), and (iii) per- τ ML fit of TKF92 to the Gillespie data via the conditional WFST (cyan $\blacktriangledown$). The dark-orange dash-dotted vertical line marks the fragment coherence half-life $t_{1/2}^{\text{frag}}$ (A.88). Bottom panel: per- τ conditional KL $D(P_{\text{GGI}}(y \mid x) \parallel P_{\tau_T''}(y \mid x))$ per cherry, in bits, on linear y axis; the ancestor distribution is factored out analytically. The per- τ Gillespie cherry budget is adapted ($\sigma = 0.005$, $N_{\text{floor}} = 3000$; see surrounding text); error bars are over 5 independent simulations.